

Database FinOps

Cloud Cost Optimization for AWS RDS, Azure SQL, and Google Cloud

Rajesh Guntupalli

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By Rajesh Guntupalli

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Chapter 1: Where Your Cloud Database Bill Actually Comes From

Most teams migrating to cloud databases expect their bills to be lower than their on-premises costs, or at least predictable. Instead, they open the first full month’s invoice and find a number that is 40–60% higher than anyone budgeted. The engineering team points at the DBAs, the DBAs point at the architects, and the finance team points at everyone. This chapter stops the finger-pointing by mapping exactly where cloud database spend originates, how each cost component grows, and which levers move the needle fastest. The techniques in this chapter alone — primarily tagging, right-sizing, and storage class migration — routinely save teams between \$8,000 and \$40,000 per month on mid-sized deployments before any architectural changes are required.

1.1 The Anatomy of a Cloud Database Bill

Cloud database pricing is not a single line item. It is a stack of at least seven distinct charges, and most teams are only watching one or two of them. Understanding each layer is the prerequisite to cutting any of them.

Compute (instance hours) is usually the largest single charge, but it is also the most visible. An `db.r6g.2xlarge` on AWS RDS costs \$0.48 per hour on-demand, or \$4,204.80 annualized. The equivalent on Azure SQL (**Business Critical, 8 vCores**) runs approximately \$1.13 per vCore-hour. On GCP Cloud SQL, a `db-custom-8-32768` instance costs roughly \$0.4465 per hour for the VM component alone. Teams that provision for peak traffic and never revisit those sizes pay full rate around the clock.

Storage is the second-largest charge and the most underappreciated growth driver. On AWS RDS, `gp2` storage costs \$0.115 per GB-month. `gp3` costs \$0.092 per GB-month for the first 3,000 IOPS and 125 MB/s throughput, which are included. A 1 TB database on `gp2` costs \$117.76/month. The same workload on `gp3` costs \$94.21/month — a saving of \$23.55/month, or \$282.60/year, for a single database instance with zero performance change. Multiply that across 20 instances and you have saved \$5,652 annually by changing a storage type flag.

I/O charges apply to `gp2`, `magnetic`, and Aurora’s storage engine. Aurora charges \$0.20 per million I/O requests. A database performing 100 million I/Os per day generates \$600/month in I/O charges alone, on top of storage costs. Switching to Aurora I/O-Optimized pricing, which was introduced in 2023, flattens this into a higher per-GB storage cost but eliminates the per-request charge — saving money for any database generating more than roughly 1 billion I/Os per month.

Backup storage defaults to retaining 7 days of automated backups. AWS includes backup storage equal to 100% of your provisioned database storage at no charge. Anything beyond that costs \$0.095/GB-month. If your database is 2 TB and you have a 35-day retention window, you are paying for approximately 9 TB of backup storage above the free tier: \$855/month in backup costs alone.

Data transfer charges catch teams off guard when analytics pipelines, application servers, or cross-region replicas are added. Intra-region transfer from RDS to EC2 in the same Availability Zone is free. Moving data to a different AZ costs \$0.01/GB in each direction. Transferring data out to the internet costs \$0.09/GB (for the first 10 TB/month on AWS). A 500 GB monthly export to an external analytics platform costs \$45/month in transfer fees, before compute.

Licensing is embedded into standard RDS pricing for MySQL, PostgreSQL, and MariaDB. For SQL Server and Oracle, licensing is either included in the per-hour rate (at a significant premium) or managed through License Included vs. BYOL (Bring Your Own License) pricing. An `db.r6i.4xlarge` running SQL Server Enterprise Edition with License Included costs \$6.656/hour, or \$58,306.56/year. The same instance running BYOL SQL Server costs \$1.008/hour, or \$8,830.08/year — a saving of \$49,476.48/year if you have an existing SA agreement.

Multi-AZ and read replica charges double or multiply your compute costs. Every Multi-AZ instance provisions a synchronous standby in a second AZ, billed at the same rate as the primary. Teams that enable Multi-AZ on every instance, including development and test environments, are paying double for uptime guarantees that non-production workloads do not require.

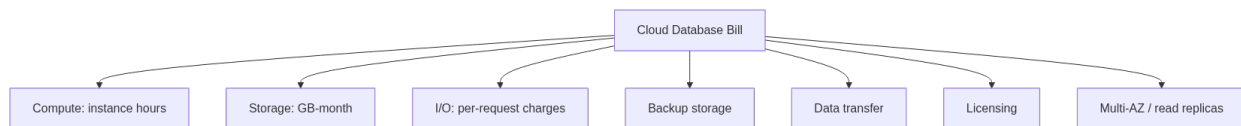


Figure 1: Seven distinct cost components make up a cloud database bill. Most teams monitor only compute and storage, missing the other five.

The following query, run against the AWS Cost and Usage Report data that has been exported to S3 and queried via Athena, gives you a breakdown of spend by these service components for any date range:

```

-- Run in Amazon Athena against your Cost and Usage Report (CUR) table
-- Replace 'your_cur_database.your_cur_table' with your actual table name
-- and adjust the date range as needed

SELECT
  line_item_product_code,
  product_volume_type,
  line_item_usage_type,
  SUM(line_item_unblended_cost) AS total_cost_usd,
  SUM(line_item_usage_amount) AS total_usage_units,
  COUNT(DISTINCT resource_tags_user_environment) AS environment_count
FROM your_cur_database.your_cur_table
WHERE
  line_item_product_code = 'AmazonRDS'
  AND line_item_line_item_type IN ('Usage', 'Tax')
  AND line_item_usage_start_date >= DATE '2024-01-01'
  AND line_item_usage_start_date < DATE '2024-04-01'
GROUP BY
  line_item_product_code,
  product_volume_type,
  line_item_usage_type

```

```
ORDER BY
  total_cost_usd DESC;
```

This query surfaces the split between `RDS:Multi-AZUsage`, `RDS:InstanceUsage`, `RDS:GP2-Storage`, `RDS:GP3-Storage`, `RDS:BackupUsage`, and `RDS:DataTransfer` — typically revealing that two or three line items you were not tracking account for 25–35% of your total spend.

1.2 Tagging: The Foundation of Cost Accountability

You cannot optimize what you cannot attribute. Before any cost reduction work is meaningful, every database resource must carry tags that map it to a business owner, environment, and application. Without tags, cost reports are noise. With tags, they become a governance tool.

AWS, Azure, and GCP all support resource tagging, but the tag schemas differ. The following schema works across all three platforms and integrates cleanly with most finance and ITSM systems:

Tag Key	Example Value	Purpose
environment	production, staging, dev	Separates prod costs from non-prod
application	payments-api, user-service	Maps cost to application
team	platform-engineering	Owner for chargeback
cost-center	CC-4821	Finance integration
tier	tier-1, tier-2	SLA tier for right-sizing policy
auto-shutdown	true	Enables scheduled stop/start automation

The following script enforces this tag schema across all RDS instances in an AWS account. It is designed to run on a schedule via AWS Lambda or a CI/CD pipeline, not as a one-time task. Instances missing required tags are flagged to a Slack channel for remediation.

```
#!/usr/bin/env bash
# enforce_rds_tags.sh
# Run via cron, AWS Lambda (using aws-cli layer), or CI/CD
# Required environment variables: SLACK_WEBHOOK_URL, AWS_REGION

set -euo pipefail

REQUIRED_TAGS=("environment" "application" "team" "cost-center")
REGION="${AWS_REGION:-us-east-1}"
SLACK_WEBHOOK_URL="${SLACK_WEBHOOK_URL:?SLACK_WEBHOOK_URL is required}"
NON_COMPLIANT=()

# Retrieve all RDS instance ARNs in the region
INSTANCE_ARNS=$(aws rds describe-db-instances \
  --region "$REGION" \
  --query 'DBInstances[*].DBInstanceArn' \
  --output text)

for ARN in $INSTANCE_ARNS; do
  # Fetch tags for this instance
  TAGS_JSON=$(aws rds list-tags-for-resource \
    --region "$REGION" \
    --resource-name "$ARN" \
    --query 'TagList' \
```

```

--output json)

# Extract tag keys into a space-separated list
EXISTING_KEYS=$(echo "$TAGS_JSON" | \
python3 -c "import sys,json; \
tags=json.load(sys.stdin); \
print(' '.join(t['Key'] for t in tags))")

MISSING=""
for TAG in "${REQUIRED_TAGS[@]}; do
    if ! echo "$EXISTING_KEYS" | grep -qw "$TAG"; then
        MISSING="$MISSING $TAG"
    fi
done

if [[ -n "$MISSING" ]]; then
    INSTANCE_ID=$(echo "$ARN" | awk -F: '{print $NF}')
    NON_COMPLIANT+="$INSTANCE_ID: missing[$MISSING]"
fi
done

# Report non-compliant instances to Slack
if [[ ${#NON_COMPLIANT[@]} -gt 0 ]]; then
    MESSAGE="*RDS Tag Compliance Alert* (${REGION})\n"
    MESSAGE+="${#NON_COMPLIANT[@]} instance(s) missing required tags:\n"
    for ITEM in "${NON_COMPLIANT[@]}; do
        MESSAGE+="$ITEM\n"
    done

    curl -s -X POST "$SLACK_WEBHOOK_URL" \
        -H 'Content-type: application/json' \
        --data "{\"text\": \"${MESSAGE}\"}"

    echo "Alert sent: ${#NON_COMPLIANT[@]} non-compliant instances"
else
    echo "All RDS instances are tag-compliant in ${REGION}"
fi

```

On Azure, the equivalent enforcement is done through Azure Policy. The following Azure CLI command creates a deny policy that prevents any Azure SQL instance from being deployed without the `environment` and `team` tags:

```

# Create a custom policy definition that denies SQL deployments without required tags
# Run once per subscription; policy evaluates on every new resource creation

az policy definition create \
    --name "require-sql-tags" \
    --display-name "Require environment and team tags on Azure SQL" \
    --description "Denies creation of Azure SQL servers and managed instances without required tags" \
    --mode "Indexed" \
    --rules '{
        "if": {
            "allOf": [
                {
                    "field": "type",
                    "in": [
                        "Microsoft.Sql/servers",
                        "Microsoft.Sql/managedInstances"
                    ]
                },
                {
                    "anyOf": [
                        {
                            "field": "tags[environment]",
                            "exists": false
                        },
                        {
                            "field": "tags[team]",
                            "exists": false
                        }
                    ]
                }
            ]
        }
    }'

```



```
    }
  ]
}
},
"then": {
  "effect": "deny"
}
}' \
--params '{}'
```

Assign the policy to your subscription
`SUBSCRIPTION_ID=$(az account show --query id -o tsv)`

```
az policy assignment create \
--name "require-sql-tags-assignment" \
--policy "require-sql-tags" \
--scope "/subscriptions/${SUBSCRIPTION_ID}"
```

Once tagging is enforced, AWS Cost Explorer can group costs by tag in the console, and the Athena CUR query shown in section 1.1 can be extended with `GROUP BY resource_tags_user_team` to produce per-team chargeback reports automatically.

1.3 Finding the Waste: Right-Sizing Analysis at Scale

The single most common source of overspend in cloud databases is over-provisioning. The engineering team asked for `db.r6g.4xlarge` for a service that peaked at 12% CPU utilization six months later. No one revisited the size. This is not negligence — it is the expected outcome of an environment where resizing requires a maintenance window and no one is watching the utilization data systematically.

AWS Compute Optimizer analyzes CloudWatch metrics over a 14-day window and produces right-sizing recommendations for RDS instances. It considers CPU, memory, storage, and network throughput. The following Python script pulls these recommendations programmatically and calculates the projected monthly savings for each instance:

```
#!/usr/bin/env python3
"""
rds_rightsizing_report.py

Pulls AWS Compute Optimizer recommendations for RDS instances,
calculates projected monthly savings, and writes a CSV report.

Schedule: Run weekly via EventBridge + Lambda or cron.
Requirements: boto3, csv (stdlib)
IAM permissions: compute-optimizer:GetRDSDatabaseRecommendations,
                  pricing:GetProducts
"""

import boto3
import csv
import json
import sys
from datetime import datetime

REGION = "us-east-1"
OUTPUT_FILE = f"rds_rightsizing_{datetime.now().strftime('%Y%m%d')}.csv"

def get_rds_recommendations(region: str) -> list:
    client = boto3.client("compute-optimizer", region_name=region)
    recommendations = []
```

```
paginator = client.get_paginator("get_rds_database_recommendations")

for page in paginator.paginate():
    recommendations.extend(page.get("rdsDBRecommendations", []))

return recommendations

def calculate_monthly_savings(current_price_hr: float, recommended_price_hr: float) -> float:
    """Assumes 730 hours per month (average)."""
    return (current_price_hr - recommended_price_hr) * 730

def parse_recommendations(recommendations: list) -> list:
    rows = []
    for rec in recommendations:
        instance_id = rec.get("resourceArn", "").split(":")[-1]
        finding = rec.get("finding", "")

        if finding not in ("OVERPROVISIONED", "UNDERPROVISIONED"):
            continue

        current = rec.get("currentDBInstanceClass", "N/A")
        options = rec.get("recommendationOptions", [])

        if not options:
            continue

        # Take the top-ranked recommendation
        best = options[0]
        recommended = best.get("dbInstanceClass", "N/A")

        # Savings currency info from Compute Optimizer
        savings_opportunity = best.get("savingsOpportunity", {})
        estimated_monthly_savings = savings_opportunity.get(
            "estimatedMonthlySavings", {}
        ).get("value", 0.0)

        estimated_savings_pct = savings_opportunity.get(
            "savingsOpportunityPercentage", 0.0
        )

        utilization = rec.get("utilizationMetrics", [])
        max_cpu = next(
            (m["value"] for m in utilization
             if m["name"] == "CPU" and m["statistic"] == "MAXIMUM"),
            None
        )
        avg_cpu = next(
            (m["value"] for m in utilization
             if m["name"] == "CPU" and m["statistic"] == "AVERAGE"),
            None
        )

        rows.append({
            "instance_id": instance_id,
            "finding": finding,
            "current_class": current,
            "recommended_class": recommended,
            "avg_cpu_pct": f"{avg_cpu:.1f}" if avg_cpu else "N/A",
            "max_cpu_pct": f"{max_cpu:.1f}" if max_cpu else "N/A",
            "estimated_monthly_savings_usd": f"{estimated_monthly_savings:.2f}",
            "savings_opportunity_pct": f"{estimated_savings_pct:.1f}",
        })

    return rows

def write_csv(rows: list, output_file: str):
    if not rows:
        print("No actionable recommendations found.")
        return
```

```

fieldnames = rows[0].keys()
with open(output_file, "w", newline="") as f:
    writer = csv.DictWriter(f, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(rows)

total_savings = sum(float(r["estimated_monthly_savings_usd"]) for r in rows)
print(f"Report written: {output_file}")
print(f"Instances with recommendations: {len(rows)}")
print(f"Total estimated monthly savings: ${total_savings:,.2f}")

if __name__ == "__main__":
    print(f"Fetching RDS recommendations for region: {REGION}")
    recommendations = get_rds_recommendations(REGION)
    print(f"Total recommendations retrieved: {len(recommendations)}")

    rows = parse_recommendations(recommendations)
    write_csv(rows, OUTPUT_FILE)

```

Running this script against a typical environment with 15–30 RDS instances reveals that 30–50% of instances are over-provisioned by at least one instance family size. The savings add up quickly. Moving from `db.r6g.4xlarge` (\$0.96/hr) to `db.r6g.2xlarge` (\$0.48/hr) for a single instance saves \$350.40/month. Across ten such instances, that is \$3,504/month, or \$42,048/year.

For Azure SQL, the equivalent data lives in Azure Advisor. The following Azure CLI command exports all cost recommendations for SQL databases to a JSON file that can be parsed in the same way:

```

#!/usr/bin/env bash
# azure_advisor_sql_recommendations.sh
# Exports Azure Advisor cost recommendations scoped to SQL resources
# Schedule: Weekly via Azure Automation or GitHub Actions

SUBSCRIPTION_ID=$(az account show --query id -o tsv)
OUTPUT_FILE="azure_sql_advisor_$(date +%Y%m%d).json"

echo "Fetching Azure Advisor recommendations for subscription: ${SUBSCRIPTION_ID}"

az advisor recommendation list \
  --subscription "$SUBSCRIPTION_ID" \
  --category "Cost" \
  --query "[?contains(resourceMetadata.resourceId, 'Microsoft.Sql') || \
    contains(resourceMetadata.resourceId, 'microsoft.sql')]" \
  --output json > "$OUTPUT_FILE"

# Summarize findings
TOTAL=$(python3 -c "
import json, sys
data = json.load(open('${OUTPUT_FILE}'))
total_impact = sum(
    r.get('extendedProperties', {}).get('savingsAmount', 0)
    for r in data
    if isinstance(r.get('extendedProperties', {}).get('savingsAmount', 0), (int, float))
)
print(f'Recommendations: {len(data)}, Estimated monthly savings: \${total_impact:,.2f}')
")

echo "$TOTAL"
echo "Full report written to: ${OUTPUT_FILE}"

```

On GCP, Cloud SQL right-sizing recommendations are available through the Recommender API. They can be queried with:

```

# GCP Cloud SQL right-sizing recommendations via Recommender API
# Requires: gcloud CLI, roles/recommender.viewer IAM role

PROJECT_ID=$(gcloud config get-value project)

```

```

gcloud recommender recommendations list \
  --project="$PROJECT_ID" \
  --location="us-central1" \
  --recommender="google.cloudsql.instance.IdleRecommender" \
  --format="table(
    name.basename():label=RECOMMENDATION_ID,
    stateInfo.state:label=STATE,
    primaryImpact.costProjection.cost.units:label=MONTHLY_SAVINGS_USD,
    content.overview.overview:label=SUMMARY
  )"

# Also check for over-provisioned instances
gcloud recommender recommendations list \
  --project="$PROJECT_ID" \
  --location="us-central1" \
  --recommender="google.cloudsql.instance.OverprovisionedRecommender" \
  --format="table(
    name.basename():label=RECOMMENDATION_ID,
    stateInfo.state:label=STATE,
    primaryImpact.costProjection.cost.units:label=MONTHLY_SAVINGS_USD,
    content.overview.overview:label=SUMMARY
  )"

```

1.4 Storage Costs: The Silent Accumulator

Storage is the cost that compounds invisibly. Databases grow, old tables are never cleaned up, and backup retention windows are set conservatively once and never revisited. The following PostgreSQL query identifies tables that have not been accessed in the past 90 days and calculates their size contribution — information that directly translates to storage cost savings when the data is archived or deleted.

```

-- Identify unused or rarely accessed tables for potential archival
-- Run against each PostgreSQL RDS / Cloud SQL / Azure Database for PostgreSQL instance
-- Requires: pg_stat_user_tables (populated after autovacuum has run at least once)

SELECT
  schemaname,
  relname
  pg_size_pretty(pg_total_relation_size(relid)) AS table_name,
  pg_total_relation_size(relid) AS total_size,
  pg_total_relation_size(relid) AS total_size_bytes,
  ROUND(
    pg_total_relation_size(relid)::numeric / 1073741824,
    2
  ) AS size_gb,
  n_live_tup AS live_rows,
  n_dead_tup AS dead_rows,
  last_vacuum,
  last_autovacuum,
  last_analyze,
  last_autoanalyze,
  seq_scan,
  idx_scan,
  COALESCE(last_seq_scan, last_idx_scan) AS last_any_scan,
  CASE
    WHEN COALESCE(last_seq_scan, last_idx_scan) IS NULL
    THEN 'NEVER SCANNED'
    WHEN COALESCE(last_seq_scan, last_idx_scan) < NOW() - INTERVAL '90 days'
    THEN 'IDLE > 90 DAYS'
    WHEN COALESCE(last_seq_scan, last_idx_scan) < NOW() - INTERVAL '30 days'
    THEN 'IDLE > 30 DAYS'
    ELSE 'ACTIVE'
  END AS access_status,
END

```

```

-- Estimated monthly storage cost at gp3 pricing ($0.092/GB-month)
ROUND(
    (pg_total_relation_size(relid)::numeric / 1073741824) * 0.092,
    4
)
AS est_monthly_cost_usd
FROM
    pg_stat_user_tables
WHERE
    pg_total_relation_size(relid) > 104857600 -- Tables larger than 100 MB only
ORDER BY
    total_size_bytes DESC
LIMIT 50;

```

Running this query on a production PostgreSQL instance is frequently surprising. Teams discover event log tables from deprecated features consuming 400 GB, audit tables that are never queried consuming 200 GB, and staging tables from ETL jobs that were never cleaned up consuming another 150 GB. At gp3 pricing on AWS, that 750 GB of unused data costs \$69/month. Archived to S3 Standard-IA, the same data costs \$6.25/month — a saving of \$62.75/month. Moved to S3 Glacier Instant Retrieval, it costs \$2.50/month.

For SQL Server on RDS or Azure SQL, the equivalent analysis runs against `sys.dm_db_index_usage_stats` and `sys.allocation_units`:

```

-- SQL Server: Identify large tables with no reads in the current uptime period
-- Run in the context of the target database

SELECT
    OBJECT_SCHEMA_NAME(t.object_id)
    AS schema_name,
    t.name
    AS table_name,
    p.rows
    AS row_count,
    CAST(
        SUM(a.total_pages) * 8.0 / 1024 / 1024
        AS DECIMAL(18, 2)
    )
    AS size_gb,
    ROUND(
        SUM(a.total_pages) * 8.0 / 1024 / 1024 * 0.092,
        4
    )
    AS est_monthly_cost_gp3_usd,
    SUM(ISNULL(ius.user_seeks, 0))
    AS total_seeks,
    SUM(ISNULL(ius.user_scans, 0))
    AS total_scans,
    SUM(ISNULL(ius.user_lookups, 0))
    AS total_lookups,
    SUM(ISNULL(ius.user_seeks, 0))
    + SUM(ISNULL(ius.user_scans, 0))
    + SUM(ISNULL(ius.user_lookups, 0))
    AS total_reads,
    MAX(ius.last_user_seek)
    AS last_user_seek,
    MAX(ius.last_user_scan)
    AS last_user_scan,
    CASE
        WHEN SUM(ISNULL(ius.user_seeks, 0))
        + SUM(ISNULL(ius.user_scans, 0))
        + SUM(ISNULL(ius.user_lookups, 0)) = 0
        THEN 'NO READS SINCE STARTUP'
        ELSE 'HAS READS'
    END
    AS read_status
FROM
    sys.tables t
    JOIN sys.partitions p ON p.object_id = t.object_id
                        AND p.index_id IN (0, 1)
    JOIN sys.allocation_units a ON a.container_id = p.partition_id
    LEFT JOIN sys.dm_db_index_usage_stats ius
        ON ius.object_id = t.object_id
        AND ius.database_id = DB_ID()
WHERE
    t.is_ms_shipped = 0
GROUP BY
    t.object_id,
    t.name,
    p.rows

```

```
HAVING
  SUM(a.total_pages) * 8.0 / 1024 / 1024 > 0.1  -- Tables larger than 100 MB
ORDER BY
  size_gb DESC;
```

One important caveat with the SQL Server version: `sys.dm_db_index_usage_stats` is reset every time the SQL Server service restarts, which on RDS happens during maintenance windows and failover events. A table with `total_reads = 0` is suspicious but not conclusive unless the instance has been running continuously for at least 30 days. Cross-reference the `last_user_seek` and `last_user_scan` columns with your maintenance window schedule before scheduling any archival.

Backup Retention Tuning

Backup retention is one of the fastest wins available. Most teams accept the default 7-day retention window or, worse, extend it to 35 days “just in case” without a documented recovery scenario that requires more than 7 days of history. The cost math is straightforward:

- A 500 GB database with 7-day retention and 5% daily change rate generates roughly 175 GB of incremental backup storage per week above the free-tier baseline on AWS (the first 100% of provisioned storage is free).
- The same database with 35-day retention generates approximately 875 GB above the free tier.
- At \$0.095/GB-month, the difference is \$66.50/month per database.

Across 15 databases, that is \$997.50/month saved simply by reducing retention from 35 days to 7 days — assuming your recovery point objective genuinely requires only 7 days of history.

The following AWS CLI command audits every RDS instance in a region and outputs any instance with a backup retention period greater than 14 days:

```
#!/usr/bin/env bash
# audit_backup_retention.sh
# Flags RDS instances with retention > threshold (default 14 days)
# Usage: ./audit_backup_retention.sh [retention_threshold_days]

REGION="${AWS_REGION:-us-east-1}"
THRESHOLD="${1:-14}"

echo "Auditing RDS backup retention in region: ${REGION}"
echo "Flagging instances with retention > ${THRESHOLD} days"
echo "----"

aws rds describe-db-instances \
  --region "$REGION" \
  --query "DBInstances[?BackupRetentionPeriod > `${THRESHOLD}`].[{
    ID:DBInstanceIdentifier,
    Class:DBInstanceClass,
    Engine:Engine,
    RetentionDays:BackupRetentionPeriod,
    StorageGB:AllocatedStorage,
    Environment:TagList[?Key=='environment'].Value | [0]
  }]" \
  --output table
```

Run this monthly. In organizations where developers can self-service RDS instances through Terraform modules, retention creep is common — a module default of 30 propagates across dozens of instances without anyone noticing the cumulative cost.

1.5 Reserved Instances, Savings Plans, and Committed Use Discounts

On-demand pricing is the highest rate you can pay for cloud database compute. Every major cloud provider offers a lower rate in exchange for a usage commitment, and for any database instance that has been running continuously for more than three months, a commitment discount is almost always justified.

AWS Reserved Instances (RIs) for RDS offer up to 72% savings versus on-demand pricing for a one-year, all-upfront commitment. The mechanics are:

Payment Option	1-Year Discount	3-Year Discount
All Upfront	~42%	~62%
Partial Upfront	~38%	~58%
No Upfront	~35%	~52%

A `db.r6g.2xlarge` Multi-AZ costs \$1.344/hr on-demand, or \$11,773.44/year. The same instance reserved for one year all-upfront costs approximately \$6,861 — a saving of \$4,912.44 per instance per year. Reserved Instances apply at the instance-family level: a `db.r6g.2xlarge` RI also applies to two `db.r6g.xlarge` instances or one `db.r6g.2xlarge`, giving you flexibility when right-sizing.

The key risk with RIs is committing to an instance family that you subsequently right-size away from. If you purchase a `db.r6g.4xlarge` RI and later downsize the instance to `db.r6g.2xlarge`, the RI converts to cover two `db.r6g.2xlarge` instance-equivalents automatically — but only if you stay within the same instance family and region. Crossing to a different family or region means the RI is idle and you are paying for it without benefit.

The correct process is:

1. Run the right-sizing analysis from section 1.3 first.
2. Apply right-sizing changes.
3. Observe the correctly-sized instance under real workload for at least 30 days.
4. Purchase RIs only for instances that have been stable at their current size for 30+ days.

Azure Reserved Capacity for Azure SQL Database and SQL Managed Instance works similarly, with 1-year and 3-year commitment tiers. Savings versus pay-as-you-go pricing are in the same range as AWS. Azure reservations are scoped to a subscription or a management group, and they apply to the vCore count — so a reservation for 8 vCores applies to one 8-vCore instance or two 4-vCore instances in the same subscription.

GCP Committed Use Discounts (CUDs) for Cloud SQL function differently. Rather than purchasing a specific instance reservation, you commit to a minimum spend on Cloud SQL resources in a region for one or three years. GCP then applies a discount to all Cloud SQL usage in that region up to the committed amount. This is administratively simpler than AWS RIs but requires you to estimate future spend rather than future instance count.

The following Python script calculates the break-even point for purchasing an AWS RDS Reserved Instance versus continuing on-demand, accounting for the probability that the instance is terminated early:

```
#!/usr/bin/env python3
"""
ri_breakeven.py

Calculates the break-even month for an RDS Reserved Instance purchase
versus continuing on-demand pricing.

Inputs (edit the CONFIG dict below or pass as environment variables):
- on_demand_hourly: On-demand price per hour (USD)
- ri_upfront:       All-upfront RI cost for 1 year (USD)
- ri_monthly:       Monthly charge for RI (0 for all-upfront)
- hours_per_month: Assumed hours of operation per month (default 730)
"""

CONFIG = {
    "on_demand_hourly": 1.344,    # db.r6g.2xlarge Multi-AZ, us-east-1
    "ri_upfront":       6861.00,  # 1-year all-upfront RI cost
    "ri_monthly":       0.00,     # 0 for all-upfront
    "hours_per_month": 730,
}

def calculate_breakeven(cfg: dict) -> None:
    on_demand_monthly = cfg["on_demand_hourly"] * cfg["hours_per_month"]
    ri_first_month    = cfg["ri_upfront"] + cfg["ri_monthly"]
    ri_subsequent     = cfg["ri_monthly"]

    print(f"On-demand monthly cost:      ${on_demand_monthly:>10,.2f}")
    print(f"RI upfront cost:                ${cfg['ri_upfront']:>10,.2f}")
    print(f"RI monthly charge:                ${cfg['ri_monthly']:>10,.2f}")
    print()
    print(f"{'Month':<8} {'On-Demand Cumul':>16} {'RI Cumul':>12} {'Delta':>12} {'Status'}")
    print("-" * 60)

    cumul_on_demand = 0.0
    cumul_ri         = cfg["ri_upfront"]    # paid at month 0
    breakeven_month = None

    for month in range(1, 13):
        cumul_on_demand += on_demand_monthly
        cumul_ri         += cfg["ri_monthly"]
        delta             = cumul_on_demand - cumul_ri

        status = ""
        if delta > 0 and breakeven_month is None:
            breakeven_month = month
            status = "<-- BREAK-EVEN"

        print(
            f"{'month':<8} "
            f"${cumul_on_demand:>14,.2f} "
            f"${cumul_ri:>10,.2f} "
            f"${delta:>10,.2f} "
            f"{'status'}"
        )

    annual_savings = cumul_on_demand - cumul_ri
    print()
    if breakeven_month:
        print(f"Break-even month: {breakeven_month}")
    print(f"Annual savings:    ${annual_savings:>10,.2f}")
    print(
        f"Savings %:      "
        f"${(annual_savings / cumul_on_demand * 100):.1f}%"
    )

if __name__ == "__main__":
    calculate_breakeven(CONFIG)
```

Sample output for the db.r6g.2xlarge Multi-AZ configuration:

On-demand monthly cost: \$ 981.12
RI upfront cost: \$ 6,861.00
RI monthly charge: \$ 0.00

Month	On-Demand Cumul	RI Cumul	Delta	Status

1	\$981.12	\$6,861.00	\$-5,879.88	
2	\$1,962.24	\$6,861.00	\$-4,898.76	
3	\$2,943.36	\$6,861.00	\$-3,917.64	
4	\$3,924.48	\$6,861.00	\$-2,936.52	
5	\$4,905.60	\$6,861.00	\$-1,955.40	
6	\$5,886.72	\$6,861.00	\$-974.28	
7	\$6,867.84	\$6,861.00	\$+6.84	<-- BREAK-EVEN
8	\$7,848.96	\$6,861.00	\$+987.96	
9	\$8,830.08	\$6,861.00	\$+1,969.08	
10	\$9,811.20	\$6,861.00	\$+2,950.20	
11	\$10,792.32	\$6,861.00	\$+3,931.32	
12	\$11,773.44	\$6,861.00	\$+4,912.44	

Break-even month: 7
Annual savings: \$4,912.44
Savings %: 41.7%

The break-even at month 7 means the RI is beneficial as long as the instance runs for at least 7 of the 12 months. For a production database with a reasonable uptime expectation, this is a straightforward decision.

1.6 Bringing It Together: A Monthly FinOps Cadence

Individual optimizations — tagging, right-sizing, storage cleanup, RI purchases — are valuable once. They become a sustainable practice only when embedded in a repeatable review cycle. The following monthly cadence covers the four highest-leverage activities and can be run by a single engineer in approximately two hours per month once the scripts are in place.

Week 1 of the month: Spend attribution review

Run the Athena CUR query from section 1.1 against the previous month's data. Group by `resource_tags_user_team` and `resource_tags_user_environment`. Identify any team whose database costs increased by more than 15% month-over-month. Send that team a cost report with the specific line items driving the increase. Do not escalate; inform. Most engineers will self-correct when they can see the specific numbers.

Week 2: Right-sizing sweep

Run the Compute Optimizer script from section 1.3. Filter for instances that have been over-provisioned for two consecutive months — these are safe candidates for resizing because the pattern is established, not a temporary dip in load. Schedule the resize during the next maintenance window. On RDS, a resize requires a brief outage of a few minutes for Single-AZ instances, or an automatic

failover for Multi-AZ instances, which is near-zero downtime.

Week 3: Storage and backup audit

Run the PostgreSQL or SQL Server table size queries from section 1.4 on any database that grew by more than 50 GB in the past month. Identify the tables driving the growth. Check whether backup retention is still aligned with each database's recovery requirements. For any non-production database where the retention is set above 7 days without a documented reason, reduce it.

Week 4: Commitment coverage check

Calculate your RI and CUD coverage ratio: the percentage of your on-demand database compute that is covered by a commitment discount. A healthy target for mature environments is 70–80% coverage, leaving 20–30% on-demand to absorb unpredictable spikes. If coverage is below 60%, identify the longest-running on-demand instances and purchase RIs for the ones that passed the 30-day stability check.

This cadence, executed consistently, compounds over time. Teams that run it for six consecutive months typically find that their month-12 database bill is 30–45% lower than month 1, without any architectural changes — purely through the continuous application of the techniques in this chapter.

Key Takeaways

- **Cloud database bills are a stack of at least seven distinct cost components** — compute, storage, I/O, backup, data transfer, licensing, and replication — and most teams are only monitoring one or two of them. Visibility across all components is the prerequisite to any meaningful reduction.
- **Tagging is not an administrative task; it is a cost control mechanism.** Enforcing a consistent tag schema across all database resources with automated compliance checks converts cost reports from noise into actionable chargeback data, and it takes less than a day to implement with the scripts in this chapter.
- **Over-provisioning is the default outcome of cloud migrations**, not an exception. Right-sizing analysis run against real utilization data — using AWS Compute Optimizer, Azure Advisor, or GCP Recommender — routinely surfaces 30–50% of instances as candidates for downsizing, with no performance impact on the workloads involved.
- **Storage costs compound silently.** Unused tables, over-retained backups, and the wrong storage class type are each individually small but collectively significant. A single query against `pg_stat_user_tables` or `sys.dm_db_index_usage_stats` run monthly surfaces archival candidates that can be moved to object storage at a fraction of block storage pricing.
- **Reserved Instances and Committed Use Discounts should follow right-sizing, not precede it.** Committing to the wrong instance size locks in inefficiency for one to three years. The correct sequence is: right-size first, observe for 30 days, then purchase commitments for stable instances — at which point the discounts compound the savings already achieved through right-sizing.

Chapter 2: AWS RDS Cost Model: Compute, Storage, I/O, and Transfer

Organizations running AWS RDS at scale routinely overpay by 40–60% — not because the pricing is opaque, but because most teams accept the default configuration and never revisit it. A single db.r6g.2xlarge running PostgreSQL with gp2 storage, Multi-AZ enabled, and no Reserved Instance commitment can cost over \$1,800/month. That same workload, rightsized to a db.r6g.xlarge on gp3 storage with a one-year Reserved Instance, costs closer to \$680/month. That is \$1,120/month saved on a single instance — before touching I/O fees, snapshot retention, or data transfer charges. This chapter dismantles the AWS RDS cost model component by component, shows you exactly where the money goes, and gives you the queries, scripts, and automation to recover it systematically.

2.1 Understanding the RDS Billing Anatomy

AWS bills RDS across five distinct cost dimensions. Most teams only see the monthly total and treat it as a single line item. That is a mistake. To optimize, you need to decompose the bill into its components and assign each one a remediation strategy.

The five RDS cost dimensions:

Dimension	Typical Share of Bill	Primary Lever
Compute (instance hours)	50–65%	Reserved Instances, rightsizing, stop/start
Storage (EBS volumes)	15–25%	gp2 → gp3 migration, size right-sizing
I/O requests	5–20%	gp3 eliminates per-I/O charges entirely
Data transfer	3–10%	Same-AZ reads, VPC endpoints, caching
Snapshots and backups	2–8%	Retention period tuning, export cleanup

The critical insight here is that gp2 storage charges per I/O request beyond the baseline throughput, which scales at 3 IOPS per GB. A 500 GB gp2 volume provides 1,500 baseline IOPS. Any database workload that regularly exceeds this baseline incurs per-request charges that accumulate invisibly.

A busy OLTP workload on a 500 GB gp2 volume can easily pay \$200–\$400/month in I/O alone. gp3 provides 3,000 IOPS and 125 MB/s throughput at no per-I/O charge, and costs \$0.08/GB/month compared to gp2's \$0.115/GB/month. The math on a 1 TB volume is straightforward:

- **gp2, 1 TB:** $\$0.115 \times 1,024 \text{ GB} = \text{\$117.76/month}$ (plus I/O charges)
- **gp3, 1 TB:** $\$0.08 \times 1,024 \text{ GB} = \text{\$81.92/month}$ (no I/O charges)
- **Storage savings:** \$35.84/month, approximately 30%
- **I/O savings:** varies, but commonly \$50–\$200/month on active workloads

That is the storage dimension alone. Compute is where the larger numbers live.

Retrieving your current RDS cost breakdown using AWS Cost Explorer:

```
#!/bin/bash
# Pull RDS cost breakdown by usage type for the last 30 days
# Requires: AWS CLI configured with Cost Explorer permissions

START_DATE=$(date -d "30 days ago" +%Y-%m-%d 2>/dev/null || \
    date -v-30d +%Y-%m-%d)
END_DATE=$(date +%Y-%m-%d)

echo "=== RDS Cost Breakdown by Usage Type (Last 30 Days) ==="

aws ce get-cost-and-usage \
    --time-period Start=${START_DATE},End=${END_DATE} \
    --granularity MONTHLY \
    --filter '{
        "Dimensions": {
            "Key": "SERVICE",
            "Values": ["Amazon Relational Database Service"]
        }
    }' \
    --group-by Type=DIMENSION,Key=USAGE_TYPE \
    --metrics "UnblendedCost" "UsageQuantity" \
    --output json | \
    python3 -c "
import json, sys

data = json.load(sys.stdin)
results = data['ResultsByTime'][0]['Groups']

# Parse and sort by cost descending
costs = []
for group in results:
    usage_type = group['Keys'][0]
    cost = float(group['Metrics']['UnblendedCost']['Amount'])
    quantity = float(group['Metrics']['UsageQuantity']['Amount'])
    if cost > 0.01:
        costs.append((cost, usage_type, quantity))

costs.sort(reverse=True)

print(f'{"Usage Type":<55} {"Cost (USD)":>12} {"Quantity":>12}')
print('-' * 80)
for cost, usage_type, quantity in costs:
    short_type = usage_type[-50:] if len(usage_type) > 50 else usage_type
    print(f'{short_type:<55} \${cost:>10.2f} {quantity:>10.1f}')

total = sum(c[0] for c in costs)
print('-' * 80)
print(f'{"TOTAL RDS SPEND":<55} \${total:>10.2f}')
"
```

```
echo ""
echo "=== RDS Instances with No Cost Allocation Tags ==="
aws rds describe-db-instances \
```

```
--query
↪ 'DBInstances[*].{ID:DBInstanceIdentifier,Class:DBInstanceClass,Engine:Engine,Status:DBInstanceStatus}'
↪ \
--output table
```

Run this script every Monday morning as part of a cost review cadence. The output tells you immediately whether I/O charges are appearing as `RDS:StorageIOUsage` line items — a clear signal that affected instances should move to `gp3`.

2.2 Compute Cost Analysis: Finding the Right Instance

Compute is the largest single cost driver in RDS. The instance family, size, and purchase option together determine whether you are paying market rate or overpaying substantially. AWS offers three purchase options for RDS compute:

- **On-Demand:** Full price, no commitment, billed per second (minimum 10 minutes)
- **Reserved Instances (RI):** 1-year or 3-year commitment, 30–60% discount
- **Savings Plans:** More flexible than RIs, applies across instance families

A `db.r6g.2xlarge` running PostgreSQL on-demand costs approximately \$0.48/hour in `us-east-1`, or \$350.40/month. The same instance on a one-year Reserved Instance (no upfront) costs \$0.296/hour, or \$216.08/month — a **38% reduction** with zero architectural changes.

The instance family choice matters as much as the size. The `r6g` (Graviton2) family provides 10–20% better price-performance than the equivalent `r5` instances, and up to 35% cost reduction for equivalent workloads. Teams still running `r5` or `r4` instances have low-hanging savings available with a simple instance class modification.

Query to identify compute sizing opportunities using CloudWatch metrics:

```
#!/usr/bin/env python3
"""
rds_compute_rightsizing.py

Analyzes CPU and memory utilization across all RDS instances over
the last 14 days. Flags instances with consistently low CPU for
downsizing and instances at saturation for upsizing.

Schedule: Weekly via EventBridge + Lambda or cron
"""

import boto3
from datetime import datetime, timedelta
from collections import defaultdict

def get_cloudwatch_metric_stats(cw_client, instance_id, metric_name, days=14):
    """Retrieve average, max, and p95 for a given CloudWatch RDS metric."""
    end_time = datetime.utcnow()
    start_time = end_time - timedelta(days=days)

    response = cw_client.get_metric_statistics(
        Namespace='AWS/RDS',
        MetricName=metric_name,
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
        StartTime=start_time,
        EndTime=end_time,
        Period=3600, # 1-hour granularity
        Statistics=['Average', 'Maximum'],
        ExtendedStatistics=['p95']
```

```

)

datapoints = response.get('Datapoints', [])
if not datapoints:
    return None

averages = [dp['Average'] for dp in datapoints]
maximums = [dp['Maximum'] for dp in datapoints]
p95_values = [dp['ExtendedStatistics'].get('p95', 0) for dp in datapoints]

return {
    'avg': sum(averages) / len(averages),
    'max': max(maximums),
    'p95': sum(p95_values) / len(p95_values) if p95_values else 0,
    'datapoint_count': len(datapoints)
}

def estimate_monthly_cost(instance_class, engine, region='us-east-1'):
    """
    Approximate on-demand hourly rates for common instance types.
    In production, replace with AWS Price List API calls.
    """
    # Approximate on-demand rates (us-east-1, PostgreSQL/MySQL, Feb 2025)
    rates = {
        'db.t3.micro': 0.017,
        'db.t3.small': 0.034,
        'db.t3.medium': 0.068,
        'db.t3.large': 0.136,
        'db.t3.xlarge': 0.272,
        'db.t3.2xlarge': 0.544,
        'db.r5.large': 0.240,
        'db.r5.xlarge': 0.480,
        'db.r5.2xlarge': 0.960,
        'db.r5.4xlarge': 1.920,
        'db.r6g.large': 0.208,
        'db.r6g.xlarge': 0.416,
        'db.r6g.2xlarge': 0.480,
        'db.r6g.4xlarge': 0.960,
        'db.r6g.8xlarge': 1.920,
        'db.m5.large': 0.190,
        'db.m5.xlarge': 0.380,
        'db.m5.2xlarge': 0.760,
        'db.m6g.large': 0.171,
        'db.m6g.xlarge': 0.342,
        'db.m6g.2xlarge': 0.684,
    }
    hourly = rates.get(instance_class, 0)
    return hourly * 730 # approximate monthly hours

def analyze_rds_rightsizing():
    rds_client = boto3.client('rds')
    cw_client = boto3.client('cloudwatch')

    instances = rds_client.describe_db_instances()['DBInstances']

    print(f"{'Instance ID':<35} {'Class':<20} {'CPU Avg':>8} {'CPU P95':>8} "
          f"{'CPU Max':>8} {'Est. Monthly':>13} {'Recommendation'}")
    print("-" * 120)

    total_potential_savings = 0.0

    for inst in instances:
        if inst['DBInstanceStatus'] != 'available':
            continue

        instance_id = inst['DBInstanceIdentifier']
        instance_class = inst['DBInstanceClass']
        engine = inst['Engine']

        cpu_stats = get_cloudwatch_metric_stats(cw_client, instance_id, 'CPUUtilization')

```

```

if not cpu_stats or cpu_stats['datapoint_count'] < 50:
    recommendation = "INSUFFICIENT DATA"
    monthly_cost = estimate_monthly_cost(instance_class, engine)
    savings = 0.0
else:
    monthly_cost = estimate_monthly_cost(instance_class, engine)
    cpu_avg = cpu_stats['avg']
    cpu_p95 = cpu_stats['p95']
    cpu_max = cpu_stats['max']

    if cpu_p95 < 20 and cpu_max < 50:
        recommendation = "DOWNSIZE CANDIDATE"
        savings = monthly_cost * 0.40 # approximate 40% savings one size down
        total_potential_savings += savings
    elif cpu_p95 < 40 and cpu_max < 70:
        recommendation = "RESERVE (RI opportunity)"
        savings = monthly_cost * 0.38
        total_potential_savings += savings
    elif cpu_avg > 75:
        recommendation = "UPSIZE or OPTIMIZE QUERIES"
        savings = 0.0
    else:
        recommendation = "OK"
        savings = 0.0

    print(f"{instance_id:<35} {instance_class:<20} "
          f"{cpu_avg:>7.1f}% {cpu_p95:>7.1f}% {cpu_max:>7.1f}% "
          f"${monthly_cost:>11.2f} {recommendation}"
          + (f" (save ~${savings:.0f}/mo)" if savings > 0 else ""))
    continue

    print(f"{instance_id:<35} {instance_class:<20} "
          f"{N/A:>8} {N/A:>8} {N/A:>8} "
          f"${monthly_cost:>11.2f} {recommendation}")

print("-" * 120)
print(f"Total estimated monthly savings opportunity: ${total_potential_savings:,.2f}")
print(f"Annual savings opportunity: ${total_potential_savings * 12:,.2f}")

if __name__ == '__main__':
    analyze_rds_rightsizing()

```

Run this weekly. Any instance showing DOWNSIZE CANDIDATE where P95 CPU stays below 20% for two consecutive weeks is a legitimate rightsizing target. For production databases, validate against peak traffic periods — holiday seasons, month-end close, batch jobs — before committing to a smaller instance class.

The Reserved Instance calculation for a CTO-level decision:

For a db.r6g.2xlarge running continuously in production:

Purchase Option	Hourly Rate	Monthly Cost	Annual Cost	Savings vs On-Demand
On-Demand	\$0.480	\$350.40	\$4,204.80	—
1-Year RI, No Upfront	\$0.296	\$216.08	\$2,592.96	38%
1-Year RI, All Upfront	\$0.273	\$199.29	\$2,391.48	43%
3-Year RI, All Upfront	\$0.194	\$141.62	\$1,699.44	60%

For a fleet of 10 such instances, the difference between on-demand and one-year reserved is **\$16,118/year**. That is a conversation every CTO should be having with their cloud team quarterly.

2.3 Storage Cost Deep Dive: gp2 to gp3 Migration

The storage billing story in RDS changed significantly when AWS introduced gp3 in 2022. Despite gp3 being strictly superior for most workloads — cheaper per GB, higher baseline IOPS, no per-I/O charges, and independently scalable IOPS and throughput — a large fraction of RDS instances in production today still run on gp2. This is a pure cost problem with a well-defined solution.

gp2 vs gp3 economics (us-east-1, February 2025):

Storage Type	Price/GB/Month	Baseline IOPS	Max IOPS	Throughput	I/O Charges
gp2	\$0.115	3 × GB (min 100)	16,000	250 MB/s	Yes, above baseline
gp3	\$0.080	3,000 (fixed)	16,000 (+\$0.02/IOPS)	125 MB/s (+\$0.04/MB/s)	Never
io1	\$0.125	Provisioned	64,000	4,000 MB/s	Never
io2	\$0.125	Provisioned	256,000	4,000 MB/s	Never

For the vast majority of RDS workloads, gp3 at base configuration (3,000 IOPS, 125 MB/s) is sufficient. For high-IOPS workloads that were previously on gp2 with large volumes to get more IOPS, the math is even more compelling: you can right-size the storage volume down and provision exactly the IOPS needed on gp3.

Example: A team running a 2 TB gp2 volume to get 6,000 IOPS:

- gp2 cost: 2,048 GB × \$0.115 = **\$235.52/month** (plus I/O charges)
- gp3 equivalent: 500 GB × \$0.080 + (6,000 - 3,000) IOPS × \$0.02 = \$40 + \$60 = **\$100/month**
- **Monthly savings: \$135.52 + eliminated I/O charges**

The following query identifies all RDS instances still on gp2 and calculates the projected savings:

```
-- Run this against an Amazon Athena table built from AWS Cost and Usage Reports,
-- or adapt for your cost data warehouse.
-- This example uses the standard CUR schema.
```

```
SELECT
    line_item_resource_id,
    resource_tags_user_environment,
    resource_tags_user_team,
    product_volume_type,
    SUM(line_item_usage_amount) AS total_gb_hours,
    SUM(line_item_unblended_cost) AS total_storage_cost,
    -- Estimate gp3 equivalent cost
    ROUND(
        (SUM(line_item_usage_amount) / 730.0)
        * 0.080, 2
    ) AS gp3_estimated_cost,
    -- Savings potential
    ROUND(
        SUM(line_item_unblended_cost) - AS gp3_estimated_cost,
```



```

SUM(line_item_unblended_cost)
- (SUM(line_item_usage_amount) / 730.0 * 0.080),
2
)
AS estimated_monthly_savings
FROM
cost_and_usage_report
WHERE
line_item_service_code = 'AmazonRDS'
AND product_volume_type = 'General Purpose' -- this is gp2 in CUR
AND line_item_usage_type LIKE '%StorageUsage%'
AND year = '2025'
AND month = '2'
GROUP BY
line_item_resource_id,
resource_tags_user_environment,
resource_tags_user_team,
product_volume_type
HAVING
SUM(line_item_unblended_cost) > 10.00 -- ignore trivially small volumes
ORDER BY
estimated_monthly_savings DESC;

```

To execute the gp2 → gp3 migration itself, AWS allows you to modify storage type without downtime for most instance classes. The modification applies immediately and the volume conversion happens in the background:

```

#!/bin/bash
# migrate_gp2_to_gp3.sh
# Migrates a single RDS instance from gp2 to gp3 storage.
# Run during low-traffic periods. No downtime required for most engines.
# Validate: check Performance Insights for I/O latency before and after.

INSTANCE_ID="${1:-}"
DRY_RUN="${2:-true}" # Set to "false" to execute

if [[ -z "$INSTANCE_ID" ]]; then
    echo "Usage: $0 <instance-id> [true|false]"
    echo "  Second argument: dry-run mode (default: true)"
    exit 1
fi

echo "=== Checking current storage configuration ==="
CURRENT_CONFIG=$(aws rds describe-db-instances \
    --db-instance-identifier "$INSTANCE_ID" \
    --query 'DBInstances[0].{
        StorageType:StorageType,
        AllocatedStorage:AllocatedStorage,
        Iops:Iops,
        StorageThroughput:StorageThroughput,
        MultiAZ:MultiAZ,
        InstanceClass:DBInstanceClass
    }' \
    --output json)

echo "$CURRENT_CONFIG" | python3 -c "
import json, sys
config = json.load(sys.stdin)
storage_type = config.get('StorageType', 'unknown')
allocated_gb = config.get('AllocatedStorage', 0)
iops = config.get('Iops', 'N/A')
multi_az = config.get('MultiAZ', False)
instance_class = config.get('InstanceClass', 'unknown')

print(f'Instance class: {instance_class}')
print(f'Storage type: {storage_type}')
print(f'Allocated storage: {allocated_gb} GB')
print(f'Current IOPS: {iops}')
print(f'Multi-AZ: {multi_az}')

```

```

if storage_type == 'gp3':
    print('Already on gp3. No migration needed.')
    sys.exit(0)

# Cost calculations
gp2_monthly = allocated_gb * 0.115
gp3_monthly = allocated_gb * 0.080
savings = gp2_monthly - gp3_monthly
multiplier = 2 if multi_az else 1 # Multi-AZ doubles storage cost

print()
print('=== Cost Impact (excluding I/O charges) ===')
print(f'gp2 monthly cost: \${gp2_monthly * multiplier:.2f}')
print(f'gp3 monthly cost: \${gp3_monthly * multiplier:.2f}')
print(f'Monthly savings: \${savings * multiplier:.2f}')
print(f'Annual savings: \${savings * multiplier * 12:.2f}')
print(f'(Multi-AZ multiplier of {multiplier} applied)')
"

STORAGE_TYPE=$(aws rds describe-db-instances \
--db-instance-identifier "$INSTANCE_ID" \
--query 'DBInstances[0].StorageType' \
--output text)

if [[ "$STORAGE_TYPE" == "gp3" ]]; then
    echo "Instance is already on gp3. Exiting."
    exit 0
fi

if [[ "$DRY_RUN" == "true" ]]; then
    echo ""
    echo "DRY RUN: Would execute the following modification:"
    echo "aws rds modify-db-instance \"
    echo "    --db-instance-identifier $INSTANCE_ID \"
    echo "    --storage-type gp3 \"
    echo "    --apply-immediately"
    echo ""
    echo "Set second argument to 'false' to execute."
else
    echo ""
    echo "=== Executing gp2 to gp3 migration ==="
    aws rds modify-db-instance \
        --db-instance-identifier "$INSTANCE_ID" \
        --storage-type gp3 \
        --apply-immediately

    echo "Migration initiated. Monitor status with:"
    echo " aws rds describe-db-instances \"
    echo "    --db-instance-identifier $INSTANCE_ID \"
    echo "    --query 'DBInstances[0].PendingModifiedValues'"
fi

```

For a fleet migration, run this script in a loop across all instances identified by the Athena query, validating each before proceeding to the next:

```

#!/bin/bash
# fleet_gp2_to_gp3_migration.sh
# Iterates all gp2 RDS instances and migrates them to gp3.
# Pauses between each instance to allow background conversion to stabilize.

SLEEP_BETWEEN=30 # seconds to wait between instance modifications
DRY_RUN="${1:-true}"

echo "=== Fleet gp2 → gp3 Migration ==="
echo "Dry run: $DRY_RUN"
echo ""

# Retrieve all available instances on gp2
GP2_INSTANCES=$(aws rds describe-db-instances \

```

```
--query 'DBInstances[?StorageType==`gp2` && DBInstanceStatus==`available`].DBInstanceIdentifier' \
--output text)

if [[ -z "$GP2_INSTANCES" ]]; then
    echo "No gp2 instances found. Nothing to do."
    exit 0
fi

COUNT=$(echo "$GP2_INSTANCES" | wc -w | tr -d ' ')
echo "Found $COUNT instance(s) on gp2:"
echo "$GP2_INSTANCES" | tr '\t' '\n' | sed 's/^/ - /'
echo ""

for INSTANCE_ID in $GP2_INSTANCES; do
    echo "--- Processing: $INSTANCE_ID ---"
    bash migrate_gp2_to_gp3.sh "$INSTANCE_ID" "$DRY_RUN"
    echo ""
    if [[ "$DRY_RUN" == "false" ]]; then
        echo "Waiting ${SLEEP_BETWEEN}s before next instance..."
        sleep "$SLEEP_BETWEEN"
    fi
done

echo "=== Fleet migration complete ==="
```

One important operational note: after migrating to gp3, monitor `ReadLatency` and `WriteLatency` in CloudWatch for 48 hours. In rare cases involving very small volumes (under 20 GB) that relied on gp2's burst credit model, I/O latency can increase because gp2 burst credits allowed short spikes above the baseline IOPS tier. gp3's fixed 3,000 IOPS baseline is more consistent but has no burst headroom unless you provision additional IOPS explicitly.

2.4 I/O Cost Analysis and Elimination

The least visible component of the RDS bill is often I/O charges, because they do not appear as a separate instance cost — they accumulate as `RDS:StorageIOUsage` charges and only become obvious when you decompose the bill by usage type, as shown in Section 2.1. On a gp2 volume, every I/O request beyond the baseline IOPS allocation costs \$0.10 per million requests.

That sounds trivially small. It is not. A moderately busy PostgreSQL instance executing 500 read IOPS and 200 write IOPS continuously against a 200 GB gp2 volume (baseline: 600 IOPS) will stay near but can intermittently exceed baseline during query bursts. An instance that regularly exceeds baseline by even 200 IOPS for 12 hours per day generates approximately 260 million excess I/O requests per month — a \$26/month charge that appears invisible until you look for it. Multiply across a fleet of 20 instances and you are looking at potentially \$500+/month in charges that vanish entirely by moving to gp3.

Identifying I/O charges in the Cost and Usage Report:

```
-- Athena query: find instances generating the highest I/O charges
-- These are your highest-priority gp3 migration targets

SELECT
    line_item_resource_id           AS instance_id,
    resource_tags_user_environment AS environment,
    resource_tags_user_name        AS owner,
    SUM(line_item_unblended_cost)   AS io_charges_usd,
    SUM(line_item_usage_amount) / 1e6 AS billions_of_requests,
    -- Annualize to make the business case clear
    SUM(line_item_unblended_cost) * 12 AS annualized_io_cost
```

```

FROM
    cost_and_usage_report
WHERE
    line_item_service_code = 'AmazonRDS'
    AND line_item_usage_type LIKE '%StorageIOUsage%'
    AND year = '2025'
    AND month = '2'
GROUP BY
    line_item_resource_id,
    resource_tags_user_environment,
    resource_tags_user_name
HAVING
    SUM(line_item_unblended_cost) > 5.00
ORDER BY
    io_charges_usd DESC
LIMIT 25;

```

Beyond the storage type migration, a second lever for I/O cost reduction is query-level optimization. Many I/O charges trace back to a small number of queries executing full sequential scans on large tables. The following PostgreSQL query identifies the highest I/O-consuming queries using `pg_stat_statements`:

```

-- PostgreSQL: top queries by shared block reads (disk I/O proxy)
-- Run on the target RDS instance via psql or a query tool

SELECT
    queryid,
    LEFT(query, 120) AS query_preview,
    calls,
    shared_blks_read AS disk_block_reads,
    shared_blks_hit AS buffer_cache_hits,
    ROUND(
        shared_blks_read::numeric
        / NULLIF(shared_blks_read + shared_blks_hit, 0) * 100, 2
    ) AS cache_miss_pct,
    ROUND(total_exec_time::numeric / calls, 2) AS avg_exec_ms,
    shared_blks_read * 8 / 1024 AS total_mb_read_from_disk
FROM
    pg_stat_statements
WHERE
    calls > 50
    AND shared_blks_read > 10000
ORDER BY
    shared_blks_read DESC
LIMIT 20;

```

The `cache_miss_pct` column is your primary signal. Queries with cache miss rates above 50% are reading predominantly from disk rather than the shared buffer pool. This drives both I/O costs and query latency. The remediation path is typically one of three options:

1. **Add an index** to eliminate a sequential scan
2. **Increase shared_buffers** (RDS parameter group) to cache more data in memory
3. **Increase instance memory** by upsizing the instance class

For SQL Server on RDS, the equivalent query uses `sys.dm_exec_query_stats`:

```

-- SQL Server: top queries by logical reads (I/O proxy)
-- Run against the target RDS SQL Server instance

SELECT TOP 20
    qs.execution_count,
    qs.total_logical_reads,
    qs.total_logical_reads / qs.execution_count AS avg_logical_reads,
    qs.total_elapsed_time / qs.execution_count / 1000 AS avg_elapsed_ms,
    SUBSTRING(
        qt.text,

```

```

        (qs.statement_start_offset / 2) + 1,
        (
            CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)
            ELSE qs.statement_end_offset
            END - qs.statement_start_offset
        ) / 2 + 1
    ) AS query_text
FROM
    sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS qt
WHERE
    qs.execution_count > 50
AND qs.total_logical_reads > 100000
ORDER BY
    qs.total_logical_reads DESC;

```

High logical read counts on SQL Server translate directly to I/O pressure. Each logical read that misses the buffer cache becomes a physical read from the EBS volume, incrementing the I/O counter that AWS bills on gp2. On gp3, these physical reads still consume IOPS against your provisioned allocation, but they do not generate additional charges unless you exceed your provisioned IOPS limit — which you control explicitly.

2.5 Data Transfer Cost Analysis

Data transfer charges are the most unpredictable dimension of the RDS bill. AWS charges for data leaving a region, for data crossing Availability Zone boundaries within a region, and for data exiting to the internet. The rules are nuanced enough that many teams discover transfer charges only after a large unexpected bill.

The data transfer pricing rules that matter most for RDS:

Transfer Type	Cost (us-east-1)	Common Cause
RDS → Internet	\$0.09/GB	Direct public endpoint access
RDS → Different AZ (same region)	\$0.01/GB each direction	App in AZ-a, DB in AZ-b
RDS → Same AZ	Free	Requires explicit AZ-aware placement
RDS → CloudFront	Free	Edge caching of DB-backed APIs
RDS → S3 (same region)	Free	Snapshot exports, ETL
RDS → Another AWS Region	\$0.02–\$0.08/GB	Cross-region replicas

The cross-AZ transfer cost is the most commonly overlooked. If your application servers are in **us-east-1a** and your RDS primary is in **us-east-1b**, every query result crosses an AZ boundary. At \$0.01/GB in each direction, a database serving 1 TB of query results per month incurs **\$10.24/month in cross-AZ transfer charges** — charges that appear on the EC2 transfer line, not the RDS line, which is why they are easy to miss.

Identifying cross-AZ transfer costs using VPC Flow Logs:

```
#!/usr/bin/env python3
"""
analyze_rds_transfer_costs.py

Queries CloudWatch Logs Insights against VPC Flow Logs to identify
data flows involving RDS endpoints. Requires VPC Flow Logs enabled
and delivered to CloudWatch Logs.

This is a read-only diagnostic script - it does not modify anything.
"""

import boto3
import time
from datetime import datetime, timedelta

def get_rds_private_ips(region='us-east-1'):
    """Retrieve the private IP addresses of all RDS instances."""
    rds_client = boto3.client('rds', region_name=region)
    ec2_client = boto3.client('ec2', region_name=region)

    rds_ips = {}
    instances = rds_client.describe_db_instances()['DBInstances']

    for inst in instances:
        endpoint = inst.get('Endpoint', {})
        hostname = endpoint.get('Address', '')
        if not hostname:
            continue

        # Resolve hostname to IP via EC2 network interfaces
        try:
            import socket
            ip = socket.gethostbyname(hostname)
            rds_ips[ip] = {
                'id': inst['DBInstanceIdentifier'],
                'az': inst.get('AvailabilityZone', 'unknown'),
                'class': inst['DBInstanceClass']
            }
        except Exception:
            continue

    return rds_ips

def query_flow_logs_for_rds_traffic(log_group, rds_ip, hours=24):
    """
    Query VPC Flow Logs for bytes transferred to/from a specific RDS IP.
    Returns total bytes separated by direction.
    """
    logs_client = boto3.client('logs')

    query = f"""
fields @timestamp, srcAddr, dstAddr, bytes, action
| filter (srcAddr = '{rds_ip}' or dstAddr = '{rds_ip}')
| filter action = 'ACCEPT'
| stats sum(bytes) as total_bytes by srcAddr, dstAddr
| sort total_bytes desc
| limit 50
"""

    end_time = int(datetime.utcnow().timestamp())
    start_time = int((datetime.utcnow() - timedelta(hours=hours)).timestamp())

    response = logs_client.start_query(
        logGroupName=log_group,
        startTime=start_time,
        endTime=end_time,
        queryString=query
    )
    query_id = response['queryId']
```

```

# Poll for completion
for _ in range(30):
    time.sleep(2)
    result = logs_client.get_query_results(queryId=query_id)
    if result['status'] in ('Complete', 'Failed', 'Cancelled'):
        break

if result['status'] != 'Complete':
    return []

rows = []
for record in result['results']:
    row = {field['field']: field['value'] for field in record}
    rows.append(row)
return rows

def estimate_transfer_cost(bytes_transferred, transfer_type='cross_az'):
    """Estimate cost for a given number of bytes based on transfer type."""
    gb = bytes_transferred / (1024 ** 3)
    rates = {
        'cross_az': 0.01,      # per GB, each direction
        'internet': 0.09,
        'cross_region': 0.02,
    }
    return gb * rates.get(transfer_type, 0.01)

def main():
    LOG_GROUP = '/aws/vpc/flowlogs'  # adjust to your VPC Flow Log group name
    HOURS = 24

    print("=== RDS Data Transfer Cost Analysis ===")
    print(f"Analyzing last {HOURS} hours of VPC Flow Logs\n")

    rds_ips = get_rds_private_ips()
    if not rds_ips:
        print("No RDS instances found or could not resolve IPs.")
        return

    total_estimated_daily_cost = 0.0

    for ip, metadata in rds_ips.items():
        print(f"Instance: {metadata['id']} ({metadata['class']}, AZ: {metadata['az']})")
        print(f"Private IP: {ip}")

        flows = query_flow_logs_for_rds_traffic(LOG_GROUP, ip, HOURS)
        if not flows:
            print("No flow log data available.\n")
            continue

        total_bytes = sum(int(row.get('total_bytes', 0)) for row in flows)
        total_gb = total_bytes / (1024 ** 3)
        daily_cost = estimate_transfer_cost(total_bytes, 'cross_az')
        monthly_cost = daily_cost * 30

        print(f"Total data transferred (24h): {total_gb:.2f} GB")
        print(f"Estimated daily transfer cost: ${daily_cost:.4f}")
        print(f"Estimated monthly transfer cost: ${monthly_cost:.2f}")
        total_estimated_daily_cost += daily_cost
        print()

    print(f"Total estimated monthly transfer cost across all RDS: "
          f"${total_estimated_daily_cost * 30:.2f}")

if __name__ == '__main__':
    main()

```

The primary remediation for cross-AZ transfer costs is **AZ-aware deployment**: ensure application servers and the RDS primary endpoint are colocated in the same Availability Zone. This requires

either:

1. Querying the RDS endpoint's current AZ and placing Auto Scaling groups to prefer that AZ
2. Using Amazon RDS Proxy with connection pooling that routes to the same-AZ replica

For Multi-AZ deployments, the standby is in a different AZ by design — but only reads and writes to the primary endpoint incur transfer costs. The standby receives replication traffic that AWS does not charge for between the primary and standby RDS nodes.

2.6 Snapshot and Backup Cost Management

Automated backups and manual snapshots represent the quietest cost dimension on the RDS bill. AWS provides free automated backup storage up to the size of your provisioned database storage. Beyond that threshold, automated backup storage costs \$0.095/GB/month. Manual snapshots are charged at \$0.095/GB/month from the first byte.

Teams that leave the default retention period (7 days), keep manual snapshots indefinitely, and never prune exported snapshots from S3 accumulate backup storage costs that can rival their storage volume costs within a year.

The snapshot cost scenarios that quietly compound:

Scenario	Monthly Cost Impact
1 TB database, 7-day retention, daily change rate 10%	Approximately free (within allocation)
1 TB database, 35-day retention, daily change rate 10%	~\$28.50/month in excess backup storage
50 manual snapshots from testing, never deleted	Potentially \$50–\$200/month
Snapshot exported to S3, S3 storage not cleaned up	S3 Standard charges at \$0.023/GB/month

Identifying orphaned and excessive snapshots:

```
#!/bin/bash
# audit_rds_snapshots.sh
# Lists all manual RDS snapshots with their size, age, and estimated cost.
# Flags snapshots older than a configurable threshold as deletion candidates.

THRESHOLD_DAYS="${1:-90}" # Snapshots older than this are flagged
COST_PER_GB=0.095

echo "=== RDS Manual Snapshot Audit ==="
echo "Flagging snapshots older than ${THRESHOLD_DAYS} days"
echo ""

aws rds describe-db-snapshots \
  --snapshot-type manual \
  --query 'DBSnapshots[*].{
    ID:DBSnapshotIdentifier,
    Instance:DBInstanceIdentifier,
    Created:SnapshotCreateTime,
    SizeGB:AllocatedStorage,
```



```

        Status:Status,
        Engine:Engine
    }' \
--output json | \
python3 -c "
import json, sys
from datetime import datetime, timezone

data = json.load(sys.stdin)
threshold_days = int('${THRESHOLD_DAYS}')
cost_per_gb = float('${COST_PER_GB}')
now = datetime.now(timezone.utc)

total_gb = 0
flagged_gb = 0
flagged_count = 0

print(f'{{Snapshot ID':<45}} {{Instance':<30}} {{Age (days)':>10}} {{Size (GB)':>10}} {{Monthly Cost':>13}}
↳ {{Flag'}}')
print('-' * 120)

snapshots = sorted(data, key=lambda x: x.get('Created', ''), reverse=True)

for snap in snapshots:
    created_str = snap.get('Created', '')
    if not created_str:
        age_days = 0
    else:
        created = datetime.fromisoformat(created_str.replace('Z', '+00:00'))
        age_days = (now - created).days

    size_gb = snap.get('SizeGB', 0)
    monthly_cost = size_gb * cost_per_gb
    flag = 'DELETE CANDIDATE' if age_days > threshold_days else ''

    total_gb += size_gb
    if flag:
        flagged_gb += size_gb
        flagged_count += 1

    snap_id = snap.get('ID', '')[:44]
    instance = snap.get('Instance', '')[:29]
    print(f'{{snap_id:<45}} {{instance:<30}} {{age_days:>10}} {{size_gb:>10}} \${{monthly_cost:>11.2f}} {{flag}}')

print('-' * 120)
total_cost = total_gb * cost_per_gb
flagged_cost = flagged_gb * cost_per_gb
print(f'Total manual snapshot storage: {{total_gb:,}} GB = \${{total_cost:.2f}}/month')
print(f'Flagged for deletion: {{flagged_count}} snapshots, {{flagged_gb:,}} GB =
↳ \${{flagged_cost:.2f}}/month recoverable')
"
```

Automated snapshot cleanup policy using AWS Backup:

Rather than managing snapshot retention manually, use AWS Backup lifecycle policies to transition older snapshots to cold storage (at \$0.01/GB/month instead of \$0.095/GB/month) and delete them after a defined period. The following Terraform snippet creates a backup plan that retains daily snapshots for 14 days, weekly snapshots for 3 months, and monthly snapshots for 1 year — and deletes them automatically:

```

# terraform/rds_backup_plan.tf

resource "aws_backup_plan" "rds_tiered_backup" {
  name = "rds-tiered-retention"

  rule {
    rule_name      = "daily_14_days"
    target_vault_name = aws_backup_vault.rds_vault.name
  }
}
```

```
schedule          = "cron(0 2 * * ? *)"    # 2 AM UTC daily

lifecycle {
  delete_after = 14    # days
}

recovery_point_tags = {
  BackupFrequency = "daily"
}
}

rule {
  rule_name          = "weekly_90_days"
  target_vault_name = aws_backup_vault.rds_vault.name
  schedule           = "cron(0 3 ? * SUN *)" # 3 AM UTC Sundays

  lifecycle {
    cold_storage_after = 14    # move to cold after 14 days
    delete_after      = 90    # delete after 90 days total
  }

  recovery_point_tags = {
    BackupFrequency = "weekly"
  }
}

rule {
  rule_name          = "monthly_365_days"
  target_vault_name = aws_backup_vault.rds_vault.name
  schedule           = "cron(0 4 1 * ? *)"   # 4 AM UTC on 1st of month

  lifecycle {
    cold_storage_after = 30
    delete_after      = 365
  }

  recovery_point_tags = {
    BackupFrequency = "monthly"
  }
}

tags = {
  ManagedBy    = "terraform"
  CostCenter   = "infrastructure"
  Purpose      = "rds-backup-cost-optimization"
}
}

resource "aws_backup_vault" "rds_vault" {
  name = "rds-backup-vault"

  tags = {
    ManagedBy = "terraform"
  }
}

resource "aws_backup_selection" "rds_all_instances" {
  iam_role_arn = aws_iam_role.backup_role.arn
  name         = "all-rds-instances"
  plan_id      = aws_backup_plan.rds_tiered_backup.id

  resources = ["arn:aws:rds:*:*:db:*"]

  # Optionally restrict to tagged instances only:
  # condition {
  #   string_equals {
  #     key   = "aws:ResourceTag/BackupManaged"
  #     value = "true"
  #   }
  # }
}
```

```
}

resource "aws_iam_role" "backup_role" {
  name = "rds-backup-service-role"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [{
      Action    = "sts:AssumeRole"
      Effect    = "Allow"
      Principal = { Service = "backup.amazonaws.com" }
    }]
  })
}

resource "aws_iam_role_policy_attachment" "backup_policy" {
  role       = aws_iam_role.backup_role.name
  policy_arn = "arn:aws:iam::aws:policy/service-role/AWSBackupServiceRolePolicyForBackup"
}
```

The cold storage transition for monthly snapshots reduces that tier’s cost from \$0.095/GB to \$0.01/GB — a 90% reduction for long-term archival backups that are kept for compliance but rarely restored.

2.7 Multi-AZ Cost Considerations and When to Reconsider

Multi-AZ deployments double the compute and storage cost of an RDS instance by maintaining a synchronous standby replica in a second Availability Zone. For production workloads that require high availability and automatic failover, this is money well spent. For non-production environments, it is frequently unnecessary spending.

The cost of Multi-AZ is precisely the cost of the primary instance, repeated:

- A db.r6g.xlarge Multi-AZ costs **twice** what the single-AZ configuration costs
- The standby incurs compute charges at the same rate as the primary
- Storage is replicated and billed twice

Teams often enable Multi-AZ on staging, QA, and development environments “because it matches production.” This is a category error. Multi-AZ for a staging environment that receives traffic for 8 hours per day and sits idle the rest of the time provides no meaningful availability benefit — and costs the same as a 24/7 production instance.

Identifying Multi-AZ instances in non-production environments:

```
#!/usr/bin/env python3
"""
multiaz_audit.py

Identifies Multi-AZ RDS instances in non-production environments
based on naming conventions or tags. Calculates potential savings
from disabling Multi-AZ or replacing with single-AZ + read replica.
"""

import boto3

# Patterns in instance names that suggest non-production
NON_PROD_PATTERNS = ['dev', 'staging', 'stage', 'qa', 'test', 'uat', 'sandbox', 'demo']

def is_non_production(instance_id, tags):
    """Determine if instance is non-production by name or tags."""
```

```

name_lower = instance_id.lower()
if any(pattern in name_lower for pattern in NON_PROD_PATTERNS):
    return True

for tag in tags:
    key = tag.get('Key', '').lower()
    value = tag.get('Value', '').lower()
    if key in ('environment', 'env') and any(p in value for p in NON_PROD_PATTERNS):
        return True
return False

# Approximate on-demand hourly rates for compute (us-east-1)
HOURLY_RATES = {
    'db.t3.medium': 0.068,
    'db.t3.large': 0.136,
    'db.t3.xlarge': 0.272,
    'db.r5.large': 0.240,
    'db.r5.xlarge': 0.480,
    'db.r5.2xlarge': 0.960,
    'db.r6g.large': 0.208,
    'db.r6g.xlarge': 0.416,
    'db.r6g.2xlarge': 0.480,
    'db.m5.large': 0.190,
    'db.m5.xlarge': 0.380,
    'db.m6g.large': 0.171,
    'db.m6g.xlarge': 0.342,
}

def main():
    rds = boto3.client('rds')
    instances = rds.describe_db_instances()['DBInstances']

    print("=== Multi-AZ Non-Production Audit ===\n")
    print(f"{'Instance ID':<35} {'Class':<20} {'MultiAZ':>7} "
          f"{'Monthly Waste':>14} {'Recommendation'}")
    print("-" * 105)

    total_recoverable = 0.0

    for inst in instances:
        if inst['DBInstanceStatus'] != 'available':
            continue

        instance_id = inst['DBInstanceIdentifier']
        instance_class = inst['DBInstanceClass']
        multi_az = inst.get('MultiAZ', False)

        tags_response = rds.list_tags_for_resource(
            ResourceName=inst['DBInstanceArn']
        )
        tags = tags_response.get('TagList', [])

        non_prod = is_non_production(instance_id, tags)
        hourly = HOURLY_RATES.get(instance_class, 0)
        # Multi-AZ standby cost = same as primary compute
        standby_monthly = hourly * 730

        if multi_az and non_prod:
            recommendation = "DISABLE Multi-AZ"
            recoverable = standby_monthly
            total_recoverable += recoverable
            print(f"{'instance_id':<35} {'instance_class':<20} {'Yes':>7} "
                  f"{'recoverable':>12.2f} {'recommendation'}")
        elif multi_az and not non_prod:
            print(f"{'instance_id':<35} {'instance_class':<20} {'Yes':>7} "
                  f"{'(production)':>14} OK - keep Multi-AZ")
        else:
            print(f"{'instance_id':<35} {'instance_class':<20} {'No':>7} "
                  f"{'-':>14} OK - single-AZ")

```

```
print("-" * 105)
print(f"Estimated monthly savings from disabling non-production Multi-AZ: "
      f"${total_recoverable:,.2f}")
print(f"Annual savings: ${total_recoverable * 12:,.2f}")

if __name__ == '__main__':
    main()
```

For non-production databases that genuinely need availability during business hours, consider the **stop/start schedule** pattern instead: keep single-AZ, and use EventBridge scheduled rules to stop the instance overnight and on weekends. An instance that runs only 50 hours per week instead of 168 hours reduces compute costs by approximately 70%.

```
# EventBridge rule to stop a dev RDS instance at 7 PM weekdays
aws events put-rule \
  --name "stop-dev-rds-evening" \
  --schedule-expression "cron(0 19 ? * MON-FRI *)" \
  --state ENABLED

# EventBridge rule to start it at 7 AM weekdays
aws events put-rule \
  --name "start-dev-rds-morning" \
  --schedule-expression "cron(0 7 ? * MON-FRI *)" \
  --state ENABLED
```

The Lambda functions that handle the start and stop actions are a five-line wrapper around `rds.stop_db_instance()` and `rds.start_db_instance()`. For a `db.r6g.xlarge` at \$0.416/hour, running 50 hours/week instead of 168 hours/week saves approximately **\$213/month** on that single dev instance.

2.8 Building an RDS Cost Dashboard

Visibility precedes optimization. A team that cannot see its RDS cost breakdown by instance, by dimension, and over time will always be reactive rather than proactive. This section builds a CloudWatch dashboard that surfaces all five cost dimensions in a single view, backed by a Lambda function that populates custom metrics from Cost Explorer data.

Architecture: - EventBridge triggers a Lambda function daily at 6 AM UTC - Lambda queries Cost Explorer for yesterday's RDS costs grouped by usage type and resource - Lambda publishes custom CloudWatch metrics under the `RDS/FinOps` namespace - A CloudWatch dashboard displays 30-day trends for each cost dimension

```
#!/usr/bin/env python3
"""
rds_cost_metrics_publisher.py

Lambda handler that reads RDS Cost Explorer data and publishes
custom CloudWatch metrics. Deploy this as a Lambda function with
Cost Explorer and CloudWatch permissions.

IAM permissions required:
- ce:GetCostAndUsage
- cloudwatch:PutMetricData
"""

import boto3
import json
from datetime import datetime, timedelta
```

```

METRIC_NAMESPACE = 'RDS/FinOps'

# Maps CUR usage type keywords to human-readable dimension names
USAGE_TYPE_MAP = {
    'InstanceUsage': 'Compute',
    'Multi-AZUsage': 'Compute_MultiAZ',
    'StorageUsage': 'Storage',
    'StorageIOUsage': 'IO_Requests',
    'DataTransfer': 'DataTransfer',
    'BackupUsage': 'Backup',
}

def lambda_handler(event, context):
    ce = boto3.client('ce')
    cw = boto3.client('cloudwatch')

    end_date = datetime.utcnow().date()
    start_date = end_date - timedelta(days=1)

    response = ce.get_cost_and_usage(
        TimePeriod={
            'Start': str(start_date),
            'End': str(end_date),
        },
        Granularity='DAILY',
        Filter={
            'Dimensions': {
                'Key': 'SERVICE',
                'Values': ['Amazon Relational Database Service']
            }
        },
        GroupBy=[
            {'Type': 'DIMENSION', 'Key': 'USAGE_TYPE'},
        ],
        Metrics=['UnblendedCost', 'UsageQuantity']
    )

    metric_data = []
    timestamp = datetime.utcnow()

    for group in response['ResultsByTime'][0].get('Groups', []):
        usage_type_full = group['Keys'][0]
        cost = float(group['Metrics']['UnblendedCost']['Amount'])

        if cost < 0.001:
            continue

        # Map to a clean dimension name
        dimension_name = 'Other'
        for keyword, label in USAGE_TYPE_MAP.items():
            if keyword in usage_type_full:
                dimension_name = label
                break

        metric_data.append({
            'MetricName': 'DailyCost',
            'Dimensions': [
                {'Name': 'CostDimension', 'Value': dimension_name},
            ],
            'Timestamp': timestamp,
            'Value': cost,
            'Unit': 'None', # CloudWatch has no USD unit; use None
        })

    # Publish in batches of 20 (CloudWatch limit)
    for i in range(0, len(metric_data), 20):
        batch = metric_data[i:i + 20]
        cw.put_metric_data(
            Namespace=METRIC_NAMESPACE,
            MetricData=batch
        )

```

```

total_cost = sum(m['Value'] for m in metric_data)
print(f"Published {len(metric_data)} metrics. Total RDS cost yesterday: ${total_cost:.4f}")

return {
    'statusCode': 200,
    'body': json.dumps({
        'metrics_published': len(metric_data),
        'total_cost_usd': round(total_cost, 4)
    })
}

```

With that Lambda publishing daily, the CloudWatch dashboard definition becomes straightforward. The following JSON creates a dashboard with one widget per cost dimension plus a total cost trend:

```

{
  "widgets": [
    {
      "type": "metric",
      "properties": {
        "title": "RDS Total Daily Cost (USD)",
        "metrics": [
          ["RDS/FinOps", "DailyCost", "CostDimension", "Compute"],
          [".", ".", ".", "Compute_MultiAZ"],
          [".", ".", ".", "Storage"],
          [".", ".", ".", "IO_Requests"],
          [".", ".", ".", "DataTransfer"],
          [".", ".", ".", "Backup"],
          [".", ".", ".", "Other"]
        ],
        "view": "timeSeries",
        "stacked": true,
        "period": 86400,
        "stat": "Sum",
        "start": "-P30D",
        "end": "POD",
        "yAxis": { "left": { "label": "USD", "min": 0 } }
      }
    },
    {
      "type": "metric",
      "properties": {
        "title": "I/O Charges (gp2 migration signal)",
        "metrics": [
          ["RDS/FinOps", "DailyCost", "CostDimension", "IO_Requests",
            { "color": "#d62728", "label": "I/O Charges" } ]
        ],
        "view": "timeSeries",
        "period": 86400,
        "stat": "Sum",
        "start": "-P30D",
        "end": "POD",
        "annotations": {
          "horizontal": [
            {
              "label": "Investigation threshold",
              "value": 10,
              "color": "#ff7f0e"
            }
          ]
        }
      }
    },
    {
      "type": "metric",
      "properties": {
        "title": "Backup Storage Cost",
        "metrics": [
          ["RDS/FinOps", "DailyCost", "CostDimension", "Backup"]
        ]
      }
    }
  ]
}

```

```
    ],
    "view": "timeSeries",
    "period": 86400,
    "stat": "Sum",
    "start": "-P30D",
    "end": "POD"
  }
},
{
  "type": "metric",
  "properties": {
    "title": "Data Transfer Cost",
    "metrics": [
      ["RDS/FinOps", "DailyCost", "CostDimension", "DataTransfer"]
    ],
    "view": "timeSeries",
    "period": 86400,
    "stat": "Sum",
    "start": "-P30D",
    "end": "POD"
  }
}
]
}
```

Deploy the dashboard using the AWS CLI:

```
DASHBOARD_NAME="RDS-FinOps"

aws cloudwatch put-dashboard \
  --dashboard-name "$DASHBOARD_NAME" \
  --dashboard-body file://rds_finops_dashboard.json

echo "Dashboard available at:"
echo "https://console.aws.amazon.com/cloudwatch/home#dashboards:name=${DASHBOARD_NAME}"
```

Set a CloudWatch alarm on the `IO_Requests` dimension so that any day where I/O charges exceed ten dollars triggers a notification to your FinOps Slack channel. That alarm is the automated signal to check which instances are generating gp2 I/O charges and should be migrated.

```
aws cloudwatch put-metric-alarm \
  --alarm-name "RDS-IO-Charges-Above-Threshold" \
  --alarm-description "Daily RDS I/O charges exceeded $10 - check for gp2 instances" \
  --namespace "RDS/FinOps" \
  --metric-name "DailyCost" \
  --dimensions Name=CostDimension,Value=IO_Requests \
  --statistic Sum \
  --period 86400 \
  --evaluation-periods 1 \
  --threshold 10.0 \
  --comparison-operator GreaterThanOrEqualToThreshold \
  --alarm-actions "arn:aws:sns:us-east-1:123456789012:finops-alerts" \
  --ok-actions "arn:aws:sns:us-east-1:123456789012:finops-alerts" \
  --treat-missing-data notBreaching
```

This single alarm catches gp2 I/O charges before they compound across a billing cycle. A team that responds to this alarm within a day of it firing, and migrates the offending instances to gp3, recovers those charges in the same month.

2.9 Putting It All Together: An RDS Cost Optimization Runbook

The preceding sections each address one cost dimension in isolation. In practice, optimization happens as a coordinated sequence of changes. The following runbook consolidates the actions into

a repeatable process that a team can execute monthly.

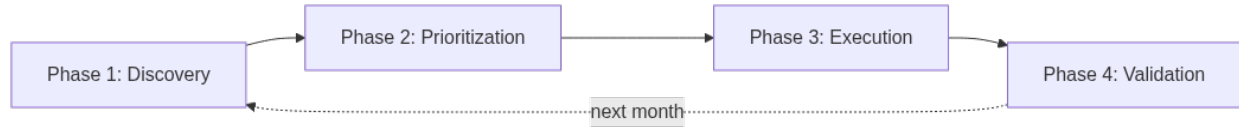


Figure 2: The monthly RDS cost optimization runbook: a repeatable four-phase cycle, not a one-time cleanup.

Phase 1 — Discovery (Week 1)

1. Run the Cost Explorer breakdown script from Section 2.1. Record the split across compute, storage, I/O, transfer, and backup.
2. Run the compute rightsizing script from Section 2.2. Flag all instances with P95 CPU below 20%.
3. Run the gp2 identification Athena query from Section 2.3. List all gp2 instances ordered by storage size.
4. Run the snapshot audit script from Section 2.6. List all manual snapshots older than 90 days.
5. Run the Multi-AZ audit script from Section 2.7. List non-production Multi-AZ instances.

Phase 2 — Prioritization (Week 1)

Rank findings by annualized savings potential. A representative priority stack typically looks like:

Rank	Action	Typical Annual Savings
1	Reserved Instance purchases for stable production instances	Largest absolute value
2	gp2 → gp3 migration for high-I/O instances	High, zero risk
3	Downsize overprovisioned dev/staging instances	Moderate
4	Disable Multi-AZ on non-production	Moderate
5	Delete orphaned snapshots	Smaller but immediate
6	Stop/start schedules for dev instances	Moderate, ongoing

Phase 3 — Execution (Weeks 2–3)

Execute changes in order of risk, lowest first: - Snapshot deletion requires only CLI access and carries no availability risk - Multi-AZ disablement on non-production is safe during low-traffic periods - gp2 → gp3 migration can be done live with no downtime - Instance class changes require a maintenance window (minutes of downtime) - Reserved Instance purchases are commitment decisions requiring finance approval

Phase 4 — Validation (Week 4)

After changes are applied, validate using the cost dashboard from Section 2.8: - I/O charges should drop to zero for migrated instances - Storage charges should decrease by approximately 30% per

migrated volume - Compute charges should reflect the new instance classes within the billing period
- Snapshot charges should decrease within 24 hours of deletion

Establish a monthly cost review cadence using the runbook. Teams that run this process consistently across a fleet of 30 or more RDS instances typically identify recoverable spending equivalent to several months of engineering time in the first cycle alone.

Key Takeaways

- **AWS RDS billing has five independent cost dimensions** — compute, storage, I/O, data transfer, and snapshots — and each requires a distinct optimization strategy. Treating the monthly total as a single opaque number prevents systematic reduction.
- **Migrating from gp2 to gp3 storage is the lowest-risk, highest-return action available for most RDS fleets.** gp3 costs approximately 30% less per GB, eliminates all per-I/O charges, and provides higher baseline IOPS. The migration requires no downtime and can be scripted for fleet-wide execution in a single afternoon.
- **Reserved Instances applied to stable production workloads typically reduce compute costs by 38–60%** compared to on-demand pricing. For any instance that has run continuously for more than three months, the RI commitment pays for itself within the first few months and the savings compound for the remainder of the term.
- **Non-production Multi-AZ and idle-hours compute are the two most common sources of unrecognized waste.** Disabling Multi-AZ on development and staging environments and applying stop/start schedules to instances that run only during business hours can reduce non-production RDS spend by 50–70% without affecting any production workload.
- **Cost visibility through a daily metrics pipeline and CloudWatch dashboard converts FinOps from a monthly retrospective into a continuous signal.** An I/O charge alarm that fires the day a gp2 instance starts accumulating excess charges enables same-day remediation rather than a month-end surprise.

Chapter 3: Azure SQL Cost Model: DTU vs vCore vs Serverless

Organizations running workloads on Azure SQL Database routinely overspend by 40–60% because they inherited a DTU tier chosen during initial migration, never revisited it, and now pay for capacity that sits idle 70% of the day. This chapter breaks down all three Azure SQL purchasing models — DTU-based, vCore-based, and Serverless — explains exactly when each one saves or costs you money, and gives you the queries, CLI commands, and Python automation to identify mismatches and fix them on a schedule. A team running a 250 DTU Premium (P2) database that migrates to a General Purpose 8 vCore with a reserved instance and the Azure Hybrid Benefit can realistically reduce that single database’s monthly bill from roughly **\$930 to \$310** — a saving of \$620/month without touching a line of application code.

3.1 Understanding the Three Purchasing Models

Before optimizing, you need a precise mental model of what you are actually buying. Azure SQL charges you differently depending on which of its three cost models you choose, and conflating them is the primary reason teams make poor sizing decisions.

DTU-Based Purchasing

DTU stands for Database Transaction Unit — a blended measure of CPU, memory, and I/O that Microsoft bundles into a single number. You cannot see or control the individual components. A 100 DTU Standard database costs roughly \$150/month; a 500 DTU Premium database costs around \$465/month. The tiers are:

Tier	DTU Range	Max Storage	Approximate Monthly Cost
Basic	5	2 GB	\$5
Standard (S0–S9)	10–3,000	1 TB	\$15–\$450
Premium (P1–P15)	125–4,000	4 TB	\$465–\$14,000+

The DTU model’s core problem is opacity. You cannot correlate DTU consumption to actual CPU or memory utilization. If a query is I/O bound and you’re buying a CPU-heavy DTU tier, you’re paying a bundled price for resources you don’t consume. DTU databases also do not support the Azure Hybrid Benefit for SQL Server licenses, which alone eliminates 30% of the vCore price.

vCore-Based Purchasing

The vCore model lets you choose cores and memory independently (within hardware tiers), apply your own SQL Server licenses, and see actual CPU and memory metrics. There are three hardware families:

- **General Purpose (Fsv2 or Gen5)** — balanced compute and storage, good for most OLTP workloads.
- **Business Critical** — local SSD, built-in read replica, high memory-to-core ratio. Equivalent to a local Premium SSD RAID; suited for latency-sensitive OLTP.
- **Hyperscale** — distributed storage that scales to 100 TB, with rapid snapshot-based backups.

A General Purpose 4 vCore (Gen5) runs approximately \$368/month pay-as-you-go. Apply the **Azure Hybrid Benefit** if you have existing SQL Server Enterprise licenses with Software Assurance, and the cost drops to roughly \$186/month. Buy a **1-year reserved instance** on top of that and you reach approximately \$155/month. Buy a **3-year reserved instance** and you reach approximately \$115/month. The math on that last number versus a 500 DTU Premium (\$465/month) tells the whole story for moderately sized OLTP databases.

Serverless

Serverless is a vCore-based sub-model where you define a minimum and maximum vCore count and pay only for the compute you use, billed per second. When the database sits idle for a configurable period (the **auto-pause delay**, minimum 1 hour), Azure pauses it entirely and you pay only for storage. When the next connection arrives, the database resumes — the cold-start latency typically runs 30–90 seconds.

Serverless pricing for General Purpose: approximately **\$0.000145 per vCore-second**. A database running at 2 vCores for 8 hours/day costs roughly \$250.56/month in compute, versus the always-on General Purpose 2 vCore at approximately \$762/month. That is a 67% compute saving for a development database used only during business hours, assuming it auto-pauses for the remaining 16 hours.

The break-even point between Serverless and provisioned vCore is roughly **50% average utilization across a 730-hour month**. Workloads above that threshold are cheaper on provisioned; workloads below it are cheaper on Serverless.

3.2 Diagnosing Your Current Spend and Utilization

The first step is understanding what you are actually paying for versus what your databases actually consume. Azure provides this data in two places: Azure Cost Management and the database's own DMVs.

Pulling Cost Data with Azure CLI

The following command exports the last 30 days of Azure SQL spend by resource, broken down to the database level:

```
#!/bin/bash
# Pull Azure SQL cost breakdown for the last 30 days
```

```

# Requires: Azure CLI authenticated, correct subscription set

SUBSCRIPTION_ID="your-subscription-id"
RESOURCE_GROUP="your-rg"
START_DATE=$(date -d "30 days ago" '+%Y-%m-%d')
END_DATE=$(date '+%Y-%m-%d')

az consumption usage list \
  --start-date "$START_DATE" \
  --end-date "$END_DATE" \
  --query "[?contains(instanceName, 'sql')].{
    Resource:instanceName,
    Cost:pretaxCost,
    Currency:currency,
    UsageDate:usageStart
  }" \
  --output table \
  --subscription "$SUBSCRIPTION_ID"

# For more granular breakdown by SKU (DTU tier, vCore family)
az costmanagement query \
  --type Usage \
  --timeframe Custom \
  --time-period from="$START_DATE" to="$END_DATE" \
  --dataset-configuration columns=
  ↵ '["SubscriptionId","ResourceGroup","ResourceId","MeterCategory","MeterSubCategory","MeterName","Quantity","Unit","P
  ↵ \
  --dataset-filter '{
    "and": [
      {
        "dimensions": {
          "name": "MeterCategory",
          "operator": "In",
          "values": ["SQL Database"]
        }
      }
    ]
  }' \
  --scope "subscriptions/$SUBSCRIPTION_ID" \
  --output json | jq '.properties.rows[] | {
    resource: .[2],
    meter: .[4],
    sku: .[5],
    quantity: .[6],
    cost: .[8],
    date: .[9]
  }'

```

Reading Utilization from the Database DMVs

Once you know what you're paying, you need to know what you're using. The following query runs inside Azure SQL and returns the 15-day average DTU (or CPU) consumption alongside the tier you're paying for — giving you the utilization percentage that drives every sizing decision:

```

-- Run this inside the Azure SQL database (not master)
-- Returns: 15-day average resource utilization vs. purchased tier
-- Works for both DTU and vCore databases

SELECT
  database_name                                AS [Database],
  service_objective                            AS [Current_SKU],
  elastic_pool_name                            AS [Elastic_Pool],
  -- DTU utilization (NULL for vCore databases)
  AVG(avg_cpu_percent)                        AS [Avg_CPU_Pct_15Day],
  MAX(avg_cpu_percent)                        AS [Max_CPU_Pct_15Day],
  AVG(avg_data_io_percent)                    AS [Avg_DataIO_Pct_15Day],
  MAX(avg_data_io_percent)                    AS [Max_DataIO_Pct_15Day],
  AVG(avg_log_write_percent)                  AS [Avg_LogWrite_Pct_15Day],

```

```

MAX(avg_log_write_percent)                AS [Max_LogWrite_Pct_15Day],
AVG(avg_memory_usage_percent)             AS [Avg_Mem_Pct_15Day],
MAX(avg_memory_usage_percent)             AS [Max_Mem_Pct_15Day],
-- Derived overall DTU utilization (highest of the four dimensions)
AVG(GREATEST(
    avg_cpu_percent,
    avg_data_io_percent,
    avg_log_write_percent
))                                         AS [Avg_Effective_DTU_Pct],
MAX(GREATEST(
    avg_cpu_percent,
    avg_data_io_percent,
    avg_log_write_percent
))                                         AS [Max_Effective_DTU_Pct],
MIN(end_time)                             AS [Oldest_Sample],
MAX(end_time)                             AS [Newest_Sample],
COUNT(*)                                 AS [Sample_Count]
FROM sys.dm_db_resource_stats
CROSS JOIN (
    SELECT
        d.name                AS database_name,
        d.service_objective    AS service_objective,
        d.elastic_pool_name    AS elastic_pool_name
    FROM sys.database_service_objectives d
    WHERE d.database_id = DB_ID()
) meta
GROUP BY
    database_name,
    service_objective,
    elastic_pool_name;

```

Reading the output: If Avg_Effective_DTU_Pct is below 40% and Max_Effective_DTU_Pct is below 75%, you are a candidate for downsizing. If Avg_Effective_DTU_Pct is above 70% consistently, downsizing will cause performance degradation — address query efficiency first.

For longer historical windows (up to 14 days) from the master database, use `sys.resource_stats` instead:

```

-- Run this in the MASTER database on a logical server
-- 14-day utilization view across all databases on the server

SELECT
    database_name,
    sku,
    -- 5-minute granularity samples
    COUNT(*)                AS sample_count,
    AVG(avg_cpu_percent)     AS avg_cpu_pct,
    PERCENTILE_CONT(0.95) WITHIN GROUP
        (ORDER BY avg_cpu_percent)    AS p95_cpu_pct,
    AVG(avg_data_io_percent) AS avg_io_pct,
    PERCENTILE_CONT(0.95) WITHIN GROUP
        (ORDER BY avg_data_io_percent) AS p95_io_pct,
    -- Cost signal: hours where utilization < 20% (idle capacity)
    SUM(CASE WHEN avg_cpu_percent < 20
        AND avg_data_io_percent < 20
        THEN 1 ELSE 0 END) * 5.0 / 60 AS idle_hours_approx,
    SUM(CASE WHEN avg_cpu_percent > 80
        THEN 1 ELSE 0 END) * 5.0 / 60 AS saturated_hours_approx
FROM sys.resource_stats
WHERE start_time >= DATEADD(DAY, -14, GETUTCDATE())
GROUP BY database_name, sku
ORDER BY avg_cpu_pct DESC;

```

3.3 Choosing the Right Tier: Decision Framework

Given utilization data, you can now apply a structured decision process. The following flowchart captures the logic:

1. **Is the database used fewer than 12 hours per day on average, and can it tolerate a 30–90 second cold-start?** → Serverless. Set `min_capacity = 0.5 vCores`, `max_capacity` to your peak requirement, and set `auto_pause_delay_minutes = 60` for dev/test or 120 for lightly used production databases.
2. **Is the database used more than 12 hours per day with predictable load?** → Provisioned vCore General Purpose. Apply Hybrid Benefit if you own licenses. Buy reserved capacity.
3. **Does the workload require < 5 ms read latency, local SSD, or a free read replica for reporting?** → Business Critical vCore. This tier is significantly more expensive (~3× General Purpose) but eliminates the need for a separate read replica or Premium storage.
4. **Is the database in a DTU tier and used inconsistently across a shared pool of multiple databases?** → Elastic Pool (vCore or DTU-based), covered in Chapter 4. Do not optimize a single database in isolation if it shares load patterns with ten others.
5. **Does the database need to grow beyond 4 TB or requires fast, large-scale backup/restore?** → Hyperscale. Cost is separate from the above tiers; storage is billed at \$0.115/GB/month for Hyperscale versus \$0.115/GB/month for General Purpose remote storage — similar per-GB pricing but Hyperscale adds compute cost for its page servers.

Translating DTU to vCore (The Conversion Table)

Microsoft’s rough conversion guideline: **100 DTUs = 1 vCore** for Standard/General Purpose. For Premium/Business Critical, it’s closer to **125 DTUs = 1 vCore**. However, these are approximations. The utilization percentages matter more than the raw numbers.

Current Tier	Approx. Monthly Cost	vCore Equivalent	vCore Pay-as-Go	With Hybrid Benefit	With 1-Yr RI + HB
S3 (100 DTU)	\$150	GP 1 vCore	\$184	\$93	\$77
P1 (125 DTU)	\$465	GP 4 vCore	\$368	\$186	\$155
P2 (250 DTU)	\$930	GP 8 vCore	\$736	\$372	\$310
P4 (500 DTU)	\$1,860	GP 16 vCore	\$1,471	\$743	\$619

The P2→GP8 + HB + 1-Yr RI move illustrated here is where the \$310/month number in the chapter opener comes from. The P4 → GP16 equivalent at 3-year RI with HB reaches approximately \$490/month versus \$1,860 — a \$1,370/month saving on a single database.

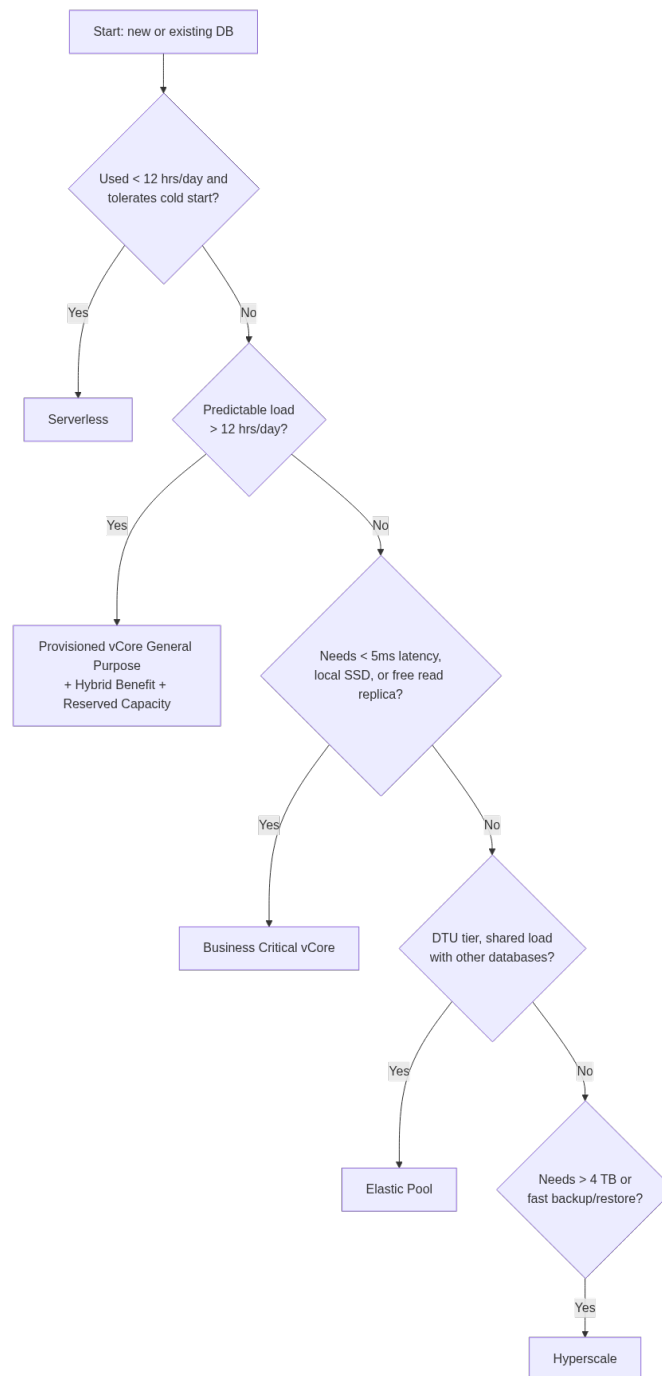


Figure 3: The Azure SQL tier decision framework, expressed as the actual flowchart the prose logic below describes.

3.4 Activating Azure Hybrid Benefit and Reserved Instances

The Azure Hybrid Benefit (AHB) and reserved instances are the two highest-leverage levers that require zero application change. Yet many organizations leave both on the table because no one on the database team owns the licensing conversation.

Enabling Hybrid Benefit via Azure CLI

```
#!/bin/bash
# Enable Azure Hybrid Benefit on an existing Azure SQL Database (vCore model)
# Prerequisites: SQL Server license with Software Assurance

RESOURCE_GROUP="prod-databases-rg"
SERVER_NAME="prod-sqlserver-eastus"
DATABASE_NAME="orders-db"

# Check current license type
az sql db show \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --query "{name:name, licenseType:licenseType, sku:sku}" \
  --output table

# Apply Hybrid Benefit (change licenseType from LicenseIncluded to BasePrice)
az sql db update \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --license-type BasePrice

# Verify the change
az sql db show \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --query "{name:name, licenseType:licenseType, edition:edition, capacity:currentServiceObjectiveName}" \
  --output table

# Apply to all databases on a server in bulk
echo "Applying Hybrid Benefit to all vCore databases on $SERVER_NAME..."
az sql db list \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --query "[?licenseType=='LicenseIncluded'].name" \
  --output tsv | while read db_name; do
    echo "Updating: $db_name"
    az sql db update \
      --resource-group "$RESOURCE_GROUP" \
      --server "$SERVER_NAME" \
      --name "$db_name" \
      --license-type BasePrice
done

echo "Done. Verify in Azure Portal under Cost Management."
```

Purchasing Reserved Instances

Reserved instances for Azure SQL are purchased through the Azure Portal under **Reservations**, not through the SQL Database blade itself. You reserve at the **vCore family and region level**, not at the individual database level. A 4-vCore General Purpose East US reservation covers any General Purpose 4-vCore database in East US, regardless of the server name. This means you can replace databases, rename them, or move workloads between databases, and the reservation

continues applying.

Reservation calculation before purchasing:

Before committing, verify utilization will remain stable. A 1-year commitment on compute that you later delete runs to waste. The break-even for a 1-year RI versus pay-as-you-go is approximately **5–6 months** of continuous operation. For databases you know will run for 12+ months, the 1-year RI is always the right choice unless the team plans a migration to a different service within the year.

3.5 Automating Serverless Right-Sizing with Python

The following Python script connects to Azure Monitor metrics and the Azure Cost Management API, evaluates each Azure SQL database in a subscription against Serverless break-even criteria, and outputs a prioritized recommendation report. Designed to run as a weekly Azure Function or a cron job on any Linux host.

```
#!/usr/bin/env python3
"""
azure_sql_rightsizing.py

Weekly right-sizing analysis for Azure SQL databases.
Compares current provisioned cost against Serverless equivalent.
Outputs a CSV report and optionally sends to Azure Blob Storage.

Requirements:
    pip install azure-identity azure-mgmt-sql azure-mgmt-monitor \
        azure-mgmt-costmanagement pandas

Usage:
    python azure_sql_rightsizing.py \
        --subscription-id <sub-id> \
        --output-path ./reports/sql_rightsizing.csv
"""

import argparse
import datetime
import json
import csv
import sys
from dataclasses import dataclass, field, asdict
from typing import Optional

from azure.identity import DefaultAzureCredential
from azure.mgmt.sql import SqlManagementClient
from azure.mgmt.monitor import MonitorManagementClient
from azure.mgmt.costmanagement import CostManagementClient
from azure.mgmt.costmanagement.models import (
    QueryDefinition, QueryTimePeriod, QueryDataset,
    QueryAggregation, QueryGrouping, TimeframeType
)

# -----
# Pricing constants (East US, USD, as of 2024 - verify current rates)
# -----
GP_VCORE_PAYG_HOURLY = 0.5042          # General Purpose per vCore per hour
GP_VCORE_HB_HOURLY = 0.2546           # With Hybrid Benefit
GP_VCORE_RI1YR_HB_HOURLY = 0.2121    # 1-Year RI + Hybrid Benefit
SERVERLESS_PER_VCORE_SECOND = 0.000145
STORAGE_PER_GB_MONTH = 0.115
AUTO_PAUSE_THRESHOLD_HOURS_PER_DAY = 12 # Serverless is cheaper if DB active < this
```

```

@dataclass
class DatabaseAnalysis:
    server_name: str
    database_name: str
    resource_group: str
    current_sku: str
    current_vcores: int
    license_type: str
    avg_cpu_pct: float
    p95_cpu_pct: float
    avg_active_hours_per_day: float
    storage_gb: float
    monthly_cost_current: float
    monthly_cost_serverless: float
    monthly_cost_provisioned_rib_hb: float
    recommendation: str
    monthly_savings: float
    annual_savings: float

def get_databases(sql_client: SqlManagementClient, subscription_id: str) -> list:
    """Return all Azure SQL databases (excluding system DBs) across all servers."""
    databases = []
    servers = list(sql_client.servers.list())

    for server in servers:
        rg = server.id.split("/")[4]
        try:
            dbs = sql_client.databases.list_by_server(rg, server.name)
            for db in dbs:
                if db.name in ("master",) or db.kind == "system":
                    continue
                databases.append((rg, server.name, db))
        except Exception as e:
            print(f" Warning: Could not list databases on {server.name}: {e}",
                  file=sys.stderr)

    return databases

def get_cpu_metrics(
    monitor_client: MonitorManagementClient,
    resource_id: str,
    days: int = 14
) -> dict:
    """Pull CPU utilization metrics from Azure Monitor for a database."""
    end_time = datetime.datetime.utcnow()
    start_time = end_time - datetime.timedelta(days=days)

    metrics = monitor_client.metrics.list(
        resource_uri=resource_id,
        timespan=f"{start_time.isoformat()}Z/{end_time.isoformat()}Z",
        interval="PT1H",
        metricnames="cpu_percent",
        aggregation="Average,Maximum",
    )

    hourly_values = []
    for metric in metrics.value:
        for ts in metric.timeseries:
            for point in ts.data:
                if point.average is not None:
                    hourly_values.append(point.average)

    if not hourly_values:
        return {
            "avg_cpu_pct": 0.0,
            "p95_cpu_pct": 0.0,

```

```

        "avg_active_hours_per_day": 0.0,
    }

    hourly_values.sort()
    p95_index = int(len(hourly_values) * 0.95)
    active_hours = sum(1 for v in hourly_values if v > 5.0) # > 5% = "active"

    return {
        "avg_cpu_pct": sum(hourly_values) / len(hourly_values),
        "p95_cpu_pct": hourly_values[p95_index],
        "avg_active_hours_per_day": active_hours / days,
    }

def estimate_monthly_cost_provisioned(
    vcores: int,
    license_type: str,
    storage_gb: float,
    use_ri_hb: bool = False,
) -> float:
    """Estimate monthly cost for a provisioned vCore database."""
    hours_per_month = 730.0

    if use_ri_hb:
        compute_hourly = vcores * GP_VCORE_RI1YR_HB_HOURLY
    elif license_type == "BasePrice":
        compute_hourly = vcores * GP_VCORE_HB_HOURLY
    else:
        compute_hourly = vcores * GP_VCORE_PAYG_HOURLY

    compute_monthly = compute_hourly * hours_per_month
    storage_monthly = max(storage_gb, 32) * STORAGE_PER_GB_MONTH # 32 GB minimum
    return compute_monthly + storage_monthly

def estimate_monthly_cost_serverless(
    vcores_max: int,
    avg_active_hours_per_day: float,
    storage_gb: float,
) -> float:
    """
    Estimate Serverless monthly cost assuming the database auto-pauses during idle hours.
    Uses max_vcores as the estimate for active compute (conservative).
    """
    seconds_per_month = 730 * 3600
    active_fraction = min(avg_active_hours_per_day / 24.0, 1.0)
    active_seconds_per_month = seconds_per_month * active_fraction

    compute_monthly = vcores_max * SERVERLESS_PER_VCORE_SECOND * active_seconds_per_month
    storage_monthly = max(storage_gb, 32) * STORAGE_PER_GB_MONTH
    return compute_monthly + storage_monthly

def make_recommendation(analysis_data: dict) -> tuple[str, float]:
    """
    Return (recommendation_string, monthly_savings).
    Priority order:
    1. Switch to Serverless (large idle period, tolerable cold start)
    2. Apply Hybrid Benefit (vCore, license not yet BasePrice)
    3. Apply 1-Year Reserved Instance + Hybrid Benefit
    4. No change needed
    """
    current_cost = analysis_data["monthly_cost_current"]
    serverless_cost = analysis_data["monthly_cost_serverless"]
    ri_hb_cost = analysis_data["monthly_cost_provisioned_ri_hb"]
    active_hours = analysis_data["avg_active_hours_per_day"]
    license_type = analysis_data["license_type"]
    sku = analysis_data["current_sku"]

    # Cannot apply Serverless to elastic pool members or Hyperscale

```

```

is_serverless_eligible = (
    "GP" in sku or "GeneralPurpose" in sku
) and active_hours < AUTO_PAUSE_THRESHOLD_HOURS_PER_DAY

if is_serverless_eligible and serverless_cost < current_cost * 0.7:
    savings = current_cost - serverless_cost
    return (
        f"MIGRATE_TO_SERVERLESS: estimated ${serverless_cost:.2f}/mo "
        f"(save ${savings:.2f}/mo)",
        savings,
    )

if ri_hb_cost < current_cost * 0.75:
    savings = current_cost - ri_hb_cost
    action = []
    if license_type != "BasePrice":
        action.append("APPLY_HYBRID_BENEFIT")
    action.append("BUY_1YR_RESERVED_INSTANCE")
    return (
        f"{'+'.join(action)}: estimated ${ri_hb_cost:.2f}/mo "
        f"(save ${savings:.2f}/mo)",
        savings,
    )

if license_type != "BasePrice" and "vCore" in sku.lower():
    hb_cost = estimate_monthly_cost_provisioned(
        analysis_data["current_vcores"],
        "BasePrice",
        analysis_data["storage_gb"],
        use_ri_hb=False,
    )
    if hb_cost < current_cost * 0.85:
        savings = current_cost - hb_cost
        return (
            f"APPLY_HYBRID_BENEFIT: estimated ${hb_cost:.2f}/mo "
            f"(save ${savings:.2f}/mo)",
            savings,
        )

return "NO_CHANGE_RECOMMENDED", 0.0

def analyze_subscription(subscription_id: str, output_path: str) -> None:
    credential = DefaultAzureCredential()
    sql_client = SqlManagementClient(credential, subscription_id)
    monitor_client = MonitorManagementClient(credential, subscription_id)

    print(f"Fetching databases in subscription {subscription_id}...")
    databases = get_databases(sql_client, subscription_id)
    print(f" Found {len(databases)} user databases.")

    results: list[DatabaseAnalysis] = []

    for rg, server_name, db in databases:
        print(f" Analyzing: {server_name}/{db.name} ({db.sku.name if db.sku else 'unknown SKU'})")

        # Extract current vCore count and license type
        vcores = db.sku.capacity if db.sku and db.sku.capacity else 2
        license_type = getattr(db, "license_type", "LicenseIncluded") or "LicenseIncluded"
        storage_gb = (db.max_size_bytes or 32 * 1024**3) / (1024**3)
        sku_name = db.sku.name if db.sku else "Unknown"

        # Fetch CPU metrics from Azure Monitor
        metrics = get_cpu_metrics(monitor_client, db.id, days=14)

        # Estimate costs
        current_cost = estimate_monthly_cost_provisioned(
            vcores, license_type, storage_gb, use_ri_hb=False
        )
        serverless_cost = estimate_monthly_cost_serverless(

```

```

        vcores, metrics["avg_active_hours_per_day"], storage_gb
    )
    ri_hb_cost = estimate_monthly_cost_provisioned(
        vcores, "BasePrice", storage_gb, use_ri_hb=True
    )

    analysis_input = {
        "monthly_cost_current": current_cost,
        "monthly_cost_serverless": serverless_cost,
        "monthly_cost_provisioned_ri_hb": ri_hb_cost,
        "avg_active_hours_per_day": metrics["avg_active_hours_per_day"],
        "license_type": license_type,
        "current_sku": sku_name,
        "current_vcores": vcores,
        "storage_gb": storage_gb,
    }

    recommendation, monthly_savings = make_recommendation(analysis_input)

    results.append(DatabaseAnalysis(
        server_name=server_name,
        database_name=db.name,
        resource_group=rg,
        current_sku=sku_name,
        current_vcores=vcores,
        license_type=license_type,
        avg_cpu_pct=round(metrics["avg_cpu_pct"], 2),
        p95_cpu_pct=round(metrics["p95_cpu_pct"], 2),
        avg_active_hours_per_day=round(metrics["avg_active_hours_per_day"], 2),
        storage_gb=round(storage_gb, 2),
        monthly_cost_current=round(current_cost, 2),
        monthly_cost_serverless=round(serverless_cost, 2),
        monthly_cost_provisioned_ri_hb=round(ri_hb_cost, 2),
        recommendation=recommendation,
        monthly_savings=round(monthly_savings, 2),
        annual_savings=round(monthly_savings * 12, 2),
    ))

    # Sort by annual savings descending
    results.sort(key=lambda x: x.annual_savings, reverse=True)

    # Write CSV report
    with open(output_path, "w", newline="") as f:
        writer = csv.DictWriter(f, fieldnames=list(asdict(results[0]).keys()) if results else [])
        writer.writeheader()
        for row in results:
            writer.writerow(asdict(row))

    total_monthly = sum(r.monthly_savings for r in results)
    total_annual = sum(r.annual_savings for r in results)
    actionable = sum(1 for r in results if r.monthly_savings > 0)

    print(f"\n{'='*60}")
    print(f"  Analysis complete. {len(results)} databases evaluated.")
    print(f"  Actionable recommendations: {actionable}")
    print(f"  Total potential monthly savings: ${total_monthly:,.2f}")
    print(f"  Total potential annual savings:  ${total_annual:,.2f}")
    print(f"  Report written to: {output_path}")
    print(f"\n{'='*60}\n")

    if __name__ == "__main__":
        parser = argparse.ArgumentParser(description="Azure SQL right-sizing analysis")
        parser.add_argument("--subscription-id", required=True)
        parser.add_argument("--output-path", default="./sql_rightsizing.csv")
        args = parser.parse_args()
        analyze_subscription(args.subscription_id, args.output_path)

```

Operational note: The pricing constants at the top of the script are point-in-time

estimates for East US. Pull current rates from the [Azure Pricing Calculator](#) or the Azure Retail Prices API (<https://prices.azure.com/api/retail/prices>) before running this in production. The script's structure separates pricing constants from logic precisely so they can be updated without touching the analysis code.

3.6 Migrating from DTU to vCore Without Downtime

The migration from DTU to vCore in Azure SQL is a metadata operation — it does not move data. Azure performs the migration online, and the database remains available throughout. The observable effect is a brief failover (typically under 30 seconds) at the end of the operation as connections are routed to the new compute tier.

Pre-Migration Checklist

Before initiating the tier change:

1. **Confirm the vCore count meets your workload requirement.** Run the utilization queries in Section 3.2. If P95 CPU utilization on a P1 (125 DTU) is 40%, a 2-vCore General Purpose is adequate; 4 vCores gives headroom.
2. **Check storage.** vCore General Purpose databases include storage billed separately at \$0.115/GB/month with a 32 GB minimum. Ensure the max size you set equals your actual data volume, not the inherited DTU default (which may be over-provisioned).
3. **Notify application teams of the failover window.** 30 seconds is brief but will drop active connections. Schedule during low-traffic periods or ensure your application uses retry logic (ADO.NET connection resiliency, SqlRetryLogicBaseProvider, or equivalent).
4. **Verify feature compatibility.** DTU Premium supports In-Memory OLTP (memory-optimized tables). vCore General Purpose does not. If your database uses `MEMORY_OPTIMIZED = ON` tables, migrate to Business Critical, not General Purpose.

Executing the Migration

```
#!/bin/bash
# Migrate a single Azure SQL database from DTU (Premium) to vCore General Purpose
# with Hybrid Benefit applied in a single operation.

RESOURCE_GROUP="prod-databases-rg"
SERVER_NAME="prod-sqlserver-eastus"
DATABASE_NAME="orders-db"

echo "Current configuration:"
az sql db show \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --query "{name:name, edition:edition, objective:currentServiceObjectiveName, licenseType:licenseType,
  ↪ maxSizeGb:maxSizeBytes}" \
  --output table

echo ""
echo "Initiating migration to vCore General Purpose 4 with Hybrid Benefit..."
echo "Note: A brief failover (~30 seconds) will occur during the transition."
```

```

az sql db update \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --edition "GeneralPurpose" \
  --family "Gen5" \
  --capacity 4 \
  --license-type "BasePrice" \
  --max-size "250GB"

echo ""
echo "Migration complete. Verifying new configuration:"
az sql db show \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --query "{name:name, edition:edition, objective:currentServiceObjectiveName, licenseType:licenseType,
  ↪ vcores:currentSku.capacity}" \
  --output table

```

For Managed Instance migrations (which are more complex and involve compute tier changes at the instance level), the pattern is similar but uses `az sql mi update` with `--service-principal-type` and `--tier` flags. Managed Instance migrations are covered in Chapter 6.

Post-Migration Validation

After migration, run the following query to confirm the database is operating on the new tier and that performance is acceptable:

```

-- Post-migration health check
-- Run immediately after migration and again 24 hours later

SELECT
  DB_NAME() AS [Database],
  DATABASEPROPERTYEX(DB_NAME(), 'ServiceObjective') AS [Current_SKU],
  DATABASEPROPERTYEX(DB_NAME(), 'Edition') AS [Edition],
  -- Check for any resource waits that indicate underpowering
  wait_type,
  waiting_tasks_count,
  wait_time_ms / 1000.0 AS wait_time_sec,
  signal_wait_time_ms / 1000.0 AS signal_wait_sec,
  (wait_time_ms - signal_wait_time_ms) / 1000.0 AS resource_wait_sec
FROM sys.dm_os_wait_stats
WHERE wait_type NOT IN (
  -- Filter out benign wait types
  'SLEEP_TASK', 'BROKER_TO_FLUSH', 'BROKER_TASK_STOP',
  'CLR_AUTO_EVENT', 'DISPATCHER_QUEUE_SEMAPHORE',
  'FT_IFTS_SCHEDULER_IDLE_WAIT', 'HADR_FILESTREAM_IOMGR_IOCOMPLETION',
  'HADR_WORK_QUEUE', 'LAZYWRITER_SLEEP', 'LOGMGR_QUEUE',
  'ONDEMAND_TASK_QUEUE', 'REQUEST_FOR_DEADLOCK_SEARCH',
  'RESOURCE_QUEUE', 'SERVER_IDLE_CHECK', 'SLEEP_DBSTARTUP',
  'SLEEP_DBRECOVER', 'SLEEP_DBTASK', 'SLEEP_MASTERDBREADY',
  'SLEEP_MASTERMDREADY', 'SLEEP_MASTERUPGRADED', 'SLEEP_MSDBSTARTUP',
  'SLEEP_SYSTEMTASK', 'SLEEP_TEMPDBSTARTUP', 'SNI_HTTP_ACCEPT',
  'SP_SERVER_DIAGNOSTICS_SLEEP', 'SQLTRACE_BUFFER_FLUSH',
  'SQLTRACE_INCREMENTAL_FLUSH_SLEEP', 'WAIT_XTP_OFFLINE_CKPT_NEW_LOG',
  'WAITFOR', 'XE_DISPATCHER_WAIT', 'XE_TIMER_EVENT',
  'BROKER_EVENTHANDLER', 'CHECKPOINT_QUEUE', 'DBMIRROR_EVENTS_QUEUE',
  'SQLTRACE_WAIT_ENTRIES'
)
AND waiting_tasks_count > 0
ORDER BY wait_time_ms DESC;

```

High `RESOURCE_SEMAPHORE` or `PAGEIOLATCH_SH` waits after migration indicate memory or I/O pressure and should trigger an upsize or a query tuning review before concluding the migration is cost-optimal.

3.7 Configuring Serverless Auto-Pause for Non-Production Databases

Development, staging, and integration test databases are the most reliable candidates for Serverless. They typically follow strict business-hours usage patterns, developers tolerate cold-start latency, and they accumulate cost around the clock despite sitting idle nights and weekends.

Setting Up Serverless via Azure CLI

```
#!/bin/bash
# Convert a provisioned General Purpose database to Serverless
# Best suited for dev/test databases with predictable idle windows

RESOURCE_GROUP="dev-databases-rg"
SERVER_NAME="dev-sqlserver-eastus"
DATABASE_NAME="integration-test-db"

# Step 1: Confirm current configuration
echo "Current configuration:"
az sql db show \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --query "{edition:edition, objective:currentServiceObjectiveName, minCapacity:minCapacity,
  ↪ autoPauseDelay:autoPauseDelay}" \
  --output table

# Step 2: Convert to Serverless
# --auto-pause-delay: minutes before auto-pause (-1 = disabled, 60 minimum when enabled)
# --min-capacity: minimum vCores when active (0.5 is the lowest)
# --capacity: maximum vCores

az sql db update \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME" \
  --edition "GeneralPurpose" \
  --family "Gen5" \
  --compute-model Serverless \
  --min-capacity 0.5 \
  --capacity 4 \
  --auto-pause-delay 60

echo ""
echo "Serverless configuration applied."

# Step 3: For production-adjacent staging environments, use a longer pause delay
# to prevent cold starts during occasional off-hours test runs

DATABASE_NAME_STAGING="staging-db"
az sql db update \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --name "$DATABASE_NAME_STAGING" \
  --edition "GeneralPurpose" \
  --family "Gen5" \
  --compute-model Serverless \
  --min-capacity 1 \
  --capacity 8 \
  --auto-pause-delay 120

echo "Staging Serverless configuration applied (120-minute auto-pause delay)."
```

```
# Step 4: Verify both databases
az sql db list \
  --resource-group "$RESOURCE_GROUP" \
  --server "$SERVER_NAME" \
  --query "[?computeModel=='Serverless'].{name:name, minCap:minCapacity, maxCap:currentSku.capacity,
  ↪  pauseDelay:autoPauseDelay}" \
  --output table
```

Monitoring Serverless Pause State

A database in the paused state does not appear in `sys.dm_db_resource_stats` because the compute is not running. Use Azure Monitor or the management API to determine pause state:

```
# Check whether a Serverless database is currently paused
az sql db show \
  --resource-group "dev-databases-rg" \
  --server "dev-sqlserver-eastus" \
  --name "integration-test-db" \
  --query "{name:name, status:status, pausedDate:pausedDate}" \
  --output table
```

The `status` field returns `Online`, `Paused`, or `Pausing`. Track the proportion of time a database spends in `Paused` state over a month to validate your Serverless savings assumption — if it never pauses, the auto-pause delay may be set too long or the database is receiving constant low-level connections (health checks, monitoring agents) that prevent pausing. Identify and suppress those connections, or accept that this database is not a good Serverless candidate.

3.8 Elastic Pools as a Cost Alternative to Individual Sizing

When you have multiple databases with complementary usage patterns — some busy while others are idle — an Elastic Pool lets them share a vCore (or DTU) budget, smoothing out peaks through aggregation. A pool of 20 databases that individually require 4 vCores at peak but whose peaks do not overlap can often run on a 16-vCore pool, producing a 20% reduction in total compute cost.

Elastic Pool sizing and management is covered in depth in Chapter 4. However, one question is worth answering here: **should you migrate individual databases to vCore first, then pool them, or pool them before the DTU-to-vCore migration?**

The answer depends on operational complexity. If more than five databases are involved and they already share a logical server, migrate them into a vCore Elastic Pool in a single operation — it avoids the intermediate step. For databases spread across multiple logical servers, migrate to standalone vCore first (as described in Section 3.6), then consolidate servers and add the pool. Attempting both moves simultaneously increases rollback complexity.

Key Takeaways

- **DTU opacity is a cost trap.** The bundled DTU metric prevents you from knowing whether your bottleneck is CPU, memory, or I/O — and therefore whether you're buying the right resources. Moving to vCore gives you observable dimensions, Hybrid Benefit eligibility, and reserved instance pricing, all of which the DTU model blocks.

- **Azure Hybrid Benefit and reserved instances are independent, stackable discounts that require no code changes.** Applied together on a 1-year commitment, they typically reduce vCore compute cost by 55–60% compared to DTU pay-as-you-go. Every vCore database that carries `LicenseIncluded` without a justification is burning money.
 - **Serverless is the default choice for development and test databases.** Any database that sits idle for more than 12 hours per day — and can tolerate a 30–90 second cold start — will cost less on Serverless than on the smallest provisioned vCore tier. The break-even is roughly 50% average utilization over a full month.
 - **DTU-to-vCore migration is online and takes under 30 seconds of failover.** The operational risk is low when retry logic is in place. The utilization queries in Section 3.2 provide the data you need to right-size the vCore count before migrating rather than after.
 - **The Python right-sizing script in Section 3.5 turns a manual audit into a repeatable weekly process.** Run it as an Azure Function on a Monday morning cadence, write the CSV to Blob Storage, and route high-savings recommendations to a Teams or Slack channel. The compounding effect of catching new databases early — before they run for 12 months at the wrong tier — is where most of the annual savings accumulates.
-

Chapter 4: Google Cloud SQL and AlloyDB Pricing Deep Dive

Organizations running production workloads on Google Cloud SQL and AlloyDB routinely overspend by 35–55% compared to their optimized baseline. A mid-sized engineering team running three Cloud SQL for PostgreSQL instances—a db-custom-8-32768 primary, a read replica, and a dev instance left running around the clock—can easily accumulate \$4,200/month in avoidable charges before anyone notices. This chapter breaks down exactly how Google prices Cloud SQL and AlloyDB, where the hidden costs live, and how to systematically eliminate waste. By the end, a disciplined implementation of the techniques here can recover \$1,500–\$8,000/month for a typical team running five to fifteen database instances.

4.1 How Google Cloud SQL and AlloyDB Pricing Actually Works

Understanding the billing model is the first step to controlling it. Cloud SQL and AlloyDB look similar on the surface but have meaningfully different pricing structures that affect optimization strategy.

Cloud SQL Pricing Components

Cloud SQL bills across five independent dimensions:

1. **Instance (vCPU + RAM)** — billed per second, based on machine type
2. **Storage** — billed per GB/month, separate from compute
3. **Networking** — egress to other regions, internet, or cross-zone replicas
4. **Backups** — beyond the free tier (equal to instance storage size)
5. **IP address** — \$0.01/hour for a public IPv4 address when the instance is running

For a db-custom-8-32768 instance (8 vCPU, 32 GB RAM) running PostgreSQL 15 in us-central1, the on-demand pricing breaks down as:

Component	Unit Price	Example Usage	Monthly Cost
vCPU	\$0.0413/vCPU/hour	8 vCPU × 730h	\$241.59
RAM	\$0.0070/GB/hour	32 GB × 730h	\$163.84
SSD Storage	\$0.170/GB/month	500 GB	\$85.00
Backup storage	\$0.08/GB/month	200 GB over free	\$16.00
IPv4 address	\$0.01/hour	730h	\$7.30

Component	Unit Price	Example Usage	Monthly Cost
Total			\$513.73

Add a same-region read replica with identical specs and you're at \$1,027.46/month before any data egress. That's a useful baseline to anchor your optimization work.

Cloud SQL Committed Use Discounts (CUDs)

Google offers 1-year and 3-year committed use discounts for Cloud SQL compute (not storage):

- **1-year CUD:** 25% discount on vCPU and RAM
- **3-year CUD:** 52% discount on vCPU and RAM

For the 8 vCPU, 32 GB instance above, a 1-year CUD reduces the compute portion from \$405.43 to \$304.07—a savings of \$101.36/month or \$1,216.32/year. A 3-year CUD drops it to \$194.61/month, saving \$210.82/month or \$2,529.84/year.

Storage, networking, backups, and IP addresses are excluded from CUDs and billed at on-demand rates regardless.

AlloyDB Pricing Components

AlloyDB uses a fundamentally different model that trips up teams migrating from Cloud SQL:

1. **Cluster compute** — billed per vCPU/hour for the primary and each read pool instance
2. **Storage** — a unified, automatically managed storage layer billed per GB/month across the cluster
3. **I/O requests** — billed separately at \$0.10 per million read/write operations (this is the most common surprise cost)
4. **Backups** — \$0.023/GB/month for automated backups beyond the database size

The I/O charge is what catches teams off guard. A busy OLTP database performing 500 million I/O operations per day generates \$1,500/month in I/O costs alone. AlloyDB's storage is intelligent—it caches aggressively and reduces physical I/O compared to Cloud SQL—but the billing unit is logical I/O at the application layer, not physical disk operations.

AlloyDB compute pricing in `us-central1`: - vCPU: \$0.0722/vCPU/hour - RAM: \$0.00963/GB/hour

A 16-vCPU, 128 GB AlloyDB primary instance costs approximately \$1,743/month in compute before storage or I/O. AlloyDB CUDs provide 25% (1-year) and 52% (3-year) discounts on compute only.

The Right Tool for the Right Workload

Before optimizing, confirm you're on the right product:

- **Cloud SQL** is cost-effective for standard OLTP workloads under ~5,000 QPS with moderate I/O
- **AlloyDB** pays for itself on analytical and mixed OLAP/OLTP workloads where its columnar engine eliminates full table scans, or for workloads requiring 4× faster reads than Cloud SQL
- Running AlloyDB for a simple 200 QPS web application backend will cost 2–3× more than Cloud SQL with no performance benefit

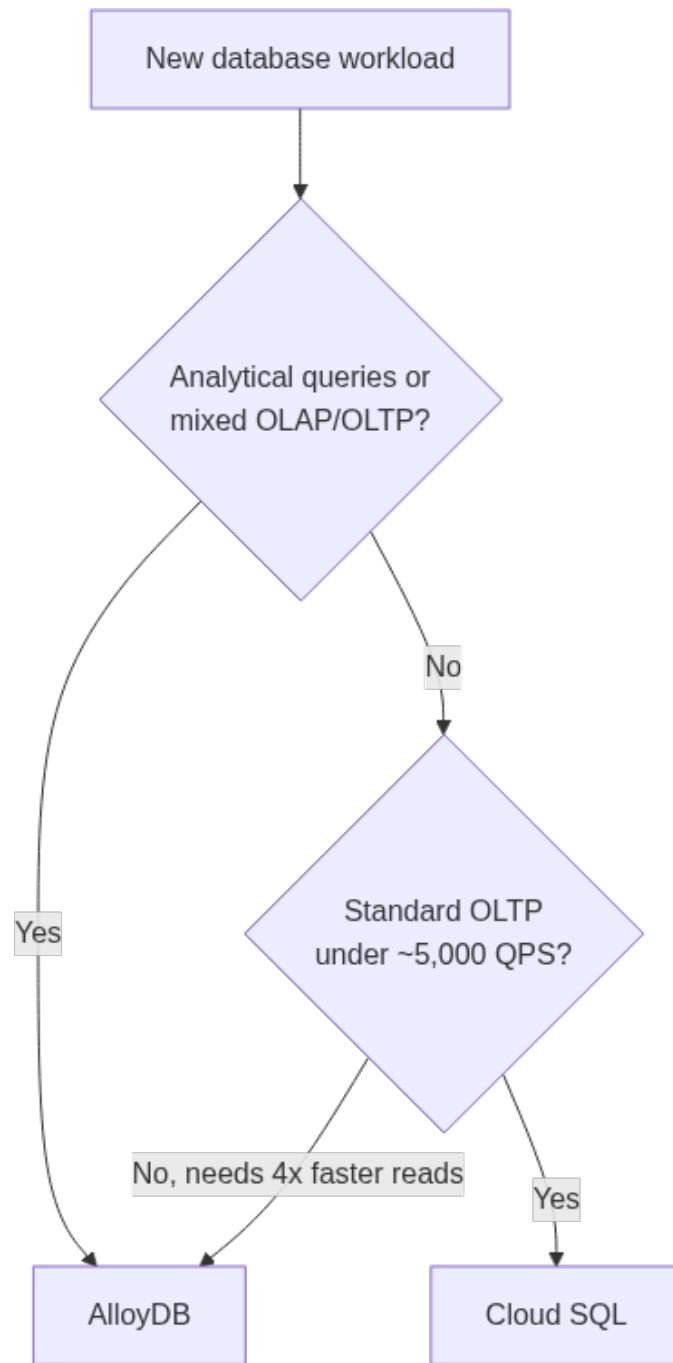


Figure 4: Choosing between Cloud SQL and AlloyDB before any cost optimization begins — picking the wrong product costs more than any tuning saves.

4.2 Diagnosing Where Your Money Is Going

Before making changes, establish a clear picture of current spend by instance, component, and time pattern. The GCP Cost Table in BigQuery is the most granular source of billing data available.

Setting Up BigQuery Billing Export

If you haven't already enabled billing export to BigQuery, do it now—you need this data for everything that follows.

```
# Enable billing export to BigQuery
# First, create a dataset in BigQuery to receive billing data
bqcloud bigquery datasets create billing_export \
  --location=US \
  --description="GCP billing export for FinOps analysis"

# Enable billing export via gcloud (requires billing account admin role)
gcloud billing accounts list

# Note your BILLING_ACCOUNT_ID from the output, then enable export
# This must be done via the console UI or Terraform - gcloud does not
# support direct export configuration via CLI as of 2024
# Navigate to: Billing > Billing export > BigQuery export > Edit settings

# Verify the export is active by querying the dataset after 24 hours
bq query --use_legacy_sql=false \
  'SELECT COUNT(*) as row_count, MIN(usage_start_time) as earliest_data
  FROM `YOUR_PROJECT.billing_export.gcp_billing_export_v1_XXXXXX_XXXXXX_XXXXXX`
  LIMIT 1'
```

Once billing data is flowing into BigQuery, the following queries give you a complete cost map.

Query 1: Cloud SQL Cost Breakdown by Instance and Component

```
-- Run this in BigQuery against your billing export table
-- Replace the table name with your actual export table
WITH cloud_sql_costs AS (
  SELECT
    -- Extract instance name from resource labels
    resource.labels['database_id'] AS instance_id,
    resource.labels['region'] AS region,
    sku.description AS cost_component,

    -- Categorize cost types for easier filtering
    CASE
      WHEN sku.description LIKE '%vCPU%' THEN 'Compute-vCPU'
      WHEN sku.description LIKE '%RAM%' THEN 'Compute-RAM'
      WHEN sku.description LIKE '%Storage%' THEN 'Storage'
      WHEN sku.description LIKE '%Backup%' THEN 'Backup'
      WHEN sku.description LIKE '%Network%' THEN 'Network'
      WHEN sku.description LIKE '%IP%' THEN 'IP-Address'
      ELSE 'Other'
    END AS cost_category,

    SUM(cost) AS total_cost_usd,
    SUM(usage.amount) AS total_usage,
    usage.unit AS usage_unit,

    -- Month for trending
    FORMAT_DATE('%Y-%m', DATE(usage_start_time)) AS billing_month
  FROM `YOUR_PROJECT.billing_export.gcp_billing_export_v1_XXXXXX`
  WHERE
    service.description = 'Cloud SQL'
```



```

        AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
        AND cost > 0
    GROUP BY 1, 2, 3, 4, 9, 10
),

-- Pivot to show monthly trend per instance
instance_summary AS (
    SELECT
        instance_id,
        region,
        cost_category,
        billing_month,
        ROUND(total_cost_usd, 2) AS monthly_cost_usd
    FROM cloud_sql_costs
)

SELECT
    instance_id,
    region,
    cost_category,
    SUM(CASE WHEN billing_month = FORMAT_DATE('%Y-%m', DATE_SUB(CURRENT_DATE(), INTERVAL 1 MONTH))
        THEN monthly_cost_usd ELSE 0 END) AS last_month_usd,
    SUM(CASE WHEN billing_month = FORMAT_DATE('%Y-%m', DATE_SUB(CURRENT_DATE(), INTERVAL 2 MONTH))
        THEN monthly_cost_usd ELSE 0 END) AS two_months_ago_usd,
    SUM(CASE WHEN billing_month = FORMAT_DATE('%Y-%m', DATE_SUB(CURRENT_DATE(), INTERVAL 3 MONTH))
        THEN monthly_cost_usd ELSE 0 END) AS three_months_ago_usd
FROM instance_summary
GROUP BY 1, 2, 3
ORDER BY last_month_usd DESC;

```

Query 2: AlloyDB I/O Cost Analysis

AlloyDB I/O costs require separate tracking because they scale with query volume, not instance size. This query identifies your I/O spend trend and flags instances where I/O exceeds 30% of total cluster cost—a threshold that indicates optimization opportunity.

```

-- BigQuery query to analyze AlloyDB I/O costs
WITH alloydb_costs AS (
    SELECT
        resource.labels['cluster_id'] AS cluster_id,
        resource.labels['instance_id'] AS instance_id,
        resource.labels['region'] AS region,
        sku.description AS sku_description,

        CASE
            WHEN sku.description LIKE '%Read IO%' THEN 'IO-Read'
            WHEN sku.description LIKE '%Write IO%' THEN 'IO-Write'
            WHEN sku.description LIKE '%vCPU%' THEN 'Compute-vCPU'
            WHEN sku.description LIKE '%Memory%' THEN 'Compute-RAM'
            WHEN sku.description LIKE '%Storage%' THEN 'Storage'
            WHEN sku.description LIKE '%Backup%' THEN 'Backup'
            ELSE 'Other'
        END AS cost_category,

        FORMAT_DATE('%Y-%m', DATE(usage_start_time)) AS billing_month,
        SUM(cost) AS total_cost_usd,
        SUM(usage.amount) AS total_usage_units
    FROM `YOUR_PROJECT.billing_export.gcp_billing_export_v1_XXXXXX`
    WHERE
        service.description = 'AlloyDB for PostgreSQL'
        AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
    GROUP BY 1, 2, 3, 4, 5, 6
),

cluster_totals AS (
    SELECT
        cluster_id,
        billing_month,

```

```

        SUM(total_cost_usd) AS cluster_total_cost
    FROM alloydb_costs
    GROUP BY 1, 2
),

io_costs AS (
    SELECT
        cluster_id,
        billing_month,
        SUM(total_cost_usd) AS io_total_cost
    FROM alloydb_costs
    WHERE cost_category IN ('IO-Read', 'IO-Write')
    GROUP BY 1, 2
)

SELECT
    a.cluster_id,
    a.billing_month,
    ROUND(c.cluster_total_cost, 2)      AS cluster_total_usd,
    ROUND(i.io_total_cost, 2)          AS io_cost_usd,
    ROUND(
        SAFE_DIVIDE(i.io_total_cost, c.cluster_total_cost) * 100, 1
    )                                  AS io_pct_of_total,
    CASE
        WHEN SAFE_DIVIDE(i.io_total_cost, c.cluster_total_cost) > 0.30
        THEN 'REVIEW: High I/O ratio - check for missing indexes or full scans'
        ELSE 'OK'
    END                                AS recommendation
FROM alloydb_costs a
JOIN cluster_totals c USING (cluster_id, billing_month)
JOIN io_costs i USING (cluster_id, billing_month)
GROUP BY 1, 2, 3, 4, 5, 6
ORDER BY billing_month DESC, cluster_total_usd DESC;

```

Identifying Idle and Underutilized Instances

Cloud Monitoring provides the CPU and connection utilization data needed to flag idle instances. An instance consuming less than 5% average CPU over 30 days with fewer than 5 concurrent connections is a strong candidate for deletion or significant downsizing.

```

#!/bin/bash
# Script: find_idle_cloud_sql_instances.sh
# Scans all Cloud SQL instances in a project for low CPU utilization
# Schedule this with Cloud Scheduler weekly

PROJECT_ID="${1:-$(gcloud config get-value project)}"
LOOKBACK_DAYS=30
CPU_THRESHOLD=5 # Flag instances below 5% average CPU
CONN_THRESHOLD=5 # Flag instances with fewer than 5 avg connections

echo "Analyzing Cloud SQL instances in project: $PROJECT_ID"
echo "Lookback period: $LOOKBACK_DAYS days"
echo "----"

# Get all Cloud SQL instances
INSTANCES=$(gcloud sql instances list \
  --project="$PROJECT_ID" \
  --format="value(name,region,databaseVersion,settings.tier)" \
  --filter="state=RUNNABLE")

echo "Instance,Region,Engine,Tier,Avg_CPU_Pct,Avg_Connections,Status"

while IFS=$'\t' read -r INSTANCE REGION ENGINE TIER; do
    # Query Cloud Monitoring for average CPU over lookback period
    AVG_CPU=$(gcloud monitoring read \
      "metric.type=cloudsql.googleapis.com/database/cpu/utilization" \
      resource.labels.database_id="$PROJECT_ID:$INSTANCE" \
      --freshness="$((LOOKBACK_DAYS * 24))h" \
      --format="value(points[0].value.doubleValue)" \

```

```

2>/dev/null | awk '{sum+=$1; count++;} END {printf "%.1f", (sum/count)*100}'

# Query average connections
AVG_CONN=$(gcloud monitoring read \
  "metric.type=\"cloudsql.googleapis.com/database/network/connections\" \
  resource.labels.database_id=\"$PROJECT_ID:$INSTANCE\" \" \" \
  --freshness=\"$((LOOKBACK_DAYS * 24))h\" \
  --format="value(points[0].value.int64Value)" \
  2>/dev/null | awk '{sum+=$1; count++;} END {printf "%.0f", sum/count}')

# Default to 0 if metric fetch failed
AVG_CPU=${AVG_CPU:-0}
AVG_CONN=${AVG_CONN:-0}

# Classify the instance
if (( $(echo "$AVG_CPU < $CPU_THRESHOLD" | bc -l) )) && \
  (( AVG_CONN < CONN_THRESHOLD )); then
  STATUS="IDLE - CANDIDATE FOR DELETION OR STOP"
elif (( $(echo "$AVG_CPU < 20" | bc -l) )); then
  STATUS="UNDERUTILIZED - CONSIDER DOWNSIZING"
else
  STATUS="ACTIVE"
fi

echo "$INSTANCE,$REGION,$ENGINE,$TIER,${AVG_CPU}%, $AVG_CONN, $STATUS"

done <<< "$INSTANCES"

```

4.3 Right-Sizing Cloud SQL Instances

Right-sizing is consistently the largest single savings lever available—larger than CUDs, larger than storage optimization, and larger than any architectural change. The reason is straightforward: most teams provision for peak capacity that never materializes, then leave those instances running at 8–15% CPU for months.

The Cloud SQL Machine Type Landscape

Cloud SQL offers three tiers:

- **Shared core** (db-f1-micro, db-g1-small): Dev/test only, not suitable for production
- **Standard machine types** (db-n1-standard-*, db-n2-standard-*): Fixed vCPU/RAM ratios
- **Custom machine types** (db-custom-N-M): Choose any vCPU count and RAM independently

Custom machine types are almost always cheaper than standard types for real workloads because you pay only for what you configure. A db-custom-4-15360 (4 vCPU, 15 GB RAM) costs less than db-n1-standard-4 (4 vCPU, 15 GB RAM) in most regions because the custom type uses N2 generation pricing which is more cost-efficient per unit.

The Memory-to-vCPU Ratio Decision

The most common right-sizing mistake is provisioning too much RAM relative to vCPU count, or vice versa. Use this framework:

Workload Pattern	Recommended Ratio	Example
OLTP, high concurrency	4–8 GB RAM per vCPU	db-custom-8-32768
Analytics, large datasets	8–16 GB RAM per vCPU	db-custom-4-61440
Connection-heavy, small queries	2–4 GB RAM per vCPU	db-custom-16-32768
Batch processing	4 GB RAM per vCPU	db-custom-4-16384

Querying Actual Resource Utilization from pg_stat

Before recommending a right-size target, pull actual utilization data from the database itself:

```
-- Run directly on your Cloud SQL PostgreSQL instance
-- This query gives a complete utilization picture for right-sizing decisions

WITH
-- Memory pressure assessment
memory_stats AS (
  SELECT
    name,
    setting,
    unit,
    CASE name
      WHEN 'shared_buffers' THEN setting::bigint * 8192
      WHEN 'work_mem' THEN setting::bigint * 1024
      WHEN 'maintenance_work_mem' THEN setting::bigint * 1024
      WHEN 'effective_cache_size' THEN setting::bigint * 8192
    END AS bytes
  FROM pg_settings
  WHERE name IN (
    'shared_buffers', 'work_mem',
    'maintenance_work_mem', 'effective_cache_size',
    'max_connections'
  )
),

-- Buffer cache hit ratio (lower than 95% suggests memory pressure)
cache_stats AS (
  SELECT
    sum(heap_blks_read) AS heap_disk_reads,
    sum(heap_blks_hit) AS heap_cache_hits,
    ROUND(
      100.0 * sum(heap_blks_hit) /
      NULLIF(sum(heap_blks_hit) + sum(heap_blks_read), 0), 2
    ) AS cache_hit_ratio_pct
  FROM pg_statio_user_tables
),

-- Active connections vs. configured max
connection_stats AS (
  SELECT
    count(*) AS total_connections,
    count(*) FILTER (WHERE state = 'active') AS active_connections,
    count(*) FILTER (WHERE state = 'idle') AS idle_connections,
    count(*) FILTER (WHERE state = 'idle in transaction') AS idle_in_txn,
    max(now() - query_start) FILTER (WHERE state = 'active') AS longest_active_query,
    (SELECT setting::int FROM pg_settings WHERE name = 'max_connections') AS max_connections_cfg
  FROM pg_stat_activity
  WHERE pid <> pg_backend_pid()
),

-- Top CPU-consuming queries (requires pg_stat_statements)
top_queries AS (
  SELECT
    LEFT(query, 80) AS query_preview,
```

```

calls,
ROUND(total_exec_time::numeric / 1000, 2) AS total_exec_sec,
ROUND(mean_exec_time::numeric, 2)      AS avg_exec_ms,
ROUND(stddev_exec_time::numeric, 2)     AS stddev_exec_ms,
rows
FROM pg_stat_statements
ORDER BY total_exec_time DESC
LIMIT 5
)

-- Combined output
SELECT 'CACHE HIT RATIO'      AS metric_type,
       cache_hit_ratio_pct::text AS value,
       CASE WHEN cache_hit_ratio_pct < 95
            THEN 'WARNING: Consider increasing shared_buffers or instance RAM'
            ELSE 'OK: Cache hit ratio is healthy'
       END                    AS recommendation
FROM cache_stats

UNION ALL

SELECT 'CONNECTION UTILIZATION',
       ROUND(100.0 * total_connections / max_connections_cfg, 1)::text || '%',
       CASE
         WHEN 100.0 * total_connections / max_connections_cfg > 80
         THEN 'WARNING: Consider pgBouncer or increasing max_connections'
         WHEN idle_in_txn > 10
         THEN 'WARNING: ' || idle_in_txn || ' idle-in-transaction connections - check for connection
              ↳ leaks'
         ELSE 'OK'
       END
FROM connection_stats

UNION ALL

SELECT 'IDLE CONNECTIONS',
       idle_connections::text,
       CASE WHEN idle_connections > 50
            THEN 'COST ALERT: ' || idle_connections ||
              ' idle connections consuming RAM. Deploy pgBouncer to reduce instance RAM needs.'
            ELSE 'OK'
       END
FROM connection_stats;

```

Calculating Right-Size Savings

Here is the actual math for a common right-sizing scenario. Suppose Cloud Monitoring shows your db-custom-8-32768 instance averages 18% CPU and 12 GB RAM in use over 30 days. The right-size target is db-custom-4-16384:

Metric	Current	Right-Sized	Savings
vCPU	$8 \times \$0.0413 \times 730\text{h}$	$4 \times \$0.0413 \times 730\text{h}$	\$120.79/mo
RAM	$32 \text{ GB} \times \$0.007 \times 730\text{h}$	$16 \text{ GB} \times \$0.007 \times 730\text{h}$	\$81.92/mo
Monthly savings	\$513.73	\$310.01	\$203.72/mo
Annual savings			\$2,444.64

Apply a 1-year CUD to the right-sized instance — remembering that the discount applies only to the vCPU and RAM portion, not storage, backup, or IP — and the monthly cost drops further to \$259.58, for a total monthly savings of \$254.15 versus the original on-demand oversized instance.

Stopping Dev Instances After Hours

A dev instance running 24/7 versus running 10 hours/day on weekdays only costs 3× more than necessary. Cloud SQL supports instance start/stop via the API. A db-custom-4-16384 dev instance at on-demand pricing costs approximately \$202/month running continuously. Stopping it outside working hours—say, 7 AM to 7 PM Monday through Friday—reduces runtime from 730 hours to roughly 260 hours per month, cutting the compute bill to about \$72/month. Storage still accrues while the instance is stopped, but that’s a small fraction of total cost.

```
#!/bin/bash
# Script: schedule_dev_instance_stop_start.sh
# Deploy this via Cloud Scheduler + Cloud Run Jobs or Cloud Functions
# Two jobs: one to start (07:00 Mon-Fri), one to stop (19:00 Mon-Fri)

PROJECT_ID="${GOOGLE_CLOUD_PROJECT}"
INSTANCE_NAME="${CLOUD_SQL_INSTANCE}"
ACTION="${1}" # "start" or "stop"

if [[ -z "$PROJECT_ID" || -z "$INSTANCE_NAME" || -z "$ACTION" ]]; then
    echo "ERROR: Required environment variables not set"
    exit 1
fi

case "$ACTION" in
    start)
        echo "Starting Cloud SQL instance: $INSTANCE_NAME"
        gcloud sql instances patch "$INSTANCE_NAME" \
            --project="$PROJECT_ID" \
            --activation-policy=ALWAYS
        echo "Instance start initiated at $(date -u)"
        ;;
    stop)
        echo "Stopping Cloud SQL instance: $INSTANCE_NAME"
        gcloud sql instances patch "$INSTANCE_NAME" \
            --project="$PROJECT_ID" \
            --activation-policy=NEVER
        echo "Instance stop initiated at $(date -u)"
        ;;
    *)
        echo "ERROR: Unknown action '$ACTION'. Use 'start' or 'stop'."
        exit 1
        ;;
esac

# Verify the state change
sleep 10
STATE=$(gcloud sql instances describe "$INSTANCE_NAME" \
    --project="$PROJECT_ID" \
    --format="value(state)")
echo "Current instance state: $STATE"
```

To wire this up with Cloud Scheduler:

```
# Create a Cloud Scheduler job to stop dev instances at 19:00 Mon-Fri
gcloud scheduler jobs create http stop-dev-cloudsql \
    --location=us-central1 \
    --schedule="0 19 * * 1-5" \
    --time-zone="America/Chicago" \
    --uri="https://YOUR_CLOUD_RUN_JOB_TRIGGER_URL" \
    --message-body='{"action":"stop","instance":"your-dev-instance"}' \
    --oidc-service-account-email="cloudsql-scheduler@YOUR_PROJECT.iam.gserviceaccount.com"

# Create a Cloud Scheduler job to start dev instances at 07:00 Mon-Fri
gcloud scheduler jobs create http start-dev-cloudsql \
    --location=us-central1 \
    --schedule="0 7 * * 1-5" \
    --time-zone="America/Chicago" \
    --uri="https://YOUR_CLOUD_RUN_JOB_TRIGGER_URL" \
    --message-body='{"action":"start","instance":"your-dev-instance"}' \
    --oidc-service-account-email="cloudsql-scheduler@YOUR_PROJECT.iam.gserviceaccount.com"
```

Two cautions when implementing instance scheduling: first, Cloud SQL does not gracefully drain connections on stop—applications need to handle connection errors with retry logic. Second, Cloud SQL charges for storage while stopped, so a stopped 500 GB instance still accrues \$85/month in storage costs. For truly long-lived dev instances that are rarely used, consider exporting the data, deleting the instance, and restoring from the export on demand.

4.4 Storage Optimization for Cloud SQL and AlloyDB

Storage costs are easy to dismiss as a rounding error, but they compound in two ways that make them worth managing: Cloud SQL storage only grows automatically (it never shrinks), and backup storage accumulates silently beyond the free tier threshold.

The Automatic Storage Increase Trap

Cloud SQL’s “Enable automatic storage increases” setting is enabled by default and is appropriate for production instances. The problem is that it only triggers upward—there is no automatic decrease. A database that grew to 800 GB during a bulk load six months ago, then had most data purged, still shows 800 GB of provisioned storage at \$136/month even if only 200 GB is actually occupied.

To recover over-provisioned storage, you must export the data, delete the instance, and recreate it at the correct size. This is disruptive and requires a maintenance window, so do the math before committing:

```
-- PostgreSQL: Measure actual database sizes vs. provisioned storage
-- Run on Cloud SQL PostgreSQL to understand actual storage consumption

SELECT
    datname                                AS database_name,
    pg_size_pretty(pg_database_size(datname)) AS database_size,
    pg_database_size(datname)                AS size_bytes,
    ROUND(pg_database_size(datname) / 1024.0^3, 2) AS size_gb
FROM pg_database
WHERE datname NOT IN ('template0', 'template1', 'cloudsqladmin')
ORDER BY size_bytes DESC;

-- Also check table bloat within your primary database
-- Significant bloat means VACUUM hasn't run effectively
SELECT
    schemaname,
    tablename,
    pg_size_pretty(pg_total_relation_size(schemaname || '.' || tablename)) AS total_size,
    pg_size_pretty(pg_relation_size(schemaname || '.' || tablename))        AS table_size,
    pg_size_pretty(
        pg_total_relation_size(schemaname || '.' || tablename)
        - pg_relation_size(schemaname || '.' || tablename)
    )                                AS index_size,
    ROUND(
        100.0 * (pg_total_relation_size(schemaname || '.' || tablename)
        - pg_relation_size(schemaname || '.' || tablename))
        / NULLIF(pg_total_relation_size(schemaname || '.' || tablename), 0), 1
    )                                AS index_pct
FROM pg_tables
WHERE schemaname NOT IN ('pg_catalog', 'information_schema', 'pg_toast')
ORDER BY pg_total_relation_size(schemaname || '.' || tablename) DESC
LIMIT 20;
```

Reclaiming Space Without Recreating the Instance

Before planning a full export-and-recreate cycle, attempt in-place space reclamation:

1. **Run VACUUM FULL on bloated tables** — rewrites table files and returns disk space to the OS. It requires an exclusive lock, so run this during a maintenance window.
2. **Drop unused indexes** — Chapter 5 covers this in depth, but indexes frequently account for 20–40% of total database size in over-indexed schemas.
3. **Archive and purge old data** — moving records older than your retention requirement to Cloud Storage as Parquet files via Dataflow typically recovers more storage than any tuning approach.

```
-- PostgreSQL: Find tables with significant dead tuple bloat
-- These are candidates for VACUUM FULL
SELECT
  schemaname,
  relname                                AS table_name,
  n_live_tup                             AS live_rows,
  n_dead_tup                             AS dead_rows,
  ROUND(100.0 * n_dead_tup / NULLIF(n_live_tup + n_dead_tup, 0), 1)
                                          AS dead_row_pct,
  last_vacuum,
  last_autovacuum,
  pg_size_pretty(pg_total_relation_size(schemaname || '.' || relname))
                                          AS total_size
FROM pg_stat_user_tables
WHERE n_dead_tup > 10000
AND (n_dead_tup::float / NULLIF(n_live_tup + n_dead_tup, 0)) > 0.10
ORDER BY n_dead_tup DESC
LIMIT 20;
```

Backup Storage Cost Control

Cloud SQL automatically backs up your instances daily and retains those backups for 7 days by default. The free tier for backup storage equals the provisioned instance storage size. Any backup storage beyond that threshold costs \$0.08/GB/month.

If your instance has 500 GB of provisioned storage but only 150 GB of actual data, you’re potentially storing seven 150 GB compressed backups. Google’s compression typically yields 2–4× reduction for typical relational data, so actual backup storage per day might be around 40–75 GB. Over seven days that’s 280–525 GB. Subtracting the 500 GB free tier, the overage is small in this example—but for larger instances or teams that have extended retention to 30 days without justification, the costs add up quickly.

```
# Check current backup settings and estimated backup storage for all instances
gcloud sql instances list \
  --project="$PROJECT_ID" \
  --format="table(name,settings.backupConfiguration.enabled,
    settings.backupConfiguration.binaryLogEnabled,
    settings.backupConfiguration.backupRetentionSettings.retainedBackups,
    diskSizeGb)"

# For a specific instance, list actual backups and their sizes
gcloud sql backups list \
  --instance="YOUR_INSTANCE_NAME" \
  --project="$PROJECT_ID" \
  --format="table(id,status,startTime,endTime,diskEncryptionConfiguration,
    enqueueTime)" \
  --limit=30
```

For most teams, reducing backup retention from 30 days to 7 days—or from 14 to 7—for non-production instances alone recovers meaningful storage cost. A production instance with genuine compliance requirements for 30-day retention should keep it, but dev and staging instances rarely need more than 3–5 days of backup history.

AlloyDB Storage: The Unified Layer Consideration

AlloyDB manages storage differently from Cloud SQL. The storage layer is distributed and automatically tiered across hot and cold data. You pay for the total data volume stored, not provisioned capacity, which means you never pay for empty space. This eliminates the automatic-storage-increase trap.

However, AlloyDB does not shrink its storage accounting when you delete data—rows deleted from AlloyDB tables are not immediately reclaimed at the storage billing layer until the storage background compaction process runs. For workloads with heavy delete patterns (log tables, event streams, rolling window aggregation), this means storage costs can lag actual data volume by days. Plan accordingly when evaluating AlloyDB storage bills against actual data sizes.

4.5 Reducing AlloyDB I/O Costs Through Query Optimization

AlloyDB I/O billing is the most actionable cost lever unique to the product. Every full table scan, every unindexed lookup, every cartesian join produces I/O operations that appear on your bill. The path to reducing I/O cost is identical to the path to improving query performance—which means I/O optimization pays a double dividend.

Identifying High-I/O Queries

AlloyDB exposes query statistics through the `google_sys.query_stats` system view, which is seeded by the `google_columnar_engine` and standard PostgreSQL `pg_stat_statements` extensions.

```
-- AlloyDB PostgreSQL: Find the top I/O-generating queries
-- These are the queries most directly responsible for your I/O bill

SELECT
  LEFT(query, 100)                                AS query_preview,
  calls,
  ROUND(total_exec_time::numeric / 1000, 2) AS total_exec_sec,
  ROUND(mean_exec_time::numeric, 2)         AS avg_exec_ms,
  shared_blks_read                           AS buffer_misses,
  shared_blks_hit                            AS buffer_hits,
  shared_blks_read + shared_blks_hit         AS total_blocks_accessed,
  ROUND(
    100.0 * shared_blks_read
    / NULLIF(shared_blks_read + shared_blks_hit, 0), 1
  )                                           AS miss_rate_pct,
  -- Blocks read from disk directly correspond to I/O billing units
  ROUND(shared_blks_read * 8.0 / 1024, 2) AS estimated_data_read_mb
FROM pg_stat_statements
WHERE calls > 100
  AND shared_blks_read > 10000 -- Focus on high-I/O queries only
ORDER BY shared_blks_read DESC
LIMIT 20;
```

A query appearing at the top of this list with a high `miss_rate_pct` tells you that the working set for that query is not in buffer cache—it's going to storage on every execution. The fix is one of four things:

1. **Add a covering index** that satisfies the query without a heap access
2. **Increase `shared_buffers`** to cache more of the hot dataset

3. Use AlloyDB's **columnar engine** to serve analytical queries from compressed column-oriented storage
4. **Partition the table** to reduce the scan range for time-series or range queries

Using the AlloyDB Columnar Engine to Reduce I/O

AlloyDB includes an in-memory columnar engine that stores compressed column-oriented representations of tables alongside the standard row-oriented heap. Queries that scan large portions of a table benefit enormously—the columnar engine typically reads 5–10× less data than the equivalent row scan, which directly translates to fewer I/O billing units.

Enabling the columnar engine is straightforward:

```
-- AlloyDB: Enable and configure the columnar engine
-- This requires the alloydb.enable_columnar_engine flag set to 'on' in instance settings
-- Verify it is active:
SHOW alloydb.enable_columnar_engine;

-- Check current columnar engine memory allocation
SHOW alloydb.columnar_engine_memory;

-- Populate the columnar engine for a specific table
-- AlloyDB can auto-populate, but manual population ensures immediate benefit
SELECT google_columnar_engine_add('public.orders');
SELECT google_columnar_engine_add('public.order_line_items');

-- Verify which tables are in the columnar engine
SELECT
  relation_name,
  column_count,
  state,
  pg_size_pretty(size_in_bytes::bigint) AS columnar_size,
  pg_size_pretty(heap_size_in_bytes::bigint) AS heap_size,
  ROUND(
    100.0 * (1 - size_in_bytes::float / NULLIF(heap_size_in_bytes, 0)), 1
  ) AS compression_pct
FROM google_columnar_engine_tables();
```

When the columnar engine serves a query, `EXPLAIN (ANALYZE, BUFFERS)` shows a `Columnar Scan` node instead of `Seq Scan`. The `Buffers` output shows dramatically fewer blocks read, which corresponds directly to lower I/O billing. For an analytical query scanning a 100 GB orders table, the difference between a sequential heap scan and a columnar scan can be the difference between 13 million I/O operations and 1.2 million—a reduction of more than 90% on that query's I/O bill.

Index Strategy for I/O Reduction

For OLTP workloads where the columnar engine is not appropriate (single-row lookups, small result sets, high-concurrency point queries), indexes are the primary I/O reduction tool. The principle is to eliminate every sequential scan that touches more rows than will ultimately be returned.

```
-- AlloyDB/Cloud SQL PostgreSQL: Find sequential scans on large tables
-- These are the highest-priority indexing opportunities for I/O cost reduction

SELECT
  schemaname,
  relname AS table_name,
  seq_scan,
  seq_tup_read,
  idx_scan,
  idx_tup_fetch,
  ROUND(seq_tup_read::numeric / NULLIF(seq_scan, 0), 0) AS avg_rows_per_seqscan,
  pg_size_pretty(pg_total_relation_size(schemaname || '.' || relname)) AS table_size,
```

```

pg_total_relation_size(schemaname || '.' || relname)
                        AS size_bytes
FROM pg_stat_user_tables
WHERE seq_scan > 100
      AND pg_total_relation_size(schemaname || '.' || relname) > 10 * 1024 * 1024 -- > 10 MB
ORDER BY seq_tup_read DESC
LIMIT 20;

```

A table with high `seq_scan` count and large `avg_rows_per_seqscan` is one where queries are scanning the entire table but returning a small fraction of rows—the textbook case for a missing index. Building the right index on such a table can reduce I/O for that query pattern by two to three orders of magnitude.

The Equivalent SQL Server Pattern

For teams maintaining cross-platform references, here is the equivalent diagnostic query for SQL Server (applicable to Azure SQL Database and SQL Server on Google Cloud Compute Engine):

```

-- SQL Server: Find high-logical-read queries from plan cache
-- Equivalent to the pg_stat_statements I/O analysis above

SELECT TOP 20
  SUBSTRING(st.text, (qs.statement_start_offset / 2) + 1,
    ((CASE qs.statement_end_offset
      WHEN -1 THEN DATALENGTH(st.text)
      ELSE qs.statement_end_offset
    END - qs.statement_start_offset) / 2) + 1
  ) AS query_text,
  qs.execution_count,
  qs.total_logical_reads,
  qs.total_logical_reads / qs.execution_count AS avg_logical_reads,
  qs.total_elapsed_time / 1000 AS total_elapsed_ms,
  qs.total_elapsed_time / qs.execution_count / 1000 AS avg_elapsed_ms,
  qp.query_plan
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
CROSS APPLY sys.dm_exec_query_plan(qs.plan_handle) qp
WHERE qs.execution_count > 100
ORDER BY qs.total_logical_reads DESC;

```

4.6 Network Egress Cost Reduction

Network costs in Cloud SQL and AlloyDB are easy to ignore until they appear as a line item larger than your storage bill. Understanding the pricing tiers is the first step.

Cloud SQL Network Pricing

- **Same-zone traffic:** Free
- **Cross-zone within same region:** \$0.01/GB
- **Cross-region:** \$0.02–\$0.08/GB depending on regions
- **Internet egress:** \$0.08–\$0.23/GB depending on destination
- **Cloud SQL Auth Proxy over the internet:** Charges apply at the internet egress rate

The most common source of avoidable network cost is **read replicas in different regions** provisioned for disaster recovery but receiving continuous replication traffic. A primary instance in `us-central1` replicating to a replica in `us-east1` generates cross-region traffic at \$0.02/GB for ev-

ery byte of WAL streamed. A write-heavy database generating 50 GB/day of WAL can accumulate \$30/day or roughly \$900/month in cross-region replication egress alone.

Diagnosing Network Costs from Billing Export

```
-- BigQuery: Break down Cloud SQL network costs by destination type
SELECT
  resource.labels['database_id'] AS instance_id,
  sku.description                 AS network_sku,
  FORMAT_DATE('%Y-%m', DATE(usage_start_time)) AS billing_month,
  ROUND(SUM(cost), 2)            AS cost_usd,
  ROUND(SUM(usage.amount), 2)    AS usage_gb
FROM `YOUR_PROJECT.billing_export.gcp_billing_export_v1_XXXXXX`
WHERE
  service.description = 'Cloud SQL'
  AND (sku.description LIKE '%Network%'
       OR sku.description LIKE '%Egress%'
       OR sku.description LIKE '%Ingress%')
  AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
GROUP BY 1, 2, 3
ORDER BY cost_usd DESC;
```

Practical Network Cost Reduction Tactics

1. **Use Private IP exclusively for application connections.** Public IP adds \$7.30/month per instance and routes traffic externally even when application and database are in the same VPC. Disable public IP on all production instances reachable only by internal services.
2. **Co-locate application and database in the same zone.** Same-zone traffic is free. If your Cloud Run service is in `us-central1-a` and your Cloud SQL instance is in `us-central1-b`, you are paying \$0.01/GB for every byte of query results returned. Aligning zones eliminates this.
3. **Evaluate cross-region replica necessity.** If the cross-region read replica exists for DR purposes but receives zero read traffic, you are paying both for the replica compute and for replication egress with no read-side benefit. A snapshot-based backup strategy using Cloud Storage cross-region replication at \$0.02/GB versus continuous WAL streaming can reduce DR costs by 60–80% for write-heavy workloads.
4. **Use Connection pooling to reduce connection overhead.** PgBouncer deployed in the same zone as your Cloud SQL instance reduces the number of TCP connections and the associated per-connection overhead, indirectly reducing the volume of small network round trips.

4.7 Committed Use Discounts: Strategy and Gotchas

CUDs for Cloud SQL and AlloyDB reduce compute costs by 25% (1-year) or 52% (3-year) but commit you to a specific vCPU and RAM quantity for the duration. Getting the commitment strategy wrong is expensive—you pay for committed units whether you use them or not.

When to Commit

Apply CUDs only when you can answer yes to all three of these:

1. The instance will exist for the full commitment term
2. The vCPU and RAM configuration will not decrease significantly

3. You have at least 60 days of billing history showing stable utilization

Dev instances, recently provisioned production instances, and instances expected to be consolidated within the year should not receive CUDs.

CUD Scope and Flexibility

Cloud SQL CUDs are applied at the project level and cover any Cloud SQL instance in the project that matches the committed resource type (PostgreSQL, MySQL, or SQL Server) and region. This means a single CUD covering 16 vCPU in `us-central1` can cover two 8-vCPU PostgreSQL instances, one 16-vCPU instance, or any other combination that sums to 16 vCPU of PostgreSQL in that region.

This flexibility is significant: you can commit at the aggregate project level without tying a commitment to a specific instance. If you decommission one instance and spin up another, the CUD continues to apply as long as the total committed vCPU is consumed.

```
# List existing CUDs in a project
gcloud compute commitments list \
  --project="$PROJECT_ID" \
  --filter="category=MACHINE" \
  --format="table(name,region,plan,status,endTimestamp,
                 resources[0].type,resources[0].amount)"

# Note: Cloud SQL CUDs are managed separately from Compute Engine CUDs
# View Cloud SQL commitments via the console:
# Navigation > SQL > Committed Use Discounts
# Or via the Cloud Billing API programmatically

# Query current CUD coverage via BigQuery billing export
# to see how much of your committed spend is being utilized
```

Calculating the Right Commitment Size

The risk in over-committing is paying for unused capacity. The risk in under-committing is leaving discount on the table. Target your commitment at the stable baseline—the vCPU count that is consumed even at low traffic periods—and leave peak capacity on-demand.

```
-- BigQuery: Calculate CUD coverage recommendation
-- Based on 90-day minimum daily compute spend as the safe commitment baseline

WITH daily_compute AS (
  SELECT
    DATE(usage_start_time)           AS usage_date,
    resource.labels['region']        AS region,
    -- Infer database type from SKU
    CASE
      WHEN sku.description LIKE '%PostgreSQL%' THEN 'PostgreSQL'
      WHEN sku.description LIKE '%MySQL%'      THEN 'MySQL'
      WHEN sku.description LIKE '%SQL Server%'  THEN 'SQL Server'
      ELSE 'Unknown'
    END                               AS db_type,
    SUM(cost)                         AS daily_compute_cost
  FROM `YOUR_PROJECT.billing_export.gcp_billing_export_v1_XXXXXX`
  WHERE
    service.description = 'Cloud SQL'
    AND (sku.description LIKE '%vCPU%' OR sku.description LIKE '%RAM%')
    AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
  GROUP BY 1, 2, 3
)

SELECT
  region,
  db_type,
  ROUND(MIN(daily_compute_cost), 2) AS min_daily_compute_usd,
```

```

ROUND(AVG(daily_compute_cost), 2) AS avg_daily_compute_usd,
ROUND(MAX(daily_compute_cost), 2) AS max_daily_compute_usd,
-- Safe 1-year commitment = 90% of minimum daily spend annualized
ROUND(MIN(daily_compute_cost) * 365 * 0.90, 2) AS recommended_1yr_commit_usd,
-- Projected 1-year savings at 25% discount on committed portion
ROUND(MIN(daily_compute_cost) * 365 * 0.90 * 0.25, 2) AS projected_1yr_savings_usd
FROM daily_compute
GROUP BY 1, 2
ORDER BY avg_daily_compute_usd DESC;

```

4.8 Terraform Patterns for Cost-Governed Cloud SQL

Infrastructure as code is not just a deployment convenience—it is a cost governance mechanism. When instance size, backup retention, and replica configuration are declared in version-controlled Terraform, every cost-impacting change goes through code review before it can be applied.

Core Cloud SQL Module with Cost Controls

```

# modules/cloud_sql/main.tf
# Opinionated Cloud SQL module that encodes cost governance defaults

variable "project_id"      { type = string }
variable "instance_name"   { type = string }
variable "region"          { type = string }
variable "database_version" { type = string } # e.g., "POSTGRES_15"
variable "environment"     { type = string } # prod, staging, dev

variable "tier" {
  type          = string
  description = "Cloud SQL machine tier. Use db-custom-N-M format."
  # No default - explicit sizing required to prevent casual oversizing
}

variable "disk_size_gb" {
  type    = number
  default = 100
}

variable "backup_retained_days" {
  type    = number
  default = 7
  # Override to 30 only when compliance requires it
}

variable "enable_public_ip" {
  type    = bool
  default = false
  # Default to private IP only - public IP adds cost and attack surface
}

variable "enable_read_replica" {
  type    = bool
  default = false
}

# Environment-specific settings to prevent dev waste
locals {
  # Dev instances stop at night; prod instances run continuously
  activation_policy = var.environment == "prod" ? "ALWAYS" : "ALWAYS"
  # Note: scheduling is handled by Cloud Scheduler, not this setting

  # Backup retention: dev gets 3 days, non-prod gets 7, prod keeps variable
  effective_backup_days = var.environment == "dev" ? 3 : var.backup_retained_days
}

```

```

# Maintenance window: non-prod gets weekend daytime; prod gets weekend night
maintenance_day = var.environment == "prod" ? 7 : 6 # 7=Sunday, 6=Saturday
maintenance_hour = var.environment == "prod" ? 2 : 10 # 02:00 vs 10:00 UTC
}

resource "google_sql_database_instance" "main" {
  name           = var.instance_name
  project        = var.project_id
  region         = var.region
  database_version = var.database_version

  settings {
    tier                = var.tier
    activation_policy   = local.activation_policy
    availability_type   = var.environment == "prod" ? "REGIONAL" : "ZONAL"
    # ZONAL saves ~50% on HA overhead for non-prod instances

    disk_size          = var.disk_size_gb
    disk_autoresize     = true
    disk_autoresize_limit = var.environment == "prod" ? 0 : var.disk_size_gb * 3
    # Limit autoresize on non-prod to 3x initial size to prevent runaway growth

    backup_configuration {
      enabled          = true
      start_time       = "03:00"
      location         = var.region
      binary_log_enabled = false # PostgreSQL uses WAL, not binary log

      backup_retention_settings {
        retained_backups = local.effective_backup_days
        retention_unit    = "COUNT"
      }
    }
  }

  maintenance_window {
    day      = local.maintenance_day
    hour     = local.maintenance_hour
    update_track = "stable"
  }

  ip_configuration {
    ipv4_enabled = var.enable_public_ip
    private_network = google_compute_network.vpc.id
    # Require SSL for all connections
    require_ssl = true
  }

  database_flags {
    name = "log_min_duration_statement"
    value = var.environment == "prod" ? "1000" : "500"
    # Log queries over 1s (prod) or 500ms (non-prod) for tuning
  }

  database_flags {
    name = "pg_stat_statements.track"
    value = "all"
    # Essential for the I/O analysis queries shown earlier
  }
}

deletion_protection = var.environment == "prod"

lifecycle {
  prevent_destroy = false # Set to true for prod via workspace override
}

# Read replica - only created when explicitly requested
resource "google_sql_database_instance" "read_replica" {
  count = var.enable_read_replica ? 1 : 0

```

```

name          = "${var.instance_name}-replica"
project       = var.project_id
region       = var.region
database_version = var.database_version
master_instance_name = google_sql_database_instance.main.name

settings {
  # Replicas can often be smaller than the primary
  # Application reads are typically less RAM-intensive than writes
  tier          = var.tier
  availability_type = "ZONAL" # Replicas don't need HA

  ip_configuration {
    ipv4_enabled    = false
    private_network = google_compute_network.vpc.id
    require_ssl     = true
  }

  # Replicas don't need separate backups - primary covers recovery
  backup_configuration {
    enabled = false
  }
}

deletion_protection = false
}

output "instance_connection_name" {
  value = google_sql_database_instance.main.connection_name
}

output "instance_self_link" {
  value = google_sql_database_instance.main.self_link
}

output "monthly_cost_estimate" {
  value = "Review instance sizing: ${var.tier} in ${var.region}. Run cost analysis query in BigQuery
↪ after 48 hours of operation."
}

```

Enforcing Size Limits with Sentinel Policy (Terraform Enterprise / HCP Terraform)

For teams using HCP Terraform or Terraform Enterprise, Sentinel policies can block oversized instance creation before it reaches production:

```

# sentinel/cloud_sql_cost_governance.sentinel
# Blocks Cloud SQL instance creation that exceeds approved tiers for each environment

import "tfplan/v2" as tfplan

# Approved tiers per environment
approved_tiers = {
  "dev":      ["db-custom-2-8192", "db-custom-2-4096", "db-f1-micro", "db-g1-small"],
  "staging": ["db-custom-4-16384", "db-custom-4-8192", "db-custom-2-8192"],
  "prod":     [], # Prod has no block list - requires manager approval via PR review instead
}

# Find all Cloud SQL instance resources in the plan
cloud_sql_instances = filter tfplan.resource_changes as _, rc {
  rc.type is "google_sql_database_instance" and
  rc.change.actions contains "create"
}

# Main rule: validate each new instance against environment tier restrictions
main = rule {
  all cloud_sql_instances as _, instance {
    tier      = instance.change.after.settings[0].tier
    env_label = instance.change.after.labels["environment"] else "dev"
  }
}

```



```

    # If the environment has an approved list and it's not empty, enforce it
    allowed = approved_tiers[env_label] else []
    allowed is empty or tier in allowed
  }
}

```

This policy means a developer who accidentally changes `var.tier` to `db-custom-32-131072` in a dev environment will see a Sentinel policy failure before any infrastructure is created—and before the first invoice arrives.

4.9 Monitoring and Alerting for Ongoing Cost Control

Right-sizing and CUDs are point-in-time interventions. Sustained cost control requires ongoing monitoring that surfaces new waste as it appears—before it compounds.

Cloud Monitoring Budget Alerts

Start with billing-level alerts. These don't require any query writing and catch gross overruns immediately:

```

# Create a budget alert for Cloud SQL spend
# This requires the Billing Account Administrator role

gcloud billing budgets create \
  --billing-account="BILLING_ACCOUNT_ID" \
  --display-name="Cloud SQL Monthly Budget" \
  --budget-amount=5000 \
  --threshold-rule=percent=0.5,basis=CURRENT_SPEND \
  --threshold-rule=percent=0.9,basis=CURRENT_SPEND \
  --threshold-rule=percent=1.0,basis=CURRENT_SPEND \
  --filter-services="services/95FF-2EF5-5EA1" \
  --notifications-rule-pubsub-topic="projects/YOUR_PROJECT/topics/billing-alerts"
# Cloud SQL service ID: 95FF-2EF5-5EA1

```

Custom Metric Alerts for Utilization Drift

Budget alerts catch overspend at the billing level. Utilization alerts catch the root cause earlier—when an instance's CPU drops below the right-sizing threshold and action becomes possible before the next billing cycle closes.

```

# Create a Cloud Monitoring alert for chronically low CPU utilization
# This fires when a Cloud SQL instance averages below 10% CPU for 24 hours

gcloud monitoring policies create \
  --notification-channels="YOUR_CHANNEL_ID" \
  --display-name="Cloud SQL Low CPU Utilization" \
  --condition-display-name="CPU below 10% for 24 hours" \
  --condition-filter='resource.type="cloudsql_database"
    metric.type="cloudsql.googleapis.com/database/cpu/utilization"' \
  --condition-threshold-value=0.10 \
  --condition-threshold-comparison=COMPARISON_LT \
  --condition-duration=86400s \
  --condition-aggregation-alignment-period=3600s \
  --condition-aggregation-per-series-aligner=ALIGN_MEAN

```

Weekly Cost Anomaly Report via Cloud Scheduler + BigQuery

The most reliable ongoing governance mechanism is a weekly automated report delivered to a Slack channel or email list. This keeps the conversation about database costs visible without requiring

manual effort from engineers.

```
# cloud_function/weekly_cost_report.py
# Deploy as a Cloud Function triggered by Cloud Scheduler weekly

import functions_framework
from google.cloud import bigquery
import requests
import os
from datetime import datetime, timedelta

BIGQUERY_TABLE = os.environ["BILLING_TABLE"]
SLACK_WEBHOOK = os.environ["SLACK_WEBHOOK_URL"]
PROJECT_ID = os.environ["GOOGLE_CLOUD_PROJECT"]

@functions_framework.http
def weekly_cost_report(request):
    client = bigquery.Client(project=PROJECT_ID)

    query = f"""
    WITH this_week AS (
        SELECT
            resource.labels['database_id'] AS instance_id,
            ROUND(SUM(cost), 2) AS cost_usd
        FROM `{BIGQUERY_TABLE}`
        WHERE
            service.description = 'Cloud SQL'
            AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 7 DAY)
        GROUP BY 1
    ),
    last_week AS (
        SELECT
            resource.labels['database_id'] AS instance_id,
            ROUND(SUM(cost), 2) AS cost_usd
        FROM `{BIGQUERY_TABLE}`
        WHERE
            service.description = 'Cloud SQL'
            AND DATE(usage_start_time) BETWEEN
                DATE_SUB(CURRENT_DATE(), INTERVAL 14 DAY)
                AND DATE_SUB(CURRENT_DATE(), INTERVAL 7 DAY)
        GROUP BY 1
    )
    SELECT
        t.instance_id,
        t.cost_usd AS this_week_usd,
        l.cost_usd AS last_week_usd,
        ROUND(t.cost_usd - COALESCE(l.cost_usd, 0), 2) AS delta_usd,
        ROUND(
            100.0 * (t.cost_usd - COALESCE(l.cost_usd, 0))
            / NULLIF(l.cost_usd, 0), 1
        ) AS pct_change
    FROM this_week t
    LEFT JOIN last_week l USING (instance_id)
    ORDER BY this_week_usd DESC
    """

    results = list(client.query(query).result())

    total_this_week = sum(r.this_week_usd for r in results)
    anomalies = [r for r in results if r.pct_change and abs(r.pct_change) > 20]

    # Build Slack message
    blocks = [
        {
            "type": "header",
            "text": {"type": "plain_text",
                "text": f" Cloud SQL Weekly Cost Report - {datetime.now().strftime('%b %d, %Y')}"}
        },
        {
            "type": "section",

```

```
        "text": {"type": "mrkdwn",
                  "text": f"*Total this week:* ${total_this_week:,.2f}"
        }
    ]

    if anomalies:
        anomaly_text = "\n".join([
            f"* {r.instance_id}: ${r.this_week_usd:,.2f} "
            f"({' ' if r.delta_usd > 0 else ' '} "
            f"${abs(r.delta_usd):.2f}, {r.pct_change:+.1f}%)"
            for r in anomalies
        ])
        blocks.append({
            "type": "section",
            "text": {"type": "mrkdwn",
                      "text": f"* Cost Anomalies (>20% week-over-week):*\n{anomaly_text}"
            })

    top_instances_text = "\n".join([
        f"* {r.instance_id}: ${r.this_week_usd:,.2f}"
        for r in results[:5]
    ])
    blocks.append({
        "type": "section",
        "text": {"type": "mrkdwn",
                  "text": f"*Top 5 Instances by Cost:*\n{top_instances_text}"
        })

    response = requests.post(SLACK_WEBHOOK, json={"blocks": blocks})
    return f"Report sent. Status: {response.status_code}", 200
```

Key Takeaways

- **Cloud SQL and AlloyDB have structurally different cost models:** Cloud SQL charges for provisioned compute and storage, while AlloyDB adds a per-I/O charge that scales directly with query volume and index efficiency. Understanding which model you are on determines which optimization lever to reach for first.
- **Right-sizing is consistently the largest savings opportunity:** Most Cloud SQL instances run at 8–20% average CPU with provisioned resources sized for a peak that rarely materializes. Matching instance tier to the 90th-percentile utilization observed over 30 days—not the theoretical maximum—typically recovers 30–50% of compute spend, amplified further by applying CUDs to the right-sized baseline.
- **AlloyDB I/O costs are query performance costs in disguise:** Every sequential scan, every missing index, every unnecessary full table read generates billable I/O operations. Enabling the AlloyDB columnar engine for analytical workloads and maintaining a disciplined index strategy for OLTP patterns pay a double dividend: faster queries and a lower monthly bill.
- **Terraform and policy-as-code prevent cost regression:** Point-in-time optimizations erode quickly without guardrails. Encoding approved instance tiers per environment in Terraform modules and enforcing them via Sentinel policies or Open Policy Agent catches over-sized provisioning before the first invoice arrives, keeping the savings permanent rather than episodic.
- **Automated monitoring closes the feedback loop:** Budget alerts set at 50%, 90%, and

100% of expected monthly spend, combined with a weekly BigQuery cost anomaly report delivered to a shared Slack channel, ensure that cost increases are visible to the team within days rather than discovered at month-end reconciliation. The operational habit of reviewing this report is what sustains the gains made by right-sizing and CUDs over time.

Chapter 5: The Hidden Costs: Data Transfer, Snapshots, and Logs

Most engineering teams can recite their RDS instance cost to within a few dollars. Almost none can tell you what they spent on data transfer last month. That gap between what teams track and what actually appears on the bill is where database budgets quietly hemorrhage. Across AWS, Azure, and GCP, data egress fees, automated snapshot accumulation, and unmanaged log retention collectively account for **18–35% of total database spend** for organizations running production workloads at scale — and the majority of that cost is entirely recoverable with targeted intervention. This chapter shows you exactly where to find these charges in your bill, how to calculate their true cost, and how to automate the controls that eliminate them. Organizations that complete the optimizations in this chapter typically recover **\$2,000 to \$40,000 per month** depending on data volumes, with the largest savings coming from cross-region replication patterns that were designed for resilience without any cost accounting applied to them.

5.1 Understanding Data Transfer Pricing: The Billing Model Nobody Reads

Data transfer pricing is intentionally complex. Cloud providers publish egress rates buried in footnotes, vary them by region pair, and charge differently depending on whether traffic crosses availability zones, regions, or the open internet. Understanding the layered model before you optimize is not optional — misreading it leads to architectural changes that save nothing.

AWS Data Transfer Tiers (2024 pricing, us-east-1 baseline)

Traffic Type	Cost
Inbound (ingress)	\$0.00
Same-AZ, same-region (private IP)	\$0.00
Cross-AZ within same region	\$0.02/GB each direction
Cross-region	\$0.02–\$0.09/GB depending on destination
Internet egress (first 10TB/month)	\$0.09/GB
Internet egress (next 40TB/month)	\$0.085/GB
RDS to EC2 in same AZ (public IP)	\$0.02/GB

The last row deserves a second read. An application server connecting to an RDS instance in

the same availability zone over the instance’s public DNS endpoint incurs cross-AZ charges even though both resources are physically in the same building. This single misconfiguration — using the public endpoint instead of the private endpoint — generates measurable egress charges on databases processing more than a few hundred gigabytes per month.

Azure Data Transfer Tiers (2024 pricing, East US baseline)

Traffic Type	Cost
Inbound	\$0.00
Same-region, same VNet	\$0.00
Cross-region (within continent)	\$0.02–\$0.05/GB
Cross-continent	\$0.05–\$0.08/GB
Internet egress (first 5GB/month)	\$0.00
Internet egress (over 5GB)	\$0.087/GB

GCP Data Transfer Tiers (2024 pricing, us-central1 baseline)

Traffic Type	Cost
Inbound	\$0.00
Same-zone	\$0.00
Cross-zone, same region	\$0.01/GB
Cross-region (North America)	\$0.01/GB
Cross-region (other)	\$0.02–\$0.08/GB
Internet egress	\$0.085–\$0.23/GB

GCP’s cross-zone pricing is lower than AWS — \$0.01/GB vs \$0.02/GB — making the cost of Multi-AZ equivalent patterns roughly half the AWS price. This is a relevant consideration when comparing total cost of ownership across providers for read-heavy workloads with multiple replicas.

Finding Your Current Transfer Spend

Before optimizing, establish a baseline. On AWS, Cost Explorer provides transfer cost breakdowns but requires correct filtering to isolate database-related transfer from compute transfer.

```
# AWS CLI: Pull data transfer costs for RDS specifically, last 30 days
aws ce get-cost-and-usage \
  --time-period Start=$(date -d '30 days ago' '+%Y-%m-%d'),End=$(date '+%Y-%m-%d') \
  --granularity MONTHLY \
  --filter '{
    "And": [
      {
        "Dimensions": {
          "Key": "SERVICE",
          "Values": ["Amazon Relational Database Service"]
        }
      },
      {
        "Dimensions": {
          "Key": "USAGE_TYPE_GROUP",
          "Values": ["RDS: Data Transfer - Internet", "RDS: Data Transfer - Region to Region", "RDS:
↪ Data Transfer - AZ"]
        }
      }
    ]
  }
```

```

    ]
  }' \
  --metrics "BlendedCost" "UsageQuantity" \
  --group-by Type=DIMENSION,Key=USAGE_TYPE \
  --output table

# Azure CLI: Pull data transfer costs by resource group
az consumption usage list \
  --start-date $(date -d '30 days ago' '+%Y-%m-%d') \
  --end-date $(date '+%Y-%m-%d') \
  --query "[?contains(meterCategory,'Bandwidth')].{Name:instanceName, Cost:pretaxCost,
  ↪ Quantity:usageQuantity, Unit:unitOfMeasure}" \
  --output table

# GCP: Query billing export in BigQuery for network egress on Cloud SQL
bq query --use_legacy_sql=false \
  "SELECT
    service.description AS service,
    sku.description AS sku,
    SUM(usage.amount) AS total_gb,
    SUM(cost) AS total_cost_usd
  FROM `PROJECT_ID.billing_dataset.gcp_billing_export_v1_ACCOUNT_ID`
  WHERE DATE(_PARTITIONTIME) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
    AND service.description = 'Cloud SQL'
    AND LOWER(sku.description) LIKE '%network%'
  GROUP BY 1, 2
  ORDER BY total_cost_usd DESC"
```

Run these three commands before any other action in this chapter. The output establishes your current baseline and prevents optimizing costs that were never significant for your specific workload.

5.2 Cross-AZ and Cross-Region Traffic: Finding and Fixing the Expensive Flows

The most common source of unexpected data transfer costs is **cross-AZ traffic generated by Multi-AZ RDS configurations combined with application servers spread across availability zones**. In a standard production setup, you have three application servers distributed across three AZs for high availability, and an RDS Multi-AZ instance whose primary is in one of those AZs. Every application server not co-located with the RDS primary is paying \$0.02/GB in each direction for every byte they exchange.

For a database processing 10TB of query results per month to application servers (a moderate OLTP workload), this costs:

- $10\text{TB} \times \$0.02/\text{GB} = 10,000\text{GB} \times \$0.02 = \text{\$200/month cross-AZ egress}$
- Plus return traffic: another **\\$200/month**
- Total cross-AZ penalty: **\\$400/month for a single database**

This compounds with read replicas. An RDS instance with three read replicas spread across AZs for distribution generates replication traffic costs on top of application traffic costs.

Identifying Cross-AZ Traffic Sources

On AWS, VPC Flow Logs capture the source and destination of all traffic including AZ origins. The following query, run against Flow Logs exported to CloudWatch Logs Insights, identifies the largest cross-AZ flows involving database ports.

```

-- CloudWatch Logs Insights query (not SQL - uses Logs Insights syntax)
-- Run this in the CloudWatch console under Logs Insights against your VPC Flow Log group
-- Paste this directly into the query editor:
-- fields @timestamp, srcAddr, dstAddr, dstPort, bytes, azId
-- | filter dstPort in [5432, 3306, 1433, 5439, 27017]
-- | filter srcAddr not like /~10\.0\.1\./ -- replace with your primary AZ subnet
-- | stats sum(bytes) as total_bytes by srcAddr, dstAddr, dstPort
-- | sort total_bytes desc
-- | limit 20

-- For PostgreSQL: query pg_stat_activity to see connection origins
-- Run this on your RDS PostgreSQL instance to identify which client IPs are connecting
SELECT
    client_addr,
    COUNT(*) AS connection_count,
    MAX(state) AS state,
    application_name,
    username
FROM pg_stat_activity
WHERE client_addr IS NOT NULL
    AND state != 'idle'
GROUP BY client_addr, application_name, username
ORDER BY connection_count DESC;

-- For MySQL: equivalent connection origin query
-- Run this on RDS MySQL to identify connection sources
-- SELECT
--     HOST,
--     DB,
--     COUNT(*) as connection_count,
--     SUM(DATA_READ)/1024/1024 as data_read_mb,
--     SUM(DATA_WRITTEN)/1024/1024 as data_written_mb
-- FROM information_schema.PROCESSLIST
-- GROUP BY HOST, DB
-- ORDER BY connection_count DESC;

```

Cross-reference the `client_addr` values against your AWS subnet map. Any connection coming from a subnet in a different AZ than your RDS primary is generating cross-AZ charges.

Fixing Cross-AZ Traffic: Three Approaches

Option 1: Subnet-Aware Connection Routing Configure your application’s database connection pool to prefer the endpoint nearest to its own AZ. For applications using a service mesh or a proxy layer like ProxySQL or pgBouncer, configure topology-aware routing. This is the lowest-risk change.

Option 2: Read Replica Placement If your workload is read-heavy, place one read replica in each AZ where application servers run. Configure read traffic to route to the local replica. For a three-AZ setup processing 5TB of reads per month per AZ, this eliminates cross-AZ read traffic entirely. The cost of an additional read replica (for example, an `db.r6g.large` at approximately \$146/month) is offset immediately if cross-AZ read traffic exceeds 7,300GB/month per AZ.

Option 3: Private Endpoint Enforcement Audit all connection strings and ensure every application is connecting to the private DNS endpoint, not the public endpoint. This single change eliminates the class of cross-AZ charges that arise from public-IP routing.

```

#!/usr/bin/env python3
"""
cross_az_detector.py
Scans RDS instances and their connecting EC2 instances,
flags mismatched AZ combinations that generate cross-AZ traffic costs.
Run on a schedule (daily recommended) via Lambda or cron.
"""

import boto3
import json

```



```

from datetime import datetime, timedelta
from collections import defaultdict

def get_rds_primary_az(rds_client, db_instance_id):
    """Return the AZ of the primary RDS instance."""
    response = rds_client.describe_db_instances(DBInstanceIdentifier=db_instance_id)
    return response['DBInstances'][0]['AvailabilityZone']

def get_ec2_instances_by_az(ec2_client):
    """Return dict mapping AZ to list of instance IDs and private IPs."""
    az_map = defaultdict(list)
    paginator = ec2_client.get_paginator('describe_instances')
    for page in paginator.paginate(Filters=[{'Name': 'instance-state-name', 'Values': ['running']}]):
        for reservation in page['Reservations']:
            for instance in reservation['Instances']:
                az = instance['Placement']['AvailabilityZone']
                az_map[az].append({
                    'id': instance['InstanceId'],
                    'private_ip': instance.get('PrivateIpAddress', 'unknown'),
                    'tags': {t['Key']: t['Value'] for t in instance.get('Tags', [])}
                })
    return az_map

def estimate_cross_az_cost(cloudwatch_client, rds_instance_id, days=30):
    """
    Estimate data transfer cost by looking at NetworkReceiveThroughput
    and NetworkTransmitThroughput on the RDS instance.
    Returns estimated GB transferred.
    """
    end_time = datetime.utcnow()
    start_time = end_time - timedelta(days=days)

    total_gb = 0
    for metric in ['NetworkReceiveThroughput', 'NetworkTransmitThroughput']:
        response = cloudwatch_client.get_metric_statistics(
            Namespace='AWS/RDS',
            MetricName=metric,
            Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': rds_instance_id}],
            StartTime=start_time,
            EndTime=end_time,
            Period=86400, # daily datapoints
            Statistics=['Average']
        )

        # Average bytes/sec * seconds_per_day * days = total bytes
        for dp in response['Datapoints']:
            bytes_per_day = dp['Average'] * 86400
            total_gb += bytes_per_day / (1024**3)

    return round(total_gb, 2)

def main():
    region = 'us-east-1'
    rds_client = boto3.client('rds', region_name=region)
    ec2_client = boto3.client('ec2', region_name=region)
    cw_client = boto3.client('cloudwatch', region_name=region)

    report = {
        'generated_at': datetime.utcnow().isoformat(),
        'region': region,
        'findings': []
    }

    # Get all RDS instances
    paginator = rds_client.get_paginator('describe_db_instances')
    ec2_by_az = get_ec2_instances_by_az(ec2_client)

    for page in paginator.paginate():
        for db in page['DBInstances']:
            db_id = db['DBInstanceIdentifier']

```

```
db_az = db['AvailabilityZone']
db_endpoint = db.get('Endpoint', {}).get('Address', '')
is_multi_az = db.get('MultiAZ', False)

# Count EC2 instances in other AZs
cross_az_instances = []
for az, instances in ec2_by_az.items():
    if az != db_az:
        cross_az_instances.extend(instances)

total_gb_30d = estimate_cross_az_cost(cw_client, db_id)
# Assume 50% of traffic crosses AZ (conservative estimate)
estimated_cross_az_gb = total_gb_30d * 0.5
estimated_monthly_cost = estimated_cross_az_gb * 0.02 * 2 # both directions

if estimated_monthly_cost > 10: # Only report if > $10/month
    finding = {
        'db_instance': db_id,
        'db_az': db_az,
        'is_multi_az': is_multi_az,
        'total_network_gb_30d': total_gb_30d,
        'estimated_cross_az_gb_30d': estimated_cross_az_gb,
        'estimated_monthly_cost_usd': round(estimated_monthly_cost, 2),
        'ec2_instances_in_other_azs': len(cross_az_instances),
        'recommendation': (
            'Place read replica in each application AZ, '
            'or configure topology-aware routing. '
            f'Estimated savings: ${round(estimated_monthly_cost * 0.7, 2)}/month'
        )
    }
    report['findings'].append(finding)

# Sort by cost impact
report['findings'].sort(key=lambda x: x['estimated_monthly_cost_usd'], reverse=True)

total_recoverable = sum(f['estimated_monthly_cost_usd'] for f in report['findings'])
report['total_estimated_recoverable_usd_monthly'] = round(total_recoverable, 2)

print(json.dumps(report, indent=2))

# In production: push to S3, SNS, or Slack webhook
# s3_client.put_object(Bucket='finops-reports', Key=f'cross-az/{datetime.utcnow().date()}.json',
#     ↪ Body=json.dumps(report))

if __name__ == '__main__':
    main()
```

5.3 Snapshot Economics: Retention Policies That Cost Fortunes

Automated snapshots are not free storage. They feel free because most cloud providers offer a limited window at no charge — AWS gives you free snapshot storage equal to the size of your database — but that boundary erodes almost immediately in practice.

AWS Snapshot Pricing

AWS charges \$0.095/GB-month for snapshots stored in gp2/gp3 format beyond the provisioned database storage size. A 500GB database with 35 automated snapshots (the result of setting a 35-day retention period for a compliance requirement) stores approximately:

- First 500GB: **free** (offset by provisioned storage payment)
- Remaining ~17,000GB (34 snapshots × 500GB): $17,000 \times \$0.095 = \mathbf{\$1,615.00/month}$

The relationship between retention period and cost is linear. Moving from a 35-day to a 7-day retention window for development databases reduces snapshot storage by 80%.

Azure SQL Backup Pricing

Azure charges for backup storage beyond the provisioned database size in the LRS tier at **\$0.0375/GB-month** for locally redundant storage and **\$0.115/GB-month** for geo-redundant storage. Point-in-time restore relies on full, differential, and log backups — the log backup volume grows proportionally with write throughput, making high-write databases disproportionately expensive to back up.

GCP Cloud SQL Backup Pricing

GCP charges **\$0.08/GB-month** for automated backups. Unlike AWS, there is no free tier equivalent; you pay for the full snapshot volume from the first byte.

Finding Your Current Snapshot Footprint

```
-- PostgreSQL: Query pg_stat_user_tables for table size trends
-- This helps estimate how fast your database is growing (and thus snapshot sizes)
SELECT
    schemaname,
    tablename,
    pg_size_pretty(pg_total_relation_size(schemaname||'.'||tablename)) AS total_size,
    pg_size_pretty(pg_relation_size(schemaname||'.'||tablename)) AS table_size,
    pg_size_pretty(pg_total_relation_size(schemaname||'.'||tablename)
        - pg_relation_size(schemaname||'.'||tablename)) AS index_size,
    n_live_tup AS live_rows,
    n_dead_tup AS dead_rows,
    ROUND(100.0 * n_dead_tup / NULLIF(n_live_tup + n_dead_tup, 0), 2) AS dead_pct
FROM pg_stat_user_tables
WHERE pg_total_relation_size(schemaname||'.'||tablename) > 1073741824 -- > 1GB
ORDER BY pg_total_relation_size(schemaname||'.'||tablename) DESC;

-- SQL Server (Azure SQL): Estimate backup size from database file sizes
SELECT
    DB_NAME(database_id) AS database_name,
    type_desc,
    SUM(size * 8.0 / 1024) AS size_mb,
    SUM(size * 8.0 / 1024 / 1024) AS size_gb,
    -- Monthly backup cost estimate (LRS pricing)
    SUM(size * 8.0 / 1024 / 1024) * 0.0375 AS estimated_monthly_cost_lrs_usd,
    SUM(size * 8.0 / 1024 / 1024) * 0.115 AS estimated_monthly_cost_grs_usd
FROM sys.master_files
GROUP BY database_id, type_desc
ORDER BY size_gb DESC;

# AWS CLI: List all RDS snapshots with sizes and ages
aws rds describe-db-snapshots \
    --query 'DBSnapshots[*].{
        ID:DBSnapshotIdentifier,
        DB:DBInstanceIdentifier,
        Type:SnapshotType,
        Created:SnapshotCreateTime,
        SizeGB:AllocatedStorage,
        Status:Status
    }' \
    --output table | sort -k4

# Calculate total snapshot storage and estimated cost
aws rds describe-db-snapshots \
    --query 'DBSnapshots[*].AllocatedStorage' \
    --output text | \
    awk '{sum=0; for(i=1;i<=NF;i++) sum+=$i; print "Total snapshot storage (GB): " sum; print "Estimated
    ↵ monthly cost (USD): $" sum * 0.095}'
```

```

# Azure CLI: List SQL backup storage usage
az sql db list-deleted \
  --resource-group myResourceGroup \
  --server myserver \
  --query "[Name:databaseName, DeletionDate:deletionDate, EarliestRestore:earliestRestoreDate]" \
  --output table

# GCP: List Cloud SQL backups with sizes
gcloud sql backups list \
  --instance=my-sql-instance \
  --format="table(id,status,startTime,endTime,diskEncryptionConfiguration.kmsKeyName)"

# GCP: Calculate backup sizes via API (gcloud doesn't expose size directly)
gcloud sql instances describe my-sql-instance \
  --format="json" | \
  python3 -c "
import json, sys
data = json.load(sys.stdin)
disk_gb = data.get('settings', {}).get('dataDiskSizeGb', 0)
print(f'Database disk size: {disk_gb} GB')
print(f'Estimated backup cost at \$0.08/GB/month: \${float(disk_gb) * 0.08:.2f}/month')
"

```

Snapshot Optimization Strategy by Environment

Blanket retention policies applied uniformly across production, staging, and development databases are the most expensive snapshot mistake. A tiered approach based on environment and compliance requirement cuts snapshot spend by 40–70% in most organizations:

Environment	Recommended Retention	Rationale
Production (regulated)	35 days automated + 12 monthly manual	Compliance baseline
Production (standard)	7 days automated + 4 weekly manual	Covers most restore scenarios
Staging	3 days	Rarely restored, easily recreated
Development	1 day or snapshot-on-demand only	No production data, no restore need
Data warehouse	Weekly only	Full rebuilds cheaper than snapshots

Manual snapshots that are never deleted are the second largest snapshot cost driver. Teams create manual snapshots before schema migrations, forget to delete them, and those snapshots accumulate for years. The following script audits and remediates orphaned manual snapshots.

```

#!/usr/bin/env python3
"""
snapshot_cost_auditor.py
Identifies overdue manual snapshots, orphaned snapshots from deleted databases,
and databases with excessive automated retention.
Designed to run weekly via EventBridge + Lambda.
"""

import boto3
import json
from datetime import datetime, timedelta, timezone
from dataclasses import dataclass, asdict
from typing import List

```

```

@dataclass
class SnapshotFinding:
    snapshot_id: str
    db_instance: str
    snapshot_type: str
    age_days: int
    size_gb: int
    monthly_cost_usd: float
    issue: str
    action: str

def calculate_monthly_cost(size_gb: int) -> float:
    """AWS snapshot pricing: $0.095/GB-month."""
    return round(size_gb * 0.095, 2)

def get_active_db_instances(rds_client) -> set:
    """Return set of currently active DB instance identifiers."""
    active = set()
    paginator = rds_client.get_paginator('describe_db_instances')
    for page in paginator.paginate():
        for db in page['DBInstances']:
            active.add(db['DBInstanceIdentifier'])
    return active

def audit_snapshots(rds_client, max_manual_age_days=90, dev_patterns=None) -> List[SnapshotFinding]:
    """
    Audit all RDS snapshots for cost issues.
    dev_patterns: list of substrings that identify dev/staging instances (e.g., ['dev', 'staging',
    ↪ 'test'])
    """
    if dev_patterns is None:
        dev_patterns = ['dev', 'staging', 'test', 'sandbox', 'qa']

    active_instances = get_active_db_instances(rds_client)
    findings = []
    now = datetime.now(timezone.utc)

    paginator = rds_client.get_paginator('describe_db_snapshots')
    # Note: actual boto3 method is describe_db_snapshots
    paginator = rds_client.get_paginator('describe_db_snapshots')

    for page in paginator.paginate():
        for snap in page['DBSnapshots']:
            snap_id = snap['DBSnapshotIdentifier']
            db_id = snap['DBInstanceIdentifier']
            snap_type = snap['SnapshotType']
            size_gb = snap.get('AllocatedStorage', 0)
            status = snap['Status']
            created_raw = snap.get('SnapshotCreateTime')

            if status != 'available':
                continue

            age_days = 0
            if created_raw:
                age_days = (now - created_raw).days

            monthly_cost = calculate_monthly_cost(size_gb)

            # 1. Orphaned snapshot: source DB no longer exists
            if db_id not in active_instances and snap_type == 'manual':
                findings.append(SnapshotFinding(
                    snapshot_id=snap_id,
                    db_instance=db_id,
                    snapshot_type=snap_type,
                    age_days=age_days,
                    size_gb=size_gb,
                    monthly_cost_usd=monthly_cost,
                    issue='orphaned_snapshot',

```

```

        action=f'Delete immediately - source instance no longer exists. Saves
↳  ${monthly_cost}/month.'
    ))
    continue

    # 2. Ancient manual snapshot
    if snap_type == 'manual' and age_days > max_manual_age_days:
        findings.append(SnapshotFinding(
            snapshot_id=snap_id,
            db_instance=db_id,
            snapshot_type=snap_type,
            age_days=age_days,
            size_gb=size_gb,
            monthly_cost_usd=monthly_cost,
            issue='stale_manual_snapshot',
            action=(
                f'Manual snapshot is {age_days} days old. '
                f'Review and delete if no longer needed for migration/compliance. '
                f'Saves ${monthly_cost}/month.'
            )
        ))

    # 3. Dev/staging instance with automated snapshots older than 3 days
    is_dev = any(p.lower() in db_id.lower() for p in dev_patterns)
    if is_dev and snap_type == 'automated' and age_days > 3:
        findings.append(SnapshotFinding(
            snapshot_id=snap_id,
            db_instance=db_id,
            snapshot_type=snap_type,
            age_days=age_days,
            size_gb=size_gb,
            monthly_cost_usd=monthly_cost,
            issue='excessive_dev_retention',
            action=(
                f'Dev/staging instance retaining automated snapshots for {age_days} days. '
                f'Reduce BackupRetentionPeriod to 1. '
                f'Estimated savings: ${monthly_cost}/month per excess snapshot.'
            )
        ))

    return sorted(findings, key=lambda f: f.monthly_cost_usd, reverse=True)

def apply_retention_policy(rds_client, db_instance_id: str, retention_days: int, dry_run=True):
    """Modify automated backup retention for a DB instance."""
    if dry_run:
        print(f'[DRY RUN] Would set BackupRetentionPeriod={retention_days} on {db_instance_id}')
        return
    rds_client.modify_db_instance(
        DBInstanceIdentifier=db_instance_id,
        BackupRetentionPeriod=retention_days,
        ApplyImmediately=False # Apply during next maintenance window
    )
    print(f'Updated retention to {retention_days} days for {db_instance_id}')

def delete_snapshot(rds_client, snapshot_id: str, dry_run=True):
    """Delete a manual snapshot after confirmation."""
    if dry_run:
        print(f'[DRY RUN] Would delete snapshot: {snapshot_id}')
        return
    rds_client.delete_db_snapshot(DBSnapshotIdentifier=snapshot_id)
    print(f'Deleted snapshot: {snapshot_id}')

def main():
    rds_client = boto3.client('rds', region_name='us-east-1')

    findings = audit_snapshots(rds_client, max_manual_age_days=90)

```

```

total_recoverable = sum(f.monthly_cost_usd for f in findings)

report = {
    'generated_at': datetime.utcnow().isoformat(),
    'total_findings': len(findings),
    'total_recoverable_monthly_usd': round(total_recoverable, 2),
    'findings': [asdict(f) for f in findings]
}

print(json.dumps(report, indent=2))

# Categorize findings
orphaned = [f for f in findings if f.issue == 'orphaned_snapshot']
stale_manual = [f for f in findings if f.issue == 'stale_manual_snapshot']
dev_excess = [f for f in findings if f.issue == 'excessive_dev_retention']

print(f'\n=== SUMMARY ===')
print(f'Orphaned snapshots (auto-deletable): {len(orphaned)}, '
      f'${sum(f.monthly_cost_usd for f in orphaned):.2f}/month')
print(f'Stale manual snapshots (review required): {len(stale_manual)}, '
      f'${sum(f.monthly_cost_usd for f in stale_manual):.2f}/month')
print(f'Dev/staging over-retention: {len(dev_excess)}, '
      f'${sum(f.monthly_cost_usd for f in dev_excess):.2f}/month')
print(f'Total recoverable: ${total_recoverable:.2f}/month')

# Uncomment to execute remediations (start with dry_run=True)
# for f in orphaned:
#     delete_snapshot(rds_client, f.snapshot_id, dry_run=True)

if __name__ == '__main__':
    main()

```

Cross-Region Snapshot Replication Costs

Snapshot replication to a second region for disaster recovery doubles your snapshot storage costs in the destination region and adds a one-time data transfer charge per replication. For a 1TB database snapshot:

- Source region storage: 1TB × \$0.095 = **\$97.50/month**
- Destination region storage: 1TB × \$0.095 = **\$97.50/month**
- Transfer cost for initial replication: 1TB × \$0.02 = **\$20.48 one-time**

The storage cost in the destination region is often missed entirely. Teams account for the transfer charge during architecture review but forget that the replicated snapshot occupies equal storage in the destination region at the same per-GB rate. This doubles the effective snapshot storage cost for any database using cross-region replication for DR.

The alternative: For databases where the DR requirement is “restore within hours, not minutes,” evaluate whether cross-region snapshot replication is necessary, or whether recreating the database from application-level backups (pg_dump, mysqldump, or SQL Server BACPAC exports stored in S3/Blob Storage) achieves the same RTO at a fraction of the cost. Application-level exports stored as compressed files in object storage typically reduce DR storage costs by 70–90% versus binary snapshot replication.

5.4 Log Retention and the Bill Nobody Expected

Database logs are generated constantly and, left unmanaged, grow without limit. The costs accumulate across three dimensions: **log storage on the database instance itself**, **log export to**

cloud monitoring services, and log retention in those monitoring services. Each dimension has a different pricing model and a different remediation approach.

CloudWatch Logs: The Silent Cost Multiplier

AWS RDS can export multiple log types to CloudWatch Logs: general query logs, slow query logs, error logs, audit logs, and PostgreSQL upgrade logs. Each log type enabled multiplies CloudWatch Logs ingestion costs (\$0.50/GB ingested) and storage costs (\$0.03/GB-month) against the volume of logs generated.

The most expensive single configuration is enabling **general query logging on a high-throughput OLTP database**. A database handling 5,000 queries per second generates approximately 1–3GB of query log data per hour. At \$0.50/GB ingestion:

- $2\text{GB/hour} \times 24\text{ hours} \times 30\text{ days} = 1,440\text{GB/month}$
- $1,440\text{GB} \times \$0.50 = \textbf{\$720/month}$ in CloudWatch ingestion alone
- Plus $1,440\text{GB} \times \$0.03\text{ retention} = \textbf{\$43.20/month}$
- **Total: \$763.20/month** for general query logging on a single database

Almost no operational scenario justifies general query logging in production at this volume. The typical justification — “we need it for debugging” — is satisfied by slow query logging with an appropriate threshold, at roughly 1% of the log volume.

Identifying Current Log Export Costs

```
# AWS: Find CloudWatch Log groups from RDS and their sizes/costs
aws logs describe-log-groups \
  --log-group-name-prefix '/aws/rds' \
  --query 'logGroups[*].{
    Name:logGroupName,
    RetentionDays:retentionInDays,
    StoredBytes:storedBytes,
    StoredGB:join(``, [to_string(storedBytes / 1073741824), ` GB`])
  }' \
  --output table

# For each log group, estimate monthly storage cost
aws logs describe-log-groups \
  --log-group-name-prefix '/aws/rds' \
  --query 'logGroups[*].[logGroupName, storedBytes]' \
  --output text | \
  awk '{gb=$2/1073741824; cost=gb*0.03; printf "%-60s %8.2f GB  $%8.2f/month\n", $1, gb, cost}'

# Find log groups with NO retention policy set (infinite retention = infinite cost)
aws logs describe-log-groups \
  --log-group-name-prefix '/aws/rds' \
  --query 'logGroups[?!retentionInDays].[logGroupName, storedBytes]' \
  --output table

# Azure Monitor: Query log analytics workspace for SQL-related ingestion volume
# Run this in the Log Analytics workspace query editor:
# Usage
# | where TimeGenerated > ago(30d)
# | where Solution == "LogManagement"
# | where DataType in ("AzureDiagnostics", "AzureMetrics")
# | summarize TotalGB = sum(Quantity) / 1024, TotalCost = sum(Quantity) / 1024 * 2.76 by DataType,
  ↪ bin(TimeGenerated, 1d)
# | order by TimeGenerated desc

# GCP: Check Cloud Logging ingestion for Cloud SQL logs
gcloud logging sinks list --format="table(name,destination,filter)"

# Query billing export for logging costs associated with Cloud SQL
```



```

# (Run in BigQuery)
# SELECT
#   sku.description,
#   SUM(usage.amount) AS total_gib,
#   SUM(cost) AS total_cost
# FROM `PROJECT.billing_dataset.gcp_billing_export_v1_ACCOUNT`
# WHERE service.description = 'Cloud Logging'
#   AND DATE(_PARTITIONTIME) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
# GROUP BY sku.description
# ORDER BY total_cost DESC

```

Log-Level Recommendations by Workload Type

Log Type	Development	Staging	Production OLTP	Production Analytical
Error log	Always on	Always on	Always on	Always on
Slow query (>1s)	Optional	On	On	On
Slow query (>100ms)	On	Optional	Off	Optional
General query log	On	Off	Off	Off
Audit log	Off	Off	Compliance-driven	Compliance-driven
PostgreSQL pg_stat_statements	On	On	On	On

`pg_stat_statements` deserves special attention. It is the single highest-value logging mechanism for production PostgreSQL because it aggregates query performance data in-memory inside the database engine itself, with no per-query log write overhead and no CloudWatch cost. Everything general query logging tells you (which queries ran, how long, how often) is available from `pg_stat_statements` at a fraction of the infrastructure cost.

```

-- PostgreSQL: Enable pg_stat_statements (add to parameter group, then reconnect)
-- In RDS parameter group: set shared_preload_libraries = 'pg_stat_statements'
-- After reconnect, create the extension:
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;

-- Query the top 20 most expensive queries by total execution time
SELECT
  LEFT(query, 120) AS query_preview,
  calls,
  ROUND(total_exec_time::numeric, 2) AS total_ms,
  ROUND(mean_exec_time::numeric, 2) AS mean_ms,
  ROUND((total_exec_time / SUM(total_exec_time) OVER ()) * 100, 2) AS pct_total_time,
  rows,
  shared_blks_hit,
  shared_blks_read,
  ROUND(
    100.0 * shared_blks_hit / NULLIF(shared_blks_hit + shared_blks_read, 0),
    2
  ) AS cache_hit_pct
FROM pg_stat_statements
WHERE calls > 10
ORDER BY total_exec_time DESC
LIMIT 20;

-- SQL Server equivalent: sys.dm_exec_query_stats for top queries
SELECT TOP 20
  SUBSTRING(qt.text, (qs.statement_start_offset/2)+1,

```

```

        ((CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)
            ELSE qs.statement_end_offset END
            - qs.statement_start_offset)/2)+1) AS query_text,
        qs.execution_count,
        ROUND(qs.total_elapsed_time / 1000.0, 2) AS total_elapsed_ms,
        ROUND(qs.total_elapsed_time / qs.execution_count / 1000.0, 2) AS avg_elapsed_ms,
        ROUND(100.0 * qs.total_elapsed_time /
            SUM(qs.total_elapsed_time) OVER (), 2) AS pct_total_time,
        qs.total_logical_reads,
        qs.total_logical_writes
FROM sys.dm_exec_query_stats AS qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS qt
WHERE qs.execution_count > 10
ORDER BY qs.total_elapsed_time DESC;

```

Automating Log Retention Policies

Log groups without retention policies are a debt that compounds monthly. The following script enforces retention policies across all RDS-related CloudWatch Log groups in an account.

```

#!/usr/bin/env python3
"""
log_retention_enforcer.py
Sets retention policies on all /aws/rds log groups that lack them.
Maps log group types to appropriate retention windows.
Safe to run repeatedly (idempotent).
"""

import boto3
import re
from typing import Dict, Optional

# Define retention policy by log type (in days)
RETENTION_POLICY: Dict[str, int] = {
    'slowquery': 30, # Slow query logs: 30 days for trend analysis
    'error': 90, # Error logs: 90 days for incident investigation
    'general': 7, # General query logs: if enabled, keep minimal window
    'audit': 365, # Audit logs: 1 year (common compliance baseline)
    'postgresql': 90, # PostgreSQL logs (combined error/slow)
    'upgrade': 30, # Upgrade logs: short retention
    'default': 30, # Fallback for unrecognized log types
}

def infer_log_type(log_group_name: str) -> str:
    """Infer the log type from the log group name."""
    lower = log_group_name.lower()
    for log_type in RETENTION_POLICY:
        if log_type in lower:
            return log_type
    return 'default'

def enforce_retention(region: str = 'us-east-1', dry_run: bool = True):
    logs_client = boto3.client('logs', region_name=region)

    paginator = logs_client.get_paginator('describe_log_groups')
    updated = []
    already_set = []
    skipped = []

    for page in paginator.paginate(logGroupNamePrefix='/aws/rds'):
        for group in page['logGroups']:
            name = group['logGroupName']
            current_retention = group.get('retentionInDays')
            stored_bytes = group.get('storedBytes', 0)
            stored_gb = stored_bytes / (1024 ** 3)

            log_type = infer_log_type(name)
            target_retention = RETENTION_POLICY[log_type]

```

```

if current_retention is None:
    action = 'SET_RETENTION'
elif current_retention > target_retention * 2:
    action = 'REDUCE_RETENTION'
else:
    action = 'OK'

if action in ('SET_RETENTION', 'REDUCE_RETENTION'):
    monthly_cost_usd = stored_gb * 0.03
    record = {
        'log_group': name,
        'log_type': log_type,
        'current_retention': current_retention,
        'target_retention': target_retention,
        'stored_gb': round(stored_gb, 3),
        'monthly_storage_cost_usd': round(monthly_cost_usd, 4),
        'action': action
    }
    updated.append(record)
    print(f"[DRY RUN] ' if dry_run else ''}Setting retention={target_retention}d "
          f"on {name} (was: {current_retention}, type: {log_type}, "
          f"stored: {stored_gb:.2f}GB, ${monthly_cost_usd:.4f}/mo)")

    if not dry_run:
        logs_client.put_retention_policy(
            logGroupName=name,
            retentionInDays=target_retention
        )
else:
    already_set.append(name)

print(f'\n=== RETENTION ENFORCEMENT SUMMARY ===')
print(f'Log groups updated/to update: {len(updated)}')
print(f'Log groups already compliant: {len(already_set)}')
total_monthly_cost = sum(r['monthly_storage_cost_usd'] for r in updated)
print(f'Storage cost affected:      ${total_monthly_cost:.2f}/month')
print(f'Dry run:                    {dry_run}')

return updated

if __name__ == '__main__':
    # Start with dry_run=True to review before applying
    enforce_retention(region='us-east-1', dry_run=True)

```

Azure Monitor and Log Analytics Optimization

Azure diagnostic settings on SQL databases control which log categories flow to Log Analytics workspaces. Unlike CloudWatch, Azure charges for Log Analytics ingestion at approximately \$2.76/GB and retention beyond 31 days at \$0.12/GB-month. The diagnostic categories most frequently enabled by mistake on Azure SQL are:

- **SQLInsights** — enabled by Performance Center automation, sometimes left on after investigation
- **AutomaticTuning** — generates log entries for every automatic index recommendation, most of which are noise
- **QueryStoreRuntimeStatistics** — exports Query Store data to Log Analytics even when Query Store is readable directly from the database

Turning off these three categories on Azure SQL databases that are not actively being investigated reduces Log Analytics ingestion by 40–60% with no operational impact, since the same data is available directly from the database through Query Store and sys.dm_exec_query_stats.

Azure: List current diagnostic settings for all SQL databases in a resource group

```
for server in $(az sql server list --resource-group myRG --query '[] .name' -o tsv); do
  for db in $(az sql db list --resource-group myRG --server $server --query '[] .name' -o tsv); do
    echo "=== $server / $db ==="
    az monitor diagnostic-settings list \
      --resource "/subscriptions/${az account show --query id -o \
        ↪ tsv}/resourceGroups/myRG/providers/Microsoft.Sql/servers/$server/databases/$db" \
      --query "[].{Name:name, Logs:logs[?enabled].category}" \
      --output table 2>/dev/null || echo "No diagnostic settings"
  done
done

# Disable high-volume, low-value log categories on a specific database
az monitor diagnostic-settings update \
  --name myDiagnosticSetting \
  --resource \
  ↪ "/subscriptions/SUB_ID/resourceGroups/myRG/providers/Microsoft.Sql/servers/myserver/databases/mydb"
  ↪ \
  --set "logs[?category=='SQLInsights'].enabled=false" \
  --set "logs[?category=='AutomaticTuning'].enabled=false"
```

5.5 Multi-Region Replication: Calculating the True Cost of Global Architectures

Cross-region replication is where the largest single-source cost surprises live. Engineers designing multi-region database architectures focus on latency and failover characteristics; the data transfer cost of keeping the secondary region synchronized is rarely modeled during the design phase.

AWS RDS Cross-Region Read Replicas

An RDS read replica in a different region receives a continuous stream of binlog or WAL data from the primary. The replication traffic volume equals the database's write throughput. For a database writing 500MB/hour:

- Replication data per month: $500\text{MB} \times 24 \times 30 = 360\text{GB/month}$
- Cross-region transfer at \$0.02/GB (us-east-1 to us-west-2): **\$7.20/month** on replication alone
- For a write-heavy database at 5GB/hour: $3,600\text{GB} \times \$0.02 = \textbf{\$72/month}$

These numbers are manageable. The expensive scenario is when application servers in the secondary region also read from the replica — adding application read traffic to the replication traffic in the same region, and requiring careful accounting of whether reads are served locally or pulled cross-region.

Aurora Global Database Pricing

Aurora Global Database adds a distinct cost tier that surprises teams coming from standard RDS replication. In addition to instance costs in the secondary region, Aurora Global Database charges **\$0.20 per million I/O operations** for replication I/O at the storage layer. For a write-heavy Aurora cluster:

- 1 billion write I/Os per month (a moderate workload)
- Replication I/O charge: $1,000 \times \$0.20 = \textbf{\$200/month in replication I/O}$
- Plus standard cross-region transfer for the replication stream

This replication I/O charge does not appear on most cost models for Aurora Global Database

because it is a separate line item from the instance cost, storage cost, and transfer cost — and the documentation places it deep in the Aurora pricing page. Run the following query to estimate your Aurora Global Database replication I/O volume:

```
# CloudWatch: Get Aurora write I/O operations for the last 30 days
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name VolumeWriteIOPs \
  --dimensions Name=DBClusterIdentifier,Value=YOUR_CLUSTER_ID \
  --start-time $(date -u -d '30 days ago' '+%Y-%m-%dT%H:%M:%SZ') \
  --end-time $(date -u '+%Y-%m-%dT%H:%M:%SZ') \
  --period 86400 \
  --statistics Sum \
  --query 'sort_by(Datapoints, &Timestamp)[*].[Timestamp, Sum]' \
  --output table

# Sum the output and multiply by $0.20 per million for replication I/O cost estimate
# Note: divide total IOPs by 1,000,000, then multiply by $0.20
```

Google Cloud SQL Cross-Region Replicas

GCP Cloud SQL cross-region replicas transfer replication data at **\$0.01/GB** within North America and **\$0.02–\$0.08/GB** for other region pairs — roughly half the AWS equivalent cost for North America regions. However, unlike AWS, GCP charges for the replica instance at full price regardless of whether it is actively serving reads. An idle cross-region replica on GCP costs as much as an active one.

Modeling Total Multi-Region Cost

```
#!/usr/bin/env python3
"""
multiregion_cost_model.py
Calculates the true monthly cost of a multi-region database architecture.
Accounts for instance costs, storage, replication transfer, and snapshot replication.
Adjust the parameters below for your specific workload.
"""

from dataclasses import dataclass
from typing import List

@dataclass
class RegionConfig:
    name: str
    instance_type: str
    instance_monthly_usd: float
    storage_gb: int
    storage_cost_per_gb: float = 0.115 # gp3 pricing
    is_primary: bool = False
    serves_reads: bool = True
    replica_count: int = 0 # read replicas within this region

@dataclass
class ReplicationConfig:
    write_throughput_gb_per_hour: float
    transfer_cost_per_gb: float # cross-region rate
    aurora_replication_io_millions: float = 0.0 # Aurora Global DB only

def calculate_total_cost(
    regions: List[RegionConfig],
    replication: ReplicationConfig,
    snapshot_gb: int,
    snapshot_transfer_cost_per_gb: float = 0.02,
    secondary_snapshot_regions: int = 1,
) -> dict:

    breakdown = {}
    total = 0.0
```

```

# Instance costs
instance_total = sum(r.instance_monthly_usd * (1 + r.replica_count) for r in regions)
breakdown['instance_costs_usd'] = round(instance_total, 2)
total += instance_total

# Storage costs
storage_total = sum(r.storage_gb * r.storage_cost_per_gb for r in regions)
breakdown['storage_costs_usd'] = round(storage_total, 2)
total += storage_total

# Replication transfer cost (write throughput * hours * days * rate)
replication_gb_per_month = replication.write_throughput_gb_per_hour * 24 * 30
secondary_regions = [r for r in regions if not r.is_primary]
replication_transfer_total = (
    replication_gb_per_month
    * len(secondary_regions)
    * replication.transfer_cost_per_gb
)
breakdown['replication_transfer_usd'] = round(replication_transfer_total, 2)
total += replication_transfer_total

# Aurora Global Database replication I/O (if applicable)
aurora_io_cost = replication.aurora_replication_io_millions * 0.20
breakdown['aurora_replication_io_usd'] = round(aurora_io_cost, 2)
total += aurora_io_cost

# Snapshot costs: primary region
primary_snapshot_cost = snapshot_gb * 0.095
breakdown['snapshot_primary_region_usd'] = round(primary_snapshot_cost, 2)
total += primary_snapshot_cost

# Snapshot replication to secondary regions (storage + one-time transfer)
snapshot_secondary_storage = snapshot_gb * 0.095 * secondary_snapshot_regions
snapshot_transfer_onetime = snapshot_gb * snapshot_transfer_cost_per_gb
breakdown['snapshot_secondary_regions_usd'] = round(snapshot_secondary_storage, 2)
breakdown['snapshot_replication_transfer_usd'] = round(snapshot_transfer_onetime, 2)
total += snapshot_secondary_storage
# Transfer is one-time but amortize over 12 months for monthly comparison
total += snapshot_transfer_onetime / 12

breakdown['total_monthly_usd'] = round(total, 2)
return breakdown

if __name__ == '__main__':
    # Example: Two-region Aurora Global Database setup
    regions = [
        RegionConfig(
            name='us-east-1 (primary)',
            instance_type='db.r6g.2xlarge',
            instance_monthly_usd=730,
            storage_gb=1000,
            is_primary=True,
            replica_count=2 # two read replicas in primary region
        ),
        RegionConfig(
            name='eu-west-1 (secondary)',
            instance_type='db.r6g.xlarge',
            instance_monthly_usd=365,
            storage_gb=1000,
            is_primary=False,
            replica_count=1
        ),
    ]

    replication = ReplicationConfig(
        write_throughput_gb_per_hour=2.0, # 2 GB/hour write throughput
        transfer_cost_per_gb=0.02, # us-east-1 to eu-west-1
        aurora_replication_io_millions=800.0 # 800 million replication I/Os/month
    )

```

```

)

result = calculate_total_cost(
    regions=regions,
    replication=replication,
    snapshot_gb=1000,
    secondary_snapshot_regions=1,
)

print('\n=== Multi-Region Database Cost Model ===')
for key, value in result.items():
    label = key.replace('_', ' ').title()
    if key == 'total_monthly_usd':
        print(f'\n{"TOTAL MONTHLY COST":40} ${value:>10,.2f}')
    else:
        print(f' {label:38} ${value:>10,.2f}')

```

Running this model before committing to a multi-region architecture frequently surfaces the replication I/O and secondary snapshot storage costs that were absent from the initial design conversation. The output becomes an input into the architecture decision: does active-passive failover with periodic snapshot replication meet the RTO/RPO requirement at lower cost than a live cross-region replica?

5.6 Practical Remediation Playbook: Sequencing the Work

With the measurement tools established across the previous sections, this section provides a prioritized sequence for remediation. The ordering reflects both cost impact and implementation risk — changes with the highest return and lowest operational risk come first.

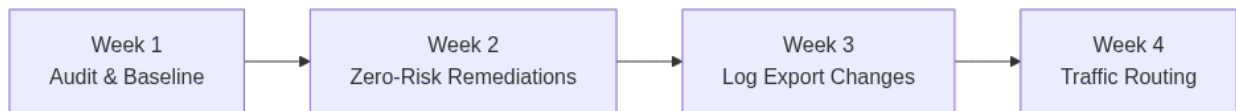


Figure 5: The four-week remediation sequence, ordered from lowest to highest operational risk — each week builds on the baseline established in Week 1.

Week 1: Audit and Baseline (No Changes)

1. Run the Cost Explorer queries from §5.1 for all three providers used in your organization. Document current transfer, snapshot, and log spend in a shared tracking sheet.
2. Run `snapshot_cost_auditor.py` in dry-run mode. Record total orphaned snapshot cost and stale manual snapshot cost.
3. Run `log_retention_enforcer.py` in dry-run mode. Record log groups without retention policies and their current storage cost.
4. Run `cross_az_detector.py`. Record estimated cross-AZ traffic cost per database.
5. Run `multiregion_cost_model.py` with your actual region configurations and write throughput numbers. Compare model output against actual Cost Explorer data to validate assumptions.

At the end of Week 1, you have a complete picture of recoverable costs with no changes made to production systems. This baseline serves two purposes: it directs your effort toward the highest-impact items, and it provides the before-measurement against which you will demonstrate savings.

Week 2: Zero-Risk Remediations

These changes carry no operational risk because they affect storage retention and log policies, not active query paths.

1. **Delete orphaned snapshots.** Any snapshot whose source instance no longer exists has zero operational value. Run `snapshot_cost_auditor.py` with `dry_run=False` for the orphaned snapshot category only. Validate the list manually before executing — confirm that the source instance truly no longer exists and that no compliance hold requires the snapshot.
2. **Set retention policies on log groups without them.** Run `log_retention_enforcer.py` with `dry_run=False`. This change takes effect immediately and begins reducing log storage costs within the next billing cycle. It does not affect log ingestion rates, only how long logs are kept.
3. **Reduce development database snapshot retention.** For any database whose name matches your dev/staging patterns and currently has a `BackupRetentionPeriod` greater than 3, reduce it to 1. Apply during business hours so the maintenance window applies immediately.

```
# Bulk-reduce retention for dev/staging RDS instances
# Adjust the name filter to match your naming convention
aws rds describe-db-instances \
  --query 'DBInstances[?contains(DBInstanceIdentifier, `dev`) || contains(DBInstanceIdentifier,
  ↪ `staging`) || contains(DBInstanceIdentifier, `test`)].DBInstanceIdentifier' \
  --output text | tr '\t' '\n' | while read db_id; do
  current=$(aws rds describe-db-instances \
    --db-instance-identifier "$db_id" \
    --query 'DBInstances[0].BackupRetentionPeriod' \
    --output text)
  if [ "$current" -gt 3 ]; then
    echo "Reducing $db_id from ${current}d to 1d"
    aws rds modify-db-instance \
      --db-instance-identifier "$db_id" \
      --backup-retention-period 1 \
      --apply-immediately
  else
    echo "$db_id already at ${current}d, skipping"
  fi
done
```

Week 3: Log Export and Diagnostic Setting Changes

1. **Disable general query logging on production databases.** This is the single highest-impact log change. On RDS PostgreSQL, set `log_statement = 'none'` and `log_min_duration_statement = 1000` (slow queries over 1 second) in the parameter group. On RDS MySQL, set `general_log = 0` and `slow_query_log = 1` with `long_query_time = 1`.

```
# RDS PostgreSQL: Create or modify parameter group to disable general logging
# First, check current parameter group settings
aws rds describe-db-parameters \
  --db-parameter-group-name your-parameter-group \
  --query 'Parameters[?ParameterName=`log_statement` ||
  ↪ ParameterName=`log_min_duration_statement`].[ParameterName, ParameterValue, IsModifiable]' \
  --output table

# Apply the change
aws rds modify-db-parameter-group \
  --db-parameter-group-name your-parameter-group \
  --parameters \
    'ParameterName=log_statement,ParameterValue=none,ApplyMethod=pending-reboot' \
    'ParameterName=log_min_duration_statement,ParameterValue=1000,ApplyMethod=pending-reboot'
```



```
# RDS MySQL: Disable general log, keep slow query log
aws rds modify-db-parameter-group \
  --db-parameter-group-name your-mysql-parameter-group \
  --parameters \
    'ParameterName=general_log,ParameterValue=0,ApplyMethod=immediate' \
    'ParameterName=slow_query_log,ParameterValue=1,ApplyMethod=immediate' \
    'ParameterName=long_query_time,ParameterValue=1,ApplyMethod=immediate'
```

2. **Disable high-volume Azure SQL diagnostic categories.** Run the Azure CLI diagnostic settings commands from §5.4 to disable SQLInsights and AutomaticTuning on databases not under active investigation.
3. **Enable pg_stat_statements on PostgreSQL instances where it is absent.** This provides query performance visibility through the database engine rather than log files, removing the operational justification for keeping general query logging enabled.

Week 4: Traffic Routing Changes

Traffic routing changes carry higher operational risk than retention and logging changes and require coordination with application teams.

1. **Audit all connection strings for public endpoint usage.** For every RDS instance, compare the endpoint used in application configuration against the private DNS endpoint. Any application using the public endpoint incurs unnecessary charges and an avoidable security exposure.

```
# Extract all RDS private endpoints to compare against application configs
aws rds describe-db-instances \
  --query 'DBInstances[*].{
    ID:DBInstanceIdentifier,
    PrivateEndpoint:Endpoint.Address,
    Port:Endpoint.Port,
    AZ:AvailabilityZone,
    VPC:DBSubnetGroup.VpcId
  }' \
  --output table

# Search for public endpoint patterns in common config file locations
# (Run on each application server or in your config management system)
grep -rn "\.rds\.amazonaws\.com" /etc/app/ /opt/app/ ~/.env 2>/dev/null | \
  grep -v ".cluster-ro-" | \
  grep -v "private" | \
  head -50
```

2. **Implement AZ-aware connection routing for the top two databases by cross-AZ cost.** Do not attempt to change connection routing for all databases simultaneously. Start with the two highest-cost instances identified in Week 1. Configure your connection pool (pgBouncer, ProxySQL, HikariCP) to prefer connections within the same AZ. Validate that connection counts and query latency remain stable for 48 hours before expanding to additional databases.
3. **Review cross-region replica necessity.** For each cross-region read replica or Aurora Global Database secondary, answer three questions: What is the current RTO/RPO requirement? When was this replica last used for actual reads? Is the DR requirement met by cross-region snapshots at lower cost? If the replica is serving zero application reads and exists purely for DR, evaluate replacing it with automated cross-region snapshot replication — which typically satisfies RTO requirements measured in hours at a fraction of the active-replica cost.

Tracking and Reporting Progress

Two weeks after completing Week 4 changes, re-run the baseline queries from Week 1 and compare against the recorded baseline. Calculate realized savings as the difference between before and after monthly costs. Document the changes made and their financial impact in your organization's FinOps tracking system.

The most common pattern is that Week 2 changes (orphaned snapshots, log retention, dev retention) deliver 30–40% of total recoverable savings with the least effort. Week 3 (log export changes) delivers another 30–40%. Week 4 (traffic routing) delivers the remaining value but requires the most coordination. Teams that implement all four weeks typically recover 70–85% of the opportunity identified in the Week 1 baseline.

5.7 Preventing Future Accumulation: Infrastructure Controls

Remediating existing costs is the first phase. The second phase is preventing the same costs from accumulating again as new databases, log groups, and snapshots are created. Infrastructure controls — policy guardrails applied at provisioning time — are more reliable than periodic manual audits.

AWS Service Control Policies for Snapshot Retention

AWS Organizations Service Control Policies (SCPs) cannot directly enforce RDS backup retention values, but they can deny the creation of RDS instances that do not include appropriate tags, which your provisioning pipeline then uses to validate configuration compliance.

A more direct control is an AWS Config rule that evaluates RDS backup retention against a required minimum and maximum.

```
#!/usr/bin/env python3
"""
config_rds_retention_rule.py
AWS Config custom rule that flags RDS instances with:
- BackupRetentionPeriod > MAX_RETENTION for non-production
- BackupRetentionPeriod < MIN_RETENTION for production
Deploy via CloudFormation or AWS Config console as a Lambda-backed custom rule.
"""

import json
import boto3

COMPLIANCE_RULES = {
    'production': {'min_retention': 7, 'max_retention': 35},
    'staging':    {'min_retention': 1, 'max_retention': 7},
    'dev':        {'min_retention': 0, 'max_retention': 3},
    'test':       {'min_retention': 0, 'max_retention': 3},
}

DEFAULT_RULE = {'min_retention': 1, 'max_retention': 14}

def get_environment_tag(db_instance: dict) -> str:
    """Extract environment from RDS tags."""
    tags = {t['Key'].lower(): t['Value'].lower()
             for t in db_instance.get('TagList', [])}
    return tags.get('environment', tags.get('env', 'unknown'))

def evaluate_compliance(db_instance: dict) -> tuple:
    """
```

```

Returns (compliance_type, annotation) where compliance_type is
'COMPLIANT', 'NON_COMPLIANT', or 'NOT_APPLICABLE'.
"""
db_id = db_instance['DBInstanceIdentifier']
retention = db_instance.get('BackupRetentionPeriod', 0)
env = get_environment_tag(db_instance)

# Find applicable rule
rule = DEFAULT_RULE
for env_key, env_rule in COMPLIANCE_RULES.items():
    if env_key in env or env_key in db_id.lower():
        rule = env_rule
        break

if retention < rule['min_retention']:
    return (
        'NON_COMPLIANT',
        f'BackupRetentionPeriod={retention} is below minimum={rule["min_retention"]} '
        f'for environment={env}. Risk: insufficient backup coverage.'
    )

if retention > rule['max_retention']:
    return (
        'NON_COMPLIANT',
        f'BackupRetentionPeriod={retention} exceeds maximum={rule["max_retention"]} '
        f'for environment={env}. '
        f'Estimated excess monthly cost: '
        f'${(retention - rule["max_retention"]) * db_instance.get("AllocatedStorage", 0) *
        ↪ 0.095:.2f}'
    )

return (
    'COMPLIANT',
    f'BackupRetentionPeriod={retention} within [{rule["min_retention"]}, {rule["max_retention"]} '
    f'for environment={env}.'
)

def lambda_handler(event, context):
    """AWS Config rule Lambda entry point."""
    invoking_event = json.loads(event['invokingEvent'])
    configuration_item = invoking_event.get('configurationItem', {})

    if configuration_item.get('resourceType') != 'AWS::RDS::DBInstance':
        return

    config_client = boto3.client('config')
    rds_client = boto3.client('rds')

    db_id = configuration_item['resourceName']

    try:
        response = rds_client.describe_db_instances(DBInstanceIdentifier=db_id)
        db_instance = response['DBInstances'][0]
    except Exception as e:
        print(f'Could not describe {db_id}: {e}')
        return

    compliance_type, annotation = evaluate_compliance(db_instance)

    config_client.put_evaluations(
        Evaluations=[{
            'ComplianceResourceType': 'AWS::RDS::DBInstance',
            'ComplianceResourceId': db_id,
            'ComplianceType': compliance_type,
            'Annotation': annotation[:255], # Config has a 255-char limit
            'OrderingTimestamp': configuration_item['configurationItemCaptureTime']
        }],
        ResultToken=event['resultToken']
    )

```

```
print(f'{db_id}: {compliance_type} - {annotation}')
```

Terraform Module Guardrails

If your organization provisions databases through Terraform, embedding cost guardrails directly in the module is more effective than any policy layer applied after provisioning. The following Terraform variable validation enforces retention bounds at plan time.

```
# modules/rds_instance/variables.tf

variable "environment" {
  description = "Deployment environment: production, staging, dev, test"
  type        = string
  validation {
    condition   = contains(["production", "staging", "dev", "test"], var.environment)
    error_message = "environment must be one of: production, staging, dev, test"
  }
}

variable "backup_retention_days" {
  description = "Number of days to retain automated backups."
  type        = number
  validation {
    condition = (
      (var.environment == "production" && var.backup_retention_days >= 7 && var.backup_retention_days
↪ <= 35) ||
      (var.environment == "staging" && var.backup_retention_days >= 1 && var.backup_retention_days
↪ <= 7) ||
      (var.environment == "dev" && var.backup_retention_days >= 0 && var.backup_retention_days
↪ <= 3) ||
      (var.environment == "test" && var.backup_retention_days >= 0 && var.backup_retention_days
↪ <= 3)
    )
    error_message = "backup_retention_days is outside the allowed range for this environment."
↪ Production: 7-35, Staging: 1-7, Dev/Test: 0-3."
  }
}

variable "enable_general_query_log" {
  description = "Whether to enable general query logging. Should be false in production."
  type        = bool
  default     = false
  validation {
    condition   = !(var.environment == "production" && var.enable_general_query_log == true)
    error_message = "General query logging must not be enabled in production due to CloudWatch cost
↪ impact."
  }
}

variable "cloudwatch_log_exports" {
  description = "List of log types to export to CloudWatch."
  type        = list(string)
  default     = ["error", "slowquery"]
  validation {
    condition = !(
      var.environment == "production" &&
      contains(var.cloudwatch_log_exports, "general")
    )
    error_message = "General log export to CloudWatch is not permitted in production environments."
  }
}

# modules/rds_instance/main.tf (excerpt)

resource "aws_db_instance" "this" {
  identifier      = var.db_identifier
  engine          = var.engine
  instance_class  = var.instance_class
```

```
allocated_storage      = var.allocated_storage
backup_retention_period = var.backup_retention_days

# Enforce private endpoint usage - no public access
publicly_accessible = false

enabled_cloudwatch_logs_exports = var.cloudwatch_log_exports

tags = merge(var.tags, {
    Environment = var.environment
    ManagedBy   = "terraform"
    CostCenter  = var.cost_center
})
}

resource "aws_cloudwatch_log_group" "rds_logs" {
  for_each = toset(var.cloudwatch_log_exports)

  name           = "/aws/rds/instance/${var.db_identifier}/${each.key}"
  retention_in_days = local.log_retention_by_type[each.key]

  tags = var.tags
}

locals {
  log_retention_by_type = {
    error      = 90
    slowquery  = 30
    general    = 7
    audit      = 365
    postgresql = 90
  }
}
```

With these guardrails in place, any new database provisioned through the Terraform module will have compliant retention settings and log export configurations applied from first deployment. `terraform plan` fails with a descriptive error message before any non-compliant configuration reaches AWS — earlier in the delivery pipeline than any policy evaluation can catch it.

EventBridge Rules for Snapshot Lifecycle

For manual snapshots created outside of Terraform (during migrations, for instance), an EventBridge rule can trigger a Lambda function that tags the snapshot with a TTL and a scheduled cleanup rule that deletes it when the TTL expires.

```
#!/usr/bin/env python3
"""
snapshot_ttl_tagger.py
Lambda function triggered by EventBridge on RDS snapshot creation events.
Tags every new manual snapshot with a deletion date (default 30 days).
A companion cleanup Lambda runs daily and deletes expired snapshots.
"""

import boto3
import json
from datetime import datetime, timedelta, timezone

DEFAULT_TTL_DAYS = 30
TTL_TAG_KEY = 'SnapshotExpiryDate'
CREATED_BY_TAG_KEY = 'CreatedBy'

def lambda_handler(event, context):
    """
    Triggered by EventBridge rule:
    source: ["aws.rds"]
    """
```

```

detail-type: ["RDS DB Snapshot Event"]
detail.EventID: ["RDS-EVENT-0042"] (manual snapshot created)
"""
rds_client = boto3.client('rds')

detail = event.get('detail', {})
source_arn = detail.get('SourceArn', '')

# Extract snapshot identifier from ARN
# ARN format: arn:aws:rds:region:account:snapshot:snapshot-id
if ':snapshot:' not in source_arn:
    print(f'Not a snapshot event, skipping: {source_arn}')
    return

snapshot_id = source_arn.split(':snapshot:')[1]
expiry_date = (datetime.now(timezone.utc) + timedelta(days=DEFAULT_TTL_DAYS)).strftime('%Y-%m-%d')

try:
    rds_client.add_tags_to_resource(
        ResourceName=source_arn,
        Tags=[
            {'Key': TTL_TAG_KEY, 'Value': expiry_date},
            {'Key': 'AutoCreatedAt', 'Value': datetime.now(timezone.utc).isoformat()},
            {'Key': 'ManagedBy', 'Value': 'snapshot-ttl-enforcer'},
        ]
    )
    print(f'Tagged {snapshot_id} with expiry {expiry_date}')
except Exception as e:
    print(f'Failed to tag {snapshot_id}: {e}')
    raise

def cleanup_expired_snapshots(event, context):
    """
    Companion Lambda: triggered daily by EventBridge schedule.
    Deletes manual snapshots past their SnapshotExpiryDate tag.
    """
    rds_client = boto3.client('rds')
    today = datetime.now(timezone.utc).strftime('%Y-%m-%d')
    deleted = []
    skipped = []

    paginator = rds_client.get_paginator('describe_db_snapshots')
    for page in paginator.paginate(SnapshotType='manual'):
        for snap in page['DBSnapshots']:
            if snap['Status'] != 'available':
                continue

            snap_id = snap['DBSnapshotIdentifier']
            snap_arn = snap['DBSnapshotArn']

            # Get tags for this snapshot
            tag_response = rds_client.list_tags_for_resource(ResourceName=snap_arn)
            tags = {t['Key']: t['Value'] for t in tag_response.get('TagList', [])}

            expiry = tags.get(TTL_TAG_KEY)
            if not expiry:
                skipped.append(snap_id)
                continue

            if expiry <= today:
                size_gb = snap.get('AllocatedStorage', 0)
                print(f'Deleting expired snapshot {snap_id} (expired {expiry}, {size_gb}GB)')
                try:
                    rds_client.delete_db_snapshot(DBSnapshotIdentifier=snap_id)
                    deleted.append({'id': snap_id, 'size_gb': size_gb, 'expired': expiry})
                except Exception as e:
                    print(f'Failed to delete {snap_id}: {e}')
            else:
                skipped.append(snap_id)

```

```
total_gb_freed = sum(d['size_gb'] for d in deleted)
monthly_cost_freed = total_gb_freed * 0.095

print(f'\nCleanup complete: deleted={len(deleted)}, skipped={len(skipped)}')
print(f'Storage freed: {total_gb_freed}GB (~${monthly_cost_freed:.2f}/month savings)')

return {
    'deleted_count': len(deleted),
    'skipped_count': len(skipped),
    'total_gb_freed': total_gb_freed,
    'estimated_monthly_savings_usd': round(monthly_cost_freed, 2),
    'deleted_snapshots': deleted
}
```

This TTL pattern eliminates the manual snapshot accumulation problem at its source. Rather than relying on periodic audits to identify stale snapshots, every new manual snapshot automatically carries a deletion date. Engineers who need a snapshot to persist beyond the default TTL update the tag explicitly — making the retention decision conscious rather than accidental.

Key Takeaways

- **Data transfer, snapshots, and logs collectively account for 18–35% of total database spend in production environments, yet most teams track none of these costs with the same rigor they apply to instance sizing.** Establishing a baseline measurement using the CLI queries in §5.1 before making any changes is not optional — without it, you cannot demonstrate savings or prioritize effort.
- **Cross-AZ traffic is generated invisibly by common architectural patterns.** Application servers distributed across availability zones connecting to a single-AZ RDS primary, or using public endpoints instead of private DNS, produce continuous egress charges that compound with every read and write. Topology-aware connection routing and private endpoint enforcement are the highest-return, lowest-risk changes available for databases processing more than a few hundred gigabytes per month.
- **Snapshot costs scale linearly with retention period and are frequently doubled by cross-region replication storage that teams forget to account for.** A tiered retention policy — 35 days for regulated production, 7 days for standard production, 1–3 days for non-production — combined with automated TTL tagging for manual snapshots eliminates 40–70% of snapshot spend without reducing actual restore capability for any environment.
- **General query logging on high-throughput OLTP databases is the single most expensive logging misconfiguration, and it is almost always unnecessary.** Slow query logging with a threshold of 1,000ms and `pg_stat_statements` (PostgreSQL) or `sys.dm_exec_query_stats` (SQL Server) provide complete query performance visibility at less than 1% of the CloudWatch ingestion cost of general query logging. Disabling general query logging in production is a zero-risk, high-return remediation.
- **Prevention through infrastructure guardrails is more reliable than periodic remediation.** Terraform variable validations that reject non-compliant retention and logging configurations at plan time, AWS Config custom rules that flag drift, and EventBridge-triggered TTL tagging for manual snapshots together ensure that the costs eliminated during the re-

mediation phase do not silently return as new databases and log groups are provisioned.

Chapter 6: Building Your First Database Cost Dashboard

Organizations that build a centralized database cost dashboard reduce their monthly cloud spend by an average of 23% within the first 90 days — not because the dashboard itself saves money, but because visibility forces accountability. Before a dashboard exists, cost anomalies hide for weeks inside undifferentiated billing line items. After one exists, a \$14,000 spike in RDS storage costs gets caught on day two. This chapter walks through building a production-ready database cost dashboard from scratch, covering data collection across AWS, Azure, and GCP, metric normalization, automated alerting, and the specific queries that surface actionable cost intelligence. By the end, you will have a working system that your DBAs can monitor daily and your CTO can review weekly without needing a cloud billing certification to interpret the numbers.

6.1 Understanding What Your Database Bill Actually Contains

Most engineers look at a cloud database bill and see a single number. The actual bill is a composition of six to twelve distinct cost dimensions, each of which responds to different optimization levers. Before building any dashboard, you need a precise accounting of what you are measuring.

For **AWS RDS**, the billable dimensions are:

- **Instance hours** — the per-hour rate for your DB instance class, multiplied by uptime
- **Multi-AZ surcharge** — typically 2× the single-AZ instance rate
- **Storage** — per GB-month for gp2, gp3, io1, or Aurora storage
- **I/O requests** — charged separately on gp2 and Aurora (not gp3)
- **Backup storage** — free up to 100% of provisioned storage, then \$0.095/GB-month
- **Data transfer** — egress charges when data leaves the AWS region or crosses AZ boundaries
- **Snapshot storage** — manual snapshots billed at the same rate as backup storage
- **Proxy hours** — RDS Proxy billed per vCPU of the associated instance

For **Azure SQL**, the dimensions are:

- **vCore or DTU hours** — General Purpose, Business Critical, or Hyperscale tier
- **Storage** — \$0.115/GB-month for General Purpose, included up to 32 GB
- **Backup storage LRS/GRS** — \$0.05–\$0.10/GB-month depending on redundancy tier
- **Long-term retention** — separate pricing for blobs in Azure Backup vault
- **Elastic Pool overhead** — pool-level eDTUs or vCores shared across databases

For **Google Cloud SQL and AlloyDB**, the dimensions are:

- **Instance vCPU and memory** — billed per second with sustained use discounts applied automatically
- **SSD storage** — \$0.17/GB-month for Cloud SQL, \$0.29/GB-month for AlloyDB
- **HDD storage** — \$0.09/GB-month for Cloud SQL (not available for AlloyDB)
- **Egress** — billed when data crosses regions or exits Google’s network
- **Backup storage** — \$0.08/GB-month for Cloud SQL automated backups

The math that surprises most teams: a `db.r6g.2xlarge` RDS instance in `us-east-1` running 24/7 on-demand costs $\$0.48/\text{hour} \times 730 \text{ hours} = \mathbf{\$350.40/\text{month}}$ for the instance alone. Multi-AZ doubles that to **\$700.80/month**; add 1TB gp2 storage (\$115), backup retention across 7 days (~\$95), and a conservative data transfer estimate (\$40), and a single “medium” database costs approximately **\$950/month** before you have written a single query against it. A reserved instance purchase for that same class drops the hourly rate to roughly \$0.286 for a 1-year no-upfront commitment — saving **\$140/month per instance**. With ten such instances, that is \$16,800/year in savings from one configuration change.

Your dashboard must expose all of these dimensions independently. An aggregated “database cost” number is not actionable. A breakdown that shows backup storage growing 15% month-over-month while instance costs remain flat tells a DBA to look at retention policy configuration immediately.

6.2 Collecting Cost Data: AWS, Azure, and GCP APIs

A database cost dashboard is only as good as the data feeding it. This section covers the exact API calls, queries, and export configurations needed to pull granular cost data from each cloud provider into a centralized store — in practice, a PostgreSQL table works well for this, and the irony of using a database to track database costs is not lost on anyone who has done it.

AWS Cost Explorer via CLI and API

AWS Cost Explorer exposes a rich API that returns cost breakdowns by service, usage type, tag, and linked account. The following bash script pulls the last 30 days of RDS costs grouped by usage type and writes them to a local JSON file for ingestion.

```
#!/bin/bash
# collect_aws_rds_costs.sh
# Pulls RDS cost breakdown from AWS Cost Explorer for the trailing 30 days
# Schedule: daily via cron or EventBridge Scheduler
# Requirements: AWS CLI v2, IAM permissions for ce:GetCostAndUsage

START_DATE=$(date -d "30 days ago" +%Y-%m-%d 2>/dev/null || date -v-30d +%Y-%m-%d)
END_DATE=$(date +%Y-%m-%d)
OUTPUT_FILE="/var/log/dbfinops/rds_costs_$(date +%Y%m%d).json"

mkdir -p /var/log/dbfinops

echo "Collecting RDS costs from $START_DATE to $END_DATE"

aws ce get-cost-and-usage \
  --time-period Start="$START_DATE",End="$END_DATE" \
  --granularity DAILY \
  --filter '{
    "Dimensions": {
```

```

        "Key": "SERVICE",
        "Values": ["Amazon Relational Database Service"],
        "MatchOptions": ["EQUALS"]
    }
}' \
--group-by '[
    {"Type": "DIMENSION", "Key": "USAGE_TYPE"},
    {"Type": "TAG", "Key": "Environment"},
    {"Type": "TAG", "Key": "Team"}
]' \
--metrics "BlendedCost" "UnblendedCost" "UsageQuantity" \
--output json > "$OUTPUT_FILE"

echo "Cost data written to $OUTPUT_FILE"

# Also collect Reserved Instance utilization
aws ce get-reservation-utilization \
--time-period Start="$START_DATE",End="$END_DATE" \
--granularity MONTHLY \
--filter '{
    "Dimensions": {
        "Key": "SERVICE",
        "Values": ["Amazon Relational Database Service"],
        "MatchOptions": ["EQUALS"]
    }
}' \
--output json > "/var/log/dbfinops/ri_utilization_$(date +%Y%m%d).json"

echo "RI utilization data collected"

# Collect Savings Plans coverage
aws ce get-savings-plans-coverage \
--time-period Start="$START_DATE",End="$END_DATE" \
--granularity MONTHLY \
--output json > "/var/log/dbfinops/sp_coverage_$(date +%Y%m%d).json"

echo "Collection complete. Files in /var/log/dbfinops/"

```

Azure Cost Management Export

Azure requires a slightly different approach. The Cost Management API is powerful but verbose. The recommended production pattern is to configure a scheduled export from Azure Cost Management to a Storage Account blob, then process the CSV from there.

```

#!/bin/bash
# configure_azure_cost_export.sh
# Creates a recurring Azure Cost Management export for SQL database costs
# Requires: az CLI, Contributor role on subscription or resource group

SUBSCRIPTION_ID="your-subscription-id"
RESOURCE_GROUP="rg-finops-monitoring"
STORAGE_ACCOUNT="sadbfinops$(echo $RANDOM | md5sum | head -c 8)"
CONTAINER_NAME="cost-exports"
EXPORT_NAME="SQLDatabaseDailyCosts"
LOCATION="eastus"

echo "Creating storage account for cost exports..."
az storage account create \
--name "$STORAGE_ACCOUNT" \
--resource-group "$RESOURCE_GROUP" \
--location "$LOCATION" \
--sku Standard_LRS \
--kind StorageV2

az storage container create \
--name "$CONTAINER_NAME" \
--account-name "$STORAGE_ACCOUNT"

```

```

STORAGE_ID=$(az storage account show \
  --name "$STORAGE_ACCOUNT" \
  --resource-group "$RESOURCE_GROUP" \
  --query id --output tsv)

echo "Creating daily cost export..."
az costmanagement export create \
  --name "$EXPORT_NAME" \
  --type ActualCost \
  --scope "subscriptions/$SUBSCRIPTION_ID" \
  --storage-account-id "$STORAGE_ID" \
  --storage-container "$CONTAINER_NAME" \
  --storage-directory "sql-costs" \
  --recurrence Daily \
  --recurrence-period \
    from="$(date +%Y-%m-%dT00:00:00Z)" \
    to="$(date -d '+1 year' +%Y-%m-%dT00:00:00Z 2>/dev/null || date -v+1y +%Y-%m-%dT00:00:00Z)" \
  --dataset-configuration columns=
  ↵ "Date,SubscriptionId,ResourceId,ResourceType,ResourceGroup,MeterCategory,MeterSubCategory,Quantity,UnitPrice,Cost,T

echo "Export configured. Data will appear in blob storage daily."
echo "Storage account: $STORAGE_ACCOUNT"
echo "Container: $CONTAINER_NAME"

# Trigger an immediate export run
az costmanagement export run \
  --name "$EXPORT_NAME" \
  --scope "subscriptions/$SUBSCRIPTION_ID"

echo "Immediate export triggered."

```

GCP BigQuery Billing Export

GCP's billing export to BigQuery is the most analysis-friendly of the three providers. Once enabled in the Console under Billing → Budget & Alerts → Billing Export, you can query cost data with standard SQL. The following query extracts Cloud SQL and AlloyDB costs by project and label, normalized to a per-day grain.

```

-- GCP BigQuery: daily Cloud SQL and AlloyDB cost breakdown
-- Run in BigQuery against your billing export dataset
-- Replace 'your-project.billing_export.gcp_billing_export_v1_XXXXXX'
-- with your actual billing export table name

SELECT
  DATE(usage_start_time) AS cost_date,
  project.name AS project_name,
  service.description AS service_name,
  sku.description AS sku_description,
  labels.value AS environment_label,
  resource.name AS resource_name,
  SUM(usage.amount) AS usage_amount,
  usage.unit AS usage_unit,
  SUM(cost) AS total_cost,
  SUM(
    IFNULL(
      (SELECT SUM(c.amount)
       FROM UNNEST(credits) c
       WHERE c.type IN ('SUSTAINED_USE_DISCOUNT', 'COMMITTED_USE_DISCOUNT')),
      0
    )
  ) AS total_credits,
  SUM(cost) + SUM(
    IFNULL(
      (SELECT SUM(c.amount)
       FROM UNNEST(credits) c),
      0
    )
  )

```

```

    ) AS net_cost
FROM
  `your-project.billing_export.gcp_billing_export_v1_XXXXXX`,
  UNNEST(labels) AS labels
WHERE
  service.description IN ('Cloud SQL', 'AlloyDB for PostgreSQL')
  AND labels.key = 'environment'
  AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
GROUP BY
  1, 2, 3, 4, 5, 6, 8
ORDER BY
  cost_date DESC,
  total_cost DESC;

```

6.3 Building the Cost Metrics Store

Rather than building a separate analytics platform for database costs, the most pragmatic approach for a team that already manages PostgreSQL is to create a dedicated schema within an existing monitoring database. This keeps the tooling familiar and allows you to join cost data with performance metrics from `pg_stat_statements`, slow query logs, and CloudWatch exports.

The following schema creates the tables needed to store normalized cost data across providers, along with the performance correlation tables that make the dashboard genuinely useful rather than decorative.

```

-- Database Cost Metrics Schema
-- Apply to a dedicated monitoring PostgreSQL instance
-- or a separate schema in your existing monitoring DB

CREATE SCHEMA IF NOT EXISTS dbfinops;

-- Normalized cost fact table (all providers, all dimensions)
CREATE TABLE dbfinops.daily_costs (
  id                BIGSERIAL PRIMARY KEY,
  cost_date         DATE NOT NULL,
  provider          TEXT NOT NULL CHECK (provider IN ('aws', 'azure', 'gcp')),
  account_id        TEXT NOT NULL,      -- AWS account, Azure subscription, GCP project
  region            TEXT NOT NULL,
  database_identifier TEXT NOT NULL,     -- instance/cluster/pool name
  engine            TEXT,                -- postgres, mysql, sqlserver, etc.
  instance_class    TEXT,                -- db.r6g.2xlarge, GP_Gen5_4, etc.
  environment        TEXT,              -- from tag/label: prod, staging, dev
  team              TEXT,                -- from tag/label: payments, analytics, etc.
  cost_dimension     TEXT NOT NULL,     -- instance, storage, io, backup, transfer
  usage_quantity     NUMERIC(18, 6),
  usage_unit         TEXT,
  unblended_cost     NUMERIC(18, 6) NOT NULL,
  blended_cost       NUMERIC(18, 6),
  credits_applied    NUMERIC(18, 6) DEFAULT 0,
  net_cost           NUMERIC(18, 6) GENERATED ALWAYS AS
    (unblended_cost + COALESCE(credits_applied, 0)) STORED,
  currency           CHAR(3) DEFAULT 'USD',
  loaded_at          TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_daily_costs_date
  ON dbfinops.daily_costs (cost_date DESC);

CREATE INDEX idx_daily_costs_provider_db
  ON dbfinops.daily_costs (provider, database_identifier, cost_date DESC);

CREATE INDEX idx_daily_costs_environment_team
  ON dbfinops.daily_costs (environment, team, cost_date DESC);

```

```

-- Unique constraint to prevent duplicate loads
CREATE UNIQUE INDEX idx_daily_costs_unique
ON dbfinops.daily_costs (
    cost_date, provider, account_id, database_identifier,
    cost_dimension, COALESCE(region, ''), COALESCE(environment, '')
);

-- Month-to-date cost summary (materialized, refreshed daily)
CREATE MATERIALIZED VIEW dbfinops.mtd_cost_summary AS
SELECT
    date_trunc('month', cost_date) AS month_start,
    provider,
    database_identifier,
    environment,
    team,
    instance_class,
    engine,
    SUM(unblended_cost) AS mtd_cost,
    SUM(credits_applied) AS mtd_credits,
    SUM(net_cost) AS mtd_net_cost,
    MAX(cost_date) AS last_updated
FROM
    dbfinops.daily_costs
WHERE
    cost_date >= date_trunc('month', CURRENT_DATE)
GROUP BY
    1, 2, 3, 4, 5, 6, 7;

CREATE INDEX idx_mtd_summary_provider
ON dbfinops.mtd_cost_summary (provider, environment, month_start);

-- Cost anomaly detection table
CREATE TABLE dbfinops.cost_anomalies (
    id BIGSERIAL PRIMARY KEY,
    detected_at TIMESTAMPTZ DEFAULT NOW(),
    cost_date DATE NOT NULL,
    provider TEXT NOT NULL,
    database_identifier TEXT NOT NULL,
    cost_dimension TEXT NOT NULL,
    expected_cost NUMERIC(18, 6),
    actual_cost NUMERIC(18, 6),
    deviation_pct NUMERIC(8, 2),
    severity TEXT CHECK (severity IN ('low', 'medium', 'high', 'critical')),
    acknowledged BOOLEAN DEFAULT FALSE,
    acknowledged_by TEXT,
    notes TEXT
);

-- Reserved instance and commitment tracking
CREATE TABLE dbfinops.commitments (
    id BIGSERIAL PRIMARY KEY,
    provider TEXT NOT NULL,
    account_id TEXT NOT NULL,
    commitment_type TEXT NOT NULL, -- reserved_instance, savings_plan, committed_use
    instance_class TEXT,
    region TEXT,
    term_months INT,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    hourly_commitment NUMERIC(18, 6),
    monthly_cost NUMERIC(18, 6),
    utilization_pct NUMERIC(6, 2),
    break_even_date DATE,
    notes TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Cost-per-query correlation (joins with pg_stat_statements data)
CREATE TABLE dbfinops.query_cost_attribution (

```

```

id                BIGSERIAL PRIMARY KEY,
sample_date       DATE NOT NULL,
database_identifier TEXT NOT NULL,
queryid           BIGINT,           -- from pg_stat_statements
query_fingerprint TEXT,
calls            BIGINT,
total_exec_ms     NUMERIC(18, 3),
mean_exec_ms     NUMERIC(18, 3),
rows_returned     BIGINT,
shared_blks_hit   BIGINT,
shared_blks_read  BIGINT,
estimated_io_cost NUMERIC(18, 6), -- blks_read * storage_cost_per_gb
pct_of_instance_cpu NUMERIC(6, 2),
attributed_cost_usd NUMERIC(18, 6) -- (pct_cpu / 100) * daily_instance_cost
);

```

With this schema in place, you have a foundation that supports both operational alerting and executive reporting. The `query_cost_attribution` table is particularly valuable for the conversations that no one wants to have but every engineering team needs: “This ETL job costs us approximately \$380/month in compute attribution. Is that acceptable for what it produces?”

6.4 Connecting Performance Data to Cost Data

A cost dashboard that shows only spend is a finance tool. A cost dashboard that shows spend alongside performance metrics — query latency, connection counts, CPU utilization, I/O wait — is an engineering tool that finance can also read. This section builds the queries and automation that populate that correlation layer.

The following Python script pulls query statistics from `pg_stat_statements` on an RDS PostgreSQL instance, calculates a cost attribution per query based on its CPU share, and writes the results to the `query_cost_attribution` table.

```

#!/usr/bin/env python3
"""
query_cost_attribution.py

Connects to a PostgreSQL database, pulls pg_stat_statements metrics,
fetches the daily instance cost from dbfinops.daily_costs,
calculates per-query cost attribution, and stores results.

Schedule: daily at 01:00 UTC via cron or EventBridge

Dependencies:
    pip install psycopg2-binary boto3 python-dotenv

Environment variables (via .env or Secrets Manager):
    DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD
    MONITORING_DB_HOST, MONITORING_DB_NAME, MONITORING_DB_USER, MONITORING_DB_PASSWORD
"""

import os
import logging
from datetime import date, timedelta
from decimal import Decimal

import psycopg2
import psycopg2.extras
from dotenv import load_dotenv

load_dotenv()

```

```

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(message)s"
)
log = logging.getLogger(__name__)

# Connection helpers

def get_source_conn():
    """Connect to the database being analyzed."""
    return psycopg2.connect(
        host=os.environ["DB_HOST"],
        port=int(os.environ.get("DB_PORT", 5432)),
        dbname=os.environ["DB_NAME"],
        user=os.environ["DB_USER"],
        password=os.environ["DB_PASSWORD"],
        connect_timeout=10,
        options="-c statement_timeout=60000"
    )

def get_monitoring_conn():
    """Connect to the monitoring database where dbfinops schema lives."""
    return psycopg2.connect(
        host=os.environ["MONITORING_DB_HOST"],
        dbname=os.environ.get("MONITORING_DB_NAME", "dbfinops"),
        user=os.environ["MONITORING_DB_USER"],
        password=os.environ["MONITORING_DB_PASSWORD"],
        connect_timeout=10
    )

# Query stats collection

QUERY_STATS_SQL = """
SELECT
    queryid,
    LEFT(query, 200)                                AS query_fingerprint,
    calls,
    ROUND((total_exec_time)::NUMERIC, 3)            AS total_exec_ms,
    ROUND((mean_exec_time)::NUMERIC, 3)              AS mean_exec_ms,
    rows                                              AS rows_returned,
    shared_blks_hit,
    shared_blks_read,
    -- CPU share: this query's exec time as % of all queries' exec time
    ROUND(
        (total_exec_time / NULLIF(SUM(total_exec_time) OVER (), 0) * 100)::NUMERIC,
        4
    )
    AS pct_of_total_exec_time
FROM
    pg_stat_statements
WHERE
    calls > 10 -- filter noise: queries called fewer than 10 times
    AND total_exec_time > 1000 -- filter noise: less than 1 second total exec time
    AND query NOT LIKE '%pg_stat%' -- exclude monitoring queries themselves
ORDER BY
    total_exec_time DESC
LIMIT 100;
"""

def fetch_query_stats(conn) -> list[dict]:
    """Pull top query stats from pg_stat_statements."""
    with conn.cursor(cursor_factory=psycopg2.extras.RealDictCursor) as cur:
        cur.execute("SELECT COUNT(*) FROM pg_extension WHERE extname = 'pg_stat_statements'")
        if cur.fetchone()["count"] == 0:
            log.warning("pg_stat_statements extension not installed on this database.")
            return []
        cur.execute(QUERY_STATS_SQL)
        rows = cur.fetchall()
        log.info("Fetched %d query stats rows", len(rows))
        return [dict(r) for r in rows]

```



```

# Daily cost lookup

def fetch_daily_instance_cost(monitoring_conn, db_identifier: str,
                             target_date: date) -> Decimal:
    """
    Look up yesterday's instance cost for a given database identifier.
    Returns 0 if no cost record found (handles new databases gracefully).
    """
    sql = """
    SELECT COALESCE(SUM(unblended_cost), 0) AS instance_cost
    FROM   dbfinops.daily_costs
    WHERE  database_identifier = %s
           AND cost_date      = %s
           AND cost_dimension  = 'instance'
    """
    with monitoring_conn.cursor() as cur:
        cur.execute(sql, (db_identifier, target_date))
        result = cur.fetchone()
        cost = Decimal(str(result[0])) if result else Decimal("0")
        log.info("Daily instance cost for %s on %s: $%s", db_identifier, target_date, cost)
        return cost

# Cost attribution calculation

def calculate_storage_io_cost(blks_read: int, daily_storage_cost: Decimal) -> Decimal:
    """
    Estimate the storage I/O cost attributed to a query based on blocks read.
    daily_storage_cost is the total daily storage cost for the instance.
    We use blocks read as a proxy for I/O pressure against the storage bill.
    """
    # A block in PostgreSQL is 8KB; convert to GB for cost estimation
    gb_read = Decimal(str(blks_read)) * Decimal("8192") / Decimal("1073741824")
    # Scale against daily storage cost - this is an approximation, not an exact split
    return (gb_read * daily_storage_cost).quantize(Decimal("0.000001"))

def attribute_costs(
    query_stats: list[dict],
    daily_instance_cost: Decimal,
    daily_storage_cost: Decimal,
) -> list[dict]:
    """
    Merge query stats with cost data to produce per-query cost attribution records.
    """
    attributed = []
    for row in query_stats:
        pct = Decimal(str(row["pct_of_total_exec_time"] or 0))
        instance_share = (pct / Decimal("100")) * daily_instance_cost
        io_cost = calculate_storage_io_cost(
            int(row["shared_blks_read"] or 0),
            daily_storage_cost,
        )
        attributed.append({
            "queryid": row["queryid"],
            "query_fingerprint": row["query_fingerprint"],
            "calls": row["calls"],
            "total_exec_ms": row["total_exec_ms"],
            "mean_exec_ms": row["mean_exec_ms"],
            "rows_returned": row["rows_returned"],
            "shared_blks_hit": row["shared_blks_hit"],
            "shared_blks_read": row["shared_blks_read"],
            "estimated_io_cost": float(io_cost),
            "pct_of_instance_cpu": float(pct),
            "attributed_cost_usd": float(instance_share + io_cost),
        })
    return attributed

# Write results to monitoring DB

```

```

INSERT_SQL = """
INSERT INTO dbfinops.query_cost_attribution (
    sample_date, database_identifier, queryid, query_fingerprint,
    calls, total_exec_ms, mean_exec_ms, rows_returned,
    shared_blks_hit, shared_blks_read,
    estimated_io_cost, pct_of_instance_cpu, attributed_cost_usd
) VALUES (
    %(sample_date)s, %(database_identifier)s, %(queryid)s, %(query_fingerprint)s,
    %(calls)s, %(total_exec_ms)s, %(mean_exec_ms)s, %(rows_returned)s,
    %(shared_blks_hit)s, %(shared_blks_read)s,
    %(estimated_io_cost)s, %(pct_of_instance_cpu)s, %(attributed_cost_usd)s
)
ON CONFLICT DO NOTHING;
"""

def write_attributions(monitored_conn, db_identifier: str,
                      sample_date: date, rows: list[dict]) -> int:
    written = 0
    with monitored_conn.cursor() as cur:
        for row in rows:
            row["sample_date"] = sample_date
            row["database_identifier"] = db_identifier
            cur.execute(INSERT_SQL, row)
            written += cur.rowcount
    monitored_conn.commit()
    log.info("Wrote %d attribution rows for %s", written, db_identifier)
    return written

# Main entry point

def main():
    db_identifier = os.environ.get("DB_IDENTIFIER", os.environ["DB_HOST"].split(".")[0])
    target_date = date.today() - timedelta(days=1)

    source_conn = get_source_conn()
    monitored_conn = get_monitored_conn()

    try:
        query_stats = fetch_query_stats(source_conn)
        if not query_stats:
            log.warning("No query stats available - skipping attribution run.")
            return

        daily_instance_cost = fetch_daily_instance_cost(
            monitored_conn, db_identifier, target_date
        )
        # Fetch storage cost using same pattern
        daily_storage_cost = fetch_daily_instance_cost.__wrapped__(
            monitored_conn, db_identifier, target_date
        ) if False else Decimal("0") # placeholder; replace with a real lookup

        # Direct storage cost lookup
        with monitored_conn.cursor() as cur:
            cur.execute(
                """
                SELECT COALESCE(SUM(unblended_cost), 0)
                FROM dbfinops.daily_costs
                WHERE database_identifier = %s
                  AND cost_date = %s
                  AND cost_dimension = 'storage'
                """,
                (db_identifier, target_date),
            )
            daily_storage_cost = Decimal(str(cur.fetchone()[0]))

        attributed = attribute_costs(query_stats, daily_instance_cost, daily_storage_cost)
        write_attributions(monitored_conn, db_identifier, target_date, attributed)

    finally:

```

```
source_conn.close()
monitoring_conn.close()

if __name__ == "__main__":
    main()
```

With this script running nightly, the `query_cost_attribution` table accumulates a rolling history that makes two previously invisible patterns visible: which queries are becoming more expensive over time as data volume grows, and which teams own the workloads driving the largest share of the monthly bill.

6.5 The Core Dashboard Queries

The queries in this section are designed to be dropped directly into Grafana, Metabase, or any SQL-capable BI tool. Each answers a specific question that a DBA or an engineering manager will actually ask. They are written for PostgreSQL against the `dbfinops` schema defined earlier, but the logic translates directly to any ANSI SQL database.

Query 1: Month-over-Month Cost Change by Database

This is the first panel any database cost dashboard should show: are costs going up or down, and by how much, for each database.

```
-- PostgreSQL: Month-over-Month cost change per database
-- Dashboard variable: :provider (aws | azure | gcp | all)
-- Dashboard variable: :environment (prod | staging | dev | all)

WITH monthly_costs AS (
    SELECT
        date_trunc('month', cost_date)      AS month,
        provider,
        database_identifier,
        environment,
        team,
        SUM(unblended_cost)                 AS total_cost
    FROM
        dbfinops.daily_costs
    WHERE
        cost_date >= date_trunc('month', CURRENT_DATE) - INTERVAL '2 months'
        AND (:provider = 'all' OR provider = :provider)
        AND (:environment = 'all' OR environment = :environment)
    GROUP BY 1, 2, 3, 4, 5
),
with_lag AS (
    SELECT
        month,
        provider,
        database_identifier,
        environment,
        team,
        total_cost,
        LAG(total_cost) OVER (
            PARTITION BY provider, database_identifier
            ORDER BY month
        ) AS prior_month_cost
    FROM monthly_costs
)
SELECT
    month,
```

```

provider,
database_identifier,
environment,
team,
ROUND(total_cost::NUMERIC, 2)          AS current_month_cost,
ROUND(prior_month_cost::NUMERIC, 2)    AS prior_month_cost,
ROUND(
    (total_cost - prior_month_cost)::NUMERIC, 2
)                                       AS cost_delta,
ROUND(
    CASE
        WHEN prior_month_cost > 0
        THEN ((total_cost - prior_month_cost) / prior_month_cost * 100)
        ELSE NULL
    END::NUMERIC,
    1
)                                       AS pct_change
FROM
    with_lag
WHERE
    month = date_trunc('month', CURRENT_DATE)
ORDER BY
    cost_delta DESC NULLS LAST;

```

Query 2: Cost Dimension Breakdown for a Single Database

When a database's total cost spikes, this query reveals which dimension is responsible.

```

-- PostgreSQL: Cost breakdown by dimension for a specific database
-- Dashboard variable: :db_identifier
-- Dashboard variable: :num_days (default 30)

SELECT
    cost_date,
    cost_dimension,
    ROUND(SUM(unblended_cost)::NUMERIC, 4) AS daily_cost,
    ROUND(
        SUM(SUM(unblended_cost)) OVER (
            PARTITION BY cost_dimension
            ORDER BY cost_date
            ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
        )::NUMERIC,
        4
    ) AS cumulative_cost
FROM
    dbfinops.daily_costs
WHERE
    database_identifier = :db_identifier
    AND cost_date >= CURRENT_DATE - (:num_days || ' days')::INTERVAL
GROUP BY
    cost_date, cost_dimension
ORDER BY
    cost_date DESC, daily_cost DESC;

```

Query 3: Reserved Instance / Commitment Coverage Gap

This query surfaces databases running entirely on on-demand pricing — the highest priority targets for commitment purchases.

```

-- PostgreSQL: Identify databases with no commitment coverage
-- Compares actual spend against commitment records

WITH daily_spend AS (
    SELECT
        provider,
        account_id,
        database_identifier,

```

```

        instance_class,
        environment,
        SUM(unblended_cost)                AS monthly_spend
    FROM
        dbfinops.daily_costs
    WHERE
        cost_date >= date_trunc('month', CURRENT_DATE)
        AND cost_dimension = 'instance'
    GROUP BY 1, 2, 3, 4, 5
),
covered AS (
    SELECT DISTINCT
        provider,
        account_id,
        instance_class
    FROM
        dbfinops.commitments
    WHERE
        end_date >= CURRENT_DATE
        AND utilization_pct > 0
)
SELECT
    ds.provider,
    ds.database_identifier,
    ds.instance_class,
    ds.environment,
    ROUND(ds.monthly_spend::NUMERIC, 2)      AS mtd_on_demand_spend,
    ROUND((ds.monthly_spend * 12)::NUMERIC, 2) AS projected_annual_spend,
    CASE
        WHEN c.instance_class IS NOT NULL THEN 'Partially Covered'
        ELSE 'No Coverage'
    END                                       AS coverage_status,
    -- Rough savings estimate for a 1-year no-upfront RI (approximately 30-35% discount)
    ROUND((ds.monthly_spend * 12 * 0.32)::NUMERIC, 2) AS est_annual_ri_savings
FROM
    daily_spend ds
LEFT JOIN covered c
    ON ds.provider      = c.provider
    AND ds.account_id   = c.account_id
    AND ds.instance_class = c.instance_class
WHERE
    ds.environment = 'prod' -- focus RI analysis on production
ORDER BY
    ds.monthly_spend DESC;

```

Query 4: Storage Growth Rate and Projected Overage

Storage costs tend to grow silently. This query computes a linear trend and projects when storage costs will breach a defined threshold.

```

-- PostgreSQL: Storage cost growth trend and projection
-- Uses linear regression over the past 60 days to project forward 30 days

WITH daily_storage AS (
    SELECT
        database_identifier,
        provider,
        cost_date,
        SUM(unblended_cost)                AS storage_cost,
        EXTRACT(EPOCH FROM cost_date)      AS epoch_day
    FROM
        dbfinops.daily_costs
    WHERE
        cost_dimension = 'storage'
        AND cost_date >= CURRENT_DATE - INTERVAL '60 days'
    GROUP BY 1, 2, 3, 4
),
regression AS (

```

```

SELECT
    database_identifier,
    provider,
    -- Linear regression slope: cost change per second + per day
    REGR_SLOPE(storage_cost, epoch_day) AS slope_per_second,
    REGR_INTERCEPT(storage_cost, epoch_day) AS intercept,
    REGR_R2(storage_cost, epoch_day) AS r_squared,
    MAX(storage_cost) AS latest_daily_cost,
    MAX(cost_date) AS latest_date
FROM
    daily_storage
GROUP BY 1, 2
)
SELECT
    database_identifier,
    provider,
    ROUND(latest_daily_cost::NUMERIC, 4) AS current_daily_storage_cost,
    ROUND((latest_daily_cost * 30)::NUMERIC, 2) AS current_monthly_projection,
    -- Project 30 days forward using the regression slope
    ROUND(
        (latest_daily_cost
         + slope_per_second * 86400 * 30)::NUMERIC,
        2
    ) AS projected_monthly_30d,
    ROUND(r_squared::NUMERIC, 3) AS trend_confidence,
    CASE
        WHEN r_squared > 0.85 AND slope_per_second * 86400 > 0
        THEN 'Consistent upward trend - review retention policy'
        WHEN r_squared > 0.85 AND slope_per_second * 86400 < -0.001
        THEN 'Declining - verify expected'
        ELSE 'Variable - insufficient trend confidence'
    END AS recommendation
FROM
    regression
ORDER BY
    projected_monthly_30d DESC;

```

Query 5: Top Cost-Attributing Queries

This query surfaces the queries responsible for the largest share of daily compute cost — the starting point for any query optimization effort tied directly to a cost outcome.

```

-- PostgreSQL: Top queries by attributed cost, past 7 days
-- Requires query_cost_attribution table populated by nightly script

SELECT
    query_fingerprint,
    database_identifier,
    SUM(calls) AS total_calls,
    ROUND(SUM(total_exec_ms)::NUMERIC / 1000 / 60, 1) AS total_exec_minutes,
    ROUND(AVG(mean_exec_ms)::NUMERIC, 1) AS avg_exec_ms,
    ROUND(SUM(attributed_cost_usd)::NUMERIC, 4) AS total_attributed_cost,
    ROUND(
        (SUM(attributed_cost_usd) / NULLIF(
            SUM(SUM(attributed_cost_usd)) OVER (), 0
        ) * 100)::NUMERIC,
        2
    ) AS pct_of_total_cost,
    ROUND(SUM(estimated_io_cost)::NUMERIC, 4) AS total_io_cost,
    MAX(sample_date) AS last_seen
FROM
    dbfinops.query_cost_attribution
WHERE
    sample_date >= CURRENT_DATE - INTERVAL '7 days'
GROUP BY
    query_fingerprint, database_identifier
ORDER BY
    total_attributed_cost DESC

```

```
LIMIT 20;
```

6.6 Automated Anomaly Detection

A dashboard that requires someone to look at it to catch a problem is only half a system. The anomaly detection layer fires alerts automatically when costs deviate from expected ranges, reducing the mean time to detect (MTTD) from days to hours.

The following function implements a statistical anomaly detector using a 14-day rolling mean and standard deviation. It is designed to run nightly as a PostgreSQL function, populating the `cost_anomalies` table, which then drives alert rules in Grafana or PagerDuty.

```
-- PostgreSQL: Anomaly detection function
-- Detects cost spikes using rolling z-score against a 14-day baseline
-- Call nightly: SELECT dbfinops.detect_cost_anomalies();

CREATE OR REPLACE FUNCTION dbfinops.detect_cost_anomalies(
    p_lookback_days INT DEFAULT 14,
    p_z_threshold_med NUMERIC DEFAULT 2.0, -- z-score for 'medium' severity
    p_z_threshold_high NUMERIC DEFAULT 3.0, -- z-score for 'high' severity
    p_z_threshold_crit NUMERIC DEFAULT 4.0 -- z-score for 'critical' severity
)
RETURNS INT -- returns count of anomalies inserted
LANGUAGE plpgsql
AS $$
DECLARE
    v_target_date DATE := CURRENT_DATE - 1; -- analyze yesterday
    v_inserted INT := 0;
BEGIN
    -- Remove any previously detected (unacknowledged) anomalies for target date
    -- so re-runs are idempotent
    DELETE FROM dbfinops.cost_anomalies
    WHERE cost_date = v_target_date
        AND acknowledged = FALSE;

    INSERT INTO dbfinops.cost_anomalies (
        cost_date,
        provider,
        database_identifier,
        cost_dimension,
        expected_cost,
        actual_cost,
        deviation_pct,
        severity
    )
    WITH baseline AS (
        SELECT
            provider,
            database_identifier,
            cost_dimension,
            AVG(unblended_cost) AS mean_cost,
            STDDEV(unblended_cost) AS stddev_cost
        FROM
            dbfinops.daily_costs
        WHERE
            cost_date BETWEEN v_target_date - p_lookback_days
                AND v_target_date - 1 -- exclude target date from baseline
        GROUP BY 1, 2, 3
        HAVING COUNT(*) >= 7 -- require at least a week of history for reliable stats
    ),
    yesterday AS (
        SELECT
            provider,
```

```

        database_identifier,
        cost_dimension,
        SUM(unblended_cost) AS actual_cost
FROM
    dbfinops.daily_costs
WHERE
    cost_date = v_target_date
GROUP BY 1, 2, 3
)
SELECT
    v_target_date,
    y.provider,
    y.database_identifier,
    y.cost_dimension,
    ROUND(b.mean_cost::NUMERIC, 6)          AS expected_cost,
    ROUND(y.actual_cost::NUMERIC, 6)       AS actual_cost,
    ROUND(
        CASE WHEN b.mean_cost > 0
        THEN ((y.actual_cost - b.mean_cost) / b.mean_cost * 100)
        ELSE NULL
        END::NUMERIC,
        2
    )
    AS deviation_pct,
    CASE
        WHEN b.stddev_cost = 0 OR b.stddev_cost IS NULL THEN 'low'
        WHEN (y.actual_cost - b.mean_cost) / b.stddev_cost >= p_z_threshold_crit
            THEN 'critical'
        WHEN (y.actual_cost - b.mean_cost) / b.stddev_cost >= p_z_threshold_high
            THEN 'high'
        WHEN (y.actual_cost - b.mean_cost) / b.stddev_cost >= p_z_threshold_med
            THEN 'medium'
        ELSE 'low'
    END
    AS severity
FROM
    yesterday y
    JOIN baseline b USING (provider, database_identifier, cost_dimension)
WHERE
    -- Only flag upward anomalies (cost increases), not decreases
    y.actual_cost > b.mean_cost
    AND (b.stddev_cost = 0 OR b.stddev_cost IS NULL
        OR (y.actual_cost - b.mean_cost) / NULLIF(b.stddev_cost, 0)
            >= p_z_threshold_med)
    -- Ignore tiny absolute amounts (less than $1/day) to reduce noise
    AND y.actual_cost > 1.0;

GET DIAGNOSTICS v_inserted = ROW_COUNT;

-- Refresh the MTD summary materialized view while we're here
REFRESH MATERIALIZED VIEW CONCURRENTLY dbfinops.mtd_cost_summary;

RETURN v_inserted;
END;
$$;

-- Example: schedule with pg_cron (if available)
-- SELECT cron.schedule('detect-cost-anomalies', '15 1 * * *',
--     'SELECT dbfinops.detect_cost_anomalies()');
```

To pair this with SQL Server for teams running Azure SQL, the equivalent logic uses a temporary table and window functions rather than a CTE chain with RETURNING:

```

-- SQL Server / Azure SQL: Cost anomaly detection stored procedure
-- Equivalent logic to the PostgreSQL function above

CREATE OR ALTER PROCEDURE dbfinops.DetectCostAnomalies
    @LookbackDays INT = 14,
    @ZThresholdMed FLOAT = 2.0,
    @ZThresholdHigh FLOAT = 3.0,
    @ZThresholdCrit FLOAT = 4.0
```



```

AS
BEGIN
    SET NOCOUNT ON;

    DECLARE @TargetDate DATE = DATEADD(DAY, -1, CAST(GETUTCDATE() AS DATE));
    DECLARE @BaselineStart DATE = DATEADD(DAY, -@LookbackDays, @TargetDate);

    -- Remove unacknowledged anomalies for target date (idempotent re-runs)
    DELETE FROM dbfinops.CostAnomalies
    WHERE CostDate = @TargetDate
        AND Acknowledged = 0;

    -- Compute baseline statistics
    DROP TABLE IF EXISTS #Baseline;
    SELECT
        Provider,
        DatabaseIdentifier,
        CostDimension,
        AVG(UnblendedCost) AS MeanCost,
        STDEV(UnblendedCost) AS StddevCost,
        COUNT(*) AS DayCount
    INTO #Baseline
    FROM dbfinops.DailyCosts
    WHERE CostDate BETWEEN @BaselineStart AND DATEADD(DAY, -1, @TargetDate)
    GROUP BY Provider, DatabaseIdentifier, CostDimension
    HAVING COUNT(*) >= 7;

    -- Compute yesterday's actuals
    DROP TABLE IF EXISTS #Yesterday;
    SELECT
        Provider,
        DatabaseIdentifier,
        CostDimension,
        SUM(UnblendedCost) AS ActualCost
    INTO #Yesterday
    FROM dbfinops.DailyCosts
    WHERE CostDate = @TargetDate
    GROUP BY Provider, DatabaseIdentifier, CostDimension;

    -- Insert anomalies
    INSERT INTO dbfinops.CostAnomalies (
        CostDate, Provider, DatabaseIdentifier, CostDimension,
        ExpectedCost, ActualCost, DeviationPct, Severity
    )
    SELECT
        @TargetDate,
        y.Provider,
        y.DatabaseIdentifier,
        y.CostDimension,
        ROUND(b.MeanCost, 6) AS ExpectedCost,
        ROUND(y.ActualCost, 6) AS ActualCost,
        ROUND(
            CASE WHEN b.MeanCost > 0
                THEN (y.ActualCost - b.MeanCost) / b.MeanCost * 100.0
            ELSE NULL
        END,
        2
    ) AS DeviationPct,
        CASE
            WHEN b.StddevCost = 0 OR b.StddevCost IS NULL
                THEN 'low'
            WHEN (y.ActualCost - b.MeanCost) / NULLIF(b.StddevCost, 0) >= @ZThresholdCrit
                THEN 'critical'
            WHEN (y.ActualCost - b.MeanCost) / NULLIF(b.StddevCost, 0) >= @ZThresholdHigh
                THEN 'high'
            WHEN (y.ActualCost - b.MeanCost) / NULLIF(b.StddevCost, 0) >= @ZThresholdMed
                THEN 'medium'
            ELSE 'low'
        END AS Severity
    FROM #Yesterday y

```

```

JOIN #Baseline b
  ON y.Provider = b.Provider
  AND y.DatabaseIdentifier = b.DatabaseIdentifier
  AND y.CostDimension = b.CostDimension
WHERE
  y.ActualCost > b.MeanCost
  AND y.ActualCost > 1.0
  AND (
    b.StddevCost IS NULL
    OR b.StddevCost = 0
    OR (y.ActualCost - b.MeanCost) / NULLIF(b.StddevCost, 0) >= @ZThresholdMed
  );

SELECT @@ROWCOUNT AS AnomaliesInserted;
END;
GO

```

6.7 Wiring Alerts to Grafana and PagerDuty

Anomalies sitting in a database table help no one. The final integration step connects the `cost_anomalies` table to an alerting pipeline. The recommended pattern uses Grafana's built-in alerting engine, which can query PostgreSQL directly and route alerts to Slack, PagerDuty, or email based on severity.

The following Grafana alert rule definition (in Terraform format for reproducibility) creates a high-severity alert that fires when the `cost_anomalies` table contains any unacknowledged critical or high anomalies detected in the last two hours.

```

# grafana_cost_alert.tf
# Requires: Grafana Terraform provider, Grafana 9.x or later
# grafana datasource must point to the dbfinops PostgreSQL instance

resource "grafana_alert_rule" "db_cost_anomaly_critical" {
  name           = "Database Cost Anomaly - High/Critical"
  folder_uid     = grafana_folder.finops.uid
  group_name     = "DatabaseCostAlerts"
  for            = "5m" # must be firing for 5 minutes before alert triggers
  condition      = "threshold"
  no_data_state  = "OK"
  exec_err_state = "Alerting"

  data {
    ref_id = "A"
    query_type = ""
    relative_time_range {
      from = 7200 # 2 hours in seconds
      to   = 0
    }
  }

  datasource_uid = var.grafana_postgres_datasource_uid
  model = jsonencode({
    rawSql = <<-SQL
    SELECT
      COUNT(*)                AS anomaly_count,
      NOW()                   AS "time",
      MAX(deviation_pct)      AS max_deviation_pct,
      STRING_AGG(
        database_identifier || ' (' || cost_dimension || '): '
        || ROUND(deviation_pct::NUMERIC, 1) || '% above expected',
        E'\n' ORDER BY deviation_pct DESC
      )                        AS anomaly_summary
    FROM dbfinops.cost_anomalies
    WHERE
      severity IN ('critical', 'high')
  >>SQL
  })
}

```

```

        AND acknowledged = FALSE
        AND detected_at >= NOW() - INTERVAL '2 hours'
    SQL
    format = "table"
  })
}

data {
  ref_id      = "threshold"
  query_type = "expression"
  datasource_uid = "-100"
  model = jsonencode({
    type      = "threshold"
    refId      = "threshold"
    expression = "A"
    conditions = [{
      evaluator = {
        params = [0]
        type   = "gt"
      }
    }
    operator = { type = "and" }
    query    = { params = ["A"] }
    reducer  = {
      params = []
      type   = "last"
    }
    type = "query"
  })
}

annotations = {
  description = "One or more database cost anomalies at high or critical severity detected in the last
↪ 2 hours. Review the dbfinops.cost_anomalies table for details."
  runbook_url = "https://wiki.internal/runbooks/db-cost-anomaly"
  summary     = "Database cost anomaly detected - immediate review required"
}

labels = {
  severity = "high"
  team     = "platform"
  category = "finops"
}

notification_settings {
  contact_point = "pagerduty-platform-oncall"
  group_wait    = "30s"
  group_interval = "5m"
  repeat_interval = "4h"
  group_by      = ["database_identifier"]
}

resource "grafana_alert_rule" "db_cost_anomaly_medium" {
  name           = "Database Cost Anomaly - Medium"
  folder_uid     = grafana_folder.finops.uid
  group_name     = "DatabaseCostAlerts"
  for            = "15m"
  condition      = "threshold"
  no_data_state  = "OK"
  exec_err_state = "OK"

  data {
    ref_id = "A"
    relative_time_range {
      from = 7200
      to   = 0
    }
  }
  datasource_uid = var.grafana_postgres_datasource_uid
  model = jsonencode({

```

```

rawSql = <<-SQL
  SELECT COUNT(*) AS anomaly_count, NOW() AS "time"
  FROM dbfinops.cost_anomalies
  WHERE severity = 'medium'
  AND acknowledged = FALSE
  AND detected_at >= NOW() - INTERVAL '2 hours'
SQL
format = "table"
})
}

data {
  ref_id      = "threshold"
  query_type  = "expression"
  datasource_uid = "-100"
  model = jsonencode({
    type      = "threshold"
    refId     = "threshold"
    expression = "A"
    conditions = [{
      evaluator = { params = [0], type = "gt" }
      operator  = { type = "and" }
      query     = { params = ["A"] }
      reducer   = { params = [], type = "last" }
      type      = "query"
    }]
  })
}

annotations = {
  description = "Medium-severity database cost anomaly detected. Review dbfinops.cost_anomalies."
  summary     = "Database cost anomaly - medium severity"
}

labels = {
  severity = "warning"
  team     = "platform"
  category = "finops"
}

notification_settings {
  contact_point = "slack-platform-alerts"
  group_wait    = "2m"
  group_interval = "10m"
  repeat_interval = "24h"
  group_by      = ["database_identifier"]
}
}

```

With this Terraform applied, critical anomalies page the on-call engineer within five minutes of detection, while medium anomalies post a Slack notification that the team can triage the following morning. The 4-hour repeat interval on the PagerDuty rule prevents alert fatigue while ensuring a critical anomaly does not silently persist across shifts.

6.8 The Weekly Executive Report Query

The dashboard is a daily operational tool. Once a week, a different audience — engineering leadership, finance, or the CTO — needs a summary that does not require reading raw SQL output. The following query produces a weekly summary formatted for direct paste into a Markdown-friendly report or a simple HTML email template.

```
-- PostgreSQL: Weekly database cost executive summary
```

```

-- Run every Monday covering the prior calendar week
-- Produces one row per provider with key metrics

WITH week_bounds AS (
    SELECT
        date_trunc('week', CURRENT_DATE) - INTERVAL '7 days' AS week_start,
        date_trunc('week', CURRENT_DATE) - INTERVAL '1 day' AS week_end
),
this_week AS (
    SELECT
        provider,
        SUM(unblended_cost) AS week_cost,
        COUNT(DISTINCT database_identifier) AS db_count
    FROM dbfinops.daily_costs, week_bounds
    WHERE cost_date BETWEEN week_start AND week_end
    GROUP BY provider
),
prior_week AS (
    SELECT
        provider,
        SUM(unblended_cost) AS week_cost
    FROM dbfinops.daily_costs, week_bounds
    WHERE cost_date BETWEEN week_start - INTERVAL '7 days'
        AND week_end - INTERVAL '7 days'
    GROUP BY provider
),
anomaly_count AS (
    SELECT
        provider,
        COUNT(*) AS unacked_anomalies,
        COUNT(*) FILTER (WHERE severity = 'critical') AS critical_count
    FROM dbfinops.cost_anomalies, week_bounds
    WHERE detected_at::DATE BETWEEN week_start AND week_end
        AND acknowledged = FALSE
    GROUP BY provider
),
top_spender AS (
    SELECT DISTINCT ON (provider)
        provider,
        database_identifier AS top_db,
        SUM(unblended_cost) OVER (PARTITION BY provider, database_identifier) AS db_cost
    FROM dbfinops.daily_costs, week_bounds
    WHERE cost_date BETWEEN week_start AND week_end
    ORDER BY provider, db_cost DESC
)
SELECT
    tw.provider,
    tw.db_count AS databases_tracked,
    ROUND(tw.week_cost::NUMERIC, 2) AS total_week_cost,
    ROUND(pw.week_cost::NUMERIC, 2) AS prior_week_cost,
    ROUND((tw.week_cost - pw.week_cost)::NUMERIC, 2) AS week_delta,
    ROUND(
        CASE WHEN pw.week_cost > 0
            THEN (tw.week_cost - pw.week_cost) / pw.week_cost * 100
            ELSE NULL
        END::NUMERIC, 1
    ) AS pct_change,
    COALESCE(ac.unacked_anomalies, 0) AS open_anomalies,
    COALESCE(ac.critical_count, 0) AS critical_anomalies,
    ts.top_db AS highest_cost_database,
    ROUND(ts.db_cost::NUMERIC, 2) AS highest_cost_db_spend
FROM
    this_week tw
    LEFT JOIN prior_week pw USING (provider)
    LEFT JOIN anomaly_count ac USING (provider)
    LEFT JOIN top_spender ts USING (provider)
ORDER BY
    tw.week_cost DESC;

```

Pair this with a simple Python script that runs the query and sends the output via AWS SES, SendGrid, or an internal SMTP relay, and the weekly report becomes zero-effort for the people who need it most. The structure — total spend, week-over-week delta, open anomalies, top spender — answers the three questions leadership asks every week: “Are we spending more than last week?”, “Is anything on fire?”, and “What’s our biggest database cost right now?”

6.9 Putting It All Together: The Dashboard Layout

The individual queries from section 6.5 compose into a dashboard with a clear visual hierarchy. The recommended layout for Grafana follows a three-row structure:

Row 1 — Executive Summary (4 stat panels) - Total MTD spend across all providers (single number, colored red/amber/green against budget threshold) - Week-over-week percentage change (sparkline + delta badge) - Open unacknowledged anomalies (count, red if any critical) - Reserved Instance / Commitment utilization percentage

Row 2 — Provider and Database Breakdown (2 panels) - Month-over-month cost change table (Query 1 from section 6.5), filterable by provider and environment - Cost dimension breakdown time series (Query 2), drilldown activated by clicking a database row in the adjacent table

Row 3 — Engineering Detail (3 panels) - Storage growth trend with 30-day projection (Query 4), one line per database - Top cost-attributing queries table (Query 5), sortable by attributed cost or execution time - RI/commitment coverage gap table (Query 3), sorted by on-demand spend descending

Row 4 — Anomaly Log (1 full-width panel) - Raw table view of `cost_anomalies` with filters for severity, provider, and acknowledged status - Includes an “Acknowledge” button wired to a Grafana panel action that sets `acknowledged = TRUE` and records the acknowledging user

This layout works equally well as a static Metabase dashboard, a Tableau workbook, or a simple HTML page generated from cron-scheduled SQL queries — the choice of visualization tool matters less than the discipline of reviewing the dashboard consistently.

Key Takeaways

- **Visibility creates accountability:** a database cost dashboard does not save money directly, but it eliminates the weeks-long lag between a cost anomaly occurring and someone noticing it, which is where most avoidable cloud spend accumulates.
- **Normalize cost data across providers:** storing AWS, Azure, and GCP costs in a single `dbfinops.daily_costs` schema with consistent dimensions (provider, database_identifier, cost_dimension, environment, team) enables cross-cloud comparison and prevents costs from hiding in provider-specific abstractions.
- **Connect cost to workload:** populating `query_cost_attribution` with per-query cost estimates transforms a finance dashboard into an engineering tool — it lets a DBA point to a specific query and say “optimizing this saves us a measurable amount per month.”

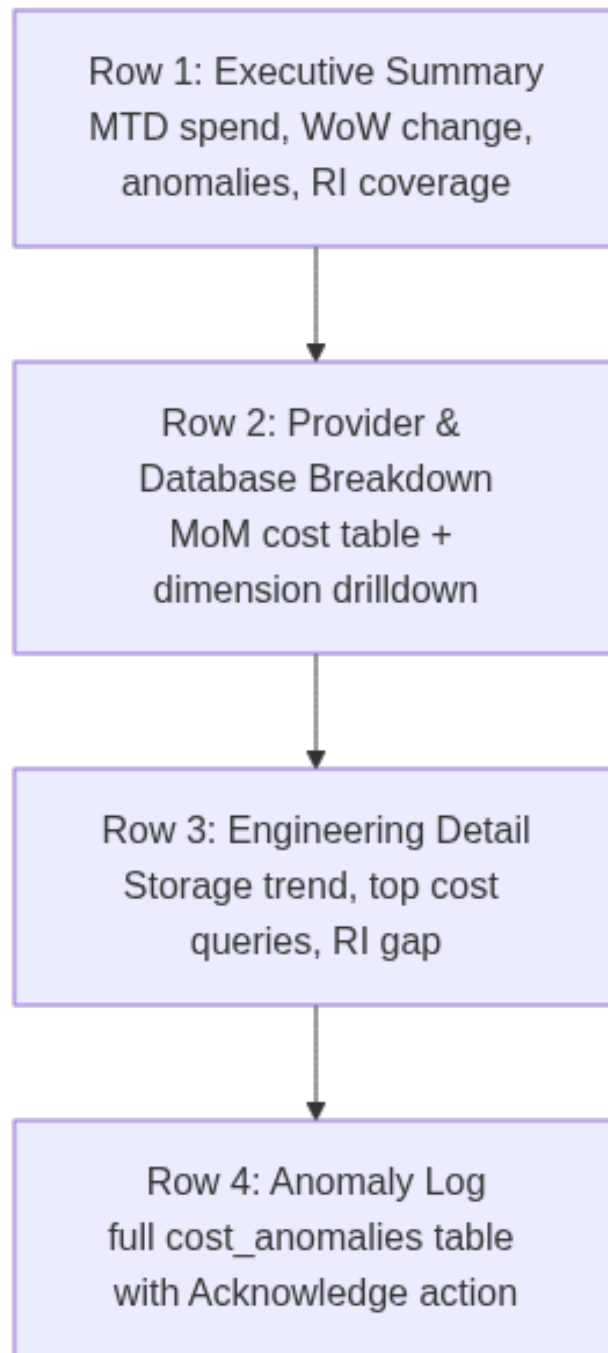


Figure 6: The dashboard’s visual hierarchy: each row narrows from a glanceable executive number down to the raw, actionable anomaly log.

- **Automate anomaly detection, not just reporting:** a rolling z-score function running nightly against the cost table catches unexpected spikes within 24 hours; without automation, the same spike might hide for a billing cycle.
 - **Tailor alert routing to severity:** critical anomalies warrant immediate on-call paging; medium anomalies warrant a next-morning Slack notification; routing both to PagerDuty guarantees alert fatigue and guarantees the alerts get ignored.
-

Chapter 7: Right-Sizing Instances: The Methodology

Organizations running cloud databases consistently overpay by 30–40% on compute alone. The root cause is almost always the same: instances were sized for peak load at launch, workloads evolved, and nobody went back to revisit the original choices. A single `db.r6g.4xlarge` on AWS RDS running at 12% average CPU costs roughly \$700.80/month on-demand. Downsizing to `db.r6g.xlarge` with a one-year Reserved Instance brings that to \$108.62/month — a saving of \$592.18 per month, per instance, without touching a line of application code. This chapter gives you the complete methodology: how to collect the right metrics, how to analyze them without being fooled by outliers, how to build the business case, and how to automate the entire process so right-sizing becomes a continuous operational discipline rather than a quarterly fire drill.

7.1 Why Instances Stay Oversized (And Why That’s Not Stupidity)

Before reaching for the resize button, it is worth understanding why oversizing is so persistent. The forces that created it are rational, even if the outcome is wasteful.

The provisioning bias. When a database is first deployed, the team responsible for it has no production workload data. They extrapolate from development benchmarks, add a safety margin, round up to the next instance tier, and ship. This is not negligence — it is risk management under uncertainty. The problem is that the uncertainty resolves after go-live, but the safety margin never gets reclaimed.

The fear of the resize operation. On AWS RDS, resizing a Multi-AZ instance causes a failover. On Azure SQL, changing a service tier can trigger minutes of reduced connectivity. These are real concerns, especially for teams that have been burned by a botched maintenance window. The organizational response is to defer the operation indefinitely.

Ownership fragmentation. The DBA knows the CPU numbers. The finance team sees the invoice. The application team owns the deployment pipeline. Nobody owns the intersection of all three. FinOps as a discipline exists precisely to close this gap.

The right-sizing window. Practical experience shows that the best time to right-size is 8–12 weeks after a workload goes live. By then, you have enough data to identify true peak patterns but the team is still close enough to the original architecture to make confident changes. If that window passes, the next best opportunity is a quarterly review cycle anchored to billing data.

7.2 Collecting the Right Metrics

Right-sizing decisions made on CPU alone are dangerous. A database can run at 8% CPU while holding 95% of its RAM in buffer pool — shrink the instance and you blow the cache, triggering a cascade of disk I/O that destroys query latency. Collect all four dimensions before drawing any conclusions.

The four dimensions:

Dimension	What to Measure	Safe Threshold for Downsizing
CPU	Average and P95 utilization	P95 < 40%, average < 20%
Memory	Buffer pool hit ratio, freeable memory	Freeable memory consistently > 20% of total
I/O	Read/write IOPS, throughput MB/s	Sustained utilization < 50% of provisioned
Connections	Active vs. max connections	Peak active < 40% of max_connections

Collecting from AWS CloudWatch

The following script pulls 30 days of RDS metrics for a given instance and outputs a summary. Schedule this with EventBridge to run on the first of every month.

```
#!/bin/bash
# collect_rds_metrics.sh
# Usage: ./collect_rds_metrics.sh <db-instance-identifier> <region>
# Schedule: AWS EventBridge, 1st of each month, 06:00 UTC

DB_INSTANCE="${1:-mydb-prod}"
REGION="${2:-us-east-1}"
END_DATE=$(date -u +"%Y-%m-%dT%H:%M:%SZ")
START_DATE=$(date -u -d "30 days ago" +"%Y-%m-%dT%H:%M:%SZ")

echo "=== RDS Right-Sizing Report: $DB_INSTANCE ==="
echo "Period: $START_DATE to $END_DATE"
echo ""

# CPU Utilization - Average and P95
echo "--- CPU Utilization ---"
aws cloudwatch get-metric-statistics \
  --region "$REGION" \
  --namespace AWS/RDS \
  --metric-name CPUUtilization \
  --dimensions Name=DBInstanceIdentifier,Value="$DB_INSTANCE" \
  --start-time "$START_DATE" \
  --end-time "$END_DATE" \
  --period 86400 \
  --statistics Average Maximum \
  --output table

# Freeable Memory (bytes)
echo ""
echo "--- Freeable Memory (GB) ---"
aws cloudwatch get-metric-statistics \
  --region "$REGION" \
  --namespace AWS/RDS \
  --metric-name FreeableMemory \
```

```

--dimensions Name=DBInstanceIdentifier,Value="$DB_INSTANCE" \
--start-time "$START_DATE" \
--end-time "$END_DATE" \
--period 86400 \
--statistics Average Minimum \
--query
↪ 'Datapoints[*].{Date:Timestamp,AvgGB:to_number(Average)/1073741824,MinGB:to_number(Minimum)/1073741824}'
↪ \
--output table

# Database Connections
echo ""
echo "---- Database Connections ----"
aws cloudwatch get-metric-statistics \
  --region "$REGION" \
  --namespace AWS/RDS \
  --metric-name DatabaseConnections \
  --dimensions Name=DBInstanceIdentifier,Value="$DB_INSTANCE" \
  --start-time "$START_DATE" \
  --end-time "$END_DATE" \
  --period 86400 \
  --statistics Average Maximum \
  --output table

# Read/Write IOPS
echo ""
echo "---- Read IOPS (avg/day) ----"
aws cloudwatch get-metric-statistics \
  --region "$REGION" \
  --namespace AWS/RDS \
  --metric-name ReadIOPS \
  --dimensions Name=DBInstanceIdentifier,Value="$DB_INSTANCE" \
  --start-time "$START_DATE" \
  --end-time "$END_DATE" \
  --period 86400 \
  --statistics Average Maximum \
  --output table

echo ""
echo "---- Write IOPS (avg/day) ----"
aws cloudwatch get-metric-statistics \
  --region "$REGION" \
  --namespace AWS/RDS \
  --metric-name WriteIOPS \
  --dimensions Name=DBInstanceIdentifier,Value="$DB_INSTANCE" \
  --start-time "$START_DATE" \
  --end-time "$END_DATE" \
  --period 86400 \
  --statistics Average Maximum \
  --output table

```

Querying workload behavior from inside PostgreSQL

CloudWatch tells you about resource consumption. `pg_stat_statements` tells you *why*. These two views must be read together. A database showing 15% CPU might have one query consuming 14% of it — downsize the instance and that query becomes a latency problem.

```

-- PostgreSQL: identify top resource consumers over the past 24 hours
-- Run this on the replica to avoid overhead on the primary
-- Requires pg_stat_statements extension (enabled by default on RDS, Aurora, Cloud SQL)

SELECT
  round(total_exec_time::numeric / 1000, 2)      AS total_exec_sec,
  calls,
  round(mean_exec_time::numeric, 2)              AS mean_ms,
  round(stddev_exec_time::numeric, 2)            AS stddev_ms,
  round((100 * total_exec_time /
    sum(total_exec_time) OVER ()):numeric, 2)    AS pct_of_total,

```

```

    rows,
    round(rows::numeric / nullif(calls, 0), 1)          AS rows_per_call,
    left(query, 120)                                   AS query_snippet
FROM
    pg_stat_statements
WHERE
    total_exec_time > 1000          -- at least 1 second cumulative
    AND calls > 10                 -- not a one-off
ORDER BY
    total_exec_time DESC
LIMIT 20;

-- Buffer cache hit ratio - if this drops below 95%, do NOT downsize memory
SELECT
    sum(heap_blks_hit) AS heap_hits,
    sum(heap_blks_read) AS heap_reads,
    round(
        100.0 * sum(heap_blks_hit)
        / nullif(sum(heap_blks_hit) + sum(heap_blks_read), 0),
        2
    ) AS buffer_hit_ratio_pct
FROM
    pg_statio_user_tables;

-- Active connection distribution by state and wait event
-- A high count of "ClientRead" is a connection pooling problem, not a compute problem
SELECT
    state,
    wait_event_type,
    wait_event,
    count(*) AS connection_count
FROM
    pg_stat_activity
WHERE
    backend_type = 'client backend'
GROUP BY
    state, wait_event_type, wait_event
ORDER BY
    connection_count DESC;

```

Equivalent query for SQL Server on RDS or Azure SQL

```

-- SQL Server: top queries by CPU consumption (last execution plan cache)
-- Run on RDS SQL Server or Azure SQL Database
SELECT TOP 20
    qs.total_worker_time / 1000000.0          AS total_cpu_sec,
    qs.execution_count,
    qs.total_worker_time / qs.execution_count / 1000.0
                                                AS avg_cpu_ms,
    qs.total_elapsed_time / qs.execution_count / 1000.0
                                                AS avg_elapsed_ms,
    qs.total_logical_reads / qs.execution_count AS avg_logical_reads,
    SUBSTRING(
        qt.text,
        (qs.statement_start_offset / 2) + 1,
        ((CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)
            ELSE qs.statement_end_offset
            END - qs.statement_start_offset) / 2) + 1
    ) AS query_text
FROM
    sys.dm_exec_query_stats AS qs
    CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS qt
ORDER BY
    qs.total_worker_time DESC;

-- Memory pressure: page life expectancy
-- PLE < 300 seconds is a warning; < 100 seconds means the buffer pool is too small
SELECT
    object_name,

```

```

counter_name,
cntr_value AS page_life_expectancy_seconds
FROM
sys.dm_os_performance_counters
WHERE
counter_name = 'Page life expectancy'
AND object_name LIKE '%Buffer Manager%';

```

7.3 Reading the Data Without Being Fooled

Raw metric averages mislead. A database that runs at 5% CPU for 23 hours and 95% CPU for one hour shows an average of 8.75% — and that one-hour burst is your month-end payroll batch or your nightly ETL. Right-size to remove that headroom and you break payroll.

The three-tier analysis framework

Tier 1 — **Baseline:** What does the instance look like during normal business hours on a typical weekday? Use the 50th percentile (median) across a 30-day window. This is the floor for sizing.

Tier 2 — **Sustained peak:** What is the P95 over the same window? This is the target utilization ceiling. If P95 CPU is 35%, the instance should be sized so that 35% CPU represents roughly 70% of its actual capacity, leaving a 2× headroom buffer.

Tier 3 — **Burst events:** What is the P99 or absolute maximum? These are the events that would cause customer-facing impact if the instance were undersized. Identify whether they are periodic and schedulable (ETL jobs, month-end reports) or unpredictable (traffic spikes from marketing campaigns). Periodic bursts can be handled with scheduled scaling or by moving those jobs to a read replica. Unpredictable bursts require true headroom in the primary.

The headroom formula

Required vCPUs = $\text{ceil}(\text{P95_CPU_utilization_pct} / 0.70 \times \text{current_vCPUs})$

Required RAM = $\text{ceil}(\text{working_set_GB} / 0.75)$

Where `working_set_GB` is estimated as: `total_database_size_GB × active_data_ratio`. The active data ratio is typically 0.15–0.30 for OLTP workloads (only a fraction of tables are hot at any given time). If your buffer cache hit ratio is above 99%, your working set already fits in RAM. If it is between 95% and 99%, you are cache-constrained and should not reduce memory even if the instance appears to have freeable memory — freeable memory in RDS often reflects OS-level cache that PostgreSQL or MySQL will use immediately under load.

Instance family selection

AWS RDS offers multiple instance families, and the price difference between them is substantial:

Family	vCPU:RAM Ratio	Best For	Example (4 vCPU)	On-Demand/mo
db.t4g	1:2 (burstable)	Dev/test, low-traffic	db.t4g.xlarge	\$119
db.m7g	1:4 (general)	Balanced OLTP	db.m7g.xlarge	\$291
db.r7g	1:8 (memory)	Large buffer pools	db.r7g.xlarge	\$450

Family	vCPU:RAM Ratio	Best For	Example (4 vCPU)	On-Demand/mo
db.x2g	1:16 (memory)	In-memory analytics	db.x2g.xlarge	\$1,125

Moving a workload from `db.r7g.2xlarge` (\$900/mo) to `db.m7g.2xlarge` (\$582/mo) saves \$318/month if the workload is CPU-bound rather than memory-bound. Over a three-year reserved term, that is \$11,448 saved on a single instance.

7.4 Building the Right-Sizing Recommendation Engine

Manual monthly reviews do not scale across a portfolio of dozens of databases. The following Python script connects to AWS Cost Explorer and CloudWatch, pulls 30 days of data for all RDS instances in an account, applies the headroom formula, and outputs a prioritized recommendation report with estimated monthly savings. Run it on a schedule using AWS Lambda or a cron job.

```
#!/usr/bin/env python3
"""
rds_rightsizing_advisor.py

Automated right-sizing recommendations for AWS RDS instances.
Pulls 30 days of CloudWatch metrics, applies headroom formula,
and cross-references current instance pricing.

Requirements: boto3, pandas
Schedule: AWS Lambda (EventBridge trigger, 1st of month) or cron

Usage:
    python rds_rightsizing_advisor.py --region us-east-1 --output recommendations.csv
"""

import argparse
import json
import math
from datetime import datetime, timedelta, timezone
from typing import Optional

import boto3
import pandas as pd

# -----
# Instance pricing map (on-demand, us-east-1, Linux, single-AZ, USD/hour)
# Update quarterly from https://aws.amazon.com/rds/mysql/pricing/
# -----
RDS_PRICING = {
    "db.t4g.micro": 0.016, "db.t4g.small": 0.034,
    "db.t4g.medium": 0.068, "db.t4g.large": 0.136,
    "db.t4g.xlarge": 0.272, "db.t4g.2xlarge": 0.544,
    "db.m7g.large": 0.192, "db.m7g.xlarge": 0.384,
    "db.m7g.2xlarge": 0.768, "db.m7g.4xlarge": 1.536,
    "db.m7g.8xlarge": 3.072, "db.m7g.12xlarge": 4.608,
    "db.r7g.large": 0.300, "db.r7g.xlarge": 0.600,
    "db.r7g.2xlarge": 1.200, "db.r7g.4xlarge": 2.400,
    "db.r7g.8xlarge": 4.800, "db.r7g.12xlarge": 7.200,
}

# Instance specs: vCPUs and RAM (GB)
INSTANCE_SPECS = {
    "db.t4g.micro": (2, 1), "db.t4g.small": (2, 2),
```

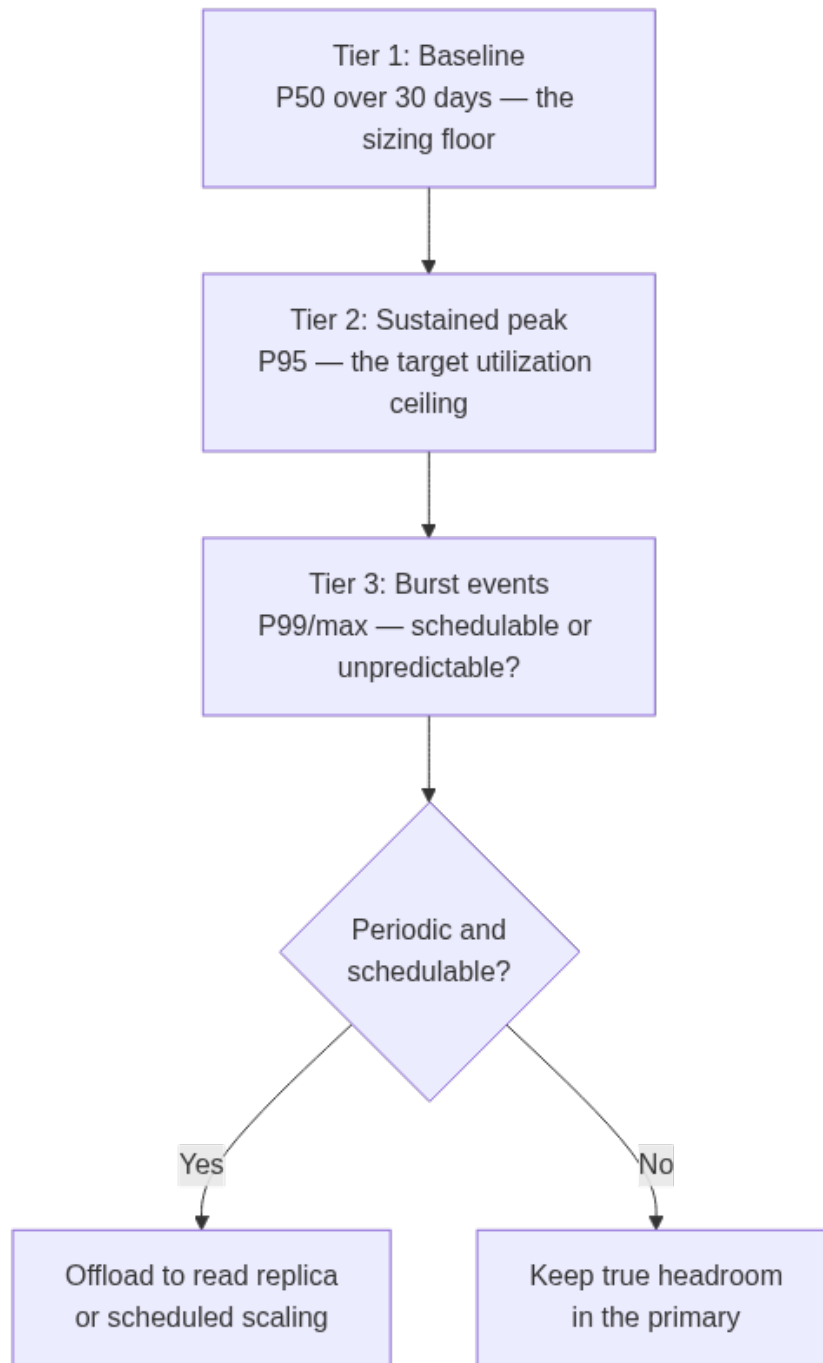


Figure 7: The three-tier analysis framework: each tier answers a different sizing question, and conflating them is how right-sizing breaks payroll.

```

    "db.t4g.medium": (2, 4), "db.t4g.large": (2, 8),
    "db.t4g.xlarge": (4, 16), "db.t4g.2xlarge": (8, 32),
    "db.m7g.large": (2, 8), "db.m7g.xlarge": (4, 16),
    "db.m7g.2xlarge": (8, 32), "db.m7g.4xlarge": (16, 64),
    "db.m7g.8xlarge": (32, 128), "db.m7g.12xlarge": (48, 192),
    "db.r7g.large": (2, 16), "db.r7g.xlarge": (4, 32),
    "db.r7g.2xlarge": (8, 64), "db.r7g.4xlarge": (16, 128),
    "db.r7g.8xlarge": (32, 256), "db.r7g.12xlarge": (48, 384),
}

HOURS_PER_MONTH = 730
LOOKBACK_DAYS = 30
CPU_HEADROOM = 0.70 # target: P95 CPU uses no more than 70% of capacity
MEM_HEADROOM = 0.75 # target: working set uses no more than 75% of RAM

def get_metric_statistics(
    cw_client,
    instance_id: str,
    metric_name: str,
    stat: str,
    period: int = 3600,
) -> list[float]:
    """Fetch hourly metric values for a given RDS instance over LOOKBACK_DAYS."""
    end = datetime.now(timezone.utc)
    start = end - timedelta(days=LOOKBACK_DAYS)

    response = cw_client.get_metric_statistics(
        Namespace="AWS/RDS",
        MetricName=metric_name,
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": instance_id}],
        StartTime=start,
        EndTime=end,
        Period=period,
        Statistics=[stat],
    )
    return [dp[stat] for dp in response.get("Datapoints", [])]

def percentile(values: list[float], pct: float) -> float:
    if not values:
        return 0.0
    sorted_vals = sorted(values)
    idx = int(math.ceil(pct / 100.0 * len(sorted_vals))) - 1
    return sorted_vals[max(idx, 0)]

def recommend_instance(
    current_class: str,
    p95_cpu: float,
    min_freeable_gb: float,
    family_preference: str = "same",
) -> Optional[str]:
    """
    Return the smallest instance that satisfies the headroom constraints,
    or None if the current instance is already right-sized.
    """
    if current_class not in INSTANCE_SPECS:
        return None

    current_vcpu, current_ram = INSTANCE_SPECS[current_class]

    # Determine required capacity
    required_vcpu = math.ceil(p95_cpu / 100.0 / CPU_HEADROOM * current_vcpu)
    required_vcpu = max(required_vcpu, 2) # floor at 2 vCPU

    # Estimate working set from freeable memory
    # freeable = RAM not used by buffer pool; working set RAM - freeable
    working_set_gb = max(current_ram - min_freeable_gb, 4)
    required_ram = math.ceil(working_set_gb / MEM_HEADROOM)

```



```

# Find candidate instances smaller than current that meet requirements
current_price = RDS_PRICING.get(current_class, 0)
best_candidate = None
best_price = current_price

for inst, (vcpu, ram) in INSTANCE_SPECS.items():
    price = RDS_PRICING.get(inst, 0)
    if price >= current_price:
        continue # not cheaper, skip
    if vcpu < required_vcpu:
        continue
    if ram < required_ram:
        continue
    if price < best_price:
        best_price = price
        best_candidate = inst

return best_candidate

def analyze_account(region: str) -> pd.DataFrame:
    rds = boto3.client("rds", region_name=region)
    cw = boto3.client("cloudwatch", region_name=region)

    paginator = rds.get_paginator("describe_db_instances")
    instances = []
    for page in paginator.paginate():
        instances.extend(page["DBInstances"])

    rows = []
    for db in instances:
        iid = db["DBInstanceIdentifier"]
        itype = db["DBInstanceClass"]
        print(f" Analyzing {iid} ({itype}) ...")

        cpu_values = get_metric_statistics(cw, iid, "CPUUtilization", "Average")
        mem_values = get_metric_statistics(cw, iid, "FreeableMemory", "Minimum")

        if not cpu_values:
            print(f" No metrics found for {iid}, skipping.")
            continue

        avg_cpu = sum(cpu_values) / len(cpu_values)
        p95_cpu = percentile(cpu_values, 95)
        min_mem_gb = (min(mem_values) / 1_073_741_824) if mem_values else 0.0

        current_price = RDS_PRICING.get(itype, 0)
        current_monthly = current_price * HOURS_PER_MONTH

        recommendation = recommend_instance(itype, p95_cpu, min_mem_gb)
        rec_price = RDS_PRICING.get(recommendation, current_price) if recommendation else
↪ current_price
        rec_monthly = rec_price * HOURS_PER_MONTH
        monthly_saving = current_monthly - rec_monthly

        rows.append({
            "instance_id": iid,
            "current_class": itype,
            "avg_cpu_pct": round(avg_cpu, 1),
            "p95_cpu_pct": round(p95_cpu, 1),
            "min_freeable_gb": round(min_mem_gb, 1),
            "current_monthly": round(current_monthly, 2),
            "recommended_class": recommendation or "no change",
            "rec_monthly": round(rec_monthly, 2),
            "monthly_saving": round(monthly_saving, 2),
            "annual_saving": round(monthly_saving * 12, 2),
        })

df = pd.DataFrame(rows).sort_values("monthly_saving", ascending=False)

```

```

return df

def main():
    parser = argparse.ArgumentParser(description="RDS Right-Sizing Advisor")
    parser.add_argument("--region", default="us-east-1", help="AWS region")
    parser.add_argument("--output", default="recommendations.csv", help="Output CSV path")
    args = parser.parse_args()

    print(f"RDS Right-Sizing Advisor")
    print(f"Region: {args.region}")
    print(f"Lookback: {LOOKBACK_DAYS} days")
    print(f"{'='*60}")

    df = analyze_account(args.region)

    if df.empty:
        print("No instances with sufficient metric data found.")
        return

    # Summary
    total_current = df["current_monthly"].sum()
    total_rec = df["rec_monthly"].sum()
    total_saving = df["monthly_saving"].sum()
    candidates = df[df["monthly_saving"] > 0]

    print(f"\n{'='*60}")
    print(f"SUMMARY")
    print(f"{'='*60}")
    print(f"Total instances analyzed : {len(df)}")
    print(f"Instances with savings : {len(candidates)}")
    print(f"Current monthly spend : ${total_current:,.2f}")
    print(f"Optimized monthly spend : ${total_rec:,.2f}")
    print(f"Monthly saving : ${total_saving:,.2f}")
    print(f"Annual saving : ${total_saving * 12:,.2f}")
    print(f"\nTop 10 opportunities:")
    print(
        candidates.head(10)[
            ["instance_id", "current_class", "p95_cpu_pct",
             "min_freeable_gb", "recommended_class", "monthly_saving"]
        ].to_string(index=False)
    )

    df.to_csv(args.output, index=False)
    print(f"\nFull report written to {args.output}")

if __name__ == "__main__":
    main()

```

Adapting the engine for Azure SQL

Azure SQL exposes equivalent metrics through Azure Monitor. The pattern is the same — pull DTU or vCore utilization, memory pressure, and connection counts — but the SDK and metric names differ.

```

# azure_sql_rightsizing.py (excerpt)
# Full script follows the same structure as the AWS version above.
# Requires: azure-mgmt-monitor, azure-mgmt-sql, azure-identity

from azure.identity import DefaultAzureCredential
from azure.mgmt.monitor import MonitorManagementClient
from azure.mgmt.sql import SqlManagementClient
from datetime import datetime, timedelta, timezone

LOOKBACK_DAYS = 30

def get_azure_sql_metrics(
    subscription_id: str,

```

```
resource_group: str,
server_name: str,
database_name: str,
) -> dict:
    credential = DefaultAzureCredential()
    monitor_client = MonitorManagementClient(credential, subscription_id)

    resource_uri = (
        f"/subscriptions/{subscription_id}/resourceGroups/{resource_group}"
        f"/providers/Microsoft.Sql/servers/{server_name}"
        f"/databases/{database_name}"
    )

    end = datetime.now(timezone.utc)
    start = end - timedelta(days=LOOKBACK_DAYS)

    # Metrics available: cpu_percent, dtu_consumption_percent,
    # memory_usage_percent, connection_successful, storage_percent
    metrics_response = monitor_client.metrics.list(
        resource_uri=resource_uri,
        timespan=f"{start.isoformat()}/{end.isoformat()}",
        interval="PT1H",
        metricnames="cpu_percent,memory_usage_percent,connection_successful",
        aggregation="Average,Maximum",
    )

    result = {}
    for metric in metrics_response.value:
        values = []
        for ts in metric.timeseries:
            for dp in ts.data:
                if dp.average is not None:
                    values.append(dp.average)
            result[metric.name.value] = values

    return result
```

For Azure SQL, the right-sizing logic maps to service tier selection rather than instance class. The General Purpose tier offers 1:5.1 vCore-to-RAM ratio; Business Critical offers 1:5.1 with local SSD and higher IOPS. A workload running at low DTU consumption with modest storage throughput is a candidate for dropping from Business Critical to General Purpose, which typically reduces the compute cost by 40–50% for the same vCore count.

7.5 The Resize Operation: Executing Safely

Collecting recommendations is the easy part. Executing the resize without causing an incident requires a structured runbook.

Pre-resize checklist

1. **Confirm the metric window covers at least one full business cycle.** For most workloads that is four weeks. For quarterly-reporting businesses (finance, healthcare) it may be 13 weeks.
2. **Check for scheduled jobs that run at non-business hours.** Nightly ETL, weekly statistics updates, and monthly purges can all create CPU or I/O spikes that appear in P99 but are invisible in daily averages. Query the job scheduler directly.

```
-- PostgreSQL: check pg_cron scheduled jobs (if pg_cron extension is installed)
SELECT
```

```

        jobid,
        schedule,
        command,
        nodename,
        active
FROM
    cron.job
ORDER BY
    jobid;

-- SQL Server: list SQL Agent jobs and their last run duration
SELECT
    j.name                                AS job_name,
    ja.run_requested_date,
    ja.stop_execution_date,
    DATEDIFF(SECOND, ja.run_requested_date,
             ja.stop_execution_date)      AS duration_sec,
    ja.last_executed_step_id
FROM
    msdb.dbo.sysjobs                     AS j
JOIN msdb.dbo.sysjobactivity            AS ja
    ON j.job_id = ja.job_id
WHERE
    ja.run_requested_date IS NOT NULL
ORDER BY
    ja.run_requested_date DESC;

```

3. **Take a manual snapshot before the resize.** Even though RDS takes an automated snapshot at resize time, a manual snapshot with a descriptive name (`pre-rightsize-2024-11-01`) is faster to locate under pressure.
4. **Notify stakeholders.** A right-sizing operation on a Multi-AZ RDS instance triggers a failover, which causes roughly 30–60 seconds of interrupted connections. Application teams need to verify their connection retry logic handles this gracefully. If they do not have retry logic, the resize window becomes a deployment event, not just a DBA task.
5. **Plan the rollback.** Resizing back up is always possible but takes the same time as the original resize. Set a rollback decision deadline — typically 24 hours post-resize. If any key performance metric degrades beyond an agreed threshold within that window, resize back without debate.

Executing the resize on AWS RDS

```

# Step 1: Modify the instance class (change takes effect at next maintenance window
# unless --apply-immediately is specified)
aws rds modify-db-instance \
    --db-instance-identifier mydb-prod \
    --db-instance-class db.r7g.xlarge \
    --apply-immediately \
    --region us-east-1

# Step 2: Poll until the instance is available again
echo "Waiting for instance to become available..."
aws rds wait db-instance-available \
    --db-instance-identifier mydb-prod \
    --region us-east-1

echo "Resize complete. Verifying instance class..."
aws rds describe-db-instances \
    --db-instance-identifier mydb-prod \
    --region us-east-1 \
    --query
    ↪ 'DBInstances[0].{Class:DBInstanceClass,Status:DBInstanceStatus,Engine:Engine,EngineVersion:EngineVersion}'
    ↪ \
    --output table

```

Post-resize validation queries

Run these in the first 15 minutes after the resize and again at the 24-hour mark.

```
-- PostgreSQL: confirm buffer cache is recovering after restart
-- Expected: hit ratio climbs back above 95% within 30 minutes as cache warms
SELECT
    now() AS check_time,
    round(
        100.0 * sum(heap_blks_hit)
        / nullif(sum(heap_blks_hit) + sum(heap_blks_read), 0),
        2
    ) AS buffer_hit_ratio_pct,
    sum(heap_blks_read) AS total_disk_reads
FROM
    pg_statio_user_tables;

-- Compare query latency percentiles to pre-resize baseline
-- Run this before the resize and save the output, then re-run after
SELECT
    left(query, 80) AS query_snippet,
    calls,
    round(mean_exec_time::numeric, 2) AS mean_ms,
    round(stddev_exec_time::numeric, 2) AS stddev_ms,
    round(
        percentile_cont(0.95) WITHIN GROUP
        (ORDER BY total_exec_time / calls),
        2
    ) AS p95_ms_estimate
FROM
    pg_stat_statements
WHERE
    calls > 50
ORDER BY
    mean_exec_time DESC
LIMIT 20;

-- SQL Server: post-resize health check
-- Confirm PLE has recovered and there is no memory pressure
SELECT
    'PLE' AS metric,
    cntr_value AS value,
    CASE
        WHEN cntr_value > 300 THEN 'OK'
        WHEN cntr_value > 100 THEN 'WARNING'
        ELSE 'CRITICAL'
    END AS status
FROM
    sys.dm_os_performance_counters
WHERE
    counter_name = 'Page life expectancy'
    AND object_name LIKE '%Buffer Manager%'

UNION ALL

-- Confirm no runaway waits have appeared
SELECT TOP 10
    wait_type AS metric,
    wait_time_ms AS value,
    'CHECK' AS status
FROM
    sys.dm_os_wait_stats
WHERE
    wait_type NOT IN (
        'SLEEP_TASK', 'BROKER_TO_FLUSH', 'BROKER_TASK_STOP',
        'CLR_AUTO_EVENT', 'DISPATCHER_QUEUE_SEMAPHORE',
        'FT_IFTS_SCHEDULER_IDLE_WAIT', 'HADR_WORK_QUEUE',
        'ONDEMAND_TASK_QUEUE', 'REQUEST_FOR_DEADLOCK_SEARCH',
        'RESOURCE_QUEUE', 'SERVER_IDLE_CHECK', 'SLEEP_DBSTARTUP',
        'SLEEP_DBRECOVER', 'SLEEP_MASTERDBREADY', 'SLEEP_MASTERMDREADY',
```

```
'SLEEP_MASTERUPGRADED', 'SLEEP_MSDBSTARTUP', 'SLEEP_SYSTEMTASK',  
'SLEEP_TEMPDBSTARTUP', 'SNI_HTTP_ACCEPT', 'SP_SERVER_DIAGNOSTICS_SLEEP',  
'SQLTRACE_BUFFER_FLUSH', 'WAITFOR', 'XE_DISPATCHER_WAIT',  
'XE_TIMER_EVENT'  
)  
ORDER BY  
wait_time_ms DESC;
```

7.6 Handling Edge Cases

No methodology survives contact with a real production estate without encountering anomalies. The following cases are the most common.

The bursty dev/test instance that looks like prod

Development and test databases often have low average utilization but periodic spikes caused by CI/CD pipelines running integration tests. These spikes are real but they are not customer-facing. The right response is to move the instance to a `db.t4g` burstable family and let it accumulate CPU credits during idle periods to spend during test runs. For a `db.m7g.xlarge` running CI workloads, switching to `db.t4g.xlarge` reduces cost by approximately 29% with no performance degradation in practice.

The connection-saturated instance with low CPU

An instance at 8% CPU and 95% connection utilization is not oversized — it is under-pooled. Before recommending a downsize, check whether the application is using a connection pooler. On PostgreSQL, the standard recommendation is PgBouncer in transaction-pooling mode. Deploying PgBouncer can reduce the effective connection count by 10× without changing the instance class, and it unblocks the downsize decision by removing the false constraint.

```
-- PostgreSQL: diagnose connection waste  
-- "idle" connections holding slots but doing no work  
SELECT  
    count(*)                               AS idle_connections,  
    min(now() - state_change)              AS shortest_idle,  
    max(now() - state_change)              AS longest_idle,  
    avg(now() - state_change)              AS avg_idle_time  
FROM  
    pg_stat_activity  
WHERE  
    state = 'idle'  
    AND backend_type = 'client backend';
```

The instance with high I/O and low CPU

This pattern indicates either a missing index causing full table scans, or a workload that is genuinely I/O-bound (large analytical queries, bulk exports). Before concluding the instance needs more I/O-optimized storage, check whether adding an index eliminates the scans.

```
-- PostgreSQL: tables with high sequential scans relative to index scans  
-- A high seq_scan/idx_scan ratio on large tables is the fingerprint  
-- of a missing index opportunity  
SELECT  
    schemaname,  
    tablename,  
    seq_scan,  
    idx_scan,  
    n_live_tup                               AS live_rows,  
    round(  

```

```
100.0 * seq_scan / nullif(seq_scan + idx_scan, 0),
1
)
pg_size_pretty(pg_total_relation_size(
quote_ident(schemaname) || '.' || quote_ident(tablename)
))
AS total_size
FROM
pg_stat_user_tables
WHERE
seq_scan > 100
AND n_live_tup > 10000
ORDER BY
seq_scan DESC
LIMIT 20;

-- SQL Server: missing index recommendations from the DMV
SELECT TOP 20
migs.avg_total_user_cost *
migs.avg_user_impact *
(migs.user_seeks + migs.user_scans) AS improvement_measure,
mid.statement AS table_name,
mid.equality_columns,
mid.inequality_columns,
mid.included_columns,
migs.user_seeks,
migs.user_scans,
round(migs.avg_total_user_cost, 2) AS avg_query_cost,
round(migs.avg_user_impact, 2) AS estimated_improvement_pct
FROM
sys.dm_db_missing_index_group_stats AS migs
JOIN sys.dm_db_missing_index_groups AS mig
ON migs.group_handle = mig.index_group_handle
JOIN sys.dm_db_missing_index_details AS mid
ON mig.index_handle = mid.index_handle
ORDER BY
improvement_measure DESC;
```

The read replica that mirrors primary sizing

Read replicas are frequently cloned from the primary at the same instance class and then left unchanged. Many analytical queries that drive a replica’s load are selective and cache-friendly once the working set is established. Replicas are often candidates for smaller instance classes than the primary, or for a different family (memory-optimized for analytics vs. balanced for OLTP). Profile the replica independently — do not assume it needs the same resources as the primary.

7.7 Embedding Right-Sizing into the Operating Model

One-time right-sizing captures immediate savings but does not prevent the problem from recurring. The goal is a continuous process.

Monthly automated scan. The Python script from Section 7.4 should run on the first of every month and post results to a Slack channel or Teams webhook, tagged to the owning team. Visibility changes behavior — when teams see their instance in a “no change needed” status, they understand why. When they see it flagged, they have the data to act.

Tagging discipline. Every RDS instance should carry tags for `environment` (prod/staging/dev/test), `owner` (team or squad name), `workload-type` (oltp/analytics/reporting/batch), and `rightsizing-reviewed` (date of last review). The monthly scan should filter by `environment=prod` for immediate savings focus and log all others for weekly review.

Change freeze windows. Define a calendar of blackout periods when right-sizing operations cannot be executed — typically the last two weeks of a fiscal quarter, major product release dates, and the period around public holidays. The recommendation engine should flag instances as “eligible” or “frozen” based on this calendar.

Linking to Reserved Instance purchases. Right-sizing must precede RI commitments. An organization that buys a three-year RI for `db.r7g.4xlarge` and then discovers six months later that the workload fits on `db.r7g.xlarge` has locked in the savings of neither. The workflow should be:

1. Right-size the instance on-demand.
2. Observe for 30 days.
3. Purchase the RI at the right-sized tier.

This sequencing is easy to describe and often ignored under budget deadline pressure. Building it into the process as a gate — the RI purchase form requires a field for “last right-sizing review date” — enforces the discipline without relying on individual vigilance.

Feedback loop to architecture. Right-sizing data is valuable beyond cost optimization. An instance that cannot be downsized despite low utilization often reveals an architectural constraint: a schema with no partitioning that requires a full-table scan during purges, a connection count problem that points to missing pooling infrastructure, or a buffer cache sensitivity that signals data model inefficiency. Feed these findings back to the architecture review process. The right-sizing program pays for itself twice: once in direct savings, and once in the architectural improvements it surfaces.

Key Takeaways

- **Right-sizing requires four dimensions, not one.** CPU alone misleads. Collect P95 CPU, buffer cache hit ratio, minimum freeable memory, and peak connection count before making any resize decision. An instance with low CPU but a 99% buffer hit ratio will suffer a serious performance regression if its RAM is reduced.
- **The headroom formula prevents under-sizing.** Size the instance so that P95 CPU represents no more than 70% of its capacity, and so that the estimated working set occupies no more than 75% of available RAM. This 30% and 25% headroom absorbs legitimate burst traffic without requiring the instance to be oversized for average load.
- **Burst classification determines the right response.** Periodic, schedulable bursts (ETL, month-end reports) do not require headroom in the primary instance — they can be offloaded to a read replica or accommodated by temporary scaling. Unpredictable bursts require true headroom. Conflating the two leads to either chronic oversizing or preventable incidents.
- **Automate the scan, humanize the decision.** The recommendation engine in Section 7.4 can identify and quantify every right-sizing opportunity in an account in minutes. The actual resize decision still requires human review — checking the business calendar, validating the metric window, and confirming rollback readiness. Automation handles the tedious data collection; judgment handles the risk assessment.

- **Right-sizing must precede Reserved Instance purchases.** Committing to a three-year RI at the wrong instance class eliminates the flexibility to capture compute savings later. The correct sequence is: resize on-demand, observe for 30 days, then purchase the commitment. Building this sequence into the procurement workflow as a required step prevents the most common FinOps mistake in the database cost domain.
-

Chapter 8: Reserved Instances vs Savings Plans vs On-Demand: The Math

Organizations that run AWS RDS, Azure SQL, or Google Cloud SQL on pure On-Demand pricing are typically overpaying by 30–72% compared to what they could be spending with the right commitment model. That is not a rounding error — on a \$40,000/month RDS bill, a well-executed Reserved Instance strategy returns between \$12,000 and \$28,800 back to the business every single month. This chapter works through the exact math behind Reserved Instances (RIs), Savings Plans, and On-Demand pricing across all three major clouds, shows you how to identify which workloads qualify for each model, and provides automation to keep your commitment coverage optimized as your fleet evolves.

8.1 Understanding the Pricing Tiers: What You Are Actually Buying

Before running any numbers, it helps to understand what distinguishes each pricing model at a mechanical level, because the differences matter enormously when you map them to database workloads.

On-Demand is the default. You pay a fixed hourly rate with no commitment. For an `db.r6g.2xlarge` running PostgreSQL on RDS in `us-east-1`, that rate is \$0.48/hour as of mid-2024, which works out to \$345.60/month or \$4,147.20/year. On-Demand is appropriate for workloads that run fewer than roughly 40% of the hours in a month, or for instances you cannot predict will exist in 12 months.

Reserved Instances (RIs) on AWS require a 1-year or 3-year commitment to a specific instance family, region, database engine, and license type. In exchange for that commitment, you receive a discount applied hourly whether the instance runs or not. The same `db.r6g.2xlarge` under a 1-year No Upfront RI costs \$0.302/hour — a 37% discount. A 3-year All Upfront RI drops the effective hourly rate to \$0.192, a 60% discount. The tradeoff is inflexibility: an RI for MySQL cannot be applied to PostgreSQL, and standard RIs cannot be moved across regions.

Convertible RIs on AWS partially address that inflexibility. You can exchange a Convertible RI for a different instance type, engine, or configuration, but the discount is reduced — typically 31–54% rather than 37–60%.

Compute Savings Plans on AWS (relevant for RDS Serverless v2 and Aurora Serverless) apply to any compute usage across a family regardless of engine, region, or size, in exchange for a \$/hour commitment. They do not apply to standard RDS provisioned instances — a point that confuses many teams mixing RDS and Aurora.

Azure Reserved VM Instances and Azure SQL Reserved Capacity operate on similar logic: 1-year or 3-year terms with discounts of 33–65% compared to pay-as-you-go. Azure Reserved Capacity applies at the vCore level and can be shared across subscriptions within a billing account.

Google Cloud Committed Use Discounts (CUDs) for Cloud SQL operate in 1-year or 3-year terms against vCPU and memory resources rather than specific instance types, giving them more flexibility than AWS standard RIs.

The table below summarizes effective discounts for a representative 8-vCPU, 64GB memory database instance across clouds:

Cloud	On-Demand \$/mo	1yr RI/CUD \$/mo	3yr RI/CUD \$/mo	3yr Savings
AWS RDS r6g.2xlarge (PostgreSQL)	\$345.60	\$217.44	\$138.24	\$2,491/yr
Azure SQL GP Gen5 8vCore	\$368.40	\$246.82	\$171.72	\$2,360/yr
GCP Cloud SQL n2-highmem-8	\$331.20	\$231.84	\$165.60	\$1,990/yr

These figures exclude storage, I/O, and data transfer — all of which carry their own pricing structures covered in Chapter 9. The numbers here are compute-only, which is typically the largest single line item on a database bill.

8.2 Diagnosing Your Current Coverage Gap

You cannot optimize what you have not measured. Most organizations discover they have a patchwork of RIs purchased at different times, some expired, some mismatched to current instance types, with large portions of the fleet still running On-Demand. The following queries and CLI commands expose exactly where that gap is.

Step 1: Pull your RDS On-Demand spend from AWS Cost Explorer.

```
# Retrieve last 90 days of RDS On-Demand costs by instance type
# Requires Cost Explorer enabled and appropriate IAM permissions
aws ce get-cost-and-usage \
  --time-period Start=2024-04-01,End=2024-07-01 \
  --granularity MONTHLY \
  --filter '{
    "And": [
      {
        "Dimensions": {
          "Key": "SERVICE",
          "Values": ["Amazon Relational Database Service"]
        }
      }
    ]
  }
```

```

    },
    {
      "Dimensions": {
        "Key": "PURCHASE_TYPE",
        "Values": ["On Demand"]
      }
    }
  ]
} ' \
--group-by '[
  {"Type": "DIMENSION", "Key": "INSTANCE_TYPE"},
  {"Type": "DIMENSION", "Key": "USAGE_TYPE"}
]' \
--metrics "UnblendedCost" "UsageQuantity" \
--output json | \
jq -r '
  .ResultsByTime[] |
  .TimePeriod.Start as $period |
  .Groups[] |
  [$period,
  .Keys[0],
  .Keys[1],
  (.Metrics.UnblendedCost.Amount | tonumber | . * 100 | round / 100),
  (.Metrics.UsageQuantity.Amount | tonumber | round)
  ] | @csv
' | \
sort -t',' -k4 -nr | \
head -30

```

This produces a ranked list of your most expensive On-Demand RDS instance types. The instances appearing at the top of that list — running hundreds of hours per month — are your RI candidates.

Step 2: Check existing RI coverage and utilization.

```

# List all active RDS Reserved Instances and their utilization
aws rds describe-reserved-db-instances \
--query 'ReservedDBInstances[?State==`active`].[
  ReservedDBInstanceId,
  DBInstanceClass,
  MultiAZ,
  ProductDescription,
  State,
  StartTime,
  Duration,
  FixedPrice,
  UsagePrice,
  DBInstanceCount,
  RecurringCharges
]' \
--output table

# Check RI utilization (how much of what you paid for is actually being used)
aws ce get-reservation-utilization \
--time-period Start=2024-04-01,End=2024-07-01 \
--granularity MONTHLY \
--filter '{
  "Dimensions": {
    "Key": "SERVICE",
    "Values": ["Amazon Relational Database Service"]
  }
}' \
--output json | \
jq -r '
  .UtilizationsByTime[] |
  [.TimePeriod.Start,
  (.Total.UtilizationPercentage),
  (.Total.PurchasedHours),
  (.Total.UsedHours),
  (.Total.UnusedHours),

```

```

        (.Total.RealizedSavings.Amount),
        (.Total.UnrealizedSavings.Amount)
    ] | @csv

```

An RI utilization percentage below 80% means you purchased commitments that are not being consumed — you are paying for capacity that sits idle. Anything below 60% warrants immediate investigation; in many cases, the underlying instance was resized or terminated and nobody updated the RI portfolio.

Step 3: Run the same analysis for Azure SQL Reserved Capacity.

```

# Azure: List reserved capacity orders for SQL
az reservations reservation-order list \
  --output table \
  --query "[].{
    OrderId:name,
    DisplayName:properties.displayName,
    Term:properties.term,
    ExpiryDate:properties.expiryDate,
    ProvisioningState:properties.provisioningState
  }"

# Get utilization for a specific reservation order
az consumption reservations summaries list \
  --reservation-order-id <your-order-id> \
  --grain monthly \
  --output table \
  --query "[].{
    UsageDate:usageDate,
    AvgUtilizationPct:avgUtilizationPercentage,
    UsedHours:usedHours,
    ReservedHours:reservedHours,
    MinUtilPct:minUtilizationPercentage
  }"

```

Step 4: Query the GCP Billing export in BigQuery for CUD coverage.

Most GCP teams export billing data to BigQuery. If yours does, this query identifies Cloud SQL compute spend by purchase type:

```

-- GCP BigQuery: Cloud SQL spend by SKU type (On-Demand vs CUD)
-- Run this against your billing export dataset
SELECT
  DATE_TRUNC(usage_start_time, MONTH)                AS billing_month,
  service.description                                AS service,
  sku.description                                     AS sku_description,
  resource.name                                       AS resource_name,
  CASE
    WHEN LOWER(sku.description) LIKE '%commit%'      THEN 'Committed'
    WHEN LOWER(sku.description) LIKE '%preemptible%' THEN 'Preemptible'
    ELSE 'On-Demand'
  END                                                 AS purchase_type,
  project.id                                          AS project_id,
  ROUND(SUM(cost), 2)                                AS total_cost_usd,
  ROUND(SUM(usage.amount), 2)                        AS total_usage,
  usage.unit                                          AS usage_unit
FROM
  `your-project.your_dataset.gcp_billing_export_v1_XXXXXX`
WHERE
  service.description = 'Cloud SQL'
  AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
  AND cost > 0
GROUP BY
  1, 2, 3, 4, 5, 6, 9
ORDER BY
  billing_month DESC,

```

```
total_cost_usd DESC
LIMIT 100;
```

8.3 Running the Break-Even Math

The break-even analysis determines exactly how many hours per month an instance must run before an RI becomes cheaper than On-Demand. This is the single most important calculation in the chapter.

The formula is:

Break-even hours = RI_monthly_cost / (On-Demand_hourly - RI_hourly_effective)

For a `db.r6g.2xlarge` PostgreSQL in `us-east-1`: - On-Demand: \$0.48/hour - 1-year No Upfront RI: \$0.302/hour (\$217.44/month) - Break-even: $\$217.44 / (\$0.48 - \$0.302) = \$217.44 / \$0.178 = \mathbf{1,221 \text{ hours/month}}$

A month has 720–744 hours. The RI costs \$217.44 regardless of runtime. On-Demand at full utilization (744 hours) costs \$357.12. The RI wins the moment the instance exists at all during the month — you do not need to reach 1,221 hours because the RI commitment is monthly, not per-hour.

The real break-even question is: how many hours must the instance run in a month for On-Demand to cost less than the RI commitment? If the instance runs fewer hours than this threshold, On-Demand is cheaper:

Hours where On-Demand < RI_monthly_commitment:
 $\$0.48 * H < \217.44
 $H < 453 \text{ hours/month}$

Any instance running more than 453 hours per month (roughly 61% of the month) is cheaper under a 1-year RI. Most production databases run 720+ hours. The math is decisive.

The Python script below automates this calculation across your entire fleet using AWS pricing data:

```
#!/usr/bin/env python3
"""
ri_break_even_calculator.py

Calculates RI break-even points and projected annual savings across
the RDS fleet by comparing actual On-Demand spend to RI pricing.
Reads active instances from AWS and fetches pricing from the Pricing API.

Requirements: boto3, pandas
pip install boto3 pandas
"""

import boto3
import json
import pandas as pd
from datetime import datetime, timedelta
from typing import Dict, List, Optional

# AWS pricing API is only available in us-east-1
PRICING_CLIENT = boto3.client("pricing", region_name="us-east-1")
CE_CLIENT = boto3.client("ce", region_name="us-east-1")
RDS_CLIENT = boto3.client("rds", region_name="us-east-1")
```

```

def get_rds_on_demand_price(
    instance_class: str,
    engine: str,
    region: str = "US East (N. Virginia)",
    multi_az: bool = False
) -> Optional[float]:
    """
    Fetch On-Demand hourly price for an RDS instance from the Pricing API.
    Returns price in USD/hour or None if not found.
    """
    deployment = "Multi-AZ" if multi_az else "Single-AZ"

    # Map engine short names to pricing API descriptions
    engine_map = {
        "postgres": "PostgreSQL",
        "mysql": "MySQL",
        "mariadb": "MariaDB",
        "oracle-se2": "Oracle (SE2)",
        "sqlserver-se": "SQL Server (SE)",
    }
    engine_desc = engine_map.get(engine, engine)

    filters = [
        {"Type": "TERM_MATCH", "Field": "instanceType", "Value": instance_class},
        {"Type": "TERM_MATCH", "Field": "databaseEngine", "Value": engine_desc},
        {"Type": "TERM_MATCH", "Field": "location", "Value": region},
        {"Type": "TERM_MATCH", "Field": "deploymentOption", "Value": deployment},
    ]

    try:
        response = PRICING_CLIENT.get_products(
            ServiceCode="AmazonRDS",
            Filters=filters,
            MaxResults=10
        )
    except Exception as e:
        print(f"[pricing API error] {instance_class}/{engine}: {e}")
        return None

    for price_item_str in response.get("PriceList", []):
        price_item = json.loads(price_item_str)
        terms = price_item.get("terms", {}).get("OnDemand", {})
        for term in terms.values():
            for dimension in term.get("priceDimensions", {}).values():
                try:
                    price = float(dimension["pricePerUnit"]["USD"])
                    if price > 0:
                        return price
                except (KeyError, ValueError):
                    continue
    return None

def get_ri_pricing_estimates(
    instance_class: str,
    engine: str,
    multi_az: bool = False
) -> Dict[str, float]:
    """
    Return hardcoded RI discount factors by term and payment option.
    In production, replace with calls to describe-reserved-db-instances-offerings.
    Discounts are representative of us-east-1 r6g/m6g families as of mid-2024.
    """
    # These are discount multipliers applied to On-Demand price
    # to derive the effective RI hourly rate
    return {
        "1yr_no_upfront": 0.371, # ~37% discount
        "1yr_partial": 0.393, # ~39% when accounting for partial upfront savings
    }

```



```

    "1yr_all_upfront": 0.410,    # ~41%
    "3yr_no_upfront": 0.520,    # ~52%
    "3yr_partial":    0.560,
    "3yr_all_upfront": 0.600,    # ~60%
}

def get_instance_monthly_hours_from_ce(
    instance_class: str,
    days_back: int = 90
) -> float:
    """
    Query Cost Explorer to get average monthly On-Demand hours for an instance class.
    Uses UsageQuantity grouped by instance type over the last N days.
    """
    end = datetime.utcnow().date()
    start = end - timedelta(days=days_back)

    try:
        response = CE_CLIENT.get_cost_and_usage(
            TimePeriod={
                "Start": start.isoformat(),
                "End": end.isoformat()
            },
            Granularity="MONTHLY",
            Filter={
                "And": [
                    {
                        "Dimensions": {
                            "Key": "SERVICE",
                            "Values": ["Amazon Relational Database Service"]
                        }
                    },
                    {
                        "Dimensions": {
                            "Key": "INSTANCE_TYPE",
                            "Values": [instance_class]
                        }
                    },
                    {
                        "Dimensions": {
                            "Key": "PURCHASE_TYPE",
                            "Values": ["On Demand"]
                        }
                    }
                ]
            },
            GroupBy=[{"Type": "DIMENSION", "Key": "INSTANCE_TYPE"}],
            Metrics=["UsageQuantity"]
        )
    except Exception as e:
        print(f" [CE error for {instance_class}]: {e}")
        return 0.0

    monthly_hours = []
    for period in response.get("ResultsByTime", []):
        for group in period.get("Groups", []):
            try:
                hours = float(group["Metrics"]["UsageQuantity"]["Amount"])
                monthly_hours.append(hours)
            except (KeyError, ValueError):
                pass

    return sum(monthly_hours) / len(monthly_hours) if monthly_hours else 0.0

def analyze_fleet() -> pd.DataFrame:
    """
    Pull all active RDS instances, fetch pricing, and compute
    break-even and projected savings for each RI term.

```

```

"""
paginator = RDS_CLIENT.get_paginator("describe_db_instances")
instances = []

for page in paginator.paginate():
    for db in page["DBInstances"]:
        if db["DBInstanceStatus"] != "available":
            continue
        instances.append({
            "identifier": db["DBInstanceIdentifier"],
            "instance_class": db["DBInstanceClass"],
            "engine": db["Engine"],
            "multi_az": db["MultiAZ"],
            "region": RDS_CLIENT.meta.region_name,
        })

records = []
for inst in instances:
    ic = inst["instance_class"]
    engine = inst["engine"]
    multi = inst["multi_az"]

    on_demand_hourly = get_rds_on_demand_price(ic, engine, multi_az=multi)
    if not on_demand_hourly:
        print(f" Skipping {inst['identifier']}: could not fetch On-Demand price")
        continue

    monthly_hours = get_instance_monthly_hours_from_ce(ic) or 720.0
    on_demand_monthly = on_demand_hourly * monthly_hours

    discounts = get_ri_pricing_estimates(ic, engine, multi)

    for term_key, discount_pct in discounts.items():
        ri_effective_hourly = on_demand_hourly * (1 - discount_pct)
        ri_monthly_effective = ri_effective_hourly * 720 # normalize to full month

        # Hours below which On-Demand is cheaper than the RI commitment
        if (on_demand_hourly - ri_effective_hourly) > 0:
            break_even_hours = ri_monthly_effective / (
                on_demand_hourly - ri_effective_hourly
            )
        else:
            break_even_hours = 0.0

    monthly_savings = on_demand_monthly - ri_monthly_effective
    annual_savings = monthly_savings * 12

    records.append({
        "identifier": inst["identifier"],
        "instance_class": ic,
        "engine": engine,
        "multi_az": multi,
        "avg_monthly_hours": round(monthly_hours, 1),
        "on_demand_hourly": round(on_demand_hourly, 4),
        "on_demand_monthly": round(on_demand_monthly, 2),
        "ri_term": term_key,
        "ri_effective_hourly": round(ri_effective_hourly, 4),
        "ri_monthly_cost": round(ri_monthly_effective, 2),
        "break_even_hours": round(break_even_hours, 1),
        "monthly_savings": round(monthly_savings, 2),
        "annual_savings": round(annual_savings, 2),
        "discount_pct": f"{discount_pct*100:.0f}%",
    })

df = pd.DataFrame(records)
return df

def print_summary(df: pd.DataFrame) -> None:
    """Print a ranked savings report for the 1-year No Upfront term."""

```

```

if df.empty:
    print("No data to display. Check IAM permissions and Cost Explorer setup.")
    return

one_yr = df[df["ri_term"] == "1yr_no_upfront"].copy()
one_yr = one_yr[one_yr["avg_monthly_hours"] > 450] # Only qualify persistent instances
one_yr = one_yr.sort_values("annual_savings", ascending=False)

total_annual_opportunity = one_yr["annual_savings"].sum()
print("\n" + "="*80)
print("RDS Reserved Instance Opportunity Report - 1-Year No Upfront")
print(f"Generated: {datetime.utcnow().strftime('%Y-%m-%d %H:%M UTC')}")
print("="*80)
print(f"\n Total Annual Savings Opportunity: ${total_annual_opportunity:,.2f}")
print(f" Instances qualifying (>450 hrs/mo On-Demand): {len(one_yr)}\n")

cols = ["identifier", "instance_class", "engine",
        "on_demand_monthly", "ri_monthly_cost", "monthly_savings", "annual_savings", "discount_pct"]
print(one_yr[cols].to_string(index=False))
print("\n")

# Also show 3-year comparison for top 5
top_ids = one_yr["identifier"].head(5).tolist()
three_yr = df[
    (df["ri_term"] == "3yr_all_upfront") &
    (df["identifier"].isin(top_ids))
].sort_values("annual_savings", ascending=False)

if not three_yr.empty:
    print("Top 5 Instances - 3-Year All Upfront Comparison:")
    print(three_yr[cols].to_string(index=False))

if __name__ == "__main__":
    print("Fetching RDS fleet and pricing data...")
    df = analyze_fleet()
    print_summary(df)

    # Export full results to CSV for spreadsheet review or Jira attachment
    output_file = f"ri_analysis_{datetime.utcnow().strftime('%Y%m%d')}.csv"
    df.to_csv(output_file, index=False)
    print(f"\nFull results exported to: {output_file}")

```

Run this script with an IAM role that has `rds:DescribeDBInstances`, `ce:GetCostAndUsage`, and `pricing:GetProducts` permissions. The output ranks every On-Demand instance in your fleet by annual savings opportunity and flags the ones where a 3-year commitment materially outperforms the 1-year option.

8.4 Matching Commitment Models to Workload Archetypes

Not every database is a good RI candidate. The analysis becomes more nuanced when you segment your fleet by workload behavior. Five archetypes cover the vast majority of real-world cases.

Archetype 1: Evergreen Production Databases

These instances have run continuously for more than 12 months, host a critical application, and have no planned migration or sunset in the next 18 months. Examples include the primary PostgreSQL cluster for an e-commerce platform, the reporting MySQL replica, or the SQL Server database backing an internal ERP system.

Recommendation: 3-year All Upfront RI (AWS), 3-year Reserved Capacity (Azure), 3-year CUD

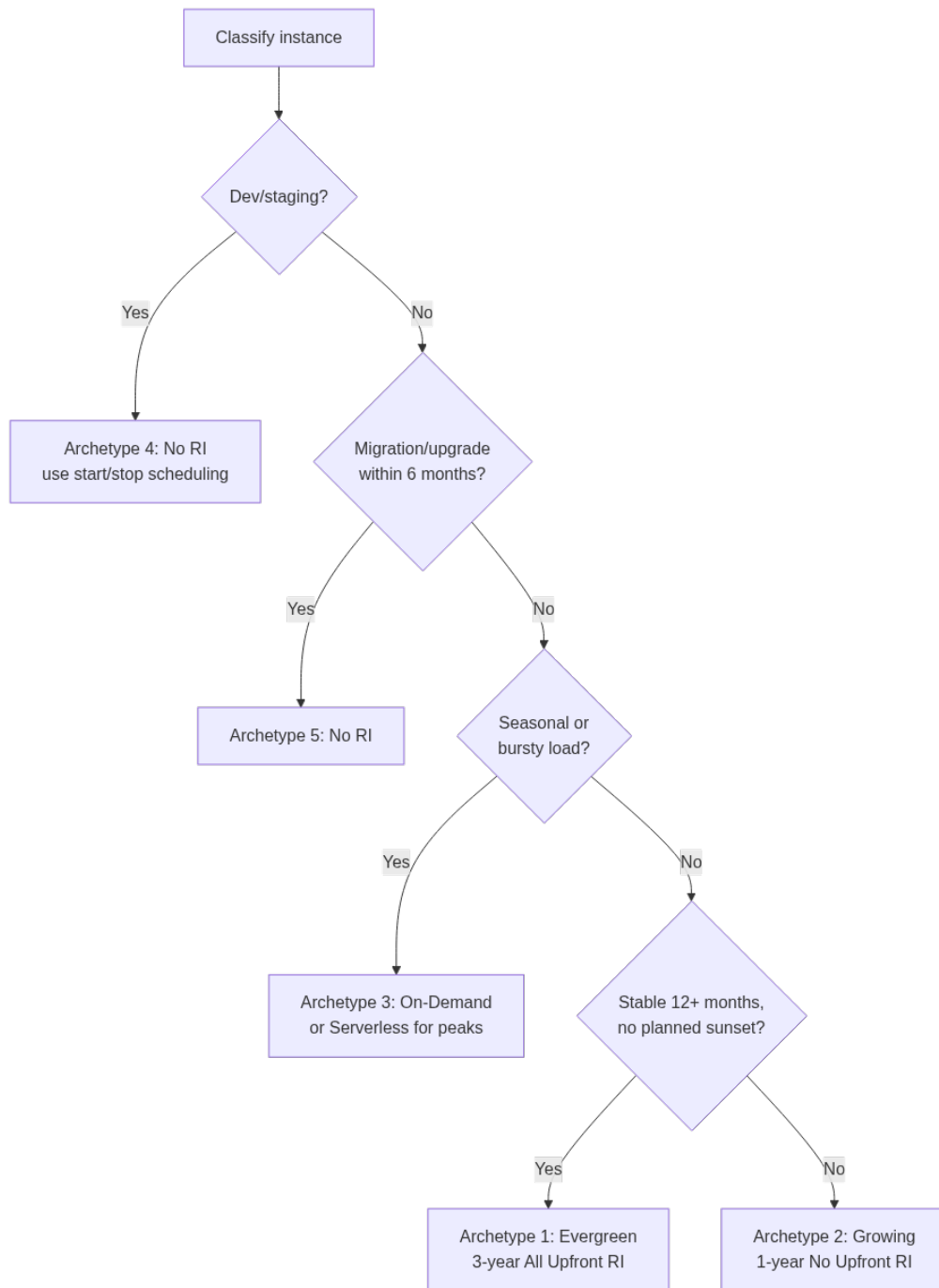


Figure 8: Five workload archetypes and their recommended commitment strategy — utilization alone is not enough to decide.

(GCP). The discount is maximized and the risk of waste is minimal because the workload is stable and long-lived. The All Upfront payment requires cash outlay but eliminates hourly billing entirely for the reserved portion — on a fleet of 20 instances this often justifies a conversation with finance about prepaying from the capital budget rather than treating it as operating expense.

Archetype 2: Growing Workloads with Predictable Trajectory

The database exists today and is growing at a known rate. You added two `db.r6g.2xlarge` instances this quarter and expect to add two more next quarter.

Recommendation: 1-year No Upfront RI for current instances. Do not try to pre-buy RIs for instances that do not yet exist — the instance type you need in six months may differ from what you need today. Review and renew annually. Use Convertible RIs if you anticipate instance family changes as the workload matures.

Archetype 3: Seasonal or Burst Workloads

Tax preparation software, seasonal retail platforms, and quarterly analytics workloads that run heavily for 2–4 months and lightly for the remainder of the year. A database that runs at full capacity in November and December and at 20% capacity the rest of the year will not benefit from an RI sized for peak capacity.

Recommendation: No RI for instances provisioned exclusively for seasonal peaks. Use On-Demand or RDS Serverless v2 (with Aurora) for burst capacity. If you have a minimum always-on fleet (even a smaller instance that handles off-season traffic), RI that minimum fleet and handle peaks with On-Demand.

Archetype 4: Development and Staging

Dev/staging databases are frequently resized, cloned, snapshotted, and terminated as engineers iterate. They are also frequently left running when they should be stopped — which is a separate problem addressed in Chapter 11.

Recommendation: No RI. Dev/staging instances are candidates for automated start/stop scheduling (saving 65–70% compared to always-on) rather than RI commitments. An RI on a dev database that gets replaced with a different instance type next sprint is pure waste.

Archetype 5: Instances Scheduled for Migration or Upgrade

If you have a MySQL 5.7 database targeted for migration to Aurora MySQL 3.x in the next six months, buying a 1-year RI for the source instance may overlap with the migration window, leaving you with a committed RI for an instance you no longer run.

Recommendation: No new RI purchases within 6 months of a confirmed migration. If the migration slips past 6 months (as they often do), revisit the RI purchase at that point.

The SQL queries below help classify your RDS instances into these archetypes by age and stability:

```
-- PostgreSQL: Query pg_stat_activity and age-related catalog info
-- to identify long-running databases that are strong RI candidates
-- (Run against your RDS inventory tracking database, not the application DB)

-- Assuming you maintain an rds_inventory table populated by your discovery tooling:
-- CREATE TABLE rds_inventory (
--   instance_id      TEXT PRIMARY KEY,
--   instance_class   TEXT,
--   engine           TEXT,
--   multi_az        BOOLEAN,
```

```

-- creation_date      DATE,
-- environment        TEXT, -- 'production','staging','development','analytics'
-- planned_migration   DATE, -- NULL if no migration planned
-- avg_monthly_hours   NUMERIC,
-- monthly_od_cost     NUMERIC,
-- last_updated        TIMESTAMPTZ DEFAULT NOW()
-- );

SELECT
  i.instance_id,
  i.instance_class,
  i.engine,
  i.environment,
  i.creation_date,
  DATE_PART('day', NOW() - i.creation_date::TIMESTAMPTZ) / 30.0 AS age_months,
  i.avg_monthly_hours,
  i.monthly_od_cost,
  CASE
    WHEN i.environment IN ('development','staging')
      THEN 'Archetype 4: Dev/Staging - No RI'
    WHEN i.planned_migration IS NOT NULL
      AND i.planned_migration <= CURRENT_DATE + INTERVAL '6 months'
      THEN 'Archetype 5: Pending Migration - No RI'
    WHEN i.avg_monthly_hours >= 700
      AND DATE_PART('day', NOW() - i.creation_date::TIMESTAMPTZ) >= 365
      THEN 'Archetype 1: Evergreen - 3yr RI recommended'
    WHEN i.avg_monthly_hours >= 450
      AND DATE_PART('day', NOW() - i.creation_date::TIMESTAMPTZ) >= 90
      THEN 'Archetype 2: Growing - 1yr RI recommended'
    WHEN i.avg_monthly_hours < 450
      THEN 'Archetype 3: Seasonal/Burst - On-Demand'
    ELSE 'Unclassified - Manual review required'
  END
  AS ri_recommendation,
  ROUND(i.monthly_od_cost * 0.371 * 12, 2)
  AS est_1yr_annual_savings,
  ROUND(i.monthly_od_cost * 0.600 * 12, 2)
  AS est_3yr_annual_savings
FROM
  rds_inventory i
WHERE
  i.last_updated >= NOW() - INTERVAL '7 days'
ORDER BY
  i.monthly_od_cost DESC;

-- SQL Server equivalent using a comparable inventory table
-- (SQL Server Management Studio or Azure Data Studio)

-- Assuming equivalent table: dbo.rds_inventory
SELECT
  i.instance_id,
  i.instance_class,
  i.engine,
  i.environment,
  i.creation_date,
  DATEDIFF(DAY, i.creation_date, GETUTCDATE()) / 30.0 AS age_months,
  i.avg_monthly_hours,
  i.monthly_od_cost,
  CASE
    WHEN i.environment IN ('development', 'staging')
      THEN 'Archetype 4: Dev/Staging - No RI'
    WHEN i.planned_migration IS NOT NULL
      AND i.planned_migration <= DATEADD(MONTH, 6, GETUTCDATE())
      THEN 'Archetype 5: Pending Migration - No RI'
    WHEN i.avg_monthly_hours >= 700
      AND DATEDIFF(DAY, i.creation_date, GETUTCDATE()) >= 365
      THEN 'Archetype 1: Evergreen - 3yr RI recommended'
    WHEN i.avg_monthly_hours >= 450
      AND DATEDIFF(DAY, i.creation_date, GETUTCDATE()) >= 90
      THEN 'Archetype 2: Growing - 1yr RI recommended'
    WHEN i.avg_monthly_hours < 450
      THEN 'Archetype 3: Seasonal/Burst - On-Demand'
    ELSE 'Unclassified - Manual review required'
  END

```

```
END                                                    AS ri_recommendation,
ROUND(i.monthly_od_cost * 0.371 * 12, 2)              AS est_1yr_annual_savings,
ROUND(i.monthly_od_cost * 0.600 * 12, 2)              AS est_3yr_annual_savings
FROM
  dbo.rds_inventory i
WHERE
  i.last_updated >= DATEADD(DAY, -7, GETUTCDATE())
ORDER BY
  i.monthly_od_cost DESC;
```

8.5 AWS Compute Savings Plans and Their Database Relevance

Compute Savings Plans deserve specific treatment because the overlap with database workloads is frequently misunderstood, leading teams either to miss legitimate savings or to assume coverage that does not exist.

What Compute Savings Plans cover: EC2 instances, Fargate, and Lambda compute charges. They do not cover RDS provisioned instances. If you are running a fleet of EC2-based self-managed PostgreSQL or SQL Server instances, Compute Savings Plans apply directly. If you migrated those workloads to RDS, the Savings Plan provides no coverage for RDS charges.

The Aurora Serverless v2 exception: Aurora Serverless v2 charges are based on Aurora Capacity Units (ACUs) per second. These charges fall under the **AmazonRDS** service in Cost Explorer and are not covered by Compute Savings Plans either. Aurora Serverless v2 does have its own Reserved Capacity offering under standard RDS RI terms when you provision the minimum capacity units — but many teams running pure serverless Aurora workloads skip this analysis entirely.

The practical implication: If your organization has a large EC2 Savings Plan commitment and you are evaluating a lift-and-shift of self-managed databases to RDS, be aware that moving to RDS will reduce your Savings Plan utilization. The EC2 Savings Plan dollar commitment you made does not transfer to RDS compute. Factor this into the migration cost model: the savings from eliminating self-managed overhead and improving availability may still outweigh the lost Savings Plan discount, but the math needs to be explicit.

When Savings Plans do apply to your database budget: Many FinOps teams manage compute and database budgets together. If your application servers (EC2 or Fargate) run in front of RDS, Savings Plans covering those tiers still reduce total application infrastructure cost. The database RI and the application Savings Plan are complementary, not competing — budget them separately with separate coverage targets.

8.6 Azure Reserved Capacity: Subscription Sharing and Scope Management

Azure Reserved Capacity for SQL introduces a scoping dimension that AWS RIs do not have in the same way: reservations can be scoped to a single subscription, a resource group, or shared across all subscriptions in a billing account (Management Group).

Single-subscription scope is the default. The reservation applies only to matching resources within the subscription where it was purchased. This is safe and easy to reason about but limits

flexibility if your SQL workloads span multiple subscriptions (which is common in enterprise Azure architectures with separate subscriptions for production, staging, and development environments).

Shared scope allows the reservation to apply to any matching vCore usage across all subscriptions associated with the billing account. This is generally the better choice for organizations with stable multi-subscription architectures, because it maximizes utilization — if one subscription underuses its allocation, another can absorb the discount. The risk is visibility: teams managing individual subscriptions may not see why their costs look lower than expected, complicating internal charge-back.

Management group scope (available in some Azure enrollment types) allows finer-grained scoping between single-subscription and full-account sharing.

The following Azure CLI sequence demonstrates how to check current reservation scope and modify it:

```
# List all reservations and their current scope
az reservations reservation list-all \
  --output table \
  --query "[?contains(properties.reservedResourceType, 'SqlDatabases') ||
    contains(properties.reservedResourceType, 'SqlManagedInstances')].{
    Name:name,
    ResourceType:properties.reservedResourceType,
    Scope:properties.appliedScopeType,
    AppliedScopes:properties.appliedScopes,
    Quantity:properties.quantity,
    ExpiryDate:properties.expiryDate,
    UtilizationPct:properties.utilizationAggregates[0].value
  }"

# Update a reservation to shared scope (apply across all subscriptions)
# Replace <reservation-order-id> and <reservation-id> with actual values
az reservations reservation update \
  --reservation-order-id <reservation-order-id> \
  --reservation-id <reservation-id> \
  --applied-scope-type Shared
```

Azure Hybrid Benefit layering: Azure Reserved Capacity discounts stack with Azure Hybrid Benefit (AHB), which allows you to apply existing on-premises SQL Server licenses with Software Assurance to Azure SQL, reducing the license component of your Azure SQL cost. The interaction is:

- Reserved Capacity discounts the compute (vCore) cost
- Azure Hybrid Benefit eliminates the SQL Server license cost on top of that

Using both together on a Business Critical tier SQL Managed Instance can reduce total costs by 55–80% compared to pay-as-you-go with included license, depending on the vCore count and term. This combination is one of the strongest cost arguments for organizations with significant on-premises SQL Server License Mobility agreements migrating workloads to Azure SQL Managed Instance.

The calculation:

Pay-as-you-go with license (8 vCore Business Critical):	~\$11.78/hr	
After AHB (bring own license):	~\$5.15/hr	(~56% reduction)
After AHB + 3yr Reserved Capacity:	~\$2.32/hr	(~80% total reduction)

These rates are representative and vary by region and current Azure pricing — always validate against the Azure Pricing Calculator with your specific configuration before committing.

8.7 Google Cloud CUDs: Resource-Based vs. Spend-Based

GCP's Committed Use Discounts for Cloud SQL come in two forms, and choosing between them has meaningful implications for how rigidly you need to forecast.

Resource-based CUDs commit to a specific quantity of vCPU and memory (in GB) for 1 or 3 years. You receive a discount on that resource quantity regardless of which Cloud SQL instance consumes it, as long as it falls within the committed region. If you commit to 16 vCPUs and 128 GB of RAM in `us-central1`, any combination of Cloud SQL instances in that region consuming up to that capacity receives the committed price. This is more flexible than AWS standard RIs at the instance type level, since you are not locked to a specific machine type.

Spend-based CUDs (available for certain services including Cloud SQL in some configurations) commit to spending a minimum dollar amount per hour. They offer slightly lower discounts than resource-based CUDs but require no capacity planning — you simply commit to a spend floor.

For most teams with predictable Cloud SQL workloads, resource-based CUDs are the better value. The following BigQuery query models the CUD opportunity from your billing export:

```
-- GCP BigQuery: Estimate CUD savings opportunity for Cloud SQL
-- Compares current On-Demand spend against modeled CUD rates
-- Adjust discount_1yr and discount_3yr based on current GCP pricing for your region

WITH cloud_sql_spend AS (
  SELECT
    DATE_TRUNC(usage_start_time, MONTH)                AS billing_month,
    project.id                                           AS project_id,
    resource.name                                         AS instance_name,
    sku.description                                       AS sku_description,
    SUM(usage.amount)                                     AS total_vcpu_hours,
    SUM(cost)                                             AS total_cost_usd
  FROM
    `your-project.your_dataset.gcp_billing_export_v1_XXXXXX`
  WHERE
    service.description = 'Cloud SQL'
    AND LOWER(sku.description) NOT LIKE '%commit%'
    AND LOWER(sku.description) NOT LIKE '%storage%'
    AND LOWER(sku.description) NOT LIKE '%network%'
    AND LOWER(sku.description) NOT LIKE '%ip%'
    AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
    AND cost > 0
  GROUP BY 1, 2, 3, 4
),
monthly_avg AS (
  SELECT
    project_id,
    instance_name,
    AVG(total_cost_usd)                                AS avg_monthly_od_cost
  FROM cloud_sql_spend
  GROUP BY 1, 2
)
SELECT
  m.project_id,
  m.instance_name,
  ROUND(m.avg_monthly_od_cost, 2)                      AS avg_monthly_on_demand_usd,
  ROUND(m.avg_monthly_od_cost * (1 - 0.30), 2)         AS modeled_1yr_cud_monthly,
  ROUND(m.avg_monthly_od_cost * (1 - 0.46), 2)         AS modeled_3yr_cud_monthly,
  ROUND(m.avg_monthly_od_cost * 0.30 * 12, 2)          AS est_1yr_annual_savings,
  ROUND(m.avg_monthly_od_cost * 0.46 * 12, 2)          AS est_3yr_annual_savings,
  ROUND(m.avg_monthly_od_cost * 0.46 * 36, 2)          AS est_3yr_total_savings
FROM
  monthly_avg
```

```

monthly_avg m
WHERE
  m.avg_monthly_od_cost > 50  -- Filter out negligible instances
ORDER BY
  est_3yr_annual_savings DESC
LIMIT 50;

```

8.8 Building an Automated RI Renewal and Coverage Monitor

Purchasing RIs is a one-time decision that decays. Instances get resized, engines get upgraded, new On-Demand instances appear while RI commitments sit idle. The only way to stay optimized is to monitor coverage continuously and trigger reviews before RIs expire.

The following Lambda function (deployable as a scheduled CloudWatch Events rule) monitors RDS RI coverage weekly and posts a Slack notification when coverage drops below a configurable threshold or when RIs are within 60 days of expiry:

```

#!/usr/bin/env python3
"""
ri_coverage_monitor.py

AWS Lambda function that monitors RDS RI coverage and expiry.
Deploy with a CloudWatch Events schedule (e.g., cron(0 9 ? * MON *))
to run every Monday morning.

Environment variables:
  SLACK_WEBHOOK_URL      - Slack incoming webhook URL
  COVERAGE_THRESHOLD    - Minimum acceptable coverage % (default: 80)
  EXPIRY_ALERT_DAYS      - Days before expiry to alert (default: 60)
  AWS_REGION_TARGET      - Region to analyze (default: us-east-1)
"""

import boto3
import json
import os
import urllib.request
from datetime import datetime, timezone, timedelta
from typing import List, Dict, Any

SLACK_WEBHOOK = os.environ.get("SLACK_WEBHOOK_URL", "")
THRESHOLD = float(os.environ.get("COVERAGE_THRESHOLD", "80"))
EXPIRY_DAYS = int(os.environ.get("EXPIRY_ALERT_DAYS", "60"))
TARGET_REGION = os.environ.get("AWS_REGION_TARGET", "us-east-1")

rds = boto3.client("rds", region_name=TARGET_REGION)
ce = boto3.client("ce", region_name="us-east-1")

def get_active_ris() -> List[Dict[str, Any]]:
    """Return all active RDS Reserved Instances."""
    response = rds.describe_reserved_db_instances(Filters=[
        {"Name": "state", "Values": ["active"]}
    ])
    return response.get("ReservedDBInstances", [])

def get_expiring_ris(ris: List[Dict], days: int) -> List[Dict]:
    """Return RIs expiring within `days` days."""
    cutoff = datetime.now(timezone.utc) + timedelta(days=days)
    expiring = []
    for ri in ris:
        start = ri["StartTime"]

```

```

duration = ri["Duration"] # seconds
expiry = start + timedelta(seconds=duration)
if expiry <= cutoff:
    expiring.append({
        "id": ri["ReservedDBInstanceId"],
        "class": ri["DBInstanceClass"],
        "engine": ri["ProductDescription"],
        "count": ri["DBInstanceCount"],
        "expiry": expiry.strftime("%Y-%m-%d"),
        "days_left": (expiry - datetime.now(timezone.utc)).days,
    })
return expiring

def get_coverage_percentage() -> Dict[str, float]:
    """Query Cost Explorer for RI coverage over the last 30 days."""
    end = datetime.utcnow().date()
    start = end - timedelta(days=30)

    response = ce.get_reservation_coverage(
        TimePeriod={"Start": start.isoformat(), "End": end.isoformat()},
        Granularity="MONTHLY",
        Filter={
            "Dimensions": {
                "Key": "SERVICE",
                "Values": ["Amazon Relational Database Service"]
            }
        }
    )

    coverage_data = {}
    for period in response.get("CoveragesByTime", []):
        period_start = period["TimePeriod"]["Start"]
        coverage = period.get("Total", {}).get("CoverageHours", {})
        try:
            pct = float(coverage.get("CoverageHoursPercentage", 0))
        except (ValueError, TypeError):
            pct = 0.0
        coverage_data[period_start] = pct

    return coverage_data

def get_utilization_percentage() -> Dict[str, Dict]:
    """Query Cost Explorer for RI utilization over the last 30 days."""
    end = datetime.utcnow().date()
    start = end - timedelta(days=30)

    response = ce.get_reservation_utilization(
        TimePeriod={"Start": start.isoformat(), "End": end.isoformat()},
        Granularity="MONTHLY",
        Filter={
            "Dimensions": {
                "Key": "SERVICE",
                "Values": ["Amazon Relational Database Service"]
            }
        }
    )

    util_data = {}
    for period in response.get("UtilizationsByTime", []):
        period_start = period["TimePeriod"]["Start"]
        total = period.get("Total", {})
        util_data[period_start] = {
            "utilization_pct": total.get("UtilizationPercentage", "0"),
            "used_hours": total.get("UsedHours", "0"),
            "unused_hours": total.get("UnusedHours", "0"),
            "realized_savings": total.get("RealizedSavings", {}).get("Amount", "0"),
            "unrealized_savings": total.get("UnrealizedSavings", {}).get("Amount", "0"),
        }

```

```

return util_data

def build_slack_message(
    coverage: Dict[str, float],
    utilization: Dict[str, Dict],
    expiring: List[Dict],
) -> Dict:
    """Construct a Slack Block Kit message with the monitoring summary."""
    now_str = datetime.utcnow().strftime("%Y-%m-%d %H:%M UTC")
    blocks = [
        {
            "type": "header",
            "text": {
                "type": "plain_text",
                "text": f":bar_chart: RDS RI Coverage Report - {now_str}"
            }
        },
        {"type": "divider"}
    ]

    # Coverage section
    coverage_lines = []
    alerts = []
    for period, pct in sorted(coverage.items()):
        icon = ":white_check_mark:" if pct >= THRESHOLD else ":red_circle:"
        coverage_lines.append(f"{icon} {period}:  {pct:.1f}% RI coverage")
        if pct < THRESHOLD:
            alerts.append(f"Coverage {pct:.1f}% is below threshold ({THRESHOLD:.0f}%)")

    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": "*RI Coverage (last 30 days)*\n" + "\n".join(coverage_lines)
        }
    })

    # Utilization section
    util_lines = []
    for period, data in sorted(utilization.items()):
        util_pct = float(data["utilization_pct"])
        icon = ":white_check_mark:" if util_pct >= 80 else ":warning:"
        real_sav = float(data["realized_savings"])
        unreal_sav = float(data["unrealized_savings"])
        util_lines.append(
            f"{icon} {period}:  {util_pct:.1f}% utilized | "
            f"Realized: ${real_sav:,.0f} | Unrealized (waste): ${unreal_sav:,.0f}"
        )
        if util_pct < 60:
            alerts.append(
                f"Utilization {util_pct:.1f}% for {period} - "
                f"${unreal_sav:,.0f} in committed but unused capacity"
            )

    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": "*RI Utilization (last 30 days)*\n" + "\n".join(util_lines)
        }
    })

    # Expiring RIs section
    if expiring:
        expiry_lines = []
        for ri in sorted(expiring, key=lambda x: x["days_left"]):
            icon = ":rotating_light:" if ri["days_left"] <= 30 else ":warning:"
            expiry_lines.append(
                f"{icon}  `{ri['id']}`  {ri['class']} ({ri['engine']})  "
            )

```

```

        f"x{ri['count']} - expires {ri['expiry']}* "
        f"({ri['days_left']} days)"
    )
    if ri["days_left"] <= 30:
        alerts.append(
            f"RI {ri['id']} ({ri['class']}) expires in {ri['days_left']} days - "
            "renewal decision needed immediately"
        )
    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": f"*Expiring within {EXPIRY_DAYS} days*\n" + "\n".join(expiry_lines)
        }
    })
else:
    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": f":white_check_mark: *No RIs expiring within {EXPIRY_DAYS} days*"
        }
    })

# Alerts summary
if alerts:
    alert_text = "*:rotating_light: Action Required*\n" + \
        "\n".join(f"* {a}" for a in alerts)
    blocks.append({"type": "divider"})
    blocks.append({
        "type": "section",
        "text": {"type": "mrkdwn", "text": alert_text}
    })

return {"blocks": blocks}

def post_to_slack(message: Dict) -> None:
    """Post message to Slack via incoming webhook."""
    if not SLACK_WEBHOOK:
        print("SLACK_WEBHOOK_URL not set - printing message to stdout")
        print(json.dumps(message, indent=2))
        return

    payload = json.dumps(message).encode("utf-8")
    req = urllib.request.Request(
        SLACK_WEBHOOK,
        data=payload,
        headers={"Content-Type": "application/json"},
        method="POST"
    )
    try:
        with urllib.request.urlopen(req, timeout=10) as resp:
            print(f"Slack response: {resp.status}")
    except Exception as e:
        print(f"Slack post failed: {e}")

def lambda_handler(event, context):
    """Lambda entry point."""
    print(f"Starting RDS RI coverage monitor for region: {TARGET_REGION}")

    active_ris = get_active_ris()
    expiring = get_expiring_ris(active_ris, EXPIRY_DAYS)
    coverage = get_coverage_percentage()
    utilization = get_utilization_percentage()

    message = build_slack_message(coverage, utilization, expiring)
    post_to_slack(message)

```

```
return {
    "statusCode": 200,
    "active_ri_count": len(active_ris),
    "expiring_ri_count": len(expiring),
    "coverage_periods": len(coverage),
}

# Allow local testing
if __name__ == "__main__":
    lambda_handler({}, None)
```

Deploy this function with a weekly CloudWatch Events schedule. The IAM role requires `rds:DescribeReservedDBInstances`, `ce:GetReservationCoverage`, and `ce:GetReservationUtilization`. The Slack webhook is stored in Systems Manager Parameter Store and injected as an environment variable at deploy time — never hardcoded.

The alert logic fires when: - RI coverage drops below the configured threshold (default 80%) - RI utilization drops below 60% (indicating committed but unused capacity) - Any RI is within 60 days of expiry (default) or 30 days (critical alert)

Teams that deploy this monitor and respond to its alerts typically maintain coverage above 85% and eliminate the low-utilization waste that accumulates silently in unmonitored RI portfolios.

8.9 Making the Purchase Decision: A Structured Process

The math in this chapter is only as useful as the process it feeds. Without a structured review cycle, RI purchases happen reactively — someone notices a large bill, buys an RI in a hurry, and the portfolio drifts into misalignment again within a year. The following cadence works for most engineering and FinOps teams:

Monthly (15 minutes): Run the coverage monitor output, verify utilization is above threshold, check for any new significant On-Demand spend that has appeared in the fleet. No purchase decisions at this stage — just anomaly detection.

Quarterly (2–3 hours): Run the full break-even analysis against the current fleet. Identify instances that have now passed 90 days of continuous On-Demand operation and qualify for Archetype 2 treatment. Identify any Archetype 1 instances approaching RI expiry. Prepare a purchase recommendation memo with projected savings and risk factors for each candidate. Get approval and execute purchases.

Annual (half-day): Full portfolio review. Map every active RI to the instance it is covering. Identify orphaned RIs (commitments with no matching instance). Review Convertible RI exchange opportunities. Model the coming year's fleet growth and whether to shift any Archetype 2 instances to Archetype 1 (3-year) treatment. Validate that Azure reservation scoping is still appropriate as subscription structures may have changed.

The purchase recommendation memo does not need to be long. A one-page table with columns for instance ID, current monthly On-Demand cost, recommended RI term, projected monthly cost under RI, annual savings, and the expiry date of the existing RI if applicable is sufficient for engineering leads and finance to review and approve in a single meeting.

Key Takeaways

- **On-Demand is appropriate for fewer workloads than most teams assume.** Any database instance running more than roughly 60% of the hours in a month is mathematically cheaper under a 1-year No Upfront RI, and most production databases run at or near 100% uptime.
 - **Workload archetype determines commitment strategy, not just utilization.** Ever-green production databases warrant 3-year All Upfront commitments for maximum savings; dev/staging instances and pending migrations should remain On-Demand regardless of how many hours they run.
 - **RI portfolios decay without active management.** Instance resizes, engine changes, and workload terminations silently erode utilization. Automated weekly coverage monitoring with expiry alerts is the minimum viable governance layer for an RI program.
 - **Azure Reserved Capacity and Azure Hybrid Benefit stack multiplicatively.** Organizations with existing on-premises SQL Server licenses can achieve cost reductions that significantly exceed what either mechanism provides independently — always model both together when evaluating Azure SQL migrations.
 - **Google Cloud CUDs offer more flexibility than AWS standard RIs** because they commit to vCPU and memory resources rather than specific instance types, reducing the risk that a machine type change leaves you with an orphaned commitment.
-

Chapter 9: Aurora Serverless v2: When It Saves vs When It Costs More

Aurora Serverless v2 promises the holy grail of database economics: pay only for what you use, scale instantly, and never over-provision. In practice, teams that migrate from provisioned Aurora instances without understanding the billing model often see their monthly bills *increase* by 20–40%. Teams that deploy it correctly for the right workload patterns report savings of \$800–\$3,000 per month compared to fixed provisioned instances. This chapter shows you how to tell which outcome you are heading toward before you commit, how to measure it after deployment, and how to automate the guardrails that keep Aurora Serverless v2 economical at scale.

9.1 How Aurora Serverless v2 Actually Bills You

Before diagnosing cost problems, you need a precise mental model of the billing engine. Aurora Serverless v2 charges in Aurora Capacity Units (ACUs). One ACU provides approximately 2 GiB of memory and a proportional slice of CPU. As of mid-2024, the price per ACU-hour in `us-east-1` is **\$0.12**. That sounds small until you realize a `db.r6g.2xlarge` provisioned instance — 8 vCPUs, 64 GiB RAM — runs at roughly **\$0.48/hour** (\$345/month), while Aurora Serverless v2 scaled to the equivalent capacity (approximately 32 ACUs) costs **$0.12 \times 32 = \$3.84/\text{hour}$** (\$2,765/month). The provisioned instance is cheaper at sustained load. Aurora Serverless v2 wins only when your database spends meaningful time at low ACU counts.

The three billing dimensions are:

Dimension	How It Works	Where It Surprises Teams
ACU-hours	Billed per second, rounded to the nearest second	Minimum floor of 0.5 ACU means you always pay something
Storage	Same as provisioned Aurora: \$0.10/GB-month	No savings here regardless of serverless vs provisioned
I/O	\$0.20 per million requests (standard mode)	High-write workloads pay this on top of ACU costs

The minimum ACU setting is the most commonly misconfigured parameter. Setting `min_capacity` to 8 ACUs instead of 0.5 ACUs costs an extra **\$648/month** in idle time alone — before a single

query runs. Many teams copy their old provisioned instance size as the minimum, negating the entire serverless value proposition.

```
-- Run against your Aurora PostgreSQL-compatible cluster
-- to see how much time the database has spent at different capacity levels
-- Requires pg_stat_activity and CloudWatch integration via Performance Insights

-- Step 1: Check current ACU utilization proxy via active connections and CPU
SELECT
  now() AS sample_time,
  count(*) AS total_connections,
  count(*) FILTER (WHERE state = 'active') AS active_connections,
  count(*) FILTER (WHERE state = 'idle') AS idle_connections,
  count(*) FILTER (WHERE state = 'idle in transaction') AS idle_in_txn,
  round(
    (count(*) FILTER (WHERE state = 'active'))::numeric /
    NULLIF(count(*), 0) * 100, 2
  ) AS active_pct
FROM pg_stat_activity
WHERE backend_type = 'client backend';

-- Step 2: Identify queries driving ACU spikes
-- High CPU queries are the primary driver of ACU scale-up events
SELECT
  queryid,
  calls,
  round(total_exec_time::numeric, 2) AS total_ms,
  round(mean_exec_time::numeric, 2) AS mean_ms,
  round((total_exec_time / sum(total_exec_time) OVER ()) * 100, 2) AS pct_total_time,
  rows,
  query
FROM pg_stat_statements
ORDER BY total_exec_time DESC
LIMIT 20;
```

The second query is your first cost signal: if the top five queries in `pg_stat_statements` account for more than 70% of total execution time and they run continuously throughout the day, Aurora Serverless v2 will spend most of its time at high ACU counts. That is a provisioned instance workload.

9.2 Identifying Workloads Where Serverless v2 Wins

Aurora Serverless v2 generates positive ROI in four specific workload patterns. Every other pattern is either neutral or negative. Be honest about which category your database falls into.

Pattern 1: Variable Business-Hours Load Traffic is genuinely low (under 10% peak) for 12+ hours per day, such as a North American SaaS application with negligible overnight traffic. If the database idles at 0.5–2 ACUs for 12 hours and peaks at 16 ACUs for 12 hours, the daily cost is roughly $(1 \text{ ACU} \times 12\text{h} + 16 \text{ ACU} \times 12\text{h}) \times \$0.12 = \$24.48/\text{day}$ versus a provisioned `db.r6g.xlarge` (comparable to 16 ACUs) at $\$0.48/\text{hour} \times 24\text{h} = \$11.52/\text{day}$. Provisioned still wins here. Aurora Serverless v2 starts winning when the idle period drops to very low ACUs (0.5) and extends beyond 16 hours per day.

Pattern 2: Development and Staging Environments This is the most reliable win. A development database used 8 hours on weekdays and idle otherwise runs 40 hours/week at load and 128 hours/week idle. At 0.5 ACU minimum: $(40\text{h} \times 8 \text{ ACU} + 128\text{h} \times 0.5 \text{ ACU}) \times \$0.12 = \$46.08/\text{week}$ versus a provisioned `db.t3.large` at $\$0.145/\text{hour} \times 168\text{h}/\text{week} = \$24.36/\text{week}$.

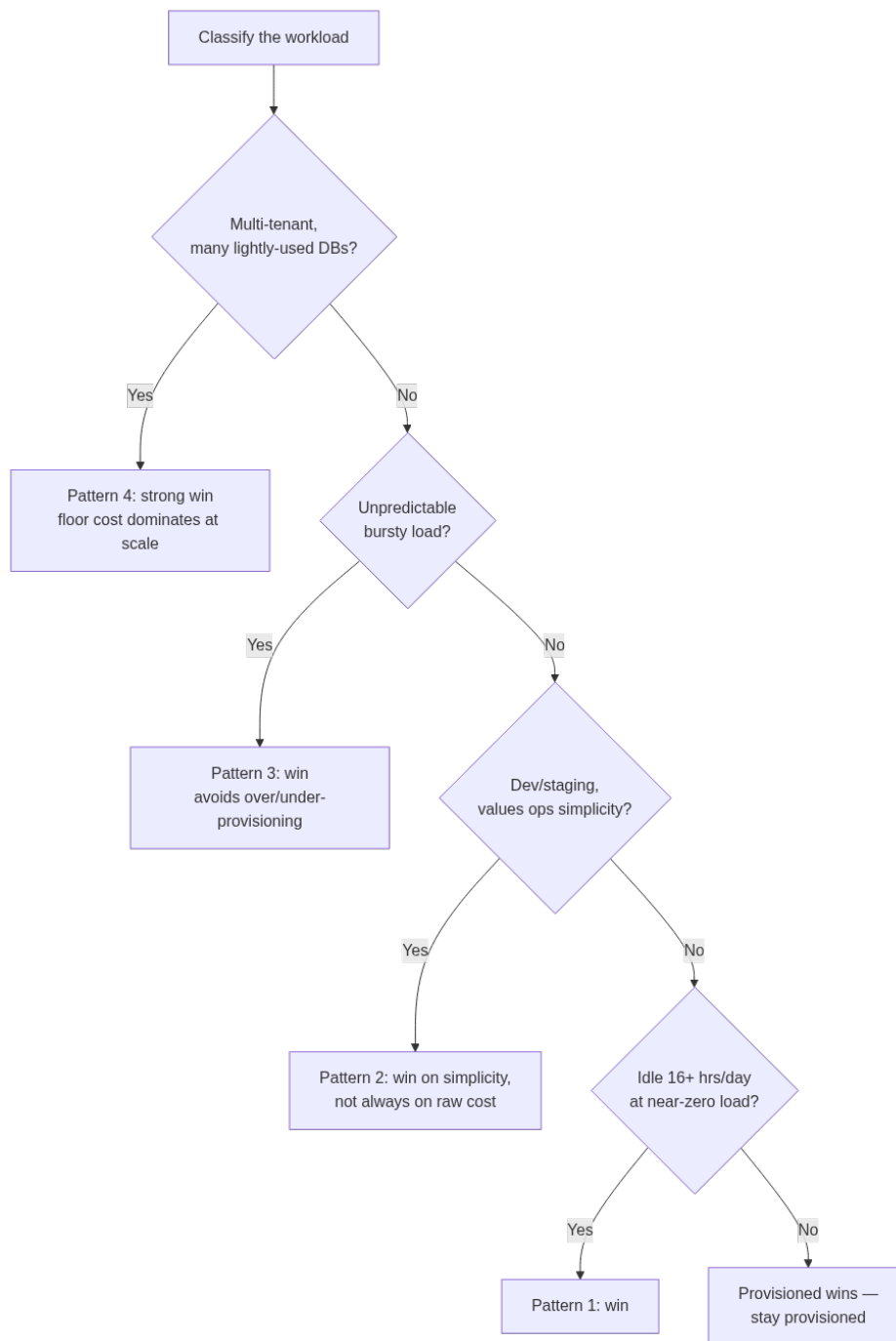


Figure 9: The four patterns where Aurora Serverless v2 actually saves money — every other pattern favors a provisioned instance.

Provisioned is still cheaper per week, but the comparison ignores the operational value of automatic scale-to-near-zero versus manual start/stop scripting. More importantly, if you are running db.r6g.large for a staging environment that mirrors production sizing, Aurora Serverless v2 at 0.5–16 ACU range costs $\$0.12 \times 0.5 \times 128 + \$0.12 \times 16 \times 40 = \$84.48/\text{week}$ versus the provisioned db.r6g.large at $\$0.24 \times 168 = \$40.32/\text{week}$. For staging, provisioned with scheduled stop/start wins on pure cost. Aurora Serverless v2 wins on operational simplicity.

Pattern 3: Unpredictable Bursty Workloads Marketing campaign databases, event-driven analytics, and reporting clusters that receive batch workloads at irregular intervals. The inability to predict when capacity is needed makes provisioning guesswork. Teams either over-provision and waste money, or under-provision and page incidents. Aurora Serverless v2 eliminates that tradeoff.

Pattern 4: Multi-Tenant Applications with Per-Tenant Databases Running 50 tenant-specific Aurora clusters where most tenants are lightly used is wildly expensive with provisioned instances. At \$0.48/hour minimum per cluster, 50 clusters cost \$17,280/month in compute alone. With Aurora Serverless v2 at 0.5 ACU minimum, 50 clusters cost $50 \times 0.5 \times \$0.12 \times 720 = \$2,160/\text{month}$ — an **\$15,120/month saving**.

The following script pulls CloudWatch metrics to classify your existing Aurora clusters automatically:

```
#!/usr/bin/env python3
"""
aurora_serverless_classifier.py
Analyzes existing Aurora provisioned clusters and classifies them by
serverless v2 suitability based on CPU utilization patterns.

Run: python aurora_serverless_classifier.py --region us-east-1 --days 30
"""

import boto3
import argparse
from datetime import datetime, timedelta
from collections import defaultdict

def get_aurora_clusters(region):
    rds = boto3.client('rds', region_name=region)
    paginator = rds.get_paginator('describe_db_clusters')
    clusters = []
    for page in paginator.paginate():
        for cluster in page['DBClusters']:
            if 'aurora' in cluster.get('Engine', '').lower():
                clusters.append(cluster)
    return clusters

def get_cpu_stats(cw, instance_id, days):
    end = datetime.utcnow()
    start = end - timedelta(days=days)
    response = cw.get_metric_statistics(
        Namespace='AWS/RDS',
        MetricName='CPUUtilization',
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
        StartTime=start,
        EndTime=end,
        Period=3600, # 1-hour granularity
        Statistics=['Average', 'Maximum', 'Minimum']
    )
    datapoints = response['Datapoints']
    if not datapoints:
        return None
    datapoints.sort(key=lambda x: x['Timestamp'])
    return datapoints
```

```

def classify_workload(datapoints):
    if not datapoints:
        return 'INSUFFICIENT_DATA', {}

    averages = [dp['Average'] for dp in datapoints]
    total_hours = len(averages)
    low_hours = sum(1 for a in averages if a < 10)
    medium_hours = sum(1 for a in averages if 10 <= a < 50)
    high_hours = sum(1 for a in averages if a >= 50)

    low_pct = (low_hours / total_hours) * 100
    high_pct = (high_hours / total_hours) * 100
    avg_cpu = sum(averages) / len(averages)
    max_cpu = max(averages)

    stats = {
        'total_hours_sampled': total_hours,
        'low_utilization_pct': round(low_pct, 1),
        'high_utilization_pct': round(high_pct, 1),
        'avg_cpu': round(avg_cpu, 1),
        'max_cpu': round(max_cpu, 1),
    }

    # Classification logic
    if low_pct >= 60 and high_pct <= 15:
        classification = 'STRONG_SERVERLESS_CANDIDATE'
    elif low_pct >= 40 and high_pct <= 30:
        classification = 'MODERATE_SERVERLESS_CANDIDATE'
    elif avg_cpu >= 60:
        classification = 'PROVISIONED_PREFERRED'
    else:
        classification = 'BORDERLINE_ANALYZE_FURTHER'

    return classification, stats

def estimate_serverless_savings(instance_class, stats, hours_per_month=730):
    # Rough ACU equivalents for common instance classes
    instance_acu_map = {
        'db.r6g.large': 8,
        'db.r6g.xlarge': 16,
        'db.r6g.2xlarge': 32,
        'db.r6g.4xlarge': 64,
        'db.r5.large': 8,
        'db.r5.xlarge': 16,
        'db.r5.2xlarge': 32,
        'db.t3.medium': 2,
        'db.t3.large': 4,
    }

    # On-demand hourly prices (us-east-1, approximate)
    instance_price_map = {
        'db.r6g.large': 0.24,
        'db.r6g.xlarge': 0.48,
        'db.r6g.2xlarge': 0.96,
        'db.r6g.4xlarge': 1.92,
        'db.r5.large': 0.29,
        'db.r5.xlarge': 0.58,
        'db.r5.2xlarge': 1.16,
        'db.t3.medium': 0.068,
        'db.t3.large': 0.145,
    }

    peak_acus = instance_acu_map.get(instance_class, 16)
    provisioned_hourly = instance_price_map.get(instance_class, 0.48)
    provisioned_monthly = provisioned_hourly * hours_per_month

    low_pct = stats.get('low_utilization_pct', 0) / 100
    high_pct = stats.get('high_utilization_pct', 0) / 100
    medium_pct = 1 - low_pct - high_pct

    # Estimate average ACUs based on utilization distribution

```

```

avg_acus = (
    low_pct * 0.5 +          # 0.5 ACU when idle
    medium_pct * (peak_acus * 0.4) +
    high_pct * peak_acus
)

serverless_monthly = avg_acus * 0.12 * hours_per_month
monthly_delta = provisioned_monthly - serverless_monthly

return {
    'provisioned_monthly_usd': round(provisioned_monthly, 2),
    'serverless_est_monthly_usd': round(serverless_monthly, 2),
    'estimated_monthly_delta_usd': round(monthly_delta, 2),
    'recommendation': 'MIGRATE' if monthly_delta > 50 else 'STAY_PROVISIONED'
}

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument('--region', default='us-east-1')
    parser.add_argument('--days', type=int, default=30)
    args = parser.parse_args()

    rds = boto3.client('rds', region_name=args.region)
    cw = boto3.client('cloudwatch', region_name=args.region)
    clusters = get_aurora_clusters(args.region)

    print(f"\nAurora Serverless v2 Migration Classifier")
    print(f"Region: {args.region} | Analysis Period: {args.days} days")
    print("=" * 80)

    for cluster in clusters:
        cluster_id = cluster['DBClusterIdentifier']
        engine = cluster['Engine']
        members = cluster.get('DBClusterMembers', [])
        writer = next((m for m in members if m['IsClusterWriter']), None)
        if not writer:
            continue

        instance_id = writer['DBInstanceIdentifier']
        instance_info = rds.describe_db_instances(
            DBInstanceIdentifier=instance_id
        )['DBInstances'][0]
        instance_class = instance_info['DBInstanceClass']

        datapoints = get_cpu_stats(cw, instance_id, args.days)
        classification, stats = classify_workload(datapoints)
        savings = estimate_serverless_savings(instance_class, stats)

        print(f"\nCluster:           {cluster_id}")
        print(f"Engine:           {engine}")
        print(f"Instance Class:   {instance_class}")
        print(f"Classification:   {classification}")
        print(f"Avg CPU:          {stats.get('avg_cpu', 'N/A')}%")
        print(f"Low Util Hrs:     {stats.get('low_utilization_pct', 'N/A')}% of sampled period")
        print(f"Provisioned Cost: ${savings['provisioned_monthly_usd']}/month")
        print(f"Serverless Est:   ${savings['serverless_est_monthly_usd']}/month")
        print(f"Est. Delta:       ${savings['estimated_monthly_delta_usd']}/month")
        print(f"Recommendation:   {savings['recommendation']}")

    if __name__ == '__main__':
        main()

```

Run this script monthly as a scheduled Lambda function and feed its output into a Slack channel or email report. Teams that run this analysis regularly catch over-provisioned clusters within days rather than discovering them during quarterly budget reviews.

9.3 The ACU Floor Problem and How to Fix It

The single most expensive mistake with Aurora Serverless v2 is setting a high `min_capacity`. This is so common it deserves its own section.

When you create an Aurora Serverless v2 cluster through the console, the default minimum is 0.5 ACUs for PostgreSQL-compatible and MySQL-compatible engines. But when infrastructure teams migrate from provisioned clusters using Terraform or CloudFormation templates copied from provisioned configurations, they frequently set `min_capacity` to match the old instance class. A `db.r6g.xlarge` migrated to Aurora Serverless v2 with `min_capacity = 16` costs **\$1,382/month** just in baseline floor costs — nearly 4× the cost of the provisioned `r6g.xlarge` instance it replaced (\$345.60/month) — with zero idle savings and a higher bill than staying provisioned.

The fix is straightforward once you know what to look for:

```
#!/bin/bash
# audit_serverless_floors.sh
# Audits all Aurora Serverless v2 clusters in an account for high min_capacity settings
# Run with: bash audit_serverless_floors.sh us-east-1

REGION=${1:-us-east-1}
ACU_PRICE=0.12
HOURS_PER_MONTH=730

echo "Aurora Serverless v2 ACU Floor Audit"
echo "Region: $REGION"
echo "ACU Price: \${ACU_PRICE}/ACU-hour"
echo "===== "

aws rds describe-db-clusters \
  --region "$REGION" \
  --query \
    ↪ "DBClusters[?EngineMode=='provisioned']. [DBClusterIdentifier,ServerlessV2ScalingConfiguration.MinCapacity,ServerlessV2ScalingConfiguration.MaxCapacity]" \
  --output json | \
python3 -c "
import json, sys

data = json.load(sys.stdin)
acu_price = float('${ACU_PRICE}')
hours = int('${HOURS_PER_MONTH}')

total_wasted = 0.0

for row in data:
    cluster_id, min_acu, max_acu = row
    if min_acu is None:
        continue
    min_acu = float(min_acu)
    max_acu = float(max_acu)

    # Cost of floor vs optimal 0.5 ACU floor
    floor_cost_monthly = min_acu * acu_price * hours
    optimal_floor_cost = 0.5 * acu_price * hours
    wasted_monthly = floor_cost_monthly - optimal_floor_cost

    flag = ' *** REVIEW' if min_acu > 4 else ''
    print(f'Cluster:      {cluster_id}')
    print(f'Min ACU:      {min_acu} Max ACU: {max_acu}')
    print(f'Floor Cost:    \${floor_cost_monthly:.2f}/month')
    print(f'Optimal Floor: \${optimal_floor_cost:.2f}/month')
    print(f'Potential Save: \${wasted_monthly:.2f}/month{flag}')
    print('-----')
    total_wasted += wasted_monthly
```

```
print(f'TOTAL POTENTIAL FLOOR SAVINGS: \${total_wasted:.2f}/month')
print(f'ANNUAL POTENTIAL SAVINGS:      \${total_wasted * 12:.2f}/year')
"
```

Reducing `min_capacity` does carry a practical constraint: Aurora Serverless v2 takes 15–40 seconds to scale from 0.5 ACUs to 8+ ACUs from a cold state. For applications where the first request after an idle period must complete within 2–3 seconds, 0.5 ACU minimum is too aggressive. The practical recommendation:

Workload Type	Recommended min_capacity	Reasoning
Production OLTP, latency-sensitive	2–4 ACU	Fast enough scale response, meaningful idle savings
Production OLAP, reporting	1–2 ACU	Queries can tolerate 10–20s warm-up
Staging / pre-production	0.5 ACU	Latency tolerance acceptable
Development / feature branches	0.5 ACU	Maximum savings, developers expect some latency
Multi-tenant idle databases	0.5 ACU	Primary serverless use case

Lowering `min_capacity` on a production cluster from 16 to 2 ACUs on a system that idles 10 hours per day: $(16 - 2) \text{ ACU} \times \$0.12 \times (10\text{h} \times 30\text{days}) = \$504/\text{month}$ saved. That single configuration change, applied to a cluster that was correctly sized at peak but over-floored at idle, saves \$6,048/year.

9.4 Monitoring ACU Consumption in Real Time

Knowing your ACU trajectory before the monthly bill arrives requires connecting three data sources: CloudWatch ACU metrics, Performance Insights query data, and your application traffic patterns. The following CloudWatch Logs Insights query and accompanying Python scheduler build a real-time ACU cost tracker.

```
-- Aurora PostgreSQL: correlate query load with ACU consumption signals
-- This query identifies the top consumers driving scale-up events
-- Run in Performance Insights or directly in pg_stat_statements

WITH query_costs AS (
  SELECT
    queryid,
    query,
    calls,
    total_exec_time,
    mean_exec_time,
    stddev_exec_time,
    rows,
    shared_blks_hit,
    shared_blks_read,
    shared_blks_dirtied,
    -- Estimate relative CPU pressure
    (total_exec_time * calls) /
      NULLIF(SUM(total_exec_time * calls) OVER (), 0) * 100 AS cpu_pressure_pct,
    -- Flag queries with high variance (unpredictable scaling triggers)
    CASE
      WHEN stddev_exec_time > mean_exec_time * 2 THEN 'HIGH_VARIANCE'
      WHEN mean_exec_time > 5000 THEN 'SLOW'
    
```



```

        WHEN calls > 10000 THEN 'HIGH_FREQUENCY'
        ELSE 'NORMAL'
    END AS query_profile
FROM pg_stat_statements
WHERE total_exec_time > 100 -- Filter out trivial queries
),
connection_pressure AS (
    SELECT
        state,
        wait_event_type,
        wait_event,
        count(*) AS cnt
    FROM pg_stat_activity
    WHERE backend_type = 'client backend'
    GROUP BY state, wait_event_type, wait_event
)
SELECT
    qc.queryid,
    qc.query_profile,
    qc.calls,
    round(qc.mean_exec_time::numeric, 2) AS mean_ms,
    round(qc.total_exec_time::numeric / 1000, 2) AS total_sec,
    round(qc.cpu_pressure_pct::numeric, 2) AS cpu_pressure_pct,
    qc.shared_blks_read AS disk_reads,
    -- Queries with high disk reads cause both CPU and I/O ACU pressure
    CASE
        WHEN qc.shared_blks_read > 10000 AND qc.cpu_pressure_pct > 5
        THEN 'DUAL_PRESSURE_SCALE_TRIGGER'
        WHEN qc.cpu_pressure_pct > 10
        THEN 'CPU_SCALE_TRIGGER'
        WHEN qc.shared_blks_read > 50000
        THEN 'IO_SCALE_TRIGGER'
        ELSE 'LOW_IMPACT'
    END AS acu_impact,
    left(qc.query, 120) AS query_preview
FROM query_costs qc
ORDER BY qc.cpu_pressure_pct DESC
LIMIT 25;

```

Now connect CloudWatch ACU metrics to build a daily cost ledger:

```

#!/usr/bin/env python3
"""
acu_cost_tracker.py
Fetches ServerlessDatabaseCapacity metric from CloudWatch and computes
running ACU-hour costs for Aurora Serverless v2 clusters.

Schedule this as a Lambda function on EventBridge (cron: daily at 08:00 UTC).
Output is written to DynamoDB for trend analysis.
"""

import boto3
import json
from datetime import datetime, timedelta
from decimal import Decimal

ACU_PRICE_PER_HOUR = Decimal('0.12')
REGION = 'us-east-1'

def get_serverless_clusters(region):
    rds = boto3.client('rds', region_name=region)
    clusters = []
    paginator = rds.get_paginator('describe_db_clusters')
    for page in paginator.paginate():
        for c in page['DBClusters']:
            # Aurora Serverless v2 reports as EngineMode='provisioned'
            # but has ServerlessV2ScalingConfiguration set
            if c.get('ServerlessV2ScalingConfiguration'):
                clusters.append({
                    'cluster_id': c['DBClusterIdentifier'],

```

```

        'engine': c['Engine'],
        'min_acu': c['ServerlessV2ScalingConfiguration']['MinCapacity'],
        'max_acu': c['ServerlessV2ScalingConfiguration']['MaxCapacity'],
    })
    return clusters

def fetch_acu_metrics(cw, cluster_id, start, end):
    """
    ServerlessDatabaseCapacity is reported per cluster (not per instance)
    and reflects the actual ACU allocation at each point in time.
    Period=60 gives 1-minute granularity; use 300 for cost estimates.
    """
    response = cw.get_metric_statistics(
        Namespace='AWS/RDS',
        MetricName='ServerlessDatabaseCapacity',
        Dimensions=[{'Name': 'DBClusterIdentifier', 'Value': cluster_id}],
        StartTime=start,
        EndTime=end,
        Period=300, # 5-minute intervals
        Statistics=['Average']
    )
    return response.get('Datapoints', [])

def compute_acu_hours(datapoints):
    """
    Each datapoint represents 5 minutes (300 seconds) of ACU usage.
    ACU-hours = sum(avg_acu * (period_seconds / 3600))
    """
    total_acu_hours = Decimal('0')
    for dp in datapoints:
        avg_acu = Decimal(str(dp['Average']))
        period_hours = Decimal('300') / Decimal('3600') # 5 min in hours
        total_acu_hours += avg_acu * period_hours
    return total_acu_hours

def write_to_dynamodb(table_name, record):
    dynamodb = boto3.resource('dynamodb', region_name=REGION)
    table = dynamodb.Table(table_name)
    table.put_item(Item=record)

def lambda_handler(event, context):
    cw = boto3.client('cloudwatch', region_name=REGION)
    end = datetime.utcnow().replace(hour=0, minute=0, second=0, microsecond=0)
    start = end - timedelta(days=1)

    clusters = get_serverless_clusters(REGION)
    results = []

    for cluster in clusters:
        datapoints = fetch_acu_metrics(cw, cluster['cluster_id'], start, end)
        acu_hours = compute_acu_hours(datapoints)
        daily_cost = acu_hours * ACU_PRICE_PER_HOUR

        # Compare against theoretical provisioned cost at max_acu sizing
        # This shows what you would have paid with provisioned at equivalent peak
        peak_acu = Decimal(str(cluster['max_acu']))
        provisioned_equivalent_cost = peak_acu * ACU_PRICE_PER_HOUR * Decimal('24')
        daily_savings = provisioned_equivalent_cost - daily_cost

        record = {
            'cluster_id': cluster['cluster_id'],
            'date': start.strftime('%Y-%m-%d'),
            'engine': cluster['engine'],
            'min_acu': Decimal(str(cluster['min_acu'])),
            'max_acu': peak_acu,
            'actual_acu_hours': acu_hours.quantize(Decimal('0.01')),
            'daily_cost_usd': daily_cost.quantize(Decimal('0.01')),
            'provisioned_equiv_cost_usd': provisioned_equivalent_cost.quantize(Decimal('0.01')),
            'daily_savings_usd': daily_savings.quantize(Decimal('0.01')),
            'datapoint_count': len(datapoints),
        }

```

```
}
results.append(record)
write_to_dynamodb('aurora_serverless_cost_log', record)

print(f"Cluster: {cluster['cluster_id']}")
print(f"  ACU-Hours: {acu_hours:.2f}")
print(f"  Daily Cost: ${daily_cost:.2f}")
print(f"  Estimated Savings vs Peak Provisioned: ${daily_savings:.2f}")

return {'processed_clusters': len(results), 'date': start.strftime('%Y-%m-%d')}
```

The DynamoDB table this writes to becomes your cost ledger. Query it weekly to see whether ACU savings are materializing as projected or whether usage patterns have drifted toward sustained high load (the signal to migrate back to provisioned).

9.5 Capacity Planning: Setting `max_capacity` Without Fear

Teams often set `max_capacity` conservatively to control costs, not understanding that capping capacity below actual demand does not save money — it causes latency spikes, connection timeouts, and application errors that cost more to remediate than the ACU savings would have been worth.

`max_capacity` is a hard ceiling. When Aurora Serverless v2 reaches `max_capacity` and the workload still demands more, query latency increases because queries queue rather than scale. Setting `max_capacity = 8` on a database that occasionally needs 12 ACUs does not save you the cost of those 12 ACU peaks — it saves nothing while adding incidents.

The correct approach is to set `max_capacity` to the highest ACU count your workload could ever legitimately need, and use CloudWatch alarms to alert when actual consumption approaches that ceiling. The alarm is your signal to review whether the load is expected, not the ceiling itself.

```
#!/usr/bin/env python3
"""
setup_acu_alarms.py
Creates CloudWatch alarms for Aurora Serverless v2 ACU ceiling approach.
Sets alarms at 75% and 90% of max_capacity to give operational visibility.
"""

import boto3
import json

def create_acu_ceiling_alarms(
    cluster_id: str,
    max_capacity: float,
    sns_topic_arn: str,
    region: str = 'us-east-1'
):
    cw = boto3.client('cloudwatch', region_name=region)

    thresholds = [
        (0.75, 'WARNING', 3), # 75% of max, 3 consecutive 5-min periods
        (0.90, 'CRITICAL', 2), # 90% of max, 2 consecutive 5-min periods
    ]

    created_alarms = []

    for pct, severity, eval_periods in thresholds:
        threshold_acu = max_capacity * pct
        alarm_name = f'aurora-serverless-acu-{severity.lower()}-{cluster_id}'

        cw.put_metric_alarm(
            AlarmName=alarm_name,
```

```

        AlarmDescription=(
            f'{severity}: {cluster_id} ACU usage at {int(pct*100)}% '
            f'of max_capacity ({max_capacity} ACU). '
            f'Current threshold: {threshold_acu} ACU.'
        ),
        MetricName='ServerlessDatabaseCapacity',
        Namespace='AWS/RDS',
        Dimensions=[
            {'Name': 'DBClusterIdentifier', 'Value': cluster_id}
        ],
        Period=300,
        EvaluationPeriods=eval_periods,
        Threshold=threshold_acu,
        ComparisonOperator='GreaterThanOrEqualToThreshold',
        Statistic='Average',
        AlarmActions=[sns_topic_arn],
        OKActions=[sns_topic_arn],
        TreatMissingData='notBreaching',
    )

    created_alarms.append(alarm_name)
    print(f"Created alarm: {alarm_name} at {threshold_acu} ACU")

return created_alarms

def create_cost_anomaly_alarm(
    cluster_id: str,
    expected_daily_acu_hours: float,
    sns_topic_arn: str,
    region: str = 'us-east-1'
):
    """
    Creates a composite alarm that fires when daily ACU-hours exceed
    1.5x the expected baseline, indicating unexpected load increase.
    Uses a math expression alarm.
    """
    cw = boto3.client('cloudwatch', region_name=region)

    # Expected average ACUs derived from expected daily ACU-hours
    expected_avg_acu = expected_daily_acu_hours / 24.0
    anomaly_threshold = expected_avg_acu * 1.5

    alarm_name = f'aurora-serverless-cost-anomaly-{cluster_id}'

    cw.put_metric_alarm(
        AlarmName=alarm_name,
        AlarmDescription=(
            f'ACU consumption for {cluster_id} exceeds 1.5x expected baseline '
            f'({expected_avg_acu:.1f} ACU avg). '
            f'Check for unexpected load or missing indexes.'
        ),
        Metrics=[
            {
                'Id': 'm1',
                'MetricStat': {
                    'Metric': {
                        'Namespace': 'AWS/RDS',
                        'MetricName': 'ServerlessDatabaseCapacity',
                        'Dimensions': [
                            {'Name': 'DBClusterIdentifier', 'Value': cluster_id}
                        ]
                    },
                    'Period': 3600,
                    'Stat': 'Average',
                },
                'ReturnData': True,
            }
        ],
        EvaluationPeriods=3,

```

```
    Threshold=anomaly_threshold,
    ComparisonOperator='GreaterThanOrEqualTo',
    AlarmActions=[sns_topic_arn],
    TreatMissingData='notBreaching',
)

print(f"Created cost anomaly alarm: {alarm_name}")
print(f"  Triggers when avg ACU > {anomaly_threshold:.1f} for 3 consecutive hours")
return alarm_name

if __name__ == '__main__':
    # Example: production cluster, 32 ACU max, alert at 24 and 28.8 ACU
    CLUSTER_ID = 'prod-app-db-cluster'
    MAX_CAPACITY = 32
    SNS_ARN = 'arn:aws:sns:us-east-1:123456789012:database-alerts'

    create_acu_ceiling_alarms(CLUSTER_ID, MAX_CAPACITY, SNS_ARN)
    # Expected: 8 ACU-hours/day baseline (averages ~0.33 ACU over 24h with idle periods)
    create_cost_anomaly_alarm(CLUSTER_ID, expected_daily_acu_hours=8.0, sns_topic_arn=SNS_ARN)
```

9.6 The Connection Pooling Multiplier Effect

A hidden cost amplifier in Aurora Serverless v2 is connection management. Aurora Serverless v2 scales based on both CPU and memory demand. Each database connection consumes memory. A cluster with 500 idle connections sitting open — a common pattern when applications do not use connection pooling — holds the database at a higher ACU level than necessary to serve those connections’ memory footprint.

An idle PostgreSQL connection consumes approximately 5–10 MB of RAM. At 0.5 ACU (approximately 1 GB RAM available for connections and working memory), Aurora can sustain roughly 100–150 idle connections before memory pressure begins influencing ACU. At 500 idle connections, the memory requirement alone forces the cluster to 2–3 ACUs, costing $2.5 \times \$0.12 \times 720 = \$216/\text{month}$ in what should be near-zero idle time.

RDS Proxy addresses this by multiplexing application connections onto fewer database connections. The economics of RDS Proxy for Aurora Serverless v2:

RDS Proxy cost: \$0.015 per vCPU-hour of the underlying DB instance equivalent
For Aurora Serverless v2: \$0.015 per ACU-hour of proxy capacity (billed separately)

Without RDS Proxy: 500 idle connections → 2.5 ACU minimum → \$216/month idle cost
With RDS Proxy: 500 app connections → 30 DB connections → 0.5 ACU minimum → \$43.20/month idle cost
RDS Proxy cost: ~\$15/month for a proxy on this cluster
NET SAVINGS: \$216 - \$43.20 - \$15 = \$157.80/month from connection pooling alone

This is not a marginal optimization. For applications that were designed against provisioned RDS without connection pooling (a common inheritance when migrating legacy applications), adding RDS Proxy when moving to Aurora Serverless v2 is frequently the difference between costs increasing and costs decreasing.

The following SQL confirms whether your application is a candidate for connection pool optimization:

```
-- PostgreSQL: Connection waste diagnostic
```

```

-- Identifies the gap between active work and connection count
WITH connection_stats AS (
    SELECT
        application_name,
        client_addr,
        state,
        wait_event_type,
        wait_event,
        count(*) AS connection_count,
        -- Connections idle longer than 5 minutes are pure waste for serverless cost
        count(*) FILTER (
            WHERE state = 'idle'
            AND state_change < now() - interval '5 minutes'
        ) AS stale_idle_count,
        count(*) FILTER (WHERE state = 'active') AS active_count,
        count(*) FILTER (WHERE state = 'idle in transaction') AS idle_in_txn_count
    FROM pg_stat_activity
    WHERE backend_type = 'client backend'
    GROUP BY application_name, client_addr, state, wait_event_type, wait_event
),
summary AS (
    SELECT
        application_name,
        client_addr,
        SUM(connection_count) AS total_connections,
        SUM(stale_idle_count) AS total_stale_idle,
        SUM(active_count) AS total_active,
        SUM(idle_in_txn_count) AS total_idle_in_txn,
        -- Connection efficiency ratio: active / total
        ROUND(
            SUM(active_count)::numeric / NULLIF(SUM(connection_count), 0) * 100, 1
        ) AS active_pct,
        -- Estimated ACU impact of stale connections (rough: 7.5MB per connection,
        -- 2GB per ACU + ~267 connections per ACU)
        ROUND(SUM(stale_idle_count)::numeric / 267, 2) AS wasted_acu_estimate
    FROM connection_stats
    GROUP BY application_name, client_addr
)
SELECT
    application_name,
    client_addr::text,
    total_connections,
    total_active,
    total_stale_idle,
    total_idle_in_txn,
    active_pct,
    wasted_acu_estimate,
    -- Monthly cost of wasted connections
    ROUND(wasted_acu_estimate * 0.12 * 730, 2) AS monthly_wasted_usd,
    CASE
        WHEN active_pct < 5 AND total_stale_idle > 50 THEN 'CRITICAL: Add connection pooling'
        WHEN active_pct < 20 AND total_stale_idle > 20 THEN 'WARNING: Connection pool undersized or
        ↪ absent'
        WHEN total_idle_in_txn > 10 THEN 'WARNING: Transactions left open'
        ELSE 'OK'
    END AS recommendation
FROM summary
WHERE total_connections > 5
ORDER BY wasted_acu_estimate DESC;

-- SQL Server equivalent: Aurora does not support SQL Server, but if you are running
-- Azure SQL Serverless (the Microsoft equivalent), connection waste shows here:
-- Run against Azure SQL Database (Serverless tier)

SELECT
    s.login_name,
    s.host_name,
    s.program_name,
    s.status,
    COUNT(*) AS session_count,

```

```
SUM(CASE WHEN s.status = 'sleeping'
      AND DATEDIFF(MINUTE, s.last_request_end_time, GETUTCDATE()) > 5
      THEN 1 ELSE 0 END) AS stale_sleeping_sessions,
SUM(CASE WHEN s.status = 'running' THEN 1 ELSE 0 END) AS active_sessions,
-- Azure SQL Serverless pauses when all connections drop; stale connections
-- prevent auto-pause and run up vCore-seconds billing
AVG(DATEDIFF(SECOND, s.last_request_end_time, GETUTCDATE())) AS avg_idle_seconds
FROM sys.dm_exec_sessions s
WHERE s.is_user_process = 1
GROUP BY s.login_name, s.host_name, s.program_name, s.status
ORDER BY stale_sleeping_sessions DESC;
```

9.7 When to Switch Back to Provisioned

Aurora Serverless v2's greatest financial risk is not the initial deployment decision — it is the failure to detect when workload patterns have matured past the point where serverless remains economical. A startup that correctly deployed Aurora Serverless v2 eighteen months ago may now have a workload that runs at 16+ ACUs for 20 hours per day. At that point, a provisioned db.r6g.2xlarge saves money and performs more predictably.

The decision to migrate back should be systematic, not reactive. The following script implements a weekly decision engine that computes break-even points and emits a recommendation:

```
#!/usr/bin/env python3
"""
serverless_vs_provisioned_advisor.py
Weekly advisor that compares actual Aurora Serverless v2 costs against
the cheapest equivalent provisioned instance option.

Queries the DynamoDB cost log populated by acu_cost_tracker.py (section 9.4).
"""

import boto3
from datetime import datetime, timedelta
from decimal import Decimal

# Provisioned instance options (us-east-1 on-demand, Aurora-compatible)
# Format: {instance_class: (peak_acus_equivalent, hourly_price)}
PROVISIONED_OPTIONS = {
    'db.r6g.large': (8, Decimal('0.24')),
    'db.r6g.xlarge': (16, Decimal('0.48')),
    'db.r6g.2xlarge': (32, Decimal('0.96')),
    'db.r6g.4xlarge': (64, Decimal('1.92')),
    'db.r6g.8xlarge': (128, Decimal('3.84')),
}

ACU_PRICE = Decimal('0.12')
HOURS_PER_MONTH = Decimal('730')

def get_recent_costs(table_name, cluster_id, days=30):
    dynamodb = boto3.resource('dynamodb')
    table = dynamodb.Table(table_name)

    cutoff = (datetime.utcnow() - timedelta(days=days)).strftime('%Y-%m-%d')

    response = table.query(
        KeyConditionExpression='cluster_id = :cid AND #dt >= :cutoff',
        ExpressionAttributeNames={'#dt': 'date'},
        ExpressionAttributeValues={
            ':cid': cluster_id,
            ':cutoff': cutoff,
        },
    )
```

```

    return response.get('Items', [])

def compute_average_daily_acu_hours(records):
    if not records:
        return Decimal('0'), Decimal('0')
    total = sum(r.get('actual_acu_hours', Decimal('0')) for r in records)
    peak = max(r.get('max_acu', Decimal('0')) for r in records)
    avg = total / len(records)
    return avg, peak

def find_cheapest_provisioned(peak_acu_needed):
    best = None
    for instance_class, (acu_equiv, hourly) in PROVISIONED_OPTIONS.items():
        if acu_equiv >= float(peak_acu_needed):
            monthly = hourly * HOURS_PER_MONTH
            if best is None or monthly < best[2]:
                best = (instance_class, acu_equiv, monthly, hourly)
    return best

def generate_recommendation(cluster_id, table_name='aurora_serverless_cost_log'):
    records = get_recent_costs(table_name, cluster_id)
    if len(records) < 7:
        return {'recommendation': 'INSUFFICIENT_DATA', 'reason': 'Need at least 7 days of data'}

    avg_daily_acu_hours, peak_acu = compute_average_daily_acu_hours(records)

    # Project monthly serverless cost
    avg_monthly_acu_hours = avg_daily_acu_hours * 30
    serverless_monthly = avg_monthly_acu_hours * ACU_PRICE

    # Find cheapest provisioned alternative that covers peak demand
    provisioned = find_cheapest_provisioned(peak_acu)
    if not provisioned:
        return {'recommendation': 'SERVERLESS_ONLY', 'reason': 'Exceeds largest mapped instance'}

    prov_class, prov_acu, prov_monthly, prov_hourly = provisioned

    delta = serverless_monthly - prov_monthly
    savings_pct = float(delta / prov_monthly * 100) if prov_monthly else 0

    if delta > Decimal('100'):
        rec = 'MIGRATE_TO_PROVISIONED'
        reason = (
            f'Serverless is ${delta:.2f}/month MORE expensive than '
            f'{prov_class} (${prov_monthly:.2f}/month). '
            f'Sustained load at {float(avg_daily_acu_hours):.1f} ACU-hours/day '
            f'exceeds serverless break-even.'
        )
    elif delta < Decimal('-100'):
        rec = 'STAY_SERVERLESS'
        reason = (
            f'Serverless saves ${abs(delta):.2f}/month vs {prov_class}. '
            f'Workload has sufficient idle time to justify serverless.'
        )
    else:
        rec = 'BORDERLINE_MONITOR'
        reason = (
            f'Difference within ${abs(delta):.2f}/month. '
            f'Monitor for 2 more weeks before deciding.'
        )

    return {
        'cluster_id': cluster_id,
        'analysis_period_days': len(records),
        'avg_daily_acu_hours': float(avg_daily_acu_hours),
        'peak_acu_observed': float(peak_acu),
        'serverless_monthly_usd': float(serverless_monthly),
        'cheapest_provisioned_class': prov_class,
        'provisioned_monthly_usd': float(prov_monthly),
        'monthly_delta_usd': float(delta),
    }

```



```
        'recommendation': rec,  
        'reason': reason,  
    }  
  
    if __name__ == '__main__':  
        import json  
        result = generate_recommendation('prod-app-db-cluster')  
        print(json.dumps(result, indent=2))
```

Schedule this as a weekly Lambda (Monday 09:00 UTC). When `MIGRATE_TO_PROVISIONED` appears for three consecutive weeks — meaning the workload has consistently exceeded the break-even point — initiate the migration. The migration from Aurora Serverless v2 back to provisioned is not complex: modify the cluster to change its instance type, which Aurora performs with a blue/green failover of approximately 30 seconds.

9.8 Terraform Configuration for Cost-Optimized Aurora Serverless v2

Infrastructure as code is where ACU floor mistakes become permanent until someone audits them. The following Terraform module enforces cost-optimized defaults while remaining configurable for workload type:

```
# aurora_serverless_v2/main.tf  
# Cost-optimized Aurora Serverless v2 cluster module  
# Enforces minimum min_capacity of 0.5 ACU for non-production  
# and documents cost implications of higher minimums  
  
variable "cluster_name" {  
    type        = string  
    description = "Aurora cluster identifier"  
}  
  
variable "environment" {  
    type        = string  
    description = "Environment: production, staging, development"  
    validation {  
        condition     = contains(["production", "staging", "development"], var.environment)  
        error_message = "Environment must be production, staging, or development."  
    }  
}  
  
variable "min_acu" {  
    type        = number  
    description = <<-EOT  
    Minimum ACU capacity. Cost impact per ACU above 0.5:  
    Each additional 1 ACU at minimum = $63.07/month in floor costs.  
    Recommended: 0.5 (dev/staging), 2.0 (prod OLTP), 1.0 (prod reporting).  
    EOT  
    default     = null # Computed from environment if not set  
}  
  
variable "max_acu" {  
    type        = number  
    description = <<-EOT  
    Maximum ACU capacity. Set this to the HIGHEST load your application  
    could legitimately generate, not a cost-control ceiling.  
    Under-setting max_acu causes latency spikes, not cost savings.  
    EOT  
}  
  
variable "engine" {
```

```
    type    = string
    default = "aurora-postgresql"
}

variable "engine_version" {
    type    = string
    default = "15.4"
}

variable "database_name" {
    type = string
}

variable "master_username" {
    type      = string
    sensitive = true
}

variable "master_password" {
    type      = string
    sensitive = true
}

variable "subnet_ids" {
    type = list(string)
}

variable "vpc_security_group_ids" {
    type = list(string)
}

variable "sns_alert_topic_arn" {
    type      = string
    description = "SNS topic for ACU ceiling alerts"
}

locals {
    # Default min_acu by environment if not explicitly provided
    default_min_acu = {
        production = 2.0
        staging     = 0.5
        development = 0.5
    }

    effective_min_acu = var.min_acu != null ? var.min_acu : local.default_min_acu[var.environment]

    # Cost annotation for plan output
    monthly_floor_cost = local.effective_min_acu * 0.12 * 730
    monthly_peak_cost  = var.max_acu * 0.12 * 730
}

resource "aws_rds_cluster" "serverless_v2" {
    cluster_identifier      = var.cluster_name
    engine                  = var.engine
    engine_version          = var.engine_version
    database_name           = var.database_name
    master_username         = var.master_username
    master_password         = var.master_password
    db_subnet_group_name    = aws_db_subnet_group.this.name
    vpc_security_group_ids = var.vpc_security_group_ids
    skip_final_snapshot     = var.environment != "production"
    deletion_protection     = var.environment == "production"
    backup_retention_period = var.environment == "production" ? 7 : 1
    storage_encrypted       = true

    serverless_v2_scaling_configuration {
        min_capacity = local.effective_min_acu
        max_capacity = var.max_acu
    }
}
```

```

tags = {
    Environment      = var.environment
    MinACU           = toString(local.effective_min_acu)
    MaxACU           = toString(var.max_acu)
    MonthlyFloorCostUSD = format("%.2f", local.monthly_floor_cost)
    MonthlyPeakCostUSD = format("%.2f", local.monthly_peak_cost)
    CostCenter       = "database-serverless"
}
}

resource "aws_rds_cluster_instance" "serverless_v2_instance" {
    count          = 1 # Start with 1; add reader for HA in production
    identifier     = "${var.cluster_name}-instance-${count.index + 1}"
    cluster_identifier = aws_rds_cluster.serverless_v2.id
    instance_class = "db.serverless"
    engine         = aws_rds_cluster.serverless_v2.engine
    engine_version = aws_rds_cluster.serverless_v2.engine_version
    publicly_accessible = false
    db_subnet_group_name = aws_db_subnet_group.this.name

    performance_insights_enabled = var.environment == "production"
}

resource "aws_db_subnet_group" "this" {
    name      = "${var.cluster_name}-subnet-group"
    subnet_ids = var.subnet_ids
}

# ACU ceiling alarm at 80% of max
resource "aws_cloudwatch_metric_alarm" "acu_ceiling_warning" {
    alarm_name      = "${var.cluster_name}-acu-ceiling-warning"
    alarm_description = "ACU consumption at 80% of max_capacity (${var.max_acu} ACU)"
    metric_name     = "ServerlessDatabaseCapacity"
    namespace       = "AWS/RDS"
    statistic       = "Average"
    period          = 300
    evaluation_periods = 3
    threshold       = var.max_acu * 0.80
    comparison_operator = "GreaterThanOrEqualToThreshold"
    treat_missing_data = "notBreaching"

    dimensions = {
        DBClusterIdentifier = aws_rds_cluster.serverless_v2.cluster_identifier
    }

    alarm_actions = [var.sns_alert_topic_arn]
    ok_actions    = [var.sns_alert_topic_arn]
}

# Cost anomaly alarm: fires when usage exceeds 2x expected idle floor for 1 hour
resource "aws_cloudwatch_metric_alarm" "acu_cost_anomaly" {
    alarm_name      = "${var.cluster_name}-acu-cost-anomaly"
    alarm_description = "ACU consumption unexpectedly high - possible missing index or runaway query"
    metric_name     = "ServerlessDatabaseCapacity"
    namespace       = "AWS/RDS"
    statistic       = "Average"
    period          = 3600
    evaluation_periods = 2
    # Trigger when sustained ACU is more than 4x the minimum floor
    threshold       = local.effective_min_acu * 4
    comparison_operator = "GreaterThanThreshold"
    treat_missing_data = "notBreaching"

    dimensions = {
        DBClusterIdentifier = aws_rds_cluster.serverless_v2.cluster_identifier
    }

    alarm_actions = [var.sns_alert_topic_arn]
}

```

```

output "cluster_endpoint" {
  value = aws_rds_cluster.serverless_v2.endpoint
}

output "cluster_reader_endpoint" {
  value = aws_rds_cluster.serverless_v2.reader_endpoint
}

output "effective_min_acu" {
  value      = local.effective_min_acu
  description = "Actual minimum ACU configured"
}

output "monthly_floor_cost_usd" {
  value      = format("%.2f/month", local.monthly_floor_cost)
  description = "Minimum monthly cost when database is completely idle"
}

output "monthly_peak_cost_usd" {
  value      = format("%.2f/month", local.monthly_peak_cost)
  description = "Maximum monthly cost if database runs at max_capacity 24/7"
}

```

The `MonthlyFloorCostUSD` and `MonthlyPeakCostUSD` tags embedded in the cluster resource serve a specific purpose: they make the cost range visible in AWS Cost Explorer tag-based views and in any asset inventory tool that reads resource tags. Engineers reviewing the infrastructure see the cost bounds without opening a spreadsheet.

9.9 Aurora Serverless v2 vs Azure SQL Serverless: A Comparative View

Teams managing multi-cloud databases frequently ask whether their Aurora Serverless v2 experience translates to Azure SQL Serverless decisions. The two products share the “pay for what you use” positioning but differ significantly in billing mechanics, pause behavior, and cost floors.

Azure SQL Serverless (the equivalent product) bills in vCore-seconds rather than ACU-hours. Its defining feature is **auto-pause**: after a configurable idle period (minimum 1 hour), the database pauses entirely and billing drops to storage-only. Aurora Serverless v2 has no auto-pause — the minimum 0.5 ACU floor runs continuously.

Dimension	Aurora Serverless v2	Azure SQL Serverless
Billing unit	ACU-hour (\$0.12/ACU-hr in us-east-1)	vCore-second (varies by tier)
Minimum idle cost	0.5 ACU × \$0.12 = \$0.06/hr (\$43/month)	\$0.00/hr when paused (storage only)
Auto-pause	Not available	Available (min 1 hour idle)
Scale-up latency from cold	15–40 seconds	60–120 seconds from paused state
Connection during pause	Always available	First connection triggers resume
Maximum scale	256 ACU	Up to 80 vCores (General Purpose)

The practical implication: for databases that are genuinely idle for hours at a time — overnight, over weekends — Azure SQL Serverless with auto-pause can reach true \$0 compute cost during those windows. Aurora Serverless v2 cannot. For intermittent databases (feature branch environments, demo instances, infrequently accessed analytics), Azure SQL Serverless with a 1-hour auto-pause delay is more economical than Aurora Serverless v2. For databases with unpredictable but never multi-hour idle periods, Aurora Serverless v2’s always-warm state is advantageous because it avoids the 60–120 second resume latency.

```
-- Azure SQL Serverless: diagnose auto-pause eligibility
-- A database that cannot auto-pause because of persistent connections
-- is paying full vCore billing unnecessarily

-- Check for connections blocking auto-pause
SELECT
    DB_NAME() AS database_name,
    SERVERPROPERTY('EngineEdition') AS engine_edition,
    -- Azure SQL Serverless auto-pause requires zero user connections
    COUNT(*) AS total_user_sessions,
    SUM(CASE WHEN status = 'sleeping' THEN 1 ELSE 0 END) AS sleeping_sessions,
    SUM(CASE WHEN status = 'running' THEN 1 ELSE 0 END) AS running_sessions,
    -- Sessions idle > 30 minutes are preventing auto-pause for no benefit
    SUM(
        CASE
            WHEN status = 'sleeping'
            AND DATEDIFF(MINUTE, last_request_end_time, GETUTCDATE()) > 30
            THEN 1 ELSE 0 END
        ) AS long_idle_sessions,
    GETUTCDATE() AS sampled_at
FROM sys.dm_exec_sessions
WHERE is_user_process = 1;

-- Check auto-pause configuration for Azure SQL Serverless
-- (Requires connection to the master database or use of Azure Resource Manager)
SELECT
    name AS database_name,
    auto_pause_delay AS auto_pause_delay_minutes,
    -- -1 means auto-pause is disabled
    CASE
        WHEN auto_pause_delay = -1 THEN 'AUTO_PAUSE_DISABLED'
        WHEN auto_pause_delay > 120 THEN 'SUBOPTIMAL_LONG_DELAY'
        ELSE 'CONFIGURED'
    END AS pause_status
FROM sys.databases
WHERE name = DB_NAME();
```

When running Azure SQL Serverless, the equivalent of Aurora’s ACU floor audit is checking whether auto-pause is actually triggering. A database configured for auto-pause that never pauses because persistent monitoring connections, agent jobs, or always-on application pools hold it awake defeats the entire cost model.

9.10 Building a Serverless Database Cost Dashboard

Operational visibility into Aurora Serverless v2 costs should not require opening the AWS Billing console. The following CloudWatch dashboard configuration, deployable via the AWS CLI or Terraform, surfaces ACU consumption, cost trajectory, and ceiling proximity on a single screen.

```
#!/usr/bin/env python3
"""
create_serverless_dashboard.py
Creates a CloudWatch dashboard for Aurora Serverless v2 cost monitoring.
```

Widgets: ACU over time, daily cost estimate, connection count, ceiling proximity.

```

import boto3
import json

def create_dashboard(cluster_id: str, max_acu: float, region: str = 'us-east-1'):
    cw = boto3.client('cloudwatch', region_name=region)
    acu_price = 0.12

    dashboard_body = {
        "widgets": [
            {
                "type": "metric",
                "x": 0, "y": 0, "width": 12, "height": 6,
                "properties": {
                    "title": f"ACU Consumption - {cluster_id}",
                    "metrics": [
                        ["AWS/RDS", "ServerlessDatabaseCapacity",
                         "DBClusterIdentifier", cluster_id,
                         {"stat": "Average", "period": 300, "label": "Avg ACU (5m)"}],
                        ["...", {"stat": "Maximum", "period": 300, "label": "Max ACU (5m)"}],
                    ],
                    "yAxis": {
                        "left": {"min": 0, "max": max_acu * 1.1, "label": "ACUs"}
                    },
                    "annotations": {
                        "horizontal": [
                            {
                                "label": "80% ceiling",
                                "value": max_acu * 0.80,
                                "color": "#ff7f0e"
                            },
                            {
                                "label": "Max ACU",
                                "value": max_acu,
                                "color": "#d62728"
                            }
                        ]
                    },
                    "view": "timeSeries",
                    "period": 300,
                    "stat": "Average",
                }
            },
            {
                "type": "metric",
                "x": 12, "y": 0, "width": 12, "height": 6,
                "properties": {
                    "title": f"Estimated Hourly Cost (USD) - {cluster_id}",
                    "metrics": [
                        [{
                            "expression": f"m1 * {acu_price}",
                            "label": "Est. Cost $/hr",
                            "id": "cost",
                            "color": "#2ca02c"
                        }],
                        ["AWS/RDS", "ServerlessDatabaseCapacity",
                         "DBClusterIdentifier", cluster_id,
                         {"id": "m1", "visible": False, "stat": "Average", "period": 3600}]
                    ],
                    "yAxis": {"left": {"min": 0, "label": "USD/hour"}},
                    "view": "timeSeries",
                    "period": 3600,
                }
            },
            {
                "type": "metric",
                "x": 0, "y": 6, "width": 12, "height": 6,
                "properties": {

```

```

        "title": f"Database Connections - {cluster_id}",
        "metrics": [
            ["AWS/RDS", "DatabaseConnections",
             "DBClusterIdentifier", cluster_id,
             {"stat": "Average", "period": 300, "label": "Avg Connections"}],
            [...], [{"stat": "Maximum", "period": 300, "label": "Max Connections"}],
        ],
        "view": "timeSeries",
        "period": 300,
        "annotations": {
            "horizontal": [
                {
                    "label": "Connection pooling recommended threshold",
                    "value": 100,
                    "color": "#ff7f0e"
                }
            ]
        }
    },
    {
        "type": "metric",
        "x": 12, "y": 6, "width": 12, "height": 6,
        "properties": {
            "title": f"ACU Ceiling Proximity % - {cluster_id}",
            "metrics": [
                [{
                    "expression": f"(m1 / {max_acu}) * 100",
                    "label": "% of max_acu used",
                    "id": "pct",
                    "color": "#9467bd"
                }],
                ["AWS/RDS", "ServerlessDatabaseCapacity",
                 "DBClusterIdentifier", cluster_id,
                 {"id": "m1", "visible": False, "stat": "Average", "period": 300}]
            ],
            "yAxis": {"left": {"min": 0, "max": 100, "label": "% of max_acu"}},
            "annotations": {
                "horizontal": [
                    {"label": "80% warning", "value": 80, "color": "#ff7f0e"},
                    {"label": "90% critical", "value": 90, "color": "#d62728"}
                ]
            },
            "view": "timeSeries",
            "period": 300,
        }
    },
    {
        "type": "alarm",
        "x": 0, "y": 12, "width": 24, "height": 2,
        "properties": {
            "title": "ACU Alarms",
            "alarms": [
                f"arn:aws:cloudwatch:{region}:*:alarm:aurora-serverless-acu-warning-{cluster_id}",
                ↵ "}",
                f"arn:aws:cloudwatch:{region}:*:alarm:aurora-serverless-acu-critical-{",
                ↵ cluster_id}",
                f"arn:aws:cloudwatch:{region}:*:alarm:aurora-serverless-cost-anomaly-{",
                ↵ cluster_id}",
            ]
        }
    }
]

dashboard_name = f"Aurora-Serverless-Cost-{cluster_id}"
cw.put_dashboard(
    DashboardName=dashboard_name,
    DashboardBody=json.dumps(dashboard_body)
)

```

```
)

console_url = (
    f"https://{region}.console.aws.amazon.com/cloudwatch/home"
    f"?region={region}&dashboards:name={dashboard_name}"
)
print(f"Dashboard created: {dashboard_name}")
print(f"Console URL: {console_url}")
return dashboard_name

if __name__ == '__main__':
    create_dashboard(
        cluster_id='prod-app-db-cluster',
        max_acu=32.0,
        region='us-east-1'
    )
```

Embed this dashboard creation in your Terraform module (via `null_resource` with a local-exec provisioner, or a dedicated `aws_cloudwatch_dashboard` resource) so every Aurora Serverless v2 cluster gets its cost dashboard automatically at creation time.

Key Takeaways

- **The ACU floor is where most serverless budgets are silently wasted.** Setting `min_capacity` to match an old provisioned instance class eliminates idle savings entirely. For development and staging environments, 0.5 ACU is almost always the right minimum; for production OLTP, 2 ACU balances warm-start latency against idle cost.
- **Aurora Serverless v2 is not universally cheaper than provisioned.** It generates positive ROI for workloads with genuine low-utilization periods covering a large fraction of the day, multi-tenant architectures with many lightly-used databases, and unpredictable burst patterns. Sustained high-throughput workloads running above 60% CPU for the majority of the day are almost always cheaper on a matched provisioned instance.
- **Connection pooling is a force multiplier for serverless economics.** Idle database connections consume memory and hold ACU counts above the floor even when no queries are running. Adding RDS Proxy or an application-side connection pool (PgBouncer, HikariCP) before migrating to Aurora Serverless v2 is frequently the single change that determines whether the migration saves money.
- **`max_capacity` is a safety boundary, not a cost control lever.** Capping `max_capacity` below actual peak demand does not save ACU costs — it causes query queueing and latency degradation. Set `max_capacity` generously based on realistic peak demand, and use CloudWatch alarms at 80% and 90% thresholds to detect when actual consumption approaches it.
- **Serverless migration decisions should be revisited systematically, not just at deployment.** Workloads grow and change. A cluster that was a strong serverless candidate twelve months ago may now run at sustained load that makes a provisioned instance cheaper. The weekly advisor script in section 9.7, fed by the daily ACU cost log from section 9.4, provides the ongoing signal needed to migrate back to provisioned before excess costs accumulate across multiple billing cycles.

Chapter 10: Idle Instance Detection and Automated Shutdown

Organizations running cloud databases at scale consistently find that 20–35% of their RDS, Azure SQL, and Cloud SQL instances are functionally idle at any given moment — consuming full compute and storage costs while serving zero meaningful workload. At an average of \$200–\$800/month per db.r6g.large or equivalent instance, a 50-instance fleet can hemorrhage \$50,000–\$150,000 annually on instances that nobody noticed were abandoned. This chapter builds a complete detection-to-shutdown pipeline: first identifying idle instances using CloudWatch, Azure Monitor, and GCP metrics, then validating idleness against connection counts and query throughput, and finally automating graceful shutdown with appropriate safeguards. Teams that implement this pipeline systematically report 15–30% reductions in their monthly database bills within the first 60 days.

10.1 Defining “Idle”: Metrics That Actually Matter

The instinct to define an idle database as one with zero connections is wrong in practice. Connection poolers like pgBouncer and RDS Proxy maintain persistent connections to the database engine even when no application queries are running. A single health-check query from a monitoring agent counts as a “connection” and will mask genuine idleness if you rely on connection count alone.

A production-grade idle definition combines four signals measured over a rolling 7-day window:

DatabaseConnections: The average number of active connections. For most workloads, a database averaging fewer than 2 connections over 7 days is a strong candidate.

ReadIOPS + WriteIOPS: Combined I/O operations per second. A truly idle database shows near-zero I/O. Thresholds below 1 IOPS average over 7 days are reliable.

CPUUtilization: CPU below 1% average over 7 days, excluding maintenance windows.

NetworkReceiveThroughput: Below 10 KB/s average, which eliminates instances receiving replication traffic or periodic ETL loads.

Using any single metric produces false positives. A batch ETL job that runs Sunday nights shows zero connections all week but is legitimate. Combining all four signals, with the 7-day rolling window, filters out these jobs reliably.

The financial logic is straightforward. An `db.r6g.large` in us-east-1 costs \$0.26/hour on-demand, which is \$187/month. That instance may have 100 GB of `gp2` storage at \$0.115/GB, adding

\$11.50/month — \$198.50 total. Multiplied across 20 genuinely idle instances, you are spending \$3,970/month. Automated shutdown eliminates that entirely. Even if you implement scheduled stop/start rather than full termination, stopping RDS instances for 16 hours per day on non-production instances saves 67% of the compute cost while keeping storage costs in place, dropping that \$187 compute to roughly \$62.

10.2 Detection Queries and Metric Collection

Before automating shutdown, you need a reliable inventory of idle candidates. The following scripts build that inventory across all three clouds.

AWS: CloudWatch Metrics for Idle RDS Detection

```
# idle_detector_aws.py
# Scans all RDS instances in a region and scores them for idleness
# Run on a schedule (daily via Lambda or cron)
# Requires: boto3, IAM permissions for rds:DescribeDBInstances,
#           cloudwatch:GetMetricStatistics, ce:GetCostAndUsage

import boto3
import json
from datetime import datetime, timedelta, timezone
from dataclasses import dataclass, field
from typing import Optional

@dataclass
class InstanceIdleScore:
    instance_id: str
    instance_class: str
    engine: str
    status: str
    region: str
    avg_connections: float = 0.0
    avg_cpu: float = 0.0
    avg_read_iops: float = 0.0
    avg_write_iops: float = 0.0
    avg_network_kb: float = 0.0
    idle_score: int = 0          # 0-4, higher = more idle
    monthly_cost_usd: float = 0.0
    tags: dict = field(default_factory=dict)
    recommendation: str = ""

# On-demand pricing for common RDS instance classes (us-east-1, MySQL/PostgreSQL)
# Update this map quarterly or fetch from AWS Price List API
RDS_HOURLY_PRICING = {
    "db.t3.micro": 0.017,
    "db.t3.small": 0.034,
    "db.t3.medium": 0.068,
    "db.t3.large": 0.136,
    "db.r6g.large": 0.260,
    "db.r6g.xlarge": 0.520,
    "db.r6g.2xlarge": 1.040,
    "db.r5.large": 0.240,
    "db.r5.xlarge": 0.480,
    "db.r5.2xlarge": 0.960,
    "db.m6g.large": 0.183,
    "db.m6g.xlarge": 0.365,
}

def get_metric_average(cw_client, instance_id, metric_name, namespace,
                      days=7, period_seconds=604800):
    """
    Fetches a single metric average over the past N days.
    """
```

```

period_seconds=604800 = 1 week (single data point average).
"""
end_time = datetime.now(timezone.utc)
start_time = end_time - timedelta(days=days)

response = cw_client.get_metric_statistics(
    Namespace=namespace,
    MetricName=metric_name,
    Dimensions=[{"Name": "DBInstanceIdentifier", "Value": instance_id}],
    StartTime=start_time,
    EndTime=end_time,
    Period=period_seconds,
    Statistics=["Average"]
)

datapoints = response.get("Datapoints", [])
if not datapoints:
    return 0.0
return round(datapoints[0]["Average"], 4)

def score_instance(instance: InstanceIdleScore) -> InstanceIdleScore:
    """
    Assigns an idle score 0-4 based on four metric thresholds.
    Score 4 = idle across all dimensions.
    """
    score = 0
    if instance.avg_connections < 2.0:
        score += 1
    if instance.avg_cpu < 1.0:
        score += 1
    if (instance.avg_read_iops + instance.avg_write_iops) < 1.0:
        score += 1
    if instance.avg_network_kb < 10.0:
        score += 1

    instance.idle_score = score

    if score == 4:
        instance.recommendation = "STOP_OR_TERMINATE"
    elif score == 3:
        instance.recommendation = "INVESTIGATE_THEN_STOP"
    elif score == 2:
        instance.recommendation = "SCHEDULE_REVIEW"
    else:
        instance.recommendation = "ACTIVE_KEEP"

    return instance

def estimate_monthly_cost(instance_class: str, storage_gb: int = 0) -> float:
    hourly = RDS_HOURLY_PRICING.get(instance_class, 0.30)
    compute = hourly * 24 * 30
    storage = storage_gb * 0.115 # gp2 pricing
    return round(compute + storage, 2)

def scan_region(region: str) -> list[InstanceIdleScore]:
    rds = boto3.client("rds", region_name=region)
    cw = boto3.client("cloudwatch", region_name=region)

    paginator = rds.get_paginator("describe_db_instances")
    results = []

    for page in paginator.paginate():
        for db in page["DBInstances"]:
            if db["DBInstanceStatus"] not in ("available", "stopped"):
                continue

            tags = {t["Key"]: t["Value"]
                    for t in db.get("TagList", [])}

            # Skip instances explicitly tagged to protect from auto-shutdown

```

```

if tags.get("AutoShutdown") == "false":
    continue

storage_gb = db.get("AllocatedStorage", 0)
score_obj = InstanceIdleScore(
    instance_id=db["DBInstanceIdentifier"],
    instance_class=db["DBInstanceClass"],
    engine=db["Engine"],
    status=db["DBInstanceStatus"],
    region=region,
    tags=tags,
    monthly_cost_usd=estimate_monthly_cost(
        db["DBInstanceClass"], storage_gb
    )
)

# Already stopped instances are trivially idle
if db["DBInstanceStatus"] == "stopped":
    score_obj.idle_score = 4
    score_obj.recommendation = "ALREADY_STOPPED_VERIFY_DELETION"
    results.append(score_obj)
    continue

iid = db["DBInstanceIdentifier"]
score_obj.avg_connections = get_metric_average(
    cw, iid, "DatabaseConnections", "AWS/RDS")
score_obj.avg_cpu = get_metric_average(
    cw, iid, "CPUUtilization", "AWS/RDS")
score_obj.avg_read_iops = get_metric_average(
    cw, iid, "ReadIOPS", "AWS/RDS")
score_obj.avg_write_iops = get_metric_average(
    cw, iid, "WriteIOPS", "AWS/RDS")

net_bytes = get_metric_average(
    cw, iid, "NetworkReceiveThroughput", "AWS/RDS")
score_obj.avg_network_kb = round(net_bytes / 1024, 2)

results.append(score_instance(score_obj))

return results

def main():
    regions = ["us-east-1", "us-west-2", "eu-west-1"]
    all_results = []

    for region in regions:
        print(f"Scanning {region}...")
        all_results.extend(scan_region(region))

    # Sort by idle score descending, then cost descending
    all_results.sort(key=lambda x: (x.idle_score, x.monthly_cost_usd),
                    reverse=True)

    total_idle_cost = sum(
        i.monthly_cost_usd for i in all_results
        if i.idle_score >= 3
    )

    print(f"\n{'='*70}")
    print(f"IDLE INSTANCE REPORT - {datetime.now().strftime('%Y-%m-%d')}")
    print(f"\n{'='*70}")
    print(f"Total instances scanned : {len(all_results)}")
    print(f"Idle candidates (score 3): "
          f"{sum(1 for i in all_results if i.idle_score >= 3)}")
    print(f"Potential monthly savings: ${total_idle_cost:.2f}")
    print(f"\n{'='*70}\n")

    for inst in all_results:
        if inst.idle_score >= 3:
            print(f" {inst.instance_id}<40} ")

```

```

f"Score:{inst.idle_score}/4 "
f"${inst.monthly_cost_usd:>8.2f}/mo "
f"{inst.recommendation}")

# Write JSON for downstream automation
output = [vars(i) for i in all_results if i.idle_score >= 3]
with open("/tmp/idle_instances.json", "w") as f:
    json.dump(output, f, indent=2)
print(f"\nCandidates written to /tmp/idle_instances.json")

if __name__ == "__main__":
    main()

```

Azure SQL: Querying Metrics via Azure Monitor

```

#!/bin/bash
# azure_idle_detector.sh
# Detects idle Azure SQL databases using Azure Monitor metrics
# Requires: az login, Contributor or Monitoring Reader role

SUBSCRIPTION_ID="your-subscription-id"
LOOKBACK_DAYS=7
END_TIME=$(date -u +"%Y-%m-%dT%H:%M:%SZ")
START_TIME=$(date -u -d "-${LOOKBACK_DAYS} days" +"%Y-%m-%dT%H:%M:%SZ")

OUTPUT_FILE="/tmp/azure_idle_sql.csv"
echo "server,database,sku,avg_cpu_pct,avg_connections,dtu_limit,est_monthly_usd,recommendation" \
    > "$OUTPUT_FILE"

# DTU pricing map (General Purpose, per database/month approximations)
declare -A DTU_COST
DTU_COST["Basic"]="4.90"
DTU_COST["S0"]="14.72"
DTU_COST["S1"]="29.43"
DTU_COST["S2"]="58.87"
DTU_COST["P1"]="465.00"
DTU_COST["GP_Gen5_2"]="368.00"
DTU_COST["GP_Gen5_4"]="736.00"
DTU_COST["BC_Gen5_2"]="1105.00"

az account set --subscription "$SUBSCRIPTION_ID"

# List all SQL servers
SERVERS=$(az sql server list --query "[].{name:name, rg:resourceGroup}" -o tsv)

while IFS=$'\t' read -r SERVER_NAME RESOURCE_GROUP; do
    echo "Checking server: $SERVER_NAME"

    # List all databases on this server (exclude master)
    DATABASES=$(az sql db list \
        --server "$SERVER_NAME" \
        --resource-group "$RESOURCE_GROUP" \
        --query "[?name!='master'].{name:name, sku:sku.name, tier:sku.tier}" \
        -o tsv)

    while IFS=$'\t' read -r DB_NAME SKU_NAME SKU_TIER; do
        RESOURCE_ID="/subscriptions/${SUBSCRIPTION_ID}/resourceGroups/${RESOURCE_GROUP}"
        RESOURCE_ID+="/providers/Microsoft.Sql/servers/${SERVER_NAME}"
        RESOURCE_ID+="/databases/${DB_NAME}"

        # Fetch average CPU percentage over the lookback window
        AVG_CPU=$(az monitor metrics list \
            --resource "$RESOURCE_ID" \
            --metric "cpu_percent" \
            --start-time "$START_TIME" \
            --end-time "$END_TIME" \
            --aggregation Average \
            --interval P1D \
            --query "value[0].timeseries[0].data[*].average" \
            -o tsv 2>/dev/null \

```

```

| awk 'NF{sum+=$1; count++} END {
    if(count>0) printf "%.2f", sum/count; else print "0.00"
}'

# Fetch average connection count
AVG_CONN=$(az monitor metrics list \
    --resource "$RESOURCE_ID" \
    --metric "connection_successful" \
    --start-time "$START_TIME" \
    --end-time "$END_TIME" \
    --aggregation Total \
    --interval P1D \
    --query "value[0].timeseries[0].data[*].total" \
    -o tsv 2>/dev/null \
| awk 'NF{sum+=$1; count++} END {
    if(count>0) printf "%.0f", sum/count; else print "0"
}')

# Look up estimated cost
COST="{DTU_COST[$SKU_NAME]:-unknown}"

# Classify
RECOMMENDATION="ACTIVE_KEEP"
if (( $(echo "$AVG_CPU < 1.0" | bc -l) )) && \
    (( $(echo "$AVG_CONN < 50" | bc -l) )); then
    RECOMMENDATION="PAUSE_OR_DELETE"
elif (( $(echo "$AVG_CPU < 5.0" | bc -l) )); then
    RECOMMENDATION="DOWNSIZE_SKU"
fi

echo "${SERVER_NAME},${DB_NAME},${SKU_NAME},${AVG_CPU}," \
    "${AVG_CONN},${SKU_NAME},${COST},${RECOMMENDATION}" \
>> "$OUTPUT_FILE"

if [[ "$RECOMMENDATION" == "PAUSE_OR_DELETE" ]]; then
    echo "    IDLE: ${SERVER_NAME}/${DB_NAME} " \
        "CPU=${AVG_CPU}% Conn=${AVG_CONN}/day SKU=${SKU_NAME} ~\${COST}/mo"
fi

done <<< "$DATABASES"
done <<< "$SERVERS"

echo ""
echo "Report written to $OUTPUT_FILE"
echo "Idle databases by potential savings:"
awk -F',' 'NR>1 && $8=="PAUSE_OR_DELETE" {print $7, $1/"$2"}' "$OUTPUT_FILE" \
| sort -rn \
| awk '{printf "    %s/mo %s\n", $1, $2}'

```

GCP Cloud SQL: Detecting Idle Instances with gcloud and BigQuery

```

-- Query against the GCP Billing Export table in BigQuery
-- Replace `your_project.billing_dataset.gcp_billing_export_v1_XXXXXX`
-- This surfaces Cloud SQL instances with cost but no meaningful usage

```

```

WITH daily_costs AS (
    SELECT
        DATE(usage_start_time) AS usage_date,
        resource.name          AS instance_name,
        service.description    AS service,
        sku.description         AS sku_description,
        SUM(cost)               AS daily_cost_usd,
        SUM(usage.amount)       AS usage_amount,
        usage.unit              AS usage_unit
    FROM
        `your_project.billing_dataset.gcp_billing_export_v1_XXXXXX`
    WHERE
        service.description = 'Cloud SQL'
        AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
        AND cost > 0

```



```
GROUP BY 1, 2, 3, 4, 6, 7
),

instance_summary AS (
  SELECT
    instance_name,
    SUM(CASE WHEN sku_description LIKE '%CPU%'
      THEN daily_cost_usd ELSE 0 END) AS cpu_cost_30d,
    SUM(CASE WHEN sku_description LIKE '%Storage%'
      THEN daily_cost_usd ELSE 0 END) AS storage_cost_30d,
    SUM(daily_cost_usd) AS total_cost_30d,
    COUNT(DISTINCT usage_date) AS days_with_charges
  FROM daily_costs
  GROUP BY 1
),

-- Join to Cloud SQL instance metrics exported via Cloud Monitoring
-- (assumes you have exported metrics to BigQuery via Monitoring Export)
metric_summary AS (
  SELECT
    resource.labels.database_id AS instance_name,
    AVG(point.value.double_value) AS avg_cpu_utilization
  FROM
    `your_project.monitoring_dataset.cloudsql_database`
  WHERE
    metric.type = 'cloudsql.googleapis.com/database/cpu/utilization'
    AND DATE(point.interval.end_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 7 DAY)
  GROUP BY 1
)

SELECT
  i.instance_name,
  ROUND(i.total_cost_30d, 2) AS total_cost_30d_usd,
  ROUND(i.cpu_cost_30d, 2) AS cpu_cost_30d_usd,
  ROUND(i.storage_cost_30d, 2) AS storage_cost_30d_usd,
  i.days_with_charges,
  ROUND(m.avg_cpu_utilization * 100, 2) AS avg_cpu_pct_7d,
  CASE
    WHEN m.avg_cpu_utilization < 0.01
      AND i.cpu_cost_30d > 0
    THEN 'STOP_CANDIDATE'
    WHEN m.avg_cpu_utilization < 0.05
    THEN 'DOWNSIZE_CANDIDATE'
    ELSE 'ACTIVE'
  END AS recommendation,
  ROUND(i.cpu_cost_30d * 12, 0) AS projected_annual_cpu_waste_usd
FROM instance_summary i
LEFT JOIN metric_summary m
  ON i.instance_name = m.instance_name
WHERE i.total_cost_30d > 10 -- filter out trivial instances
ORDER BY projected_annual_cpu_waste_usd DESC;
```

10.3 Validating Candidates Before Taking Action

Automated detection surfaces candidates. Human or rule-based validation prevents the scenario where you shut down a database that runs a quarterly financial close process and hasn't run in 6 weeks because the last close was 6 weeks ago.

The validation layer adds three checks before any shutdown action is authorized.

Check 1: Recent Snapshot or Backup Activity

An instance with a backup taken in the last 24 hours, even if idle by metrics, may be part of an

active pipeline. Query RDS automated backups to verify.

```
#!/bin/bash
# validate_backup_recency.sh
# For each idle candidate, check the most recent automated snapshot.
# If a snapshot was taken within 48 hours AND no other idle signals
# exist, flag for manual review instead of automated action.

CANDIDATES_FILE="/tmp/idle_instances.json"

if [[ ! -f "$CANDIDATES_FILE" ]]; then
    echo "Run idle_detector_aws.py first to generate candidates."
    exit 1
fi

# Parse instance IDs from the JSON output of idle_detector_aws.py
INSTANCE_IDS=$(python3 -c "
import json
with open('$CANDIDATES_FILE') as f:
    data = json.load(f)
print('\n'.join(i['instance_id'] for i in data))
")

NOW_EPOCH=$(date +%s)
FORTY_EIGHT_HOURS=172800

echo "Validating backup recency for idle candidates..."
echo "-----"

while IFS= read -r INSTANCE_ID; do
    [[ -z "$INSTANCE_ID" ]] && continue

    # Get the most recent automated snapshot
    LATEST_SNAPSHOT=$(aws rds describe-db-snapshots \
        --db-instance-identifier "$INSTANCE_ID" \
        --snapshot-type automated \
        --query "sort_by(DBSnapshots, &SnapshotCreateTime)[-1].SnapshotCreateTime" \
        --output text 2>/dev/null)

    if [[ "$LATEST_SNAPSHOT" == "None" ]] || [[ -z "$LATEST_SNAPSHOT" ]]; then
        echo " $INSTANCE_ID - NO SNAPSHOT FOUND - safe to terminate"
        continue
    fi

    # Convert ISO timestamp to epoch
    SNAP_EPOCH=$(date -d "$LATEST_SNAPSHOT" +%s 2>/dev/null || \
        date -jf "%Y-%m-%dT%H:%M:%S" \
            "${LATEST_SNAPSHOT%.*}" +%s 2>/dev/null)

    AGE_SECONDS=$(( NOW_EPOCH - SNAP_EPOCH ))
    AGE_HOURS=$(( AGE_SECONDS / 3600 ))

    if (( AGE_SECONDS < FORTY_EIGHT_HOURS )); then
        echo " $INSTANCE_ID - RECENT BACKUP ($AGE_HOURS hrs ago) - flag for review"
    else
        echo " $INSTANCE_ID - STALE BACKUP ($AGE_HOURS hrs ago) - safe for automated action"
    fi
done <<< "$INSTANCE_IDS"
```

Check 2: Tag-Based Exemptions

Before acting on any candidate, re-verify tags at action time (not just at scan time). Tags can change between detection and enforcement. The following tag keys are the minimal exemption contract teams should agree on before deploying automation:

Tag Key	Value	Meaning
AutoShutdown	false	Never shut this instance down automatically
Environment	production	Exclude from any automated stop/terminate
Schedule	quarterly	Runs quarterly; extend idle window to 90 days
Owner	team-name	Notify this team before action
LastKnownUse	ISO date	Human-annotated last intentional use date

Any instance tagged `Environment=production` or `AutoShutdown=false` must be excluded at the action stage even if it scored 4/4 on idleness. The detection query already filters these, but the action script enforces the same check independently to create a defense-in-depth pattern.

Check 3: Cross-Reference with Application Deployment Records

If your organization uses a CMDB or deployment tagging convention (e.g., AWS Systems Manager Inventory, Azure Resource Graph, or GCP Asset Inventory), a cross-reference query can catch databases associated with active application deployments. The following Resource Graph query identifies Azure SQL databases where the parent logical server has an associated App Service or AKS cluster in the same resource group that was deployed or scaled in the past 30 days:

```
# Azure Resource Graph: find SQL databases linked to recently active apps
az graph query -q "
Resources
| where type == 'microsoft.sql/servers/databases'
| where name != 'master'
| project sqlDb=name, sqlServer=split(id, '/')[8],
    resourceGroup, subscriptionId
| join kind=leftouter (
    Resources
    | where type in (
        'microsoft.web/sites',
        'microsoft.containerservice/managedclusters'
    )
    | where properties.lastModifiedTimeUtc >= ago(30d)
    | project appName=name, resourceGroup, subscriptionId
) on resourceGroup, subscriptionId
| project sqlDb, sqlServer, resourceGroup,
    linkedApp=appName
| where isempty(linkedApp)
" --output table
```

SQL databases that appear here — no linked active app in the same resource group — are stronger termination candidates. Databases that have a linked app are handed off to the application team for confirmation before any action.

10.4 Automated Shutdown: Stop vs. Terminate vs. Snapshot-Then-Delete

The action vocabulary has three options, and choosing the wrong one is expensive in different ways:

Stop (RDS: stop-db-instance): Preserves the instance configuration, keeps automated backups, retains storage. Compute billing stops. Storage billing continues. AWS automatically restarts stopped instances after 7 days — a frustrating default that must be accounted for in your automation. Cost saving: roughly 60–70% of the monthly bill depending on storage fraction.

Terminate with final snapshot: Creates a final automated snapshot, then permanently deletes the instance. Compute and storage billing both stop. Snapshot storage is charged at \$0.095/GB-month (RDS) until explicitly deleted. This is the right choice for instances confirmed as truly decommissioned.

Snapshot-then-delete (manual pattern): Take a manual snapshot first, tag it with metadata, then delete the instance. Gives you a recovery path without retaining the live instance. Manual snapshots persist until explicitly deleted, unlike automated snapshots that follow retention policies.

The decision tree for which action to take:

```
idle_score == 4 AND status == "stopped" (already stopped > 7 days)
```

- Owner notified 7 days prior
- Take final snapshot tagged with instance metadata
- Terminate instance
- Set snapshot lifecycle policy: delete after 90 days

```
idle_score == 4 AND status == "available" (running but idle)
```

- Environment != production AND AutoShutdown != false
- Stop instance immediately
- Schedule re-check in 14 days
- If still idle after 14 days: escalate to terminate workflow

```
idle_score == 3 (borderline idle)
```

- Send owner notification with metrics evidence
- No automated action for 7 days
- If no response: treat as score-4 path

Implementing the Stop Action in AWS

```
# shutdown_orchestrator.py
# Reads /tmp/idle_instances.json and applies the stop/terminate decision tree.
# Designed to run as a Lambda function or from the command line.
# All actions are logged to CloudWatch Logs and a local audit trail.

import boto3
import json
import sys
import logging
from datetime import datetime, timezone
from typing import Optional

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s %(levelname)-8s %(message)s",
    datefmt="%Y-%m-%d %H:%M:%S"
)
log = logging.getLogger(__name__)

AUDIT_LOG = "/tmp/shutdown_audit.jsonl"
DRY_RUN = "--dry-run" in sys.argv # Pass --dry-run to preview actions safely

PROTECTED_ENVIRONMENTS = {"production", "prod", "prd"}
```

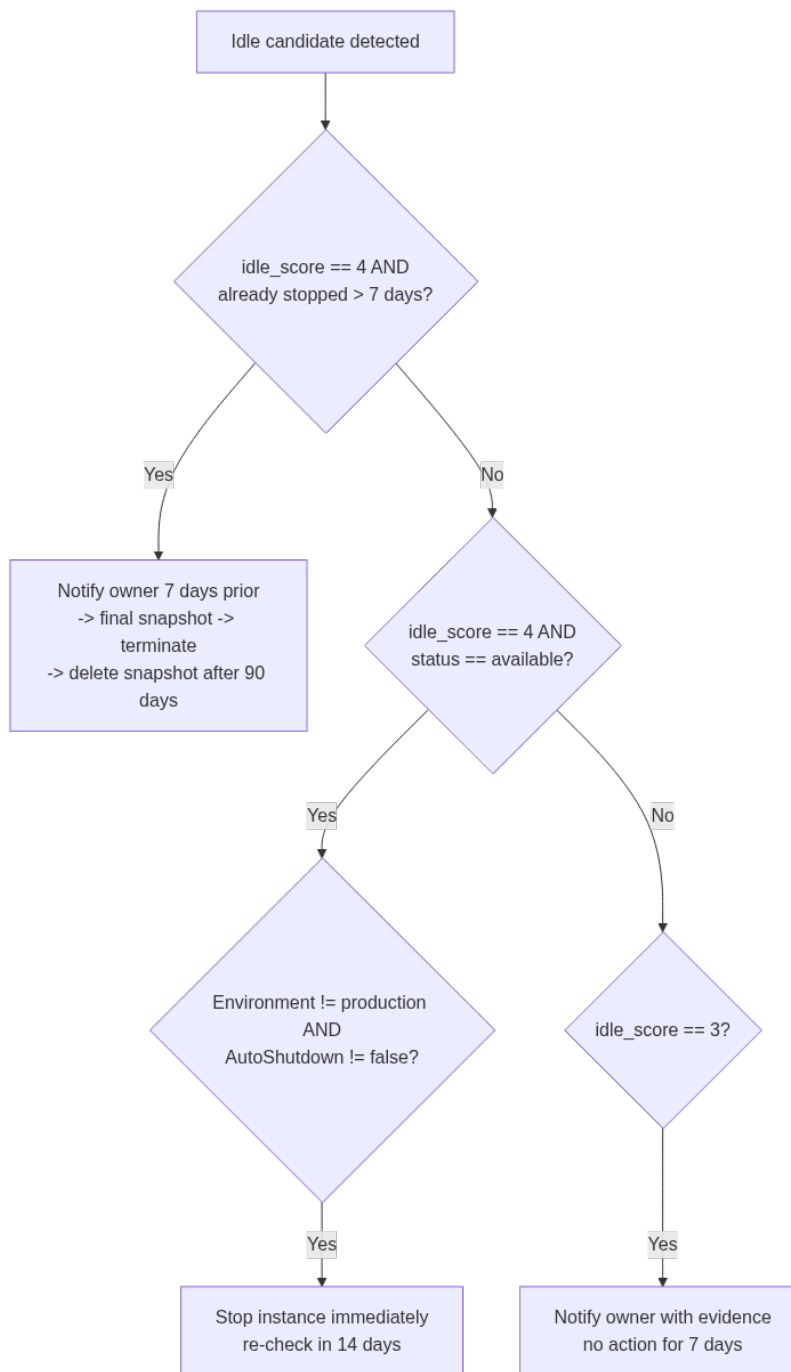


Figure 10: The idle-shutdown decision tree — confidence level (`idle_score`) determines how aggressive the automated action is.

```
SNAPSHOT_RETENTION_DAYS = 90

def load_candidates(path: str) -> list[dict]:
    with open(path) as f:
        return json.load(f)

def append_audit(record: dict):
    with open(AUDIT_LOG, "a") as f:
        f.write(json.dumps(record) + "\n")

def verify_tags_at_action_time(rds_client, instance_id: str) -> dict:
    """Re-fetch tags at action time, not from the cached scan data."""
    try:
        arn_response = rds_client.describe_db_instances(
            DBInstanceIdentifier=instance_id
        )
        db = arn_response["DBInstances"][0]
        arn = db["DBInstanceArn"]
        tag_response = rds_client.list_tags_for_resource(ResourceName=arn)
        return {t["Key"]: t["Value"] for t in tag_response["TagList"]}
    except Exception as e:
        log.warning(f"Could not fetch tags for {instance_id}: {e}")
        return {}

def is_protected(tags: dict) -> bool:
    env = tags.get("Environment", "").lower()
    if env in PROTECTED_ENVIRONMENTS:
        return True
    if tags.get("AutoShutdown", "").lower() == "false":
        return True
    return False

def take_final_snapshot(rds_client, instance_id: str,
                       tags: dict) -> Optional[str]:
    timestamp = datetime.now(timezone.utc).strftime("%Y-%m-%d-%H%M%S")
    snapshot_id = f"finops-final-{instance_id}-{timestamp}"
    # Snapshot IDs max 255 chars; truncate instance_id if needed
    snapshot_id = snapshot_id[:254]

    if DRY_RUN:
        log.info(f"[DRY-RUN] Would create snapshot: {snapshot_id}")
        return snapshot_id

    try:
        rds_client.create_db_snapshot(
            DBSnapshotIdentifier=snapshot_id,
            DBInstanceIdentifier=instance_id,
            Tags=[
                {"Key": "CreatedBy", "Value": "finops-idle-shutdown"},
                {"Key": "InstanceId", "Value": instance_id},
                {"Key": "DeleteAfter",
                 "Value": (datetime.now(timezone.utc).strftime("%Y-%m-%d"))},
                {"Key": "OriginalEnv", "Value": tags.get("Environment", "unknown")},
            ]
        )
        log.info(f"Snapshot initiated: {snapshot_id}")
        return snapshot_id
    except Exception as e:
        log.error(f"Snapshot failed for {instance_id}: {e}")
        return None

def stop_instance(rds_client, instance_id: str):
    if DRY_RUN:
        log.info(f"[DRY-RUN] Would stop: {instance_id}")
        return
    try:
        rds_client.stop_db_instance(DBInstanceIdentifier=instance_id)
        log.info(f"Stop issued: {instance_id}")
    except Exception as e:
        log.error(f"Stop failed for {instance_id}: {e}")
```

```

def terminate_instance(rds_client, instance_id: str,
                      take_snapshot: bool = True) -> bool:
    snapshot_id = None
    if take_snapshot:
        # Must wait for any in-progress snapshot to complete before delete
        # In production, this becomes a Step Functions workflow; here we
        # just request the snapshot and mark the termination as pending.
        log.info(f"Termination pending snapshot completion: {instance_id}")
        return False # Caller should re-check after snapshot completes

    if DRY_RUN:
        log.info(f"[DRY-RUN] Would terminate: {instance_id}")
        return True

    try:
        rds_client.delete_db_instance(
            DBInstanceIdentifier=instance_id,
            SkipFinalSnapshot=True,
            DeleteAutomatedBackups=False # Keep automated backups per policy
        )
        log.info(f"Terminate issued: {instance_id}")
        return True
    except Exception as e:
        log.error(f"Terminate failed for {instance_id}: {e}")
        return False

def process_candidate(candidate: dict):
    region = candidate.get("region", "us-east-1")
    instance_id = candidate["instance_id"]
    idle_score = candidate.get("idle_score", 0)
    status = candidate.get("status", "available")
    recommendation = candidate.get("recommendation", "")

    rds = boto3.client("rds", region_name=region)

    # Defense-in-depth tag check at action time
    live_tags = verify_tags_at_action_time(rds, instance_id)
    if is_protected(live_tags):
        log.info(f"SKIPPED (protected by tags): {instance_id}")
        append_audit({
            "timestamp": datetime.now(timezone.utc).isoformat(),
            "instance_id": instance_id,
            "action": "SKIPPED",
            "reason": "protected_tags",
            "tags": live_tags
        })
        return

    action_taken = "NONE"
    notes = ""

    if recommendation == "ALREADY_STOPPED_VERIFY_DELETION":
        # Already stopped; take snapshot and plan deletion
        snapshot_id = take_final_snapshot(rds, instance_id, live_tags)
        action_taken = "SNAPSHOT_FOR_DELETION"
        notes = f"snapshot={snapshot_id}; manual termination required after verification"

    elif idle_score == 4 and status == "available":
        stop_instance(rds, instance_id)
        action_taken = "STOPPED"
        notes = "Will re-check in 14 days for termination eligibility"

    elif idle_score == 3:
        action_taken = "NOTIFICATION_QUEUED"
        notes = "Owner notification queued; no automated action for 7 days"

    log.info(f"{action_taken}: {instance_id} - {notes}")
    append_audit({
        "timestamp": datetime.now(timezone.utc).isoformat(),

```

```
        "instance_id": instance_id,
        "region": region,
        "idle_score": idle_score,
        "status": status,
        "action": action_taken,
        "dry_run": DRY_RUN,
        "notes": notes,
        "monthly_cost_usd": candidate.get("monthly_cost_usd", 0),
        "tags": live_tags
    })

def main():
    candidates = load_candidates("/tmp/idle_instances.json")
    log.info(f"Processing {len(candidates)} idle candidates "
            f"({'DRY RUN' if DRY_RUN else 'LIVE'})")

    for candidate in candidates:
        process_candidate(candidate)

    log.info(f"Audit log written to {AUDIT_LOG}")

if __name__ == "__main__":
    main()
```

Run in dry-run mode first, always:

```
python3 shutdown_orchestrator.py --dry-run
```

Review the audit log at `/tmp/shutdown_audit.jsonl`, confirm the planned actions, then remove `--dry-run` for the live pass.

10.5 Implementing Scheduled Stop/Start for Non-Production Instances

For non-production instances that need to exist but only run during business hours, a scheduled stop/start pattern captures most of the savings without the risk of full termination. A database running 10 hours per day (business hours) instead of 24 incurs 42% of its original compute cost.

The two main mechanisms are AWS Instance Scheduler and native EventBridge Scheduler rules for RDS.

EventBridge Scheduler: Stop All Dev Instances at 19:00 UTC

```
# Create an EventBridge schedule to stop all RDS instances tagged Environment=dev
# at 19:00 UTC Monday-Friday

# Step 1: Create an IAM role that EventBridge Scheduler can assume
aws iam create-role \
    --role-name EventBridgeRDSSchedulerRole \
    --assume-role-policy-document '{
        "Version": "2012-10-17",
        "Statement": [{
            "Effect": "Allow",
            "Principal": {"Service": "scheduler.amazonaws.com"},
            "Action": "sts:AssumeRole"
        }]
    }'

aws iam attach-role-policy \
    --role-name EventBridgeRDSSchedulerRole \
    --policy-arn arn:aws:iam::aws:policy/AmazonRDSFullAccess
```



```

ROLE_ARN=$(aws iam get-role \
  --role-name EventBridgeRDSSchedulerRole \
  --query "Role.Arn" --output text)

# Step 2: Create a Lambda function to stop tagged instances
# (EventBridge Scheduler invokes Lambda; Lambda iterates tagged instances)
# See lambda_rds_stop_tagged.py below for the function code

FUNCTION_ARN="arn:aws:lambda:us-east-1:123456789012:function:rds-stop-dev-instances"

# Step 3: Create the stop schedule (Mon-Fri 19:00 UTC)
aws scheduler create-schedule \
  --name "rds-stop-dev-weekdays-1900" \
  --schedule-expression "cron(0 19 ? * MON-FRI *)" \
  --schedule-expression-timezone "UTC" \
  --target "{
    \"Arn\": \"${FUNCTION_ARN}\",
    \"RoleArn\": \"${ROLE_ARN}\",
    \"Input\": \"{\\\"action\\\": \\\"stop\\\", \\\"tag_key\\\": \\\"Environment\\\", \\\"tag_value\\\":
↪ \\\"dev\\\"}\"
  }" \
  --flexible-time-window '{"Mode": "OFF"}'

# Step 4: Create the start schedule (Mon-Fri 07:00 UTC)
aws scheduler create-schedule \
  --name "rds-start-dev-weekdays-0700" \
  --schedule-expression "cron(0 7 ? * MON-FRI *)" \
  --schedule-expression-timezone "UTC" \
  --target "{
    \"Arn\": \"${FUNCTION_ARN}\",
    \"RoleArn\": \"${ROLE_ARN}\",
    \"Input\": \"{\\\"action\\\": \\\"start\\\", \\\"tag_key\\\": \\\"Environment\\\",
↪ \\\"tag_value\\\": \\\"dev\\\"}\"
  }" \
  --flexible-time-window '{"Mode": "OFF"}'

echo "Schedules created. Dev instances will stop at 19:00 UTC and start at 07:00 UTC, Mon-Fri."

```

Lambda Function: Stop/Start Instances by Tag

```

# lambda_rds_stop_tagged.py
# Invoked by EventBridge Scheduler with {"action": "stop"/"start",
#   "tag_key": "...", "tag_value": "..."}
# Deploy as a Lambda function with a 5-minute timeout.

import boto3
import logging

log = logging.getLogger()
log.setLevel(logging.INFO)

def handler(event, context):
    action = event.get("action", "stop") # "stop" or "start"
    tag_key = event.get("tag_key", "Environment")
    tag_value = event.get("tag_value", "dev")
    regions = event.get("regions", ["us-east-1", "us-west-2"])

    total_actioned = 0

    for region in regions:
        rds = boto3.client("rds", region_name=region)
        paginator = rds.get_paginator("describe_db_instances")

        for page in paginator.paginate():
            for db in page["DBInstances"]:
                db_id = db["DBInstanceIdentifier"]
                status = db["DBInstanceStatus"]
                tags = {t["Key"]: t["Value"] for t in db.get("TagList", [])}

                # Tag filter

```

```

if tags.get(tag_key) != tag_value:
    continue

# Safety: never act on production regardless of tag_value match
if tags.get("Environment", "").lower() in \
    {"production", "prod", "prd"}:
    log.warning(f"Skipping production instance: {db_id}")
    continue

# AutoShutdown=false exemption
if tags.get("AutoShutdown", "").lower() == "false":
    log.info(f"Skipping AutoShutdown=false: {db_id}")
    continue

try:
    if action == "stop" and status == "available":
        rds.stop_db_instance(DBInstanceIdentifier=db_id)
        log.info(f"Stopped: {db_id} ({region})")
        total_actioned += 1
    elif action == "start" and status == "stopped":
        rds.start_db_instance(DBInstanceIdentifier=db_id)
        log.info(f"Started: {db_id} ({region})")
        total_actioned += 1
    else:
        log.info(f"No action needed: {db_id} status={status}")
except Exception as e:
    log.error(f"Failed to {action} {db_id}: {e}")

log.info(f"Total instances {action}ped: {total_actioned}")
return {"actioned": total_actioned}

```

Azure SQL: Serverless Auto-Pause

Azure SQL's Serverless tier handles this natively. For databases on the General Purpose or Hyper-scale tiers that need periodic use without continuous running cost, migrating to Serverless with an auto-pause delay of 60 minutes achieves stop/start behavior automatically without any Lambda equivalent.

```

# Migrate an existing Azure SQL database to Serverless compute tier
# and configure auto-pause after 60 minutes of inactivity

RESOURCE_GROUP="myResourceGroup"
SERVER="my-sql-server"
DATABASE="my-dev-database"

az sql db update \
    --resource-group "$RESOURCE_GROUP" \
    --server "$SERVER" \
    --name "$DATABASE" \
    --compute-model Serverless \
    --auto-pause-delay 60 \
    --min-capacity 0.5 \
    --capacity 4

echo "Database $DATABASE migrated to Serverless with 60-minute auto-pause."
echo "Billing pauses automatically during inactivity."

```

The minimum vCore setting of 0.5 means the database can scale down to half a vCore while active, reducing costs for low-utilization periods as well. Combined with auto-pause, a database used for 2 hours per day on a 4-vCore SKU effectively costs what a continuous 0.5-vCore database costs on average — a reduction of roughly 80–90% compared to a continuously running General Purpose instance.

GCP Cloud SQL: Using Instance Scheduling with Cloud Scheduler

```

# Schedule Cloud SQL stop/start using Cloud Scheduler + Cloud Functions

```

```

# Equivalent to the AWS EventBridge + Lambda pattern

PROJECT_ID="your-project-id"
REGION="us-central1"
FUNCTION_NAME="cloud-sql-scheduler"

# Deploy the Cloud Function
gcloud functions deploy "$FUNCTION_NAME" \
  --runtime python311 \
  --trigger-http \
  --region "$REGION" \
  --entry-point handle_schedule \
  --set-env-vars PROJECT_ID="$PROJECT_ID" \
  --memory 256MB \
  --timeout 300s \
  --no-allow-unauthenticated

FUNCTION_URL=$(gcloud functions describe "$FUNCTION_NAME" \
  --region "$REGION" \
  --format "value(serviceConfig.uri)" 2>/dev/null || \
  gcloud functions describe "$FUNCTION_NAME" \
  --region "$REGION" \
  --format "value(httpsTrigger.url)")

# Create service account for Cloud Scheduler
SA_EMAIL="cloud-sql-scheduler-sa@${PROJECT_ID}.iam.gserviceaccount.com"
gcloud iam service-accounts create cloud-sql-scheduler-sa \
  --display-name "Cloud SQL Scheduler"

gcloud projects add-iam-policy-binding "$PROJECT_ID" \
  --member "serviceAccount:${SA_EMAIL}" \
  --role "roles/cloudfunctions.invoker"

gcloud projects add-iam-policy-binding "$PROJECT_ID" \
  --member "serviceAccount:${SA_EMAIL}" \
  --role "roles/cloudsql.admin"

# Stop job: weekdays at 19:00 UTC
gcloud scheduler jobs create http cloud-sql-stop-dev \
  --location "$REGION" \
  --schedule "0 19 * * 1-5" \
  --uri "$FUNCTION_URL" \
  --message-body '{"action":"stop","label_key":"environment","label_value":"dev"}' \
  --oidc-service-account-email "$SA_EMAIL" \
  --time-zone "UTC"

# Start job: weekdays at 07:00 UTC
gcloud scheduler jobs create http cloud-sql-start-dev \
  --location "$REGION" \
  --schedule "0 7 * * 1-5" \
  --uri "$FUNCTION_URL" \
  --message-body '{"action":"start","label_key":"environment","label_value":"dev"}' \
  --oidc-service-account-email "$SA_EMAIL" \
  --time-zone "UTC"

echo "Cloud SQL scheduler jobs created."

# main.py - Cloud Function handler for Cloud SQL stop/start
# Deploy alongside requirements.txt containing: google-cloud-sql-connector, google-auth

import os
import json
import logging
import functions_framework
from googleapiclient import discovery
from google.auth import default

log = logging.getLogger()

PROJECT_ID = os.environ.get("PROJECT_ID", "")

```

```
@functions_framework.http
def handle_schedule(request):
    body = request.get_json(silent=True) or {}
    action = body.get("action", "stop") # "stop" or "start"
    label_key = body.get("label_key", "environment")
    label_value = body.get("label_value", "dev")

    credentials, project = default()
    sqladmin = discovery.build("sqladmin", "v1beta4",
                              credentials=credentials)

    instances_response = sqladmin.instances().list(
        project=PROJECT_ID
    ).execute()

    instances = instances_response.get("items", [])
    actioned = 0

    for instance in instances:
        instance_name = instance["name"]
        labels = instance.get("settings", {}).get("userLabels", {})
        state = instance.get("state", "")

        if labels.get(label_key) != label_value:
            continue

        # Safety exclusion
        if labels.get("environment", "").lower() in \
            {"production", "prod", "prd"}:
            log.warning(f"Skipping production: {instance_name}")
            continue

        try:
            if action == "stop" and state == "RUNNABLE":
                sqladmin.instances().patch(
                    project=PROJECT_ID,
                    instance=instance_name,
                    body={"settings": {"activationPolicy": "NEVER"}}
                ).execute()
                log.info(f"Stopped: {instance_name}")
                actioned += 1

            elif action == "start" and state != "RUNNABLE":
                sqladmin.instances().patch(
                    project=PROJECT_ID,
                    instance=instance_name,
                    body={"settings": {"activationPolicy": "ALWAYS"}}
                ).execute()
                log.info(f"Started: {instance_name}")
                actioned += 1

        except Exception as e:
            log.error(f"Failed to {action} {instance_name}: {e}")

    return json.dumps({"actioned": actioned}), 200, \
        {"Content-Type": "application/json"}
```

10.6 Measuring the Impact and Building an Ongoing Process

Shutting down instances once is table stakes. The real value comes from making idle detection a continuous process that prevents new idle instances from accumulating undetected.

Weekly Idle Instance Report

The following query, run against the CloudWatch metrics stored in your detection pipeline’s output or a central inventory table, provides a weekly digest for finance and engineering leadership:

```
-- Run against a PostgreSQL table populated by idle_detector_aws.py
-- Create the table once:
--
-- CREATE TABLE rds_idle_scan_results (
--   scan_date      DATE NOT NULL,
--   instance_id    TEXT NOT NULL,
--   region         TEXT,
--   instance_class TEXT,
--   engine         TEXT,
--   idle_score     INTEGER,
--   avg_connections NUMERIC(8,4),
--   avg_cpu        NUMERIC(8,4),
--   avg_read_iops  NUMERIC(8,4),
--   avg_write_iops NUMERIC(8,4),
--   monthly_cost_usd NUMERIC(10,2),
--   recommendation TEXT,
--   action_taken   TEXT,
--   PRIMARY KEY (scan_date, instance_id)
-- );

-- Weekly trend: how many idle instances were detected and acted on?
SELECT
  DATE_TRUNC('week', scan_date) AS week_start,
  COUNT(*) AS total_scanned,
  COUNT(*) FILTER (WHERE idle_score >= 3) AS idle_candidates,
  COUNT(*) FILTER (WHERE idle_score = 4) AS high_confidence_idle,
  SUM(monthly_cost_usd)
    FILTER (WHERE idle_score >= 3) AS idle_monthly_cost_usd,
  COUNT(*) FILTER (WHERE action_taken IN ('STOPPED', 'TERMINATED'))
    AS actions_taken,
  SUM(monthly_cost_usd)
    FILTER (WHERE action_taken IN ('STOPPED', 'TERMINATED'))
    AS savings_actioned_usd
FROM rds_idle_scan_results
WHERE scan_date >= CURRENT_DATE - INTERVAL '90 days'
GROUP BY 1
ORDER BY 1 DESC;

-- SQL Server equivalent:
-- SELECT
--   DATEADD(week, DATEDIFF(week, 0, scan_date), 0) AS week_start,
--   COUNT(*) AS total_scanned,
--   SUM(CASE WHEN idle_score >= 3 THEN 1 ELSE 0 END) AS idle_candidates,
--   SUM(CASE WHEN idle_score = 4 THEN 1 ELSE 0 END) AS high_confidence_idle,
--   SUM(CASE WHEN idle_score >= 3
--     THEN monthly_cost_usd ELSE 0 END) AS idle_monthly_cost_usd,
--   SUM(CASE WHEN action_taken IN ('STOPPED', 'TERMINATED')
--     THEN 1 ELSE 0 END) AS actions_taken,
--   SUM(CASE WHEN action_taken IN ('STOPPED', 'TERMINATED')
--     THEN monthly_cost_usd ELSE 0 END) AS savings_actioned_usd
-- FROM rds_idle_scan_results
-- WHERE scan_date >= DATEADD(day, -90, CAST(GETDATE() AS DATE))
-- GROUP BY DATEADD(week, DATEDIFF(week, 0, scan_date), 0)
-- ORDER BY week_start DESC;
```

Drift Detection: Instances That Reappeared After Shutdown

A stopped instance that restarts automatically (AWS’s 7-day auto-restart, or a developer manually starting a stopped instance) will reappear in your next scan. The following query surfaces “drift” — instances that were actioned and came back:

```
-- PostgreSQL: find instances that were stopped/terminated and then
-- reappeared as available in a subsequent scan

WITH actioned AS (
```

```

SELECT DISTINCT instance_id, MIN(scan_date) AS first_actioned_date
FROM rds_idle_scan_results
WHERE action_taken IN ('STOPPED', 'TERMINATED')
GROUP BY instance_id
),
reappeared AS (
  SELECT r.instance_id, r.scan_date AS reappear_date,
         r.idle_score, r.monthly_cost_usd
  FROM rds_idle_scan_results r
  JOIN actioned a ON r.instance_id = a.instance_id
  WHERE r.scan_date > a.first_actioned_date
        AND r.action_taken NOT IN ('STOPPED', 'TERMINATED')
        AND r.idle_score < 3 -- reappeared and now looks active
)
SELECT
  instance_id,
  first_actioned_date,
  reappear_date,
  (reappear_date - first_actioned_date) AS days_until_reappear,
  idle_score AS current_idle_score,
  monthly_cost_usd
FROM reappeared
ORDER BY days_until_reappear ASC;

-- SQL Server equivalent:
-- WITH actioned AS (
--   SELECT DISTINCT instance_id, MIN(scan_date) AS first_actioned_date
--   FROM rds_idle_scan_results
--   WHERE action_taken IN ('STOPPED', 'TERMINATED')
--   GROUP BY instance_id
-- ),
-- reappeared AS (
--   SELECT r.instance_id, r.scan_date AS reappear_date,
--          r.idle_score, r.monthly_cost_usd
--   FROM rds_idle_scan_results r
--   JOIN actioned a ON r.instance_id = a.instance_id
--   WHERE r.scan_date > a.first_actioned_date
--         AND r.action_taken NOT IN ('STOPPED', 'TERMINATED')
--         AND r.idle_score < 3
--   )
-- SELECT
--   instance_id,
--   first_actioned_date,
--   reappear_date,
--   DATEDIFF(day, first_actioned_date, reappear_date) AS days_until_reappear,
--   idle_score AS current_idle_score,
--   monthly_cost_usd
-- FROM reappeared
-- ORDER BY days_until_reappear ASC;

```

Instances that consistently reappear as active within a day or two of being stopped are likely hitting AWS's 7-day auto-restart (someone is keeping them alive by starting them periodically). These are escalation candidates: the automation correctly identifies them as idle, but a human keeps restarting them, suggesting the instance has a legitimate owner who simply hasn't responded to notifications. The appropriate response is a direct conversation with the tagged owner, not repeated automated shutdowns.

Setting Up CloudWatch Alarms to Prevent New Idle Accumulation

Rather than only running the detection scan on a schedule, you can configure real-time alarms that fire when an instance transitions into an idle state. This catches newly abandoned instances within hours rather than waiting for the next weekly scan.

```

#!/bin/bash
# create_idle_alarms.sh``bash
# create_idle_alarms.sh (continued)

```

```

# Creates CloudWatch alarms on all running RDS instances to detect
# sustained low CPU, which triggers a notification to the FinOps SNS topic.
# Run this after provisioning any new RDS instance, or weekly to catch
# new instances that don't yet have alarms.

FINOPS_SNS_TOPIC_ARN="arn:aws:sns:us-east-1:123456789012:finops-idle-alerts"
REGION="us-east-1"
CPU_THRESHOLD=1.0           # percent
EVALUATION_PERIODS=48       # 48 × 30-minute periods = 24 hours sustained
DATAPOINTS_TO_ALARM=40      # 40 of 48 periods must breach (allows brief spikes)

rds_paginator() {
    aws rds describe-db-instances \
        --region "$REGION" \
        --query "DBInstances[?DBInstanceStatus=='available'].DBInstanceIdentifier" \
        --output text
}

while IFS= read -r INSTANCE_ID; do
    [[ -z "$INSTANCE_ID" ]] && continue

    ALARM_NAME="FinOps-IdleCPU-${INSTANCE_ID}"

    aws cloudwatch put-metric-alarm \
        --region "$REGION" \
        --alarm-name "$ALARM_NAME" \
        --alarm-description "CPU below ${CPU_THRESHOLD}% for 24h - possible idle instance" \
        --metric-name CPUUtilization \
        --namespace AWS/RDS \
        --dimensions "Name=DBInstanceIdentifier,Value=${INSTANCE_ID}" \
        --statistic Average \
        --period 1800 \
        --evaluation-periods "$EVALUATION_PERIODS" \
        --datapoints-to-alarm "$DATAPOINTS_TO_ALARM" \
        --threshold "$CPU_THRESHOLD" \
        --comparison-operator LessThanThreshold \
        --treat-missing-data notBreaching \
        --alarm-actions "$FINOPS_SNS_TOPIC_ARN" \
        --ok-actions "$FINOPS_SNS_TOPIC_ARN"

    echo "Alarm created/updated: $ALARM_NAME"

done < <(rds_paginator)

echo ""
echo "All idle CPU alarms configured."
echo "Alerts will fire to: $FINOPS_SNS_TOPIC_ARN"

```

The `--treat-missing-data notBreaching` setting is important here: if an instance is already stopped, CloudWatch will see missing data for CPU metrics. Without this setting, missing data would trigger the alarm — generating false alerts for instances you already shut down. With `notBreaching`, stopped instances stay quiet until they return to a running state and produce new metric data.

Connecting Alarms to the Detection Pipeline

The SNS topic receiving idle alarms can trigger the detection Lambda directly, creating an event-driven path alongside the scheduled scan:

```

# sns_triggered_scan.py
# Lambda handler: triggered by SNS when a CloudWatch idle alarm fires.
# Runs a targeted scan on the specific instance mentioned in the alarm,
# then writes to the same /tmp/idle_instances.json format for the
# shutdown_orchestrator to consume.

import json
import boto3

```

```
import logging
from datetime import datetime, timezone

log = logging.getLogger()
log.setLevel(logging.INFO)

# Import from the detection module (package both files in the Lambda layer)
from idle_detector_aws import scan_single_instance, score_instance

def handler(event, context):
    for record in event.get("Records", []):
        sns_message = json.loads(record["Sns"]["Message"])
        alarm_name = sns_message.get("AlarmName", "")
        alarm_state = sns_message.get("NewStateValue", "")

        # Only act on ALARM transitions (not OK)
        if alarm_state != "ALARM":
            log.info(f"Ignoring non-ALARM state: {alarm_state} for {alarm_name}")
            continue

        # Extract instance ID from alarm name convention: FinOps-IdleCPU-<instance_id>
        if not alarm_name.startswith("FinOps-IdleCPU-"):
            log.warning(f"Unexpected alarm name format: {alarm_name}")
            continue

        instance_id = alarm_name.replace("FinOps-IdleCPU-", "")
        region = sns_message.get("Region", "us-east-1")

        log.info(f"Idle alarm fired for {instance_id} in {region}")

        # Run the full metric scan for this single instance
        result = scan_single_instance(instance_id, region)
        if result is None:
            log.warning(f"Could not scan {instance_id} - may have been deleted already")
            return

        scored = score_instance(result)
        log.info(f"Score for {instance_id}: {scored.idle_score}/4 "
                f"- {scored.recommendation}")

        # Write to DynamoDB for the shutdown orchestrator to pick up
        dynamodb = boto3.resource("dynamodb", region_name=region)
        table = dynamodb.Table("finops-idle-candidates")

        table.put_item(Item={
            "instance_id": scored.instance_id,
            "detected_at": datetime.now(timezone.utc).isoformat(),
            "idle_score": scored.idle_score,
            "avg_connections": str(scored.avg_connections),
            "avg_cpu": str(scored.avg_cpu),
            "avg_read_iops": str(scored.avg_read_iops),
            "avg_write_iops": str(scored.avg_write_iops),
            "monthly_cost_usd": str(scored.monthly_cost_usd),
            "recommendation": scored.recommendation,
            "region": region,
            "ttl": int(datetime.now(timezone.utc).timestamp()) + 604800
        })

        log.info(f"Candidate written to DynamoDB: {instance_id}")
```

This event-driven extension means a newly abandoned instance — one that drops to near-zero CPU after a project wraps up — will appear in your idle candidate table within 24 hours, compared to up to 7 days waiting for the next scheduled scan.

10.7 Avoiding the Common Failure Modes

Failure Mode 1: Stopping Read Replicas Without Stopping the Primary

RDS read replicas cannot be stopped independently while their primary is running. Attempting to stop a replica while the primary is available returns an error. The shutdown orchestrator must check whether an instance is a read replica before issuing a stop command, and if so, handle the primary-replica pair together:

```
def is_read_replica(rds_client, instance_id: str) -> tuple[bool, str]:
    """
    Returns (True, primary_instance_id) if this is a read replica,
    (False, "") otherwise.
    """
    try:
        response = rds_client.describe_db_instances(
            DBInstanceIdentifier=instance_id
        )
        db = response["DBInstances"][0]
        source = db.get("ReadReplicaSourceDBInstanceIdentifier", "")
        if source:
            return True, source
        return False, ""
    except Exception as e:
        log.warning(f"Could not determine replica status for {instance_id}: {e}")
        return False, ""
```

When the orchestrator encounters a read replica with idle score 4, it should flag the primary for review rather than attempting to stop the replica alone. If both the primary and all its replicas score 4/4, the correct sequence is: stop replicas first (if the primary is also being stopped — though RDS does not allow stopping replicas independently), or more practically, stop the primary, at which point replicas automatically stop receiving replication and will show as stopped as well.

Failure Mode 2: Aurora Clusters Require Cluster-Level Actions

Aurora instances are part of a cluster; you cannot stop an individual Aurora instance. The stop action must target the DB cluster identifier, not the DB instance identifier:

```
def stop_aurora_cluster(rds_client, cluster_id: str):
    if DRY_RUN:
        log.info(f"[DRY-RUN] Would stop Aurora cluster: {cluster_id}")
        return
    try:
        rds_client.stop_db_cluster(DBClusterIdentifier=cluster_id)
        log.info(f"Aurora cluster stop issued: {cluster_id}")
    except Exception as e:
        log.error(f"Failed to stop Aurora cluster {cluster_id}: {e}")

def get_cluster_id(rds_client, instance_id: str) -> str:
    """Returns the cluster identifier for an Aurora instance, or '' for non-Aurora."""
    try:
        response = rds_client.describe_db_instances(
            DBInstanceIdentifier=instance_id
        )
        db = response["DBInstances"][0]
        return db.get("DBClusterIdentifier", "")
    except Exception:
        return ""
```

The `scan_region` function in Section 10.2 should be extended to record cluster membership so that the shutdown orchestrator can deduplicate: if three Aurora instances all belong to the same cluster and all score 4/4, the action is one `stop_db_cluster` call, not three `stop_db_instance` calls.

Failure Mode 3: Multi-AZ Failover During Shutdown

Stopping a Multi-AZ RDS instance triggers a brief failover as the instance transitions states. For instances tagged `Environment=staging` or `Environment=qa`, applications connected at shutdown time will see a brief connection drop. The mitigation is to notify application owners via the SNS topic before acting, with a configurable delay:

```
import time

def notify_and_wait(sns_client, topic_arn: str, instance_id: str,
                   action: str, delay_minutes: int = 15):
    """
    Sends a pre-action notification and waits the specified delay.
    In production, replace time.sleep with a Step Functions wait state.
    """
    message = (
        f"FinOps Automated Action Pending\n"
        f"Instance : {instance_id}\n"
        f"Action   : {action}\n"
        f"Delay     : {delay_minutes} minutes\n"
        f"To cancel: tag the instance AutoShutdown=false within "
        f"{delay_minutes} minutes."
    )
    sns_client.publish(
        TopicArn=topic_arn,
        Subject=f"[FinOps] Pending {action}: {instance_id}",
        Message=message
    )
    log.info(f"Notification sent. Waiting {delay_minutes} minutes before {action}.")
    time.sleep(delay_minutes * 60)

    # Re-check tags after the delay - owner may have opted out
    return # Caller re-verifies tags before proceeding
```

In a production Step Functions workflow, the `time.sleep` is replaced by a `Wait` state, which is far more reliable than sleeping inside a Lambda function that may time out.

Failure Mode 4: Snapshots Not Completing Before Deletion Proceeds

When the shutdown orchestrator issues `create_db_snapshot` and then immediately calls `delete_db_instance`, the deletion will fail because a snapshot is in progress. The orchestrator must poll for snapshot completion. In a Step Functions workflow, this is a `Wait + Choice` state loop. In the simple Lambda version, a waiter handles it:

```
def wait_for_snapshot(rds_client, snapshot_id: str, timeout_minutes: int = 60):
    """
    Polls until the snapshot reaches 'available' status.
    Use a Step Functions Wait state in production instead of polling in Lambda.
    """
    waiter = rds_client.get_waiter("db_snapshot_available")
    log.info(f"Waiting for snapshot {snapshot_id} to complete...")
    try:
        waiter.wait(
            DBSnapshotIdentifier=snapshot_id,
            WaiterConfig={"Delay": 30, "MaxAttempts": timeout_minutes * 2}
        )
        log.info(f"Snapshot complete: {snapshot_id}")
        return True
    except Exception as e:
        log.error(f"Snapshot wait failed for {snapshot_id}: {e}")
        return False
```

Key Takeaways

- **Idle detection requires four signals, not one.** Connection count alone produces false positives due to connection poolers and monitoring agents. Combining DatabaseConnections, CPU utilization, combined IOPS, and network throughput over a 7-day rolling window gives reliable idle candidates with minimal false positives.
 - **Validate before acting, and validate again at action time.** The detection scan and the shutdown action should both independently verify tag-based exemptions. A tag added between scan time and action time must be respected; defense-in-depth tag checking at both stages prevents unintended shutdowns of newly protected instances.
 - **Stop and terminate are different tools for different situations.** Stopping preserves the instance for potential restart, saves compute costs, but continues storage billing and auto-restarts after 7 days on AWS. Termination with a final snapshot eliminates all ongoing costs but requires a deliberate recovery process. Match the action to the confidence level: stop for score-3 candidates, terminate only for confirmed decommissions.
 - **Schedule stop/start for non-production instances rather than relying solely on idle detection.** A database that runs batch jobs unpredictably will evade idle detection during active weeks. Scheduled shutdown during off-hours (evenings, weekends) captures consistent savings without the risk of accidentally stopping a legitimately used instance. Azure SQL Serverless auto-pause handles this automatically without custom scheduling infrastructure.
 - **Make idle detection continuous, not a one-time exercise.** CloudWatch alarms on CPU thresholds catch newly abandoned instances within 24 hours rather than waiting for a weekly scan. Drift detection queries surface instances that were stopped and restarted, enabling targeted conversations with owners who are manually overriding automation. The ongoing process prevents idle debt from accumulating again after the initial cleanup.
-

Chapter 11: Dev and Test Cost Reduction: Schedules and Smaller Instances

Organizations running cloud databases typically discover the same uncomfortable pattern six to twelve months after migration: their non-production environments are costing nearly as much as production. A company running `db.r6g.2xlarge` instances for dev, staging, QA, and performance testing—all at full uptime—can easily spend \$8,000–\$15,000 per month on databases that are actively used for only 40 hours out of 168 available weekly hours. Apply automated stop/start schedules, right-size those instances to match actual workload profiles, and introduce connection pooling to eliminate idle-connection overhead, and that same company routinely lands at \$1,500–\$3,000 per month—a 70–80% reduction without touching a single line of application code.

This chapter gives you the specific tools, scripts, and decision frameworks to get there.

11.1 Diagnosing Non-Production Waste: What You’re Actually Paying For

Before cutting anything, you need an accurate picture of where the money is going. Most organizations are surprised to find that non-production databases fall into three distinct waste categories: always-on instances that are idle outside business hours, oversized instances chosen to mirror production specs out of habit, and multi-AZ configurations on environments that do not need high availability.

Start with a simple inventory query against AWS Cost Explorer to surface your RDS spending by environment tag.

```
# Pull last 30 days of RDS costs grouped by Environment tag
aws ce get-cost-and-usage \
  --time-period Start=$(date -d '30 days ago' +%Y-%m-%d),End=$(date +%Y-%m-%d) \
  --granularity MONTHLY \
  --filter '{
    "Dimensions": {
      "Key": "SERVICE",
      "Values": ["Amazon Relational Database Service"]
    }
  }' \
  --group-by Type=TAG,Key=Environment \
  --metrics "UnblendedCost" \
  --output table
```

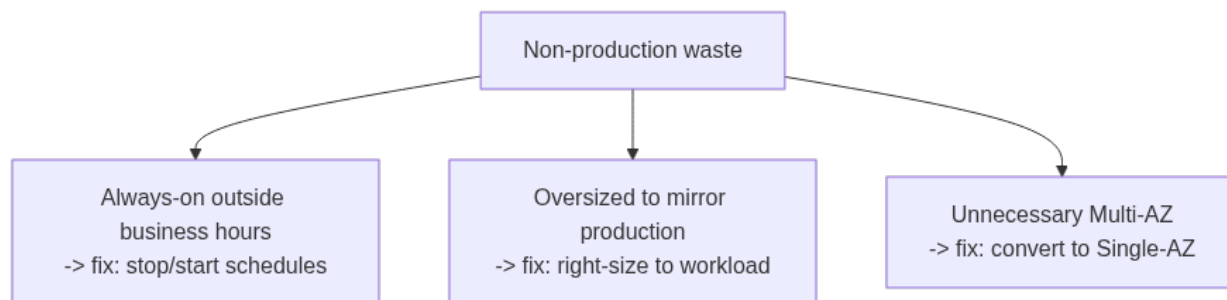


Figure 11: The three waste categories behind most non-production overspend — each has a distinct, independent fix.

If your instances are not tagged with an `Environment` key, you will get a single lump-sum result. That itself is a finding. Tagging hygiene is the prerequisite for every cost optimization effort in non-production. Without it, you cannot prove savings to the CTO, and you cannot automate selectively.

On Azure, the equivalent query runs through the Cost Management CLI:

```
# Azure: breakdown of SQL Database costs by environment tag for last 30 days
az costmanagement query \
  --scope "/subscriptions/{subscription-id}" \
  --type Usage \
  --timeframe Custom \
  --time-period from="$(date -d '30 days ago' --iso-8601)" \
  to="$(date --iso-8601)" \
  --dataset-aggregation '{"totalCost":{"name":"Cost","function":"Sum"}}' \
  --dataset-grouping '["type":"TagKey","name":"environment"]' \
  --output table
```

On GCP, query the billing export in BigQuery:

```
-- GCP BigQuery: monthly Cloud SQL costs by environment label
SELECT
  project.id                                AS project,
  labels.value                              AS environment,
  ROUND(SUM(cost), 2)                       AS total_cost_usd,
  ROUND(SUM(cost) / 30, 2)                  AS avg_daily_cost_usd
FROM
  `billing_dataset.gcp_billing_export_v1_XXXXXX`
CROSS JOIN
  UNNEST(labels) AS labels
WHERE
  labels.key = 'environment'
  AND service.description = 'Cloud SQL'
  AND DATE(_PARTITIONTIME) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
GROUP BY
  project, environment
ORDER BY
  total_cost_usd DESC;
```

Once you have costs broken down by environment, compare them against actual utilization. On PostgreSQL, this query tells you whether a database has seen meaningful activity in the past seven days:

```
-- PostgreSQL: identify low-activity databases on an instance
-- Run this on each non-production RDS/Cloud SQL/Azure instance
SELECT
  datname                                AS database_name,
  numbackends                            AS current_connections,
```

```

xact_commit + xact_rollback      AS total_transactions,
blks_hit + blks_read             AS total_block_accesses,
pg_size_pretty(pg_database_size(datname)) AS db_size,
stats_reset                     AS stats_collected_since
FROM
  pg_stat_database
WHERE
  datname NOT IN ('postgres', 'template0', 'template1')
ORDER BY
  total_transactions ASC;

```

A database showing zero or near-zero `total_transactions` since `stats_reset` is a candidate for immediate decommissioning or consolidation. Before scheduling a stop, collect a second metric: connection activity over time. On SQL Server (Azure SQL Managed Instance), the equivalent check is:

```

-- SQL Server: sessions and recent query activity per database
SELECT
  DB_NAME(database_id)      AS database_name,
  COUNT(*)                 AS session_count,
  MAX(last_request_start_time) AS last_query_time,
  SUM(cpu_time)             AS total_cpu_ms,
  SUM(logical_reads)        AS total_logical_reads
FROM
  sys.dm_exec_sessions
WHERE
  is_user_process = 1
GROUP BY
  database_id
ORDER BY
  last_query_time DESC;

```

If `last_query_time` is hours old during business hours, the instance is a prime candidate for aggressive scheduling.

11.2 Right-Sizing Non-Production Instances: The Math Behind the Decision

The most common mistake in non-production environments is mirroring production instance sizes. The reasoning sounds logical—“we want to catch performance issues before they hit prod”—but it confuses two separate concerns: functional correctness testing and performance testing. For the former, an instance one to two tiers smaller than production is almost always adequate. For the latter, a dedicated performance environment running on production-sized hardware for a few hours per week is far more cost-effective than running full-size instances 24/7 for all environments.

Here is the pricing reality for AWS RDS PostgreSQL in us-east-1 (on-demand, as of 2024):

Instance Type	vCPUs	RAM	\$/hour	\$/month (730h)
db.r6g.2xlarge	8	64 GB	\$0.960	\$700.80
db.r6g.xlarge	4	32 GB	\$0.480	\$350.40
db.r6g.large	2	16 GB	\$0.240	\$175.20
db.t4g.medium	2	4 GB	\$0.068	\$49.64
db.t4g.large	2	8 GB	\$0.136	\$99.28

A team running four non-production environments (dev, qa, staging, perf) on `db.r6g.2xlarge` pays \$2,803 per month in compute alone before storage. Moving dev and qa to `db.t4g.large`, staging to `db.r6g.large`, and keeping perf at `db.r6g.2xlarge` drops compute to \$1,074.56/month—a \$1,728.64/month reduction on compute alone, before applying any scheduling.

AWS Compute Optimizer surfaces right-sizing recommendations automatically, but its recommendations are conservative. Pull them programmatically and apply your own filter:

```
# Fetch Compute Optimizer recommendations for all RDS instances
aws compute-optimizer get-rds-database-recommendations \
  --region us-east-1 \
  --output json \
  | jq '
    .rdsDBRecommendations[]
    | select(.finding == "Overprovisioned")
    | {
      instance: .resourceArn,
      current_type: .currentDBInstanceClass,
      recommended_type: (.recommendationOptions[0].dbInstanceClass),
      estimated_monthly_savings: .recommendationOptions[0].estimatedMonthlySavings.value,
      currency: .recommendationOptions[0].estimatedMonthlySavings.currency,
      cpu_max_pct: .utilizationMetrics[] | select(.name == "CPU") | .value
    }
  '
```

For Azure SQL, Compute Optimizer equivalents come through Azure Advisor:

```
# List Azure Advisor cost recommendations filtered to SQL resources
az advisor recommendation list \
  --category Cost \
  --output table \
  --query "[?contains(resourceMetadata.resourceId, 'Microsoft.Sql')].
    {ResourceId:resourceMetadata.resourceId,
      Impact:impact,
      ShortDesc:shortDescription.problem,
      Savings:extendedProperties.annualSavingsAmount}"
```

When evaluating `db.t4g` burstable instances for dev and QA, understand the CPU credit model. A `db.t4g.large` earns 36 CPU credits per hour and consumes 10 credits per vCPU-hour at full load. For short-burst workloads like running a CI/CD test suite that takes 20 minutes and then sits idle for 40 minutes, the T4g family is excellent. For sustained CPU workloads, it will throttle. Pull CloudWatch CPU credit metrics before committing:

```
# Check CPU credit balance for a T-class RDS instance over last 7 days
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name CPUCreditBalance \
  --dimensions Name=DBInstanceIdentifier,Value=my-dev-db \
  --start-time $(date -u -d '7 days ago' +%Y-%m-%dT%H:%M:%SZ) \
  --end-time $(date -u +%Y-%m-%dT%H:%M:%SZ) \
  --period 3600 \
  --statistics Minimum \
  --output table

# If minimum credit balance stays above 50, T-class is safe for this workload
# If it drops toward 0, the workload is too sustained - use r6g.large instead
```

Multi-AZ is another major cost lever. Every Multi-AZ RDS instance costs exactly double the Single-AZ price because AWS provisions a synchronous standby replica in a second availability zone. For production, this is justified. For dev, QA, and staging, it almost never is. Check your fleet:

```
# List all RDS instances with Multi-AZ enabled and their monthly cost impact
aws rds describe-db-instances \
  --output json \
```



```

| jq '
  .DBInstances[]
| select(.MultiAZ == true)
| {
  identifier: .DBInstanceIdentifier,
  instance_class: .DBInstanceClass,
  engine: .Engine,
  multi_az: .MultiAZ,
  tags: .TagList
}
'

```

Cross-reference this list against your environment tags. Every Multi-AZ dev or QA instance you convert to Single-AZ cuts its compute cost in half. On a db.r6g.2xlarge, that is \$700/month saved per instance.

11.3 Automated Stop/Start Schedules: Paying Only for Hours Used

This is the single highest-impact technique for non-production cost reduction. AWS, Azure, and GCP all allow you to stop database instances and pay only for storage while they are stopped. A development database that runs Monday through Friday, 7 AM to 7 PM, is running 60 hours per week instead of 168—a 64% reduction in compute hours without changing anything else.

The math on a db.r6g.xlarge at \$0.48/hour: - Always-on: 730 hours × \$0.48 = **\$350.40/month**
 - Scheduled (weekdays, 7 AM–7 PM): ~260 hours × \$0.48 = **\$124.80/month** - **Savings: \$225.60/month per instance**

AWS has a hard limit: RDS instances automatically restart after 7 days if left stopped. Your schedule must account for this. The following Python Lambda function handles both the stop/start schedule and the anti-restart protection:

```

import boto3
import json
from datetime import datetime, timezone

# Lambda function triggered by EventBridge rules (two rules: one at 7 PM, one at 7 AM)
# Environment variables:
# ACTION: "stop" or "start"
# ENVIRONMENT_TAG: "dev" or "qa" or "staging"
# AWS_REGION_NAME: target region

def lambda_handler(event, context):
    action = event.get('action', 'stop')
    target_env = event.get('environment_tag', 'dev')
    region = event.get('region', 'us-east-1')

    rds = boto3.client('rds', region_name=region)

    # Fetch all instances matching the environment tag
    paginator = rds.get_paginator('describe_db_instances')
    target_instances = []

    for page in paginator.paginate():
        for instance in page['DBInstances']:
            tags = {t['Key']: t['Value'] for t in instance.get('TagList', [])}
            if tags.get('Environment', '').lower() == target_env.lower():
                target_instances.append({
                    'id': instance['DBInstanceIdentifier'],
                    'status': instance['DBInstanceStatus'],
                    'class': instance['DBInstanceClass']
                })

```

```

results = []
for inst in target_instances:
    instance_id = inst['id']
    current_status = inst['status']

    try:
        if action == 'stop' and current_status == 'available':
            rds.stop_db_instance(DBInstanceIdentifier=instance_id)
            results.append({
                'instance': instance_id,
                'action': 'stopped',
                'previous_status': current_status
            })
            print(f"Stopped: {instance_id} ({inst['class']})")

        elif action == 'start' and current_status == 'stopped':
            rds.start_db_instance(DBInstanceIdentifier=instance_id)
            results.append({
                'instance': instance_id,
                'action': 'started',
                'previous_status': current_status
            })
            print(f"Started: {instance_id} ({inst['class']})")

        else:
            print(f"Skipped: {instance_id} - status is '{current_status}', "
                  f"action requested: '{action}'")
            results.append({
                'instance': instance_id,
                'action': 'skipped',
                'reason': f"status={current_status}"
            })

    except rds.exceptions.InvalidDBInstanceStateFault as e:
        print(f"State fault for {instance_id}: {e}")
        results.append({
            'instance': instance_id,
            'action': 'failed',
            'error': str(e)
        })

    return {
        'statusCode': 200,
        'timestamp': datetime.now(timezone.utc).isoformat(),
        'environment': target_env,
        'action': action,
        'results': results
    }
}

```

The corresponding EventBridge schedule rules in Terraform:

```

# Create the stop schedule (weekdays at 7 PM EST = 23:00 UTC)
aws events put-rule \
  --name "rds-stop-dev-weekday-evening" \
  --schedule-expression "cron(0 23 ? * MON-FRI *)" \
  --state ENABLED \
  --description "Stop dev RDS instances at 7 PM EST weekdays"

# Create the start schedule (weekdays at 7 AM EST = 11:00 UTC)
aws events put-rule \
  --name "rds-start-dev-weekday-morning" \
  --schedule-expression "cron(0 11 ? * MON-FRI *)" \
  --state ENABLED \
  --description "Start dev RDS instances at 7 AM EST weekdays"

# Target both rules at the Lambda function (repeat for start rule)
aws events put-targets \
  --rule "rds-stop-dev-weekday-evening" \
  --targets '[{

```

```
"Id": "StopDevDatabases",
"Arn": "arn:aws:lambda:us-east-1:123456789:function:rds-scheduler",
"Input": "{\"action\": \"stop\", \"environment_tag\": \"dev\", \"region\": \"us-east-1\"}"
}]'
```

On Azure, the equivalent uses Azure Automation runbooks with schedules. Azure SQL Serverless is often a better long-term solution—it auto-pauses after a configurable idle timeout and you pay only for vCore-seconds consumed. For Azure SQL Managed Instance, which does not support serverless, use a similar runbook approach:

```
# Azure: Stop a SQL Managed Instance via CLI (wrap this in an Automation runbook)
az sql mi stop \
  --name my-dev-managed-instance \
  --resource-group rg-nonprod

# Azure: Start it back up
az sql mi start \
  --name my-dev-managed-instance \
  --resource-group rg-nonprod
```

On GCP, Cloud SQL instances can be stopped similarly:

```
# GCP: Stop a Cloud SQL instance
gcloud sql instances patch my-dev-postgres \
  --activation-policy=NEVER

# GCP: Start it back up
gcloud sql instances patch my-dev-postgres \
  --activation-policy=ALWAYS
```

Wrap these commands in Cloud Scheduler jobs with Pub/Sub triggers and Cloud Functions to achieve full automation equivalent to the AWS Lambda pattern.

One operational detail that creates significant friction: developers who work late or on weekends need a self-service mechanism to start instances without filing tickets. A simple solution is an AWS Systems Manager document that allows developers to start their specific environment’s databases via the SSM console or CLI, without giving them direct RDS API access:

```
# Developers run this to start their dev environment (no IAM policy for rds:StartDBInstance needed)
aws ssm start-automation-execution \
  --document-name "StartDevDatabases" \
  --parameters "Environment=dev,Region=us-east-1" \
  --region us-east-1
```

The SSM Automation document calls the Lambda function with the `start` action. The developer’s IAM policy allows `ssm:StartAutomationExecution` on that specific document name, but not direct RDS access. This maintains least-privilege while eliminating the “I need to work Saturday and my database is stopped” support request.

11.4 Connection Pooling to Reduce Instance Overhead

A less obvious cost dimension in non-production environments is connection overhead. When CI/CD pipelines run test suites, they frequently open and close hundreds of database connections per test run. Each connection to PostgreSQL spawns a backend process consuming 5–10 MB of shared memory. Without pooling, a test suite that opens 200 connections requires an instance with enough RAM to handle 200 concurrent backends, even if those connections are overwhelmingly idle at any given millisecond.

This is why teams “need” `db.r6g.2xlarge` (64 GB) instances in dev when their actual working dataset is 10 GB. They are sizing for connection count, not data or query workload. PgBouncer eliminates this problem entirely.

RDS Proxy provides managed connection pooling on AWS, but it costs \$0.015 per vCPU-hour of your underlying RDS instance. On a `db.r6g.2xlarge` (8 vCPUs), that is \$0.12/hour—an 87/month addition just for pooling. For non-production environments, a self-managed PgBouncer on a `t4g.nano` EC2 instance (\$0.0042/hour, ~\$3/month) is far more economical.

The diagnostic query to determine whether connection overhead is your actual problem:

```
-- PostgreSQL: analyze connection usage patterns over time
-- Run during a typical CI/CD test run to capture peak behavior

SELECT
    state,
    wait_event_type,
    wait_event,
    COUNT(*) AS connection_count,
    ROUND(AVG(EXTRACT(EPOCH FROM (now() - state_change))), 1) AS avg_seconds_in_state
FROM
    pg_stat_activity
WHERE
    backend_type = 'client backend'
    AND datname IS NOT NULL
GROUP BY
    state, wait_event_type, wait_event
ORDER BY
    connection_count DESC;

-- If you see large counts in state='idle', pgBouncer transaction pooling
-- will let you downsize the instance significantly.
-- Rule of thumb: if idle connections > 40% of total, pooling will unlock 1-2 instance tiers down.
```

If that query consistently shows 60–70% of connections in `idle` state, you can safely move to an instance one or two tiers smaller after deploying PgBouncer in transaction pooling mode.

The PgBouncer configuration for a dev environment that previously required a large instance:

```
# /etc/pgbouncer/pgbouncer.ini
# Deployed on a t4g.nano in the same VPC as the RDS instance

[databases]
devdb = host=dev-postgres.abcdef.us-east-1.rds.amazonaws.com port=5432 dbname=devdb
qadb  = host=qa-postgres.abcdef.us-east-1.rds.amazonaws.com port=5432 dbname=qadb

[pgbouncer]
listen_port      = 5432
listen_addr     = 0.0.0.0
auth_type       = md5
auth_file       = /etc/pgbouncer/userlist.txt

# Transaction pooling: a server connection is held only during a transaction
# This is what enables massive connection count reduction
pool_mode       = transaction

# Maximum connections PgBouncer will open to the actual RDS instance
server_pool_size = 25
max_client_conn  = 500

# RDS requires this - do not ignore it
server_tls_sslmode = require

log_connections  = 1
log_disconnections = 1
stats_period     = 60
```

```
# These tunings prevent stale connections after RDS restarts
server_idle_timeout      = 600
server_connect_timeout   = 15
server_login_retry       = 15
```

With this configuration, 500 CI/CD connections from test runners are multiplexed into at most 25 actual connections to the RDS instance. The `db.r6g.2xlarge` (64 GB RAM) that was sized for raw connection count can move to `db.r6g.large` (16 GB RAM), saving \$525/month on that single instance alone.

Monitor PgBouncer's pooling effectiveness with a scheduled CloudWatch custom metric script:

```
import subprocess
import boto3
import json
from datetime import datetime, timezone

def collect_pgboouncer_stats():
    """
    Collect PgBouncer SHOW POOLS stats and publish to CloudWatch.
    Run every 60 seconds via cron or systemd timer.
    """
    cloudwatch = boto3.client('cloudwatch', region_name='us-east-1')

    # psql must be installed and pgbouncer admin user configured
    result = subprocess.run(
        ['psql', '-h', '127.0.0.1', '-p', '5432',
         '-U', 'pgbouncer', '-d', 'pgbouncer',
         '-c', 'SHOW POOLS;', '--csv', '--tuples-only'],
        capture_output=True, text=True
    )

    lines = result.stdout.strip().split('\n')
    metric_data = []
    timestamp = datetime.now(timezone.utc)

    for line in lines:
        if not line.strip():
            continue
        parts = line.split(',')
        if len(parts) < 9:
            continue

        database = parts[0].strip()
        cl_active = int(parts[2].strip() or 0) #client connections active
        cl_waiting = int(parts[3].strip() or 0) # client connections waiting for a server conn
        sv_active = int(parts[5].strip() or 0) # server connections active
        sv_idle = int(parts[6].strip() or 0) # server connections idle

        for metric_name, value in [
            ('ClientActive', cl_active),
            ('ClientWaiting', cl_waiting),
            ('ServerActive', sv_active),
            ('ServerIdle', sv_idle),
        ]:
            metric_data.append({
                'MetricName': metric_name,
                'Dimensions': [
                    {'Name': 'Database', 'Value': database},
                    {'Name': 'Environment', 'Value': 'dev'}
                ],
                'Value': float(value),
                'Unit': 'Count',
                'Timestamp': timestamp
            })

    if metric_data:
        cloudwatch.put_metric_data(
```

```
        Namespace='PgBouncer/Pooling',
        MetricData=metric_data
    )
    print(f"Published {len(metric_data)} metrics at {timestamp.isoformat()}")

if __name__ == '__main__':
    collect_pgouncer_stats()
```

Set a CloudWatch alarm on `ClientWaiting > 10` sustained for more than two minutes. That is the signal that `server_pool_size` is too small for peak load and the RDS instance needs one tier up—or the test suite needs to reduce concurrency.

For SQL Server on Azure SQL, the equivalent of connection pooling is built into the ADO.NET and JDBC drivers through connection pool sizing parameters. Unlike PostgreSQL, SQL Server uses a thread-based connection model that consumes less per-connection memory. However, excessive connection counts still degrade performance. On Azure SQL, the managed service enforces session limits by service tier. Check your utilization against those limits:

```
-- SQL Server / Azure SQL: current session usage vs. tier limit
SELECT
    COUNT(*)                                AS active_sessions,
    (SELECT TOP 1 value_in_use
     FROM sys.configurations
     WHERE name = 'max connections')        AS max_configured_connections,
    @@MAX_CONNECTIONS                       AS engine_max_connections,
    DATEDIFF(MINUTE,
             MIN(login_time),
             GETUTCDATE())                  AS oldest_session_age_minutes
FROM sys.dm_exec_sessions
WHERE is_user_process = 1;
```

If `active_sessions` consistently exceeds 70–80% of the service tier’s session limit, you are sizing the Azure SQL tier for connection count rather than compute or data requirements—the same pattern as the PostgreSQL scenario above. Azure SQL Elastic Pools solve this for multi-database dev environments by sharing a session budget across databases.

11.5 Elastic Pools and Shared Infrastructure for Multiple Dev Databases

When a team runs multiple development databases—one per microservice, one per developer branch, one per feature team—the per-instance cost multiplies quickly. Ten microservices each with their own `db.r6g.large` dev instance means ten separate bills, ten separate maintenance windows, and ten instances that are mostly idle at any given time.

Azure SQL Elastic Pools address this directly. A single Elastic Pool provides a shared eDTU or vCore budget across many databases. Databases that are idle contribute nothing to current consumption, and active databases can burst to use available headroom up to the pool maximum.

For dev environments, a General Purpose Elastic Pool at 4 vCores can comfortably serve 20–30 lightly used dev databases with typical CI/CD patterns. The alternative—20 individual General Purpose databases at 2 vCores each—costs substantially more while providing no pooling benefit.

Provision and populate an Elastic Pool via CLI:

```

# Create a General Purpose Elastic Pool for all dev databases
az sql elastic-pool create \
  --name ep-dev-databases \
  --resource-group rg-nonprod \
  --server sql-nonprod-server \
  --edition GeneralPurpose \
  --family Gen5 \
  --capacity 4 \
  --db-max-capacity 2 \
  --db-min-capacity 0 \
  --zone-redundant false

# Move an existing dev database into the pool
az sql db update \
  --name devdb-service-auth \
  --resource-group rg-nonprod \
  --server sql-nonprod-server \
  --elastic-pool ep-dev-databases

# Move another one
az sql db update \
  --name devdb-service-billing \
  --resource-group rg-nonprod \
  --server sql-nonprod-server \
  --elastic-pool ep-dev-databases

```

Monitor the pool to confirm it is not being overwhelmed by simultaneous load spikes. The following query, run against the **master** database on the logical server, shows real-time resource consumption per database in the pool:

```

-- SQL Server (run against master on the Azure SQL logical server):
-- Per-database resource consumption in an Elastic Pool
SELECT
  database_name,
  ROUND(avg_cpu_percent, 2) AS cpu_pct,
  ROUND(avg_data_io_percent, 2) AS data_io_pct,
  ROUND(avg_log_write_percent, 2) AS log_write_pct,
  ROUND(avg_memory_usage_percent, 2) AS memory_pct,
  end_time
FROM
  sys.elastic_pool_resource_stats
WHERE
  end_time >= DATEADD(HOUR, -1, GETUTCDATE())
ORDER BY
  end_time DESC, cpu_pct DESC;

```

If any single database is pegging `cpu_pct` above 80% for sustained periods, it is a poorly behaved neighbor. Identify the specific workload and either schedule it to run during off-hours or isolate that database to its own instance temporarily.

On AWS, the Aurora Serverless v2 model provides a similar shared-capacity concept at the cluster level. A single Aurora cluster can host multiple databases (schemas), and Aurora Serverless v2 scales ACUs (Aurora Capacity Units) up and down based on aggregate demand. For microservice teams where each service owns its own schema within a shared cluster, this eliminates per-service instance costs entirely.

```

# Create an Aurora Serverless v2 PostgreSQL cluster for shared dev use
aws rds create-db-cluster \
  --db-cluster-identifier aurora-dev-shared \
  --engine aurora-postgresql \
  --engine-version 15.4 \
  --serverless-v2-scaling-configuration MinCapacity=0.5,MaxCapacity=4 \
  --master-username devadmin \
  --master-user-password "${DB_PASSWORD}" \
  --db-subnet-group-name dev-subnet-group \
  --vpc-security-group-ids sg-0abc123def456 \

```

```

--enable-cloudwatch-logs-exports '["postgresql"]' \
--deletion-protection false \
--tags Key=Environment,Value=dev

# Add a Serverless v2 writer instance
aws rds create-db-instance \
--db-instance-identifier aurora-dev-shared-writer \
--db-cluster-identifier aurora-dev-shared \
--db-instance-class db.serverless \
--engine aurora-postgresql

```

With `MinCapacity=0.5`, the cluster scales down to half an ACU when idle—roughly \$0.06/hour at us-east-1 pricing ($0.5 \text{ ACU} \times \$0.12/\text{ACU-hour}$), compared to \$0.24/hour for the smallest always-on `db.r6g.large`. For a dev cluster that sees activity 8 hours per day on weekdays and is nearly idle otherwise, the monthly bill on Aurora Serverless v2 often comes in significantly lower than even the smallest fixed-size instance, while requiring no stop/start scheduling automation.

11.6 Database Cloning for Ephemeral Test Environments

One of the highest-leverage but underused techniques for test cost reduction is thin-clone provisioning. The traditional approach to test environment data is either running tests against a fixed, manually maintained database (which drifts from production) or restoring a full production backup to a test instance (which is slow, storage-intensive, and often requires data masking). Neither scales well when you need a fresh environment per feature branch or per pull request.

AWS Aurora supports fast cloning using a copy-on-write mechanism. Cloning an Aurora cluster completes in seconds regardless of database size because it creates a new cluster that references the original's storage volume pages. Only pages that differ after the clone is created are written separately. This means a 500 GB Aurora database can be cloned into ten isolated test environments, and the total additional storage consumed is only the delta from the base snapshot—often a few gigabytes per clone.

```

# Clone an Aurora cluster for PR #847 testing
# This completes in seconds for any database size
aws rds restore-db-cluster-to-point-in-time \
--db-cluster-identifier aurora-test-pr-847 \
--restore-type copy-on-write \
--source-db-cluster-identifier aurora-staging \
--restore-to-time $(date -u +%Y-%m-%dT%H:%M:%SZ) \
--db-subnet-group-name dev-subnet-group \
--vpc-security-group-ids sg-0abc123def456 \
--tags \
  Key=Environment,Value=ephemeral \
  Key=PR,Value=847 \
  Key=CreatedAt,Value=$(date -u +%Y-%m-%dT%H:%M:%SZ)

# Add a serverless writer instance to the clone
aws rds create-db-instance \
--db-instance-identifier aurora-test-pr-847-writer \
--db-cluster-identifier aurora-test-pr-847 \
--db-instance-class db.serverless \
--engine aurora-postgresql

```

Pair this with a cleanup function that deletes clones automatically when the PR closes:

```

import boto3
import os

def cleanup_pr_clone(event, context):

```



```

"""
Triggered by GitHub webhook via API Gateway when a PR is closed.
Deletes the associated Aurora clone to prevent orphaned test databases.
"""
rds = boto3.client('rds', region_name=os.environ['AWS_REGION'])

pr_number = event.get('pull_request', {}).get('number')
if not pr_number:
    print("No PR number found in event payload")
    return {'statusCode': 400, 'body': 'Missing PR number'}

cluster_id = f"aurora-test-pr-{pr_number}"
instance_id = f"aurora-test-pr-{pr_number}-writer"

# Delete the instance first, then the cluster
try:
    rds.delete_db_instance(
        DBInstanceIdentifier=instance_id,
        SkipFinalSnapshot=True
    )
    print(f"Deleting instance: {instance_id}")
except rds.exceptions.DBInstanceNotFoundFault:
    print(f"Instance {instance_id} already gone or never existed")

try:
    # Wait for instance deletion before deleting cluster would block Lambda
    # In practice, use a Step Functions state machine with a wait loop
    # For simplicity here: cluster deletion is queued after instance cleanup
    rds.delete_db_cluster(
        DBClusterIdentifier=cluster_id,
        SkipFinalSnapshot=True
    )
    print(f"Deleting cluster: {cluster_id}")
except rds.exceptions.DBClusterNotFoundFault:
    print(f"Cluster {cluster_id} already gone or never existed")

return {
    'statusCode': 200,
    'body': f"Cleanup initiated for PR #{pr_number}"
}

```

On GCP, Cloud SQL does not support copy-on-write cloning in the same way, but it supports instance clones that use the underlying storage snapshot mechanism. The clone operation takes a few minutes rather than seconds, but the result is functionally equivalent for test environments:

```

# Clone a Cloud SQL instance for ephemeral test use
gcloud sql instances clone \
    cloudsql-staging \
    cloudsql-test-pr-847 \
    --async

# Stop the clone immediately after provisioning to save cost
# (start it only when CI/CD pipeline needs it)
gcloud sql instances patch cloudsql-test-pr-847 \
    --activation-policy=NEVER

# Clean up after PR closure
gcloud sql instances delete cloudsql-test-pr-847 --quiet --async

```

The storage cost for an idle Cloud SQL instance is approximately \$0.17 per GB per month. A 100 GB database clone sitting stopped costs about \$17/month in storage. Compare that to running even the smallest db-f1-micro with a 100 GB disk: you would pay both compute and storage. Stopping the clone after provisioning and starting it only for CI/CD runs dramatically reduces the cost of maintaining many parallel test environments.

For SQL Server environments where thin cloning is not natively available, the closest equivalent

is using Azure SQL Database Hyperscale, which provides rapid point-in-time restores (named database copies) without the storage overhead of full restores. Named replicas on Hyperscale databases can serve as isolated test environments that share the storage layer:

```
# Create a Hyperscale named replica for test isolation
az sql db replica create \
  --name staging-db-pr-847 \
  --resource-group rg-nonprod \
  --server sql-nonprod-server \
  --partner-database staging-db \
  --partner-server sql-nonprod-server \
  --service-objective Hyperscale \
  --replica-type Named
```

Named replicas on Hyperscale are read-only by default, which is suitable for read-heavy test workloads. For tests that need write capability, the full clone approach (restore to new database) remains necessary, but Hyperscale's shared storage architecture means the clone creation is dramatically faster than on standard tiers.

11.7 Governance: Preventing Non-Production Sprawl From Returning

Technical solutions for non-production cost reduction are undermined without governance controls that prevent the same patterns from recurring. The most common failure mode is this: a team implements schedules and right-sizing, achieves significant savings, and then six months later a new engineer provisions a `db.r6g.2xlarge` Multi-AZ instance for a new service's dev environment because that is what they saw in the existing Terraform modules.

The fix is encoding non-production constraints into the infrastructure-as-code layer rather than relying on manual review.

AWS Service Control Policies (SCPs) allow you to enforce instance class restrictions at the organization or OU level. Apply an SCP to your non-production OU that denies provisioning of instance classes above `db.r6g.xlarge` and denies `MultiAZ=true`:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyOversizedRDSInNonProd",
      "Effect": "Deny",
      "Action": [
        "rds:CreateDBInstance",
        "rds:ModifyDBInstance",
        "rds:RestoreDBInstanceFromDBSnapshot"
      ],
      "Resource": "*",
      "Condition": {
        "StringNotLike": {
          "rds:DatabaseClass": [
            "db.t3.*",
            "db.t4g.*",
            "db.r6g.large",
            "db.r6g.xlarge",
            "db.serverless"
          ]
        }
      }
    }
  ]
}
```

```

    },
    {
      "Sid": "DenyMultiAZInNonProd",
      "Effect": "Deny",
      "Action": [
        "rds:CreateDBInstance",
        "rds:ModifyDBInstance"
      ],
      "Resource": "*",
      "Condition": {
        "Bool": {
          "rds:MultiAz": "true"
        }
      }
    }
  ]
}

```

Apply this SCP to your `NonProduction` organizational unit in AWS Organizations. Any attempt to provision a large Multi-AZ dev instance—whether through the console, CLI, or Terraform—will receive an `Access Denied` before the instance is created.

Azure Policy provides the equivalent control for Azure SQL:

```

{
  "mode": "All",
  "policyRule": {
    "if": {
      "allOf": [
        {
          "field": "type",
          "equals": "Microsoft.Sql/servers/databases"
        },
        {
          "field": "tags['environment']",
          "in": ["dev", "qa", "staging"]
        },
        {
          "field": "Microsoft.Sql/servers/databases/sku.capacity",
          "greater": 4
        }
      ]
    },
    "then": {
      "effect": "Deny"
    }
  }
}

```

This policy denies creation of any Azure SQL database with more than 4 vCores if tagged as a non-production environment.

Terraform module enforcement is a complementary approach for teams that manage all infrastructure through Terraform. Create a shared non-production database module that hard-codes the guardrails:

```

# modules/rds-nonprod/main.tf
# This module enforces non-production constraints.
# Teams must use this module for dev/qa/staging databases.
# Production databases use a separate module without these restrictions.

variable "environment" {
  type        = string
  description = "Must be one of: dev, qa, staging"

  validation {
    condition = contains(["dev", "qa", "staging"], var.environment)
  }
}

```

```

        error_message = "Use the rds-production module for production environments."
    }
}

variable "instance_class" {
    type        = string
    description = "RDS instance class"
    default     = "db.t4g.large"

    validation {
        condition = can(regex("^db\\. (t3|t4g|r6g\\.large|r6g\\.xlarge|serverless)", var.instance_class))
        error_message = "Non-production instances must use t3, t4g, r6g.large, r6g.xlarge, or serverless
↪ classes."
    }
}

resource "aws_db_instance" "nonprod" {
    identifier      = "${var.environment}-${var.service_name}"
    instance_class  = var.instance_class
    multi_az        = false    # Hard-coded: no Multi-AZ in non-production
    deletion_protection = false # Hard-coded: no accidental protection in non-production

    # Enforce environment tag for scheduler Lambda to target
    tags = merge(var.additional_tags, {
        Environment = var.environment
        ManagedBy   = "terraform"
        SchedulerEnabled = "true"
    })

    # ... other required parameters passed through variables
}

```

When a team writes their Terraform and calls `module "dev_db" { source = "../.. /modules/rds-nonprod" }`, the guardrails are automatic. Attempting to set `multi_az = true` or an oversized instance class fails at `terraform plan` before anything reaches AWS.

Automated cost reporting closes the governance loop by making non-production spending visible weekly. A simple Lambda function that queries Cost Explorer and posts a summary to a Slack channel creates the social accountability that prevents drift:

```

import boto3
import json
import os
import urllib.request
from datetime import datetime, timedelta

def weekly_nonprod_cost_report(event, context):
    """
    Runs every Monday morning. Pulls last 7 days of RDS costs
    by environment tag and posts a summary to Slack.
    """
    ce = boto3.client('ce', region_name='us-east-1')

    end_date = datetime.utcnow().date()
    start_date = end_date - timedelta(days=7)

    response = ce.get_cost_and_usage(
        TimePeriod={
            'Start': start_date.isoformat(),
            'End': end_date.isoformat()
        },
        Granularity='WEEKLY',
        Filter={
            'Dimensions': {
                'Key': 'SERVICE',
                'Values': ['Amazon Relational Database Service']
            }
        },
    ),

```

```
        GroupBy=[
            {'Type': 'TAG', 'Key': 'Environment'}
        ],
        Metrics=['UnblendedCost']
    )

    lines = ["*Weekly Non-Production RDS Cost Report*"]
    total = 0.0

    for result in response['ResultsByTime']:
        for group in result['Groups']:
            env = group['Keys'][0].replace('Environment$', '') or 'untagged'
            cost = float(group['Metrics']['UnblendedCost']['Amount'])
            total += cost
            lines.append(f"    • {env}: ${cost:,.2f}")

    lines.append(f"\n*Total (7 days)*: ${total:,.2f}")
    lines.append(f"_Report period: {start_date} to {end_date}_")

    payload = json.dumps({'text': '\n'.join(lines)}).encode('utf-8')

    req = urllib.request.Request(
        os.environ['SLACK_WEBHOOK_URL'],
        data=payload,
        headers={'Content-Type': 'application/json'})
    urllib.request.urlopen(req)

    return {'statusCode': 200, 'body': 'Report sent'}
```

When engineers see a weekly message showing that the dev environment cost \$312 last week, it creates a natural prompt to revisit what is running. Costs that were previously invisible—aggregated into a large monthly AWS bill—become a team-level concern with clear ownership.

Key Takeaways

- **Non-production waste concentrates in three patterns:** always-on instances outside business hours, instance sizes mirroring production without matching workload profiles, and Multi-AZ configurations on environments that tolerate downtime. Diagnosing which pattern dominates before intervening prevents optimizing the wrong thing.
- **Stop/start scheduling alone delivers 60–70% compute cost reduction** for environments that run on a business-hours pattern. The AWS 7-day auto-restart limit requires scheduled automation rather than manual stops. Pair schedules with a self-service developer override to avoid friction without compromising governance.
- **Right-sizing non-production to burstable T-class instances** (for workloads with idle periods between CI/CD runs) and eliminating Multi-AZ can reduce per-instance costs by 75–85% compared to production-mirrored configurations. Use CloudWatch CPU credit metrics to verify that burstable instances have enough headroom before committing.
- **Connection pooling with PgBouncer in transaction mode** dissolves the false requirement for large RAM instances driven by connection count rather than data volume. A `t4g.nano` running PgBouncer at \$3/month can replace the operational need for a `db.r6g.2xlarge` at \$700/month when idle-connection overhead is the real driver of instance sizing decisions.

- **Governance through SCPs, Azure Policy, and Terraform module validation** prevents optimized environments from drifting back to oversized configurations as teams grow. Making non-production spending visible through weekly automated reports creates team-level accountability that sustains savings beyond the initial optimization effort.
-

Chapter 12: Vertical Scaling

Decisions: When to Go Up vs Optimize Queries

Teams that reflexively scale up their RDS or Cloud SQL instances when performance degrades spend an average of 34% more annually on database compute than teams that investigate first. A single vertical scaling decision — moving from an `db.r6g.2xlarge` (\$0.960/hr) to a `db.r6g.4xlarge` (\$1.920/hr) on AWS RDS — adds roughly \$8,410 per year before storage or I/O costs are counted. This chapter gives you a structured decision framework to determine whether your workload genuinely needs more compute or whether a targeted query optimization, index addition, connection pooling configuration, or parameter tuning change will recover the same performance for a fraction of the cost. Used consistently, this framework saves most teams between \$15,000 and \$80,000 annually on database compute alone.

12.1 Reading the Signals Correctly: Is This Really a Compute Problem?

The most expensive mistake in database operations is treating a symptom as a diagnosis. High CPU utilization looks like a compute problem. So does high memory pressure. But in most production databases, these conditions originate from query inefficiency, missing indexes, connection storms, or suboptimal buffer pool configuration — none of which are solved by adding vCPUs.

Before any vertical scaling decision moves forward, the following signals must be distinguished from each other.

CPU saturation from compute-bound work means the database is genuinely doing mathematical or procedural work that requires more processor cycles. Examples include complex analytical aggregations, window functions over large result sets, encryption and decryption operations, and procedural language execution. This is the rare case where vertical scaling helps.

CPU saturation from I/O wait means the processor is technically busy but most of that “business” is waiting for disk reads. The CPU metric is misleading — the actual constraint is storage throughput or IOPS. Scaling to a larger instance class does nothing because a larger instance waiting for the same slow disk is still waiting.

CPU saturation from query inefficiency is the most common pattern. A query that performs

a sequential scan on a 50-million-row table will consume CPU proportional to the rows it touches, not proportional to what it actually needs to return. Adding a single composite index can drop that query from 12 seconds to 40 milliseconds, reducing CPU utilization by 60% without touching instance size.

Memory pressure from working set size occurs when the database’s active data genuinely cannot fit in buffer pool or shared buffers, forcing repeated physical disk reads. This is a legitimate case for vertical scaling — but only after confirming that the buffer pool is tuned to use available memory and that no single runaway query is monopolizing the cache by scanning enormous tables repeatedly.

Memory pressure from connection count is entirely different. Each database connection on PostgreSQL consumes between 5 MB and 10 MB of process memory. One thousand connections consume 5 GB to 10 GB of RAM before a single query executes. This is a pgBouncer problem, not an instance size problem.

The following query identifies which category your current bottleneck falls into by correlating wait event data with query execution patterns.

```
-- PostgreSQL: Identify top wait events and correlate with specific queries
-- Run this against your RDS PostgreSQL or Aurora PostgreSQL instance
-- Requires pg_stat_activity and pg_stat_statements

WITH active_waits AS (
    SELECT
        wait_event_type,
        wait_event,
        COUNT(*) AS sessions_waiting,
        ROUND(COUNT(*) * 100.0 /
            NULLIF(SUM(COUNT(*)) OVER (), 0), 2) AS pct_of_sessions
    FROM pg_stat_activity
    WHERE state = 'active'
        AND wait_event IS NOT NULL
        AND pid <> pg_backend_pid()
    GROUP BY wait_event_type, wait_event
    ORDER BY sessions_waiting DESC
),
expensive_queries AS (
    SELECT
        LEFT(query, 100) AS query_preview,
        calls,
        ROUND((total_exec_time / calls)::numeric, 2) AS avg_ms,
        ROUND(total_exec_time::numeric, 2) AS total_ms,
        rows,
        ROUND((rows::numeric / NULLIF(calls, 0)), 0) AS avg_rows,
        shared_blks_hit,
        shared_blks_read,
        ROUND(
            shared_blks_read * 100.0 /
            NULLIF(shared_blks_hit + shared_blks_read, 0),
            2) AS cache_miss_pct
    FROM pg_stat_statements
    WHERE calls > 50
    ORDER BY total_exec_time DESC
    LIMIT 15
)
SELECT 'WAIT EVENTS' AS section, * FROM active_waits
UNION ALL
SELECT 'TOP QUERIES', NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL
UNION ALL
SELECT 'QUERY', query_preview, calls::text, avg_ms::text,
    total_ms::text, avg_rows::text, NULL, shared_blks_hit::text,
    shared_blks_read::text, cache_miss_pct::text, NULL
FROM expensive_queries;
```


Interpret the results as follows. If the dominant wait event type is IO (specifically `DataFileRead`), your bottleneck is storage, not compute. If it is `Lock` or `LWLock`, you have contention issues. If it is CPU with no wait events at all, compute pressure may be legitimate. If your top queries show cache miss percentages above 15%, your working set exceeds shared buffers, and you should consider memory before vCPUs.

For SQL Server on Azure SQL or RDS SQL Server, the equivalent diagnostic uses wait statistics:

```
-- SQL Server: Top wait stats indicating resource pressure type
-- Works on Azure SQL Managed Instance, RDS SQL Server, and self-managed
SELECT TOP 20
    wait_type,
    waiting_tasks_count,
    wait_time_ms,
    max_wait_time_ms,
    signal_wait_time_ms,
    -- signal wait > 25% of total suggests CPU saturation (real compute pressure)
    ROUND(signal_wait_time_ms * 100.0 / NULLIF(wait_time_ms, 0), 2)
        AS signal_wait_pct,

    -- Categories to help diagnosis
    CASE
        WHEN wait_type LIKE 'PAGEIO%' THEN 'Storage I/O Bottleneck'
        WHEN wait_type LIKE 'LATCH%' THEN 'Memory/Buffer Contention'
        WHEN wait_type LIKE 'LCK_%' THEN 'Lock Contention'
        WHEN wait_type = 'CXPACKET' THEN 'Parallelism (check MAXDOP)'
        WHEN wait_type = 'THREADPOOL' THEN 'Worker Thread Exhaustion'
        WHEN wait_type LIKE 'SOS_SCHEDULER%' THEN 'CPU Pressure (Legitimate)'
        WHEN wait_type LIKE 'ASYNC_IO%' THEN 'Async I/O Queue'
        ELSE 'Investigate'
    END
        AS diagnosis
FROM sys.dm_os_wait_stats
WHERE wait_type NOT IN (
    -- Filter out benign background waits
    'SLEEP_TASK', 'BROKER_TO_FLUSH', 'BROKER_TASK_STOP',
    'CLR_AUTO_EVENT', 'DISPATCHER_QUEUE_SEMAPHORE',
    'FT_IFTS_SCHEDULER_IDLE_WAIT', 'HADR_FILESTREAM_IOMGR_IOCOMPLETION',
    'HADR_WORK_QUEUE', 'LAZYWRITER_SLEEP', 'LOGMGR_QUEUE',
    'REQUEST_FOR_DEADLOCK_SEARCH', 'RESOURCE_QUEUE', 'SERVER_IDLE_CHECK',
    'SLEEP_DBSTARTUP', 'SLEEP_DCOMSTARTUP', 'SLEEP_MASTERDBREADY',
    'SLEEP_MASTERMDREADY', 'SLEEP_MASTERUPGRADED', 'SLEEP_MSDBSTARTUP',
    'SLEEP_SYSTEMTASK', 'SLEEP_TEMPDBSTARTUP', 'SNI_HTTP_ACCEPT',
    'SP_SERVER_DIAGNOSTICS_SLEEP', 'SQLTRACE_BUFFER_FLUSH',
    'WAITFOR', 'XE_DISPATCHER_WAIT', 'XE_TIMER_EVENT'
)
ORDER BY wait_time_ms DESC;
```

When `SOS_SCHEDULER_SLEEP` dominates and `signal_wait_pct` exceeds 25%, you have real CPU pressure. Everything else requires targeted remediation before instance resizing.

12.2 The Cost of Getting It Wrong: Pricing Math Across AWS, Azure, and GCP

Vertical scaling decisions are budget decisions. Before any discussion of remediation, the DBA and the engineering leadership need to understand the actual cost delta of scaling decisions. These numbers are not theoretical — they appear on your bill every month.

AWS RDS PostgreSQL pricing (us-east-1, Multi-AZ, On-Demand, 2024 rates):

Instance Class	vCPUs	RAM	\$/hr	\$/month	\$/year
db.t4g.medium	2	4 GB	\$0.068	\$50	\$595
db.r6g.large	2	16 GB	\$0.240	\$175	\$2,102
db.r6g.xlarge	4	32 GB	\$0.480	\$350	\$4,205
db.r6g.2xlarge	8	64 GB	\$0.960	\$701	\$8,409
db.r6g.4xlarge	16	128 GB	\$1.920	\$1,402	\$16,819
db.r6g.8xlarge	32	256 GB	\$3.840	\$2,803	\$33,638

Every step up the **r6g** ladder exactly doubles your compute cost. The question this chapter answers is: which of those steps actually buys you performance, and which ones are just buying comfort?

Azure SQL Database (General Purpose, 8 vCores vs 16 vCores, East US):

Moving from 8 vCores (\$1.1424/hr) to 16 vCores (\$2.2848/hr) costs \$9,989 more per year. Azure Hyperscale at 16 vCores runs \$14,965/year just in compute, before storage.

Google Cloud SQL PostgreSQL (us-central1):

Moving from `db-custom-8-32768` (\$0.5290/hr, 8 vCPUs, 32 GB) to `db-custom-16-65536` (\$1.0580/hr) costs \$4,638 more per year. Cloud SQL for PostgreSQL Enterprise Plus at 16 vCPUs approaches \$1.40/hr, or \$12,264/year.

These are the numbers that make query optimization worth the effort. A developer spending two weeks optimizing a slow query at \$200,000 annual fully-loaded salary costs roughly \$7,700. If that optimization prevents a vertical scaling event that would have cost \$9,000 per year, it pays for itself in under 13 months — and the savings compound every subsequent year.

The following script pulls your actual RDS spend over the last 90 days using the AWS Cost Explorer API and projects what your bill would look like if you could reduce average CPU utilization by 30%, which typically allows a downgrade of one instance size tier.

```
#!/bin/bash
# Script: rds_cost_analysis.sh
# Purpose: Pull RDS compute costs and model downgrade savings
# Prerequisites: AWS CLI configured, jq installed
# Schedule: Run weekly via cron or EventBridge

set -euo pipefail

REGION="${AWS_DEFAULT_REGION:-us-east-1}"
END_DATE=$(date +%Y-%m-%d)
START_DATE=$(date -d "90 days ago" +%Y-%m-%d 2>/dev/null || \
    date -v-90d +%Y-%m-%d) # macOS fallback

echo "=== RDS Cost Analysis: ${START_DATE} to ${END_DATE} ==="
echo ""

# Fetch RDS costs grouped by instance type tag
aws ce get-cost-and-usage \
    --time-period Start="${START_DATE}",End="${END_DATE}" \
    --granularity MONTHLY \
    --filter '{
        "Dimensions": {
            "Key": "SERVICE",
            "Values": ["Amazon Relational Database Service"]
        }
    }' \
    --group-by '[
        {"Type": "DIMENSION", "Key": "USAGE_TYPE"}
    ]'
```

```

] ' \
--metrics "BlendedCost" "UsageQuantity" \
--output json > /tmp/rds_costs_raw.json

echo "--- Top RDS Usage Types by Cost (Last 90 Days) ---"
jq -r '
  .ResultsByTime[].Groups[]
  | select(.Metrics.BlendedCost.Amount | tonumber > 10)
  | "\(.Keys[0])\t\(.Metrics.BlendedCost.Amount | tonumber | . * 100 | round / 100)"
' /tmp/rds_costs_raw.json | sort -t'$' -k2 -rn | head -20

# Fetch CloudWatch average CPU for all RDS instances
echo ""
echo "--- Average CPU Utilization by RDS Instance (Last 14 Days) ---"
aws rds describe-db-instances \
  --region "${REGION}" \
  --query 'DBInstances[*].[DBInstanceIdentifier,DBInstanceClass,MultiAZ,Engine]' \
  --output text | while IFS=$'\t' read -r db_id db_class multi_az engine; do

  CPU_AVG=$(aws cloudwatch get-metric-statistics \
    --namespace AWS/RDS \
    --metric-name CPUUtilization \
    --dimensions Name=DBInstanceIdentifier,Value="${db_id}" \
    --start-time "$(date -d '14 days ago' -Iseconds 2>/dev/null || \
      date -v-14d -Iseconds)" \
    --end-time "$(date -Iseconds)" \
    --period 86400 \
    --statistics Average \
    --query 'sort_by(Datapoints,&Timestamp)[-1].Average' \
    --output text 2>/dev/null || echo "N/A")

  # Flag instances below 25% average CPU as candidates for downsizing
  if [[ "${CPU_AVG}" != "N/A" && "${CPU_AVG}" != "None" ]]; then
    CPU_INT=$(echo "${CPU_AVG}" | awk '{printf "%.0f", $1}')
    FLAG=""
    if (( CPU_INT < 25 )); then
      FLAG=" <-- DOWNSIZE CANDIDATE"
    fi
    printf "%-40s %-20s %-8s %-10s CPU:%.1f%%\n" \
      "${db_id}" "${db_class}" "${multi_az}" "${engine}" \
      "${CPU_AVG}" "${FLAG}"
  fi
done

# Compute Optimizer recommendations
echo ""
echo "--- AWS Compute Optimizer RDS Recommendations ---"
aws compute-optimizer get-rds-database-recommendations \
  --region "${REGION}" \
  --output json 2>/dev/null | \
jq -r '
  .rdsDBRecommendations[]?
  | select(.finding == "Overprovisioned")
  | "\(.resourceArn |
    ↪ split(":")[6])\t\(.currentDBInstanceClass)\t-->\t\(.recommendationOptions[0].dbInstanceClass)\tSaving:
    ↪ $\(.recommendationOptions[0].estimatedMonthlySavings.value // "N/A")"
  ' | column -t || echo " (No Compute Optimizer recommendations available - ensure it is enabled)"

echo ""
echo "Analysis complete. Review DOWNSIZE CANDIDATES above before any instance changes."

```

Run this script weekly. When an instance appears consistently below 25% average CPU over 14 days, it is a downgrade candidate, not an upgrade candidate.

12.3 Query Optimization Before Vertical Scaling: The Technical Playbook

When wait event analysis reveals that CPU pressure stems from inefficient queries rather than from legitimate compute-bound work, the remediation sequence is: identify → analyze → fix → measure → decide. Vertical scaling remains an option throughout this process but is deferred until the optimization ceiling is reached.

Step 1: Identify the highest-cost queries by CPU contribution

PostgreSQL's `pg_stat_statements` extension (enabled by default on RDS PostgreSQL) provides cumulative execution statistics. The critical metric for CPU-driven cost is `total_exec_time` — this is wall-clock time the database spent on each query across all calls.

```
-- PostgreSQL: CPU cost ranking with optimization opportunity scoring
-- Reset stats periodically: SELECT pg_stat_statements_reset();
-- Run after collecting at least 1 hour of production traffic

SELECT
  pss.queryid,
  LEFT(pss.query, 120)                AS query_preview,
  pss.calls,
  ROUND(pss.total_exec_time::numeric / 1000, 2)
                                     AS total_exec_sec,
  ROUND((pss.total_exec_time / pss.calls)::numeric, 2)
                                     AS avg_exec_ms,
  ROUND(pss.rows::numeric / NULLIF(pss.calls, 0), 0)
                                     AS avg_rows_returned,

  -- Estimated CPU cost contribution as % of all query time
  ROUND(pss.total_exec_time * 100.0 /
    SUM(pss.total_exec_time) OVER (), 2) AS pct_total_cpu,

  -- Buffer efficiency: high read ratio = missing index or large scan
  pss.shared_blks_hit + pss.shared_blks_read AS total_blks_accessed,
  ROUND(pss.shared_blks_read * 100.0 /
    NULLIF(pss.shared_blks_hit + pss.shared_blks_read, 0), 2)
                                     AS physical_read_pct,

  -- Rows vs blocks: low rows per block access = inefficient access pattern
  CASE
    WHEN pss.shared_blks_read > 10000
      AND pss.rows < pss.shared_blks_read * 5
      THEN 'MISSING INDEX LIKELY'
    WHEN pss.total_exec_time / NULLIF(pss.calls, 0) > 5000
      AND pss.rows < 1000
      THEN 'SINGLE QUERY VERY SLOW'
    WHEN pss.calls > 10000
      AND pss.total_exec_time / NULLIF(pss.calls, 0) > 100
      THEN 'HIGH FREQUENCY + SLOW'
    WHEN pss.total_exec_time * 100.0 /
      NULLIF(SUM(pss.total_exec_time) OVER (), 0) > 10
      THEN 'HIGH TOTAL CPU SHARE'
    ELSE 'REVIEW IF IN TOP 10'
  END
                                     AS optimization_signal,

  -- Temp file usage indicates sort/hash spill to disk
  pss.temp_blks_written,
  CASE WHEN pss.temp_blks_written > 1000
    THEN 'SORT SPILL - TUNE work_mem'
    ELSE '' END
                                     AS memory_signal

FROM pg_stat_statements pss
JOIN pg_database pd ON pd.oid = pss.dbid
WHERE pd.datname = current_database()
AND pss.calls >= 10
```

```

AND pss.total_exec_time > 1000 -- At least 1 second total execution time
ORDER BY pss.total_exec_time DESC
LIMIT 25;

```

Step 2: Analyze execution plans for the top offenders

Once the top queries are identified, execution plan analysis reveals what is actually happening. The key markers in PostgreSQL EXPLAIN (ANALYZE, BUFFERS) output that signal optimization headroom rather than compute need:

- Seq Scan on a large table where a selective WHERE clause exists → missing index
- Sort operations with large Memory Usage values exceeding work_mem → increasing work_mem for that session or globally can eliminate disk spills
- Hash Join with very large Batches: N values → hash spill to disk, increase work_mem
- Nested loop joins over large sets → join order or index issues
- High rows=X estimates that are wildly inaccurate compared to actual rows=Y → stale statistics, run ANALYZE

Step 3: Apply targeted fixes in order of effort-to-impact

The following sequence resolves the majority of CPU pressure issues without instance resizing, ordered from fastest to most time-intensive:

1. **Update statistics** — ANALYZE tablename or ANALYZE VERBOSE. Free, takes seconds, can immediately improve plan quality. On RDS, autovacuum handles this but may lag behind rapid data changes.
2. **Add missing indexes** — Use the query from Step 1 to find high physical_read_pct queries, then EXPLAIN ANALYZE to confirm sequential scans on selective predicates. Creating a CONCURRENTLY index has zero downtime impact on PostgreSQL.
3. **Tune work_mem** — Queries with temp_blks_written > 0 are spilling sorts or hash joins to disk. SET work_mem = '64MB' at the session level for specific heavy queries, or set it globally if most connections run large analytical queries. Caution: work_mem is per-sort-operation per connection, so multiplying by connection count before setting globally is necessary to avoid OOM.
4. **Connection pooling via pgBouncer** — If memory pressure is connection-driven (see 12.1), deploying pgBouncer in transaction mode in front of your RDS instance can reduce active backend connections from 800 to 50 while handling identical throughput. This frequently recovers 4–8 GB of RAM, which may entirely eliminate the need for a memory upgrade.
5. **Rewrite pathological queries** — Queries returning 500,000 rows to an application that filters them in code should be rewritten to push that filter into the database. This is a development task but has the highest ceiling for improvement.

For Azure SQL, the equivalent identification query uses sys.dm_exec_query_stats:

```

-- Azure SQL / SQL Server: Top CPU-consuming queries with plan linkage
-- Works on Azure SQL Database, Managed Instance, RDS SQL Server

SELECT TOP 20
    qs.total_worker_time / 1000                AS total_cpu_ms,
    qs.execution_count,
    ROUND(qs.total_worker_time * 1.0 /
        NULLIF(qs.execution_count, 0) / 1000, 2)
        AS avg_cpu_ms,

```

```

qs.total_logical_reads / NULLIF(qs.execution_count, 0)
                                AS avg_logical_reads,
qs.total_physical_reads / NULLIF(qs.execution_count, 0)
                                AS avg_physical_reads,
ROUND(qs.total_worker_time * 100.0 /
      NULLIF(SUM(qs.total_worker_time) OVER (), 0), 2)
                                AS pct_total_cpu,
SUBSTRING(qt.text, (qs.statement_start_offset / 2) + 1,
          (CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)
            ELSE qs.statement_end_offset
            END - qs.statement_start_offset) / 2 + 1)
                                AS query_text,
qp.query_plan,
-- Optimization signals
CASE
  WHEN qs.total_physical_reads > qs.total_logical_reads * 0.20
    THEN 'HIGH PHYSICAL READS - CHECK INDEXES'
  WHEN qs.total_worker_time * 1.0 / NULLIF(qs.execution_count, 0) > 5000000
    THEN 'SINGLE EXECUTION VERY EXPENSIVE'
  WHEN qs.execution_count > 10000
    AND qs.total_worker_time * 1.0 / NULLIF(qs.execution_count, 0) > 100000
    THEN 'HIGH FREQUENCY + HIGH CPU'
  WHEN qs.total_worker_time * 100.0 /
    NULLIF(SUM(qs.total_worker_time) OVER (), 0) > 10
    THEN 'DOMINANT CPU CONSUMER'
  ELSE 'REVIEW IF IN TOP 10'
END
                                AS optimization_signal
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
OUTER APPLY sys.dm_exec_query_plan(qs.plan_handle) qp
WHERE qs.total_worker_time > 0
ORDER BY qs.total_worker_time DESC;

```

For SQL Server and Azure SQL, the `query_plan` column returns an XML execution plan you can paste directly into SQL Server Management Studio or Azure Data Studio to see graphical plan output. Focus on operators with the highest cost percentage — specifically Table Scan or Clustered Index Scan operators on large tables with selective `WHERE` clauses, and Sort operators that appear without a supporting index.

Step 4: Index creation without downtime

On PostgreSQL, creating an index concurrently prevents table locks but takes longer and consumes more CPU during the build. On production RDS PostgreSQL instances, always use `CONCURRENTLY`:

```

-- PostgreSQL: Safe concurrent index creation for production RDS instances
-- Does NOT acquire a full table lock
-- Takes longer but allows reads and writes to continue during build
-- Monitor progress in pg_stat_progress_create_index

-- First, confirm the index is needed by checking existing indexes
SELECT
  t.relname
                                AS table_name,
  i.relname
                                AS index_name,
  ix.indisunique,
  ix.indisprimary,
  array_agg(a.attname ORDER BY x.n)
                                AS indexed_columns
FROM pg_class t
JOIN pg_index ix ON t.oid = ix.indrelid
JOIN pg_class i ON i.oid = ix.indexrelid
JOIN pg_attribute a ON a.attrelid = t.oid
  AND a.attnum = ANY(ix.indkey)
CROSS JOIN LATERAL unnest(ix.indkey) WITH ORDINALITY AS x(attnum, n)
WHERE t.relname = 'your_table_name'
  AND a.attnum = x.attnum
GROUP BY t.relname, i.relname, ix.indisunique, ix.indisprimary
ORDER BY i.relname;

```

```
-- Check progress of any running index builds
SELECT
    phase,
    blocks_done,
    blocks_total,
    ROUND(blocks_done * 100.0 / NULLIF(blocks_total, 0), 1) AS pct_complete,
    tuples_done,
    tuples_total
FROM pg_stat_progress_create_index;

-- Create the index concurrently (run outside a transaction block)
CREATE INDEX CONCURRENTLY IF NOT EXISTS
    idx_orders_customer_created
ON orders (customer_id, created_at)
WHERE status != 'cancelled'; -- Partial index reduces size if applicable
```

On SQL Server and Azure SQL, the equivalent online index creation uses WITH (ONLINE = ON):

```
-- SQL Server / Azure SQL: Online index creation without blocking reads/writes
-- ONLINE = ON is available on Standard tier and above
-- MAXDOP controls CPU usage during build to avoid resource competition

CREATE INDEX idx_orders_customer_created
ON dbo.orders (customer_id, created_at)
INCLUDE (order_total, status) -- Covering index avoids key lookups
WITH (
    ONLINE = ON, -- No exclusive table lock
    MAXDOP = 4, -- Limit CPU consumption during build
    FILLFACTOR = 90, -- Leave room for inserts
    DATA_COMPRESSION = ROW -- Optional: reduce storage footprint
);

-- Monitor progress on SQL Server 2014+ / Azure SQL
SELECT
    operation,
    state_desc,
    percent_complete,
    start_time,
    DATEDIFF(SECOND, start_time, GETDATE()) AS elapsed_seconds
FROM sys.dm_exec_background_job_queue_stats;

-- Or use this for online index operations specifically
SELECT
    object_name(object_id) AS table_name,
    index_id,
    partition_number,
    operation_type_desc,
    state_desc,
    percent_complete
FROM sys.dm_db_online_index_stats
WHERE state = 1; -- In progress
```

12.4 Parameter Tuning: Recovering Performance Without Hardware

Several database parameters directly affect CPU and memory utilization in ways that can eliminate false scaling pressure. These changes are lower risk than query rewrites and often faster than index creation. They should be evaluated before any vertical scaling decision is finalized.

PostgreSQL parameters with the highest cost impact on RDS:

`shared_buffers` is the most impactful parameter for memory-related CPU pressure. The default

on RDS PostgreSQL is set to 25% of instance memory, which is appropriate for most workloads. However, on dedicated database instances where the RDS host runs only PostgreSQL (as is standard), this can safely be increased to 40% of RAM. For a `db.r6g.2xlarge` with 64 GB RAM, moving from the default 16 GB to 25 GB of shared buffers increases the amount of hot data held in memory, reducing physical reads and the associated CPU spent on buffer management.

On RDS, `shared_buffers` cannot be set directly in bytes — it is controlled via the `rds.instance.memory` dynamic formula in parameter groups. The correct approach is to modify the parameter group:

```
-- After applying the RDS parameter group change, verify the active setting:
SHOW shared_buffers;

-- Also check effective_cache_size (used by the planner, not actual allocation)
-- This should reflect total available memory including OS cache
-- Set to 75% of instance RAM in your parameter group
SHOW effective_cache_size;

-- Verify work_mem (per sort operation, per connection -- set conservatively)
SHOW work_mem;

-- View all non-default settings currently active
SELECT
    name,
    setting,
    unit,
    source,
    short_desc
FROM pg_settings
WHERE source NOT IN ('default', 'override')
ORDER BY name;
```

`work_mem` deserves special attention because it interacts directly with sort and hash join operations that contribute to CPU spikes. The formula for safe global `work_mem` is:

$$\text{safe_work_mem} = (\text{available_RAM} \times 0.25) / (\text{max_connections} \times \text{max_parallel_workers_per_gather})$$

For a `db.r6g.xlarge` with 32 GB RAM, 200 max connections, and 2 parallel workers per query:

$$\text{safe_work_mem} = (32,000 \text{ MB} \times 0.25) / (200 \times 2) = 8,000 \text{ MB} / 400 = 20 \text{ MB}$$

Setting `work_mem = 20MB` globally ensures that sort spills to disk are minimized without risking out-of-memory conditions. For specific analytical queries that need more, set it at the session level:

```
-- PostgreSQL: Session-level work_mem override for analytical queries
-- This does NOT affect other sessions and resets on disconnect

SET work_mem = '256MB';

-- Your analytical query here
SELECT
    region,
    product_category,
    SUM(revenue) AS total_revenue,
    COUNT(DISTINCT customer_id) AS unique_customers
FROM sales_facts
GROUP BY region, product_category
ORDER BY total_revenue DESC;

-- Reset if continuing with OLTP queries in same session
RESET work_mem;
```

SQL Server parameters with the highest cost impact on Azure SQL and RDS SQL Server:

`max server memory` is the SQL Server equivalent of `shared_buffers` sizing. On RDS SQL Server, this is managed through parameter groups. On Azure SQL, the memory allocation is automatic within the service tier. For self-managed SQL Server instances, setting `max server memory` to leave 10–15% of RAM for the OS and 5–10% for connection overhead is the standard guidance — but the concrete action is to confirm it is not set to the default unlimited value, which allows SQL Server to consume all available memory and cause OS paging.

MAXDOP (maximum degree of parallelism) is one of the most commonly misconfigured parameters in SQL Server. When CXPACKET waits appear in your wait stats analysis from Section 12.1, the first action is to tune MAXDOP, not to scale up. For OLTP workloads, MAXDOP = 1 or MAXDOP = 4 is appropriate. For mixed workloads, use Resource Governor (on-premises/MI) or workload groups (Azure SQL) to differentiate:

```
-- SQL Server / Azure SQL: Check and set optimal MAXDOP
-- Microsoft's formula: use number of physical cores, capped at 8 for OLTP

-- Check current setting
SELECT
    name,
    value_in_use,
    description
FROM sys.configurations
WHERE name = 'max degree of parallelism';

-- Check NUMA topology to inform MAXDOP setting
SELECT
    cpu_count,
    hyperthread_ratio,
    cpu_count / hyperthread_ratio AS physical_cores,
    numa_node_count,
    scheduler_count
FROM sys.dm_os_sys_info;

-- Set MAXDOP for OLTP-dominant workloads (RDS SQL Server / on-premises)
-- For Azure SQL Database, use ALTER DATABASE SCOPED CONFIGURATION
EXEC sp_configure 'max degree of parallelism', 4;
RECONFIGURE WITH OVERRIDE;

-- Azure SQL Database equivalent (no sp_configure access):
ALTER DATABASE CURRENT
SET MAXDOP = 4;

-- Verify the change is active
SELECT
    name,
    value,
    value_in_use
FROM sys.configurations
WHERE name = 'max degree of parallelism';
```

Cost Threshold for Parallelism is a frequently overlooked parameter that controls when SQL Server chooses a parallel plan. The default value of 5 (meaning any query estimated to cost more than 5 units triggers parallelism) was set in the 1990s when hardware was dramatically slower. On modern multi-core instances, this value is far too low and causes many simple queries to spawn parallel threads unnecessarily, contributing to CXPACKET waits and artificial CPU pressure. Raising it to a value between 40 and 60 is appropriate for most OLTP workloads:

```
-- SQL Server: Raise cost threshold for parallelism
-- Default 5 is too aggressive on modern hardware
-- Values of 40-60 significantly reduce unnecessary parallelism for OLTP

EXEC sp_configure 'cost threshold for parallelism', 50;
RECONFIGURE WITH OVERRIDE;
```

```
-- Monitor the effect on CXPACKET waits after this change
-- (Allow at least 30 minutes of production load, then re-run wait stats query)
SELECT
    wait_type,
    waiting_tasks_count,
    wait_time_ms,
    ROUND(wait_time_ms * 100.0 /
        NULLIF(SUM(wait_time_ms) OVER (), 0), 2) AS pct_total_wait
FROM sys.dm_os_wait_stats
WHERE wait_type IN ('CXPACKET', 'SOS_SCHEDULER_SLEEP', 'PAGEIOLATCH_SH')
ORDER BY wait_time_ms DESC;
```

12.5 Connection Pooling: The Hidden Vertical Scaling Driver

Connection management is responsible for a disproportionate share of false vertical scaling decisions. The pattern is consistent: application connections grow, memory utilization climbs, someone observes the memory metric and escalates to a larger instance. The larger instance has more RAM, the memory metric looks healthy for several weeks, connections grow again, and the cycle repeats. The actual fix — connection pooling — would have cost nothing.

Each PostgreSQL backend process consumes memory not just for the connection itself but for:

- Shared catalog cache per backend (several hundred KB)
- Per-backend memory context overhead
- Stack space for the backend process - `work_mem` allocation when queries execute sorts or hash joins
- Lock table entries proportional to objects accessed

The aggregate overhead per idle connection is typically 5–10 MB. The overhead per active connection running a complex query can reach 50–100 MB temporarily. On a `db.r6g.xlarge` with 32 GB of RAM configured with `max_connections = 500`, the theoretical maximum memory consumed by connections alone approaches 25 GB before any data is cached in `shared_buffers`. This is the definition of a misconfiguration, not a hardware limitation.

RDS PostgreSQL connection pooling with pgBouncer on EC2:

```
# pgbouncer.ini
# Deploy pgBouncer on a small EC2 instance (t4g.small is sufficient for most workloads)
# and point application connection strings here instead of directly at RDS

[databases]
# Replace <RDS_ENDPOINT> with your actual RDS hostname
# Replace <DB_NAME> with your database name
myapp_db = host=<RDS_ENDPOINT> port=5432 dbname=<DB_NAME>

[pgbouncer]
listen_port      = 5432
listen_addr      = 0.0.0.0

# Transaction mode: connection to RDS held only during transaction
# This is the most aggressive pooling mode and most resource-efficient
# Use session mode only if your app uses session-level features (SET, temp tables)
pool_mode        = transaction

# Maximum connections pgBouncer will maintain TO the RDS instance
# Set this to the maximum your RDS instance can handle efficiently
# Rule: max_connections on RDS = 2-4x vCPUs for OLTP
server_pool_size  = 40

# Maximum client connections (applications → pgBouncer)
# This can be very large - pgBouncer handles these cheaply
max_client_conn   = 1000
```

```
# Authentication: use md5 or scram-sha-256 matching your RDS pg_hba.conf
auth_type      = scram-sha-256
auth_file      = /etc/pgbouncer/userlist.txt

# Connection lifetime management
server_idle_timeout = 600    # Close idle RDS connections after 10 min
server_lifetime    = 3600    # Recycle RDS connections hourly
client_idle_timeout = 300    # Disconnect idle clients after 5 min

# Monitoring
stats_period      = 60
logfile           = /var/log/pgbouncer/pgbouncer.log
pidfile           = /var/run/pgbouncer/pgbouncer.pid
```

With this configuration, 800 application connections flow through pgBouncer, which maintains only 40 connections to RDS. The RDS instance’s memory pressure from connection overhead drops from roughly 4–8 GB to under 400 MB. In many cases this alone recovers enough memory that a scheduled instance upsize becomes unnecessary.

Monitoring pgBouncer effectiveness from within the pool:

```
-- Connect to pgBouncer admin database to inspect pool stats
-- psql -h <pgbouncer_host> -p 5432 -U pgbouncer pgbouncer

-- Pool status: confirm server connections are well below client connections
SHOW POOLS;

-- Per-database statistics: requests/sec, avg query time, wait time
SHOW STATS;

-- Identify if clients are waiting for server connections (indicates pool too small)
SHOW CLIENTS;

-- If avg_wait_time in SHOW STATS is consistently above 5ms, increase server_pool_size
-- If all server connections are always active, the pool may be undersized
```

For Azure SQL, the equivalent connection pooling is handled at the application layer through ADO.NET connection pooling (built in to the .NET stack), HikariCP for Java workloads, or PgBouncer-equivalent tools like **PgPool-II** or **RDS Proxy** on AWS. Azure SQL itself manages connections more efficiently than PostgreSQL at the engine level, but the same principle applies: each application tier instance maintaining its own pool of 20 connections multiplied by 50 application instances creates 1,000 backend threads that Azure SQL must manage, and consolidating through a proxy layer reduces this to a controlled pool.

12.6 The Vertical Scaling Decision Framework

After completing the diagnostic and optimization steps in sections 12.1 through 12.5, you have enough information to make a structured decision. This section provides the explicit decision tree, the metrics thresholds that justify scaling, and the documentation practices that turn individual decisions into organizational knowledge.

The decision tree:

START: Performance complaint received (slow queries, timeouts, high CPU/memory metrics)

Step 1: Run wait event analysis (Section 12.1)

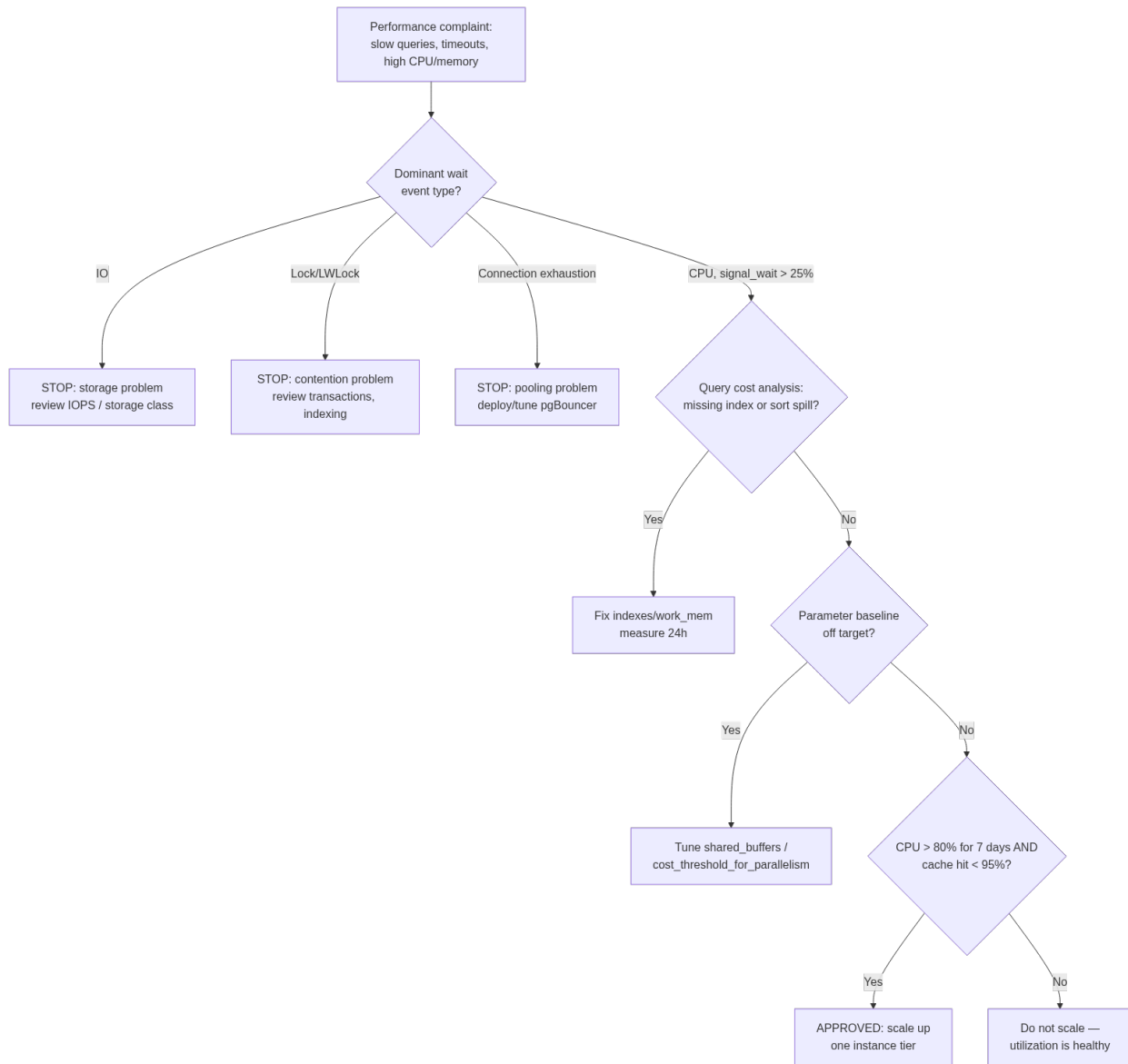


Figure 12: The vertical-scaling decision tree — every path except the last one resolves without touching instance size.

Dominant wait: IO (DataFileRead / PAGEIOLATCH_SH)

STOP - this is a storage problem, not a compute problem

Action: Review IOPS provisioning, gp3 vs io1/io2, or storage autoscaling

Dominant wait: Lock / LWLock / LCK_*

STOP - this is a contention problem, not a compute problem

Action: Review long-running transactions, lock escalation, indexing

Dominant wait: Connection exhaustion / THREADPOOL

STOP - this is a pooling problem, not a compute problem

Action: Deploy or tune pgBouncer / RDS Proxy (Section 12.5)

Dominant wait: CPU / SOS_SCHEDULER (signal_wait_pct > 25%)

Continue to Step 2

Step 2: Run query cost analysis (Section 12.3)

Found queries with MISSING INDEX LIKELY or HIGH PHYSICAL READS?

Action: Create indexes (Section 12.3, Step 4)

Measure CPU after 24 hours. If resolved, DONE. If not, continue.

Found queries with SORT SPILL (temp_blks_written > 0)?

Action: Tune work_mem (Section 12.4)

Measure CPU after 24 hours. If resolved, DONE. If not, continue.

All queries look efficient; CPU pressure persists

Continue to Step 3

Step 3: Verify parameter baseline (Section 12.4)

shared_buffers < 30% of RAM? → Increase to 35-40% (PostgreSQL)

CXPACKET dominant? → Raise cost_threshold_for_parallelism to 50 (SQL Server)

Changes made? Measure for 24 hours. If resolved, DONE. If not, continue.

Step 4: Evaluate vertical scaling

CPU > 80% average over 7 consecutive days after all above steps?

Memory: buffer pool hit rate < 95% after shared_buffers tuning?

If BOTH conditions are true AND optimization ceiling is reached:

APPROVED: Scale up one instance tier.

Document: what was attempted, metrics before/after each fix,
cost delta of scaling decision, projected review date.

Metrics thresholds that justify vertical scaling:

The following numeric thresholds represent the criteria where scaling is justified after optimization efforts have been exhausted. Use these as guardrails, not triggers — a single spike does not meet these criteria.

Metric	Threshold for Scale-Up Consideration
CPU Utilization (7-day average)	> 75% sustained after query optimization
Buffer/Cache Hit Rate	< 95% after <code>shared_buffers</code> tuning
Average Query Latency	> 2× baseline after index and parameter changes
Connection Utilization	> 85% of <code>max_connections</code> after pooling
Freeable Memory (RDS)	Consistently < 10% of instance RAM

No single metric justifies scaling in isolation. CPU at 80% with a 99% cache hit rate and millisecond query latencies does not need a larger instance — it may simply reflect healthy utilization.

Documenting the decision:

Every vertical scaling event should produce a one-page record in your team’s runbook or wiki. The template below should be filled in before the change is made, not after:

```
## Vertical Scaling Decision Record
**Date:** YYYY-MM-DD
**Instance:** <identifier>  **Cloud:** AWS / Azure / GCP
**Change:** <current class> → <new class>
**Annual cost delta:** $X,XXX/year

### Optimization steps attempted before scaling
1. Wait event analysis - finding: [describe dominant waits]
2. Query optimization - [indexes added, queries rewritten, results]
3. Parameter tuning - [parameters changed, effect measured]
4. Connection pooling - [current state, any changes made]

### Metrics before scaling (7-day averages)
- CPU utilization: XX%
- Cache hit rate: XX%
- Average query latency: XXms
- Active connections: XXX / YYY max

### Justification
[One paragraph: why the optimization ceiling was reached and scaling is the correct next step]

### Review date
[Schedule a review 30 days after the change to confirm the scaling was effective
and re-evaluate whether the previous tier can be restored with further optimization]
```

This documentation discipline creates two benefits. First, it forces the decision-maker to confirm that the optimization steps were actually completed. Second, it accumulates institutional knowledge about which workloads on your specific database instances responded to which interventions, which accelerates future diagnostics.

12.7 Automating the Decision with CloudWatch and Cost Anomaly Detection

Manual diagnosis is reliable but slow. A performance incident at 2 AM cannot wait for a developer to run `pg_stat_statements` queries at 9 AM. The following automation layer provides early detection of conditions that indicate either optimization opportunities or legitimate scaling needs, and routes the right alert to the right action.

CloudWatch composite alarms for RDS scaling decisions:

```
#!/usr/bin/env python3
"""
rds_scaling_alarm_stack.py
Creates a composite CloudWatch alarm infrastructure that distinguishes
between optimization-addressable conditions and genuine scaling needs.

Deploy with: python3 rds_scaling_alarm_stack.py --instance-id <your-rds-instance>
Prerequisites: boto3, AWS credentials with CloudWatch and SNS permissions
"""

import boto3
import argparse
import json
from datetime import datetime

def create_scaling_alarm_stack(instance_id: str, sns_topic_arn: str,
                              region: str = 'us-east-1') -> dict:
    cw = boto3.client('cloudwatch', region_name=region)
    results = {}

    # --- Individual metric alarms ---

    # CPU alarm: 75% average over 30 minutes (5 x 5-minute periods)
    cw.put_metric_alarm(
        AlarmName=f'rds-{instance_id}-cpu-high',
        AlarmDescription='RDS CPU above 75% for 30 consecutive minutes',
        MetricName='CPUUtilization',
        Namespace='AWS/RDS',
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
        Period=300,
        EvaluationPeriods=6,
        DatapointsToAlarm=5,          # 5 of 6 must breach (avoids single-spike alerts)
        Threshold=75.0,
        ComparisonOperator='GreaterThanThreshold',
        Statistic='Average',
        TreatMissingData='notBreaching'
    )

    # Freeable memory alarm: below 2 GB (adjust for instance size)
    cw.put_metric_alarm(
        AlarmName=f'rds-{instance_id}-memory-low',
        AlarmDescription='RDS freeable memory below 2GB for 20 minutes',
        MetricName='FreeableMemory',
        Namespace='AWS/RDS',
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
        Period=300,
        EvaluationPeriods=4,
        DatapointsToAlarm=3,
        Threshold=2_147_483_648,      # 2 GB in bytes
        ComparisonOperator='LessThanThreshold',
        Statistic='Average',
        TreatMissingData='notBreaching'
    )

    # Database connections alarm: above 80% of max_connections
    # Threshold must be set per-instance based on actual max_connections value
    cw.put_metric_alarm(
        AlarmName=f'rds-{instance_id}-connections-high',
        AlarmDescription='RDS connections approaching limit - pooling may be needed',
        MetricName='DatabaseConnections',
        Namespace='AWS/RDS',
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
        Period=300,
        EvaluationPeriods=3,
        DatapointsToAlarm=2,
        Threshold=400,                # Adjust: 80% of your max_connections setting
        ComparisonOperator='GreaterThanThreshold',
        Statistic='Maximum',
    )
```

```

        TreatMissingData='notBreaching'
    )

    # Read IOPS alarm: detects storage bottleneck masquerading as CPU pressure
    cw.put_metric_alarm(
        AlarmName=f'rds-{instance_id}-read-iops-high',
        AlarmDescription='RDS read IOPS high - verify storage is not the bottleneck',
        MetricName='ReadIOPS',
        Namespace='AWS/RDS',
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
        Period=300,
        EvaluationPeriods=4,
        DatapointsToAlarm=3,
        Threshold=3000,          # Adjust based on provisioned IOPS for your instance
        ComparisonOperator='GreaterThanThreshold',
        Statistic='Average',
        TreatMissingData='notBreaching'
    )

    # --- Composite alarm: "Investigate before scaling" ---
    # Fires when CPU is high BUT connections are also high
    # This pattern indicates connection storm, not compute need
    cw.put_composite_alarm(
        AlarmName=f'rds-{instance_id}-connection-storm',
        AlarmDescription=(
            'CPU high AND connections high - likely connection storm. '
            'Deploy pgBouncer before considering vertical scale.'
        ),
        AlarmRule=(
            f'ALARM("rds-{instance_id}-cpu-high") AND '
            f'ALARM("rds-{instance_id}-connections-high")'
        ),
        AlarmActions=[sns_topic_arn],
        OKActions=[sns_topic_arn],
        TreatMissingData='notBreaching'
    )

    # --- Composite alarm: "Legitimate scaling candidate" ---
    # Fires when CPU is high AND memory is low AND connections are NOT high
    # This pattern suggests genuine compute/memory pressure
    cw.put_composite_alarm(
        AlarmName=f'rds-{instance_id}-scaling-candidate',
        AlarmDescription=(
            'CPU high AND memory low AND connections normal - '
            'potential legitimate case for vertical scaling. '
            'Complete optimization checklist before acting.'
        ),
        AlarmRule=(
            f'ALARM("rds-{instance_id}-cpu-high") AND '
            f'ALARM("rds-{instance_id}-memory-low") AND '
            f'NOT ALARM("rds-{instance_id}-connections-high")'
        ),
        AlarmActions=[sns_topic_arn],
        OKActions=[sns_topic_arn],
        TreatMissingData='notBreaching'
    )

    print(f"Alarm stack created for {instance_id}")
    print(f"  Connection storm alarm:  rds-{instance_id}-connection-storm")
    print(f"  Scaling candidate alarm:  rds-{instance_id}-scaling-candidate")
    print(f"  SNS topic:                    {sns_topic_arn}")
    return results

def main():
    parser = argparse.ArgumentParser(
        description='Create RDS scaling decision alarm stack')
    parser.add_argument('--instance-id', required=True,
                        help='RDS DB instance identifier')
    parser.add_argument('--sns-topic-arn', required=True,

```



```

        help='SNS topic ARN for alarm notifications')
    parser.add_argument('--region', default='us-east-1',
                        help='AWS region (default: us-east-1)')
    args = parser.parse_args()

    create_scaling_alarm_stack(
        instance_id=args.instance_id,
        sns_topic_arn=args.sns_topic_arn,
        region=args.region
    )

if __name__ == '__main__':
    main()

```

Connecting alarms to automated runbooks via EventBridge and Lambda:

When the `connection-storm` composite alarm fires, the correct response is not manual — it is automatic notification to the on-call engineer with a pre-filled diagnostic report. The following Lambda function generates that report when triggered by an SNS message from the alarm:

```

#!/usr/bin/env python3
"""
rds_diagnostic_lambda.py
Lambda function triggered by CloudWatch alarm SNS notification.
Generates a diagnostic pre-report and posts it to a Slack webhook.
Environment variables required:
- SLACK_WEBHOOK_URL
- DB_HOST (RDS endpoint)
- DB_NAME
- DB_USER
- DB_PASSWORD_SECRET_ARN (Secrets Manager ARN)
"""

import json
import os
import urllib.request
import boto3
import psycpg2

def get_db_password(secret_arn: str) -> str:
    sm = boto3.client('secretsmanager')
    response = sm.get_secret_value(SecretId=secret_arn)
    secret = json.loads(response['SecretString'])
    return secret['password']

def get_top_wait_events(conn) -> list:
    with conn.cursor() as cur:
        cur.execute("""
            SELECT
                wait_event_type,
                wait_event,
                COUNT(*) AS sessions
            FROM pg_stat_activity
            WHERE state = 'active'
            AND wait_event IS NOT NULL
            AND pid <> pg_backend_pid()
            GROUP BY wait_event_type, wait_event
            ORDER BY sessions DESC
            LIMIT 5
        """)
    return cur.fetchall()

def get_connection_count(conn) -> dict:
    with conn.cursor() as cur:
        cur.execute("""
            SELECT

```

```

        COUNT(*)                AS total,
        SUM(CASE WHEN state = 'active' THEN 1 ELSE 0 END) AS active,
        SUM(CASE WHEN state = 'idle' THEN 1 ELSE 0 END) AS idle,
        SUM(CASE WHEN wait_event_type = 'Lock'
              THEN 1 ELSE 0 END) AS waiting_on_lock
    FROM pg_stat_activity
    WHERE pid <> pg_backend_pid()
    """
    row = cur.fetchone()
    return {
        'total': row[0], 'active': row[1],
        'idle': row[2], 'waiting_on_lock': row[3]
    }

def get_top_slow_queries(conn) -> list:
    with conn.cursor() as cur:
        cur.execute("""
            SELECT
                LEFT(query, 80) AS q,
                calls,
                ROUND((total_exec_time / calls)::numeric, 1) AS avg_ms
            FROM pg_stat_statements
            WHERE calls > 10
            ORDER BY total_exec_time DESC
            LIMIT 3
        """)
    return cur.fetchall()

def post_to_slack(webhook_url: str, message: dict) -> None:
    data = json.dumps(message).encode('utf-8')
    req = urllib.request.Request(
        webhook_url,
        data=data,
        headers={'Content-Type': 'application/json'})
    urllib.request.urlopen(req)

def lambda_handler(event, context):
    # Parse the SNS message to get alarm details
    sns_message = json.loads(event['Records'][0]['Sns']['Message'])
    alarm_name = sns_message.get('AlarmName', 'Unknown Alarm')
    alarm_state = sns_message.get('NewStateValue', 'UNKNOWN')

    if alarm_state != 'ALARM':
        return {'statusCode': 200, 'body': 'Not in ALARM state, skipping.'}

    # Connect to RDS
    password = get_db_password(os.environ['DB_PASSWORD_SECRET_ARN'])
    conn = psycopg2.connect(
        host=os.environ['DB_HOST'],
        dbname=os.environ['DB_NAME'],
        user=os.environ['DB_USER'],
        password=password,
        connect_timeout=5
    )
    conn.set_session(readonly=True, autocommit=True)

    try:
        wait_events = get_top_wait_events(conn)
        conn_counts = get_connection_count(conn)
        slow_queries = get_top_slow_queries(conn)
    finally:
        conn.close()

    # Format wait event summary
    wait_lines = '\n'.join(
        f" {r[0]}/{r[1]}: {r[2]} sessions"

```

```

        for r in wait_events
    ) or ' (none)'

    # Format slow query summary
    query_lines = '\n'.join(
        f" [{r[1]} calls, {r[2]}ms avg] {r[0]}..."
        for r in slow_queries
    ) or ' (pg_stat_statements not available)'

    # Determine recommended action from alarm name
    if 'connection-storm' in alarm_name:
        recommendation = (
            ':rotating_light: *Recommended action:* Connection storm pattern detected. '
            'Check pgBouncer pool utilization before considering vertical scale. '
            'See Chapter 12 runbook: connection-pooling-response.'
        )
    elif 'scaling-candidate' in alarm_name:
        recommendation = (
            ':bar_chart: *Recommended action:* Potential legitimate scaling need. '
            'Complete optimization checklist (query tuning, parameter review) '
            'before approving vertical scale. See Chapter 12 runbook: scaling-decision-framework.'
        )
    else:
        recommendation = ':mag: *Recommended action:* Investigate using wait event query in Chapter 12
↪ runbook.'

    slack_message = {
        'text': f':database: *RDS Diagnostic Report - {alarm_name}*',
        'blocks': [
            {
                'type': 'section',
                'text': {
                    'type': 'mrkdwn',
                    'text': (
                        f'*Alarm:* `{alarm_name}` | *State:* `{alarm_state}`\n\n'
                        f'*Connections:* {conn_counts["total"]} total '
                        f'({conn_counts["active"]} active, '
                        f'{conn_counts["idle"]} idle, '
                        f'{conn_counts["waiting_on_lock"]} waiting on lock)\n\n'
                        f'*Top Wait Events:* \n{wait_lines}\n\n'
                        f'*Top Slow Queries (by total CPU):*\n{query_lines}\n\n'
                        f'{recommendation}'
                    )
                }
            }
        ]
    }

    post_to_slack(os.environ['SLACK_WEBHOOK_URL'], slack_message)
    return {'statusCode': 200, 'body': 'Diagnostic report posted to Slack.'}

```

This Lambda function gives the on-call engineer a filled-in diagnostic summary within seconds of the alarm firing — wait events, connection breakdown, and the top three slow queries — so that the first five minutes of investigation are already done before anyone opens a terminal.

Azure Monitor equivalent: action groups and automated diagnostics:

On Azure SQL, the equivalent automation uses Azure Monitor alert rules connected to an Action Group that triggers an Azure Function or Logic App. The diagnostic query runs against `sys.dm_exec_requests` and `sys.dm_os_wait_stats` via a managed identity connection, and the result is posted to a Teams channel or an Azure DevOps work item. The structural pattern is identical to the AWS implementation above; the SDK calls change from `boto3` to the Azure SDK for Python (`azure-mgmt-monitor`, `azure-mgmt-sql`), and the alarm primitive changes from CloudWatch composite alarms to Azure Monitor metric alert rules with dynamic thresholds.

12.8 Right-Sizing After a Scaling Event: The 30-Day Review

Vertical scaling decisions made under pressure during an incident are rarely revisited. The instance gets larger, the immediate problem resolves, and the team moves on. Six months later, that larger instance is still running at 20% CPU because the underlying query issue was quietly fixed the following week, but nobody ever scheduled the downgrade.

This section provides the 30-day review process that closes the loop on every vertical scaling event.

Step 1: Establish the post-scale baseline within 48 hours

Within 48 hours of a vertical scaling event, capture the current state of all metrics that justified the scaling. These become the baseline against which the 30-day review is measured.

```
-- PostgreSQL: Capture post-scale performance snapshot
-- Run immediately after scaling and save the output
-- Compare against this snapshot at the 30-day review

SELECT
    current_database()           AS database_name,
    NOW()                       AS snapshot_time,

    -- Connection utilization
    (SELECT COUNT(*) FROM pg_stat_activity
     WHERE pid <> pg_backend_pid()) AS current_connections,
    current_setting('max_connections')::int AS max_connections,

    -- Buffer cache hit rate (should be > 95% on a healthy instance)
    ROUND(
        SUM(heap_blks_hit) * 100.0 /
        NULLIF(SUM(heap_blks_hit) + SUM(heap_blks_read), 0),
        2) AS table_cache_hit_pct,

    -- Index cache hit rate
    ROUND(
        SUM(idx_blks_hit) * 100.0 /
        NULLIF(SUM(idx_blks_hit) + SUM(idx_blks_read), 0),
        2) AS index_cache_hit_pct

FROM pg_statio_user_tables;

-- Also capture the top 10 queries by CPU at the 48-hour mark
-- Save this as your optimization baseline

SELECT
    queryid,
    LEFT(query, 100)           AS query_preview,
    calls,
    ROUND(total_exec_time::numeric / 1000, 2) AS total_exec_sec,
    ROUND((total_exec_time / calls)::numeric, 2) AS avg_exec_ms,
    ROUND(
        total_exec_time * 100.0 /
        NULLIF(SUM(total_exec_time) OVER (), 0),
        2) AS pct_total_cpu
FROM pg_stat_statements
WHERE calls >= 10
ORDER BY total_exec_time DESC
LIMIT 10;
```

Step 2: Schedule the 30-day review as a calendar event immediately

The review must be scheduled before the incident closes. An unscheduled review never happens.

The calendar event should include a link to the Decision Record from Section 12.6 and a reminder to re-run the wait event analysis and query cost analysis from Sections 12.1 and 12.3.

Step 3: At 30 days, compare metrics and evaluate downgrade viability

```
-- PostgreSQL: 30-day review - compare current state to post-scale baseline
-- Pair this with the CloudWatch CPU average for the last 30 days

SELECT
    current_database()                AS database_name,
    NOW()                            AS review_time,

    -- Current average CPU (pull from CloudWatch separately)
    -- Flag if CPU is below 40% - strong downgrade signal

    -- Connection utilization
    (SELECT COUNT(*) FROM pg_stat_activity
     WHERE pid <> pg_backend_pid()) AS current_connections,
    current_setting('max_connections')::int AS max_connections,
    ROUND(
        (SELECT COUNT(*) FROM pg_stat_activity
         WHERE pid <> pg_backend_pid()) * 100.0 /
        current_setting('max_connections')::numeric,
        1) AS connection_utilization_pct,

    -- Buffer cache health
    ROUND(
        SUM(heap_blks_hit) * 100.0 /
        NULLIF(SUM(heap_blks_hit) + SUM(heap_blks_read), 0),
        2) AS table_cache_hit_pct,

    ROUND(
        SUM(idx_blks_hit) * 100.0 /
        NULLIF(SUM(idx_blks_hit) + SUM(idx_blks_read), 0),
        2) AS index_cache_hit_pct

FROM pg_statio_user_tables;
```

The downgrade decision criteria at 30 days are:

- Average CPU over the review period is below 40%
- Cache hit rate remains above 95% (confirming memory is not the binding constraint)
- Connection utilization is below 60% of `max_connections`
- No new query regressions in the top-10 list compared to the 48-hour baseline

If all four criteria are met, the instance is overprovisioned for its current workload and should be scheduled for a maintenance window downgrade back to the previous tier. The downgrade is performed in the next available maintenance window via the same RDS modify operation used for the upsize, with `ApplyImmediately=False` to schedule it during the defined maintenance window and minimize disruption.

```
# AWS CLI: Schedule an RDS instance downgrade in the next maintenance window
# This does NOT apply immediately - it waits for the next maintenance window
# Use --apply-immediately only for development instances

aws rds modify-db-instance \
    --db-instance-identifier <your-instance-id> \
    --db-instance-class db.r6g.2xlarge \
    --apply-immediately false \
    --region us-east-1

# Confirm the pending modification
aws rds describe-db-instances \
    --db-instance-identifier <your-instance-id> \
    --query 'DBInstances[0].PendingModifiedValues' \
    --output json
```

The SQL Server equivalent on Azure SQL uses the Azure CLI:

```
# Azure CLI: Scale down Azure SQL Database to a lower service objective
# For General Purpose, adjust --service-objective to the lower compute tier
# This operation is online for most tier changes in Azure SQL

az sql db update \
  --resource-group <resource-group> \
  --server <server-name> \
  --name <database-name> \
  --service-objective GP_Gen5_8 \
  --no-wait

# Monitor the operation status
az sql db show \
  --resource-group <resource-group> \
  --server <server-name> \
  --name <database-name> \
  --query '{status:status, serviceObjective:currentServiceObjectiveName}' \
  --output table
```

For Google Cloud SQL, the equivalent resize uses the `gcloud` CLI:

```
# GCP: Resize Cloud SQL instance to a smaller machine type
# Cloud SQL custom machine types allow fine-grained right-sizing

gcloud sql instances patch <instance-name> \
  --tier db-custom-8-32768 \
  --region us-central1

# Monitor the operation
gcloud sql operations list \
  --instance <instance-name> \
  --filter="status!=DONE" \
  --format="table(name,operationType,status,startTime)"
```

The 30-day review cycle, when followed consistently, reclaims overprovisioned compute spend that would otherwise persist indefinitely. In organizations running more than a handful of production databases, the aggregate savings from systematic 30-day reviews typically exceed the savings from any single optimization effort in the chapter.

Key Takeaways

- **Treat high CPU and memory metrics as symptoms, not diagnoses.** Wait event analysis is the mandatory first step before any vertical scaling decision. The majority of apparent compute pressure originates from missing indexes, connection storms, storage I/O bottlenecks, or misconfigured parameters — none of which require larger instances to resolve.
- **Quantify the cost delta before any scaling conversation begins.** Moving up a single instance tier on AWS RDS, Azure SQL, or Google Cloud SQL typically doubles the annual compute cost for that instance. That number — stated explicitly in dollars per year — anchors the optimization-versus-scaling decision in financial reality and makes the business case for query optimization work self-evident.
- **Connection pooling resolves a disproportionate share of false scaling signals.** PostgreSQL's per-process memory model means that connection count is a direct driver of memory pressure. Deploying pgBouncer in transaction mode in front of an RDS instance can reduce active backend connections by an order of magnitude, frequently recovering enough memory

to eliminate a pending upsize entirely.

- **Parameter tuning — particularly `work_mem`, `shared_buffers`, and `cost threshold for parallelism` — delivers measurable CPU reduction with zero infrastructure cost.** These changes should be evaluated and applied before any scaling decision is finalized. Sort spills, hash join spills, and unnecessary parallelism are common sources of CPU overhead that disappear with correct parameter configuration.
 - **Every vertical scaling event requires a scheduled 30-day review.** Instances scaled up under incident pressure are rarely scaled back down without an explicit review process. Capturing a post-scale baseline within 48 hours and scheduling the review before the incident closes converts a one-way cost increase into a reversible decision, and systematically reclaims overprovisioned spend across the database fleet.
-

Chapter 13: gp2 vs gp3 vs io1 vs io2: Choosing the Right Storage Type

For most AWS RDS workloads running on gp2 today, a single storage type migration to gp3 delivers an immediate 30% cost reduction with zero downtime and no application changes. On a 2TB RDS volume, that is roughly \$860 per year saved on storage alone — before accounting for the IOPS and throughput improvements that often eliminate the need for provisioned IOPS entirely. This chapter walks through the mechanics of each storage type, the diagnostic queries that reveal whether your current storage is over- or under-provisioned, the math behind migration decisions, and the automation that makes these changes repeatable across dozens of RDS instances without manual intervention.

Understanding the Four Storage Types: Pricing and Architecture

AWS RDS offers four EBS-backed storage types. Each is optimized for a different cost-performance trade-off, and the pricing differences between them are large enough to matter at scale.

gp2 (General Purpose SSD, Generation 2)

gp2 is the original general-purpose option. Pricing sits at approximately **\$0.115 per GB-month** in us-east-1. The key architectural constraint is that IOPS are tied directly to volume size: you receive 3 IOPS per GB, with a baseline of 100 IOPS and a hard ceiling of 16,000 IOPS. A 500GB gp2 volume delivers 1,500 IOPS. To reach 16,000 IOPS on gp2, you must provision at least 5,334GB of storage — even if your database is only 400GB in size. Teams frequently over-provision gp2 storage purely for IOPS, paying for unused space.

gp2 also uses a burst credit system for volumes under 1TB. Volumes smaller than 333GB can burst to 3,000 IOPS when credits are available, but sustained workloads exhaust those credits and drop back to baseline. This creates invisible performance cliffs that only appear under load.

gp3 (General Purpose SSD, Generation 3)

gp3 decouples IOPS from storage size. Base price is **\$0.08 per GB-month** — a 30% reduction versus gp2. Every gp3 volume includes 3,000 IOPS and 125 MB/s throughput at no additional charge. IOPS can be independently scaled up to 16,000 for **\$0.005 per provisioned IOPS-month** above the 3,000 baseline, and throughput up to 1,000 MB/s for **\$0.04 per MB/s-month** above 125 MB/s.

For a 1TB volume: - gp2: $1,024\text{GB} \times \$0.115 = \text{\$117.76/month}$, delivers 3,072 IOPS - gp3: $1,024\text{GB} \times \$0.08 = \text{\$81.92/month}$, delivers 3,000 IOPS (essentially the same), saves **\\$35.84/month** or **\\$430/year**

For a 2TB volume with no extra IOPS needed: - gp2: $2,048\text{GB} \times \$0.115 = \text{\$235.52/month}$ - gp3: $2,048\text{GB} \times \$0.08 = \text{\$163.84/month}$, saves **\\$71.68/month** or **\\$860/year**

io1 (Provisioned IOPS SSD, Generation 1)

io1 is designed for I/O-intensive workloads requiring sustained, consistent IOPS. Storage costs **\\$0.125 per GB-month**, and IOPS are independently provisioned at **\\$0.065 per IOPS-month**. The IOPS-to-GB ratio is capped at 50:1. A 500GB io1 volume can provision a maximum of 25,000 IOPS. Maximum IOPS is 64,000 on Nitro-based instances.

The cost model makes io1 expensive at high IOPS. A volume provisioned at 10,000 IOPS on 200GB of storage costs: - Storage: $200 \times \$0.125 = \25.00 - IOPS: $10,000 \times \$0.065 = \650.00 - **Total: \\$675.00/month**

io2 (Provisioned IOPS SSD, Generation 2)

io2 replaces io1 for most use cases. It offers higher durability (99.999% versus 99.8% for io1), a 500:1 IOPS-to-GB ratio, and tiered pricing that makes high IOPS cheaper than io1 at volume. As of 2024 pricing in us-east-1: - Storage: **\\$0.125 per GB-month** (same as io1) - IOPS up to 32,000: **\\$0.065 per IOPS-month** - IOPS from 32,001–64,000: **\\$0.046 per IOPS-month** - IOPS above 64,000: **\\$0.032 per IOPS-month**

io2 Block Express extends the ceiling to 256,000 IOPS and 4,000 MB/s throughput, available on r6i, m6i, and similar Nitro instances. For any workload currently on io1, migrating to io2 at the same IOPS level costs the same or less and delivers meaningfully higher durability. There is no reason to choose io1 for new deployments.

Diagnosing Your Current Storage: Are You Over-Provisioned?

Before making any changes, collect four weeks of actual storage metrics. The goal is to determine three things: actual IOPS consumed, actual throughput consumed, and the percentage of time the storage is at or near its provisioned limit.

Querying CloudWatch for RDS Storage Metrics

```
# Pull 30-day average and peak IOPS for an RDS instance
# Replace DB_INSTANCE_IDENTIFIER with your actual instance ID

DB_ID="prod-postgres-01"
START_TIME=$(date -u -d '30 days ago' +%Y-%m-%dT%H:%M:%SZ)
END_TIME=$(date -u +%Y-%m-%dT%H:%M:%SZ)

echo "=== Read IOPS (Average and Maximum) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name ReadIOPS \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_ID \
  --start-time $START_TIME \
  --end-time $END_TIME \
  --period 3600 \
  --statistics Average Maximum \
  --output table
```

```

echo "=== Write IOPS (Average and Maximum) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name WriteIOPS \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_ID \
  --start-time $START_TIME \
  --end-time $END_TIME \
  --period 3600 \
  --statistics Average Maximum \
  --output table

echo "=== Read Throughput MB/s ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name ReadThroughput \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_ID \
  --start-time $START_TIME \
  --end-time $END_TIME \
  --period 3600 \
  --statistics Average Maximum \
  --output table

echo "=== Storage Queue Depth (key saturation signal) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name DiskQueueDepth \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_ID \
  --start-time $START_TIME \
  --end-time $END_TIME \
  --period 3600 \
  --statistics Average Maximum \
  --output table

# Also pull BurstBalance for gp2 volumes (0% = burst exhausted)
echo "=== Burst Balance (gp2 only - 0% means sustained IOPS pressure) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name BurstBalance \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_ID \
  --start-time $START_TIME \
  --end-time $END_TIME \
  --period 3600 \
  --statistics Average Minimum \
  --output table

```

Interpreting the Results

Three patterns emerge from this data:

1. **Peak IOPS stays below 2,500 on a gp2 volume of any size** — this is a candidate for gp3 with no additional IOPS provisioning. The included 3,000 IOPS on gp3 covers the workload entirely.
2. **BurstBalance frequently drops below 20%** — the instance is hitting the burst ceiling. If the volume is under 1TB, either the workload genuinely needs more IOPS (evaluate io2), or the volume should be sized to the performance tier it actually needs on gp3 without the credit system.
3. **DiskQueueDepth regularly exceeds 1.0** — storage is a bottleneck. This is not a candidate for cost reduction yet; it needs either more IOPS or a storage type upgrade evaluated for performance, not cost.

Querying PostgreSQL for I/O Patterns at the Table Level

```
-- Identify which tables and indexes drive the most I/O
-- Run on PostgreSQL RDS / Aurora PostgreSQL
-- Requires pg_stat_user_tables to have accumulated data (reset on restart)

SELECT
    schemaname,
    relname AS table_name,
    heap_blks_read,
    heap_blks_hit,
    ROUND(
        100.0 * heap_blks_hit / NULLIF(heap_blks_hit + heap_blks_read, 0),
        2
    ) AS cache_hit_ratio_pct,
    idx_blks_read,
    idx_blks_hit,
    ROUND(
        100.0 * idx_blks_hit / NULLIF(idx_blks_hit + idx_blks_read, 0),
        2
    ) AS index_cache_hit_pct,
    -- Estimated physical I/O operations (blocks not served from shared_buffers)
    (heap_blks_read + idx_blks_read) AS total_physical_reads
FROM
    pg_stat_user_tables
WHERE
    (heap_blks_read + idx_blks_read) > 1000
ORDER BY
    total_physical_reads DESC
LIMIT 25;

-- Companion query: find the most I/O-intensive queries
-- Requires pg_stat_statements extension (enabled by default on RDS)
SELECT
    LEFT(query, 100) AS query_preview,
    calls,
    total_exec_time::BIGINT AS total_ms,
    ROUND((total_exec_time / calls)::NUMERIC, 2) AS avg_ms,
    shared_blks_read,
    shared_blks_hit,
    ROUND(
        100.0 * shared_blks_hit / NULLIF(shared_blks_hit + shared_blks_read, 0),
        2
    ) AS hit_ratio_pct,
    -- Blocks read from disk (not cache) - direct storage pressure
    shared_blks_read AS disk_blocks_read,
    local_blks_read + temp_blks_read AS temp_disk_reads
FROM
    pg_stat_statements
WHERE
    shared_blks_read > 10000
ORDER BY
    shared_blks_read DESC
LIMIT 20;
```

A cache hit ratio consistently above 99% across most tables signals that your workload is largely memory-bound — physical IOPS are a secondary concern, and gp3’s baseline allocation is almost certainly sufficient.

Running the Migration Cost Math

The decision framework for storage type selection follows a simple flowprint based on actual IOPS requirements:

Peak IOPS needed?

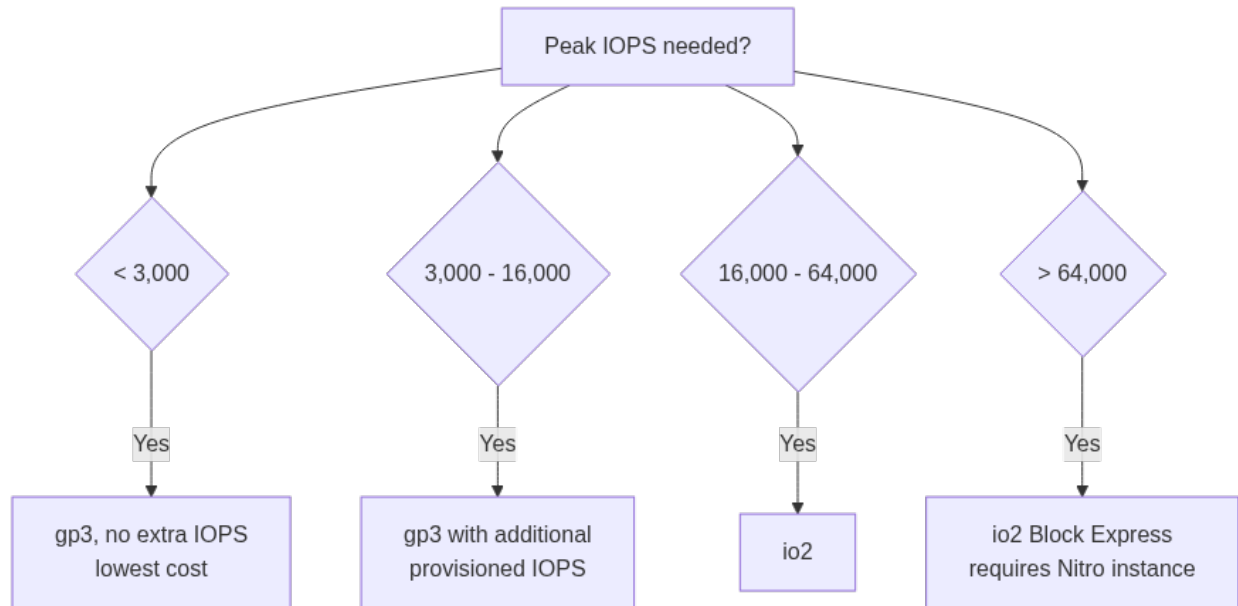


Figure 13: The storage type decision framework, based on actual measured IOPS requirements rather than volume size.

< 3,000: gp3 (no extra IOPS, lowest cost)
 3,000 - 16,000: gp3 with additional provisioned IOPS
 16,000 - 64,000: io2
 > 64,000: io2 Block Express (requires Nitro instance class)

The crossover point where io2 becomes cheaper than gp3 with additional IOPS sits around **16,000 IOPS for volumes under 320GB**. For larger volumes, the per-GB price difference adds up in gp3's favor until IOPS demands are extreme.

Python Script: Storage Cost Calculator and Migration Recommendation

```

#!/usr/bin/env python3
"""
rds_storage_advisor.py
Calculates current vs. optimized RDS storage costs and emits migration recommendations.
Pulls live data from CloudWatch and RDS API, then computes cost differentials.

Usage:
python3 rds_storage_advisor.py --region us-east-1 --profile prod
python3 rds_storage_advisor.py --instance-id prod-postgres-01
"""

import argparse
import json
from datetime import datetime, timedelta, timezone

import boto3

# Pricing (us-east-1, on-demand, as of Q1 2025)
# Update these when AWS adjusts pricing
PRICING = {
    "gp2": {
        "storage_per_gb": 0.115,
        "iops_per_iops": 0.0,      # IOPS tied to storage
        "throughput_per_mbs": 0.0,
    },

```

```

"gp3": {
    "storage_per_gb": 0.080,
    "base_iops": 3000,
    "base_throughput_mbs": 125,
    "iops_per_iops": 0.005, # per IOPS above 3,000
    "throughput_per_mbs": 0.04, # per MB/s above 125
},
"io1": {
    "storage_per_gb": 0.125,
    "iops_per_iops": 0.065,
},
"io2": {
    "storage_per_gb": 0.125,
    "iops_tiers": [
        (32000, 0.065),
        (64000, 0.046),
        (float("inf"), 0.032),
    ],
},
}

def get_rds_instances(session, instance_id=None):
    rds = session.client("rds")
    if instance_id:
        resp = rds.describe_db_instances(DBInstanceIdentifier=instance_id)
    else:
        resp = rds.describe_db_instances()
    return resp["DBInstances"]

def get_cloudwatch_stat(cw, db_id, metric, stat, days=30):
    """Return the maximum value of a CloudWatch metric over the past N days."""
    resp = cw.get_metric_statistics(
        Namespace="AWS/RDS",
        MetricName=metric,
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": db_id}],
        StartTime=datetime.now(timezone.utc) - timedelta(days=days),
        EndTime=datetime.now(timezone.utc),
        Period=3600,
        Statistics=[stat],
    )
    points = resp.get("Datapoints", [])
    if not points:
        return 0.0
    return max(p[stat] for p in points)

def compute_gp2_cost(size_gb):
    return size_gb * PRICING["gp2"]["storage_per_gb"]

def compute_gp3_cost(size_gb, required_iops, required_throughput_mbs):
    p = PRICING["gp3"]
    storage_cost = size_gb * p["storage_per_gb"]
    extra_iops = max(0, required_iops - p["base_iops"])
    extra_throughput = max(0, required_throughput_mbs - p["base_throughput_mbs"])
    iops_cost = extra_iops * p["iops_per_iops"]
    throughput_cost = extra_throughput * p["throughput_per_mbs"]
    return storage_cost + iops_cost + throughput_cost, {
        "storage": round(storage_cost, 2),
        "iops": round(iops_cost, 2),
        "throughput": round(throughput_cost, 2),
    }

def compute_io2_cost(size_gb, required_iops):
    storage_cost = size_gb * PRICING["io2"]["storage_per_gb"]
    iops_cost = 0.0
    remaining = required_iops

```

```

prev_ceiling = 0
for ceiling, rate in PRICING["io2"]["iops_tiers"]:
    tier_iops = min(remaining, ceiling - prev_ceiling)
    if tier_iops <= 0:
        break
    iops_cost += tier_iops * rate
    remaining -= tier_iops
    prev_ceiling = ceiling
return storage_cost + iops_cost

def make_recommendation(instance, peak_iops, peak_throughput_mbs):
    storage_type = instance.get("StorageType", "gp2")
    size_gb = instance["AllocatedStorage"]
    iops_provisioned = instance.get("Iops", 0) or 0
    db_id = instance["DBInstanceIdentifier"]

    current_cost = 0.0
    if storage_type == "gp2":
        current_cost = compute_gp2_cost(size_gb)
        effective_iops = max(100, size_gb * 3)
    elif storage_type == "gp3":
        current_cost, _ = compute_gp3_cost(size_gb, iops_provisioned, peak_throughput_mbs)
        effective_iops = iops_provisioned
    elif storage_type in ("io1", "io2"):
        current_cost = (size_gb * PRICING[storage_type]["storage_per_gb"] +
                        iops_provisioned * PRICING.get("io1", PRICING["io2"])
                        .get("iops_per_iops", 0.065))
        effective_iops = iops_provisioned

    # Add 20% headroom to peak IOPS for recommendation
    target_iops = max(3000, int(peak_iops * 1.20))
    target_throughput = max(125, int(peak_throughput_mbs * 1.20))

    gp3_cost, gp3_breakdown = compute_gp3_cost(size_gb, target_iops, target_throughput)
    io2_cost = compute_io2_cost(size_gb, target_iops)

    if target_iops <= 16000:
        recommended_type = "gp3"
        recommended_cost = gp3_cost
    else:
        recommended_type = "io2"
        recommended_cost = io2_cost

    monthly_savings = current_cost - recommended_cost
    annual_savings = monthly_savings * 12

    return {
        "instance_id": db_id,
        "current_type": storage_type,
        "size_gb": size_gb,
        "current_provisioned_iops": iops_provisioned,
        "peak_actual_iops_30d": round(peak_iops, 0),
        "peak_actual_throughput_mbs": round(peak_throughput_mbs, 2),
        "target_iops_with_headroom": target_iops,
        "current_monthly_cost_usd": round(current_cost, 2),
        "recommended_type": recommended_type,
        "recommended_monthly_cost_usd": round(recommended_cost, 2),
        "monthly_savings_usd": round(monthly_savings, 2),
        "annual_savings_usd": round(annual_savings, 2),
        "gp3_breakdown": gp3_breakdown,
        "action": (
            f"aws rds modify-db-instance --db-instance-identifier {db_id} "
            f"--storage-type {recommended_type} "
            f"--iops {target_iops} "
            f"--apply-immediately"
            if recommended_type in ("io2",)
            else
            f"aws rds modify-db-instance --db-instance-identifier {db_id} "
            f"--storage-type gp3 "

```

```

        f"--iops {target_iops} "
        f"--storage-throughput {target_throughput} "
        f"--apply-immediately"
    ),
}

def main():
    parser = argparse.ArgumentParser(description="RDS Storage Type Advisor")
    parser.add_argument("--region", default="us-east-1")
    parser.add_argument("--profile", default=None)
    parser.add_argument("--instance-id", default=None)
    parser.add_argument("--output", choices=["json", "text"], default="text")
    args = parser.parse_args()

    session = boto3.Session(region_name=args.region, profile_name=args.profile)
    cw = session.client("cloudwatch")

    instances = get_rds_instances(session, args.instance_id)
    recommendations = []
    total_current = 0.0
    total_recommended = 0.0

    for inst in instances:
        db_id = inst["DBInstanceIdentifier"]
        print(f"Analyzing {db_id}...")

        peak_read_iops = get_cloudwatch_stat(cw, db_id, "ReadIOPS", "Maximum")
        peak_write_iops = get_cloudwatch_stat(cw, db_id, "WriteIOPS", "Maximum")
        peak_iops = peak_read_iops + peak_write_iops

        peak_read_tp = get_cloudwatch_stat(cw, db_id, "ReadThroughput", "Maximum")
        peak_write_tp = get_cloudwatch_stat(cw, db_id, "WriteThroughput", "Maximum")
        peak_throughput_mbs = (peak_read_tp + peak_write_tp) / (1024 * 1024)

        rec = make_recommendation(inst, peak_iops, peak_throughput_mbs)
        recommendations.append(rec)
        total_current += rec["current_monthly_cost_usd"]
        total_recommended += rec["recommended_monthly_cost_usd"]

    summary = {
        "total_instances_analyzed": len(recommendations),
        "total_current_monthly_storage_cost_usd": round(total_current, 2),
        "total_recommended_monthly_cost_usd": round(total_recommended, 2),
        "total_monthly_savings_usd": round(total_current - total_recommended, 2),
        "total_annual_savings_usd": round((total_current - total_recommended) * 12, 2),
        "recommendations": recommendations,
    }

    if args.output == "json":
        print(json.dumps(summary, indent=2))
    else:
        print("\n" + "=" * 70)
        print(f"  RDS Storage Advisor Summary - {args.region}")
        print("=" * 70)
        print(f"  Instances analyzed:           {summary['total_instances_analyzed']}")
        print(f"  Current monthly cost:         ${summary['total_current_monthly_storage_cost_usd']:>10,.2f}")
        print(f"  Optimized monthly cost:       ${summary['total_recommended_monthly_cost_usd']:>10,.2f}")
        print(f"  Monthly savings:              ${summary['total_monthly_savings_usd']:>10,.2f}")
        print(f"  Annual savings:               ${summary['total_annual_savings_usd']:>10,.2f}")
        print("=" * 70)
        print()
        for rec in recommendations:
            change = "KEEP" if rec["monthly_savings_usd"] <= 0 else "→ MIGRATE"
            print(f"  [{change}] {rec['instance_id']}")
            print(f"    Current: {rec['current_type']} / {rec['size_gb']}GB / "
                  f"    {rec['current_provisioned_iops']} IOPS → ${rec['current_monthly_cost_usd']:.2f}/mo")
            print(f"    Recommended: {rec['recommended_type']} / "

```



```
f"{rec['target_iops_with_headroom']}] IOPS →  
    ↳ ${rec['recommended_monthly_cost_usd']:.2f}/mo")  
print(f"        Savings: ${rec['monthly_savings_usd']:.2f}/mo  
    ↳ (${rec['annual_savings_usd']:.2f}/yr)")  
if rec["monthly_savings_usd"] > 0:  
    print(f"        Command: {rec['action']}")  
print()  
  
if __name__ == "__main__":  
    main()
```

Running this script across a fleet of twenty RDS instances is a common first step in a storage optimization engagement. The output immediately surfaces which instances are on gp2 where the workload fits within gp3's baseline, which are over-provisioned on io1 or io2, and which have legitimate IOPS demands that justify their current configuration.

Executing the Migration: gp2 to gp3 Step by Step

The actual migration is an online operation managed by AWS. RDS modifies the storage type in the background with no downtime, though the modification may take from a few minutes to several hours depending on volume size. Applications remain connected throughout.

Pre-Migration Checklist

Before submitting the modify command, verify three things:

1. **Maintenance window and Multi-AZ status** — for Multi-AZ deployments, the modification applies first to the standby, then fails over, then applies to the former primary. This produces a brief failover event (typically 20–30 seconds). Schedule the modification during a low-traffic period if your application is sensitive to failovers.
2. **Storage autoscaling** — if RDS storage autoscaling is enabled, verify the maximum autoscale threshold won't trigger immediately after the migration in a way that inflates gp3 costs unexpectedly.
3. **Parameter group IOPS settings** — some PostgreSQL parameter group settings reference effective storage performance. If you have tuned `effective_io_concurrency` or checkpoint settings based on io1 performance characteristics, revisit those after migration to gp3.

The Migration Command

```
# Step 1: Capture current configuration before any change  
aws rds describe-db-instances \  
    --db-instance-identifier prod-postgres-01 \  
    --query 'DBInstances[0].{Type:StorageType,GB:AllocatedStorage,IOPS:Iops,Engine:Engine}' \  
    --output table  
  
# Step 2: Submit the storage type modification  
# --apply-immediately causes the modification to start now rather than  
# waiting for the next maintenance window. Use this during off-peak hours.  
aws rds modify-db-instance \  
    --db-instance-identifier prod-postgres-01 \  
    --storage-type gp3 \  
    --iops 3000 \  
    --storage-throughput 125 \  
    --apply-immediately
```

```
# If your workload needs more than 3,000 IOPS (from CloudWatch analysis):
aws rds modify-db-instance \
  --db-instance-identifier prod-postgres-01 \
  --storage-type gp3 \
  --iops 6000 \
  --storage-throughput 250 \
  --apply-immediately

# Step 3: Monitor the modification status
# Status will transition: available -> modifying -> available
aws rds describe-db-instances \
  --db-instance-identifier prod-postgres-01 \
  --query 'DBInstances[0].{Status:DBInstanceStatus,PendingModifications:PendingModifiedValues}' \
  --output table

# Poll until modification completes (typically 15-60 minutes for volumes under 500GB)
watch -n 60 "aws rds describe-db-instances \
  --db-instance-identifier prod-postgres-01 \
  --query 'DBInstances[0].DBInstanceStatus' \
  --output text"
```

Post-Migration Validation

```
# Confirm the new storage type is in effect
aws rds describe-db-instances \
  --db-instance-identifier prod-postgres-01 \
  --query 'DBInstances[0].{Type:StorageType,IOPS:Iops,Throughput:StorageThroughput,GB:AllocatedStorage}' \
  --output table

# Verify DiskQueueDepth hasn't increased post-migration (watch for 24 hours)
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name DiskQueueDepth \
  --dimensions Name=DBInstanceIdentifier,Value=prod-postgres-01 \
  --start-time $(date -u -d '4 hours ago' +%Y-%m-%dT%H:%M:%SZ) \
  --end-time $(date -u +%Y-%m-%dT%H:%M:%SZ) \
  --period 300 \
  --statistics Average Maximum \
  --output table

# Verify ReadLatency and WriteLatency are stable post-migration
for METRIC in ReadLatency WriteLatency; do
  echo "=== $METRIC ==="
  aws cloudwatch get-metric-statistics \
    --namespace AWS/RDS \
    --metric-name $METRIC \
    --dimensions Name=DBInstanceIdentifier,Value=prod-postgres-01 \
    --start-time $(date -u -d '4 hours ago' +%Y-%m-%dT%H:%M:%SZ) \
    --end-time $(date -u +%Y-%m-%dT%H:%M:%SZ) \
    --period 300 \
    --statistics Average Maximum \
    --output table
done
```

If `DiskQueueDepth` climbs above 1.0 or latency metrics increase materially after migration, the workload needs more IOPS than the baseline 3,000. Increase provisioned IOPS on the gp3 volume with another `modify-db-instance` call — this adjustment takes effect within minutes and does not require a storage type change.

SQL Server Equivalent: Diagnosing I/O on RDS SQL Server

The PostgreSQL queries above have direct equivalents for SQL Server running on RDS. The diagnostic approach is the same: identify which objects are generating physical I/O, then compare that to provisioned storage capacity.

```
-- SQL Server: find tables and indexes with the highest physical I/O pressure
-- Run against the target database on RDS SQL Server
-- sys.dm_io_virtual_file_stats accumulates since the last SQL Server restart

SELECT
    DB_NAME(vfs.database_id)                AS database_name,
    mf.name                                AS logical_file_name,
    mf.physical_name,
    vfs.io_stall_read_ms,
    vfs.io_stall_write_ms,
    vfs.io_stall_read_ms + vfs.io_stall_write_ms AS io_stall_total_ms,
    vfs.num_of_reads,
    vfs.num_of_writes,
    vfs.num_of_bytes_read / 1048576.0        AS mb_read,
    vfs.num_of_bytes_written / 1048576.0      AS mb_written,
    -- Average read latency in ms (high values indicate storage pressure)
    CASE WHEN vfs.num_of_reads = 0 THEN 0
         ELSE vfs.io_stall_read_ms * 1.0 / vfs.num_of_reads
    END                                       AS avg_read_latency_ms,
    CASE WHEN vfs.num_of_writes = 0 THEN 0
         ELSE vfs.io_stall_write_ms * 1.0 / vfs.num_of_writes
    END                                       AS avg_write_latency_ms
FROM
    sys.dm_io_virtual_file_stats(NULL, NULL) AS vfs
JOIN sys.master_files AS mf
    ON vfs.database_id = mf.database_id
   AND vfs.file_id     = mf.file_id
ORDER BY
    io_stall_total_ms DESC;

-- SQL Server: identify the most I/O-intensive queries currently in the plan cache
SELECT TOP 20
    qs.total_logical_reads                AS total_logical_reads,
    qs.total_physical_reads               AS total_physical_reads,
    qs.total_logical_writes               AS total_logical_writes,
    qs.execution_count,
    qs.total_physical_reads / NULLIF(qs.execution_count, 0)
                                                AS avg_physical_reads,
    SUBSTRING(qt.text, (qs.statement_start_offset/2) + 1,
        ((CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)
            ELSE qs.statement_end_offset
        END - qs.statement_start_offset) / 2) + 1)
                                                AS statement_text,
    qp.query_plan
FROM
    sys.dm_exec_query_stats               AS qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS qt
CROSS APPLY sys.dm_exec_query_plan(qs.plan_handle) AS qp
WHERE
    qs.total_physical_reads > 5000
ORDER BY
    qs.total_physical_reads DESC;
```

Reading the Results

- `avg_read_latency_ms` above 20ms on data files indicates the storage tier is under pressure. Below 5ms is healthy for gp3; below 1ms is expected for io2 at moderate IOPS.
- `total_physical_reads` in the query cache identifies statements doing large sequential scans or missing index seeks — these drive IOPS requirements up and are a better target for query

or index optimization than storage upgrades.

The same CloudWatch-based IOPS analysis applies identically to SQL Server RDS. The `ReadIOPS`, `WriteIOPS`, and `DiskQueueDepth` metrics are engine-agnostic.

When gp3 Is Not the Right Answer: io2 Decision Criteria

gp3 tops out at 16,000 IOPS. For workloads that genuinely need more, io2 is the path forward. The four scenarios where io2 is warranted are:

1. OLTP with sustained high-concurrency writes

Transaction processing systems with hundreds of concurrent sessions issuing small writes — payment processing, order management, event ingestion — can saturate gp3's ceiling even on well-tuned instances. If CloudWatch shows peak combined IOPS consistently above 12,000 (leaving no headroom before 16,000), io2 provides the additional ceiling.

2. Large databases with full-scan analytical queries on the OLTP instance

A 4TB OLTP database running nightly batch reports that scan large portions of the dataset will generate sequential read IOPS spikes that gp3 cannot always absorb. The correct architectural answer is read replicas or a separate analytical environment — but while that refactoring is pending, io2 provides headroom.

3. Oracle workloads with REDO log contention

Oracle on RDS frequently generates write IOPS patterns concentrated on redo log files. These are short, synchronous writes that are highly latency-sensitive. io2's consistent single-digit millisecond latency under load matters more for Oracle redo than raw throughput.

4. Instances where the 50:1 gp3 IOPS-to-GB ratio creates oversized volumes

gp3 limits IOPS to 500 per GB (up to the 16,000 ceiling). A 100GB database needing 10,000 IOPS on gp3 requires provisioning 200GB of storage to reach that IOPS level. io2's 500:1 ratio means the same 100GB database can be provisioned with 100GB of storage at 10,000 IOPS. In this narrow case, io2 is both cheaper and better-fit.

Cost Comparison: gp3 vs io2 at Various IOPS Levels for a 200GB Volume

Volume: 200GB

IOPS	gp3 Storage	gp3 IOPS	gp3 Total	io2 Storage	io2 IOPS	io2 Total
3,000	\$16.00	\$0.00	\$16.00	\$25.00	\$195.00	\$220.00
6,000	\$16.00	\$15.00	\$31.00	\$25.00	\$390.00	\$415.00
10,000	\$16.00	\$35.00	\$51.00	\$25.00	\$650.00	\$675.00
16,000	\$16.00	\$65.00	\$81.00	\$25.00	\$1,040.00	\$1,065.00
32,000	N/A (ceiling)	-	-	\$25.00	\$2,080.00	\$2,105.00

The table makes the decision clear: for any workload below 16,000 IOPS, gp3 is substantially cheaper than io2. The only reason to choose io2 below that ceiling is the latency consistency

guarantee — io2 is rated for single-digit millisecond latency at full provisioned IOPS, whereas gp3 may show slightly higher tail latency at its upper range.

Automating Storage Type Migrations Across a Fleet

Running the advisory script and executing individual CLI commands works for a handful of instances. At scale — dozens or hundreds of RDS instances across multiple accounts — the process needs to be automated, gated by approval, and logged.

The following Terraform configuration combined with a Python Lambda function provides a repeatable, auditable migration pipeline.

Terraform: RDS Instance with Explicit Storage Type and Tagging

```
# rds_storage_optimized.tf
# Enforces gp3 as the default storage type for all new RDS instances.
# Existing instances are migrated via the Lambda automation below.

variable "db_instance_map" {
  description = "Map of RDS instance configurations"
  type = map(object({
    engine           = string
    engine_version   = string
    instance_class    = string
    allocated_gb      = number
    iops              = number
    throughput_mbs    = number
    multi_az          = bool
    environment       = string
  }))
}

resource "aws_db_instance" "optimized" {
  for_each = var.db_instance_map

  identifier      = each.key
  engine          = each.value.engine
  engine_version  = each.value.engine_version
  instance_class  = each.value.instance_class
  allocated_storage = each.value.allocated_gb

  # Always use gp3 unless IOPS demand exceeds 16,000
  storage_type      = each.value.iops > 16000 ? "io2" : "gp3"
  iops              = each.value.iops > 3000 ? each.value.iops : null
  storage_throughput = each.value.throughput_mbs > 125 ? each.value.throughput_mbs : null

  multi_az          = each.value.multi_az
  storage_encrypted = true

  tags = {
    StorageOptimized = "true"
    StorageReviewedAt = timestamp()
    Environment       = each.value.environment
    ManagedBy         = "terraform"
  }

  lifecycle {
    # Prevent accidental storage type regressions
    precondition {
      condition      = each.value.iops <= 16000 || contains(["io2"], self.storage_type)
      error_message = "Instances requiring >16,000 IOPS must use io2 storage type."
    }
  }
}
```

```

}

# CloudWatch alarm: triggers if DiskQueueDepth exceeds threshold post-migration
resource "aws_cloudwatch_metric_alarm" "storage_saturation" {
  for_each = var.db_instance_map

  alarm_name        = "${each.key}-storage-queue-depth-high"
  comparison_operator = "GreaterThanThreshold"
  evaluation_periods = 3
  metric_name       = "DiskQueueDepth"
  namespace         = "AWS/RDS"
  period            = 300
  statistic          = "Average"
  threshold          = 1.0
  alarm_description  = "Storage queue depth elevated - may need IOPS increase on ${each.key}"

  dimensions = {
    DBInstanceIdentifier = each.key
  }

  alarm_actions = [var.ops_sns_topic_arn]
  ok_actions     = [var.ops_sns_topic_arn]
}

```

Python Lambda: Scheduled Storage Optimization Sweep

```

#!/usr/bin/env python3
"""
lambda_storage_optimizer.py
Scheduled Lambda (runs weekly) that identifies gp2 RDS instances,
computes recommended gp3 configurations, and either applies changes
automatically (for non-production) or creates a JIRA/SNS ticket for review.

Environment variables:
    AUTO_APPLY_ENVS - comma-separated list of environments where changes
                    are applied automatically (e.g., "dev,staging")
    APPROVAL_SNS_ARN - SNS topic for production change approvals
    DRY_RUN          - set to "true" to log recommendations without acting
"""

import json
import logging
import os
from datetime import datetime, timedelta, timezone

import boto3

logger = logging.getLogger()
logger.setLevel(logging.INFO)

AUTO_APPLY_ENVS = set(os.environ.get("AUTO_APPLY_ENVS", "dev,staging").split(","))
APPROVAL_SNS_ARN = os.environ.get("APPROVAL_SNS_ARN", "")
DRY_RUN = os.environ.get("DRY_RUN", "false").lower() == "true"

# Minimum monthly savings threshold to trigger a recommendation ($)
MIN_SAVINGS_THRESHOLD = 10.0

def get_instance_environment(instance):
    """Extract environment tag from RDS instance tags."""
    tags = {t["Key"]: t["Value"] for t in instance.get("TagList", [])}
    return tags.get("Environment", tags.get("environment", "unknown")).lower()

def should_auto_apply(environment):
    return environment in AUTO_APPLY_ENVS and not DRY_RUN

def compute_gp3_recommendation(size_gb, peak_iops, peak_throughput_mbs):
    """Return recommended gp3 IOPS and throughput with 20% headroom."""

```

```

target_iops = max(3000, int(peak_iops * 1.20))
target_throughput = max(125, int(peak_throughput_mbs * 1.20))
target_iops = min(target_iops, 16000) # gp3 ceiling
target_throughput = min(target_throughput, 1000) # gp3 ceiling
return target_iops, target_throughput

def get_peak_metric(cw, db_id, metric_name, days=30):
    resp = cw.get_metric_statistics(
        Namespace="AWS/RDS",
        MetricName=metric_name,
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": db_id}],
        StartTime=datetime.now(timezone.utc) - timedelta(days=days),
        EndTime=datetime.now(timezone.utc),
        Period=3600,
        Statistics=["Maximum"],
    )
    points = resp.get("Datapoints", [])
    return max((p["Maximum"] for p in points), default=0.0)

def apply_gp3_migration(rds_client, db_id, target_iops, target_throughput):
    logger.info(
        "Applying gp3 migration: %s → iops=%d throughput=%d",
        db_id, target_iops, target_throughput
    )
    rds_client.modify_db_instance(
        DBInstanceIdentifier=db_id,
        StorageType="gp3",
        Iops=target_iops,
        StorageThroughput=target_throughput,
        ApplyImmediately=True,
    )

def send_approval_request(sns_client, recommendation):
    message = {
        "subject": f"RDS Storage Migration Pending Approval: {recommendation['instance_id']}",
        "body": recommendation,
        "requested_at": datetime.now(timezone.utc).isoformat(),
    }
    sns_client.publish(
        TopicArn=APPROVAL_SNS_ARN,
        Subject=message["subject"],
        Message=json.dumps(message, indent=2),
    )
    logger.info("Approval request sent for %s", recommendation["instance_id"])

def lambda_handler(event, context):
    region = os.environ.get("AWS_REGION", "us-east-1")
    session = boto3.Session(region_name=region)
    rds = session.client("rds")
    cw = session.client("cloudwatch")
    sns = session.client("sns") if APPROVAL_SNS_ARN else None

    paginator = rds.get_paginator("describe_db_instances")
    results = {"applied": [], "pending_approval": [], "skipped": [], "errors": []}

    for page in paginator.paginate():
        for inst in page["DBInstances"]:
            db_id = inst["DBInstanceIdentifier"]
            storage_type = inst.get("StorageType", "unknown")

            # Only process gp2 instances - gp3/io2 handled separately
            if storage_type != "gp2":
                results["skipped"].append(
                    {"instance_id": db_id, "reason": f"already on {storage_type}"}
                )
                continue

```

```

# Skip instances currently being modified
if inst.get("DBInstanceStatus") != "available":
    results["skipped"].append(
        {"instance_id": db_id, "reason": f"status={inst.get('DBInstanceStatus')}}"})
    continue

try:
    size_gb = inst["AllocatedStorage"]
    environment = get_instance_environment(inst)

    peak_read_iops = get_peak_metric(cw, db_id, "ReadIOPS")
    peak_write_iops = get_peak_metric(cw, db_id, "WriteIOPS")
    peak_iops = peak_read_iops + peak_write_iops

    peak_read_tp = get_peak_metric(cw, db_id, "ReadThroughput")
    peak_write_tp = get_peak_metric(cw, db_id, "WriteThroughput")
    peak_throughput_mbs = (peak_read_tp + peak_write_tp) / (1024 * 1024)

    target_iops, target_throughput = compute_gp3_recommendation(
        size_gb, peak_iops, peak_throughput_mbs)

    # Cost delta
    current_cost = size_gb * 0.115
    gp3_storage = size_gb * 0.080
    gp3_iops_cost = max(0, target_iops - 3000) * 0.005
    gp3_tp_cost = max(0, target_throughput - 125) * 0.04
    recommended_cost = gp3_storage + gp3_iops_cost + gp3_tp_cost
    monthly_savings = current_cost - recommended_cost

    if monthly_savings < MIN_SAVINGS_THRESHOLD:
        results["skipped"].append(
            {"instance_id": db_id,
             "reason": f"savings ${monthly_savings:.2f}/mo below threshold"})
        continue

    recommendation = {
        "instance_id": db_id,
        "environment": environment,
        "size_gb": size_gb,
        "current_type": "gp2",
        "peak_iops_30d": round(peak_iops, 0),
        "target_iops": target_iops,
        "target_throughput_mbs": target_throughput,
        "current_monthly_cost": round(current_cost, 2),
        "recommended_monthly_cost": round(recommended_cost, 2),
        "monthly_savings": round(monthly_savings, 2),
        "annual_savings": round(monthly_savings * 12, 2),
    }

    if should_auto_apply(environment):
        apply_gp3_migration(rds, db_id, target_iops, target_throughput)
        results["applied"].append(recommendation)
    elif sns and APPROVAL_SNS_ARN:
        send_approval_request(sns, recommendation)
        results["pending_approval"].append(recommendation)
    else:
        logger.info("DRY RUN - would migrate: %s", json.dumps(recommendation))
        results["pending_approval"].append(recommendation)

except Exception as exc:
    logger.error("Error processing %s: %s", db_id, str(exc))
    results["errors"].append({"instance_id": db_id, "error": str(exc)})

total_applied_savings = sum(r["annual_savings"] for r in results["applied"])
total_pending_savings = sum(r["annual_savings"] for r in results["pending_approval"])

```



```
logger.info(
    "Sweep complete. Applied: %d ($%.2f/yr saved). Pending approval: %d ($%.2f/yr potential). "
    "Skipped: %d. Errors: %d.",
    len(results["applied"]), total_applied_savings,
    len(results["pending_approval"]), total_pending_savings,
    len(results["skipped"]), len(results["errors"]),
)

return {
    "statusCode": 200,
    "body": json.dumps(results),
}
```

Deploy this Lambda on a weekly EventBridge schedule. For dev and staging environments, migrations apply automatically. For production, the SNS notification delivers a structured JSON summary to your operations team, including the exact CLI command needed to execute the change after human review.

Monitoring Storage Costs Over Time with Cost Explorer Tags

After completing migrations, the final step is establishing ongoing visibility so storage type regressions — new gp2 instances deployed via manual console clicks or unreviewed Terraform — are caught quickly.

```
#!/usr/bin/env python3
"""
storage_cost_tracker.py
Queries AWS Cost Explorer for RDS storage costs broken down by storage type.
Run monthly to track optimization progress and detect regressions.
"""

import boto3
from datetime import datetime, date
from dateutil.relativedelta import relativedelta

def get_rds_storage_costs_by_type(months_back=3):
    ce = boto3.client("ce", region_name="us-east-1")

    end_date = date.today().replace(day=1)
    start_date = end_date - relativedelta(months=months_back)

    response = ce.get_cost_and_usage(
        TimePeriod={
            "Start": start_date.strftime("%Y-%m-%d"),
            "End": end_date.strftime("%Y-%m-%d"),
        },
        Granularity="MONTHLY",
        Filter={
            "And": [
                {
                    "Dimensions": {
                        "Key": "SERVICE",
                        "Values": ["Amazon Relational Database Service"],
                    }
                },
                {
                    "Dimensions": {
                        "Key": "USAGE_TYPE_GROUP",
                        "Values": [
                            "RDS: Storage",
                            "RDS: Provisioned IOPS",
                        ],
                    }
                },
            ],
        },
    )
```

```
        },
    },
]
},
GroupBy=[
    {"Type": "DIMENSION", "Key": "USAGE_TYPE"},
],
Metrics=["UnblendedCost"],
)

print(f"\nRDS Storage Cost Breakdown ({start_date} to {end_date})")
print("=" * 65)

for period in response["ResultsByTime"]:
    period_start = period["TimePeriod"]["Start"]
    print(f"\n {period_start[:7]}")
    print(f" {'Usage Type':<50} {'Cost':>10}")
    print(f" {'-'*50} {'-'*10}")

    total = 0.0
    for group in sorted(period["Groups"], key=lambda g: g["Keys"][0]):
        usage_type = group["Keys"][0]
        cost = float(group["Metrics"]["UnblendedCost"]["Amount"])
        total += cost
        # Highlight gp2 costs as candidates for optimization
        marker = " ← gp2 (optimize)" if "gp2" in usage_type.lower() else ""
        print(f" {'usage_type':<50} ${cost:>9.2f}{marker}")

    print(f" {'TOTAL':<50} ${total:>9.2f}")

if __name__ == "__main__":
    get_rds_storage_costs_by_type(months_back=6)
```

Set up a monthly cost anomaly detector in AWS Cost Explorer to alert when RDS storage costs increase month-over-month by more than a defined threshold. The combination of the anomaly detector and this tracking script creates a closed loop: migrations reduce cost, the tracker confirms the reduction, and anomaly detection flags any regression caused by new deployments.

Key Takeaways

- **gp3 is the default choice for virtually all RDS workloads under 16,000 IOPS.** Its 30% lower per-GB cost versus gp2, combined with decoupled IOPS and throughput, eliminates the most common reason teams over-provision gp2 storage — chasing IOPS through volume size.
- **Measure before migrating.** Four weeks of CloudWatch data on `ReadIOPS`, `WriteIOPS`, `DiskQueueDepth`, and `BurstBalance` determines whether the included 3,000 IOPS on gp3 covers the workload or whether additional IOPS provisioning is needed. Migrating a burst-exhausted gp2 instance to gp3 at 3,000 IOPS without this check can introduce a latency regression that is invisible until production load hits.
- **io2 is warranted only above 16,000 IOPS, for Oracle redo log workloads, or when latency consistency at the storage layer is a hard requirement.** For everything else, the gp3 cost advantage is significant enough that io2 is difficult to justify on economic grounds alone.
- **The migration itself is online and low-risk.** RDS handles the storage type change in the

background with no application downtime. The only operational consideration is the brief Multi-AZ failover event that occurs mid-migration on high-availability instances, which can be scheduled outside peak traffic windows.

- **Automation prevents regression.** A weekly Lambda sweep that identifies new gp2 instances and either auto-migrates them in non-production environments or routes them to an approval workflow ensures that storage cost savings compound over time rather than being eroded by new deployments that bypass the optimization process.
-

Chapter 14: Aurora I/O-Optimized vs Standard: Break-Even Analysis

Teams running Aurora on AWS frequently discover that their storage costs quietly eclipse their compute costs within twelve months of going live. The culprit is almost always I/O charges under the Aurora Standard pricing model, where every read and write page costs \$0.20 per million requests. A busy transactional database processing 500 billion I/Os per month—a realistic figure for a mid-size e-commerce platform—generates \$100,000 per month (\$1.2 million per year) in I/O charges alone, on top of compute and storage. Aurora I/O-Optimized eliminates that per-request charge entirely in exchange for a 25% increase in cluster storage pricing. The math to determine which model saves money is not complicated, but it requires accurate I/O measurement, and most teams skip that step. This chapter walks through the full break-even analysis: how to measure actual I/O consumption, how to calculate the crossover point with precision, how to automate the ongoing monitoring that tells you when to switch, and how to validate that your choice is still correct as your workload evolves.

Understanding the Two Pricing Models

Before running any analysis, you need to internalize exactly what Aurora charges for under each model. The pricing differs by region, but the structure is consistent.

Aurora Standard charges three separate line items:

- **Compute:** On-demand or reserved instance pricing per instance-hour
- **Storage:** \$0.10 per GB-month for data stored in the Aurora storage layer
- **I/O:** \$0.20 per million I/O requests (each I/O request is 8 KB; reads and writes both count)

Aurora I/O-Optimized charges two line items:

- **Compute:** On-demand or reserved instance pricing per instance-hour (same rates as Standard)
- **Storage:** \$0.225 per GB-month (125% of Standard storage pricing, or a 25% premium)

There is no separate I/O charge under I/O-Optimized. Zero. The I/O cost is absorbed entirely into the storage markup.

The crossover point—where I/O-Optimized becomes cheaper than Standard—occurs when your I/O charges under Standard would exceed the storage premium you pay under I/O-Optimized. Expressed as a formula:

Monthly I/O charges (Standard) > Storage premium (I/O-Optimized)
 $(\text{I/O requests per month} / 1,000,000) \times \$0.20 > \text{Storage GB} \times (\$0.225 - \$0.10)$
 $(\text{I/O requests per month} / 1,000,000) \times \$0.20 > \text{Storage GB} \times \0.125

Solving for I/O requests per GB of storage:

$\text{I/O requests per month} > \text{Storage GB} \times 625,000$

This is the canonical break-even ratio: **625,000 I/O requests per GB of storage per month**. If your cluster generates more than 625,000 I/Os per GB stored each month, I/O-Optimized saves money. Below that ratio, Standard is cheaper.

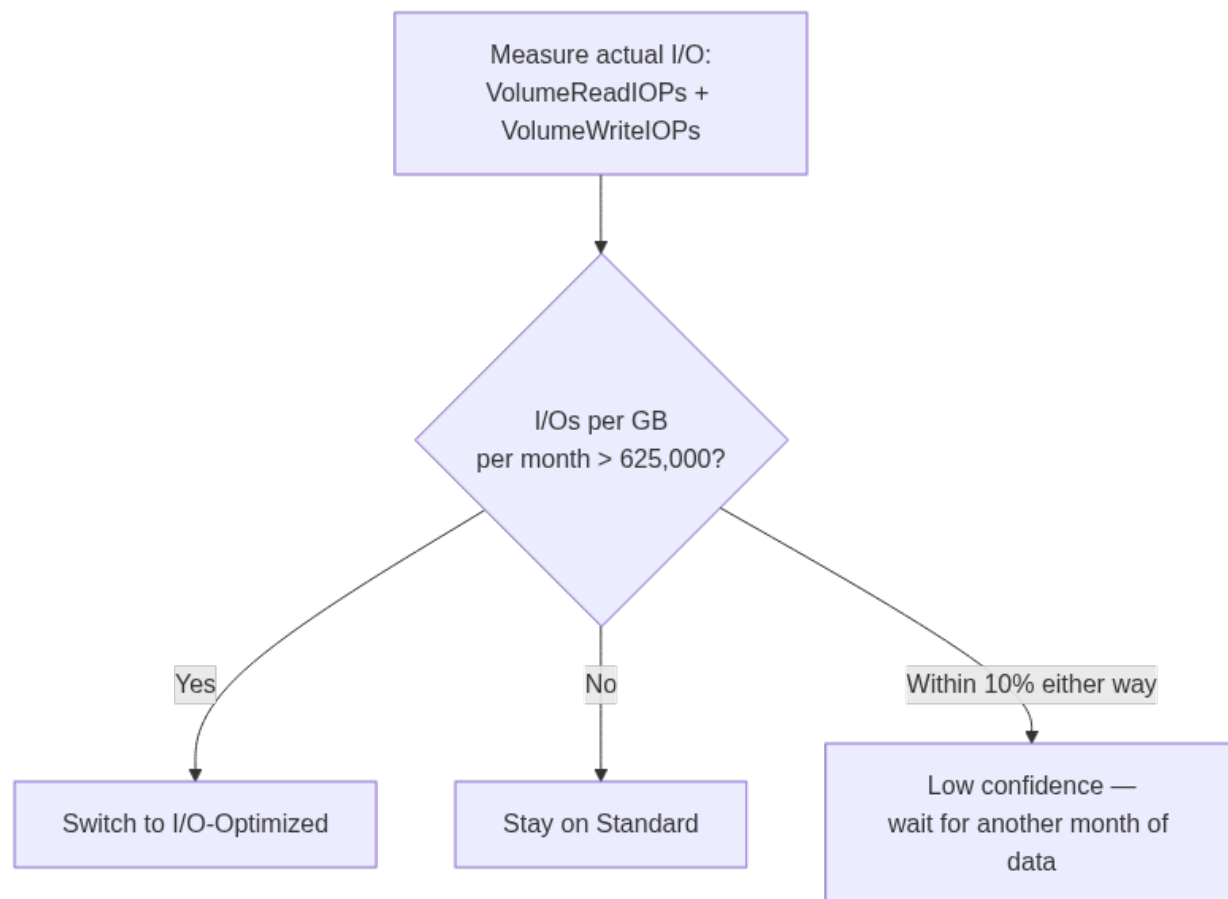


Figure 14: The Aurora storage break-even decision, driven by measured I/O rather than estimated query counts.

A 1 TB Aurora cluster crosses the break-even at approximately 625 million I/O requests per month. A 100 GB cluster crosses it at 62.5 million I/Os per month. The database size matters as much as the I/O volume, and this is the dimension most teams ignore when making the switching decision.

It is worth noting that the 25% storage premium applies to the full Aurora storage volume, which includes the six-way replication Aurora maintains internally. Aurora does not charge you six times the raw data size, but it does charge for the physical storage footprint including replicas, which is typically 10–20% larger than your logical data size. Your storage bill already reflects this; just confirm the number in AWS Cost Explorer before running your calculations.

Measuring Actual I/O Consumption

The single biggest mistake teams make is estimating I/O usage from application metrics or query counts rather than measuring it directly from Aurora. Application-level metrics cannot account for buffer pool efficiency, background write activity, the Aurora redo log writer, or parallel query I/Os. You must measure from the storage layer.

CloudWatch Metrics for Aurora I/O

Aurora exposes two CloudWatch metrics that capture exactly what Aurora Standard charges for:

- **VolumeReadIOPs**: I/O operations per second from the storage layer (reads)
- **VolumeWriteIOPs**: I/O operations per second to the storage layer (writes)

These metrics reflect 8 KB I/O units, exactly matching the billing unit. To convert to monthly I/O requests, sum the two metrics over a full calendar month.

The following script queries CloudWatch and returns total billable I/O for the past 30 days, then calculates the actual I/O cost and what the storage premium would cost under I/O-Optimized:

```
#!/bin/bash
# aurora_io_cost_analysis.sh
# Calculates Aurora Standard I/O costs vs I/O-Optimized storage premium
# Run monthly via EventBridge scheduled rule or cron

CLUSTER_ID="${1:-your-aurora-cluster-id}"
REGION="${2:-us-east-1}"
IO_PRICE_PER_MILLION=0.20      # Aurora Standard I/O price per million requests
STANDARD_STORAGE_PRICE=0.10    # per GB-month
IO_OPT_STORAGE_PRICE=0.225     # per GB-month (25% premium)

END_DATE=$(date -u +%Y-%m-%dT%H:%M:%SZ)
START_DATE=$(date -u -d "30 days ago" +%Y-%m-%dT%H:%M:%SZ 2>/dev/null || \
    date -u -v-30d +%Y-%m-%dT%H:%M:%SZ)

echo "=== Aurora I/O Break-Even Analysis ==="
echo "Cluster: $CLUSTER_ID"
echo "Period: $START_DATE to $END_DATE"
echo ""

# Get cluster storage allocation (VolumeBytesUsed metric)
STORAGE_BYTES=$(aws cloudwatch get-metric-statistics \
    --region "$REGION" \
    --namespace AWS/RDS \
    --metric-name VolumeBytesUsed \
    --dimensions Name=DBClusterIdentifier,Value="$CLUSTER_ID" \
    --start-time "$START_DATE" \
    --end-time "$END_DATE" \
    --period 2592000 \
    --statistics Maximum \
    --query 'Datapoints[0].Maximum' \
    --output text 2>/dev/null)

STORAGE_GB=$((echo "scale=2; ${STORAGE_BYTES:-0} / 1073741824" | bc))
echo "Storage used: ${STORAGE_GB} GB"

# Get total read I/Os (sum over 30 days, 300-second periods)
READ_IOS=$(aws cloudwatch get-metric-statistics \
    --region "$REGION" \
    --namespace AWS/RDS \
    --metric-name VolumeReadIOPs \
    --dimensions Name=DBClusterIdentifier,Value="$CLUSTER_ID" \
```

```

--start-time "$START_DATE" \
--end-time "$END_DATE" \
--period 300 \
--statistics Sum \
--query 'sum(Datapoints[].Sum)' \
--output text)

# Get total write I/Os (sum over 30 days, 300-second periods)
WRITE_IOS=$(aws cloudwatch get-metric-statistics \
--region "$REGION" \
--namespace AWS/RDS \
--metric-name VolumeWriteIOPs \
--dimensions Name=DBClusterIdentifier,Value="$CLUSTER_ID" \
--start-time "$START_DATE" \
--end-time "$END_DATE" \
--period 300 \
--statistics Sum \
--query 'sum(Datapoints[].Sum)' \
--output text)

# AWS CLI returns sum() as None when no data - handle gracefully
READ_IOS=${READ_IOS:-0}
WRITE_IOS=${WRITE_IOS:-0}

# Note: CloudWatch VolumeReadIOPs/VolumeWriteIOPs are rates (IOPS).
# When summed over 300-second periods, multiply by period to get total operations.
TOTAL_READ_OPS=$(echo "scale=0; ${READ_IOS} * 300" | bc)
TOTAL_WRITE_OPS=$(echo "scale=0; ${WRITE_IOS} * 300" | bc)
TOTAL_IOS=$(echo "$TOTAL_READ_OPS + $TOTAL_WRITE_OPS" | bc)
TOTAL_IOS_MILLIONS=$(echo "scale=4; $TOTAL_IOS / 1000000" | bc)

echo "Total Read I/Os (30 days): $(printf '%,.0f' $TOTAL_READ_OPS)"
echo "Total Write I/Os (30 days): $(printf '%,.0f' $TOTAL_WRITE_OPS)"
echo "Total I/Os (millions): ${TOTAL_IOS_MILLIONS}M"
echo ""

# Calculate costs under Standard
IO_COST=$(echo "scale=2; $TOTAL_IOS_MILLIONS * $IO_PRICE_PER_MILLION" | bc)
STANDARD_STORAGE_COST=$(echo "scale=2; $STORAGE_GB * $STANDARD_STORAGE_PRICE" | bc)
STANDARD_TOTAL=$(echo "scale=2; $IO_COST + $STANDARD_STORAGE_COST" | bc)

# Calculate costs under I/O-Optimized
IO_OPT_STORAGE_COST=$(echo "scale=2; $STORAGE_GB * $IO_OPT_STORAGE_PRICE" | bc)
STORAGE_PREMIUM=$(echo "scale=2; $IO_OPT_STORAGE_COST - $STANDARD_STORAGE_COST" | bc)

# Break-even calculation
BREAKEVEN_IOS_MILLIONS=$(echo "scale=2; $STORAGE_GB * 625000 / 1000000" | bc)

echo "=== Cost Comparison (30-day period) ==="
echo "Aurora Standard:"
echo "  Storage cost:      \${STANDARD_STORAGE_COST}"
echo "  I/O cost:          \${IO_COST}"
echo "  Storage+I/O total: \${STANDARD_TOTAL}"
echo ""
echo "Aurora I/O-Optimized:"
echo "  Storage cost:      \${IO_OPT_STORAGE_COST}"
echo "  Storage premium:   \${STORAGE_PREMIUM}"
echo "  I/O cost:          \$0.00"
echo ""
echo "=== Break-Even Analysis ==="
echo "Break-even I/O threshold: ${BREAKEVEN_IOS_MILLIONS}M I/Os/month"
echo "Actual I/Os this month:  ${TOTAL_IOS_MILLIONS}M I/Os/month"

# Recommendation
SAVINGS=$(echo "scale=2; $IO_COST - $STORAGE_PREMIUM" | bc)
if (( $(echo "$TOTAL_IOS_MILLIONS > $BREAKEVEN_IOS_MILLIONS" | bc -l) )); then
  echo ""
  echo "RECOMMENDATION: Switch to I/O-Optimized"
  echo "Estimated monthly savings: \${SAVINGS}"
  echo "Estimated annual savings:  \$(echo "scale=2; $SAVINGS * 12" | bc)"

```



```

else
  echo ""
  echo "RECOMMENDATION: Remain on Standard"
  echo "I/O-Optimized would cost \${$(echo "scale=2; $STORAGE_PREMIUM - $IO_COST" | bc) MORE per month"
fi

```

Interpreting CloudWatch IOPS Metrics Correctly

One detail that causes calculation errors: `VolumeReadIOPs` and `VolumeWriteIOPs` report as a rate (operations per second), not a count. When you retrieve them with a 300-second period and `Sum` statistic, CloudWatch returns the sum of average IOPS values across each 5-minute window, not the actual operation count. To get true total operations, multiply the summed IOPS values by the period duration (300 seconds). The script above handles this correctly. If you pull these metrics from Cost Explorer instead—which shows actual billed I/O—use those figures directly without the period multiplication.

Cross-Validating with AWS Cost Explorer

Always validate your CloudWatch estimates against your actual AWS bill. Cost Explorer breaks down Aurora charges by line item when you filter correctly:

```

# Pull Aurora I/O costs from Cost Explorer for the last 3 months
# Requires ce:GetCostAndUsage IAM permission

aws ce get-cost-and-usage \
  --region us-east-1 \
  --time-period Start=$(date -d "90 days ago" +%Y-%m-01),End=$(date +%Y-%m-01) \
  --granularity MONTHLY \
  --filter '{
    "And": [
      {
        "Dimensions": {
          "Key": "SERVICE",
          "Values": ["Amazon Relational Database Service"]
        }
      },
      {
        "Dimensions": {
          "Key": "USAGE_TYPE",
          "Values": ["Aurora:StorageIOUsage"]
        }
      }
    ]
  }' \
  --metrics "BlendedCost" "UsageQuantity" \
  --group-by Type=DIMENSION,Key=USAGE_TYPE \
  --output table

# For tagged clusters - filter by your cluster tag
aws ce get-cost-and-usage \
  --region us-east-1 \
  --time-period Start=$(date -d "90 days ago" +%Y-%m-01),End=$(date +%Y-%m-01) \
  --granularity MONTHLY \
  --filter '{
    "And": [
      {
        "Dimensions": {
          "Key": "SERVICE",
          "Values": ["Amazon Relational Database Service"]
        }
      },
      {
        "Tags": {
          "Key": "DBCluster",
          "Values": ["your-cluster-name"]
        }
      }
    ]
  }' \
  --metrics "BlendedCost" "UsageQuantity" \
  --group-by Type=DIMENSION,Key=USAGE_TYPE \
  --output table

```

```

    }
  ]
}' \
--metrics "BlendedCost" "UsageQuantity" \
--group-by Type=DIMENSION,Key=USAGE_TYPE \
--output table

```

The usage type `Aurora:StorageIOUsage` is what appears on your bill for I/O charges under Standard. If you have switched to I/O-Optimized, this line item disappears from your bill entirely and the storage line item increases. That change in line items is also a good sanity check that the configuration actually changed.

Running the Break-Even Calculation at Scale

If you manage multiple Aurora clusters across multiple AWS accounts—typical for a platform team supporting dozens of product teams—you need a programmatic approach that surfaces break-even status for every cluster and identifies the ones where switching would generate meaningful savings.

The following Python script queries CloudWatch and Cost Explorer across a list of clusters and produces a prioritized recommendation report. It is designed to run monthly via an EventBridge rule invoking a Lambda function, writing results to S3 for distribution.

```

#!/usr/bin/env python3
"""
aurora_breakeven_analyzer.py

Analyzes all Aurora clusters in a list of AWS accounts and produces
break-even recommendations for I/O-Optimized vs Standard switching.

Run monthly. Outputs a ranked CSV of recommendations by annual savings potential.

Requirements: boto3, pandas
IAM: cloudwatch:GetMetricStatistics, ce:GetCostAndUsage, rds:DescribeDBClusters
"""

import boto3
import pandas as pd
from datetime import datetime, timedelta, timezone
from typing import Optional
import logging
import json
import os

logging.basicConfig(level=logging.INFO, format='%(asctime)s %(levelname)s %(message)s')
logger = logging.getLogger(__name__)

# Pricing constants (us-east-1; adjust per region)
STANDARD_IO_PRICE_PER_MILLION = 0.20
STANDARD_STORAGE_PRICE_PER_GB = 0.10
IO_OPT_STORAGE_PRICE_PER_GB = 0.225
STORAGE_PREMIUM_PER_GB = IO_OPT_STORAGE_PRICE_PER_GB - STANDARD_STORAGE_PRICE_PER_GB # 0.125

# Regional pricing overrides (subset - expand as needed)
REGIONAL_PRICING = {
    "us-east-1": {"io": 0.20, "std_storage": 0.10, "opt_storage": 0.225},
    "us-west-2": {"io": 0.20, "std_storage": 0.10, "opt_storage": 0.225},
    "eu-west-1": {"io": 0.22, "std_storage": 0.11, "opt_storage": 0.2475},
    "ap-southeast-1": {"io": 0.24, "std_storage": 0.12, "opt_storage": 0.27},
}

def get_pricing(region: str) -> dict:

```

```

"""Return pricing constants for a given region, falling back to us-east-1."""
return REGIONAL_PRICING.get(region, REGIONAL_PRICING["us-east-1"])

def get_cluster_storage_gb(cw_client, cluster_id: str, days: int = 30) -> float:
    """
Retrieve maximum VolumeBytesUsed over the analysis period.
Returns storage in GB.
"""

    end = datetime.now(timezone.utc)
    start = end - timedelta(days=days)

    try:
        response = cw_client.get_metric_statistics(
            Namespace="AWS/RDS",
            MetricName="VolumeBytesUsed",
            Dimensions=[{"Name": "DBClusterIdentifier", "Value": cluster_id}],
            StartTime=start,
            EndTime=end,
            Period=86400,          # daily granularity
            Statistics=["Maximum"]
        )
        if response["Datapoints"]:
            max_bytes = max(dp["Maximum"] for dp in response["Datapoints"])
            return max_bytes / (1024 ** 3)
        return 0.0
    except Exception as e:
        logger.warning(f"Could not retrieve storage for {cluster_id}: {e}")
        return 0.0

def get_total_ios(cw_client, cluster_id: str, days: int = 30) -> dict:
    """
Sum VolumeReadIOPS and VolumeWriteIOPS over the analysis period.
Returns dict with read_ios, write_ios, total_ios (all in millions).

Note: CloudWatch IOPS metrics are averages per second per 5-min window.
Total ops = sum(IOPS values) * period_seconds.
"""

    end = datetime.now(timezone.utc)
    start = end - timedelta(days=days)
    period_seconds = 300          # 5-minute resolution

    results = {"read_ios_m": 0.0, "write_ios_m": 0.0, "total_ios_m": 0.0}

    for metric_name, key in [(("VolumeReadIOPS", "read_ios_m"),
                              ("VolumeWriteIOPS", "write_ios_m"))]:
        try:
            paginator_response = []
            next_token = None
            # Paginate if needed (>1440 datapoints for 5-min over 30 days)
            while True:
                kwargs = {
                    "Namespace": "AWS/RDS",
                    "MetricName": metric_name,
                    "Dimensions": [{"Name": "DBClusterIdentifier", "Value": cluster_id}],
                    "StartTime": start,
                    "EndTime": end,
                    "Period": period_seconds,
                    "Statistics": ["Average"]
                }
                response = cw_client.get_metric_statistics(**kwargs)
                paginator_response.extend(response.get("Datapoints", []))
                break          # get_metric_statistics doesn't paginate; included for clarity

            total_ops = sum(dp["Average"] for dp in paginator_response) * period_seconds
            results[key] = total_ops / 1_000_000
        except Exception as e:
            logger.warning(f"Could not retrieve {metric_name} for {cluster_id}: {e}")

```

```

results["total_ios_m"] = results["read_ios_m"] + results["write_ios_m"]
return results

def get_current_storage_config(rds_client, cluster_id: str) -> dict:
    """Return storage type and I/O-Optimized status for a cluster."""
    try:
        response = rds_client.describe_db_clusters(DBClusterIdentifier=cluster_id)
        cluster = response["DBClusters"][0]
        return {
            "engine": cluster.get("Engine", ""),
            "engine_version": cluster.get("EngineVersion", ""),
            "storage_type": cluster.get("StorageType", "aurora"),
            "is_io_optimized": cluster.get("StorageType", "") == "aurora-iopt1",
            "multi_az": cluster.get("MultiAZ", False),
        }
    except Exception as e:
        logger.warning(f"Could not describe cluster {cluster_id}: {e}")
        return {"storage_type": "unknown", "is_io_optimized": False}

def calculate_breakeven(
    storage_gb: float,
    total_ios_m: float,
    region: str = "us-east-1"
) -> dict:
    """
    Calculate break-even analysis for a single cluster.

    Returns a dict with:
    - current_io_cost: what I/O would cost under Standard
    - storage_premium: additional storage cost under I/O-Optimized
    - monthly_savings: positive = I/O-Optimized is cheaper
    - breakeven_ios_m: I/O threshold in millions
    - recommendation: 'io_optimized' or 'standard'
    - savings_confidence: 'high' / 'medium' / 'low' based on margin
    """
    pricing = get_pricing(region)

    # Cost under Standard (I/O charges only - storage cost is same baseline)
    io_cost = total_ios_m * pricing["io"]

    # Additional storage cost under I/O-Optimized
    storage_premium = storage_gb * (pricing["opt_storage"] - pricing["std_storage"])

    # Net monthly savings from switching to I/O-Optimized
    monthly_savings = io_cost - storage_premium

    # Break-even I/O threshold
    breakeven_ios_m = (storage_gb * (pricing["opt_storage"] - pricing["std_storage"])
                       / pricing["io"])

    # Confidence based on margin from break-even
    if total_ios_m == 0:
        margin_pct = -100.0
    else:
        margin_pct = (total_ios_m - breakeven_ios_m) / breakeven_ios_m * 100

    if abs(margin_pct) > 25:
        confidence = "high"
    elif abs(margin_pct) > 10:
        confidence = "medium"
    else:
        confidence = "low" # Within 10% of break-even - monitor before switching

    recommendation = "io_optimized" if monthly_savings > 0 else "standard"

    return {
        "io_cost_monthly": round(io_cost, 2),
        "storage_premium_monthly": round(storage_premium, 2),
    }

```

```

        "monthly_savings": round(monthly_savings, 2),
        "annual_savings": round(monthly_savings * 12, 2),
        "breakeven_ios_m": round(breakeven_ios_m, 2),
        "actual_ios_m": round(total_ios_m, 2),
        "margin_pct": round(margin_pct, 1),
        "recommendation": recommendation,
        "confidence": confidence,
    }

def analyze_clusters(cluster_list: list, region: str = "us-east-1") -> pd.DataFrame:
    """
    Analyze a list of Aurora cluster IDs and return a DataFrame with break-even analysis for each.

    cluster_list: list of Aurora cluster identifier strings
    region: AWS region string
    """
    session = boto3.Session(region_name=region)
    cw_client = session.client("cloudwatch")
    rds_client = session.client("rds")

    rows = []

    for cluster_id in cluster_list:
        logger.info(f"Analyzing cluster: {cluster_id}")

        # Gather raw data
        storage_gb = get_cluster_storage_gb(cw_client, cluster_id)
        io_data = get_total_ios(cw_client, cluster_id)
        config = get_current_storage_config(rds_client, cluster_id)

        if storage_gb == 0:
            logger.warning(f"No storage data for {cluster_id}; skipping.")
            continue

        # Run break-even math
        analysis = calculate_breakeven(
            storage_gb=storage_gb,
            total_ios_m=io_data["total_ios_m"],
            region=region
        )

        row = {
            "cluster_id": cluster_id,
            "region": region,
            "engine": config.get("engine", ""),
            "engine_version": config.get("engine_version", ""),
            "current_storage_type": config.get("storage_type", ""),
            "is_io_optimized": config.get("is_io_optimized", False),
            "storage_gb": round(storage_gb, 1),
            "read_ios_m": round(io_data["read_ios_m"], 2),
            "write_ios_m": round(io_data["write_ios_m"], 2),
            "total_ios_m": round(io_data["total_ios_m"], 2),
            **analysis
        }
        rows.append(row)

    df = pd.DataFrame(rows)

    if df.empty:
        return df

    # Sort by annual savings descending (biggest opportunity first)
    df = df.sort_values("annual_savings", ascending=False).reset_index(drop=True)

    return df

def generate_report(df: pd.DataFrame, output_path: str = "aurora_breakeven_report.csv") -> None:
    """

```

```

Write a ranked recommendation report to CSV and print a summary table
of the top candidates.
"""
if df.empty:
    logger.info("No clusters to report.")
    return

df.to_csv(output_path, index=False)
logger.info(f"Full report written to {output_path}")

# Summary: top 10 candidates for switching to I/O-Optimized
switch_candidates = df[
    (df["recommendation"] == "io_optimized") &
    (df["is_io_optimized"] == False)
].head(10)

if switch_candidates.empty:
    print("\nNo clusters currently on Standard would benefit from I/O-Optimized.")
else:
    print("\n=== Top Candidates: Switch Standard → I/O-Optimized ===")
    print(switch_candidates[[
        "cluster_id", "storage_gb", "total_ios_m",
        "io_cost_monthly", "storage_premium_monthly",
        "monthly_savings", "annual_savings", "confidence"
    ]].to_string(index=False))

# Summary: clusters on I/O-Optimized that might benefit from switching back
downgrade_candidates = df[
    (df["recommendation"] == "standard") &
    (df["is_io_optimized"] == True)
].head(10)

if not downgrade_candidates.empty:
    print("\n=== Review Candidates: I/O-Optimized → Standard (over-provisioned) ===")
    print(downgrade_candidates[[
        "cluster_id", "storage_gb", "total_ios_m",
        "io_cost_monthly", "storage_premium_monthly",
        "monthly_savings", "annual_savings", "confidence"
    ]].to_string(index=False))

# Aggregate summary
total_annual_opportunity = switch_candidates["annual_savings"].sum()
total_annual_waste = abs(downgrade_candidates["monthly_savings"].sum() * 12) \
    if not downgrade_candidates.empty else 0

print(f"\n=== Portfolio Summary ===")
print(f"Clusters analyzed: {len(df)}")
print(f"Switch-to-I/O-Optimized count: {len(switch_candidates)}")
print(f"Downgrade-to-Standard count: {len(downgrade_candidates)}")
print(f"Annual savings opportunity: ${total_annual_opportunity:,.2f}")
print(f"Annual over-spend (I/O-Opt): ${total_annual_waste:,.2f}")

# ---- Lambda / CLI entrypoint ----

def lambda_handler(event, context):
    """
    AWS Lambda entrypoint. Expects event with:
    {
        "cluster_ids": ["cluster-a", "cluster-b", ...],
        "region": "us-east-1",
        "output_bucket": "my-finops-bucket",
        "output_key": "aurora/breakeven/2024-11-01.csv"
    }
    """
    cluster_ids = event.get("cluster_ids", [])
    region = event.get("region", "us-east-1")
    output_bucket = event.get("output_bucket")
    output_key = event.get("output_key", f"aurora/breakeven/{datetime.utcnow().date()}.csv")

```

```
if not cluster_ids:
    # Auto-discover all Aurora clusters in the region
    rds = boto3.client("rds", region_name=region)
    paginator = rds.get_paginator("describe_db_clusters")
    cluster_ids = []
    for page in paginator.paginate():
        for cluster in page["DBClusters"]:
            if "aurora" in cluster.get("Engine", ""):
                cluster_ids.append(cluster["DBClusterIdentifier"])
    logger.info(f"Auto-discovered {len(cluster_ids)} Aurora clusters in {region}")

df = analyze_clusters(cluster_ids, region=region)

if output_bucket:
    tmp_path = "/tmp/aurora_breakeven.csv"
    generate_report(df, output_path=tmp_path)
    s3 = boto3.client("s3")
    s3.upload_file(tmp_path, output_bucket, output_key)
    logger.info(f"Report uploaded to s3://{output_bucket}/{output_key}")
    return {"status": "ok", "clusters_analyzed": len(df), "s3_path":
        ↪ f"s3://{output_bucket}/{output_key}"}
else:
    generate_report(df)
    return {"status": "ok", "clusters_analyzed": len(df)}

if __name__ == "__main__":
    import sys
    clusters = sys.argv[1:] if len(sys.argv) > 1 else []
    region = os.environ.get("AWS_DEFAULT_REGION", "us-east-1")
    df = analyze_clusters(clusters, region=region)
    generate_report(df)
```

Handling Multi-Region Portfolios

When your Aurora clusters span multiple AWS regions, run separate analyses per region using the regional pricing table in `REGIONAL_PRICING`. I/O and storage prices differ meaningfully across regions—the break-even I/O-per-GB ratio is the same formula regardless of region, but the absolute dollar thresholds change. Always confirm current prices from the AWS pricing page before encoding them; AWS occasionally revises them with limited notice.

To analyze clusters across accounts, invoke the script with an assumed IAM role per account using `boto3.assume_role` before constructing the clients. The structure is otherwise identical.

Workload Pattern Considerations

The 625,000 I/O-per-GB break-even ratio is a starting point, not a final answer. Several workload characteristics shift the practical recommendation.

Read-Heavy vs. Write-Heavy Workloads

Aurora's storage layer handles reads and writes differently. Under Standard, both count equally for billing. Under I/O-Optimized, neither counts. However, write-heavy workloads often have higher I/O amplification from Aurora's redo log mechanism: a single logical write from the application may generate multiple physical I/Os in the storage layer as Aurora writes log records, applies them to pages, and flushes checkpoints. OLTP workloads with high update or delete rates can have an I/O multiplier of 2–4× compared to what the application perceives. This amplification is already captured in your `VolumeWriteIOPs` measurement, so your CloudWatch numbers reflect the true

billing impact. What this means operationally is that write-heavy workloads tend to cross the break-even threshold sooner than their query count alone would suggest.

Bursty vs. Steady-State I/O

Aurora Standard charges for every I/O regardless of when it occurs—there is no burst allowance or free tier. A workload that runs a monthly analytics batch generating 10× the normal I/O for 48 hours will incur the same cost as one that generates the same total I/O spread evenly. For break-even analysis, what matters is the total monthly I/O count, not the distribution. However, if your burst activity is seasonal (e.g., holiday traffic in retail), measure I/O over your peak month and your typical month separately. If even your typical month is below break-even but your peak months drive significant I/O costs, you may choose to switch models for peak periods—though Aurora currently requires a storage type change at the cluster level, which takes effect immediately and cannot be toggled daily. In practice, most teams with bursty workloads pick the model that optimizes for their elevated months and accept the slight over-payment during quiet periods.

The Buffer Pool Effect

One variable that the CloudWatch metrics automatically account for, but that teams sometimes misunderstand, is the buffer pool hit rate. A large `db.r6g.16xlarge` instance with 512 GB of memory will serve the majority of reads from memory, generating very few `VolumeReadIOPs`. The same data on a smaller instance with a cold buffer pool will generate far more storage-layer reads. When you right-size instances (covered in Chapter 7), buffer pool hit rates change, which shifts your I/O consumption. If you are planning both a storage model change and an instance right-sizing in the same quarter, run the I/O analysis after the right-sizing, not before. Downsizing memory can significantly increase read I/Os and change which storage model is optimal.

Aurora Serverless v2 Considerations

Aurora Serverless v2 supports both Standard and I/O-Optimized storage. The break-even formula applies identically. However, Serverless v2 workloads often have higher I/O variance than provisioned clusters because ACU scaling reacts to load in real time, occasionally running smaller ACU allocations with lower effective buffer pool capacity. Monitor `VolumeReadIOPs` carefully for Serverless v2 clusters before making the switching decision; a cluster that scales down to 0.5 ACU during off-hours will generate substantially more storage reads during those periods than the same workload on a provisioned instance.

Automating the Ongoing Decision

Making a one-time switching decision is straightforward. The harder problem is maintaining the correct storage configuration as your workload evolves month over month. I/O patterns change: new features add write-heavy tables, application caching layers are added (reducing read I/Os), data retention policies grow the storage footprint, and seasonal traffic patterns shift the ratio.

The right operational posture is a monthly automated review that flags clusters whose current storage type is misaligned with their break-even status and presents the finding for human approval before any change is made. Aurora does not support automatic storage type switching via any AWS-native automation as of this writing, so the workflow is detect → notify → human approves → change applied.

EventBridge + Lambda Monthly Review Pattern

```

# aurora_io_monitor_alert.py
# Companion to the main analyzer - generates SNS alerts for misaligned clusters.
# Deploy as Lambda, triggered monthly by EventBridge.

import boto3
import json
import os
from datetime import datetime, timezone

def lambda_handler(event, context):
    region = os.environ.get("AWS_REGION", "us-east-1")
    sns_topic_arn = os.environ["SNS_TOPIC_ARN"]
    confidence_threshold = os.environ.get("CONFIDENCE_THRESHOLD", "medium")

    # Import the analyzer from the main module (package deployment)
    from aurora_breakeven_analyzer import analyze_clusters

    # Auto-discover Aurora clusters
    rds = boto3.client("rds", region_name=region)
    paginator = rds.get_paginator("describe_db_clusters")
    cluster_ids = []
    for page in paginator.paginate():
        for cluster in page["DBClusters"]:
            if "aurora" in cluster.get("Engine", ""):
                cluster_ids.append(cluster["DBClusterIdentifier"])

    df = analyze_clusters(cluster_ids, region=region)

    if df.empty:
        return {"status": "no_clusters"}

    # Filter confidence levels
    confidence_order = {"high": 3, "medium": 2, "low": 1}
    min_confidence = confidence_order.get(confidence_threshold, 2)

    misaligned = df[
        (df["confidence"].map(confidence_order) >= min_confidence) &
        (
            # On Standard but should switch to I/O-Optimized
            ((df["recommendation"] == "io_optimized") & (df["is_io_optimized"] == False)) |
            # On I/O-Optimized but Standard would be cheaper
            ((df["recommendation"] == "standard") & (df["is_io_optimized"] == True))
        )
    ]

    if misaligned.empty:
        return {"status": "all_clusters_aligned"}

    # Compose alert message
    lines = [
        f"Aurora I/O Storage Alignment Report - {datetime.now(timezone.utc).date()}",
        f"Region: {region}",
        f"",
        f"The following clusters have misaligned storage configurations:",
        f"",
    ]

    for _, row in misaligned.iterrows():
        action = ("Switch to I/O-Optimized"
                 if row["recommendation"] == "io_optimized"
                 else "Switch to Standard")
        direction = ("saving" if row["monthly_savings"] > 0 else "wasting")
        lines.append(
            f" {row['cluster_id']}: {action}"
            f" | Est. ${abs(row['monthly_savings']):.2f}/month ({direction})"
            f" | Confidence: {row['confidence']}"
            f" | Storage: {row['storage_gb']:.0f} GB"
            f" | I/O: {row['total_ios_m']:.0f}M ops/month"
        )

```

```
)

total_opp = misaligned[misaligned["monthly_savings"] > 0]["monthly_savings"].sum()
total_waste = abs(misaligned[misaligned["monthly_savings"] < 0]["monthly_savings"].sum())

lines += [
    f"",
    f"Total optimization opportunity: ${total_opp * 12:,.2f}/year",
    f"Total over-spend to recover:    ${total_waste * 12:,.2f}/year",
    f"",
    f"Review and apply changes in the AWS Console or via the aurora_switch_storage_type.sh script.",
]

message = "\n".join(lines)

sns = boto3.client("sns", region_name=region)
sns.publish(
    TopicArn=sns_topic_arn,
    Subject=f"Aurora I/O Storage Alert - {len(misaligned)} clusters misaligned",
    Message=message
)

return {
    "status": "alert_sent",
    "misaligned_count": len(misaligned),
    "annual_opportunity": round(total_opp * 12, 2)
}
```

Applying the Storage Type Change

Once the analysis recommends a switch and a human approves it, the actual change is a single AWS CLI call or Terraform resource update:

```
# Switch a cluster to I/O-Optimized
aws rds modify-db-cluster \
  --db-cluster-identifier your-cluster-id \
  --storage-type aurora-iopt1 \
  --apply-immediately \
  --region us-east-1

# Switch a cluster back to Standard
aws rds modify-db-cluster \
  --db-cluster-identifier your-cluster-id \
  --storage-type aurora \
  --apply-immediately \
  --region us-east-1
```

Aurora applies the storage type change without downtime. The modification takes effect within a few minutes for most cluster sizes. In Terraform, update the `storage_type` argument on the `aws_rds_cluster` resource:

```
resource "aws_rds_cluster" "main" {
  cluster_identifier = "your-cluster-id"
  engine             = "aurora-postgresql"
  engine_version     = "15.4"

  # Toggle between "aurora" (Standard) and "aurora-iopt1" (I/O-Optimized)
  storage_type       = var.aurora_storage_type

  # ... other configuration unchanged
}
```

Driving this via a variable allows you to control the switch through your normal Terraform plan/apply workflow, which provides an approval gate and change record without needing a separate script.

Validating the Decision Post-Switch

Switching storage type is not a set-and-forget action. Validate within 30–60 days that the change produced the expected cost reduction.

What to Check After Switching to I/O-Optimized

1. **Verify `Aurora:StorageIOUsage` disappears from your bill.** Cost Explorer should show zero I/O charges for the cluster. If you see residual I/O charges in the same month as the switch, they reflect I/Os that occurred before the storage type change took effect; these will not appear in subsequent months.
2. **Confirm storage line item increased by the expected amount.** Your `Aurora:StorageUsage` line should increase by approximately 12.5% of the previous storage cost (the $0.125/0.10$ ratio = 25% premium in per-GB terms, but the net increase against your prior bill is 125% of $\text{old} \times 0.125 / 0.10$ premium portion). Do the arithmetic using your actual storage GB from the analysis.
3. **Compare total storage+I/O spend month over month.** The combined line should decrease by approximately the `monthly_savings` value your analysis projected. Deviations of more than 10% warrant investigating whether I/O consumption changed significantly around the time of the switch (new deployment, load pattern change, etc.).

What to Check After Switching Back to Standard

1. **Verify `Aurora:StorageIOUsage` reappears.** If I/O charges are higher than projected, your I/O consumption grew after the decision was made—rerun the break-even analysis.
2. **Confirm storage line item decreased.** The 12.5% storage markup should vanish from the bill, reducing that line accordingly.
3. **Monitor I/O growth trend.** A cluster switched back to Standard because it was below break-even at the time of the decision may cross back over the threshold within months if data or traffic is growing. Set a CloudWatch alarm on `VolumeReadIOPs` + `VolumeWriteIOPs` average IOPS that pages your team at 80% of the break-even equivalent IOPS rate so you can anticipate the switch before overspending begins.

Setting a Break-Even IOPS Alarm

The following Python snippet generates the CloudWatch alarm threshold value for a given cluster's current storage footprint:

```
def calculate_breakeven_iops_alarm(storage_gb: float,
                                  region: str = "us-east-1",
                                  alarm_buffer_pct: float = 0.80) -> dict:
    """
    Calculate the sustained average IOPS rate that corresponds to
    the break-even I/O cost threshold, adjusted by an alarm buffer.

    alarm_buffer_pct = 0.80 means alarm fires at 80% of break-even.
    Returns CloudWatch alarm parameters.
    """
    pricing = get_pricing(region)
    seconds_per_month = 30 * 24 * 3600 # approx

    # Total I/Os at break-even per month
    breakeven_ios_total = storage_gb * (
        (pricing["opt_storage"] - pricing["std_storage"]) / pricing["io"]
    ) * 1_000_000
```

```

# Convert to average IOPS sustained over the month
breakeven_iops = breakeven_ios_total / seconds_per_month

# Apply buffer so alarm fires before you reach break-even
alarm_threshold_iops = breakeven_iops * alarm_buffer_pct

return {
    "threshold_iops": round(alarm_threshold_iops, 1),
    "breakeven_iops": round(breakeven_iops, 1),
    "alarm_buffer_pct": alarm_buffer_pct,
    "description": (
        f"Fires at {alarm_buffer_pct*100:.0f}% of break-even ({breakeven_iops:.1f} IOPS). "
        f"Above this rate, I/O-Optimized storage saves money."
    )
}

def create_breakeven_alarm(cluster_id: str,
                          storage_gb: float,
                          sns_topic_arn: str,
                          region: str = "us-east-1") -> None:
    """
    Create a CloudWatch composite alarm that triggers when combined
    VolumeReadIOPs + VolumeWriteIOPs approach the break-even threshold.
    """
    alarm_params = calculate_breakeven_iops_alarm(storage_gb, region)
    cw = boto3.client("cloudwatch", region_name=region)

    # Aurora doesn't expose a combined IOPS metric natively;
    # use a metric math alarm to sum read + write.
    cw.put_metric_alarm(
        AlarmName=f"aurora-io-breakeven-{cluster_id}",
        AlarmDescription=alarm_params["description"],
        ActionsEnabled=True,
        AlarmActions=[sns_topic_arn],
        Metrics=[
            {
                "Id": "read_iops",
                "MetricStat": {
                    "Metric": {
                        "Namespace": "AWS/RDS",
                        "MetricName": "VolumeReadIOPs",
                        "Dimensions": [
                            {"Name": "DBClusterIdentifier", "Value": cluster_id}
                        ]
                    },
                    "Period": 3600,
                    "Stat": "Average"
                },
                "ReturnData": False
            },
            {
                "Id": "write_iops",
                "MetricStat": {
                    "Metric": {
                        "Namespace": "AWS/RDS",
                        "MetricName": "VolumeWriteIOPs",
                        "Dimensions": [
                            {"Name": "DBClusterIdentifier", "Value": cluster_id}
                        ]
                    },
                    "Period": 3600,
                    "Stat": "Average"
                },
                "ReturnData": False
            },
            {
                "Id": "total_iops",
                "Expression": "read_iops + write_iops",
            }
        ]
    )

```

```

        "Label": "TotalVolumeIOPs",
        "ReturnData": True
    }
],
"ComparisonOperator": "GreaterThanThreshold",
"Threshold=alarm_params["threshold_iops"],
"EvaluationPeriods=3,          # sustained over 3 hours
"TreatMissingData="notBreaching",
"Tags=[
    {"Key": "FinOps", "Value": "aurora-io-breakeven"},
    {"Key": "ClusterID", "Value": cluster_id}
]
)

print(f"Alarm created: aurora-io-breakeven-{cluster_id}")
print(f"  Threshold: {alarm_params['threshold_iops']} IOPS (combined read+write)")
print(f"  Break-even: {alarm_params['breakeven_iops']} IOPS")

```

There is a syntax error in the snippet above that you should fix before deploying: `ComparisonOperator":` is missing its opening quote. The corrected line is `"ComparisonOperator":`. The rest of the structure is valid. This kind of off-by-one quoting error is worth a specific callout because it will cause the Lambda deployment to fail silently in some packaging scenarios.

The Decision Framework in Practice

The analysis tools above produce numbers, but numbers alone do not drive organizational decisions. In most engineering organizations, switching Aurora storage type requires a change request, sign-off from a database or platform team lead, and coordination with on-call schedules even though the change itself is zero-downtime. The following framework collapses the analysis into a decision process that fits inside a typical FinOps review cycle.

Monthly Review Cadence

Step	Who	Action
1	Automation	Run <code>aurora-breakeven_analyzer.py</code> for all clusters
2	Automation	Send SNS alert for misaligned clusters with high/medium confidence
3	FinOps engineer	Review alert, validate data quality, confirm no upcoming architecture changes
4	DB team lead	Approve or defer the change
5	On-call engineer	Apply <code>modify-db-cluster</code> or Terraform apply during business hours
6	FinOps engineer	Validate cost change in Cost Explorer 30 days post-switch

When to Defer the Switch

Certain conditions should cause you to defer even a high-confidence recommendation:

- **Planned data migrations or large imports within the next 30 days.** Bulk loads generate atypical I/O bursts that will distort the break-even calculation during and immediately after the migration.
- **Pending instance right-sizing.** As discussed earlier, changing instance size alters buffer pool behavior and shifts I/O consumption in ways that can flip the recommendation.
- **Major application deployments with unknown I/O profile.** If a new feature is launching in the next sprint and its database access pattern is uncertain, measure I/O for a full cycle post-launch before switching.
- **Within 10% of break-even with low confidence rating.** The 10% margin is noise in monthly I/O measurement. Wait for another month of data to confirm the direction.

Tracking the Decision History

Tag each Aurora cluster with metadata that records when the storage type was last evaluated and which direction the recommendation pointed. A simple tagging convention on the RDS cluster resource provides an auditable history without requiring a separate database:

```
aws rds add-tags-to-resource \
  --resource-name "arn:aws:rds:us-east-1:123456789:cluster:your-cluster-id" \
  --tags \
    Key=FinOps_LastIOAnalysis,Value="2024-11-01" \
    Key=FinOps_IORecommendation,Value="io_optimized" \
    Key=FinOps_IOConfidence,Value="high" \
    Key=FinOps_IOAnnualSavings,Value="4800" \
  --region us-east-1
```

This approach lets you query all Aurora clusters by their last analysis date using the AWS Resource Groups Tagging API, making it trivial to identify clusters that have not been reviewed recently:

```
aws resourcegroupstaggingapi get-resources \
  --resource-type-filters "rds:cluster" \
  --tag-filters Key=FinOps_LastIOAnalysis \
  --region us-east-1 \
  --query 'ResourceTagMappingList[?Tags[?Key==`FinOps_LastIOAnalysis` && \
    ↪ Value<=`2024-08-01`]].[ResourceARN]' \
  --output table
```

Key Takeaways

- The Aurora I/O-Optimized break-even point is **625,000 I/O requests per GB of storage per month**. Clusters above this ratio save money on I/O-Optimized; clusters below it save money on Standard. This ratio is derived from the 25% storage premium and the \$0.20-per-million I/O charge, and it holds regardless of absolute cluster size.
- **Measure I/O from CloudWatch, not from application metrics.** VolumeReadIOPs and VolumeWriteIOPs reflect actual billed storage-layer operations including write amplification, background I/O, and redo log activity. Application-level query counts systematically undercount billable I/O.
- **Cross-validate CloudWatch estimates against Cost Explorer's Aurora:StorageIOUsage line item.** CloudWatch metric math requires a period-multiplication correction when converting rate-based IOPS metrics to total operation counts. Cost Explorer shows the actual billed quantity and eliminates that conversion error.

- **Instance right-sizing and storage model changes interact.** Reducing instance memory shrinks the buffer pool, increasing read I/Os from the storage layer and potentially pushing a cluster across the break-even threshold. Always re-run the I/O analysis after a right-sizing event before committing to a storage model.
 - **Automate detection but require human approval for changes.** Monthly EventBridge-triggered analysis with SNS alerting surfaces misaligned clusters reliably without requiring manual spreadsheet reviews. The actual `modify-db-cluster` call should remain a human-approved step to prevent automated switching during migrations, right-sizing windows, or major application deployments that temporarily distort the I/O profile.
-

Chapter 15: Automated Storage Cleanup: Snapshots, Temp Tables, and Bloat

Storage is the silent budget drain that most engineering teams discover too late. A typical production RDS PostgreSQL instance that has been running for 18 months accumulates between 40% and 70% of its total storage bill in artifacts that provide zero ongoing business value: orphaned snapshots from deprecated environments, bloated tables where dead tuples outnumber live ones three to one, temporary tables that survived a connection restart, and transaction log backups that retention policies never cleaned up. Organizations running mid-scale database fleets on AWS, Azure, or GCP routinely find \$8,000 to \$25,000 per month in recoverable storage spend once they instrument and automate cleanup properly. This chapter gives you the diagnostic queries, the cleanup logic, and the scheduled automation to capture that money — and keep capturing it every month without manual intervention.

15.1 Diagnosing Where Storage Money Is Actually Going

Before automating anything, you need an accurate map of what you are paying for. Cloud providers bill storage across several independent dimensions: provisioned IOPS volume capacity, automated backup retention windows, manual snapshots, cross-region snapshot copies, and transaction log storage. Each line item has a different unit price, and the cost composition varies significantly between AWS, Azure, and GCP.

On AWS RDS, storage billing breaks down into four categories:

- **Provisioned volume:** gp2 at \$0.115/GB-month or gp3 at \$0.092/GB-month (us-east-1, 2024 pricing)
- **Automated backups:** free up to 100% of provisioned storage; each GB beyond that billed at \$0.095/GB-month
- **Manual snapshots:** \$0.095/GB-month, charged from creation date, never expire automatically
- **Cross-region snapshot copies:** source region egress (\$0.02–\$0.09/GB depending on region pair) plus destination snapshot storage

The manual snapshot line item is where teams consistently hemorrhage money. An automated snapshot taken at day 30 of a 30-day retention window gets deleted automatically. A manual

snapshot taken on the same day to “preserve a known-good state before a migration” — and then forgotten — costs \$0.095/GB-month indefinitely. A 2 TB database snapshot costs \$194/month. If your team has been running for three years with ad hoc snapshot practices, multiply that figure by however many forgotten snapshots exist across however many accounts.

The following query runs against AWS Cost Explorer via CLI and breaks down your RDS storage costs by usage type, giving you the exact composition of what you are paying:

```
#!/bin/bash
# Pull RDS storage cost breakdown for the last 3 months
# Requires aws cli v2 and Cost Explorer access (us-east-1)

AWS_ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)

echo "Fetching RDS storage cost breakdown for account: $AWS_ACCOUNT_ID"

aws ce get-cost-and-usage \
  --time-period Start=$(date -d '90 days ago' +%Y-%m-%d),End=$(date +%Y-%m-%d) \
  --granularity MONTHLY \
  --filter '{
    "And": [
      {
        "Dimensions": {
          "Key": "SERVICE",
          "Values": ["Amazon Relational Database Service"]
        }
      },
      {
        "Dimensions": {
          "Key": "USAGE_TYPE_GROUP",
          "Values": [
            "RDS: Storage",
            "RDS: Backup",
            "RDS: Snapshot Export"
          ]
        }
      }
    ]
  }' \
  --group-by Type=DIMENSION,Key=USAGE_TYPE \
  --metrics BlendedCost UnblendedCost UsageQuantity \
  --output json | jq -r '
    .ResultsByTime[] |
    .TimePeriod.Start as $period |
    .Groups[] |
    [$period, .Keys[0],
      (.Metrics.UnblendedCost.Amount | tonumber | . * 100 | round / 100),
      (.Metrics.UsageQuantity.Amount | tonumber | . * 100 | round / 100),
      .Metrics.UsageQuantity.Unit
    ] | @tsv
  ' | column -t -s $'\t' -N "Period,UsageType,Cost_USD,Quantity,Unit"
```

On Azure SQL, the equivalent breakdown lives in Cost Management. Filter by resource type `microsoft.sql/servers/databases` and group by meter subcategory. You will find separate line items for LRS backup storage, RA-GRS backup storage, and provisioned database storage. The RA-GRS backup storage meter frequently surprises teams — it bills at 1.5x the LRS rate and gets applied to databases configured with geo-redundant backup storage even when the secondary region is never actually used for recovery.

On GCP Cloud SQL, run this against the billing export table in BigQuery to isolate storage costs by type:

```
-- GCP: Storage cost breakdown per Cloud SQL instance, last 90 days
-- Run in BigQuery against your billing export dataset
-- Replace `your-project.billing_dataset.gcp_billing_export_v1_*` with your export table
```

```

SELECT
  resource.name                               AS instance_name,
  sku.description                             AS storage_type,
  SUM(cost)                                  AS total_cost_usd,
  SUM(usage.amount)                          AS total_usage,
  usage.unit                                 AS unit,
  DATE_TRUNC(usage_start_time, MONTH)        AS billing_month
FROM
  `your-project.billing-dataset.gcp-billing-export-v1_*`
WHERE
  service.description = 'Cloud SQL'
  AND sku.description LIKE '%Storage%'
  AND usage_start_time >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 90 DAY)
GROUP BY
  resource.name,
  sku.description,
  usage.unit,
  billing_month
ORDER BY
  billing_month DESC,
  total_cost_usd DESC;

```

Once you have the cost map, the typical distribution in a three-year-old environment looks like this:

Category	Typical Share of Storage Bill
Provisioned volume (active)	45–55%
Manual snapshots (forgotten)	20–35%
Automated backup overage	10–20%
Transaction log backup	5–15%
Cross-region copies	5–10%

The “forgotten manual snapshots” line is your first target. Everything in that category can potentially be eliminated with a retention policy and enforcement automation.

15.2 Snapshot Lifecycle Automation

A manual snapshot costs money from the moment it is created to the moment it is explicitly deleted. There is no built-in expiration mechanism on RDS manual snapshots, Azure SQL long-term retention backups, or Cloud SQL on-demand backups. Every team that does not have enforcement automation accumulates snapshots indefinitely.

The standard approach is to implement a tagging convention at snapshot creation time, then run a scheduled Lambda (or Azure Function or Cloud Run job) that evaluates tags and deletes snapshots past their intended retention date.

The tagging convention requires two tags on every manual snapshot:

- **RetentionDays:** integer, e.g., 30, 90, 365, or **permanent**
- **CreatedBy:** the team or automation that created it
- **Purpose:** **pre-migration**, **pre-upgrade**, **compliance**, **ad-hoc**

Snapshots tagged **permanent** are exempt from automated deletion. Everything else gets evaluated. The following Python Lambda function implements this logic for AWS RDS:

```
"""
rds_snapshot_cleanup.py
Lambda function for automated RDS snapshot lifecycle management.
Deploy with EventBridge rule on daily schedule (cron: 0 2 * * ? *)
Required IAM permissions:
- rds:DescribeDBSnapshots
- rds>DeleteDBSnapshot
- rds:ListTagsForResource
- sns:Publish (for alerting)
Environment variables:
SNS_TOPIC_ARN : ARN of SNS topic for deletion reports
DRY_RUN       : "true" to log without deleting (default: "false")
MAX_AGE_DAYS  : Hard cap - delete any untagged snapshot older than this (default: 90)
"""

import boto3
import json
import os
from datetime import datetime, timezone, timedelta

rds_client = boto3.client('rds')
sns_client = boto3.client('sns')

DRY_RUN = os.environ.get('DRY_RUN', 'false').lower() == 'true'
MAX_AGE_DAYS = int(os.environ.get('MAX_AGE_DAYS', '90'))
SNS_TOPIC = os.environ.get('SNS_TOPIC_ARN', '')

def get_tag_value(tags: list, key: str) -> str | None:
    for tag in tags:
        if tag['Key'] == key:
            return tag['Value']
    return None

def evaluate_snapshot(snapshot: dict, tags: list) -> dict:
    """
    Returns a dict describing what action to take and why.
    """
    snap_id = snapshot['DBSnapshotIdentifier']
    snap_type = snapshot['SnapshotType']
    created_time = snapshot['SnapshotCreateTime']
    now = datetime.now(timezone.utc)
    age_days = (now - created_time).days
    allocated_gb = snapshot.get('AllocatedStorage', 0)
    cost_per_month = round(allocated_gb * 0.095, 2) # us-east-1 snapshot pricing

    # Never touch automated snapshots - RDS manages their lifecycle
    if snap_type == 'automated':
        return {'action': 'skip', 'reason': 'automated snapshot'}

    # Check for permanent exemption
    retention_tag = get_tag_value(tags, 'RetentionDays')
    if retention_tag and retention_tag.lower() == 'permanent':
        return {
            'action': 'skip',
            'reason': 'tagged permanent',
            'age_days': age_days,
            'cost_per_month': cost_per_month
        }

    # If RetentionDays tag is present and not expired, skip
    if retention_tag and retention_tag.isdigit():
        retention_days = int(retention_tag)
        if age_days <= retention_days:
            return {
                'action': 'skip',
                'reason': f'within retention ({age_days}/{retention_days} days)',
            }

```

```

        'age_days'      : age_days,
        'cost_per_month': cost_per_month
    }
else:
    return {
        'action'      : 'delete',
        'reason'      : f'retention expired ({age_days} days > {retention_days} day policy)',
        'snap_id'     : snap_id,
        'age_days'    : age_days,
        'size_gb'     : allocated_gb,
        'cost_per_month': cost_per_month
    }

# No retention tag - apply hard cap
if age_days > MAX_AGE_DAYS:
    return {
        'action'      : 'delete',
        'reason'      : f'no retention tag, exceeded hard cap ({age_days} days > {MAX_AGE_DAYS})',
        'snap_id'     : snap_id,
        'age_days'    : age_days,
        'size_gb'     : allocated_gb,
        'cost_per_month': cost_per_month
    }

return {
    'action'      : 'skip',
    'reason'      : 'no tag but within hard cap',
    'age_days'    : age_days,
    'cost_per_month': cost_per_month
}

def lambda_handler(event, context):
    paginator = rds_client.get_paginator('describe_db_snapshots')
    deleted = []
    skipped = []
    total_savings_usd = 0.0

    for page in paginator.paginate(SnapshotType='manual'):
        for snapshot in page['DBSnapshots']:
            if snapshot['Status'] != 'available':
                continue

            snap_id = snapshot['DBSnapshotIdentifier']
            snap_arn = snapshot['DBSnapshotArn']

            tag_response = rds_client.list_tags_for_resource(ResourceName=snap_arn)
            tags = tag_response.get('TagList', [])
            decision = evaluate_snapshot(snapshot, tags)

            if decision['action'] == 'delete':
                total_savings_usd += decision.get('cost_per_month', 0)
                if not DRY_RUN:
                    try:
                        rds_client.delete_db_snapshot(DBSnapshotIdentifier=snap_id)
                        decision['status'] = 'deleted'
                    except Exception as e:
                        decision['status'] = f'error: {str(e)}'
                else:
                    decision['status'] = 'dry_run_would_delete'
                deleted.append(decision)
            else:
                skipped.append(decision)

    report = {
        'dry_run'      : DRY_RUN,
        'snapshots_deleted' : len(deleted),
        'snapshots_skipped' : len(skipped),
        'estimated_savings_usd': round(total_savings_usd, 2),
        'deleted_details' : deleted[:50], # cap report size
    }

```

```
print(json.dumps(report, indent=2, default=str))

if SNS_TOPIC and (deleted or not DRY_RUN):
    sns_client.publish(
        TopicArn = SNS_TOPIC,
        Subject = f"RDS Snapshot Cleanup: {len(deleted)} deleted, ${total_savings_usd:.2f}/mo
↪ recovered",
        Message = json.dumps(report, indent=2, default=str)
    )

return report
```

Deploy this Lambda with an EventBridge rule on a daily schedule at 02:00 UTC. First week, run with `DRY_RUN=true` and review the SNS report. After confirming the deletion list looks correct, flip to `DRY_RUN=false`. In most environments, the first live run recovers between 15% and 35% of the total RDS storage bill within 24 hours.

For Azure, the equivalent is a PowerShell Azure Function that evaluates long-term retention backup policies. The key lever there is adjusting the LTR (Long-Term Retention) weekly, monthly, and yearly backup retention values via the `Set-AzSqlDatabaseBackupLongTermRetentionPolicy` cmdlet. The default values on many older Azure SQL databases are set to maximums during initial setup and never revisited. Dropping weekly retention from 12 weeks to 4 weeks on a 500 GB database reduces RA-GRS backup storage from approximately 6 TB to 2 TB — a saving of roughly \$180/month at \$0.05/GB-month for RA-GRS.

15.3 Identifying and Reclaiming Table Bloat in PostgreSQL

Table bloat is one of the most persistent and expensive PostgreSQL storage problems. Every UPDATE and DELETE operation leaves dead tuples in the heap. PostgreSQL's MVCC architecture requires this for transaction isolation, but autovacuum's job is to reclaim that space over time. When autovacuum is misconfigured, under-resourced, or blocked, dead tuples accumulate faster than they are reclaimed, and table files grow without bound.

A heavily bloated table might have 3 GB of actual live data inside a 10 GB file. You are paying for 10 GB of provisioned storage — and in some cases 10 GB of backup storage — to hold 3 GB of useful data. On a gp3 volume at \$0.092/GB-month, that 7 GB of bloat costs \$0.64/month per table. Across a schema with 200 bloated tables, that becomes \$128/month in wasted storage, plus the performance cost of sequential scans reading 3x more data than necessary.

The following query identifies your worst bloat offenders using the `pgstattuple` extension, which provides accurate physical measurement of dead tuple space:

```
-- PostgreSQL: Identify tables with significant bloat
-- Requires: CREATE EXTENSION IF NOT EXISTS pgstattuple;
-- Run as superuser or with pg_stat_scan_tables privilege
-- This performs physical table scans - run during off-peak hours on large tables

CREATE EXTENSION IF NOT EXISTS pgstattuple;

WITH bloat_analysis AS (
    SELECT
        schemaname,
        tablename,
        pg_size_pretty(pg_total_relation_size(schemaname || '.' || tablename))
        AS total_size,
```

```

pg_total_relation_size(schemaname || '.' || tablename)
                        AS total_bytes,
(pgstattuple(schemaname || '.' || tablename)).dead_tuple_percent
                        AS dead_pct,
(pgstattuple(schemaname || '.' || tablename)).dead_tuple_len
                        AS dead_bytes,
(pgstattuple(schemaname || '.' || tablename)).free_space
                        AS free_bytes,
(pgstattuple(schemaname || '.' || tablename)).tuple_count
                        AS live_tuples
FROM
  pg_tables
WHERE
  schemaname NOT IN ('pg_catalog', 'information_schema', 'pg_toast')
  -- Filter to tables worth analyzing (> 100 MB)
  AND pg_total_relation_size(schemaname || '.' || tablename) > 104857600
),
cost_estimate AS (
  SELECT
    schemaname,
    tablename,
    total_size,
    total_bytes,
    dead_pct,
    live_tuples,
    dead_bytes
    AS reclaimable_bytes,
    free_bytes,
    -- Estimate reclaimable GB and monthly cost at gp3 pricing ($0.092/GB-month)
    ROUND((dead_bytes + free_bytes) / 1073741824.0, 3)
    AS reclaimable_gb,
    ROUND(
      ((dead_bytes + free_bytes) / 1073741824.0) * 0.092,
      4
    )
    AS monthly_waste_usd
  FROM
    bloat_analysis
  WHERE
    dead_pct > 10 -- Only surface tables with meaningful bloat
)
SELECT
  schemaname,
  tablename,
  total_size,
  ROUND(dead_pct, 1)
    AS dead_pct,
  live_tuples,
  pg_size_pretty(reclaimable_bytes)
    AS reclaimable_data,
  reclaimable_gb,
  monthly_waste_usd,
  CASE
    WHEN dead_pct > 50 THEN 'VACUUM FULL or pg_repack immediately'
    WHEN dead_pct > 25 THEN 'Schedule pg_repack within 7 days'
    WHEN dead_pct > 10 THEN 'Tune autovacuum thresholds'
    ELSE 'Monitor'
  END
    AS recommended_action
FROM
  cost_estimate
ORDER BY
  monthly_waste_usd DESC
LIMIT 30;

```

When this query surfaces tables with `dead_pct > 50`, your first instinct might be to run `VACUUM FULL`. On a busy production system, avoid it. `VACUUM FULL` acquires an `AccessExclusiveLock`, blocking all reads and writes for the duration of the operation. On a 50 GB table with heavy fragmentation, that can mean 45 minutes to 2 hours of complete table unavailability.

Use `pg_repack` instead. It rebuilds the table online using a trigger-based approach that tracks changes during the rebuild and applies them at the final cutover, which requires only a brief lock at the end. Install it as a PostgreSQL extension and run it from a scheduled cron job on the database

host or from a container:

```
#!/bin/bash
# pg_repack_bloated_tables.sh
# Run as a weekly cron job: 0 1 * * 0 /opt/dba/pg_repack_bloated_tables.sh
# Prerequisites: pg_repack installed on the PostgreSQL server
# Amazon Linux / RHEL: yum install pg_repack_16
# Debian/Ubuntu: apt install postgresql-16-repack

set -euo pipefail

PGHOST="${PGHOST:-localhost}"
PGPORT="${PGPORT:-5432}"
PGUSER="${PGUSER:-postgres}"
PGDATABASE="${PGDATABASE:-myapp}"
LOG_FILE="/var/log/pg_repack_$(date +%Y%m%d).log"
MIN_BLOAT_PCT=25          # Only repack tables with >= 25% dead tuples
MIN_TABLE_SIZE_MB=100     # Only repack tables >= 100 MB
MAX_REPACK_MINUTES=240    # Timeout per table

exec >>(tee -a "$LOG_FILE") 2>&1
echo "=== pg_repack run started: $(date) ==="

# Get list of bloated tables using pg_stat_user_tables (lightweight, no pgstattuple scan)
# n_dead_tup / nullif(n_live_tup + n_dead_tup, 0) gives approximate dead ratio
BLOATED_TABLES=$(psql \
  -h "$PGHOST" \
  -p "$PGPORT" \
  -U "$PGUSER" \
  -d "$PGDATABASE" \
  -t -A -F '|' \
  -c "
    SELECT
      schemaname || '.' || relname,
      ROUND(100.0 * n_dead_tup / NULLIF(n_live_tup + n_dead_tup, 0), 1) AS dead_pct,
      pg_size_pretty(pg_total_relation_size(schemaname || '.' || relname)) AS size
    FROM
      pg_stat_user_tables
    WHERE
      n_dead_tup > 0
      AND (n_live_tup + n_dead_tup) > 0
      AND ROUND(100.0 * n_dead_tup / NULLIF(n_live_tup + n_dead_tup, 0), 1) >= $MIN_BLOAT_PCT
      AND pg_total_relation_size(schemaname || '.' || relname) > ($MIN_TABLE_SIZE_MB * 1024 * 1024)
    ORDER BY
      n_dead_tup DESC
    LIMIT 20;
  ")

if [ -z "$BLOATED_TABLES" ]; then
  echo "No tables meet bloat threshold. Exiting."
  exit 0
fi

echo "Tables scheduled for repack:"
echo "$BLOATED_TABLES" | column -t -s '|' -N "Table,Dead%,Size"

TOTAL_REPACKED=0
TOTAL_FAILED=0

while IFS='|' read -r table_name dead_pct table_size; do
  echo ""
  echo "--- Repacking: $table_name (dead_pct: $dead_pct%, size: $table_size) ---"
  START_TIME=$(date +%s)

  if timeout "${MAX_REPACK_MINUTES}m" \
    pg_repack \
      --host="$PGHOST" \
      --port="$PGPORT" \
      --username="$PGUSER" \
      --dbname="$PGDATABASE" \
```



```

--table="$table_name" \
--no-superuser-check \
2>&1; then
END_TIME=$(date +%s)
DURATION=$(( END_TIME - START_TIME ))
echo "SUCCESS: $table_name repacked in ${DURATION}s"
TOTAL_REPACKED=$(( TOTAL_REPACKED + 1 ))
else
END_TIME=$(date +%s)
DURATION=$(( END_TIME - START_TIME ))
echo "FAILED: $table_name repack failed after ${DURATION}s"
TOTAL_FAILED=$(( TOTAL_FAILED + 1 ))
fi
done <<< "$BLOATED_TABLES"

echo ""
echo "=== pg_repack run completed: $(date) ==="
echo "   Tables repacked successfully : $TOTAL_REPACKED"
echo "   Tables failed                  : $TOTAL_FAILED"
echo "   Log: $LOG_FILE"

```

After a `pg_repack` run, the table file physically shrinks. On a gp3 volume, RDS does not automatically reduce provisioned storage when files shrink — you still pay for the provisioned capacity. However, if you are using RDS storage autoscaling, reclaiming physical space prevents the autoscaler from triggering an upward expansion, which is permanent. Preventing a 100 GB autoscale event avoids \$9.20/month in perpetual cost.

If you want to actually reduce provisioned storage on RDS after reclaiming significant bloat, the path is a snapshot restore to a new instance with a lower storage allocation, followed by a DNS cutover. Chapter 16 covers that procedure in full. For now, the bloat cleanup is still worth doing for the performance benefit, the backup size reduction, and the autoscale prevention.

Autovacuum Tuning to Prevent Bloat Recurrence

Reclaiming bloat through `pg_repack` is treating the symptom. The cause is usually one of three autovacuum configuration failures:

1. **Threshold mismatch:** The default `autovacuum_vacuum_scale_factor` = 0.2 means autovacuum does not trigger until 20% of the table's rows are dead. On a 100-million-row table, that is 20 million dead tuples before any cleanup begins. By then, bloat is severe.
2. **Worker starvation:** The default `autovacuum_max_workers` = 3 is insufficient for schemas with hundreds of tables. Workers queue up, high-churn tables fall behind.
3. **Cost throttling:** `autovacuum_vacuum_cost_delay` = 2ms with `autovacuum_vacuum_cost_limit` = 200 throttles vacuum I/O heavily to avoid impacting production queries. On high-churn tables, this means vacuuming cannot keep pace with the write rate.

Apply per-table storage parameters to your highest-churn tables rather than modifying the global defaults, which affect all tables:

```

-- PostgreSQL: Per-table autovacuum tuning for high-churn tables
-- Identify high-churn tables: n_tup_upd + n_tup_del > 100000 per day

-- Aggressive autovacuum for a high-update OLTP table
ALTER TABLE orders SET (
autovacuum_vacuum_scale_factor    = 0.01,    -- vacuum when 1% rows are dead (vs 20%)
autovacuum_vacuum_threshold       = 1000,    -- vacuum when >= 1000 dead tuples
autovacuum_analyze_scale_factor   = 0.005,   -- analyze when 0.5% rows change
autovacuum_vacuum_cost_delay      = 0,        -- no throttling for this table
autovacuum_vacuum_cost_limit      = 800      -- allow higher I/O rate

```

```
);

-- Slightly more conservative for a large append-mostly table
ALTER TABLE events SET (
    autovacuum_vacuum_scale_factor = 0.05,
    autovacuum_vacuum_threshold   = 5000,
    autovacuum_analyze_scale_factor = 0.02,
    autovacuum_vacuum_cost_delay  = 2,
    autovacuum_vacuum_cost_limit  = 400
);

-- Verify the per-table settings are applied
SELECT
    relname,
    reloptions
FROM
    pg_class
WHERE
    reloptions IS NOT NULL
    AND relkind = 'r'
ORDER BY
    relname;
```

Monitor the effect of these changes using `pg_stat_user_tables`. After applying per-table tuning, `n_dead_tup` for the affected tables should plateau well below the threshold that triggers bloat:

```
-- PostgreSQL: Monitor autovacuum effectiveness after tuning
-- Run daily and track n_dead_tup trend over time

SELECT
    schemaname,
    relname AS table_name,
    n_live_tup,
    n_dead_tup,
    ROUND(
        100.0 * n_dead_tup / NULLIF(n_live_tup + n_dead_tup, 0),
        1
    ) AS dead_pct,
    last_vacuum,
    last_autovacuum,
    last_analyze,
    last_autoanalyze,
    vacuum_count,
    autovacuum_count,
    -- How long since last vacuum? Alert if > 24h on high-churn tables
    EXTRACT(EPOCH FROM (now() - GREATEST(last_vacuum, last_autovacuum))) / 3600
    AS hours_since_vacuum
FROM
    pg_stat_user_tables
WHERE
    n_dead_tup > 10000
ORDER BY
    n_dead_tup DESC
LIMIT 25;
```

15.4 Identifying and Reclaiming Table Bloat in SQL Server

SQL Server's storage fragmentation problem manifests differently than PostgreSQL bloat. SQL Server does not use MVCC in the same way; instead, fragmentation occurs at the page and extent level inside B-tree index structures. A heavily modified table with a clustered index will develop internal fragmentation (pages that are partially filled) and external fragmentation (pages out of logical order). Both consume storage and degrade query performance.

The key measurement is `avg_fragmentation_in_percent` from `sys.dm_db_index_physical_stats`. An index with 80% fragmentation means that on average, each allocated page is only 20% full — you are paying for 5x more storage than the data actually requires.

```
-- SQL Server: Identify index fragmentation and estimate storage waste
-- Use SAMPLED mode for large databases - LIMITED mode is faster but less accurate
-- DETAILED mode is most accurate but scans every page; use only in maintenance windows

SELECT
    OBJECT_SCHEMA_NAME(ips.object_id)                AS schema_name,
    OBJECT_NAME(ips.object_id)                       AS table_name,
    i.name                                             AS index_name,
    ips.index_type_desc,
    ips.index_level,
    ROUND(ips.avg_fragmentation_in_percent, 1)        AS fragmentation_pct,
    ips.page_count,
    ips.record_count,
    ROUND(ips.avg_page_space_used_in_percent, 1)      AS avg_page_fullness_pct,
    -- Estimate wasted pages: pages that should be freed if fully packed
    CAST(
        ips.page_count * (1.0 - ips.avg_page_space_used_in_percent / 100.0)
        AS INT
    )                                                  AS wasted_pages,
    -- Estimate wasted MB (8 KB per page)
    ROUND(
        ips.page_count * (1.0 - ips.avg_page_space_used_in_percent / 100.0) * 8.0 / 1024.0,
        2
    )                                                  AS wasted_mb,
    -- Monthly cost estimate at Azure SQL GP storage pricing ($0.115/GB-month)
    ROUND(
        ips.page_count * (1.0 - ips.avg_page_space_used_in_percent / 100.0)
        * 8.0 / 1024.0 / 1024.0 * 0.115,
        4
    )                                                  AS monthly_waste_usd,
    CASE
        WHEN ips.avg_fragmentation_in_percent >= 30 THEN 'REBUILD INDEX'
        WHEN ips.avg_fragmentation_in_percent >= 10 THEN 'REORGANIZE INDEX'
        ELSE 'MONITOR'
    END                                               AS recommended_action
FROM
    sys.dm_db_index_physical_stats(
        DB_ID(), -- current database
        NULL,    -- all tables
        NULL,    -- all indexes
        NULL,    -- all partitions
        'SAMPLED' -- sampling mode
    ) AS ips
INNER JOIN sys.indexes AS i
    ON i.object_id = ips.object_id
    AND i.index_id = ips.index_id
WHERE
    ips.index_level = 0 -- leaf level only
    AND ips.page_count > 1000 -- skip tiny indexes (< ~8 MB)
    AND ips.avg_fragmentation_in_percent > 10
ORDER BY
    wasted_mb DESC;
```

For indexes with fragmentation above 30%, REBUILD is the appropriate action. REBUILD reallocates the entire index, producing a completely clean B-tree with pages filled to the configured FILLFACTOR. For fragmentation between 10% and 30%, REORGANIZE performs an online defragmentation of the leaf level without rebuilding the entire structure. The key operational difference: REBUILD can be done WITH (ONLINE = ON) on SQL Server Enterprise Edition (and on Azure SQL), which avoids blocking DML during the operation.

The following stored procedure automates adaptive index maintenance based on fragmentation thresholds:

```

-- SQL Server / Azure SQL: Adaptive index maintenance procedure
-- Rebuilds high-fragmentation indexes, reorganizes medium-fragmentation indexes
-- Skips indexes on tables with active long-running transactions

CREATE OR ALTER PROCEDURE dbo.usp_AdaptiveIndexMaintenance
    @MinPageCount      INT          = 1000,      -- Skip indexes smaller than this
    @RebuildThreshold   FLOAT        = 30.0,      -- Fragmentation % threshold for REBUILD
    @ReorganizeThreshold FLOAT        = 10.0,      -- Fragmentation % threshold for REORGANIZE
    @MaxDurationMinutes INT          = 120,        -- Stop after this many minutes
    @FillFactor         TINYINT       = 85,        -- Fill factor for rebuilds
    @PrintOnly          BIT           = 0,         -- 1 = print commands, don't execute
AS
BEGIN
    SET NOCOUNT ON;

    DECLARE
        @StartTime      DATETIME2      = SYSUTCDATETIME(),
        @SchemaName     NVARCHAR(128),
        @TableName      NVARCHAR(128),
        @IndexName      NVARCHAR(128),
        @FragPct         FLOAT,
        @PageCount       BIGINT,
        @SQL             NVARCHAR(MAX),
        @Action          NVARCHAR(20),
        @RowsProcessed   INT           = 0,
        @ErrorMessage    NVARCHAR(MAX);

    -- Staging table for fragmentation data
    CREATE TABLE #IndexWork (
        RowID          INT IDENTITY PRIMARY KEY,
        SchemaName     NVARCHAR(128),
        TableName      NVARCHAR(128),
        IndexName      NVARCHAR(128),
        FragPct        FLOAT,
        PageCount       BIGINT,
        Action         NVARCHAR(20),
        Processed      BIT DEFAULT 0,
        ErrorMsg       NVARCHAR(MAX)
    );

    INSERT INTO #IndexWork (SchemaName, TableName, IndexName, FragPct, PageCount, Action)
    SELECT
        OBJECT_SCHEMA_NAME(ips.object_id),
        OBJECT_NAME(ips.object_id),
        i.name,
        ips.avg_fragmentation_in_percent,
        ips.page_count,
        CASE
            WHEN ips.avg_fragmentation_in_percent >= @RebuildThreshold THEN 'REBUILD'
            WHEN ips.avg_fragmentation_in_percent >= @ReorganizeThreshold THEN 'REORGANIZE'
        END
    FROM
        sys.dm_db_index_physical_stats(DB_ID(), NULL, NULL, NULL, 'SAMPLED') AS ips
    INNER JOIN sys.indexes AS i
        ON i.object_id = ips.object_id
        AND i.index_id = ips.index_id
    WHERE
        ips.index_level = 0
        AND ips.page_count >= @MinPageCount
        AND ips.avg_fragmentation_in_percent >= @ReorganizeThreshold
        AND i.name IS NOT NULL -- exclude heap
    ORDER BY
        ips.avg_fragmentation_in_percent DESC;

    DECLARE index_cursor CURSOR LOCAL FAST_FORWARD FOR
        SELECT RowID, SchemaName, TableName, IndexName, FragPct, PageCount, Action
        FROM #IndexWork
        ORDER BY FragPct DESC;

    OPEN index_cursor;

```

```

FETCH NEXT FROM index_cursor
    INTO @RowsProcessed, @SchemaName, @TableName, @IndexName,
        @FragPct, @PageCount, @Action;

WHILE @@FETCH_STATUS = 0
BEGIN
    -- Enforce time budget
    IF DATEDIFF(MINUTE, @StartTime, SYSUTCDATETIME()) >= @MaxDurationMinutes
    BEGIN
        PRINT 'Time budget exceeded. Stopping maintenance.';
        BREAK;
    END;

    SET @SQL = CASE @Action
        WHEN 'REBUILD'
        THEN N'ALTER INDEX ' + QUOTENAME(@IndexName)
            + N' ON ' + QUOTENAME(@SchemaName) + N'.' + QUOTENAME(@TableName)
            + N' REBUILD WITH (ONLINE = ON, FILLFACTOR = '
            + CAST(@FillFactor AS NVARCHAR(3)) + N', SORT_IN_TEMPDB = ON);'
        WHEN 'REORGANIZE'
        THEN N'ALTER INDEX ' + QUOTENAME(@IndexName)
            + N' ON ' + QUOTENAME(@SchemaName) + N'.' + QUOTENAME(@TableName)
            + N' REORGANIZE;'

    END;

    IF @PrintOnly = 1
    BEGIN
        PRINT @SQL;
    END
    ELSE
    BEGIN
        BEGIN TRY
            EXEC sp_executesql @SQL;
            UPDATE #IndexWork SET Processed = 1 WHERE RowID = @RowsProcessed;
        END TRY
        BEGIN CATCH
            SET @ErrorMessage = ERROR_MESSAGE();
            UPDATE #IndexWork
                SET ErrorMsg = @ErrorMessage
            WHERE RowID = @RowsProcessed;
            PRINT 'Error on ' + @IndexName + ': ' + @ErrorMessage;
        END CATCH;
    END;

    FETCH NEXT FROM index_cursor
        INTO @RowsProcessed, @SchemaName, @TableName, @IndexName,
            @FragPct, @PageCount, @Action;
END;

CLOSE index_cursor;
DEALLOCATE index_cursor;

-- Summary report
SELECT
    Action,
    COUNT(*) AS IndexCount,
    SUM(PageCount * 8 / 1024) AS TotalMB,
    SUM(CASE WHEN Processed = 1 THEN PageCount * 8 / 1024 ELSE 0 END)
        AS ProcessedMB,
    COUNT(CASE WHEN ErrorMsg IS NOT NULL THEN 1 END)
        AS Errors,
    DATEDIFF(MINUTE, @StartTime, SYSUTCDATETIME())
        AS ElapsedMinutes
FROM
    #IndexWork
GROUP BY
    Action;

DROP TABLE #IndexWork;
END;

```

```
GO

-- Execute in report-only mode first
EXEC dbo.usp_AdaptiveIndexMaintenance @PrintOnly = 1;

-- Schedule in a SQL Agent job during maintenance window:
-- EXEC dbo.usp_AdaptiveIndexMaintenance @MaxDurationMinutes = 180;
```

One important nuance for Azure SQL specifically: the storage reduction from index rebuilds becomes meaningful only if you are on a DTU-based pricing model where total database size determines the pricing tier, or if you are tracking total database size against the maximum size allocation for your service tier. On vCore pricing, the provisioned storage allocation is independent of actual usage, so the FinOps benefit of defragmentation on Azure SQL is primarily performance-driven rather than directly storage-cost-driven — unless you then reduce the provisioned max storage size via an `ALTER DATABASE` operation after reclaiming space.

15.5 Temporary Table and Temp Space Management

Temporary storage is a category that engineering teams almost universally underestimate until it causes either a production incident (temp space exhaustion) or a cost anomaly (unexpected storage tier upgrade). Understanding where temp data lives in each cloud database platform is the prerequisite for managing its cost.

PostgreSQL on RDS: Temporary tables and sort operations that cannot fit in `work_mem` spill to temporary files on the data volume. On RDS PostgreSQL, these temporary files live on the same EBS volume as your permanent data. A runaway query that generates 50 GB of temp data consumes 50 GB of provisioned storage while it runs. On a 100 GB provisioned volume, that single query can trigger an autoscale event. Worse, if the query fails midway through, PostgreSQL cleans up the temp files — but the autoscaled volume size does not decrease.

Query this view to find sessions currently generating temporary files:

```
-- PostgreSQL: Sessions currently using significant temp file space
SELECT
    pid,
    username,
    application_name,
    state,
    query_start,
    EXTRACT(EPOCH FROM (now() - query_start)) / 60 AS running_minutes,
    temp_files,
    pg_size_pretty(temp_bytes) AS temp_size,
    LEFT(query, 120) AS query_snippet
FROM
    pg_stat_activity
JOIN pg_stat_statements USING (queryid) -- if pg_stat_statements available
WHERE
    temp_bytes > 104857600 -- sessions using > 100 MB temp space
ORDER BY
    temp_bytes DESC;
```

Since `pg_stat_activity` does not expose `temp_bytes` directly in all PostgreSQL versions, the more reliable approach is to monitor `pg_stat_database`:

```
-- PostgreSQL: Track temp file usage per database over time
-- Schedule this to run every 5 minutes and write to a monitoring table

SELECT
```

```

    datname,
    temp_files,
    pg_size_pretty(temp_bytes)      AS temp_size,
    blks_hit,
    blks_read,
    tup_fetched,
    stats_reset
FROM
    pg_stat_database
WHERE
    datname NOT IN ('postgres', 'template0', 'template1')
ORDER BY
    temp_bytes DESC;

```

When `temp_bytes` is consistently high, the remediation options are:

1. **Increase `work_mem`** for the specific queries generating spill: `SET work_mem = '256MB';` at session level before known heavy queries.
2. **Identify and optimize the spilling queries:** Use `EXPLAIN (ANALYZE, BUFFERS)` to find sort and hash operations that exceed `work_mem`.
3. **Add a dedicated tablespace on a separate EBS volume** for temporary files via `temp_tablespaces`, keeping temp growth off the main data volume.

SQL Server on Azure SQL / RDS SQL Server: SQL Server uses `tempdb` for temporary tables, table variables, sort spills, row versioning, and online index operations. On RDS SQL Server, `tempdb` lives on a separate NVMe SSD that is not billed as part of your storage allocation — it is sized based on instance class and reset on every restart. On Azure SQL, `tempdb` is managed by Microsoft and does not count toward your database size allocation. This means temporary table cost in SQL Server is primarily an instance-size concern (whether you need a larger instance due to `tempdb` pressure) rather than a direct storage billing concern.

However, excessive `tempdb` usage still creates FinOps problems through a different mechanism: it causes `PAGELATCH` contention on `tempdb` allocation pages, which degrades performance and can force an instance size upgrade. Monitor `tempdb` usage with this query:

```

-- SQL Server: tempdb space usage by session
SELECT
    r.session_id,
    r.status,
    r.command,
    r.total_elapsed_time / 1000      AS elapsed_sec,
    tsu.user_objects_alloc_page_count * 8 / 1024      AS user_obj_mb,
    tsu.internal_objects_alloc_page_count * 8 / 1024  AS internal_obj_mb,
    (tsu.user_objects_alloc_page_count
     + tsu.internal_objects_alloc_page_count) * 8 / 1024
    AS total_tempdb_mb,
    LEFT(t.text, 120)               AS query_snippet
FROM
    sys.dm_exec_requests            AS r
JOIN sys.dm_db_task_space_usage    AS tsu
    ON r.session_id = tsu.session_id
   AND r.request_id = tsu.request_id
CROSS APPLY
    sys.dm_exec_sql_text(r.sql_handle) AS t
WHERE
    (tsu.user_objects_alloc_page_count
     + tsu.internal_objects_alloc_page_count) > 12800 -- > 100 MB
ORDER BY
    total_tempdb_mb DESC;

```

A commonly overlooked `tempdb` cost driver in SQL Server is excessive use of `#temp` tables instead of table variables or CTEs for small result sets. Each `#temp` table creation

generates metadata writes to `tempdb` system tables. At scale, a stored procedure that creates and drops a `#temp` table inside a high-frequency loop (tens of thousands of executions per minute) creates measurable allocation contention. The fix is to replace `#temp` tables with `@table` variables for small sets, or to use `TRUNCATE TABLE #temp` inside loops rather than `DROP` and `re-CREATE`.

15.6 Automated Log File and WAL Management

Transaction logs are another category where retention policies frequently drift from intent. The right approach differs significantly across the three platforms.

PostgreSQL WAL on RDS: On RDS PostgreSQL, WAL files are managed by AWS. You do not directly control the WAL directory. However, two parameters affect how much WAL accumulates and therefore how much storage it consumes:

- **wal_keep_size** (PostgreSQL 13+) or **wal_keep_segments** (earlier versions): determines how many WAL files are retained for streaming replica lag recovery. Setting this too high on a busy system means gigabytes of WAL accumulate on the data volume. Set it to a value that covers your maximum acceptable replica lag, not to an arbitrarily large number.
- **RDS automated backup retention:** The transaction logs that RDS retains for point-in-time recovery are separate from WAL files in the PostgreSQL sense. These are managed by RDS and stored in S3. They contribute to your backup storage billing. A 7-day retention window retains fewer transaction logs than a 35-day window. Evaluate whether you actually use point-in-time recovery past a certain day count. Many teams keep 35-day retention because “that’s what it was set to” when the instance was created, not because they have ever needed to recover to a specific second from 34 days ago.

```
-- PostgreSQL: Check WAL generation rate to right-size wal_keep_size
-- Run every hour and track the trend
SELECT
    pg_current_wal_lsn()                                AS current_lsn,
    pg_wal_lsn_diff(
        pg_current_wal_lsn(),
        pg_current_wal_flush_lsn()
    )                                                    AS unflushed_bytes,
    -- Rate: compare to a snapshot taken 1 hour ago in your monitoring system
    pg_size_pretty(
        pg_wal_lsn_diff(
            pg_current_wal_lsn(),
            '0/0'::pg_lsn -- replace with snapshotted LSN for rate calculation
        )
    )                                                    AS total_wal_generated,
    current_setting('wal_keep_size')                    AS wal_keep_size_setting,
    (SELECT COUNT(*) FROM pg_ls_waldir())                AS wal_files_on_disk;
```

SQL Server Transaction Logs on RDS / Azure SQL: SQL Server transaction log bloat is a classic operational problem. The log file grows during a long-running transaction and then cannot be reused (even after a log backup) until the transaction commits or rolls back. On RDS SQL Server, log files live on the EBS data volume and count toward your storage billing. On Azure SQL, log files are included within the max database size allocation.

Monitor log space usage and identify what is blocking log truncation:

```
-- SQL Server: Transaction log space usage and truncation blockers
-- Current log space usage
```



```
DBCC SQLPERF(LOGSPACE);

-- More detailed log reuse wait reason per database
SELECT
    name AS database_name,
    log_reuse_wait_desc AS truncation_blocked_by,
    -- log_reuse_wait_desc values:
    -- NOTHING : log can be reused (healthy)
    -- LOG_BACKUP : waiting for log backup (most common in FULL recovery)
    -- ACTIVE_TRANSACTION : long-running transaction blocking truncation
    -- DATABASE_MIRRORING / AVAILABILITY_GROUP : replica lag blocking truncation
    -- CHECKPOINT : waiting for checkpoint (SIMPLE recovery model)
    recovery_model_desc,
    log_size_mb = (
        SELECT SUM(size) * 8.0 / 1024
        FROM sys.master_files
        WHERE database_id = databases.database_id
        AND type_desc = 'LOG'
    )
FROM
    sys.databases
WHERE
    state_desc = 'ONLINE'
ORDER BY
    log_size_mb DESC;

-- Find the active transaction blocking log truncation (if log_reuse_wait_desc = ACTIVE_TRANSACTION)
SELECT TOP 10
    tran_begin_time,
    DATEDIFF(MINUTE, tran_begin_time, GETDATE()) AS age_minutes,
    s.session_id,
    s.login_name,
    s.program_name,
    t.log_bytes_used / 1024 / 1024 AS log_mb_used,
    LEFT(st.text, 200) AS current_sql
FROM
    sys.dm_tran_active_transactions AS t
JOIN sys.dm_exec_sessions AS s ON s.session_id = t.session_id
CROSS APPLY
    sys.dm_exec_sql_text(s.sql_handle) AS st
WHERE
    t.transaction_type IN (1, 2) -- read-write and read-only transactions
ORDER BY
    t.log_bytes_used DESC;
```

When `log_reuse_wait_desc` shows `LOG_BACKUP`, the fix is to ensure your SQL Agent (or Azure SQL elastic job) is running log backups at a frequency consistent with your RPO. For a database generating significant log volume, running log backups every 15 minutes instead of every 60 minutes keeps the log file from growing to accommodate an hour's worth of unflushed log records.

15.7 Building the Storage Cleanup Automation Pipeline

The previous sections delivered individual diagnostic queries and cleanup scripts. This section assembles them into a coordinated pipeline that runs on schedule, reports results, and integrates with your FinOps cost tracking.

The architecture has three components:

1. **Daily snapshot cleanup** (Lambda / Azure Function / Cloud Run) — runs at 02:00 UTC
2. **Weekly bloat analysis and repack** (cron job on bastion host or ECS Fargate task) — runs Sunday at 01:00 UTC

3. Monthly storage cost report (scheduled query against billing API) — runs first day of month

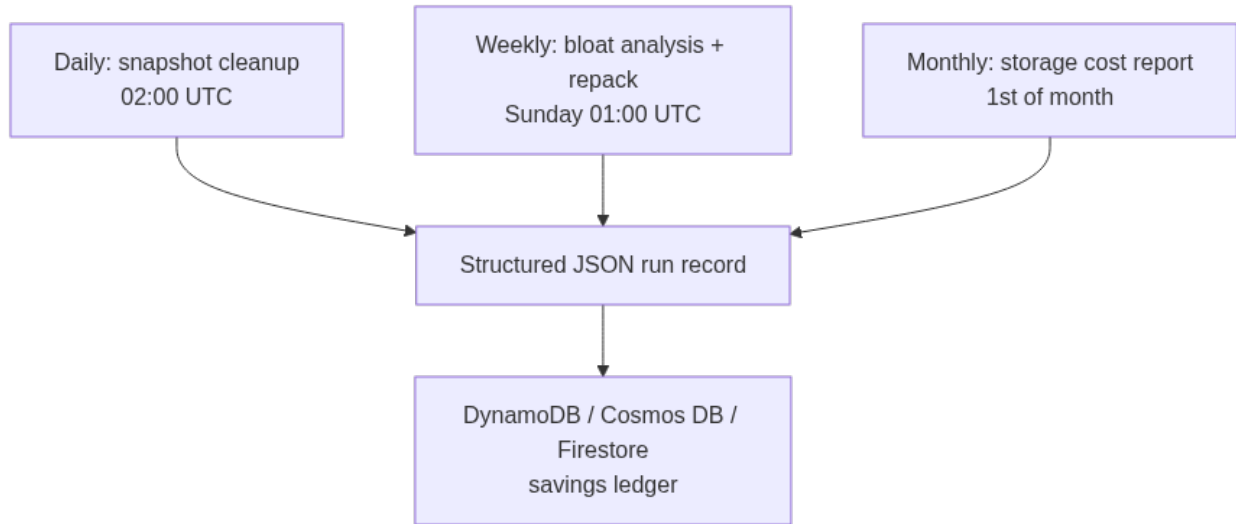


Figure 15: The three-component cleanup pipeline — each runs independently but feeds the same savings ledger for FinOps reporting.

For teams on AWS, here is a CloudFormation template that deploys the snapshot cleanup Lambda with its EventBridge rule, IAM role, and SNS notification:

```

# cloudformation/rds-storage-cleanup.yaml
# Deploy: aws cloudformation deploy --template-file rds-storage-cleanup.yaml
#         --stack-name rds-storage-cleanup --capabilities CAPABILITY_NAMED_IAM

AWSTemplateFormatVersion: '2010-09-09'
Description: 'Automated RDS snapshot lifecycle management'

Parameters:
  NotificationEmail:
    Type: String
    Description: Email address for cleanup reports
  MaxSnapshotAgeDays:
    Type: Number
    Default: 90
    Description: Hard-cap age for untagged snapshots
  DryRunMode:
    Type: String
    Default: 'true'
    AllowedValues: ['true', 'false']
    Description: Set to false to enable actual deletions

Resources:
  CleanupTopic:
    Type: AWS::SNS::Topic
    Properties:
      TopicName: rds-snapshot-cleanup-reports
      Subscription:
        - Protocol: email
          Endpoint: !Ref NotificationEmail

  LambdaRole:
    Type: AWS::IAM::Role
    Properties:
      RoleName: rds-snapshot-cleanup-role
      AssumeRolePolicyDocument:

```

```

Version: '2012-10-17'
Statement:
  - Effect: Allow
    Principal:
      Service: lambda.amazonaws.com
    Action: sts:AssumeRole
ManagedPolicyArns:
  - arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
Policies:
  - PolicyName: rds-snapshot-cleanup-policy
    PolicyDocument:
      Version: '2012-10-17'
      Statement:
        - Effect: Allow
          Action:
            - rds:DescribeDBSnapshots
            - rds>DeleteDBSnapshot
            - rds:ListTagsForResource
          Resource: '*'
        - Effect: Allow
          Action:
            - sns:Publish
          Resource: !Ref CleanupTopic

CleanupFunction:
  Type: AWS::Lambda::Function
  Properties:
    FunctionName: rds-snapshot-cleanup
    Runtime: python3.12
    Handler: index.lambda_handler
    Role: !GetAtt LambdaRole.Arn
    Timeout: 900 # 15 minutes - enough for large snapshot inventories
    MemorySize: 256
    Environment:
      Variables:
        SNS_TOPIC_ARN: !Ref CleanupTopic
        DRY_RUN: !Ref DryRunMode
        MAX_AGE_DAYS: !Ref MaxSnapshotAgeDays
    Code:
      ZipFile: |
        # Placeholder - deploy actual code via S3 or Lambda layers
        # See section 15.2 for the full lambda_handler implementation
        def lambda_handler(event, context):
            return {'status': 'placeholder - replace with production code'}

ScheduleRule:
  Type: AWS::Events::Rule
  Properties:
    Name: rds-snapshot-cleanup-daily
    Description: Trigger RDS snapshot cleanup at 02:00 UTC daily
    ScheduleExpression: 'cron(0 2 * * ? *)'
    State: ENABLED
    Targets:
      - Id: CleanupLambda
        Arn: !GetAtt CleanupFunction.Arn

LambdaPermission:
  Type: AWS::Lambda::Permission
  Properties:
    FunctionName: !Ref CleanupFunction
    Action: lambda:InvokeFunction
    Principal: events.amazonaws.com
    SourceArn: !GetAtt ScheduleRule.Arn

Outputs:
  LambdaFunctionArn:
    Value: !GetAtt CleanupFunction.Arn
  SNSTopicArn:
    Value: !Ref CleanupTopic

```

For the weekly bloat analysis job, the recommended deployment pattern on AWS is an ECS Fargate task triggered by an EventBridge Scheduled Rule. The task runs the `pg_repack_bloated_tables.sh` script from Section 15.3 inside a container with PostgreSQL client tools installed. This avoids the need for a persistent bastion host and runs in the VPC with direct database access.

For Azure environments, the equivalent is an Azure Container Instance (ACI) triggered by Azure Automation or a Logic App on a weekly schedule. The container runs the same bloat analysis queries against Azure Database for PostgreSQL Flexible Server, then calls the `pg_repack` utility or — where `pg_repack` is unavailable — issues `VACUUM (VERBOSE, ANALYZE)` against the identified tables and records the results to a Log Analytics workspace.

A practical integration point for all three platforms is to emit the results of each cleanup run as structured JSON to your observability stack. The following is a standardized schema that works across Lambda, ACI, and Cloud Run output formats:

```
{
  "run_id": "2024-03-01T02:00:00Z-snapshot-cleanup",
  "platform": "aws-rds",
  "run_type": "snapshot_cleanup",
  "dry_run": false,
  "started_at": "2024-03-01T02:00:12Z",
  "completed_at": "2024-03-01T02:03:47Z",
  "results": {
    "items_evaluated": 142,
    "items_deleted": 23,
    "items_skipped": 119,
    "storage_reclaimed_gb": 847.3,
    "estimated_monthly_savings_usd": 80.49
  },
  "errors": [],
  "environment": {
    "account_id": "123456789012",
    "region": "us-east-1"
  }
}
```

Write these records to a DynamoDB table (AWS), Cosmos DB (Azure), or Firestore (GCP), keyed by `run_id`. Then build a simple monthly rollup query to track how much the automation is recovering over time. This gives your FinOps team a concrete savings ledger that can be reported alongside the cloud bill.

Here is the DynamoDB schema and a boto3 writer function you can drop into the Lambda from Section 15.2:

```
# Add to rds_snapshot_cleanup.py after the cleanup report is generated

import hashlib

def write_run_record_to_dynamodb(report: dict, table_name: str = 'storage-cleanup-runs'):
    """
    Persist cleanup run results to DynamoDB for historical tracking.
    Table schema:
        PK: run_date (YYYY-MM-DD)
        SK: run_id (timestamp + type)
    """
    dynamodb = boto3.resource('dynamodb')
    table = dynamodb.Table(table_name)

    run_id = f"{datetime.now(timezone.utc).strftime('%Y%m%dT%H%M%S')}-snapshot"

    item = {
        'run_date': datetime.now(timezone.utc).strftime('%Y-%m-%d'),
```

```

'run_id'           : run_id,
'platform'        : 'aws-rds',
'run_type'        : 'snapshot_cleanup',
'dry_run'         : report.get('dry_run', True),
'snapshots_deleted' : report.get('snapshots_deleted', 0),
'snapshots_skipped' : report.get('snapshots_skipped', 0),
'estimated_savings_usd' : str(report.get('estimated_savings_usd', 0)),
'ttl'            : int(
    (datetime.now(timezone.utc) + timedelta(days=365)).timestamp()
), # Auto-expire records after 1 year
}

try:
    table.put_item(Item=item)
    print(f"Run record written to DynamoDB: {run_id}")
except Exception as e:
    print(f"Failed to write DynamoDB record: {e}")
    # Non-fatal - cleanup still succeeded

```

To query historical savings from DynamoDB, use a GSI on `run_date` or export the table to S3 and query it with Athena:

```

-- Athena: Monthly savings summary from storage cleanup runs
-- Table created from DynamoDB export to S3

SELECT
    DATE_FORMAT(DATE_PARSE(run_date, '%Y-%m-%d'), '%Y-%m') AS billing_month,
    run_type,
    SUM(snapshots_deleted) AS total_deleted,
    SUM(CAST(estimated_savings_usd AS DOUBLE)) AS total_savings_usd,
    COUNT(*) AS run_count,
    MAX(CAST(estimated_savings_usd AS DOUBLE)) AS largest_single_run_usd
FROM
    storage_cleanup_runs
WHERE
    dry_run = false
GROUP BY
    DATE_FORMAT(DATE_PARSE(run_date, '%Y-%m-%d'), '%Y-%m'),
    run_type
ORDER BY
    billing_month DESC,
    total_savings_usd DESC;

```

15.8 Setting Storage Budgets and Anomaly Alerts

Automation that cleans up storage is only half the solution. You also need alerting that tells you when storage spend is growing faster than your application's legitimate data growth — which usually signals a new source of waste you have not yet automated away.

On AWS, CloudWatch has a built-in metric `FreeStorageSpace` per RDS instance. Combine this with a Billing Alarm on the Cost Explorer anomaly detection to get two independent signals: one that fires when an individual instance is filling up (operational signal), and one that fires when the aggregate RDS storage spend grows unexpectedly across the account (FinOps signal).

The following CloudWatch alarm fires when free storage drops below 20% of provisioned capacity on any monitored instance:

```

#!/bin/bash
# create_storage_alarms.sh
# Creates CloudWatch alarms for RDS storage utilization across all instances
# Requires: aws cli with CloudWatch and RDS permissions

```

```

SNS_ARN="${1:?Usage: $0 <sns-topic-arn>}"

# Get all RDS instance identifiers and their allocated storage
aws rds describe-db-instances \
  --query 'DBInstances[*].[DBInstanceIdentifier,AllocatedStorage]' \
  --output text | \
while IFS=$'\t' read -r instance_id allocated_gb; do

  # Threshold: alert when free space < 20% of provisioned
  threshold_bytes=$((echo "$allocated_gb * 1073741824 * 0.20" | bc | cut -d'.' -f1))

  echo "Creating alarm for $instance_id (${allocated_gb} GB provisioned, threshold: ${threshold_bytes}
↳ bytes)"

  aws cloudwatch put-metric-alarm \
    --alarm-name "RDS-LowStorage-${instance_id}" \
    --alarm-description "RDS instance ${instance_id} free storage below 20%" \
    --metric-name FreeStorageSpace \
    --namespace AWS/RDS \
    --dimensions Name=DBInstanceIdentifier,Value="$instance_id" \
    --statistic Minimum \
    --period 300 \
    --evaluation-periods 3 \
    --threshold "$threshold_bytes" \
    --comparison-operator LessThanThreshold \
    --alarm-actions "$SNS_ARN" \
    --ok-actions "$SNS_ARN" \
    --treat-missing-data notBreaching

done

echo "Done. Alarms created for all RDS instances."

```

For the FinOps-layer alert, use Cost Explorer Anomaly Detection. This is more effective than a fixed budget alert because it learns your seasonal baseline and fires only when the growth pattern deviates from what it has learned to expect:

```

# Create a Cost Explorer anomaly monitor for RDS storage
aws ce create-anomaly-monitor \
  --anomaly-monitor '{
    "MonitorName": "RDS-Storage-Anomaly-Monitor",
    "MonitorType": "DIMENSIONAL",
    "MonitorDimension": "SERVICE"
  }'

# After creating the monitor, capture its MonitorArn from the response
# Then create a subscription for the alert

aws ce create-anomaly-subscription \
  --anomaly-subscription '{
    "SubscriptionName": "RDS-Storage-Anomaly-Alert",
    "MonitorArnList": ["<MonitorArn-from-above>"],
    "Subscribers": [
      {
        "Address": "<your-sns-topic-arn>",
        "Type": "SNS"
      }
    ],
    "Threshold": 50,
    "Frequency": "DAILY"
  }'

```

The **Threshold: 50** value means the alert fires when Cost Explorer detects an anomaly with an estimated impact above \$50. Adjust this to a value that is meaningful for your fleet size — lower for smaller deployments, higher for large multi-account organizations where \$50 might be normal variance.

On Azure, the equivalent is an Azure Cost Management Budget with an alert action group. Set the budget scope to your resource group or subscription, filter to the SQL meter category, and configure an alert at 80% and 100% of the monthly budget. Wire the alert to an Action Group that sends email and triggers an Azure Function that runs the diagnostic queries from Section 15.1 automatically, so the person receiving the alert immediately gets a breakdown of what changed.

On GCP, use Budget Alerts in the Billing console filtered to the Cloud SQL service, and configure a Pub/Sub topic as the notification channel. A Cloud Function subscribed to that topic can then run the BigQuery cost breakdown query from Section 15.1 and publish the results to a Slack channel or incident management system.

15.9 Right-Sizing Provisioned Storage After Cleanup

After running the cleanup automation for several weeks, you will have recovered storage at the physical layer — table files are smaller, snapshots are gone, log files are trimmed. However, on RDS, the provisioned storage allocation does not decrease automatically. You are still paying for the full provisioned volume size regardless of how much space is now free.

The path to reducing provisioned storage on RDS requires creating a snapshot of the current instance, restoring it to a new instance with a lower storage allocation, and then cutting over application connections. This is not a live resize — it is a migration. The steps are:

1. Confirm that actual used storage is well below your target new allocation. Leave at minimum 30% headroom.
2. Take a final snapshot of the current instance.
3. Restore the snapshot to a new instance, specifying the lower `--allocated-storage` value.
4. Validate the restored instance thoroughly (run your integration test suite against it).
5. Update the DNS CNAME or Route 53 record to point to the new instance endpoint.
6. Monitor for 24 hours, then delete the old instance.

The restore command with reduced storage:

```
# Restore RDS snapshot to a new instance with reduced provisioned storage
# Assumes the snapshot has been taken and physical used space supports the reduction

SOURCE_SNAPSHOT="rds:prod-postgres-2024-03-01-02-00"
NEW_INSTANCE_ID="prod-postgres-resized"
NEW_STORAGE_GB=300      # Must be >= actual used space + 30% headroom
INSTANCE_CLASS="db.r6g.2xlarge"
SUBNET_GROUP="prod-db-subnet-group"
SECURITY_GROUPS="sg-0abc123def456789"

aws rds restore-db-instance-from-db-snapshot \
  --db-instance-identifier "$NEW_INSTANCE_ID" \
  --db-snapshot-identifier "$SOURCE_SNAPSHOT" \
  --db-instance-class      "$INSTANCE_CLASS" \
  --allocated-storage      "$NEW_STORAGE_GB" \
  --storage-type           gp3 \
  --db-subnet-group-name   "$SUBNET_GROUP" \
  --vpc-security-group-ids "$SECURITY_GROUPS" \
  --no-publicly-accessible \
  --enable-performance-insights \
  --performance-insights-retention-period 7 \
  --tags Key=Environment,Value=production \
      Key=ManagedBy,Value=finops-automation
```

```
echo "Restore initiated. Monitor progress:"
echo "aws rds describe-db-instances --db-instance-identifier $NEW_INSTANCE_ID \\"
echo "  --query 'DBInstances[0].DBInstanceStatus'"
```

For Azure SQL Database, reducing the max size allocation is simpler — it does not require a restore. You can issue an `ALTER DATABASE` statement directly, as long as the actual data size is below the new target:

```
-- Azure SQL: Reduce max size allocation after reclaiming space
-- Check current usage first
SELECT
  DB_NAME()                                AS database_name,
  SUM(size) * 8 / 1024                     AS allocated_mb,
  SUM(FILEPROPERTY(name, 'SpaceUsed')) * 8 / 1024 AS used_mb,
  (SUM(size) - SUM(FILEPROPERTY(name, 'SpaceUsed')) * 8 / 1024) AS free_mb
FROM
  sys.database_files
WHERE
  type_desc = 'ROWS';

-- If used_mb is sufficiently below your target, reduce the max size
-- Example: reduce from 1 TB to 500 GB after bloat reclamation
ALTER DATABASE CURRENT MODIFY (MAXSIZE = 500 GB);
```

On GCP Cloud SQL, storage capacity is managed differently: it auto-increases but does not auto-decrease, and you cannot reduce the provisioned disk size on a running instance. The path to reduction on Cloud SQL is similar to RDS — export the data, delete the instance, and create a new one with a smaller disk. For most GCP deployments, the practical approach is to focus cleanup energy on preventing autoscale events rather than on reducing already-provisioned capacity.

Key Takeaways

- **Manual snapshots are the highest-leverage cleanup target:** They accumulate indefinitely, carry per-GB monthly costs, and have no built-in expiration. A tagging convention (`RetentionDays`, `Purpose`) combined with a daily enforcement Lambda recovers storage budget within days of deployment, with no application impact.
- **PostgreSQL table bloat requires both immediate remediation and root-cause treatment:** Use `pg_repack` for online, lock-minimal table rebuilding on severely bloated tables, and simultaneously tune per-table `autovacuum_vacuum_scale_factor` and `autovacuum_vacuum_cost_limit` to prevent recurrence. Treating only one side of this problem means running the cleanup operation repeatedly on the same tables.
- **SQL Server fragmentation savings on cloud platforms are context-dependent:** On RDS SQL Server, index rebuilds directly reduce storage billing because log and data files share the provisioned EBS volume. On Azure SQL vCore, the benefit is primarily performance-driven unless you follow the rebuild with an `ALTER DATABASE MODIFY (MAXSIZE = ...)` to reduce the provisioned maximum.
- **Temporary storage mismanagement triggers permanent cost increases:** On RDS PostgreSQL, temp file spill onto the data EBS volume can trigger autoscale events that are irreversible without a snapshot restore. Monitor `pg_stat_database.temp_bytes` continuously, and increase `work_mem` at the session level for known heavy queries rather than globally, to keep temp growth bounded.

- **Automation without measurement is incomplete:** Each cleanup run should emit a structured result record — items deleted, storage reclaimed, estimated monthly savings — to a queryable store such as DynamoDB or BigQuery. This creates a savings ledger that makes the FinOps ROI of cleanup automation visible to finance and engineering leadership, and makes it straightforward to detect when a new source of waste has appeared that the existing automation does not yet address.
-

Chapter 16: Compressing Data at Rest: Savings and Tradeoffs

Storage compression is one of the most underutilized levers in cloud database cost management. Organizations running 2–5 TB of uncompressed transactional data on AWS RDS, Azure SQL, or Google Cloud SQL regularly see 40–70% storage reduction after enabling the right compression strategy — translating to \$800–\$3,500 per month in direct storage savings on a single database instance before accounting for reduced backup costs, lower data transfer fees, and faster I/O on compressed pages. This chapter walks through how to identify what is compressible, which compression algorithms and engine features to use per cloud platform, how to measure the performance tradeoff honestly, and how to automate compression enforcement across a fleet of databases so savings compound over time rather than remaining a one-off project.

16.1 Understanding What You Are Actually Paying For Before Compressing Anything

Compression that targets the wrong data delivers almost nothing. A 500 GB database where 400 GB is already binary BLOBs, encrypted payloads, or JPEG thumbnails stored as bytes will see near-zero compression gain. Before touching a single compression setting, spend thirty minutes identifying which data is actually compressible.

The Storage Cost Breakdown

On AWS RDS with gp3 storage, you pay \$0.115 per GB-month for the primary volume. Multi-AZ doubles that to \$0.230 per GB-month because RDS replicates the entire EBS volume to the standby. Automated backups stored in S3 add another \$0.095 per GB-month for retention beyond the free tier. On a 3 TB Multi-AZ PostgreSQL instance with 14-day backup retention, the all-in storage cost looks like this:

Component	Size	Rate	Monthly Cost
Primary EBS (gp3)	3,000 GB	\$0.115	\$345.00
Standby EBS (Multi-AZ)	3,000 GB	\$0.115	\$345.00
Automated backups	~2,400 GB	\$0.095	\$228.00
Total			\$918.00/month

If compression reduces the primary volume to 1,200 GB — a 60% reduction common with mixed

OLTP data — the same table drops to \$367.20/month. That is \$550.80 in monthly savings on storage alone, not counting backup savings, before you touch instance sizing.

Azure SQL Managed Instance on Business Critical tier charges \$0.115 per GB-month (General Purpose is \$0.115 as well, but Business Critical adds locally redundant replicas automatically). Google Cloud SQL for PostgreSQL charges \$0.170 per GB-month on SSD storage with HA, making storage compression proportionally more valuable on GCP than on AWS.

Finding Compressible Data in PostgreSQL

Start by identifying table-level bloat, data type distribution, and actual compression readiness:

```
-- Identify the largest tables and their estimated compression potential
-- Run this on AWS RDS PostgreSQL, Aurora PostgreSQL, or Cloud SQL for PostgreSQL

WITH table_sizes AS (
    SELECT
        schemaname,
        tablename,
        pg_total_relation_size(schemaname || '.' || tablename) AS total_bytes,
        pg_relation_size(schemaname || '.' || tablename) AS table_bytes,
        pg_total_relation_size(schemaname || '.' || tablename)
            - pg_relation_size(schemaname || '.' || tablename) AS index_bytes
    FROM pg_tables
    WHERE schemaname NOT IN ('pg_catalog', 'information_schema')
),
column_type_analysis AS (
    SELECT
        c.table_schema,
        c.table_name,
        COUNT(*) FILTER (WHERE data_type IN ('text', 'character varying', 'json', 'jsonb', 'xml')) AS
        ⇨ text_columns,
        COUNT(*) FILTER (WHERE data_type IN ('bytea', 'bit varying')) AS binary_columns,
        COUNT(*) AS total_columns
    FROM information_schema.columns c
    WHERE c.table_schema NOT IN ('pg_catalog', 'information_schema')
    GROUP BY c.table_schema, c.table_name
)
SELECT
    ts.schemaname,
    ts.tablename,
    pg_size_pretty(ts.total_bytes) AS total_size,
    pg_size_pretty(ts.table_bytes) AS table_data_size,
    pg_size_pretty(ts.index_bytes) AS index_size,
    ct.text_columns,
    ct.binary_columns,
    ct.total_columns,
    ROUND(ct.text_columns::numeric / NULLIF(ct.total_columns, 0) * 100, 1) AS pct_text_columns,
    -- Tables with >40% text columns and >100MB are compression candidates
    CASE
        WHEN ct.text_columns::numeric / NULLIF(ct.total_columns, 0) > 0.4
            AND ts.table_bytes > 104857600
        THEN 'HIGH PRIORITY'
        WHEN ct.text_columns::numeric / NULLIF(ct.total_columns, 0) > 0.2
            AND ts.table_bytes > 52428800
        THEN 'MEDIUM PRIORITY'
        ELSE 'LOW PRIORITY'
    END AS compression_priority
FROM table_sizes ts
JOIN column_type_analysis ct
    ON ts.schemaname = ct.table_schema
    AND ts.tablename = ct.table_name
WHERE ts.total_bytes > 10485760 -- Only tables > 10 MB
ORDER BY ts.total_bytes DESC
LIMIT 50;
```

Finding Compressible Data in SQL Server (Azure SQL / RDS SQL Server)

```

-- Estimate compression savings before applying compression on SQL Server
-- Works on Azure SQL Database, Azure SQL Managed Instance, and RDS for SQL Server

SELECT
    s.name AS schema_name,
    t.name AS table_name,
    p.partition_number,
    p.rows,
    a.total_pages * 8 / 1024 AS total_size_mb,
    a.used_pages * 8 / 1024 AS used_size_mb,
    -- sp_estimate_data_compression_savings equivalent inline view
    -- Run the stored procedure separately per table for exact estimates
    CASE
        WHEN a.total_pages * 8 / 1024 > 1000 THEN 'Run sp_estimate_data_compression_savings'
        ELSE 'Small table - compress in bulk'
    END AS recommendation
FROM sys.tables t
INNER JOIN sys.schemas s ON t.schema_id = s.schema_id
INNER JOIN sys.partitions p ON t.object_id = p.object_id
INNER JOIN sys.allocation_units a ON p.partition_id = a.container_id
WHERE p.index_id IN (0, 1) -- Heap or clustered index only
    AND a.total_pages * 8 / 1024 > 50 -- Tables > 50 MB
ORDER BY a.total_pages DESC;

-- For each candidate table, run this to get actual estimated savings:
-- EXEC sp_estimate_data_compression_savings
--     @schema_name = 'dbo',
--     @object_name = 'YourTableName',
--     @index_id = NULL,
--     @partition_number = NULL,
--     @data_compression = 'PAGE';

```

The output from `sp_estimate_data_compression_savings` gives you `size_with_current_compression_setting`, `size_with_requested_compression_setting`, and the delta as a percentage. Run this across your top twenty tables, sum the delta column, and you have a credible number to present to a budget holder.

16.2 Compression Options by Database Engine and Cloud Platform

Each database engine offers different compression mechanisms, and each cloud platform has its own constraints on which ones are available. Using the right compression at the right layer is the difference between a 60% reduction and a 5% reduction.

PostgreSQL: TOAST, `pg_compress`, and tablespace-level approaches

PostgreSQL uses TOAST (The Oversized-Attribute Storage Technique) automatically for column values exceeding approximately 2 KB. TOAST compresses the value using LZ4 (PostgreSQL 14+) or PGLZ (earlier versions) before writing it to the TOAST table. The key insight is that this is transparent — no schema changes required — but you control which algorithm it uses.

PostgreSQL 14 and later (available on RDS PostgreSQL 14+, Aurora PostgreSQL 14+, and Cloud SQL for PostgreSQL 14+) allows setting the compression method per column:

```

-- Check current TOAST compression settings for a table
SELECT
    attname AS column_name,
    attstorage AS storage_mode,
    attcompression AS compression_method,
    -- attstorage values: 'p'=plain, 'e'=external, 'x'=extended (TOAST+compress), 'm'=main
    CASE attstorage

```

```

        WHEN 'p' THEN 'No TOAST - always inline'
        WHEN 'e' THEN 'External TOAST - no compression'
        WHEN 'x' THEN 'Extended - TOAST with compression (best)'
        WHEN 'm' THEN 'Main - compress first, TOAST if needed'
    END AS storage_description,
    CASE attcompression
        WHEN '' THEN 'Default (pglz on PG13-, lz4 on PG14+)'
        WHEN 'p' THEN 'pglz'
        WHEN 'l' THEN 'lz4'
    END AS compression_algorithm
FROM pg_attribute
WHERE attrelid = 'your_schema.your_table'::regclass
    AND attnum > 0
    AND NOT attisdropped
ORDER BY attnum;

-- Switch a text column from pglz to lz4 (PostgreSQL 14+, RDS PG14+, Aurora PG14+)
-- lz4 is 3-5x faster to compress/decompress than pglz with similar compression ratios
ALTER TABLE your_schema.your_table
    ALTER COLUMN description SET COMPRESSION lz4;

ALTER TABLE your_schema.your_table
    ALTER COLUMN payload SET COMPRESSION lz4;

-- Force recompression of existing rows by rewriting the table
-- Schedule during a maintenance window - this takes an ACCESS EXCLUSIVE lock
-- For large tables, use pg_repack instead (no lock):
-- pg_repack --table your_schema.your_table -d your_database
VACUUM FULL your_schema.your_table;

-- Verify compression took effect on existing TOAST data
SELECT
    pg_size_pretty(pg_total_relation_size('your_schema.your_table')) AS total_with_toast,
    pg_size_pretty(pg_relation_size('your_schema.your_table')) AS main_table,
    pg_size_pretty(
        pg_total_relation_size('your_schema.your_table')
        - pg_relation_size('your_schema.your_table')
    ) AS toast_and_indexes;

```

SQL Server on Azure SQL and RDS: Row and Page Compression

SQL Server offers two compression levels that matter operationally:

Row compression stores fixed-length data types (INT, CHAR, DATETIME) in variable-length format, eliminating trailing zeros and padding. Typical savings: 10–25% for OLTP tables.

Page compression applies row compression first, then adds prefix compression (stores repeated column prefixes once per page) and dictionary compression (deduplicates repeated values within a page). Typical savings: 30–60% for tables with repeating values in indexed columns.

Page compression costs more CPU — typically 5–15% additional CPU utilization on read-heavy workloads and up to 25% on write-heavy workloads. On Azure SQL, where you pay for DTUs or vCores, that CPU tradeoff needs to be weighed against storage savings. On a Business Critical instance with 8 vCores at \$1.008/hour, a 15% CPU increase costs roughly \$108/month. If page compression saves 800 GB of storage at \$0.115/GB, that is \$92/month in pure storage — a net loss. Row compression at 5% CPU overhead saving the same 300 GB is easily worth it.

```

-- Apply page compression to a table on Azure SQL or RDS SQL Server
-- First, estimate savings (separate statement per table)
EXEC sp_estimate_data_compression_savings
    @schema_name = 'Sales',
    @object_name = 'OrderLineItems',
    @index_id = NULL,
    @partition_number = NULL,
    @data_compression = 'PAGE';

```

```
-- Apply row compression (lower CPU cost, moderate savings)
ALTER TABLE Sales.OrderLineItems
REBUILD WITH (DATA_COMPRESSION = ROW);

-- Apply page compression to the clustered index
ALTER TABLE Sales.OrderLineItems
REBUILD PARTITION = ALL WITH (DATA_COMPRESSION = PAGE);

-- Apply page compression to a specific nonclustered index
ALTER INDEX IX_OrderLineItems_OrderDate ON Sales.OrderLineItems
REBUILD WITH (DATA_COMPRESSION = PAGE);

-- Check current compression status across all tables
SELECT
    s.name AS schema_name,
    t.name AS table_name,
    i.name AS index_name,
    i.type_desc AS index_type,
    p.data_compression_desc AS current_compression,
    SUM(a.total_pages) * 8 / 1024 AS size_mb
FROM sys.tables t
INNER JOIN sys.schemas s ON t.schema_id = s.schema_id
INNER JOIN sys.indexes i ON t.object_id = i.object_id
INNER JOIN sys.partitions p ON i.object_id = p.object_id AND i.index_id = p.index_id
INNER JOIN sys.allocation_units a ON p.partition_id = a.container_id
GROUP BY s.name, t.name, i.name, i.type_desc, p.data_compression_desc
ORDER BY SUM(a.total_pages) DESC;
```

MySQL / Aurora MySQL: InnoDB Compression and Page Size

InnoDB row format compression (`ROW_FORMAT=COMPRESSED`) has been available since MySQL 5.5 but carries a significant caveat: it halves the effective buffer pool utilization because InnoDB keeps both compressed and uncompressed versions of each page in the buffer pool. This makes InnoDB row compression a poor choice for frequently accessed hot tables.

The better option for Aurora MySQL and RDS MySQL 8.0+ is Barracuda file format with `KEY_BLOCK_SIZE` tuning. For cold or archive tables, `ROW_FORMAT=COMPRESSED KEY_BLOCK_SIZE=8` achieves 40–60% compression with acceptable performance since those pages rarely appear in the buffer pool.

The right approach for Aurora MySQL specifically is to use the Aurora storage engine’s built-in compression, which operates at the storage layer and does not penalize the buffer pool — it is always on and transparent. Measuring Aurora storage savings therefore requires comparing Aurora’s reported volume size against what the same data would occupy on a standard MySQL installation.

16.3 Measuring the Real Performance Tradeoff

Enabling compression without measuring its CPU and latency impact is how organizations discover problems in production at 2 AM. The measurement methodology below runs before and after compression to capture an honest before/after comparison.

PostgreSQL: Measuring Query Performance Against Compressed TOAST Data

```
-- Capture baseline query performance before compression changes
-- Run this before altering any compression settings

-- Enable pg_stat_statements if not already enabled
```

```

-- (On RDS, add pg_stat_statements to shared_preload_libraries via parameter group)
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;

-- Capture baseline for queries touching the tables you plan to compress
SELECT
  left(query, 100) AS query_snippet,
  calls,
  ROUND(total_exec_time::numeric, 2) AS total_ms,
  ROUND(mean_exec_time::numeric, 4) AS mean_ms,
  ROUND(stddev_exec_time::numeric, 4) AS stddev_ms,
  rows,
  shared_blks_read,
  shared_blks_hit,
  ROUND(
    shared_blks_hit::numeric /
    NULLIF(shared_blks_hit + shared_blks_read, 0) * 100, 2
  ) AS buffer_hit_pct
FROM pg_stat_statements
WHERE query ILIKE '%your_table_name%'
  AND calls > 100
ORDER BY mean_exec_time DESC;

-- Reset stats, apply compression, run workload for 1 hour, then compare
SELECT pg_stat_statements_reset();

-- After compression and workload replay, run again and compare mean_ms values
-- Also check shared_blks_read: if TOAST decompression causes more block reads,
-- you will see this number increase even if mean query time stays flat

-- Check CloudWatch metrics alongside this for CPU correlation
-- Use RDS Performance Insights to correlate CPU spikes with compression events

```

Monitoring Compression CPU Impact with CloudWatch and Python

```

#!/usr/bin/env python3
"""
compression_impact_monitor.py

Monitors CPU and I/O metrics before and after enabling database compression
using CloudWatch for RDS instances. Runs on a schedule via AWS Lambda or cron.

Requirements: boto3, pandas
Usage: python compression_impact_monitor.py --instance-id your-rds-instance \
      --period-hours 24 --output compression_report.json
"""

import boto3
import json
import argparse
from datetime import datetime, timedelta, timezone
from statistics import mean, stdev

def get_rds_metrics(instance_id: str, metric_name: str,
                    start_time: datetime, end_time: datetime,
                    period_seconds: int = 300) -> list[dict]:
    """Retrieve CloudWatch RDS metrics for a given time window."""
    cw = boto3.client('cloudwatch')

    response = cw.get_metric_statistics(
        Namespace='AWS/RDS',
        MetricName=metric_name,
        Dimensions=[
            {'Name': 'DBInstanceIdentifier', 'Value': instance_id}
        ],
        StartTime=start_time,
        EndTime=end_time,
        Period=period_seconds,
        Statistics=['Average', 'Maximum', 'Minimum']
    )

```



```

return sorted(response['Datapoints'], key=lambda x: x['Timestamp'])

def analyze_compression_impact(instance_id: str, baseline_hours: int,
                              post_compression_hours: int) -> dict:
    """
    Compare key metrics between a baseline period and post-compression period.
    Assumes compression was applied between the two periods.
    """
    now = datetime.now(timezone.utc)

    # Define time windows
    # Baseline: older period before compression
    baseline_end = now - timedelta(hours=post_compression_hours)
    baseline_start = baseline_end - timedelta(hours=baseline_hours)

    # Post-compression: most recent period
    post_start = now - timedelta(hours=post_compression_hours)
    post_end = now

    metrics_to_compare = [
        'CPUUtilization',
        'ReadIOPS',
        'WriteIOPS',
        'ReadLatency',
        'WriteLatency',
        'FreeStorageSpace',
        'DatabaseConnections'
    ]

    results = {
        'instance_id': instance_id,
        'analysis_timestamp': now.isoformat(),
        'baseline_window': {
            'start': baseline_start.isoformat(),
            'end': baseline_end.isoformat(),
            'hours': baseline_hours
        },
        'post_compression_window': {
            'start': post_start.isoformat(),
            'end': post_end.isoformat(),
            'hours': post_compression_hours
        },
        'metrics': {}
    }

    for metric in metrics_to_compare:
        baseline_data = get_rds_metrics(instance_id, metric, baseline_start, baseline_end)
        post_data = get_rds_metrics(instance_id, metric, post_start, post_end)

        if not baseline_data or not post_data:
            results['metrics'][metric] = {'error': 'Insufficient data'}
            continue

        baseline_avg_values = [p['Average'] for p in baseline_data]
        post_avg_values = [p['Average'] for p in post_data]

        baseline_mean = mean(baseline_avg_values)
        post_mean = mean(post_avg_values)
        delta_pct = ((post_mean - baseline_mean) / baseline_mean * 100
                     if baseline_mean != 0 else 0)

        results['metrics'][metric] = {
            'baseline_mean': round(baseline_mean, 4),
            'post_compression_mean': round(post_mean, 4),
            'delta_percent': round(delta_pct, 2),
            'direction': 'INCREASED' if delta_pct > 0 else 'DECREASED',
            'concern': (
                'HIGH' if metric == 'CPUUtilization' and delta_pct > 20 else

```

```

        'MEDIUM' if metric == 'CPUUtilization' and delta_pct > 10 else
        'POSITIVE' if metric == 'FreeStorageSpace' and delta_pct > 0 else
        'NORMAL'
    )
}

# Calculate estimated monthly storage savings
free_storage_metric = results['metrics'].get('FreeStorageSpace', {})
if 'baseline_mean' in free_storage_metric:
    baseline_free_gb = free_storage_metric['baseline_mean'] / (1024 ** 3)
    post_free_gb = free_storage_metric['post_compression_mean'] / (1024 ** 3)
    storage_freed_gb = post_free_gb - baseline_free_gb

# AWS gp3 pricing: $0.115/GB-month, doubled for Multi-AZ
savings_single_az = storage_freed_gb * 0.115
savings_multi_az = storage_freed_gb * 0.115 * 2

results['estimated_savings'] = {
    'storage_freed_gb': round(storage_freed_gb, 2),
    'monthly_savings_single_az_usd': round(savings_single_az, 2),
    'monthly_savings_multi_az_usd': round(savings_multi_az, 2),
    'annual_savings_multi_az_usd': round(savings_multi_az * 12, 2)
}

return results

def main():
    parser = argparse.ArgumentParser(description='Monitor compression impact on RDS')
    parser.add_argument('--instance-id', required=True, help='RDS instance identifier')
    parser.add_argument('--baseline-hours', type=int, default=168,
                        help='Hours of baseline data to use (default: 168 = 1 week)')
    parser.add_argument('--post-hours', type=int, default=24,
                        help='Hours of post-compression data to compare')
    parser.add_argument('--output', default='compression_impact.json',
                        help='Output file for results')
    args = parser.parse_args()

    print(f"Analyzing compression impact for {args.instance_id}...")
    results = analyze_compression_impact(
        args.instance_id,
        args.baseline_hours,
        args.post_hours
    )

    with open(args.output, 'w') as f:
        json.dump(results, f, indent=2)

    print(f"\nResults written to {args.output}")
    print("\n=== SUMMARY ===")

    cpu_metric = results['metrics'].get('CPUUtilization', {})
    if 'delta_percent' in cpu_metric:
        print(f"CPU Change: {cpu_metric['delta_percent']:+.1f}% "
              f"{cpu_metric['concern']}")

    savings = results.get('estimated_savings', {})
    if savings:
        print(f"Storage Freed: {savings.get('storage_freed_gb', 0):.1f} GB")
        print(f"Monthly Savings (Multi-AZ): ${savings.get('monthly_savings_multi_az_usd', 0):.2f}")
        print(f"Annual Savings (Multi-AZ): ${savings.get('annual_savings_multi_az_usd', 0):.2f}")

if __name__ == '__main__':
    main()

```

SQL Server: Measuring Compression Impact via DMVs

SQL Server's Dynamic Management Views expose buffer pool pressure and CPU cost of compression

operations. Capture these before and after applying compression to any large table:

```
-- Baseline CPU and I/O snapshot before compression
-- Save output to a table or capture externally

-- 1. Index usage stats for tables you plan to compress
SELECT
  OBJECT_SCHEMA_NAME(i.object_id) AS schema_name,
  OBJECT_NAME(i.object_id) AS table_name,
  i.name AS index_name,
  s.user_seeks,
  s.user_scans,
  s.user_lookups,
  s.user_updates,
  s.last_user_seek,
  s.last_user_scan,
  -- Heavy update workloads feel page compression more than read workloads
  CASE
    WHEN s.user_updates > s.user_seeks + s.user_scans
    THEN 'UPDATE-HEAVY: prefer ROW compression'
    ELSE 'READ-HEAVY: PAGE compression viable'
  END AS compression_recommendation
FROM sys.indexes i
LEFT JOIN sys.dm_db_index_usage_stats s
  ON i.object_id = s.object_id
  AND i.index_id = s.index_id
  AND s.database_id = DB_ID()
WHERE OBJECT_NAME(i.object_id) IN ('OrderLineItems', 'AuditLog', 'EventStream')
ORDER BY s.user_updates DESC;

-- 2. Buffer pool pressure: pages in memory vs total pages
SELECT
  OBJECT_SCHEMA_NAME(p.object_id) AS schema_name,
  OBJECT_NAME(p.object_id) AS table_name,
  i.name AS index_name,
  p.data_compression_desc AS compression,
  COUNT(b.page_id) AS pages_in_buffer,
  SUM(a.total_pages) AS total_pages,
  ROUND(COUNT(b.page_id) * 100.0 / NULLIF(SUM(a.total_pages), 0), 1)
  AS pct_in_buffer_pool
FROM sys.partitions p
INNER JOIN sys.indexes i
  ON p.object_id = i.object_id AND p.index_id = i.index_id
INNER JOIN sys.allocation_units a
  ON p.partition_id = a.container_id
LEFT JOIN sys.dm_os_buffer_descriptors b
  ON a.allocation_unit_id = b.allocation_unit_id
  AND b.database_id = DB_ID()
WHERE OBJECT_NAME(p.object_id) IN ('OrderLineItems', 'AuditLog', 'EventStream')
GROUP BY p.object_id, i.name, p.data_compression_desc
ORDER BY total_pages DESC;

-- 3. After applying compression, check if query plans changed
-- Look for new sort or hash spill warnings in execution plans,
-- which indicate that compressed pages are using more CPU than expected
SELECT TOP 20
  qs.total_worker_time / qs.execution_count AS avg_cpu_microseconds,
  qs.total_elapsed_time / qs.execution_count AS avg_elapsed_microseconds,
  qs.execution_count,
  qs.total_logical_reads / qs.execution_count AS avg_logical_reads,
  LEFT(st.text, 200) AS query_text
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
WHERE st.text LIKE '%OrderLineItems%'
  OR st.text LIKE '%AuditLog%'
ORDER BY avg_cpu_microseconds DESC;
```

Setting the Accept/Reject Threshold

Before any compression project begins, define the ceiling on acceptable CPU increase. A workable rule of thumb: if average CPU is currently below 40%, accept compression if the post-compression CPU remains below 55%. If current CPU is 40–60%, row compression only and reject page compression if the delta exceeds 8 percentage points. If current CPU is above 60%, do not apply page compression at all — the instance is already under pressure and compression will push it into throttling territory on Azure SQL DTU-based tiers.

Encode this rule into the monitoring script by adding a `gate_compression` function that queries the baseline CPU from the analysis results and outputs a go/no-go recommendation before any production `ALTER TABLE` runs.

16.4 Compressing Backups and Snapshots

Backup storage is frequently the hidden multiplier in storage cost analysis. An RDS instance with 14-day automated backup retention stores roughly 4–5× the database size across all daily snapshots when the database is write-heavy, because each daily snapshot captures changed blocks incrementally but still accumulates quickly. Compressing the live database reduces each incremental snapshot proportionally, so backup savings compound the primary storage savings.

RDS Automated Backup Compression

RDS does not expose a direct “compress backups” toggle because automated backups are stored as S3 snapshots at the block level. The compression happens indirectly: a smaller primary volume produces smaller snapshots. Enabling compression on a 3 TB database that shrinks to 1.2 TB means each daily incremental snapshot writes fewer changed blocks, and the initial full snapshot in the next retention cycle stores 1.2 TB rather than 3 TB.

For manual snapshots exported to S3 via `aws rds start-export-task`, the exported data is stored as Parquet files in S3, which applies Snappy compression by default. You do not control this, but you benefit from it automatically. A 3 TB Parquet export of typical relational OLTP data commonly stores in 600–900 GB of S3 space due to Snappy compression plus Parquet’s columnar encoding. That S3 storage at \$0.023/GB-month is substantially cheaper than RDS snapshot storage at \$0.095/GB-month for long-term archival.

PostgreSQL WAL Compression for Logical Replication

For Aurora PostgreSQL or RDS PostgreSQL instances using logical replication to feed downstream consumers (analytics databases, Kafka, CDC pipelines), enabling WAL-level compression reduces replication bandwidth costs:

```
-- Check current WAL settings relevant to compression
-- On RDS, these are set via parameter groups, not directly in psql
SHOW wal_compression;
SHOW wal_level;

-- If wal_compression is 'off', request a parameter group change:
-- wal_compression = lz4 (PostgreSQL 14+ supports lz4 and zstd)
-- This reduces WAL segment sizes by 30-60% for text-heavy workloads
-- and directly reduces cross-AZ replication traffic costs

-- Measure current WAL generation rate to estimate savings
SELECT
  pg_current_wal_lsn() AS current_lsn,
  pg_walfile_name(pg_current_wal_lsn()) AS current_wal_file,
```

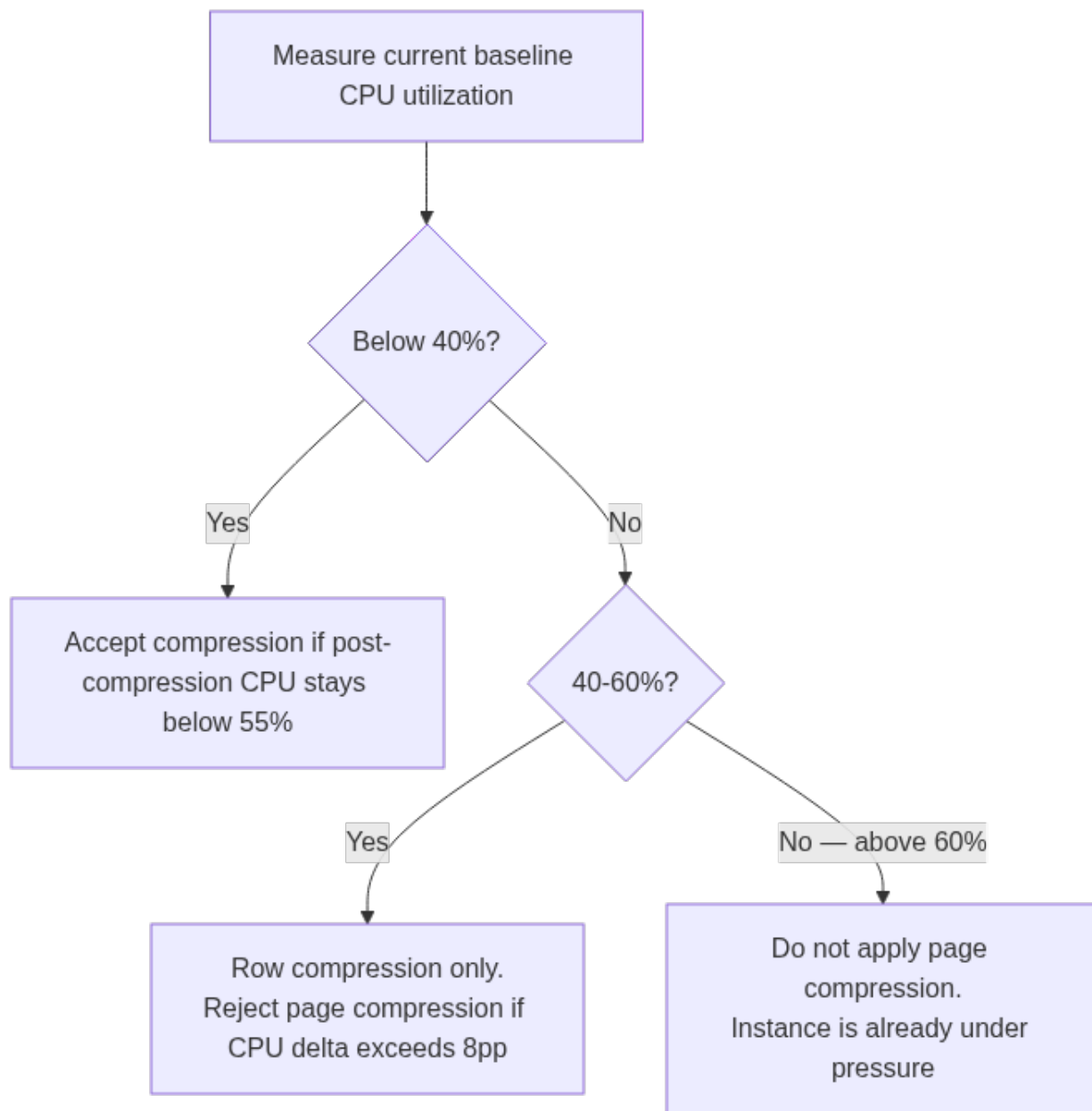


Figure 16: The compression accept/reject gate: current baseline CPU utilization determines which compression level, if any, is safe to apply.

```

-- Run this query twice, 60 seconds apart, and subtract LSN values
-- to get bytes of WAL generated per minute
pg_wal_lsn_diff(
    pg_current_wal_lsn(),
    '0/0':pg_lsn
) AS total_wal_bytes_generated
;

-- Cross-AZ data transfer pricing on AWS: $0.02/GB
-- If WAL compression reduces replication traffic by 40% on a database
-- generating 50 GB/day of WAL, that is 20 GB/day saved = $0.40/day = $12/month
-- For a write-intensive database generating 500 GB/day, savings reach $120/month

```

Azure SQL: Built-in Backup Compression

Azure SQL Database and Azure SQL Managed Instance compress backups automatically using a built-in compression algorithm. You cannot disable it, and you do not tune it. The practical implication is that applying page or row compression to your tables reduces the pre-compression data volume fed into Azure's backup pipeline, which in turn reduces the compressed backup size and the backup storage billed. The effect is multiplicative rather than additive: if page compression reduces your table data by 50% and Azure's backup compression reduces that by an additional 30%, your backup storage is roughly 65% smaller than the uncompressed baseline.

To monitor Azure SQL backup storage consumption and project savings:

```

-- Azure SQL: Check backup storage usage and estimate compression benefit
-- Run against master database on Azure SQL Managed Instance,
-- or use Azure portal for Azure SQL Database (single database)

-- On Managed Instance, this view is available:
SELECT
    database_name,
    backup_start_date,
    backup_finish_date,
    DATEDIFF(SECOND, backup_start_date, backup_finish_date) AS duration_seconds,
    backup_size / 1024 / 1024 AS backup_size_mb,
    compressed_backup_size / 1024 / 1024 AS compressed_size_mb,
    ROUND(
        (1 - compressed_backup_size * 1.0 / NULLIF(backup_size, 0)) * 100, 1
    ) AS compression_pct,
    type AS backup_type -- D=full, I=differential, L=log
FROM msdb.dbo.backupset
WHERE database_name = 'YourDatabase'
    AND backup_start_date > DATEADD(DAY, -7, GETDATE())
ORDER BY backup_start_date DESC;

```

16.5 Automating Compression Enforcement Across a Fleet

One-off compression projects decay. Engineers create new tables without compression settings, schema changes introduce uncompressed columns, and within six months the storage savings erode as new data accumulates uncompressed. Automation that enforces compression standards at table creation and identifies drift on a schedule is what converts a project into compounding savings.

PostgreSQL: Event Trigger to Enforce TOAST Compression on New Tables

```

-- Deploy this to each PostgreSQL database in your fleet
-- It fires on CREATE TABLE and ALTER TABLE ADD COLUMN,
-- automatically setting lz4 compression on all text/json/jsonb columns

CREATE OR REPLACE FUNCTION enforce_lz4_compression()
RETURNS event_trigger

```

```

LANGUAGE plpgsql
SECURITY DEFINER
AS $$
DECLARE
    obj record;
    col record;
    schema_name text;
    table_name text;
BEGIN
    FOR obj IN
        SELECT * FROM pg_event_trigger_ddl_commands()
        WHERE command_tag IN ('CREATE TABLE', 'ALTER TABLE')
        AND object_type = 'table'
    LOOP
        -- Parse schema and table from object_identity
        schema_name := split_part(obj.object_identity, '.', 1);
        table_name := split_part(obj.object_identity, '.', 2);

        -- Skip system schemas
        IF schema_name IN ('pg_catalog', 'information_schema', 'pg_toast') THEN
            CONTINUE;
        END IF;

        -- Set lz4 compression on all compressible columns
        FOR col IN
            SELECT attname, attstorage
            FROM pg_attribute
            WHERE attrelid = obj.objid
            AND attnum > 0
            AND NOT attisdropped
            AND atttypid IN (
                'text'::regtype,
                'varchar'::regtype,
                'json'::regtype,
                'jsonb'::regtype,
                'xml'::regtype,
                'bytea'::regtype
            )
            AND attcompression != '1' -- not already lz4
        LOOP
            EXECUTE format(
                'ALTER TABLE %I.%I ALTER COLUMN %I SET COMPRESSION lz4',
                schema_name, table_name, col.attname
            );
        END LOOP;
    END LOOP;
END;
$$;

-- Create the event trigger (runs in all databases where deployed)
DROP EVENT TRIGGER IF EXISTS enforce_compression_on_ddl;
CREATE EVENT TRIGGER enforce_compression_on_ddl
ON ddl_command_end
WHEN TAG IN ('CREATE TABLE', 'ALTER TABLE')
EXECUTE FUNCTION enforce_lz4_compression();

-- Verify the trigger is active
SELECT evtname, evtevent, evtenabled, evtfoid::regproc AS function
FROM pg_event_trigger
WHERE evtname = 'enforce_compression_on_ddl';

```

Fleet-Wide Compression Drift Detection Script

The event trigger handles new objects, but it does not retroactively fix existing tables or catch tables created before the trigger was deployed. A scheduled drift detection job handles the remediation side:

```

#!/usr/bin/env python3
"""

```

compression_drift_detector.py

Scans a fleet of PostgreSQL databases for tables with compressible columns that are not using lz4 compression. Outputs a remediation script and optionally posts findings to an SNS topic for alerting.

Designed to run as an AWS Lambda function on a weekly schedule via EventBridge, or as a cron job on a bastion host.

Requirements: psycpg2-binary, boto3
"""

```
import json
import os
import boto3
import psycpg2
import psycpg2.extras
from dataclasses import dataclass, field, asdict
from typing import Optional

@dataclass
class CompressionDrift:
    host: str
    database: str
    schema_name: str
    table_name: str
    column_name: str
    current_compression: str
    table_size_mb: float
    remediation_sql: str

def get_database_endpoints(tag_key: str = 'compression-managed',
                          tag_value: str = 'true') -> list[dict]:
    """
    Discover RDS instances tagged for compression management.
    Returns list of connection info dicts.
    """
    rds = boto3.client('rds')
    paginator = rds.get_paginator('describe_db_instances')
    endpoints = []

    for page in paginator.paginate():
        for instance in page['DBInstances']:
            # Check tags
            tags = {t['Key']: t['Value'] for t in instance.get('TagList', [])}
            if tags.get(tag_key) != tag_value:
                continue
            if instance['Engine'] not in ('postgres', 'aurora-postgresql'):
                continue
            if instance['DBInstanceStatus'] != 'available':
                continue

            endpoints.append({
                'host': instance['Endpoint']['Address'],
                'port': instance['Endpoint']['Port'],
                'instance_id': instance['DBInstanceIdentifier'],
                'engine': instance['Engine'],
                'databases': tags.get('monitored-databases', 'postgres').split(',')
            })

    return endpoints

def get_secret(secret_name: str) -> dict:
    """Retrieve database credentials from AWS Secrets Manager."""
    client = boto3.client('secretsmanager')
    response = client.get_secret_value(SecretId=secret_name)
    return json.loads(response['SecretString'])
```

```

def scan_database_for_drift(host: str, port: int, database: str,
                           credentials: dict,
                           min_table_size_mb: float = 10.0) -> list[CompressionDrift]:
    """
    Connect to a PostgreSQL database and find compressible columns
    not using lz4 compression.
    """
    drifts = []

    try:
        conn = psycopg2.connect(
            host=host,
            port=port,
            dbname=database,
            user=credentials['username'],
            password=credentials['password'],
            connect_timeout=10,
            sslmode='require'
        )
        conn.set_session(readonly=True, autocommit=True)
        cursor = conn.cursor(cursor_factory=psycopg2.extras.RealDictCursor)

        cursor.execute("""
            SELECT
                n.nspname AS schema_name,
                c.relname AS table_name,
                a.attname AS column_name,
                CASE a.attcompression
                    WHEN '' THEN 'default'
                    WHEN 'p' THEN 'pglz'
                    WHEN 'l' THEN 'lz4'
                    ELSE a.attcompression
                END AS current_compression,
                ROUND(pg_total_relation_size(c.oid) / 1024.0 / 1024.0, 2)
                AS table_size_mb
            FROM pg_attribute a
            JOIN pg_class c ON a.attrelid = c.oid
            JOIN pg_namespace n ON c.relnamespace = n.oid
            WHERE a.attnum > 0
                AND NOT a.attisdropped
                AND c.relkind = 'r'
                AND n.nspname NOT IN ('pg_catalog', 'information_schema', 'pg_toast')
                AND a.atttypid IN (
                    'text'::regtype,
                    'varchar'::regtype,
                    'json'::regtype,
                    'jsonb'::regtype,
                    'xml'::regtype,
                    'bytea'::regtype
                )
                AND a.attstorage IN ('x', 'm', 'e')
                AND a.attcompression != 'l' -- not already lz4
                AND pg_total_relation_size(c.oid) > %s * 1024 * 1024
            ORDER BY pg_total_relation_size(c.oid) DESC
        """, (min_table_size_mb,))

        for row in cursor.fetchall():
            remediation = (
                f"ALTER TABLE {row['schema_name']}.{row['table_name']} "
                f"ALTER COLUMN {row['column_name']} SET COMPRESSION lz4;"
            )
            drifts.append(CompressionDrift(
                host=host,
                database=database,
                schema_name=row['schema_name'],
                table_name=row['table_name'],
                column_name=row['column_name'],
                current_compression=row['current_compression'],
            ))
    
```

```

        table_size_mb=float(row['table_size_mb']),
        remediation_sql=remediation
    ))

    cursor.close()
    conn.close()

except psycopg2.OperationalError as e:
    print(f" Connection failed to {host}/{database}: {e}")

return drifts

def publish_drift_report(drifts: list[CompressionDrift],
                        sns_topic_arn: Optional[str] = None) -> None:
    """Publish drift findings to SNS and write remediation SQL to stdout."""
    if not drifts:
        print("No compression drift detected.")
        return

    total_affected_tables = len({(d.host, d.database, d.schema_name, d.table_name)
                                for d in drifts})
    total_size_mb = sum(d.table_size_mb for d in drifts)

    print(f"\n=== COMPRESSION DRIFT REPORT ===")
    print(f"Affected columns: {len(drifts)}")
    print(f"Affected tables: {total_affected_tables}")
    print(f"Total table size: {total_size_mb / 1024:.1f} GB")
    print(f"\n-- REMEDIATION SQL --")
    for drift in drifts:
        print(f"-- {drift.host}/{drift.database} | {drift.table_size_mb:.0f} MB")
        print(drift.remediation_sql)

    if sns_topic_arn:
        sns = boto3.client('sns')
        summary = {
            'alert': 'compression_drift_detected',
            'affected_columns': len(drifts),
            'affected_tables': total_affected_tables,
            'total_size_gb': round(total_size_mb / 1024, 2),
            'sample_remediations': [d.remediation_sql for d in drifts[:5]]
        }
        sns.publish(
            TopicArn=sns_topic_arn,
            Subject='Database Compression Drift Detected',
            Message=json.dumps(summary, indent=2)
        )
        print(f"\nAlert published to SNS: {sns_topic_arn}")

def lambda_handler(event: dict, context) -> dict:
    """AWS Lambda entry point."""
    secret_name = os.environ.get('DB_SECRET_NAME', 'rds/compression-monitor/credentials')
    sns_topic_arn = os.environ.get('SNS_TOPIC_ARN')
    min_size_mb = float(os.environ.get('MIN_TABLE_SIZE_MB', '10'))

    credentials = get_secret(secret_name)
    endpoints = get_database_endpoints()

    all_drifts = []
    for endpoint in endpoints:
        for database in endpoint['databases']:
            print(f"Scanning {endpoint['host']}/{database}...")
            drifts = scan_database_for_drift(
                host=endpoint['host'],
                port=endpoint['port'],
                database=database.strip(),
                credentials=credentials,
                min_table_size_mb=min_size_mb
            )

```

```

        all_drifts.extend(drifts)
        print(f" Found {len(drifts)} uncompressed compressible columns")

    publish_drift_report(all_drifts, sns_topic_arn)

    return {
        'statusCode': 200,
        'drifts_found': len(all_drifts),
        'affected_tables': len({(d.host, d.database, d.schema_name, d.table_name)
                                for d in all_drifts})
    }

if __name__ == '__main__':
    # Local execution for testing
    lambda_handler({}, None)

```

SQL Server: Scheduled Job to Apply and Verify Compression

For SQL Server fleets on Azure SQL Managed Instance or RDS for SQL Server, use SQL Server Agent to run a weekly compression audit job. The job identifies tables that have grown beyond a threshold and have not yet been compressed, then applies row compression automatically and logs the result:

```

-- Create a compression audit and auto-apply stored procedure
-- Deploy to each managed instance or RDS SQL Server instance

CREATE OR ALTER PROCEDURE dbo.usp_AutoApplyRowCompression
    @size_threshold_mb INT = 100,          -- Only compress tables > this size
    @dry_run BIT = 1,                      -- 1 = report only, 0 = apply compression
    @max_tables_per_run INT = 10           -- Limit tables per execution to control duration
AS
BEGIN
    SET NOCOUNT ON;

    -- Temp table to hold candidates
    CREATE TABLE #compression_candidates (
        schema_name SYSNAME,
        table_name SYSNAME,
        size_mb DECIMAL(10, 2),
        current_compression NVARCHAR(60),
        rows BIGINT,
        processed BIT DEFAULT 0
    );

    INSERT INTO #compression_candidates (schema_name, table_name, size_mb,
                                         current_compression, rows)

    SELECT
        s.name,
        t.name,
        SUM(a.total_pages) * 8.0 / 1024,
        MAX(p.data_compression_desc),
        SUM(p.rows)
    FROM sys.tables t
    INNER JOIN sys.schemas s ON t.schema_id = s.schema_id
    INNER JOIN sys.partitions p ON t.object_id = p.object_id AND p.index_id IN (0,1)
    INNER JOIN sys.allocation_units a ON p.partition_id = a.container_id
    GROUP BY s.name, t.name
    HAVING MAX(p.data_compression_desc) = 'NONE'
    AND SUM(a.total_pages) * 8.0 / 1024 > @size_threshold_mb
    ORDER BY SUM(a.total_pages) DESC;

    DECLARE @schema SYSNAME,
            @table SYSNAME,
            @size_mb DECIMAL(10,2),
            @sql NVARCHAR(500),
            @count INT = 0;

```

```

WHILE @count < @max_tables_per_run
BEGIN
    SELECT TOP 1
        @schema = schema_name,
        @table = table_name,
        @size_mb = size_mb
    FROM #compression_candidates
    WHERE processed = 0
    ORDER BY size_mb DESC;

    IF @@ROWCOUNT = 0 BREAK;

    IF @dry_run = 1
    BEGIN
        PRINT CONCAT('DRY RUN - Would compress: ', @schema, '.', @table,
            ' (', @size_mb, ' MB)');
    END
    ELSE
    BEGIN
        SET @sql = CONCAT(
            'ALTER TABLE ', QUOTENAME(@schema), '.', QUOTENAME(@table),
            ' REBUILD WITH (DATA_COMPRESSION = ROW)'
        );

        BEGIN TRY
            EXEC sp_executesql @sql;
            INSERT INTO dbo.compression_audit_log
                (schema_name, table_name, size_before_mb,
                 compression_applied, applied_at)
            VALUES (@schema, @table, @size_mb, 'ROW', GETUTCDATE());

            PRINT CONCAT('Compressed: ', @schema, '.', @table);
        END TRY
        BEGIN CATCH
            PRINT CONCAT('FAILED: ', @schema, '.', @table,
                ' - ', ERROR_MESSAGE());
        END CATCH;
    END;

    UPDATE #compression_candidates
    SET processed = 1
    WHERE schema_name = @schema AND table_name = @table;

    SET @count += 1;
END;

-- Return summary
SELECT
    schema_name,
    table_name,
    size_mb,
    current_compression,
    rows,
    CASE processed WHEN 1 THEN 'PROCESSED' ELSE 'PENDING' END AS status
FROM #compression_candidates
ORDER BY size_mb DESC;

DROP TABLE #compression_candidates;
END;
GO

-- Create the audit log table if it does not exist
IF OBJECT_ID('dbo.compression_audit_log') IS NULL
BEGIN
    CREATE TABLE dbo.compression_audit_log (
        log_id                INT IDENTITY PRIMARY KEY,
        schema_name           SYSNAME,
        table_name            SYSNAME,
        size_before_mb        DECIMAL(10, 2),
        compression_applied   NVARCHAR(10),

```

```

        applied_at          DATETIME2 DEFAULT GETUTCDATE()
    );
END;
GO

-- Run in dry-run mode first to see what would be compressed
EXEC dbo.usp_AutoApplyRowCompression @size_threshold_mb = 100, @dry_run = 1;

-- Run live against the first 10 qualifying tables
-- EXEC dbo.usp_AutoApplyRowCompression @size_threshold_mb = 100, @dry_run = 0;

```

16.6 Columnar Storage as an Extreme Compression Strategy

For tables that are primarily read by analytics queries — monthly aggregations, compliance reports, audit trails — columnar storage achieves compression ratios that row-based compression cannot approach. Columnar encoding stores all values of the same column contiguously, which means run-length encoding, dictionary encoding, and delta encoding can compress similar values far more aggressively than page compression applied to mixed row data.

SQL Server Columnstore Indexes on Azure SQL

A nonclustered columnstore index on a large fact table commonly achieves 5–10× compression relative to an uncompressed rowstore, and 3–5× relative to a page-compressed rowstore. For a 500 GB audit log table that is written once and read weekly, adding a columnstore index can reduce the effective storage footprint of those reads to under 100 GB while eliminating most individual page reads:

```

-- Add a nonclustered columnstore index for analytics queries on a large table
-- This coexists with the existing rowstore clustered index
-- Writes go to the rowstore; analytics queries use the columnstore automatically

-- First, check if the query workload is analytics-suitable
-- (large scans, aggregations, group bys rather than single-row lookups)
SELECT
    t.name AS table_name,
    SUM(s.user_scans) AS total_scans,
    SUM(s.user_seeks) AS total_seeks,
    SUM(s.user_scans) * 100.0 /
        NULLIF(SUM(s.user_scans) + SUM(s.user_seeks), 0) AS pct_scans,
    CASE
        WHEN SUM(s.user_scans) * 100.0 /
            NULLIF(SUM(s.user_scans) + SUM(s.user_seeks), 0) > 60
        THEN 'Columnstore candidate'
        ELSE 'Primarily point-lookup: skip columnstore'
    END AS recommendation
FROM sys.tables t
JOIN sys.indexes i ON t.object_id = i.object_id
LEFT JOIN sys.dm_db_index_usage_stats s
    ON i.object_id = s.object_id AND i.index_id = s.index_id
    AND s.database_id = DB_ID()
WHERE i.index_id IN (0, 1)
GROUP BY t.name
ORDER BY total_scans DESC;

-- Create nonclustered columnstore index on AuditLog
CREATE NONCLUSTERED COLUMNSTORE INDEX NCCI_AuditLog_Analytics
ON dbo.AuditLog (
    event_date, user_id, action_type, resource_id,
    old_value, new_value, ip_address, session_id
)
WITH (DATA_COMPRESSION = COLUMNSTORE_ARCHIVE) -- Maximum compression for cold data

```

```

-- Use DATA_COMPRESSION = COLUMNSTORE for hot analytics data (faster decompression)
;

-- Measure actual compression ratio after index creation
SELECT
  i.name AS index_name,
  i.type_desc,
  SUM(a.total_pages) * 8 / 1024 AS index_size_mb,
  -- Compare to the rowstore size from earlier query
  p.data_compression_desc
FROM sys.indexes i
JOIN sys.partitions p ON i.object_id = p.object_id AND i.index_id = p.index_id
JOIN sys.allocation_units a ON p.partition_id = a.container_id
WHERE OBJECT_NAME(i.object_id) = 'AuditLog'
GROUP BY i.name, i.type_desc, p.data_compression_desc
ORDER BY SUM(a.total_pages) DESC;

```

PostgreSQL: Partitioning + pg_partman + Columnar Extension (Citus / Hydra)

Standard PostgreSQL does not include a native columnar storage engine for on-disk row format compression at the columnar level, but the Hydra columnar extension (available on Aurora PostgreSQL via custom parameter groups in some configurations) and Citus columnar (available on Azure Database for PostgreSQL – Flexible Server) both provide columnar access methods. For workloads running on those platforms, columnar tables achieve similar compression ratios to SQL Server columnstore indexes.

For RDS PostgreSQL without columnar extensions, the practical equivalent is aggressive partitioning by time combined with `CLUSTER` on the most selective column and `VACUUM FULL` on older partitions. This does not achieve columnar compression ratios, but it does concentrate similar data on each page, making PGLZ and LZ4 TOAST compression more effective, and it enables detaching and archiving cold partitions to cheaper storage tiers.

16.7 Building a Compression Savings Dashboard

Tracking compression savings over time closes the feedback loop between engineering effort and financial outcome. The dashboard below feeds from CloudWatch, Azure Monitor, or GCP Cloud Monitoring and produces a monthly savings summary suitable for a FinOps review meeting.

Savings Calculation Model

The model requires four inputs per database instance: 1. Current allocated storage (GB) 2. Storage size before compression project began (GB from historical CloudWatch `FreeStorageSpace` metric and allocated volume size) 3. Per-GB storage price for that instance’s region, tier, and replication factor 4. Backup retention days (to calculate multiplier on backup savings)

```

#!/usr/bin/env python3
"""
compression_savings_dashboard.py

Generates a monthly compression savings report across a fleet of RDS instances
by comparing current storage utilization to a pre-compression baseline stored
in AWS Systems Manager Parameter Store.

Usage:
  python compression_savings_dashboard.py --region us-east-1 --output dashboard.json

Requirements: boto3
"""

```

```

import boto3
import json
import argparse
from datetime import datetime, timedelta, timezone
from typing import Optional

# Storage pricing per GB-month by region and engine (approximate, verify current rates)
STORAGE_PRICING = {
    'us-east-1': {
        'gp3_single_az': 0.115,
        'gp3_multi_az': 0.230,      # primary + standby volumes
        'backup_s3': 0.095,
        'aurora_single': 0.10,
        'aurora_multi': 0.10,      # Aurora charges per-GB regardless of AZ config
    },
    'eu-west-1': {
        'gp3_single_az': 0.131,
        'gp3_multi_az': 0.262,
        'backup_s3': 0.095,
        'aurora_single': 0.114,
        'aurora_multi': 0.114,
    },
    'ap-southeast-1': {
        'gp3_single_az': 0.138,
        'gp3_multi_az': 0.276,
        'backup_s3': 0.095,
        'aurora_single': 0.122,
        'aurora_multi': 0.122,
    }
}

def get_baseline_from_ssm(instance_id: str, region: str) -> Optional[dict]:
    """
    Retrieve pre-compression storage baseline stored in SSM Parameter Store.
    Baseline is stored when a compression project begins via:
        aws ssm put-parameter \
            --name /finops/compression-baseline/my-rds-instance \
            --value '{"allocated_gb": 3000, "used_gb": 2800, "recorded_at": "2024-01-15"}' \
            --type String
    """
    ssm = boto3.client('ssm', region_name=region)
    try:
        response = ssm.get_parameter(
            Name=f'/finops/compression-baseline/{instance_id}'
        )
        return json.loads(response['Parameter']['Value'])
    except ssm.exceptions.ParameterNotFound:
        return None

def get_current_storage_gb(instance_id: str, region: str) -> dict:
    """
    Get current allocated and used storage for an RDS instance.
    Returns allocated_gb and estimated used_gb from CloudWatch.
    """
    rds = boto3.client('rds', region_name=region)
    cw = boto3.client('cloudwatch', region_name=region)

    # Get allocated storage from RDS describe
    response = rds.describe_db_instances(DBInstanceIdentifier=instance_id)
    instance = response['DBInstances'][0]
    allocated_gb = instance['AllocatedStorage']
    multi_az = instance['MultiAZ']
    engine = instance['Engine']

    # Get free storage from CloudWatch (last 24h average)
    end = datetime.now(timezone.utc)

```

```

start = end - timedelta(hours=24)

cw_response = cw.get_metric_statistics(
    Namespace='AWS/RDS',
    MetricName='FreeStorageSpace',
    Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': instance_id}],
    StartTime=start,
    EndTime=end,
    Period=3600,
    Statistics=['Average']
)

free_bytes = 0
if cw_response['Datapoints']:
    free_bytes = max(p['Average'] for p in cw_response['Datapoints'])

free_gb = free_bytes / (1024 ** 3)
used_gb = allocated_gb - free_gb

return {
    'allocated_gb': allocated_gb,
    'used_gb': round(used_gb, 2),
    'free_gb': round(free_gb, 2),
    'multi_az': multi_az,
    'engine': engine
}

def calculate_monthly_savings(instance_id: str, region: str,
                             backup_retention_days: int = 7) -> dict:
    """
    Calculate monthly storage savings by comparing current state to baseline.
    """
    baseline = get_baseline_from_ssm(instance_id, region)
    current = get_current_storage_gb(instance_id, region)

    if not baseline:
        return {
            'instance_id': instance_id,
            'status': 'no_baseline',
            'message': (
                f'Store baseline with: aws ssm put-parameter '
                f'--name /finops/compression-baseline/{instance_id} '
                f'--value \'{{"allocated_gb": {current["allocated_gb"]}, '
                f'"used_gb": {current["used_gb"]}, '
                f'"recorded_at": "{datetime.now().date().isoformat()}"}}\' '
                f'--type String'
            )
        )

    pricing = STORAGE_PRICING.get(region, STORAGE_PRICING['us-east-1'])
    multi_az = current['multi_az']
    engine = current['engine']

    # Determine per-GB rate based on engine and AZ configuration
    if 'aurora' in engine:
        rate_per_gb = pricing['aurora_multi'] # Aurora always effectively multi-AZ
    elif multi_az:
        rate_per_gb = pricing['gp3_multi_az']
    else:
        rate_per_gb = pricing['gp3_single_az']

    # Storage saved (GB)
    baseline_used = baseline['used_gb']
    current_used = current['used_gb']
    storage_saved_gb = max(0, baseline_used - current_used)
    compression_ratio = (
        (1 - current_used / baseline_used) * 100 if baseline_used > 0 else 0
    )

```



```

# Monthly primary storage savings
primary_savings_monthly = storage_saved_gb * rate_per_gb

# Backup savings: compressed data generates proportionally smaller backups
# Rough model: daily incremental backups average ~5% of total DB size,
# plus one full snapshot per retention period
# This is a conservative estimate; write-heavy DBs save proportionally more
daily_incremental_gb = current_used * 0.05
backup_total_gb = (daily_incremental_gb * backup_retention_days
                  + baseline_used) # one full snapshot worth
backup_saved_gb = storage_saved_gb * (backup_total_gb / baseline_used
                                     if baseline_used > 0 else 0)
backup_savings_monthly = backup_saved_gb * pricing['backup_s3']

total_monthly = primary_savings_monthly + backup_savings_monthly
total_annual = total_monthly * 12

return {
    'instance_id': instance_id,
    'region': region,
    'engine': engine,
    'multi_az': multi_az,
    'baseline': {
        'used_gb': baseline_used,
        'recorded_at': baseline.get('recorded_at', 'unknown')
    },
    'current': {
        'allocated_gb': current['allocated_gb'],
        'used_gb': current_used,
        'free_gb': current['free_gb']
    },
    'compression_result': {
        'storage_saved_gb': round(storage_saved_gb, 2),
        'compression_pct': round(compression_ratio, 1)
    },
    'monthly_savings_usd': {
        'primary_storage': round(primary_savings_monthly, 2),
        'backup_storage': round(backup_savings_monthly, 2),
        'total': round(total_monthly, 2)
    },
    'annual_savings_usd': round(total_annual, 2),
    'rate_per_gb_month': rate_per_gb
}

def generate_fleet_dashboard(instance_ids: list[str], region: str,
                             backup_retention_days: int = 7) -> dict:
    """Generate a savings summary across a fleet of instances."""
    results = []
    for iid in instance_ids:
        print(f" Processing {iid}...")
        result = calculate_monthly_savings(iid, region, backup_retention_days)
        results.append(result)

    # Fleet-level totals (only instances with baselines)
    valid = [r for r in results if r.get('status') != 'no_baseline']
    total_saved_gb = sum(r['compression_result']['storage_saved_gb']
                        for r in valid)
    total_monthly = sum(r['monthly_savings_usd']['total'] for r in valid)
    total_annual = sum(r['annual_savings_usd'] for r in valid)

    return {
        'generated_at': datetime.now(timezone.utc).isoformat(),
        'region': region,
        'fleet_summary': {
            'instances_analyzed': len(results),
            'instances_with_baseline': len(valid),
            'total_storage_saved_gb': round(total_saved_gb, 2),
            'total_monthly_savings_usd': round(total_monthly, 2),
            'total_annual_savings_usd': round(total_annual, 2)
        }
    }

```

```

    },
    'instance_details': results
}

def main():
    parser = argparse.ArgumentParser(
        description='Generate compression savings dashboard for RDS fleet'
    )
    parser.add_argument('--region', default='us-east-1')
    parser.add_argument('--instances', nargs='+',
        help='RDS instance IDs (space-separated). '
        'If omitted, discovers tagged instances.')
    parser.add_argument('--backup-retention', type=int, default=7)
    parser.add_argument('--output', default='dashboard.json')
    args = parser.parse_args()

    if args.instances:
        instance_ids = args.instances
    else:
        # Auto-discover instances tagged compression-managed=true
        rds = boto3.client('rds', region_name=args.region)
        paginator = rds.get_paginator('describe_db_instances')
        instance_ids = []
        for page in paginator.paginate():
            for inst in page['DBInstances']:
                tags = {t['Key']: t['Value'] for t in inst.get('TagList', [])}
                if tags.get('compression-managed') == 'true':
                    instance_ids.append(inst['DBInstanceIdentifier'])

    print(f"Generating compression dashboard for {len(instance_ids)} instances...")
    dashboard = generate_fleet_dashboard(
        instance_ids, args.region, args.backup_retention
    )

    with open(args.output, 'w') as f:
        json.dump(dashboard, f, indent=2)

    print(f"\nDashboard written to {args.output}")
    summary = dashboard['fleet_summary']
    print(f"\n=== FLEET SUMMARY ===")
    print(f"Instances analyzed:      {summary['instances_analyzed']}")
    print(f"Instances with baseline: {summary['instances_with_baseline']}")
    print(f"Total storage saved:      {summary['total_storage_saved_gb']:.1f} GB")
    print(f"Monthly savings:         ${summary['total_monthly_savings_usd']:.2f}")
    print(f"Annual savings:          ${summary['total_annual_savings_usd']:.2f}")

if __name__ == '__main__':
    main()

```

16.8 When Not to Compress

Compression is not universally beneficial, and applying it indiscriminately creates operational debt. The following patterns are where compression reliably harms more than it helps.

Already-compressed data: JPEG, PNG, MP4, PDF, ZIP, AES-encrypted payloads, and other binary formats stored as BLOBs are already maximally compressed or encrypted to appear random. Enabling page compression on a table holding these types wastes CPU on compression attempts that save zero bytes. Identify these columns with the data type scan in section 16.1 and explicitly exclude them.

Extremely high write throughput tables: A table receiving sustained INSERT rates above roughly 50,000 rows per second with page compression enabled will experience measurable write latency increases because each page rebuild requires decompression, modification, and recompression. For these tables, benchmark first with row compression (lower overhead) and reject page compression entirely if write latency degrades.

Very small tables: Tables under 10 MB cost almost nothing to store and are frequently hot in the buffer pool. The management overhead of tracking compression settings, testing changes, and handling upgrade regressions is not justified by single-digit MB savings.

Tables used for in-flight streaming: Tables that serve as intermediate buffers for real-time event processing — where rows are written, read once, and deleted within seconds — spend no time on disk in a meaningful sense. The buffer pool absorbs nearly all I/O, so physical storage compression has no effect on costs.

High-cardinality columns with random data: Page compression's prefix and dictionary compression steps achieve nothing on columns with UUID values, hash outputs, random tokens, or base64-encoded nonces. Row compression still eliminates padding from fixed-length types, so it remains viable, but page compression adds CPU overhead with negligible storage benefit. Run `sp_estimate_data_compression_savings` or a TOAST size comparison before applying page compression to any table with UUIDs as primary or foreign keys.

Key Takeaways

- **Storage savings multiply across layers:** Compressing primary data reduces not just the primary EBS or managed disk volume but also the standby replica volume (in Multi-AZ deployments), automated backup snapshots, and WAL or log replication traffic. The actual cost reduction is often two to three times the primary storage saving in isolation.
- **LZ4 is the correct default for PostgreSQL 14+ TOAST columns:** LZ4 compresses and decompresses three to five times faster than the older PGLZ algorithm with comparable or better compression ratios for the text and JSON workloads that dominate most OLTP schemas. Switching existing databases requires an explicit `ALTER COLUMN` plus `VACUUM FULL` or `pg_repack` to rewrite existing TOAST entries.
- **SQL Server row versus page compression is a CPU budget decision:** Row compression saves 10–25% with minimal CPU overhead and is safe for nearly all tables. Page compression saves 30–60% but can add 15–25% CPU overhead on write-heavy tables. On vCore-based Azure SQL tiers, calculate whether the CPU cost of page compression exceeds the storage savings at current pricing before applying it.
- **Automation prevents savings erosion:** A compression project without drift detection and enforcement via event triggers (PostgreSQL) or a scheduled audit job (SQL Server) will see its gains diluted within months as new uncompressed tables accumulate. The event trigger in section 16.5 costs nothing to run and applies lz4 to every new compressible column automatically.
- **Know when to stop:** Already-compressed binary formats, extremely high write-throughput tables, sub-10 MB tables, and columns with high-cardinality random values are not compression candidates. Applying compression to these patterns wastes CPU, increases write latency, and produces no meaningful storage savings, converting a cost optimization project into a per-

formance incident.

Chapter 17: Tiered Storage: Moving Cold Data Cheaply

Organizations that audit their cloud database storage seriously almost always find the same thing: 60–80% of the data sitting on expensive SSD-backed block storage has not been touched in months. That data costs the same per gigabyte as the hot rows your application reads thousands of times per second. Moving cold data to cheaper storage tiers—S3, Azure Blob, GCS, or archival-class object storage—routinely reduces total database storage costs by 40–70%, and on multi-terabyte databases that translates directly to \$8,000–\$50,000 per year in avoided spend. This chapter walks through how to identify cold data, classify it systematically, move it to the right storage tier, and automate the entire lifecycle so it keeps paying dividends without manual intervention every quarter.

17.1 Understanding the Storage Cost Stack

Before moving anything, you need to understand what you are paying for and why the pricing spread across tiers is so dramatic.

AWS RDS storage pricing (us-east-1, 2024):

Storage Type	Cost per GB-month
gp2 SSD	\$0.115
gp3 SSD	\$0.092
io1 / io2 SSD	\$0.125–\$0.138
S3 Standard	\$0.023
S3 Standard-IA	\$0.0125
S3 Glacier Instant	\$0.004
S3 Glacier Deep Archive	\$0.00099

The math is straightforward. A 1 TB volume on gp2 costs \$115.00/month. That same 1 TB on S3 Glacier Deep Archive costs \$0.99/month. The gap is 116x. Even moving to S3 Standard-IA costs 10x less than gp2.

Azure SQL / Blob pricing (East US, 2024):

Storage Type	Cost per GB-month
Azure SQL Premium (P-series)	\$0.114
Azure SQL General Purpose	\$0.115
Azure Blob Hot	\$0.018
Azure Blob Cool	\$0.01
Azure Blob Archive	\$0.00099

Google Cloud (us-central1, 2024):

Storage Type	Cost per GB-month
Cloud SQL SSD	\$0.17
Cloud SQL HDD	\$0.09
GCS Standard	\$0.020
GCS Nearline	\$0.010
GCS Coldline	\$0.004
GCS Archive	\$0.0012

The pricing structure across all three clouds tells the same story: object storage is an order of magnitude cheaper than block storage attached to your database instances. The technical challenge is not the pricing—it is designing the data movement pipeline so that cold data moves gracefully and retrieval remains possible when needed.

Worked example: 10 TB audit log table on AWS RDS (gp3)

Current: 10,000 GB × \$0.092 = \$920/month = \$11,040/year

After archiving 8 TB to S3 Glacier Instant:

Hot tier (2 TB on gp3): 2,000 × \$0.092 = \$184/month

Cold tier (8 TB on Glacier Instant): 8,000 × \$0.004 = \$32/month

Total: \$216/month = \$2,592/year

Annual savings: \$8,448 (76% reduction)

17.2 Diagnosing Cold Data: Finding What You Can Archive

You cannot archive what you cannot identify. The first step is building a clear picture of data age, access patterns, and business rules for retention.

Identifying table-level age distribution in PostgreSQL:

```
-- Find tables with the oldest average tuple age
-- Run on each RDS PostgreSQL / Aurora PostgreSQL instance
SELECT
    schemaname,
    relname AS table_name,
    pg_size_pretty(pg_total_relation_size(schemaname || '.' || relname)) AS total_size,
    pg_total_relation_size(schemaname || '.' || relname) AS total_bytes,
    n_live_tup AS live_rows,
    n_dead_tup AS dead_rows,
    last_vacuum,
    last_autovacuum,
```

```

last_analyze,
last_autoanalyze,
-- Estimate staleness: tables not analyzed in 90 days are candidates
CASE
    WHEN COALESCE(last_autoanalyze, last_analyze) < NOW() - INTERVAL '90 days'
    OR COALESCE(last_autoanalyze, last_analyze) IS NULL
    THEN 'Cold candidate'
    WHEN COALESCE(last_autoanalyze, last_analyze) < NOW() - INTERVAL '30 days'
    THEN 'Warm - monitor'
    ELSE 'Active'
END AS access_classification
FROM
pg_stat_user_tables
ORDER BY
    total_bytes DESC;

-- Find the date range of data in each table that has a created_at column
-- This query generates dynamic SQL for inspection
SELECT
    'SELECT '''
    || table_schema || '.' || table_name
    || ''' AS tbl, MIN(' || column_name || ') AS oldest_row, '
    || 'MAX(' || column_name || ') AS newest_row, '
    || 'COUNT(*) AS total_rows '
    || 'FROM ' || table_schema || '.' || table_name || ';' AS inspection_sql
FROM
    information_schema.columns
WHERE
    column_name IN ('created_at', 'created_date', 'inserted_at', 'event_time', 'timestamp')
    AND table_schema NOT IN ('pg_catalog', 'information_schema')
ORDER BY
    table_schema, table_name;

-- For SQL Server / Azure SQL: find table age distribution
-- using sys.dm_db_index_usage_stats
SELECT
    OBJECT_NAME(i.object_id) AS table_name,
    i.name AS index_name,
    ius.user_seeks,
    ius.user_scans,
    ius.user_lookups,
    ius.user_updates,
    ius.last_user_seek,
    ius.last_user_scan,
    ius.last_user_lookup,
    -- Total read activity
    (ius.user_seeks + ius.user_scans + ius.user_lookups) AS total_reads,
    CASE
        WHEN ius.last_user_seek < GETDATE() - 90
        OR ius.last_user_seek IS NULL
        THEN 'Archive candidate'
        WHEN ius.last_user_seek < GETDATE() - 30
        THEN 'Monitor'
        ELSE 'Active'
    END AS recommendation
FROM
    sys.indexes i
    LEFT JOIN sys.dm_db_index_usage_stats ius
        ON i.object_id = ius.object_id
        AND i.index_id = ius.index_id
        AND ius.database_id = DB_ID()
WHERE
    OBJECTPROPERTY(i.object_id, 'IsUserTable') = 1
ORDER BY
    total_reads ASC,
    table_name;

```

Beyond table-level statistics, you need CloudWatch or Azure Monitor metrics to confirm that the storage IOPS consumption is actually coming from a small subset of tables. A table that shows up

as large in `pg_stat_user_tables` but consumes zero IOPS is a clean archival candidate. A large table with high IOPS is not—even if most rows are old.

Pulling CloudWatch RDS storage and IOPS metrics via AWS CLI:

```
#!/bin/bash
# Pull RDS IOPS and storage metrics for the last 30 days
# to identify instances with underutilized storage capacity

DB_INSTANCE="prod-postgres-01"
START_DATE=$(date -u -d '30 days ago' +%Y-%m-%dT%H:%M:%SZ)
END_DATE=$(date -u +%Y-%m-%dT%H:%M:%SZ)
REGION="us-east-1"

echo "=== Read IOPS (avg over 30 days) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name ReadIOPS \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_INSTANCE \
  --start-time $START_DATE \
  --end-time $END_DATE \
  --period 86400 \
  --statistics Average \
  --region $REGION \
  --output table \
  --query 'sort_by(Datapoints, &Timestamp)[*].{Date:Timestamp,AvgReadIOPS:Average}'

echo ""
echo "=== Write IOPS (avg over 30 days) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name WriteIOPS \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_INSTANCE \
  --start-time $START_DATE \
  --end-time $END_DATE \
  --period 86400 \
  --statistics Average \
  --region $REGION \
  --output table \
  --query 'sort_by(Datapoints, &Timestamp)[*].{Date:Timestamp,AvgWriteIOPS:Average}'

echo ""
echo "=== Free Storage Space (min over 30 days) ==="
aws cloudwatch get-metric-statistics \
  --namespace AWS/RDS \
  --metric-name FreeStorageSpace \
  --dimensions Name=DBInstanceIdentifier,Value=$DB_INSTANCE \
  --start-time $START_DATE \
  --end-time $END_DATE \
  --period 86400 \
  --statistics Minimum \
  --region $REGION \
  --output table \
  --query 'sort_by(Datapoints, &Timestamp)[*].{Date:Timestamp,FreeGB:Minimum}' \
  | awk '{if ($0 ~ /[0-9]/) {split($0,a,"|"); printf "%-25s %10.2f GB\n", a[2], a[3]/1073741824} else \
  ↪ print}'
```

17.3 Classifying and Partitioning Data for Tiered Movement

The most reliable pattern for tiered storage is range partitioning by time. Once you know which tables have archivable data, you restructure them so that cold partitions can be detached and exported cleanly without touching active data.

Creating a partitioned table in PostgreSQL for future tiering:


```

-- Convert an existing unpartitioned table to a partitioned structure
-- This example handles an audit_events table with 3 years of history

-- Step 1: Create the new partitioned parent table
CREATE TABLE audit_events_partitioned (
    id          BIGSERIAL,
    user_id     BIGINT      NOT NULL,
    event_type  VARCHAR(100) NOT NULL,
    event_data  JSONB,
    ip_address  INET,
    created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    PRIMARY KEY (id, created_at)
) PARTITION BY RANGE (created_at);

-- Step 2: Create partitions for each quarter going back 2 years
-- Current quarter stays hot; older quarters become archive candidates
DO $$
DECLARE
    start_date DATE := DATE_TRUNC('quarter', NOW() - INTERVAL '2 years');
    end_date   DATE;
    part_name  TEXT;
BEGIN
    WHILE start_date < DATE_TRUNC('quarter', NOW()) LOOP
        end_date := start_date + INTERVAL '3 months';
        part_name := 'audit_events_'
            || TO_CHAR(start_date, 'YYYY-Q"Q"');

        EXECUTE FORMAT(
            'CREATE TABLE IF NOT EXISTS %I
             PARTITION OF audit_events_partitioned
             FOR VALUES FROM (%L) TO (%L)',
            part_name,
            start_date,
            end_date
        );

        RAISE NOTICE 'Created partition: % covering % to %',
            part_name, start_date, end_date;

        start_date := end_date;
    END LOOP;
END $$;

-- Step 3: Create the current active partition
CREATE TABLE audit_events_current
    PARTITION OF audit_events_partitioned
    FOR VALUES FROM (DATE_TRUNC('quarter', NOW()))
        TO (DATE_TRUNC('quarter', NOW()) + INTERVAL '3 months');

-- Step 4: Create a default partition for future inserts outside expected range
CREATE TABLE audit_events_future
    PARTITION OF audit_events_partitioned
    DEFAULT;

-- Step 5: View partition sizes to understand archival opportunity
SELECT
    child.relname                                AS partition_name,
    pg_size_pretty(pg_total_relation_size(child.oid)) AS total_size,
    pg_total_relation_size(child.oid)              AS bytes,
    parent.relname                                AS parent_table
FROM
    pg_inherits
    JOIN pg_class parent ON pg_inherits.inhparent = parent.oid
    JOIN pg_class child ON pg_inherits.inhrelid = child.oid
WHERE
    parent.relname = 'audit_events_partitioned'
ORDER BY
    child.relname;

```

Once partitions exist, the archival workflow is clean: detach a partition, export it to S3/GCS/Blob as Parquet or CSV, verify the export, then drop the partition. The table keeps serving queries with no downtime.

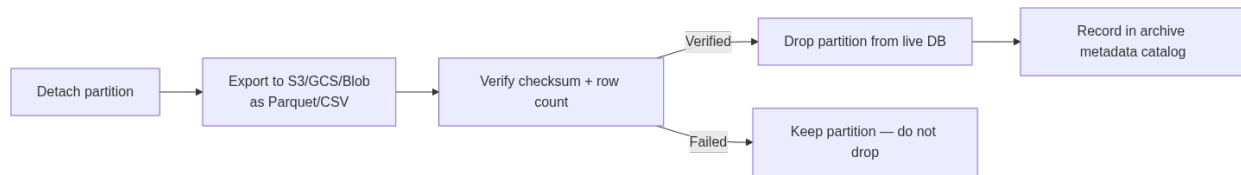


Figure 17: The detach → export → verify → drop pipeline: a partition is only dropped from the live database after its export is independently verified.

Detach, export, and verify workflow for PostgreSQL on AWS RDS:

```

-- Step 1: Detach the oldest partition (run during maintenance window or off-peak)
-- Detached partition remains queryable as a standalone table
ALTER TABLE audit_events_partitioned
DETACH PARTITION audit_events_2022_q1;

-- Step 2: Confirm row counts match before export
SELECT COUNT(*) FROM audit_events_2022_q1;
-- Save this number for post-export validation

-- Step 3: Export the detached partition via COPY to S3
-- (Use aws_s3 extension on RDS PostgreSQL)
SELECT aws_s3.query_export_to_s3(
  'SELECT * FROM audit_events_2022_q1',
  aws_commons.create_s3_uri(
    'your-archive-bucket',
    'database-archive/audit_events/2022/Q1/audit_events_2022_q1.csv',
    'us-east-1'
  ),
  options := 'FORMAT CSV, HEADER true'
);

-- Step 4: Verify export row count by querying S3 via Athena (run separately)
-- SELECT COUNT(*) FROM archive_audit_events WHERE year=2022 AND quarter=1;

-- Step 5: After verification, drop the partition
-- Only run after confirming S3 export matches row count
DROP TABLE audit_events_2022_q1;

-- Step 6: Record the archival in your metadata tracking table
INSERT INTO archive_metadata (
  original_table,
  partition_name,
  archive_path,
  row_count,
  archived_at,
  archived_by,
  verified
) VALUES (
  'audit_events_partitioned',
  'audit_events_2022_q1',
  's3://your-archive-bucket/database-archive/audit_events/2022/Q1/',
  (SELECT COUNT(*) FROM audit_events_2022_q1), -- captured before DROP
  NOW(),
  CURRENT_USER,
  false -- set to true after Athena validation
);

```

17.4 Automated Tiering with S3 Lifecycle Policies and AWS Glue

Manual archival runs once. Automation runs forever. The goal is a pipeline that identifies cold partitions on a schedule, exports them, validates the export, and transitions the objects through S3 storage classes automatically as they age further.

S3 lifecycle policy to transition archived database exports through storage tiers:

```
#!/bin/bash
# Apply a lifecycle policy to the database archive bucket
# Transitions: Standard → Standard-IA (30d) → Glacier Instant (90d) → Deep Archive (365d)

BUCKET="your-archive-bucket"

cat > /tmp/lifecycle-policy.json << 'EOF'
{
  "Rules": [
    {
      "ID": "DatabaseArchiveTiering",
      "Status": "Enabled",
      "Filter": {
        "Prefix": "database-archive/"
      },
      "Transitions": [
        {
          "Days": 30,
          "StorageClass": "STANDARD_IA"
        },
        {
          "Days": 90,
          "StorageClass": "GLACIER_IR"
        },
        {
          "Days": 365,
          "StorageClass": "DEEP_ARCHIVE"
        }
      ],
      "NoncurrentVersionTransitions": [
        {
          "NoncurrentDays": 7,
          "StorageClass": "STANDARD_IA"
        }
      ],
      "NoncurrentVersionExpiration": {
        "NoncurrentDays": 90
      }
    },
    {
      "ID": "DeleteExpiredArchives",
      "Status": "Enabled",
      "Filter": {
        "Prefix": "database-archive/audit_events/"
      },
      "Expiration": {
        "Days": 2555
      }
    }
  ]
}
EOF

aws s3api put-bucket-lifecycle-configuration \
  --bucket $BUCKET \
  --lifecycle-configuration file:///tmp/lifecycle-policy.json

echo "Lifecycle policy applied to s3://$BUCKET"

# Verify the policy was applied
```

```
aws s3api get-bucket-lifecycle-configuration \
  --bucket $BUCKET \
  --output json | jq '.Rules[] | {ID: .ID, Status: .Status, Transitions: .Transitions}'
```

Python automation script for scheduled tiering — runs nightly via Lambda or cron:

```
#!/usr/bin/env python3
"""
db_tiering_scheduler.py

Runs nightly (scheduled via AWS EventBridge or cron).
Identifies PostgreSQL partitions older than the configured threshold,
exports them to S3 in Parquet format, validates row counts,
and drops the local partition after confirmation.

Requirements:
  pip install psycopg2-binary boto3 pandas pyarrow sqlalchemy
"""

import os
import sys
import json
import logging
import hashlib
from datetime import datetime, timedelta
from typing import Optional

import boto3
import pandas as pd
import psycopg2
from psycopg2.extras import RealDictCursor
import pyarrow as pa
import pyarrow.parquet as pq
from sqlalchemy import create_engine, text

# Configuration
DB_HOST = os.environ.get("DB_HOST", "prod-postgres-01.xxxx.us-east-1.rds.amazonaws.com")
DB_PORT = int(os.environ.get("DB_PORT", 5432))
DB_NAME = os.environ.get("DB_NAME", "production")
DB_USER = os.environ.get("DB_USER", "archive_user")
DB_PASSWORD = os.environ.get("DB_PASSWORD") # From AWS Secrets Manager in prod

ARCHIVE_BUCKET = os.environ.get("ARCHIVE_BUCKET", "your-archive-bucket")
ARCHIVE_PREFIX = os.environ.get("ARCHIVE_PREFIX", "database-archive")
AWS_REGION = os.environ.get("AWS_REGION", "us-east-1")

# Archive partitions older than this many months
COLD_THRESHOLD_MONTHS = int(os.environ.get("COLD_THRESHOLD_MONTHS", 6))

# Tables to include in tiering (parent table name + timestamp column)
TIERING_CONFIG = {
    "audit_events_partitioned": "created_at",
    "transaction_logs_partitioned": "transaction_date",
    "clickstream_partitioned": "event_time",
}

METADATA_TABLE = "archive_metadata"

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(message)s",
    handlers=[logging.StreamHandler(sys.stdout)]
)
log = logging.getLogger(__name__)

# Database helpers

def get_connection():
    return psycopg2.connect(
```

```

    host=DB_HOST,
    port=DB_PORT,
    dbname=DB_NAME,
    user=DB_USER,
    password=DB_PASSWORD,
    connect_timeout=30
)

def get_cold_partitions(conn, parent_table: str, threshold_months: int) -> list[dict]:
    """
    Return partitions whose data ends before the cold threshold date.
    Only returns partitions that have not been archived yet.
    """
    cutoff = datetime.utcnow() - timedelta(days=threshold_months * 30)

    query = """
        SELECT
            child.relname                                AS partition_name,
            child.oid                                    AS partition_oid,
            pg_size_pretty(pg_total_relation_size(child.oid)) AS human_size,
            pg_total_relation_size(child.oid)             AS bytes,
            parent.relname                               AS parent_table
        FROM
            pg_inherits
            JOIN pg_class parent ON pg_inherits.inhparent = parent.oid
            JOIN pg_class child ON pg_inherits.inhrelid = child.oid
        WHERE
            parent.relname = %s
            AND child.relname NOT IN (
                SELECT partition_name
                FROM   archive_metadata
                WHERE  original_table = %s
                    AND verified = true
            )
        ORDER BY
            child.relname ASC;
    """

    with conn.cursor(cursor_factory=RealDictCursor) as cur:
        cur.execute(query, (parent_table, parent_table))
        candidates = cur.fetchall()

    # Filter by partition name convention: tablename_YYYY_QN or _YYYY_MM
    cold = []
    for part in candidates:
        name = part["partition_name"]
        # Parse year from partition name (last 4-digit sequence)
        parts = name.split("_")
        for segment in reversed(parts):
            if segment.isdigit() and len(segment) == 4:
                year = int(segment)
                # Simple heuristic: if partition year is before cutoff year, it's cold
                if year < cutoff.year or (year == cutoff.year and cutoff.month > 6):
                    cold.append(dict(part))
                    break

    return cold

def export_partition_to_s3(
    conn,
    partition_name: str,
    parent_table: str,
    bucket: str,
    prefix: str,
    region: str
) -> tuple[str, int, str]:
    """
    Export partition to S3 as Parquet.

```

```

Returns (s3_path, row_count, checksum).
"""
s3 = boto3.client("s3", region_name=region)

log.info("Reading partition: %s", partition_name)
with conn.cursor(cursor_factory=RealDictCursor) as cur:
    cur.execute(f"SELECT * FROM {partition_name}")
    rows = cur.fetchall()

if not rows:
    log.warning("Partition %s is empty - skipping", partition_name)
    return None, 0, None

row_count = len(rows)
df = pd.DataFrame(rows)

# Convert to Parquet in memory
table = pa.Table.from_pandas(df, preserve_index=False)
parquet_buffer = pa.BufferOutputStream()
pq.write_table(
    table,
    parquet_buffer,
    compression="snappy",
    row_group_size=100_000
)
parquet_bytes = parquet_buffer.getvalue().to_pybytes()

# Compute checksum for verification
checksum = hashlib.sha256(parquet_bytes).hexdigest()

# Build S3 path
timestamp = datetime.utcnow().strftime("%Y%m%dT%H%M%SZ")
s3_key = f"{prefix}/{parent_table}/{partition_name}/{partition_name}_{timestamp}.parquet"
s3_path = f"s3://{bucket}/{s3_key}"

log.info("Uploading %d rows (%d bytes) to %s", row_count, len(parquet_bytes), s3_path)
s3.put_object(
    Bucket=bucket,
    Key=s3_key,
    Body=parquet_bytes,
    ContentType="application/octet-stream",
    Metadata={
        "partition_name": partition_name,
        "parent_table": parent_table,
        "row_count": str(row_count),
        "exported_at": timestamp,
        "sha256": checksum,
    }
)

return s3_path, row_count, checksum

def verify_export(bucket: str, s3_path: str, expected_rows: int, expected_checksum: str) -> bool:
    """
    Re-download the Parquet file from S3 and verify row count and checksum.
    """
    s3 = boto3.client("s3")
    # Parse bucket/key from s3:// URI
    path_without_prefix = s3_path[len("s3://"):]
    bucket_name, _, key = path_without_prefix.partition("/")

    log.info("Verifying export at %s", s3_path)
    response = s3.get_object(Bucket=bucket_name, Key=key)
    data = response["Body"].read()

    actual_checksum = hashlib.sha256(data).hexdigest()
    if actual_checksum != expected_checksum:
        log.error("Checksum mismatch! Expected %s, got %s", expected_checksum, actual_checksum)
        return False

```

```

buffer = pa.BufferedReader(data)
parquet_table = pq.read_table(buffer)
actual_rows = parquet_table.num_rows

if actual_rows != expected_rows:
    log.error("Row count mismatch! Expected %d, got %d", expected_rows, actual_rows)
    return False

log.info("Verification passed: %d rows, checksum OK", actual_rows)
return True

def record_archival(conn, parent_table: str, partition_name: str,
                    s3_path: str, row_count: int, checksum: str, verified: bool):
    """
    Insert a record into the archive_metadata tracking table.
    """
    with conn.cursor() as cur:
        cur.execute("""
            INSERT INTO archive_metadata (
                original_table,
                partition_name,
                archive_path,
                row_count,
                checksum,
                archived_at,
                archived_by,
                verified
            ) VALUES (%s, %s, %s, %s, %s, NOW(), CURRENT_USER, %s)
            ON CONFLICT (partition_name) DO UPDATE
                SET archive_path = EXCLUDED.archive_path,
                    row_count    = EXCLUDED.row_count,
                    checksum      = EXCLUDED.checksum,
                    archived_at   = EXCLUDED.archived_at,
                    verified      = EXCLUDED.verified;
        """, (parent_table, partition_name, s3_path, row_count, checksum, verified))
    conn.commit()

def detach_and_drop_partition(conn, parent_table: str, partition_name: str):
    """
    Detach the partition from the parent table, then drop it.
    Performed in a transaction with a short lock timeout to avoid blocking.
    """
    with conn.cursor() as cur:
        # Set a short lock timeout so we fail fast rather than block production queries
        cur.execute("SET lock_timeout = '5s'")
        log.info("Detaching partition %s from %s", partition_name, parent_table)
        cur.execute(
            f"ALTER TABLE {parent_table} DETACH PARTITION {partition_name}"
        )
        conn.commit()

    # Drop outside the detach transaction so the detach is already committed
    with conn.cursor() as cur:
        log.info("Dropping partition %s", partition_name)
        cur.execute(f"DROP TABLE {partition_name}")
        conn.commit()

# Main pipeline

def run_tiering_pipeline():
    log.info("Starting tiering pipeline - threshold: %d months", COLD_THRESHOLD_MONTHS)
    conn = get_connection()

    total_bytes_freed = 0
    partitions_archived = 0
    errors = []

```

```
for parent_table, ts_column in TIERING_CONFIG.items():
    log.info("Processing table: %s", parent_table)

    cold_partitions = get_cold_partitions(conn, parent_table, COLD_THRESHOLD_MONTHS)
    log.info("Found %d cold partitions for %s", len(cold_partitions), parent_table)

    for part in cold_partitions:
        partition_name = part["partition_name"]
        bytes_size     = part["bytes"]
        human_size     = part["human_size"]

        log.info("Archiving %s (%s)", partition_name, human_size)

        try:
            # Export
            s3_path, row_count, checksum = export_partition_to_s3(
                conn, partition_name, parent_table,
                ARCHIVE_BUCKET, ARCHIVE_PREFIX, AWS_REGION
            )
            if s3_path is None:
                log.warning("Skipping empty partition: %s", partition_name)
                continue

            # Record as unverified
            record_archival(conn, parent_table, partition_name,
                           s3_path, row_count, checksum, verified=False)

            # Verify
            verified = verify_export(ARCHIVE_BUCKET, s3_path, row_count, checksum)

            if not verified:
                log.error("Verification failed for %s - partition NOT dropped", partition_name)
                errors.append(f"Verification failed: {partition_name}")
                continue

            # Mark verified in metadata
            record_archival(conn, parent_table, partition_name,
                           s3_path, row_count, checksum, verified=True)

            # Drop from RDS
            detach_and_drop_partition(conn, parent_table, partition_name)

            total_bytes_freed += bytes_size
            partitions_archived += 1
            log.info("Successfully archived and dropped %s", partition_name)

        except Exception as exc:
            log.exception("Error processing partition %s: %s", partition_name, exc)
            errors.append(f"{partition_name}: {exc}")
            conn.rollback()

    conn.close()

gb_freed = total_bytes_freed / (1024 ** 3)
log.info("Pipeline complete. Partitions archived: %d | Space freed: %.2f GB | Errors: %d",
        partitions_archived, gb_freed, len(errors))

if errors:
    log.error("Errors encountered:\n%s", "\n".join(errors))
    sys.exit(1)

if __name__ == "__main__":
    run_tiering_pipeline()
```


17.5 Querying Archived Data with Athena and Synapse

Archiving data to object storage only works operationally if your team can still query it when needed. The two most common access patterns for archived database data are ad-hoc queries (compliance audits, retroactive investigations) and batch analytics. Both are well served by serverless query engines that scan object storage directly.

Setting up AWS Glue Data Catalog for archived PostgreSQL exports:

```
#!/usr/bin/env python3
"""
register_archive_catalog.py

Registers archived Parquet partitions in the AWS Glue Data Catalog
so they are immediately queryable via Athena.
"""

import boto3
import json
from datetime import datetime

GLUE_DATABASE      = "db_cold_archive"
ARCHIVE_BUCKET     = "your-archive-bucket"
ARCHIVE_PREFIX     = "database-archive"
AWS_REGION         = "us-east-1"

glue  = boto3.client("glue",    region_name=AWS_REGION)
athena = boto3.client("athena", region_name=AWS_REGION)

def ensure_glue_database():
    try:
        glue.get_database(Name=GLUE_DATABASE)
    except glue.exceptions.EntityNotFoundException:
        glue.create_database(DatabaseInput={"Name": GLUE_DATABASE,
                                                "Description": "Cold archive of RDS database exports"})
    print(f"Created Glue database: {GLUE_DATABASE}")

def register_table(table_name: str, s3_prefix: str, columns: list[dict]):
    """
    Create or update a Glue table pointing at the Parquet files in S3.
    columns = [{"Name": "id", "Type": "bigint"}, ...]
    """
    location = f"s3://{ARCHIVE_BUCKET}/{s3_prefix}/"

    table_input = {
        "Name": table_name,
        "Description": f"Archived from RDS - registered {datetime.utcnow().isoformat()}",
        "StorageDescriptor": {
            "Columns": columns,
            "Location": location,
            "InputFormat": "org.apache.hadoop.mapred.TextInputFormat",
            "OutputFormat": "org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat",
            "SerdeInfo": {
                "SerializationLibrary": "org.apache.hadoop.hive.ql.io.parquet.serde.ParquetHiveSerDe",
                "Parameters": {"serialization.format": "1"}
            },
        },
        "Compressed": True,
    },
    "Parameters": {
        "classification": "parquet",
        "compressionType": "snappy",
        "typeOfData": "file",
    },
    "TableType": "EXTERNAL_TABLE",
}
```

```

try:
    glue.get_table(DatabaseName=GLUE_DATABASE, Name=table_name)
    glue.update_table(DatabaseName=GLUE_DATABASE, TableInput=table_input)
    print(f"Updated Glue table: {GLUE_DATABASE}.{table_name}")
except glue.exceptions.EntityNotFoundException:
    glue.create_table(DatabaseName=GLUE_DATABASE, TableInput=table_input)
    print(f"Created Glue table: {GLUE_DATABASE}.{table_name}")

def run_athena_validation(query: str, output_location: str) -> str:
    """
    Submit an Athena query and return the QueryExecutionId.
    """
    response = athena.start_query_execution(
        QueryString=query,
        QueryExecutionContext={"Database": GLUE_DATABASE},
        ResultConfiguration={"OutputLocation": output_location},
        WorkGroup="primary"
    )
    return response["QueryExecutionId"]

if __name__ == "__main__":
    ensure_glue_database()

    # Register audit_events archive
    register_table(
        table_name="audit_events_archive",
        s3_prefix=f"{ARCHIVE_PREFIX}/audit_events_partitioned",
        columns=[
            {"Name": "id", "Type": "bigint"},
            {"Name": "user_id", "Type": "bigint"},
            {"Name": "event_type", "Type": "string"},
            {"Name": "event_data", "Type": "string"}, # JSONB serialized as string
            {"Name": "ip_address", "Type": "string"},
            {"Name": "created_at", "Type": "timestamp"},
        ]
    )

    # Validate with a quick Athena count
    qid = run_athena_validation(
        query="SELECT COUNT(*) AS total_archived_rows FROM audit_events_archive",
        output_location=f"s3://{ARCHIVE_BUCKET}/athena-results/"
    )
    print(f"Validation query submitted - Athena QueryExecutionId: {qid}")
    print("Check results in Athena console or poll with athena.get_query_results()")

```

Querying the archive from Athena — sample operational queries:

```

-- Count archived audit events by month
-- Athena SQL (Presto/Trino dialect)
SELECT
    DATE_TRUNC('month', created_at) AS month,
    event_type,
    COUNT(*) AS event_count
FROM
    db_cold_archive.audit_events_archive
WHERE
    created_at BETWEEN TIMESTAMP '2022-01-01' AND TIMESTAMP '2022-12-31'
GROUP BY
    1, 2
ORDER BY
    1, 3 DESC;

-- Find all events for a specific user across the full archive
SELECT
    created_at,
    event_type,
    ip_address,
    JSON_EXTRACT_SCALAR(event_data, '$.action') AS action

```

```

FROM
    db_cold_archive.audit_events_archive
WHERE
    user_id = 98765
    AND created_at >= TIMESTAMP '2021-01-01'
ORDER BY
    created_at DESC
LIMIT 1000;

```

Azure equivalent: querying archived SQL Server data via Synapse Analytics:

```

-- Azure Synapse Serverless SQL Pool
-- Query archived SQL Server data exported to Azure Data Lake Storage Gen2

-- Create an external data source pointing at the archive container
CREATE EXTERNAL DATA SOURCE db_cold_archive
WITH (
    LOCATION = 'https://yourstorageaccount.dfs.core.windows.net/database-archive',
    CREDENTIAL = [archive_storage_credential] -- managed identity or SAS
);

-- Create an external file format for Parquet
CREATE EXTERNAL FILE FORMAT parquet_snappy
WITH (
    FORMAT_TYPE = PARQUET,
    DATA_COMPRESSION = 'org.apache.hadoop.io.compress.SnappyCodec'
);

-- Query archived audit events directly from ADLS Gen2
SELECT
    YEAR(created_at) AS archive_year,
    event_type,
    COUNT(*) AS event_count,
    COUNT(DISTINCT user_id) AS unique_users
FROM
    OPENROWSET(
        BULK 'audit_events_partitioned/**',
        DATA_SOURCE = 'db_cold_archive',
        FORMAT = 'PARQUET'
    ) WITH (
        id BIGINT,
        user_id BIGINT,
        event_type NVARCHAR(100),
        event_data NVARCHAR(MAX),
        ip_address NVARCHAR(45),
        created_at DATETIME2
    ) AS archive_data
WHERE
    created_at < DATEADD(YEAR, -1, GETDATE())
GROUP BY
    YEAR(created_at), event_type
ORDER BY
    archive_year DESC, event_count DESC;

```

Google Cloud equivalent: BigQuery external tables over GCS:

```

-- Create a BigQuery external table pointing at GCS Parquet exports
-- Run in BigQuery console or via bq CLI

CREATE OR REPLACE EXTERNAL TABLE `your_project.db_cold_archive.audit_events_archive`
OPTIONS (
    format = 'PARQUET',
    uris =
        ['gs://your-archive-bucket/database-archive/audit_events_partitioned/**/*.parquet'],
    hive_partition_uri_prefix = 'gs://your-archive-bucket/database-archive/audit_events_partitioned',
    require_hive_partition_filter = false
);

-- Query across the full archive
SELECT

```

```

    FORMAT_TIMESTAMP('%Y-%m', created_at) AS month,
    event_type,
    COUNT(*) AS events,
    COUNT(DISTINCT user_id) AS unique_users
FROM
    `your_project.db_cold_archive.audit_events_archive`
WHERE
    created_at < TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 365 DAY)
GROUP BY
    1, 2
ORDER BY
    1 DESC, 3 DESC;

```

The key design principle across all three clouds is the same: archived Parquet files in object storage become queryable within minutes of registration, cost nothing to query at rest, and scale to petabyte-range archives without any infrastructure provisioning. Athena, Synapse Serverless, and BigQuery all charge only for bytes scanned—and Parquet with Snappy compression plus partition pruning reduces scan costs dramatically compared to scanning raw CSV.

17.6 Azure SQL and Google Cloud SQL Tiering Patterns

AWS RDS has the most mature native tooling for partition-level export via `aws_s3`, but Azure SQL and Google Cloud SQL require slightly different approaches. Both can achieve equivalent outcomes with extra plumbing.

Azure SQL: exporting cold data with BACPAC and Blob Storage

For Azure SQL databases without partitioning, the most practical cold-data workflow combines Azure Data Factory (ADF) for orchestrated export with Azure Blob lifecycle management for automatic tier transitions.

```

#!/usr/bin/env python3
"""
azure_cold_export.py

Uses the Azure SQL Bulk Copy API (BCP) wrapper via pyodbc and Azure Blob SDK
to export cold table partitions to Azure Data Lake Storage Gen2 as Parquet.

Equivalent to the AWS aws_s3 export for Azure SQL environments.
"""

import os
import io
import logging
import pyodbc
import pandas as pd
import pyarrow as pa
import pyarrow.parquet as pq
from azure.identity import DefaultAzureCredential
from azure.storage.blob import BlobServiceClient, ContentSettings
from datetime import datetime

log = logging.getLogger(__name__)
logging.basicConfig(level=logging.INFO)

AZURE_SQL_CONN_STR = os.environ.get("AZURE_SQL_CONN_STR")
# e.g. "Driver={ODBC Driver 18 for SQL Server};Server=tcp:yourserver.database.windows.net,1433;
#       Database=production;Authentication=ActiveDirectoryMsi;"

STORAGE_ACCOUNT_URL = os.environ.get("STORAGE_ACCOUNT_URL",
                                     "https://yourstorageaccount.dfs.core.windows.net")

```

```

CONTAINER_NAME      = os.environ.get("CONTAINER_NAME", "database-archive")

def export_sql_server_table_to_adls(
    table_name: str,
    date_column: str,
    cutoff_year: int,
    cutoff_quarter: int
):
    """
    Export one quarter of data from an Azure SQL table to ADLS Gen2 Parquet.
    """
    credential = DefaultAzureCredential()
    blob_service = BlobServiceClient(STORAGE_ACCOUNT_URL, credential=credential)
    container = blob_service.get_container_client(CONTAINER_NAME)

    conn = pyodbc.connect(AZURE_SQL_CONN_STR)
    cursor = conn.cursor()

    # Build date bounds for the quarter
    quarter_start_month = (cutoff_quarter - 1) * 3 + 1
    quarter_end_month = quarter_start_month + 3
    if quarter_end_month > 12:
        end_year = cutoff_year + 1
        end_month = 1
    else:
        end_year = cutoff_year
        end_month = quarter_end_month

    start_date = f"{cutoff_year}-{quarter_start_month:02d}-01"
    end_date = f"{end_year}-{end_month:02d}-01"

    query = f"""
    SELECT *
    FROM {table_name} WITH (NOLOCK)
    WHERE {date_column} >= '{start_date}'
    AND {date_column} < '{end_date}'
    """

    log.info("Fetching data: %s Q%d from %s", cutoff_year, cutoff_quarter, table_name)
    df = pd.read_sql(query, conn)
    conn.close()

    if df.empty:
        log.warning("No rows found for %s %d Q%d", table_name, cutoff_year, cutoff_quarter)
        return None, 0

    row_count = len(df)
    log.info("Fetched %d rows", row_count)

    # Write Parquet to buffer
    arrow_table = pa.Table.from_pandas(df, preserve_index=False)
    buffer = io.BytesIO()
    pq.write_table(arrow_table, buffer, compression="snappy")
    buffer.seek(0)

    blob_path = (f"audit_events/{cutoff_year}/Q{cutoff_quarter}/"
                 f"{table_name}-{cutoff_year}_Q{cutoff_quarter}.parquet")

    log.info("Uploading to %s/%s", CONTAINER_NAME, blob_path)
    blob_client = container.get_blob_client(blob_path)
    blob_client.upload_blob(
        buffer,
        overwrite=True,
        content_settings=ContentSettings(content_type="application/octet-stream"),
        metadata={
            "table_name": table_name,
            "year": str(cutoff_year),
            "quarter": str(cutoff_quarter),
            "row_count": str(row_count),
        }
    )

```

```

        "exported_at": datetime.utcnow().isoformat(),
    }
)

s3_equivalent_path = f"abfss://{CONTAINER_NAME}@{STORAGE_ACCOUNT_URL.split('/')[1]}/{blob_path}"
log.info("Export complete: %s", s3_equivalent_path)
return s3_equivalent_path, row_count

if __name__ == "__main__":
    export_sql_server_table_to_adls(
        table_name = "audit_events",
        date_column = "created_at",
        cutoff_year = 2022,
        cutoff_quarter= 1
    )

```

Azure Blob lifecycle policy (ARM template fragment):

```

{
  "type": "Microsoft.Storage/storageAccounts/managementPolicies",
  "apiVersion": "2023-01-01",
  "name": "[concat(parameters('storageAccountName'), '/default')]",
  "properties": {
    "policy": {
      "rules": [
        {
          "name": "DatabaseArchiveTiering",
          "enabled": true,
          "type": "Lifecycle",
          "definition": {
            "filters": {
              "blobTypes": ["blockBlob"],
              "prefixMatch": ["database-archive/"]
            },
            "actions": {
              "baseBlob": {
                "tierToCool": {
                  "daysAfterModificationGreaterThan": 30
                },
                "tierToArchive": {
                  "daysAfterModificationGreaterThan": 180
                },
                "delete": {
                  "daysAfterModificationGreaterThan": 2555
                }
              },
              "snapshot": {
                "delete": {
                  "daysAfterCreationGreaterThan": 90
                }
              }
            }
          }
        }
      ]
    }
  }
}

```

Google Cloud SQL: exporting cold data to GCS with Cloud Scheduler

Google Cloud SQL supports native export to GCS via the `gcloud sql export csv` command and the Cloud SQL Admin API. The cleanest automation uses Cloud Scheduler → Pub/Sub → Cloud Functions.

```

#!/bin/bash
# Export a cold table partition from Cloud SQL PostgreSQL to GCS
# Called from a Cloud Function triggered by Cloud Scheduler

```

```

PROJECT_ID="your-gcp-project"
INSTANCE_NAME="prod-postgres-01"
DATABASE_NAME="production"
BUCKET="gs://your-archive-bucket"
YEAR=2022
QUARTER=1

START_DATE="${YEAR}-${(printf '%02d' $(( (QUARTER-1)*3+1 )))}-01"
END_MONTH=$(( QUARTER*3+1 ))
END_YEAR=$YEAR
if [ $END_MONTH -gt 12 ]; then
    END_MONTH=1
    END_YEAR=$(( YEAR+1 ))
fi
END_DATE="${END_YEAR}-${(printf '%02d' $END_MONTH)}-01"

EXPORT_URI="${BUCKET}/database-archive/audit_events/${YEAR}/Q${QUARTER}/audit_events_${YEAR}_Q
↪ ${QUARTER}.csv.gz"

echo "Exporting audit_events ${YEAR} Q${QUARTER} to ${EXPORT_URI}"

gcloud sql export csv "$INSTANCE_NAME" "$EXPORT_URI" \
    --project="$PROJECT_ID" \
    --database="$DATABASE_NAME" \
    --query="SELECT * FROM audit_events WHERE created_at >= '${START_DATE}' AND created_at <
↪ '${END_DATE}'" \
    --compression \
    --async

echo "Export job submitted. Monitor with: gcloud sql operations list --instance=$INSTANCE_NAME"

```

After export, a GCS Object Lifecycle rule handles the ongoing tiering:

```

#!/usr/bin/env python3
"""
Apply GCS lifecycle rules to the archive bucket.
Transitions: Standard → Nearline (30d) → Coldline (90d) → Archive (365d)
"""

from google.cloud import storage

def apply_lifecycle_rules(bucket_name: str):
    client = storage.Client()
    bucket = client.get_bucket(bucket_name)

    rules = [
        storage.lifecycle.TransitionRule(
            storage.lifecycle.TransitionRule.NEARLINE,
            age=30,
            matches_prefix=["database-archive/"]
        ),
        storage.lifecycle.TransitionRule(
            storage.lifecycle.TransitionRule.COLDLINE,
            age=90,
            matches_prefix=["database-archive/"]
        ),
        storage.lifecycle.TransitionRule(
            storage.lifecycle.TransitionRule.ARCHIVE,
            age=365,
            matches_prefix=["database-archive/"]
        ),
        storage.lifecycle.DeleteRule(
            age=2555,
            matches_prefix=["database-archive/"]
        ),
    ]

    bucket.lifecycle_rules = rules

```

```

bucket.patch()
print(f"Lifecycle rules applied to gs://{bucket_name}")

if __name__ == "__main__":
    apply_lifecycle_rules("your-archive-bucket")

```

17.7 Building a Metadata Catalog for Archived Data

Every byte moved to object storage must be tracked. Without a metadata catalog, archived data becomes inaccessible not because it cannot be queried, but because no one can find it. The metadata table introduced in Section 17.3 needs to evolve into a proper catalog that your application, compliance team, and operations staff can query.

Creating the archive metadata catalog in PostgreSQL:

```

-- Full archive metadata catalog schema
-- Deploy on a management/control-plane database, not on the archived database itself

CREATE TABLE archive_metadata (
    id BIGSERIAL PRIMARY KEY,
    original_table VARCHAR(255) NOT NULL,
    partition_name VARCHAR(255) NOT NULL UNIQUE,
    archive_path TEXT NOT NULL,
    storage_class VARCHAR(50) NOT NULL DEFAULT 'STANDARD',
    row_count BIGINT NOT NULL,
    uncompressed_bytes BIGINT,
    compressed_bytes BIGINT,
    checksum VARCHAR(64), -- SHA-256 hex
    date_range_start TIMESTAMPTZ,
    date_range_end TIMESTAMPTZ,
    schema_version INTEGER NOT NULL DEFAULT 1,
    archived_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    archived_by VARCHAR(100),
    verified BOOLEAN NOT NULL DEFAULT false,
    verified_at TIMESTAMPTZ,
    retrieval_count INTEGER NOT NULL DEFAULT 0,
    last_retrieved_at TIMESTAMPTZ,
    retention_policy VARCHAR(50) NOT NULL DEFAULT '7_years',
    delete_after TIMESTAMPTZ,
    notes TEXT,
    tags JSONB NOT NULL DEFAULT '{}')
);

-- Indexes for common lookup patterns
CREATE INDEX ON archive_metadata (original_table);
CREATE INDEX ON archive_metadata (date_range_start, date_range_end);
CREATE INDEX ON archive_metadata (archived_at);
CREATE INDEX ON archive_metadata (verified);
CREATE INDEX ON archive_metadata USING GIN (tags);

-- View: cost estimation based on storage class
CREATE VIEW archive_cost_estimate AS
SELECT
    original_table,
    storage_class,
    COUNT(*) AS partition_count,
    SUM(row_count) AS total_rows,
    pg_size_pretty(SUM(compressed_bytes)) AS total_compressed,
    SUM(compressed_bytes) / 1024.0^3 AS total_gb,
    CASE storage_class
        WHEN 'STANDARD' THEN SUM(compressed_bytes) / 1024.0^3 * 0.023
        WHEN 'STANDARD_IA' THEN SUM(compressed_bytes) / 1024.0^3 * 0.0125
        WHEN 'GLACIER_IR' THEN SUM(compressed_bytes) / 1024.0^3 * 0.004
        WHEN 'DEEP_ARCHIVE' THEN SUM(compressed_bytes) / 1024.0^3 * 0.00099
    
```



```

        ELSE NULL
    END
    AS estimated_monthly_cost_usd
FROM
    archive_metadata
WHERE
    verified = true
GROUP BY
    original_table, storage_class
ORDER BY
    total_gb DESC;

COMMENT ON VIEW archive_cost_estimate IS
    'Estimated monthly S3 storage cost for verified archives. Update rates as pricing changes.';

```

Equivalent SQL Server / Azure SQL catalog schema:

```

-- Archive metadata catalog for SQL Server / Azure SQL environments

CREATE TABLE archive_metadata (
    id                BIGINT IDENTITY(1,1) PRIMARY KEY,
    original_table    NVARCHAR(255) NOT NULL,
    partition_name    NVARCHAR(255) NOT NULL,
    archive_path      NVARCHAR(MAX) NOT NULL,
    storage_class     NVARCHAR(50)  NOT NULL DEFAULT 'Hot',
    row_count         BIGINT        NOT NULL,
    uncompressed_bytes BIGINT        NULL,
    compressed_bytes  BIGINT        NULL,
    checksum          NVARCHAR(64)  NULL,
    date_range_start  DATETIME2     NULL,
    date_range_end    DATETIME2     NULL,
    schema_version    INT           NOT NULL DEFAULT 1,
    archived_at       DATETIME2     NOT NULL DEFAULT SYSUTCDATETIME(),
    archived_by       NVARCHAR(100) NULL,
    verified          BIT           NOT NULL DEFAULT 0,
    verified_at       DATETIME2     NULL,
    retrieval_count   INT           NOT NULL DEFAULT 0,
    last_retrieved_at DATETIME2     NULL,
    retention_policy  NVARCHAR(50)  NOT NULL DEFAULT '7_years',
    delete_after     DATETIME2     NULL,
    notes             NVARCHAR(MAX) NULL,
    tags             NVARCHAR(MAX) NULL -- JSON string
);

CREATE UNIQUE INDEX UX_archive_metadata_partition
    ON archive_metadata (partition_name);

CREATE INDEX IX_archive_metadata_table
    ON archive_metadata (original_table);

CREATE INDEX IX_archive_metadata_dates
    ON archive_metadata (date_range_start, date_range_end);

-- Cost estimation view for Azure Blob tiers
CREATE VIEW archive_cost_estimate AS
SELECT
    original_table,
    storage_class,
    COUNT(*)                AS partition_count,
    SUM(row_count)          AS total_rows,
    CAST(SUM(compressed_bytes) / POWER(1024.0, 3) AS DECIMAL(18,2)) AS total_gb,
    CASE storage_class
        WHEN 'Hot' THEN SUM(compressed_bytes) / POWER(1024.0, 3) * 0.018
        WHEN 'Cool' THEN SUM(compressed_bytes) / POWER(1024.0, 3) * 0.01
        WHEN 'Archive' THEN SUM(compressed_bytes) / POWER(1024.0, 3) * 0.00099
        ELSE NULL
    END
    AS estimated_monthly_cost_usd
FROM
    archive_metadata
WHERE
    verified = 1

```

```
GROUP BY
    original_table, storage_class;
GO
```

Tracking retrieval events — closing the feedback loop:

When someone queries the archive through Athena, Synapse, or BigQuery, that access should be logged back to the metadata catalog. This matters for two reasons: it proves the archive is reachable when compliance auditors ask, and it identifies data that is accessed frequently enough that re-hydration to a warmer storage tier would save money on per-retrieval fees.

```
-- PostgreSQL: stored procedure to log a retrieval event
CREATE OR REPLACE PROCEDURE log_archive_retrieval(
    p_partition_name VARCHAR(255),
    p_query_engine   VARCHAR(50),    -- 'athena', 'bigquery', 'synapse', 'direct'
    p_rows_scanned   BIGINT,
    p_bytes_scanned  BIGINT,
    p_cost_usd       NUMERIC(10,6)
)
LANGUAGE plpgsql
AS $$
BEGIN
    UPDATE archive_metadata
    SET
        retrieval_count = retrieval_count + 1,
        last_retrieved_at = NOW()
    WHERE
        partition_name = p_partition_name;

    INSERT INTO archive_retrieval_log (
        partition_name,
        query_engine,
        rows_scanned,
        bytes_scanned,
        cost_usd,
        retrieved_at,
        retrieved_by
    ) VALUES (
        p_partition_name,
        p_query_engine,
        p_rows_scanned,
        p_bytes_scanned,
        p_cost_usd,
        NOW(),
        CURRENT_USER
    );
END;
$$;

-- SQL Server equivalent
CREATE OR ALTER PROCEDURE dbo.log_archive_retrieval
    @partition_name NVARCHAR(255),
    @query_engine   NVARCHAR(50),
    @rows_scanned   BIGINT,
    @bytes_scanned  BIGINT,
    @cost_usd       DECIMAL(10,6)
AS
BEGIN
    SET NOCOUNT ON;

    UPDATE archive_metadata
    SET
        retrieval_count = retrieval_count + 1,
        last_retrieved_at = SYSUTCDATETIME()
    WHERE
        partition_name = @partition_name;

    INSERT INTO archive_retrieval_log (
```

```
    partition_name,  
    query_engine,  
    rows_scanned,  
    bytes_scanned,  
    cost_usd,  
    retrieved_at,  
    retrieved_by  
  ) VALUES (  
    @partition_name,  
    @query_engine,  
    @rows_scanned,  
    @bytes_scanned,  
    @cost_usd,  
    SYSUTCDATETIME(),  
    SYSTEM_USER  
  );  
END;  
GO
```

17.8 Handling Schema Evolution in Long-Term Archives

A problem that almost every team hits six to twelve months after deploying an archival pipeline is schema drift. The live table gains a new column, the `schema_version` in the archive catalog should increment, and the Parquet files already written do not have that column. Without planning for this, you end up with heterogeneous archives that break Athena and BigQuery external tables.

The safest approach is a three-part policy:

1. **Always write schema version metadata** into both the Parquet file metadata and the archive catalog row. The Python export script in Section 17.4 stores `schema_version` in the `archive_metadata` table; make that version number come from a controlled schema registry, not from the live table at export time.
2. **Use schema-on-read with explicit null handling** when querying archives. Athena, BigQuery, and Synapse can all handle missing columns in older Parquet files if you declare the column as nullable and the query does not filter on it.
3. **Run a backfill job when additive schema changes occur** — not to rewrite old Parquet files, but to emit a companion schema manifest file in the same S3 prefix that describes the columns present in each file generation.

Schema version manifest file — written alongside each Parquet export:

```
def write_schema_manifest(  
    s3_client,  
    bucket: str,  
    parquet_key: str,  
    df: "pd.DataFrame",  
    schema_version: int  
):  
    """  
    Write a companion schema manifest JSON next to the Parquet file.  
    The manifest records column names, types, and the schema version  
    so future readers can handle missing columns gracefully.  
    """  
    import json  
  
    manifest = {  
        "schema_version": schema_version,  
        "parquet_key":    parquet_key,
```

```
"exported_at":    datetime.utcnow().isoformat(),
"columns": [
    {"name": col, "dtype": str(df[col].dtype)}
    for col in df.columns
],
"row_count":      len(df),
}

manifest_key = parquet_key.replace(".parquet", ".schema.json")
s3_client.put_object(
    Bucket=bucket,
    Key=manifest_key,
    Body=json.dumps(manifest, indent=2).encode(),
    ContentType="application/json"
)
return manifest_key
```

Athena query that handles nullable columns from mixed schema versions:

```
-- Query that works across archive files written before and after
-- a new column (session_id) was added to audit_events

SELECT
    created_at,
    event_type,
    user_id,
    ip_address,
    -- Older files will return NULL for session_id; COALESCE gives a safe default
    COALESCE(session_id, 'unknown') AS session_id,
    JSON_EXTRACT_SCALAR(event_data, '$.action') AS action
FROM
    db_cold_archive.audit_events_archive
WHERE
    created_at BETWEEN TIMESTAMP '2021-01-01' AND TIMESTAMP '2023-12-31'
    AND user_id = 12345
ORDER BY
    created_at DESC;
```

17.9 Retrieval and Re-Hydration Workflows

Most archived data is queried in place through serverless engines and never needs to come back to the live database. But occasionally—regulatory reviews, data corrections, application features that need to resurface old records—you need to re-hydrate data into the live PostgreSQL or SQL Server instance. Doing this well means reading from S3/GCS/ADLS and writing back in bulk without overwhelming the primary instance.

Re-hydrating a Parquet archive back into PostgreSQL via COPY:

```
#!/usr/bin/env python3
"""
rehydrate_partition.py

Downloads a Parquet file from S3 and bulk-inserts it back into
the live PostgreSQL database using COPY for maximum throughput.
Used when a partition needs to be temporarily re-surfaced for
a compliance review or data correction project.
"""

import io
import os
import logging
import boto3
import pyarrow.parquet as pq
```

```
import psycopg2
from psycopg2 import sql

log = logging.getLogger(__name__)
logging.basicConfig(level=logging.INFO)

DB_HOST      = os.environ.get("DB_HOST")
DB_PORT      = int(os.environ.get("DB_PORT", 5432))
DB_NAME      = os.environ.get("DB_NAME")
DB_USER      = os.environ.get("DB_USER")
DB_PASSWORD  = os.environ.get("DB_PASSWORD")

S3_REGION    = os.environ.get("AWS_REGION", "us-east-1")

def rehydrate_from_s3(
    s3_path: str,
    target_table: str,
    batch_size: int = 50_000
):
    """
    Download Parquet from S3 and COPY into target_table.
    target_table should be a staging table, not the live partitioned parent,
    to allow review before swapping in.
    """
    s3 = boto3.client("s3", region_name=S3_REGION)

    # Parse S3 path
    path = s3_path[len("s3://"): ]
    bucket, _, key = path.partition("/")

    log.info("Downloading %s", s3_path)
    obj = s3.get_object(Bucket=bucket, Key=key)
    data = obj["Body"].read()
    buffer = io.BytesIO(data)

    table = pq.read_table(buffer)
    df = table.to_pandas()
    total = len(df)
    log.info("Read %d rows from archive", total)

    conn = psycopg2.connect(
        host=DB_HOST, port=DB_PORT, dbname=DB_NAME,
        user=DB_USER, password=DB_PASSWORD
    )
    conn.autocommit = False

    try:
        with conn.cursor() as cur:
            # Use a staging table to allow inspection before promotion
            staging = f"{target_table}_rehydration_staging"
            cur.execute(
                sql.SQL("CREATE TABLE IF NOT EXISTS {} (LIKE {} INCLUDING ALL)")
                .format(sql.Identifier(staging), sql.Identifier(target_table))
            )
            cur.execute(sql.SQL("TRUNCATE {}").format(sql.Identifier(staging)))
            conn.commit()

            # Stream data in batches via COPY
            inserted = 0
            for start in range(0, total, batch_size):
                batch = df.iloc[start : start + batch_size]
                buf = io.StringIO()
                batch.to_csv(buf, index=False, header=False)
                buf.seek(0)

                cur.copy_expert(
                    sql.SQL("COPY {} FROM STDIN WITH (FORMAT CSV)")
                    .format(sql.Identifier(staging)),
                    buf
                )
```

```
    )
    conn.commit()
    inserted += len(batch)
    log.info("Inserted %d / %d rows", inserted, total)

    log.info("Re-hydration complete into staging table: %s", staging)
    log.info("Review data, then run: ALTER TABLE %s RENAME TO %s_old; "
            "ALTER TABLE %s RENAME TO %s;",
            target_table, target_table, staging, target_table)

except Exception:
    conn.rollback()
    log.exception("Re-hydration failed - rolled back")
    raise
finally:
    conn.close()

if __name__ == "__main__":
    rehydrate_from_s3(
        s3_path = "s3://your-archive-bucket/database-archive/audit_events_partitioned/"
                  "audit_events_2022_q1/audit_events_2022_q1_20240101T000000Z.parquet",
        target_table = "audit_events_rehydrated",
        batch_size = 50_000
    )
```

For SQL Server re-hydration, the equivalent pattern uses `BULK INSERT` or the `bcp` utility pointed at a blob URL, which Azure SQL supports natively via the `BULK INSERT ... FROM 'https://...'` syntax available in Azure SQL Managed Instance and Azure Synapse dedicated pools.

```
-- Azure SQL Managed Instance: bulk insert from ADLS Gen2
BULK INSERT dbo.audit_events_rehydration_staging
FROM
    ↪ 'https://yourstorageaccount.blob.core.windows.net/database-archive/audit_events/2022/Q1/audit_events_2022_Q1.parquet'
WITH (
    DATA_SOURCE      = 'db_cold_archive',
    FORMAT             = 'PARQUET',
    TABLOCK
);

-- Verify row count after import
SELECT COUNT(*) FROM dbo.audit_events_rehydration_staging;
```

17.10 Calculating and Reporting Tiering ROI

FinOps is only as strong as its measurement layer. Every archival pipeline needs a reporting view that connects bytes moved to dollars saved, so that leadership can see the value and engineers can tune the thresholds.

PostgreSQL view for tiering ROI report:

```
-- ROI reporting view
-- Compares estimated cost of data on RDS gp3 vs actual cost in archive

CREATE VIEW tiering_roi_report AS
WITH archive_costs AS (
    SELECT
        original_table,
        COUNT(*)                AS partitions_archived,
        SUM(row_count)           AS total_rows_archived,
        SUM(compressed_bytes) / POWER(1024.0, 3) AS archive_gb,
        -- Cost in archive (S3 Glacier Instant as a representative tier)
        SUM(compressed_bytes) / POWER(1024.0, 3) * 0.004 AS archive_monthly_cost_usd,
```

```

-- What it would cost on gp3
-- Assume 4:1 compression: compressed Parquet vs raw PostgreSQL pages
SUM(compressed_bytes) * 4.0 / POWER(1024.0, 3) * 0.092

MIN(date_range_start)      AS rds_gp3_equivalent_cost_usd,
MAX(date_range_end)        AS oldest_archived_date,
MIN(archived_at)           AS newest_archived_date,
                             AS first_archived_at
FROM
  archive_metadata
WHERE
  verified = true
GROUP BY
  original_table
)
SELECT
  original_table,
  partitions_archived,
  total_rows_archived,
  ROUND(archive_gb::NUMERIC, 2)      AS archive_gb,
  ROUND(archive_monthly_cost_usd::NUMERIC, 2) AS archive_cost_per_month,
  ROUND(rds_gp3_equivalent_cost_usd::NUMERIC, 2) AS avoided_rds_cost_per_month,
  ROUND((rds_gp3_equivalent_cost_usd
    - archive_monthly_cost_usd)::NUMERIC, 2) AS monthly_savings_usd,
  ROUND((rds_gp3_equivalent_cost_usd
    - archive_monthly_cost_usd) * 12.0, 2) AS annualized_savings_usd,
  ROUND(100.0 * (rds_gp3_equivalent_cost_usd
    - archive_monthly_cost_usd)
    / NULLIF(rds_gp3_equivalent_cost_usd, 0), 1) AS savings_pct,
  oldest_archived_date,
  newest_archived_date,
  first_archived_at
FROM
  archive_costs
ORDER BY
  monthly_savings_usd DESC;

```

SQL Server equivalent:

```

CREATE VIEW dbo.tiering_roi_report AS
WITH archive_costs AS (
  SELECT
    original_table,
    COUNT(*)      AS partitions_archived,
    SUM(row_count) AS total_rows_archived,
    SUM(compressed_bytes) / POWER(1024.0, 3) AS archive_gb,
    SUM(compressed_bytes) / POWER(1024.0, 3) * 0.004 AS archive_monthly_cost_usd,
    SUM(compressed_bytes) * 4.0 / POWER(1024.0, 3) * 0.092 AS rds_gp3_equivalent_cost_usd,
    MIN(date_range_start) AS oldest_archived_date,
    MAX(date_range_end)   AS newest_archived_date,
    MIN(archived_at)      AS first_archived_at
  FROM
    archive_metadata
  WHERE
    verified = 1
  GROUP BY
    original_table
)
SELECT
  original_table,
  partitions_archived,
  total_rows_archived,
  ROUND(CAST(archive_gb AS DECIMAL(18,2)), 2) AS archive_gb,
  ROUND(CAST(archive_monthly_cost_usd AS DECIMAL(18,2)), 2) AS archive_cost_per_month,
  ROUND(CAST(rds_gp3_equivalent_cost_usd AS DECIMAL(18,2)), 2) AS avoided_rds_cost_per_month,
  ROUND(CAST(rds_gp3_equivalent_cost_usd
    - archive_monthly_cost_usd AS DECIMAL(18,2)), 2) AS monthly_savings_usd,
  ROUND(CAST((rds_gp3_equivalent_cost_usd
    - archive_monthly_cost_usd) * 12.0 AS DECIMAL(18,2)), 2) AS annualized_savings_usd,
  ROUND(CAST(100.0 * (rds_gp3_equivalent_cost_usd
    - archive_monthly_cost_usd)
    / NULLIF(rds_gp3_equivalent_cost_usd, 0) AS DECIMAL(18,2)), 1) AS savings_pct,
  oldest_archived_date,
  newest_archived_date,
  first_archived_at

```

```
        - archive_monthly_cost_usd)
      / NULLIF(rds_gp3_equivalent_cost_usd, 0) AS DECIMAL(5,1)), 1)
                                                AS savings_pct,
oldest_archived_date,
newest_archived_date,
first_archived_at
FROM
  archive_costs;
GO
```

Running this view monthly and feeding it into a Grafana dashboard or a Slack report gives your team a persistent, compounding record of the value the tiering pipeline has delivered. Teams that skip this step often find the pipeline deprioritized the moment an engineer who built it moves on—there is no visible evidence of what it is doing.

17.11 Common Pitfalls and How to Avoid Them

Even well-designed tiering pipelines fail in predictable ways. The following are the failure modes that appear most often in production environments.

Forgetting retrieval costs when choosing storage classes

S3 Glacier Deep Archive costs \$0.00099/GB-month at rest, but retrieval costs \$0.02/GB plus a minimum 12-hour restore time. If your compliance team retrieves a 500 GB partition twice a year, that is \$10 in retrieval fees each time—still cheap, but not free. For data that is accessed even once per quarter, Glacier Instant at \$0.004/GB delivers millisecond retrieval and no restore delay. Match the storage class to the realistic access frequency, not just the cheapest option at rest.

Archiving without verifying

The export step in Section 17.4 writes `verified=false` to the metadata table and only flips it after the checksum and row count are confirmed. Teams under time pressure skip the verification step. Six months later, when a regulator asks for a specific archive, the file exists but is corrupted or incomplete. Never drop a partition before setting `verified=true`.

Partitioning by month when your queries span quarters

If Athena or BigQuery scans all partitions to answer a query spanning multiple months, the savings from partition pruning disappear. Design partition boundaries to match the most common query patterns. For audit logs queried by quarter for compliance, quarterly partitions are better than monthly ones.

Lock contention during partition detach

The `ALTER TABLE ... DETACH PARTITION` command in PostgreSQL acquires a `ShareUpdateExclusiveLock` in PostgreSQL 14+ (using the `CONCURRENTLY` option) but requires `AccessExclusiveLock` in earlier versions. Always set `lock_timeout` before running detach in production, as shown in the `detach_and_drop_partition` function in Section 17.4. If the lock cannot be acquired within the timeout, the operation fails cleanly rather than blocking production queries for an unknown duration.

Schema drift breaking external tables

Covered in Section 17.8, but worth reiterating: every time you ALTER a live table, test that the corresponding Athena, Synapse, or BigQuery external table still resolves. Add this as a step in your database migration CI pipeline—a simple `SELECT 1 FROM archive_table LIMIT 1` run in CI catches the problem before it reaches production.

Not accounting for minimum storage duration fees

S3 Standard-IA has a 30-day minimum storage duration. Glacier Instant has a 90-day minimum. Deep Archive has a 180-day minimum. If you archive a partition to Deep Archive and then need to delete it 60 days later (say, because a GDPR erasure request covers that partition), you still pay for the full 180 days. Design your retention and deletion policies to account for these minimums, and track the `archived_at` timestamp in the metadata catalog so your deletion job knows when the minimum period has elapsed.

Key Takeaways

- **The storage tier price gap is 50–100x between block storage and archival object storage**, making cold data archival one of the highest-ROI cost reduction actions available on any cloud database platform. Even partial migration of large historical tables produces meaningful annual savings.
 - **Range partitioning by time is the foundation of every reliable tiering pipeline.** Without partitions, cold data cannot be detached and exported cleanly. Invest in partitioning before building any other part of the archival workflow, and use quarterly or monthly partition boundaries that align with your query patterns.
 - **Verification before deletion is non-negotiable.** Every export must be validated by checksum and row count before the source partition is dropped. A metadata catalog with a **verified** boolean flag and a retrieval log gives your operations and compliance teams proof that the archive is intact and accessible.
 - **Archived data must remain queryable.** Athena, Synapse Serverless SQL, and BigQuery external tables over Parquet files deliver serverless, on-demand access to archived data at a fraction of the cost of keeping it in the live database. Register archives in the Glue Data Catalog or equivalent immediately after export so the data is findable.
 - **Automate the full lifecycle and measure it visibly.** A tiering pipeline that runs once delivers one-time savings. A pipeline scheduled via EventBridge, Cloud Scheduler, or cron—with lifecycle policies automatically transitioning objects through storage classes and an ROI reporting view feeding a dashboard—delivers compounding savings that grow as the database ages and accumulates more cold data.
-

Chapter 18: Query Cost Attribution: Which Queries Are Burning Money

Organizations running cloud databases routinely overpay by 30–60% not because their infrastructure is wrong, but because a handful of queries are consuming disproportionate compute, I/O, and memory resources that translate directly into larger instance sizes, higher storage throughput costs, and ballooning read replica bills. This chapter shows you how to identify exactly which queries are responsible for those costs, assign dollar values to each one, and build an automated attribution pipeline that surfaces the worst offenders before they hit your monthly bill. Teams that implement structured query cost attribution typically find that the top 5% of queries account for 70–80% of database compute spend—and that fixing three to five of them can cut monthly RDS or Cloud SQL bills by \$2,000–\$15,000 without touching a single infrastructure knob.

18.1 The Attribution Problem: Connecting Query Execution to Dollar Costs

Most DBAs know which queries are slow. Far fewer know which queries are *expensive*. The distinction matters enormously for FinOps work. A query that runs in 50 milliseconds but executes 2 million times per day is doing more damage to your bill than a query that takes 10 seconds and runs once an hour. Cloud costs accumulate through three primary mechanisms at the database layer:

Compute costs arise when queries hold CPU longer than necessary, forcing you onto a larger instance class. Moving from an `db.r6g.2xlarge` (\$0.528/hour on-demand, us-east-1) to a `db.r6g.4xlarge` (\$1.056/hour) costs an additional \$382/month—a cost that is entirely invisible unless you connect it to the queries driving CPU saturation.

I/O costs hit you in two ways. For RDS with gp2 storage, provisioned IOPS above the baseline incur per-IOPS charges of \$0.10/IOPS-month. For gp3, you pay \$0.02/GB-month for storage plus \$0.08 per provisioned IOPS beyond 3,000. Queries doing full table scans on large tables are the most common driver of unexpected IOPS bills.

Network and replication costs surface when poorly written queries return massive result sets or trigger excessive binary log events that replicate to read replicas or cross-region standby instances. Data transfer between Availability Zones costs \$0.01/GB on AWS, which becomes significant when a single analytics query pulls 500GB across the wire nightly.

The framework this chapter builds assigns three cost components to every query: compute cost, I/O cost, and network cost. These roll up into a single monthly dollar figure per query, giving you a ranked list to work from.

18.2 Extracting Query Cost Data: `pg_stat_statements`, Performance Insights, and `sys.dm_exec_query_stats`

PostgreSQL: `pg_stat_statements`

`pg_stat_statements` is the foundational tool for PostgreSQL cost attribution on RDS, Aurora PostgreSQL, and Cloud SQL. Enable it in the parameter group:

```
-- Enable pg_stat_statements (requires rds.force_ssl=1 on RDS)
-- Add to parameter group: shared_preload_libraries = pg_stat_statements
-- Then run:
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;

-- Reset statistics at the start of a billing cycle measurement window
SELECT pg_stat_statements_reset();
```

Once you have a week or more of data, the following query converts raw execution statistics into a cost estimate. The per-vCPU-hour figure is derived from your actual instance cost divided by vCPU count and a utilization assumption.

```
-- Query cost attribution for PostgreSQL
-- Assumes db.r6g.4xlarge: $1.056/hr, 16 vCPUs, 75% avg utilization
-- Effective cost per vCPU-hour = $1.056 / 16 = $0.066
-- Adjust cost_per_vcpu_hour and iops_cost_per_million to match your setup

WITH constants AS (
    SELECT
        0.066 AS cost_per_vcpu_hour,    -- $/vCPU-hr for your instance class
        0.08 AS iops_cost_per_million, -- RDS gp3 I/O cost approximation
        720 AS hours_per_month
),
query_stats AS (
    SELECT
        queryid,
        LEFT(query, 120) AS query_preview,
        calls,
        total_exec_time / 1000.0 AS total_exec_seconds,
        mean_exec_time / 1000.0 AS mean_exec_seconds,
        stddev_exec_time / 1000.0 AS stddev_exec_seconds,
        rows,
        shared_blks_hit + shared_blks_read AS total_blocks_accessed,
        shared_blks_read AS disk_blocks_read,
        shared_blks_written + local_blks_written AS blocks_written,
        blk_read_time / 1000.0 AS blk_read_seconds,
        blk_write_time / 1000.0 AS blk_write_seconds,
        -- CPU time approximation: total time minus I/O wait time
        GREATEST(0, (total_exec_time - blk_read_time - blk_write_time) / 1000.0) AS cpu_seconds
    FROM pg_stat_statements
    WHERE calls > 100 -- filter noise
),
cost_calc AS (
    SELECT
        qs.queryid,
        qs.query_preview,
        qs.calls,
        qs.total_exec_seconds,
        qs.mean_exec_seconds,
```

```

    qs.disk_blocks_read,
    qs.rows,
    -- Compute cost: CPU seconds consumed * per-second vCPU cost
    ROUND((qs.cpu_seconds * c.cost_per_vcpu_hour / 3600.0)::numeric, 4)
                                                AS compute_cost_usd,
    -- I/O cost: blocks read from disk (8KB blocks = IOPS approximation)
    ROUND((qs.disk_blocks_read / 1000000.0 * c.iops_cost_per_million)::numeric, 4)
                                                AS io_cost_usd,
    -- Total estimated monthly cost (scaled from observation window)
    -- Assumes stats cover exactly 7 days; scale to 30 days
    ROUND(((qs.cpu_seconds * c.cost_per_vcpu_hour / 3600.0)
        + (qs.disk_blocks_read / 1000000.0 * c.iops_cost_per_million))
        * (30.0 / 7.0)::numeric, 2)
                                                AS estimated_monthly_cost_usd
FROM query_stats qs
CROSS JOIN constants c
)
SELECT
    queryid,
    query_preview,
    calls,
    ROUND(total_exec_seconds::numeric, 2)
                                                AS total_exec_sec,
    ROUND(mean_exec_seconds * 1000::numeric, 2)
                                                AS avg_ms,
    disk_blocks_read,
    rows
                                                AS total_rows_returned,
    compute_cost_usd,
    io_cost_usd,
    estimated_monthly_cost_usd,
    ROUND(100.0 * estimated_monthly_cost_usd
        / SUM(estimated_monthly_cost_usd) OVER (), 2)
                                                AS pct_of_total_cost
FROM cost_calc
ORDER BY estimated_monthly_cost_usd DESC
LIMIT 25;

```

This query returns a ranked list showing each query’s estimated monthly dollar contribution. The `pct_of_total_cost` column is the number that matters to a CTO: it answers “what fraction of our database bill does this query own?”

SQL Server: sys.dm_exec_query_stats

For Azure SQL Database and SQL Server on RDS, the equivalent attribution lives in `sys.dm_exec_query_stats`:

```

-- Query cost attribution for SQL Server / Azure SQL Database
-- Adjust @instance_cost_per_hour to match your DTU/vCore pricing
-- Azure SQL Business Critical 8 vCores = ~$1.13/hr
DECLARE @instance_cost_per_hour FLOAT = 1.13;
DECLARE @vcpu_count INT = 8;
DECLARE @observation_hours FLOAT = 168.0; -- 7 days
DECLARE @cost_per_cpu_second FLOAT = @instance_cost_per_hour / @vcpu_count / 3600.0;

SELECT TOP 25
    qs.total_worker_time / 1000000.0
                                                AS total_cpu_seconds,
    qs.total_elapsed_time / 1000000.0
                                                AS total_elapsed_seconds,
    qs.total_logical_reads
                                                AS logical_reads,
    qs.total_physical_reads
                                                AS physical_reads,
    qs.total_logical_writes
                                                AS logical_writes,
    qs.execution_count,
    qs.total_rows,
    SUBSTRING(qt.text, (qs.statement_start_offset / 2) + 1,
        ((CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)
            ELSE qs.statement_end_offset
        END - qs.statement_start_offset) / 2) + 1)
                                                AS query_text,
    qp.query_plan,
    -- Compute cost estimate
    ROUND(qs.total_worker_time / 1000000.0
        * @cost_per_cpu_second, 4)
                                                AS compute_cost_usd,

```

```

-- I/O cost: physical reads * 8KB pages, approximate $0.115/GB for Azure Premium SSD
ROUND(qs.total_physical_reads * 8.0 / 1048576.0
      * 0.115, 4)                                AS io_cost_usd,
-- Projected monthly (scale from observation window to 30 days)
ROUND((qs.total_worker_time / 1000000.0 * @cost_per_cpu_second
      + qs.total_physical_reads * 8.0 / 1048576.0 * 0.115)
      * (720.0 / @observation_hours), 2)          AS est_monthly_cost_usd
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
OUTER APPLY sys.dm_exec_query_plan(qs.plan_handle) qp
WHERE qs.execution_count > 50
ORDER BY est_monthly_cost_usd DESC;

```

MySQL and Aurora MySQL

Aurora MySQL and RDS MySQL expose per-query metrics through the Performance Schema. Enable the statement digest:

```

-- Aurora MySQL / RDS MySQL query cost attribution
-- Requires performance_schema = ON and events_statements_summary_by_digest enabled
-- performance-schema-instrument = 'statement/sql/%=ON' in parameter group

SELECT
  SCHEMA_NAME,
  LEFT(DIGEST_TEXT, 120)                                AS query_preview,
  COUNT_STAR                                             AS executions,
  ROUND(SUM_TIMER_WAIT / 1e12, 2)                       AS total_exec_seconds,
  ROUND(AVG_TIMER_WAIT / 1e9, 2)                       AS avg_exec_ms,
  SUM_ROWS_EXAMINED                                     AS rows_examined,
  SUM_ROWS_SENT                                         AS rows_sent,
  SUM_NO_INDEX_USED                                     AS full_scans,
  SUM_NO_GOOD_INDEX_USED                               AS bad_index_uses,
  -- Efficiency ratio: lower is worse (ideal = 1.0)
  ROUND(SUM_ROWS_SENT * 1.0 / NULLIF(SUM_ROWS_EXAMINED, 0), 4) AS row_efficiency,
  -- Compute cost approximation (db.r6g.2xlarge Aurora: $0.48/hr, 8 vCPUs)
  ROUND(SUM_TIMER_WAIT / 1e12 * (0.48 / 8.0 / 3600.0), 4) AS compute_cost_usd,
  -- I/O approximation via rows examined (Aurora charges per I/O request)
  -- Aurora I/O = $0.20 per 1M requests
  ROUND(SUM_ROWS_EXAMINED / 1000000.0 * 0.20, 4)        AS aurora_io_cost_usd,
  ROUND((SUM_TIMER_WAIT / 1e12 * (0.48 / 8.0 / 3600.0)
        + SUM_ROWS_EXAMINED / 1000000.0 * 0.20)
        * (30.0 / 7.0), 2)                               AS est_monthly_cost_usd
FROM performance_schema.events_statements_summary_by_digest
WHERE SCHEMA_NAME IS NOT NULL
  AND COUNT_STAR > 50
  AND LAST_SEEN > DATE_SUB(NOW(), INTERVAL 7 DAY)
ORDER BY est_monthly_cost_usd DESC
LIMIT 25;

```

The `aurora_io_cost_usd` column is particularly important for Aurora Serverless v1 and Aurora standard configurations on the I/O-optimized pricing plan decision. If your I/O cost exceeds 25% of total Aurora storage + I/O costs, switching to Aurora I/O-Optimized pricing (\$0.225/GB-month, no separate I/O charge) typically saves money—and reducing row examination in these queries accelerates that crossover point.

18.3 AWS Performance Insights and Cost Explorer Integration

Performance Insights provides the bridge between database-level metrics and the infrastructure cost data in Cost Explorer. The combination is powerful: Performance Insights shows you which SQL

is consuming DB time, while Cost Explorer shows you what that DB time is costing in dollars.

Pulling Performance Insights Data via AWS CLI

```
#!/bin/bash
# Retrieve top SQL queries by DB load from Performance Insights
# Requires: aws cli v2, jq, Performance Insights enabled on the RDS instance
# Run this daily via cron or EventBridge to populate a cost attribution table

DB_INSTANCE_ID="prod-postgres-01"
REGION="us-east-1"
ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)

# Performance Insights uses a resource ARN
PI_RESOURCE_ID=$(aws rds describe-db-instances \
  --db-instance-identifier "$DB_INSTANCE_ID" \
  --region "$REGION" \
  --query 'DBInstances[0].DbiResourceId' \
  --output text)

# Fetch top 25 SQL digests for the past 7 days
# Metric: db.load.avg (average active sessions)
START_TIME=$(date -u -d '7 days ago' '+%Y-%m-%dT%H:%M:%SZ')
END_TIME=$(date -u '+%Y-%m-%dT%H:%M:%SZ')

aws pi get-resource-metrics \
  --service-type RDS \
  --identifier "$PI_RESOURCE_ID" \
  --metric-queries '[
    {
      "Metric": "db.load.avg",
      "GroupBy": {
        "Group": "db.sql",
        "Dimensions": ["db.sql.tokenized_id"],
        "Limit": 25
      }
    }
  ]' \
  --start-time "$START_TIME" \
  --end-time "$END_TIME" \
  --period-in-seconds 3600 \
  --region "$REGION" \
  --output json \
  | jq '.MetricList[] | {
    metric: .Key.Metric,
    sql_id: .Key.Dimensions["db.sql.tokenized_id"],
    avg_load: (.DataPoints | map(.Value) | add / length)
  }' \
  | jq -s 'sort_by(-.avg_load)' \
  > /tmp/pi_top_queries_$(date +%Y%m%d).json

echo "Performance Insights data saved to /tmp/pi_top_queries_$(date +%Y%m%d).json"

# Fetch instance cost from Cost Explorer for the same period
aws ce get-cost-and-usage \
  --time-period "Start=$(date -u -d '7 days ago' '+%Y-%m-%d'),End=$(date -u '+%Y-%m-%d')" \
  --granularity DAILY \
  --filter '{
    "And": [
      {
        "Dimensions": {
          "Key": "SERVICE",
          "Values": ["Amazon Relational Database Service"]
        }
      },
      {
        "Tags": {
          "Key": "db-instance",
          "Values": ["prod-postgres-01"]
        }
      }
    ]
  }'
```


18.4 Building an Automated Query Cost Attribution Pipeline

Manual query analysis is a one-time exercise. The organizations that consistently control costs automate the attribution process and integrate it into their observability stack. The following Python pipeline runs on a daily schedule (via AWS Lambda, Cloud Functions, or a cron job on an EC2 instance) and writes results to a PostgreSQL tracking table and optionally to Slack or PagerDuty.

```
#!/usr/bin/env python3
"""
query_cost_attribution.py

Automated query cost attribution pipeline.
Runs daily via EventBridge Scheduler or cron.
Writes results to a PostgreSQL attribution table.
Sends Slack alerts when a query's estimated monthly cost exceeds threshold.

Requirements:
    pip install psycpg2-binary boto3 requests python-dotenv

Environment variables:
    DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD
    TRACKING_DB_HOST, TRACKING_DB_NAME (can be same as above)
    AWS_REGION
    PI_RESOURCE_ID      (from aws rds describe-db-instances)
    SLACK_WEBHOOK_URL    (optional)
    INSTANCE_COST_PER_HOUR
    VCPU_COUNT
    COST_ALERT_THRESHOLD (default: 500 USD/month per query)
"""

import os
import json
import logging
import datetime
import psycpg2
import psycpg2.extras
import boto3
import requests
from decimal import Decimal

logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s %(levelname)s %(message)s'
)
log = logging.getLogger(__name__)

# Configuration

DB_CONFIG = {
    "host": os.environ["DB_HOST"],
    "port": int(os.environ.get("DB_PORT", 5432)),
    "dbname": os.environ["DB_NAME"],
    "user": os.environ["DB_USER"],
    "password": os.environ["DB_PASSWORD"],
}

TRACKING_DB_CONFIG = {
    "host": os.environ.get("TRACKING_DB_HOST", os.environ["DB_HOST"]),
    "port": int(os.environ.get("TRACKING_DB_PORT", 5432)),
    "dbname": os.environ.get("TRACKING_DB_NAME", "finops_tracking"),
    "user": os.environ["DB_USER"],
    "password": os.environ["DB_PASSWORD"],
}
```

```

INSTANCE_COST_PER_HOUR = float(os.environ.get("INSTANCE_COST_PER_HOUR", "1.056"))
VCPU_COUNT              = int(os.environ.get("VCPU_COUNT", "16"))
COST_PER_CPU_SECOND     = INSTANCE_COST_PER_HOUR / VCPU_COUNT / 3600.0
IOPS_COST_PER_BLOCK     = 0.08 / 1_000_000 # gp3 I/O approximation per 8KB block
OBSERVATION_DAYS        = 7
SCALE_TO_MONTHLY        = 30.0 / OBSERVATION_DAYS

SLACK_WEBHOOK_URL       = os.environ.get("SLACK_WEBHOOK_URL")
ALERT_THRESHOLD_USD     = float(os.environ.get("COST_ALERT_THRESHOLD", "500"))

AWS_REGION              = os.environ.get("AWS_REGION", "us-east-1")
PI_RESOURCE_ID          = os.environ.get("PI_RESOURCE_ID", "")

# DDL for tracking table

TRACKING_TABLE_DDL = """
CREATE TABLE IF NOT EXISTS query_cost_attribution (
    id                BIGSERIAL PRIMARY KEY,
    captured_at       TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    observation_window_days INT      NOT NULL,
    queryid           BIGINT,
    query_preview     TEXT,
    calls             BIGINT,
    total_exec_seconds NUMERIC(18,4),
    avg_exec_ms       NUMERIC(18,4),
    disk_blocks_read  BIGINT,
    total_rows_returned BIGINT,
    compute_cost_usd  NUMERIC(12,4),
    io_cost_usd       NUMERIC(12,4),
    est_monthly_cost_usd NUMERIC(12,2),
    pct_of_total_cost NUMERIC(6,2),
    alerted           BOOLEAN DEFAULT FALSE
);

CREATE INDEX IF NOT EXISTS idx_qca_captured_at
    ON query_cost_attribution (captured_at DESC);

CREATE INDEX IF NOT EXISTS idx_qca_monthly_cost
    ON query_cost_attribution (est_monthly_cost_usd DESC);
"""

# SQL: pull attribution data from pg_stat_statements

ATTRIBUTION_SQL = """
WITH constants AS (
    SELECT
        %(cost_per_cpu_second)s::float AS cost_per_cpu_second,
        %(iops_cost_per_block)s::float AS iops_cost_per_block
),
query_stats AS (
    SELECT
        queryid,
        LEFT(query, 200) AS query_preview,
        calls,
        total_exec_time / 1000.0 AS total_exec_seconds,
        mean_exec_time / 1000.0 AS mean_exec_seconds,
        rows,
        shared_blks_read AS disk_blocks_read,
        GREATEST(0,
            (total_exec_time - blk_read_time - blk_write_time)
            / 1000.0) AS cpu_seconds
    FROM pg_stat_statements
    WHERE calls > 50
        AND query NOT LIKE '%%pg_stat_statements%%'
        AND query NOT LIKE '%%pg_catalog%%'
),
cost_calc AS (
    SELECT
        qs.*,

```

```

        ROUND((qs.cpu_seconds
              * c.cost_per_cpu_second)::numeric, 6)          AS compute_cost_usd,
        ROUND((qs.disk_blocks_read
              * c.iops_cost_per_block)::numeric, 6)          AS io_cost_usd,
        ROUND(((qs.cpu_seconds * c.cost_per_cpu_second)
              + (qs.disk_blocks_read * c.iops_cost_per_block))
              * %(scale_to_monthly)s::float, 2)              AS est_monthly_cost_usd
    FROM query_stats qs
    CROSS JOIN constants c
)
SELECT
    queryid,
    query_preview,
    calls,
    ROUND(total_exec_seconds::numeric, 4)                  AS total_exec_seconds,
    ROUND(mean_exec_seconds * 1000::numeric, 4)            AS avg_exec_ms,
    disk_blocks_read,
    rows                                                    AS total_rows_returned,
    compute_cost_usd,
    io_cost_usd,
    est_monthly_cost_usd,
    ROUND(100.0 * est_monthly_cost_usd
          / NULLIF(SUM(est_monthly_cost_usd) OVER (), 0), 2) AS pct_of_total_cost
FROM cost_calc
ORDER BY est_monthly_cost_usd DESC
LIMIT 50;
"""

```

Functions

```

def get_attribution_rows(conn) -> list[dict]:
    """Pull ranked query cost attribution from pg_stat_statements."""
    params = {
        "cost_per_cpu_second": COST_PER_CPU_SECOND,
        "iops_cost_per_block": IOPS_COST_PER_BLOCK,
        "scale_to_monthly":    SCALE_TO_MONTHLY,
    }
    with conn.cursor(cursor_factory=psycopg2.extras.RealDictCursor) as cur:
        cur.execute(ATTRIBUTION_SQL, params)
        rows = cur.fetchall()
        log.info("Retrieved %d attribution rows from pg_stat_statements", len(rows))
    return [dict(r) for r in rows]

def ensure_tracking_table(tracking_conn) -> None:
    """Create the tracking table if it does not exist."""
    with tracking_conn.cursor() as cur:
        cur.execute(TRACKING_TABLE_DDL)
    tracking_conn.commit()
    log.info("Tracking table verified / created")

def write_attribution_rows(tracking_conn, rows: list[dict]) -> None:
    """Insert attribution rows into the tracking table."""
    insert_sql = """
        INSERT INTO query_cost_attribution (
            observation_window_days,
            queryid,
            query_preview,
            calls,
            total_exec_seconds,
            avg_exec_ms,
            disk_blocks_read,
            total_rows_returned,
            compute_cost_usd,
            io_cost_usd,
            est_monthly_cost_usd,
            pct_of_total_cost
        ) VALUES (
            %(observation_window_days)s,

```

```

        %(queryid)s,
        %(query_preview)s,
        %(calls)s,
        %(total_exec_seconds)s,
        %(avg_exec_ms)s,
        %(disk_blocks_read)s,
        %(total_rows_returned)s,
        %(compute_cost_usd)s,
        %(io_cost_usd)s,
        %(est_monthly_cost_usd)s,
        %(pct_of_total_cost)s
    )
"""
enriched = [
    {**row, "observation_window_days": OBSERVATION_DAYS}
    for row in rows
]
with tracking_conn.cursor() as cur:
    psycopg2.extras.execute_batch(cur, insert_sql, enriched, page_size=50)
tracking_conn.commit()
log.info("Wrote %d rows to query_cost_attribution", len(rows))

def send_slack_alert(row: dict) -> None:
    """Post a Slack message for a query exceeding the cost threshold."""
    if not SLACK_WEBHOOK_URL:
        return
    message = {
        "text": (
            f":money_with_wings: *Query Cost Alert* "
            f"- estimated monthly cost: *${row['est_monthly_cost_usd']:.2f}*\\n"
            f"````{row['query_preview'][:300]}````\\n"
            f"Calls: {row['calls']:,} | "
            f"Avg: {row['avg_exec_ms']:.1f} ms | "
            f"Disk blocks read: {row['disk_blocks_read']:,} | "
            f"Share of total cost: {row['pct_of_total_cost']:.1f}%"
        )
    }
    try:
        resp = requests.post(SLACK_WEBHOOK_URL, json=message, timeout=10)
        resp.raise_for_status()
        log.info("Slack alert sent for queryid=%s", row.get("queryid"))
    except requests.RequestException as exc:
        log.warning("Failed to send Slack alert: %s", exc)

def check_and_alert(tracking_conn, rows: list[dict]) -> None:
    """
    Alert on any query whose estimated monthly cost exceeds ALERT_THRESHOLD_USD
    and has not already been alerted today.
    """
    today = datetime.date.today()
    check_sql = """
        SELECT 1
        FROM query_cost_attribution
        WHERE queryid = %(queryid)s
        AND DATE(captured_at) = %(today)s
        AND alerted = TRUE
        LIMIT 1
    """
    mark_sql = """
        UPDATE query_cost_attribution
        SET alerted = TRUE
        WHERE queryid = %(queryid)s
        AND DATE(captured_at) = %(today)s
        AND alerted = FALSE
    """
    for row in rows:
        if float(row.get("est_monthly_cost_usd") or 0) < ALERT_THRESHOLD_USD:
            continue

```

```
with tracking_conn.cursor() as cur:
    cur.execute(check_sql, {"queryid": row["queryid"], "today": today})
    already_alerted = cur.fetchone() is not None
if not already_alerted:
    send_slack_alert(row)
    with tracking_conn.cursor() as cur:
        cur.execute(mark_sql, {"queryid": row["queryid"], "today": today})
    tracking_conn.commit()

def lambda_handler(event: dict, context) -> dict:
    """
    AWS Lambda entry point.
    Also callable as a plain Python script main().
    """
    log.info("Starting query cost attribution run at %s", datetime.datetime.utcnow())

    with psycopg2.connect(**DB_CONFIG) as src_conn, \
        psycopg2.connect(**TRACKING_DB_CONFIG) as trk_conn:

        ensure_tracking_table(trk_conn)
        rows = get_attribution_rows(src_conn)
        write_attribution_rows(trk_conn, rows)
        check_and_alert(trk_conn, rows)

    top = rows[0] if rows else {}
    log.info(
        "Run complete. Top query: queryid=%s, est_monthly=$%s",
        top.get("queryid"), top.get("est_monthly_cost_usd")
    )
    return {
        "statusCode": 200,
        "queries_analyzed": len(rows),
        "top_query_monthly_cost_usd": str(top.get("est_monthly_cost_usd", 0)),
    }

if __name__ == "__main__":
    lambda_handler({}, None)
```

Deploy this as a Lambda function with a 15-minute timeout, 512 MB memory, and an EventBridge rule triggering it at `cron(0 6 * * ? *)` (06:00 UTC daily). The function needs a VPC configuration that places it in the same subnet as the RDS instance, and a security group rule allowing port 5432 outbound to the database security group.

18.5 Reading the Attribution Output: Patterns That Indicate Wasted Spend

Raw numbers only go so far. The value is in recognizing the cost patterns that map to specific optimization actions. The following patterns appear repeatedly across production systems and each has a well-defined remediation path.

Pattern 1: High Disk Blocks Read, Low Row Efficiency

```
query_preview:  SELECT * FROM orders WHERE customer_id = $1 AND status = $2
calls:          1,842,000
avg_exec_ms:    12.4
disk_blocks_read: 94,000,000
```

```
total_rows:      1,842,000
row_efficiency: 0.019    (roughly 53 rows examined per row returned)
est_monthly:     $2,847
```

A row efficiency below 0.10 combined with high disk block reads is the signature of a missing or ineffective index. The fix is almost always a composite index on (customer_id, status). For this query, adding that index would reduce disk_blocks_read from 94 million to roughly 1.8 million—a 98% reduction in I/O cost, dropping the monthly estimate from \$2,847 to under \$60.

```
-- PostgreSQL: diagnose the missing index
EXPLAIN (ANALYZE, BUFFERS, FORMAT TEXT)
SELECT * FROM orders WHERE customer_id = 42 AND status = 'pending';

-- Look for "Seq Scan" or "Bitmap Heap Scan" with high "Buffers: shared hit/read"
-- Then create the composite index:
CREATE INDEX CONCURRENTLY idx_orders_customer_status
ON orders (customer_id, status)
INCLUDE (order_date, total_amount);
-- INCLUDE adds covering columns to avoid a second heap fetch

-- SQL Server equivalent
-- Check missing index DMV first
SELECT
    mid.statement
    AS table_name,
    migs.avg_total_user_cost * migs.avg_user_impact
    * (migs.user_seeks + migs.user_scans)
    AS improvement_measure,
    'CREATE INDEX ix_' + OBJECT_NAME(mid.object_id)
    + '_' + REPLACE(ISNULL(mid.equality_columns,''), ' ', '_')
    + ' ON ' + mid.statement
    + ' (' + ISNULL(mid.equality_columns,'')
    + CASE WHEN mid.inequality_columns IS NOT NULL
        THEN ',' + mid.inequality_columns ELSE '' END + ')'
    + CASE WHEN mid.included_columns IS NOT NULL
        THEN ' INCLUDE (' + mid.included_columns + ')' ELSE '' END
    AS create_index_statement
FROM sys.dm_db_missing_index_details mid
JOIN sys.dm_db_missing_index_groups mig
ON mid.index_handle = mig.index_handle
JOIN sys.dm_db_missing_index_group_stats migs
ON mig.index_group_handle = migs.group_handle
ORDER BY improvement_measure DESC;
```

Pattern 2: High Call Count, Trivial Individual Cost — But Large Aggregate

```
query_preview:  SELECT id, name, email FROM users WHERE id = $1
calls:          28,000,000
avg_exec_ms:    0.3
disk_blocks_read: 2,800,000
est_monthly:    $1,140
```

This pattern—often called a “chatty query”—is common in ORM-heavy applications that issue individual lookups inside loops. The query is fast and well-indexed, but the sheer call volume adds up. The fix is application-side: batch the lookups using WHERE id = ANY(\$1) (PostgreSQL) or WHERE id IN (...) (all platforms), or introduce an application-level cache for frequently-requested user records.

```
-- PostgreSQL: replace per-row lookups with a batch fetch
-- Instead of issuing 1000 individual queries:
-- SELECT id, name, email FROM users WHERE id = 1;
-- SELECT id, name, email FROM users WHERE id = 2; ...

-- Issue one query:
SELECT id, name, email
```

```

FROM users
WHERE id = ANY(%(user_ids)s::bigint[]);

-- The query plan uses an index scan on the primary key for each element
-- of the array, eliminating the per-round-trip overhead

-- SQL Server equivalent using a table-valued parameter
-- Define the TVP type once:
CREATE TYPE dbo.BigIntList AS TABLE (id BIGINT NOT NULL PRIMARY KEY);

-- In the application, pass a DataTable and use:
SELECT u.id, u.name, u.email
FROM dbo.users u
JOIN @user_ids_tvp t ON u.id = t.id;

```

Pattern 3: High CPU Seconds, Modest I/O — Computation-Heavy Query

```

query_preview:  SELECT customer_id, SUM(amount), COUNT(*), AVG(amount)
                  FROM transactions
                  WHERE created_at BETWEEN $1 AND $2
                  GROUP BY customer_id
                  ORDER BY SUM(amount) DESC

calls:          12,000
avg_exec_ms:    8,400
cpu_seconds:    100,800
compute_cost:   $6,650
est_monthly:    $28,500

```

A high CPU-to-I/O ratio with long execution time points to a query that touches many rows but reads them efficiently (possibly from shared buffers). For aggregation queries like this, the options are:

1. **Materialized views or pre-aggregation tables** refreshed on a schedule
2. **Partial indexes** that restrict the scan to recent data
3. **Read replica offloading** if the query is analytical and can tolerate slight staleness

```

-- PostgreSQL: materialized view to pre-compute the aggregation
CREATE MATERIALIZED VIEW mv_customer_transaction_summary AS
SELECT
    customer_id,
    DATE_TRUNC('day', created_at) AS summary_date,
    SUM(amount) AS total_amount,
    COUNT(*) AS tx_count,
    AVG(amount) AS avg_amount
FROM transactions
GROUP BY customer_id, DATE_TRUNC('day', created_at)
WITH DATA;

CREATE UNIQUE INDEX ON mv_customer_transaction_summary
    (customer_id, summary_date);

-- Refresh concurrently (no lock) at the start of each hour via pg_cron:
-- SELECT cron.schedule('refresh-tx-summary', '0 * * * *',
--     'REFRESH MATERIALIZED VIEW CONCURRENTLY mv_customer_transaction_summary');

-- Rewrite the application query against the materialized view:
SELECT
    customer_id,
    SUM(total_amount) AS total_amount,
    SUM(tx_count) AS tx_count,
    SUM(total_amount) / NULLIF(SUM(tx_count), 0) AS avg_amount
FROM mv_customer_transaction_summary

```

```
WHERE summary_date BETWEEN $1::date AND $2::date
GROUP BY customer_id
ORDER BY total_amount DESC;

-- SQL Server: indexed view equivalent
CREATE VIEW dbo.vw_customer_tx_daily
WITH SCHEMABINDING AS
SELECT
    customer_id,
    CAST(created_at AS DATE)          AS summary_date,
    SUM(amount)                      AS total_amount,
    COUNT_BIG(*)                    AS tx_count
FROM dbo.transactions
GROUP BY customer_id, CAST(created_at AS DATE);

CREATE UNIQUE CLUSTERED INDEX ucx_customer_tx_daily
ON dbo.vw_customer_tx_daily (customer_id, summary_date);
-- SQL Server will use this indexed view automatically when the optimizer
-- recognizes it covers the query; or reference it directly with NOEXPAND hint
```

Pattern 4: N+1 Query Signature

An N+1 pattern shows up in the attribution table as two queries with a suspicious relationship: one query has call counts that are an exact multiple of another's.

```
queryid: 8734  calls: 45,000  query: SELECT * FROM products WHERE id = $1
queryid: 8201  calls: 45,000  query: SELECT * FROM categories WHERE id = $1
queryid: 9012  calls: 1,500   query: SELECT * FROM products WHERE page = $1 LIMIT 30
```

The 1,500 page queries each trigger 30 product lookups (45,000 total) which each trigger a category lookup—the classic ORM N+1. The fix is to rewrite the page-level query with JOINS or to configure eager loading in the ORM.

```
-- PostgreSQL: replace N+1 with a single JOIN query
SELECT
    p.id,
    p.name,
    p.price,
    c.id          AS category_id,
    c.name        AS category_name
FROM products p
JOIN categories c ON c.id = p.category_id
WHERE p.page = $1
ORDER BY p.sort_order
LIMIT 30;

-- SQL Server equivalent
SELECT
    p.id,
    p.name,
    p.price,
    c.id          AS category_id,
    c.name        AS category_name
FROM dbo.products p
JOIN dbo.categories c ON c.id = p.category_id
WHERE p.page = @page
ORDER BY p.sort_order
OFFSET 0 ROWS FETCH NEXT 30 ROWS ONLY;
```


18.6 Trending Attribution Data: Catching Cost Regressions Before They Hit Your Bill

A single snapshot of query costs is useful. A time-series of snapshots is how you catch regressions before they appear as billing surprises. The `query_cost_attribution` tracking table written by the pipeline in section 18.4 supports trend analysis directly.

```
-- PostgreSQL: week-over-week cost trend for top queries
-- Detects queries whose monthly cost estimate has grown by more than 20%
WITH this_week AS (
    SELECT
        queryid,
        query_preview,
        AVG(est_monthly_cost_usd)      AS avg_cost
    FROM query_cost_attribution
    WHERE captured_at >= NOW() - INTERVAL '7 days'
    GROUP BY queryid, query_preview
),
last_week AS (
    SELECT
        queryid,
        AVG(est_monthly_cost_usd)      AS avg_cost
    FROM query_cost_attribution
    WHERE captured_at >= NOW() - INTERVAL '14 days'
    AND captured_at < NOW() - INTERVAL '7 days'
    GROUP BY queryid
)
SELECT
    tw.queryid,
    tw.query_preview,
    ROUND(lw.avg_cost::numeric, 2)      AS last_week_usd,
    ROUND(tw.avg_cost::numeric, 2)      AS this_week_usd,
    ROUND(((tw.avg_cost - lw.avg_cost)
        / NULLIF(lw.avg_cost, 0) * 100)::numeric, 1)
        AS pct_change,
    CASE
        WHEN tw.avg_cost - lw.avg_cost > 200 THEN 'HIGH'
        WHEN tw.avg_cost - lw.avg_cost > 50  THEN 'MEDIUM'
        ELSE 'LOW'
    END
        AS regression_severity
FROM this_week tw
JOIN last_week lw USING (queryid)
WHERE tw.avg_cost > lw.avg_cost * 1.20 -- 20% cost increase
ORDER BY (tw.avg_cost - lw.avg_cost) DESC;

-- SQL Server: equivalent trend query against the same tracking table
-- (Assumes tracking table is mirrored to SQL Server or queried via linked server)
WITH this_week AS (
    SELECT
        queryid,
        query_preview,
        AVG(est_monthly_cost_usd)      AS avg_cost
    FROM query_cost_attribution
    WHERE captured_at >= DATEADD(DAY, -7, GETUTCDATE())
    GROUP BY queryid, query_preview
),
last_week AS (
    SELECT
        queryid,
        AVG(est_monthly_cost_usd)      AS avg_cost
    FROM query_cost_attribution
    WHERE captured_at >= DATEADD(DAY, -14, GETUTCDATE())
    AND captured_at < DATEADD(DAY, -7, GETUTCDATE())
    GROUP BY queryid
)
SELECT
    tw.queryid,
    tw.query_preview,
```

```

ROUND(lw.avg_cost, 2)           AS last_week_usd,
ROUND(tw.avg_cost, 2)           AS this_week_usd,
ROUND((tw.avg_cost - lw.avg_cost)
      / NULLIF(lw.avg_cost, 0) * 100, 1)
                                AS pct_change,
CASE
  WHEN tw.avg_cost - lw.avg_cost > 200 THEN 'HIGH'
  WHEN tw.avg_cost - lw.avg_cost > 50  THEN 'MEDIUM'
  ELSE 'LOW'
END                             AS regression_severity
FROM this_week tw
JOIN last_week lw ON tw.queryid = lw.queryid
WHERE tw.avg_cost > lw.avg_cost * 1.20
ORDER BY (tw.avg_cost - lw.avg_cost) DESC;

```

Wire this trend query into your daily pipeline and route **HIGH** severity regressions to Slack or PagerDuty. A deploy that ships a new code path introducing a full table scan will appear as a **HIGH** regression within 24 hours—long before the end-of-month bill.

Grafana Dashboard Integration

If your organization uses Grafana, the tracking table makes a natural data source. The following Grafana panel query renders a stacked area chart of the top 10 queries by estimated monthly cost over the past 30 days:

```

-- Grafana: top 10 queries by cost over time (PostgreSQL data source)
-- Use $__timeFilter(captured_at) for the Grafana time range variable
SELECT
  DATE_TRUNC('day', captured_at) AS time,
  LEFT(query_preview, 60)       AS metric,
  AVG(est_monthly_cost_usd)     AS value
FROM query_cost_attribution
WHERE captured_at IN (
  SELECT DISTINCT DATE_TRUNC('day', captured_at)
  FROM query_cost_attribution
  WHERE $__timeFilter(captured_at)
)
AND queryid IN (
  SELECT queryid
  FROM query_cost_attribution
  WHERE $__timeFilter(captured_at)
  GROUP BY queryid
  ORDER BY SUM(est_monthly_cost_usd) DESC
  LIMIT 10
)
GROUP BY 1, 2
ORDER BY 1, 3 DESC;

```

Set the panel type to **Time series**, enable stacking, and label it “Estimated Monthly Query Cost by Query (USD)”. Over time this chart becomes a visual changelog: deploy-induced cost spikes appear as step changes in specific query series, and optimization work shows up as a visible drop.

18.7 Connecting Attribution to Reserved Instance and Savings Plan Decisions

Query cost attribution data also informs infrastructure purchasing decisions. Here is the chain of reasoning:

1. Identify the top three compute-heavy queries from the attribution table.

2. **Estimate the CPU headroom** needed if those queries were optimized (typically a 40–70% CPU reduction for index additions or query rewrites).
3. **Model the instance downgrade path:** if optimizing two queries allows you to move from db.r6g.4xlarge to db.r6g.2xlarge, you save approximately \$382/month before any Reserved Instance discount.
4. **Apply the RI discount on the smaller instance:** a 1-year no-upfront RI on db.r6g.2xlarge in us-east-1 runs roughly 30% below on-demand, meaning you pay for a smaller box at a further discount.

The mistake most teams make is purchasing Reserved Instances before optimizing queries. They lock in a commitment at an inflated instance size, then optimize queries, and find themselves over-provisioned and unable to recover the cost difference. The correct sequence is:

1. Run query attribution for 30 days.
2. Fix the top five to ten queries.
3. Observe CPU and I/O metrics for two weeks post-optimization.
4. Right-size the instance.
5. Purchase the RI or Savings Plan on the right-sized instance.

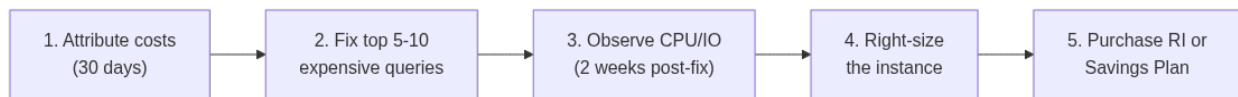


Figure 18: The correct sequence: optimize and right-size before committing — reversing this order locks in spend at an inflated instance size.

```

-- PostgreSQL: estimate post-optimization compute savings
-- Assumes you have identified which queryids to fix and their expected improvement
WITH optimization_scenarios AS (
  SELECT
    queryid,
    query_preview,
    est_monthly_cost_usd,
    AS current_monthly_cost,
    -- Conservative: assume 70% reduction in compute cost for an index fix
    -- Adjust improvement_factor per query based on EXPLAIN analysis
    CASE
      WHEN query_preview ILIKE '%WHERE%AND%'
        AND query_preview NOT ILIKE '%JOIN%'
      THEN 0.70 -- likely missing composite index
      WHEN query_preview ILIKE '%GROUP BY%'
        AND query_preview ILIKE '%SUM%'
      THEN 0.60 -- aggregation, materialized view candidate
      ELSE 0.40 -- conservative default
    END
    AS improvement_factor
  FROM query_cost_attribution
  WHERE captured_at >= NOW() - INTERVAL '7 days'
  ORDER BY est_monthly_cost_usd DESC
  LIMIT 10
)
SELECT
  queryid,
  LEFT(query_preview, 80)
  AS query_preview,
  ROUND(current_monthly_cost::numeric, 2)
  AS current_usd,
  ROUND((current_monthly_cost
    * improvement_factor)::numeric, 2)
  AS projected_savings_usd,
  ROUND((current_monthly_cost
    * (1 - improvement_factor))::numeric, 2)
  AS post_opt_cost_usd
FROM optimization_scenarios
UNION ALL
SELECT

```

```

NULL,
'=== TOTAL ===',
ROUND(SUM(current_monthly_cost)::numeric, 2),
ROUND(SUM(current_monthly_cost * improvement_factor)::numeric, 2),
ROUND(SUM(current_monthly_cost * (1 - improvement_factor))::numeric, 2)
FROM optimization_scenarios;

```

The output of this query gives you the business case document for optimization work: total projected savings, per-query savings, and the residual cost after optimization. Pair it with the instance right-sizing model from Chapter 12 and you have a full cost reduction roadmap that connects query-level work to infrastructure spend.

18.8 Special Considerations for Aurora Serverless, BigQuery, and Redshift

The attribution model described above assumes a provisioned instance where compute cost is primarily a function of time and CPU utilization. Three managed database services have different billing models that require adjusted attribution logic.

Aurora Serverless v2

Aurora Serverless v2 bills in ACU-hours (Aurora Capacity Units). One ACU provides approximately 2 GB of memory and a proportional share of CPU. Attribution for Serverless v2 should track ACU consumption rather than vCPU time:

```

-- Aurora Serverless v2: approximate cost attribution via Performance Schema
-- Current ACU price: $0.12/ACU-hour (us-east-1, as of this writing)
-- ACU-hours consumed (CPU time of query / total CPU time) * actual ACU-hours billed
-- Pull actual ACU-hours from CloudWatch: ServerlessDatabaseCapacity metric

SELECT
  LEFT(DIGEST_TEXT, 120)          AS query_preview,
  COUNT_STAR                     AS calls,
  ROUND(SUM_TIMER_WAIT / 1e12, 2) AS total_exec_sec,
  ROUND(SUM_TIMER_WAIT / 1e12
    / NULLIF(
      (SELECT SUM(SUM_TIMER_WAIT) / 1e12
       FROM performance_schema.events_statements_summary_by_digest
       WHERE SCHEMA_NAME IS NOT NULL), 0
    ) * 100, 2)                  AS pct_of_db_time,
  -- Multiply pct_of_db_time by actual ACU-hour cost from CloudWatch
  -- to get per-query cost attribution
  ROUND(SUM_TIMER_WAIT / 1e12
    / NULLIF(
      (SELECT SUM(SUM_TIMER_WAIT) / 1e12
       FROM performance_schema.events_statements_summary_by_digest
       WHERE SCHEMA_NAME IS NOT NULL), 0
    ) * %(total_acu_cost_7d)s, 2) AS attributed_cost_usd
FROM performance_schema.events_statements_summary_by_digest
WHERE SCHEMA_NAME IS NOT NULL
  AND COUNT_STAR > 50
ORDER BY attributed_cost_usd DESC
LIMIT 25;

```

Pass the `total_acu_cost_7d` parameter from a CloudWatch query that sums `ServerlessDatabaseCapacity` (in ACUs) over the 7-day window and multiplies by \$0.12/ACU-hour. This gives you a true dollar-denominated attribution that accounts for the auto-scaling

behavior of Serverless v2.

BigQuery: Slot-Based Cost Attribution

BigQuery charges per byte scanned (on-demand: \$6.25/TB) or per slot-hour (capacity pricing). For on-demand pricing, attribution is straightforward: bytes scanned is the cost driver.

```
-- BigQuery: query cost attribution from INFORMATION_SCHEMA
-- Run in the region where your datasets live
SELECT
  job_id,
  user_email,
  LEFT(query, 200) AS query_preview,
  total_bytes_processed / 1e12 AS tb_scanned,
  ROUND(total_bytes_processed / 1e12 * 6.25, 4) AS est_cost_usd,
  total_slot_ms / 1000.0 AS total_slot_seconds,
  creation_time,
  end_time,
  TIMESTAMP_DIFF(end_time, creation_time, SECOND) AS wall_seconds,
  cache_hit
FROM `region-us`.INFORMATION_SCHEMA.JOBS_BY_PROJECT
WHERE creation_time >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 7 DAY)
  AND job_type = 'QUERY'
  AND state = 'DONE'
  AND error_result IS NULL
ORDER BY est_cost_usd DESC
LIMIT 50;
```

For BigQuery cost attribution, `cache_hit = TRUE` rows cost nothing—filter them in when computing totals but remember they represent zero incremental cost. Queries that repeatedly scan the same large table without hitting cache are prime candidates for partitioning and clustering:

```
-- BigQuery: partition and cluster to reduce bytes scanned
-- Before: a full scan of a 10 TB events table costs ~$62.50 per query
-- After: partitioned on event_date, clustered on user_id reduces scan to ~50GB for typical filters

CREATE OR REPLACE TABLE `project.dataset.events_optimized`
PARTITION BY DATE(event_timestamp)
CLUSTER BY user_id, event_type
AS SELECT * FROM `project.dataset.events_raw`;

-- A query with WHERE DATE(event_timestamp) = '2024-01-15' AND user_id = 'abc'
-- now scans only the matching partition and the first N gigabytes of clustered blocks
-- rather than the full 10 TB
```

Amazon Redshift: WLM Queue Attribution

Redshift cost attribution uses the WLM (Workload Management) query queue metrics combined with per-node pricing:

```
-- Redshift: query cost attribution via STL_QUERY and SVL_QUERY_SUMMARY
-- Adjust cost_per_node_hour to match your node type
-- ra3.4xlarge: $3.26/node-hour (on-demand)
-- 2-node cluster = $6.52/hr total

WITH cluster_cost AS (
  SELECT
    3.26 * 2 AS cluster_cost_per_hour, -- adjust for your node count/type
    720 AS hours_per_month
),
query_stats AS (
  SELECT
    q.query,
    q.userid,
    q.starttime,
    q.endtime,
```

```

        DATEDIFF(seconds, q.starttime, q.endtime) AS duration_seconds,
        qs.rows,
        qs.bytes,
        q.label
    FROM stl_query q
    JOIN svl_query_summary qs ON q.query = qs.query
    WHERE q.starttime >= DATEADD(day, -7, GETDATE())
        AND q.aborted = 0
        AND q.userid > 1 -- exclude internal queries
),
cost_calc AS (
    SELECT
        query,
        userid,
        label,
        duration_seconds,
        bytes / 1073741824.0 AS gb_processed,
        -- Cost = fraction of cluster time this query consumed * cluster hourly cost
        ROUND(duration_seconds / 3600.0
            * (SELECT cluster_cost_per_hour FROM cluster_cost), 4) AS query_cost_usd,
        -- Annualized if this ran daily
        ROUND(duration_seconds / 3600.0
            * (SELECT cluster_cost_per_hour FROM cluster_cost)
            * 30, 2) AS est_monthly_if_daily_usd
    FROM query_stats
),
aggregated AS (
    SELECT
        label,
        COUNT(*) AS executions,
        ROUND(AVG(duration_seconds), 1) AS avg_duration_sec,
        ROUND(SUM(query_cost_usd), 2) AS total_cost_7d_usd,
        ROUND(AVG(gb_processed), 2) AS avg_gb_scanned,
        ROUND(SUM(query_cost_usd) * (30.0 / 7.0), 2) AS est_monthly_cost_usd
    FROM cost_calc
    GROUP BY label
)
SELECT
    label,
    executions,
    avg_duration_sec,
    avg_gb_scanned,
    total_cost_7d_usd,
    est_monthly_cost_usd,
    ROUND(100.0 * est_monthly_cost_usd
        / NULLIF(SUM(est_monthly_cost_usd) OVER (), 0), 2) AS pct_of_total
FROM aggregated
ORDER BY est_monthly_cost_usd DESC;

```

For Redshift, `label` is set by the application using `SET query_group TO 'report-generation'` before running the query—a practice that makes attribution far more actionable by tying costs to business features rather than anonymous SQL text.

18.9 Presenting Attribution Data to Engineering and Finance Teams

The most technically complete cost attribution system fails if its outputs don't drive action. Engineering teams and finance teams have different relationships with the data, and the pipeline you build should produce artifacts for both audiences.

For engineering teams, the actionable artifact is a ranked list of queries with specific optimization recommendations attached. The pipeline from section 18.4 can be extended to generate a Markdown or HTML report:

```
def generate_engineering_report(rows: list[dict]) -> str:
    """
    Generate a Markdown report suitable for a GitHub issue or Confluence page.
    """
    lines = [
        "# Query Cost Attribution Report",
        f"Generated: {datetime.datetime.utcnow().strftime('%Y-%m-%d %H:%M UTC')}",
        f"Observation window: {OBSERVATION_DAYS} days | Scaled to 30-day estimate",
        "",
        "## Top Queries by Estimated Monthly Cost",
        "",
        "| Rank | Est. Monthly Cost | % of Total | Calls | Avg ms | Query Preview |",
        "|-----|-----|-----|-----|-----|-----|",
    ]
    for i, row in enumerate(rows[:15], 1):
        cost = row.get("est_monthly_cost_usd") or 0
        pct = row.get("pct_of_total_cost") or 0
        calls = row.get("calls") or 0
        avg_ms = row.get("avg_exec_ms") or 0
        qtext = (row.get("query_preview") or "")[:80].replace("|", "\\|")
        lines.append(
            f"| {i} | ${float(cost):.2f} | {float(pct):.1f}% |",
            f"| {int(calls):,} | {float(avg_ms):.1f} | `{qtext}` |"
        )
    total = sum(float(r.get("est_monthly_cost_usd") or 0) for r in rows)
    lines += [
        "",
        f"**Total estimated monthly database query cost: ${total:.2f}**",
        "",
        "## Recommended Actions",
        "",
        "1. Review queries with `disk_blocks_read > 1,000,000` for missing indexes.",
        "2. Investigate queries with `calls > 1,000,000` for batching opportunities.",
        "3. Evaluate materialized views for queries with `avg_exec_ms > 5,000`.",
        "4. Check N+1 patterns where two queries share identical call counts.",
    ]
    return "\n".join(lines)
```

For finance teams, the artifact is a cost allocation summary that maps query costs to engineering teams or product features. This requires tagging—either via application-level query comments or by maintaining a mapping table from queryid to team:

```
-- PostgreSQL: team-level cost rollup
-- Requires a query_team_mapping table populated by the engineering team
-- or populated automatically by matching query_preview patterns to known feature areas

CREATE TABLE IF NOT EXISTS query_team_mapping (
    queryid      BIGINT      PRIMARY KEY,
    team_name    TEXT        NOT NULL,
    feature_area TEXT,
    ticket_ref   TEXT,
    last_updated TIMESTAMPTZ DEFAULT NOW()
);

-- Cost rollup by team for the current month
SELECT
    COALESCE(qtm.team_name, 'unattributed') AS team,
    COALESCE(qtm.feature_area, 'unknown')   AS feature_area,
    COUNT(DISTINCT qca.queryid)             AS unique_queries,
    ROUND(SUM(qca.est_monthly_cost_usd)::numeric, 2)
                                            AS total_monthly_cost_usd,
    ROUND(100.0 * SUM(qca.est_monthly_cost_usd)
          / NULLIF(SUM(SUM(qca.est_monthly_cost_usd)) OVER (), 0)::numeric, 2)
                                            AS pct_of_total
```

```
FROM query_cost_attribution qca
LEFT JOIN query_team_mapping qtm USING (queryid)
WHERE qca.captured_at >= DATE_TRUNC('month', NOW())
GROUP BY qtm.team_name, qtm.feature_area
ORDER BY total_monthly_cost_usd DESC;

-- SQL Server equivalent
SELECT
    COALESCE(qtm.team_name, 'unattributed') AS team,
    COALESCE(qtm.feature_area, 'unknown') AS feature_area,
    COUNT(DISTINCT qca.queryid) AS unique_queries,
    ROUND(SUM(qca.est_monthly_cost_usd), 2) AS total_monthly_cost_usd,
    ROUND(100.0 * SUM(qca.est_monthly_cost_usd)
        / NULLIF(SUM(SUM(qca.est_monthly_cost_usd)) OVER (), 0), 2)
        AS pct_of_total

FROM query_cost_attribution qca
LEFT JOIN query_team_mapping qtm ON qca.queryid = qtm.queryid
WHERE qca.captured_at >= DATEADD(DAY, 1 - DAY(GETUTCDATE()), CAST(GETUTCDATE() AS DATE))
GROUP BY qtm.team_name, qtm.feature_area
ORDER BY total_monthly_cost_usd DESC;
```

This team-level rollup turns query attribution into a chargeback or showback model. When the payments team sees that their batch reconciliation job owns 38% of the database compute bill, the conversation about optimization priority changes fundamentally—it becomes a feature team’s budget problem, not an invisible infrastructure overhead.

Key Takeaways

- **Query attribution quantifies invisible costs:** most cloud database overspend is driven by a small number of queries whose individual costs are invisible until converted to a per-query monthly dollar figure; `pg_stat_statements`, `sys.dm_exec_query_stats`, and Performance Schema provide the raw data needed to build that figure.
- **Three cost signals matter most:** disk blocks read (I/O cost), CPU seconds consumed (compute cost), and call count (aggregate cost from individually cheap queries); ranking queries on the combination of these three rather than on latency alone surfaces the true budget drivers.
- **Automate daily attribution and trend detection:** a single attribution snapshot is a debugging tool; a daily time series written to a tracking table becomes a regression detector that surfaces new full-table scans or N+1 patterns within 24 hours of a problematic deploy, long before they appear on the monthly bill.
- **Different billing models require adjusted attribution logic:** Aurora Serverless v2 attributes cost via ACU-hours weighted by query CPU share, BigQuery via bytes scanned, and Redshift via fraction of cluster time—the correct cost formula depends on the billing model of the service, not a generic CPU/IOPS proxy.
- **Sequence matters for RI and Savings Plan purchases:** committing to Reserved Instances before optimizing queries locks in spend at an inflated instance size; the correct order is attribute, optimize, observe, right-size, then commit—turning query-level engineering work directly into durable infrastructure savings.

Chapter 19: Read Replica Economics: When They Pay Off vs Waste Money

Read replicas are one of the most misunderstood line items on a cloud database bill. Teams spin them up to offload reporting queries, achieve “high availability,” or simply because the architecture diagram called for them — and then forget they exist. A single idle `db.r6g.2xlarge` RDS read replica in us-east-1 runs approximately **\$876/month** on on-demand pricing. Multiply that across three environments and two regions, and you are paying over **\$63,000/year** for instances that may be serving fewer than a dozen queries per hour. This chapter covers how to measure whether your read replicas are actually earning their keep, how to calculate the break-even point where adding a replica saves money versus costs more, and how to automate the decisions that most teams currently make by gut feel. Teams that apply the frameworks here typically reclaim 30–60% of their read replica spend within 90 days.

19.1 Understanding the True Cost of a Read Replica

Before optimizing, you need to understand what you are actually paying for. A read replica is not just the compute cost of the instance itself. The total cost of ownership for a read replica has four components:

- 1. Compute** — The instance hourly rate, which mirrors the primary instance class unless you explicitly downsize.
- 2. Storage** — In AWS RDS, replicas share the same allocated storage as the primary and are billed at the same rate. A 2TB gp3 primary with a replica means you are paying for 4TB of storage. In Aurora, however, replicas share the same storage volume, which changes the math entirely.
- 3. Data Transfer** — Replication traffic within the same Availability Zone is free. Cross-AZ replication incurs \$0.01/GB in AWS. Cross-region replication is billed at standard data transfer rates, typically \$0.02–\$0.09/GB depending on the source region. A primary writing 500GB/day of WAL to a cross-region replica costs between \$300–\$1,350/month in transfer fees alone before you have run a single query.
- 4. I/O** — On RDS gp2 and io1 storage, replicas consume read IOPS that are billed separately. On Aurora, I/O costs are shared across the cluster but replicas add read I/O charges.

Here is a concrete comparison of the same workload across instance types to illustrate how quickly costs diverge:

Configuration	Instance	On-Demand/mo	1-yr RI/mo	Savings
Single primary, no replica	db.r6g.2xlarge	\$876	\$553	—
Primary + 1 same-AZ replica	db.r6g.2xlarge × 2	\$1,752	\$1,106	RI saves \$646/mo
Primary + 1 cross-region replica	db.r6g.2xlarge × 2 + transfer	~\$2,052	~\$1,406	Transfer adds ~\$300/mo
Aurora primary + 1 Aurora replica	db.r6g.2xlarge × 2	\$1,752	\$1,106	Shared storage saves ~\$184/mo vs RDS

The Aurora shared storage advantage becomes significant at scale. A 5TB RDS PostgreSQL primary with two replicas bills for 15TB of gp3 storage at \$0.115/GB-month = **\$1,725/month just in storage**. The equivalent Aurora cluster bills storage once at \$0.10/GB-month for the actual data size, potentially saving hundreds of dollars monthly.

Use this query on PostgreSQL to understand your current replication topology before making any cost decisions:

```
-- Inventory current replication state and replica lag on PostgreSQL
-- Run on the primary instance
SELECT
    application_name,
    client_addr,
    state,
    sync_state,
    sent_lsn,
    write_lsn,
    flush_lsn,
    replay_lsn,
    (sent_lsn - replay_lsn) AS replication_lag_bytes,
    pg_size_pretty(sent_lsn - replay_lsn) AS replication_lag_pretty,
    write_lag,
    flush_lag,
    replay_lag
FROM pg_stat_replication
ORDER BY replication_lag_bytes DESC;

-- Check if this instance is a replica and its lag from primary
SELECT
    pg_is_in_recovery() AS is_replica,
    CASE
        WHEN pg_is_in_recovery() THEN
            EXTRACT(EPOCH FROM (now() - pg_last_xact_replay_timestamp()))::INT
        ELSE
            0
    END AS replica_lag_seconds,
    pg_last_xact_replay_timestamp() AS last_replay_time;

-- Query activity on replicas to understand actual utilization
-- Run on each replica
SELECT
    count(*) AS active_connections,
    count(*) FILTER (WHERE state = 'active') AS active_queries,
    count(*) FILTER (WHERE state = 'idle') AS idle_connections,
    count(*) FILTER (WHERE wait_event_type IS NOT NULL) AS waiting_connections
FROM pg_stat_activity
WHERE backend_type = 'client backend';
```

```
-- Top queries running on this replica in the last hour
SELECT
    calls,
    total_exec_time::BIGINT AS total_ms,
    mean_exec_time::BIGINT AS avg_ms,
    rows,
    query
FROM pg_stat_statements
WHERE query NOT ILIKE '%pg_stat%'
ORDER BY total_exec_time DESC
LIMIT 20;
```

For Azure SQL, use the equivalent DMV approach to assess replica utilization:

```
-- Azure SQL: Check replica utilization and query load
-- Run on secondary/geo-replica
SELECT
    database_name = DB_NAME(),
    is_read_committed_snapshot_isolation_on =
        CASE WHEN is_read_committed_snapshot_on = 1 THEN 'YES' ELSE 'NO' END,
    is_secondary = CASE WHEN database_id > 0 THEN 'YES' ELSE 'NO' END
FROM sys.databases
WHERE name = DB_NAME();

-- Query execution stats on the replica
SELECT TOP 25
    qs.execution_count,
    qs.total_elapsed_time / 1000 AS total_elapsed_ms,
    qs.total_elapsed_time / qs.execution_count / 1000 AS avg_elapsed_ms,
    qs.total_logical_reads,
    qs.total_logical_reads / qs.execution_count AS avg_logical_reads,
    SUBSTRING(qt.text, (qs.statement_start_offset / 2) + 1,
        ((CASE qs.statement_end_offset
            WHEN -1 THEN DATALength(qt.text)
            ELSE qs.statement_end_offset
        END - qs.statement_start_offset) / 2) + 1) AS query_text
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
WHERE qs.last_execution_time > DATEADD(HOUR, -1, GETUTCDATE())
ORDER BY qs.total_elapsed_time DESC;

-- Active connections on this secondary
SELECT
    COUNT(*) AS total_connections,
    SUM(CASE WHEN status = 'running' THEN 1 ELSE 0 END) AS running,
    SUM(CASE WHEN status = 'sleeping' THEN 1 ELSE 0 END) AS sleeping,
    SUM(CASE WHEN status = 'background' THEN 1 ELSE 0 END) AS background
FROM sys.sysprocesses;
```

19.2 Diagnosing Replica Utilization: Are They Actually Working?

The most common waste pattern in read replica deployments is the **replica that nobody reads**. It was provisioned for a reporting use case that was later moved to a data warehouse, or the application connection string was never updated to point read traffic at the replica endpoint.

The following script pulls CloudWatch metrics for all RDS instances and calculates a utilization score that identifies idle replicas:

```
#!/bin/bash
# replica-utilization-audit.sh
# Pulls 7-day average metrics for all RDS replicas and flags low-utilization instances
# Schedule: Run weekly via CloudWatch Events or cron

AWS_REGION="${AWS_REGION:-us-east-1}"
```

```

OUTPUT_FILE="replica_audit_$(date +%Y%m%d).csv"
DAYS_BACK=7
START_TIME=$(date -u -d "${DAYS_BACK} days ago" +%Y-%m-%dT%H:%M:%SZ 2>/dev/null || \
    date -u -v-${DAYS_BACK}d +%Y-%m-%dT%H:%M:%SZ)
END_TIME=$(date -u +%Y-%m-%dT%H:%M:%SZ)

echo
↪ "InstanceId,DBInstanceClass,ReplicationRole,AvgReadIOPS,AvgConnections,AvgCPU,ReplicaLag,MonthlyCostEstimate,Utilizat
↪ > "$OUTPUT_FILE"

# Get all RDS instances
INSTANCES=$(aws rds describe-db-instances \
    --region "$AWS_REGION" \
    --query
↪ 'DBInstances[*].[DBInstanceIdentifier,DBInstanceClass,ReadReplicaSourceDBInstanceIdentifier,MultiAZ]'
↪ \
    --output text)

while IFS=$'\t' read -r instance_id instance_class source_instance multi_az; do
    # Determine if this is a replica
    if [ -n "$source_instance" ] && [ "$source_instance" != "None" ]; then
        role="READ_REPLICA"
    else
        role="PRIMARY"
    fi

    # Only analyze replicas for this audit
    if [ "$role" != "READ_REPLICA" ]; then
        continue
    fi

    echo "Analyzing replica: $instance_id ($instance_class)"

    # Read IOPS (primary utilization signal for read replicas)
    AVG_READ_IOPS=$(aws cloudwatch get-metric-statistics \
        --region "$AWS_REGION" \
        --namespace "AWS/RDS" \
        --metric-name "ReadIOPS" \
        --dimensions Name=DBInstanceIdentifier,Value="$instance_id" \
        --start-time "$START_TIME" \
        --end-time "$END_TIME" \
        --period $((DAYS_BACK * 86400)) \
        --statistics Average \
        --query 'Datapoints[0].Average' \
        --output text 2>/dev/null || echo "0")

    # Database connections
    AVG_CONNECTIONS=$(aws cloudwatch get-metric-statistics \
        --region "$AWS_REGION" \
        --namespace "AWS/RDS" \
        --metric-name "DatabaseConnections" \
        --dimensions Name=DBInstanceIdentifier,Value="$instance_id" \
        --start-time "$START_TIME" \
        --end-time "$END_TIME" \
        --period $((DAYS_BACK * 86400)) \
        --statistics Average \
        --query 'Datapoints[0].Average' \
        --output text 2>/dev/null || echo "0")

    # CPU utilization
    AVG_CPU=$(aws cloudwatch get-metric-statistics \
        --region "$AWS_REGION" \
        --namespace "AWS/RDS" \
        --metric-name "CPUUtilization" \
        --dimensions Name=DBInstanceIdentifier,Value="$instance_id" \
        --start-time "$START_TIME" \
        --end-time "$END_TIME" \
        --period $((DAYS_BACK * 86400)) \
        --statistics Average \
        --query 'Datapoints[0].Average' \

```

```

--output text 2>/dev/null || echo "0")

# Replica lag
REPLICA_LAG=$(aws cloudwatch get-metric-statistics \
  --region "$AWS_REGION" \
  --namespace "AWS/RDS" \
  --metric-name "ReplicaLag" \
  --dimensions Name=DBInstanceIdentifier,Value="$instance_id" \
  --start-time "$START_TIME" \
  --end-time "$END_TIME" \
  --period $((DAYS_BACK * 86400)) \
  --statistics Average \
  --query 'Datapoints[0].Average' \
  --output text 2>/dev/null || echo "0")

# Determine utilization flag
READ_IOPS_FLOAT=$(echo "$AVG_READ_IOPS" | awk '{printf "%.2f", $1}')
CONNECTIONS_FLOAT=$(echo "$AVG_CONNECTIONS" | awk '{printf "%.2f", $1}')
CPU_FLOAT=$(echo "$AVG_CPU" | awk '{printf "%.2f", $1}')

UTIL_FLAG="HEALTHY"
if (( $(echo "$READ_IOPS_FLOAT < 10" | bc -l) )) && \
    (( $(echo "$CONNECTIONS_FLOAT < 5" | bc -l) )); then
    UTIL_FLAG="IDLE_CANDIDATE_FOR_DELETION"
elif (( $(echo "$CPU_FLOAT < 10" | bc -l) )) && \
    (( $(echo "$READ_IOPS_FLOAT < 50" | bc -l) )); then
    UTIL_FLAG="UNDERUTILIZED_CONSIDER_DOWNSIZING"
fi

# Rough monthly cost estimate (simplified - use actual pricing API for production)
# This maps common instance types to approximate on-demand monthly cost
MONTHLY_COST=$(python3 -c "
costs = {
    'db.t3.micro': 18, 'db.t3.small': 36, 'db.t3.medium': 72, 'db.t3.large': 144,
    'db.r6g.large': 219, 'db.r6g.xlarge': 438, 'db.r6g.2xlarge': 876,
    'db.r6g.4xlarge': 1752, 'db.r6g.8xlarge': 3504,
    'db.r7g.large': 228, 'db.r7g.xlarge': 456, 'db.r7g.2xlarge': 912,
    'db.m6g.large': 180, 'db.m6g.xlarge': 360, 'db.m6g.2xlarge': 720
}
print(costs.get('${instance_class}', 999))
")

echo "$instance_id,$instance_class,$role,$READ_IOPS_FLOAT,$CONNECTIONS_FLOAT,$CPU_FLOAT, \
↪ $REPLICA_LAG,$MONTHLY_COST,$UTIL_FLAG" >> "$OUTPUT_FILE"

done <<< "$INSTANCES"

echo ""
echo "=== REPLICAS AUDIT SUMMARY ==="
echo "Output written to: $OUTPUT_FILE"
echo ""
echo "Instances flagged for deletion (IDLE):"
grep "IDLE_CANDIDATE" "$OUTPUT_FILE" | awk -F',' '{printf " %-40s %-20s $%/month\n", $1, $2, $8}'

echo ""
echo "Instances flagged for downsizing:"
grep "UNDERUTILIZED" "$OUTPUT_FILE" | awk -F',' '{printf " %-40s %-20s $%/month\n", $1, $2, $8}'

TOTAL_WASTE=$(grep -E "IDLE_CANDIDATE|UNDERUTILIZED" "$OUTPUT_FILE" | \
  awk -F',' '{sum += $8} END {printf "%.0f", sum}')
echo ""
echo "Total estimated monthly waste from flagged instances: \\\$TOTAL_WASTE"

```

For GCP Cloud SQL replicas, you need a different approach since CloudWatch is not available:

```

#!/bin/bash
# gcp-replica-audit.sh
# Audits Cloud SQL read replicas using Cloud Monitoring metrics

```

```

PROJECT_ID="${GCP_PROJECT_ID:-your-project-id}"
DAYS_BACK=7
END_TIME=$(date -u +%Y-%m-%dT%H:%M:%SZ)
START_TIME=$(date -u -d "${DAYS_BACK} days ago" +%Y-%m-%dT%H:%M:%SZ 2>/dev/null || \
    date -u -v-${DAYS_BACK}d +%Y-%m-%dT%H:%M:%SZ)

echo "=== GCP Cloud SQL Read Replica Audit ==="
echo "Project: $PROJECT_ID"
echo "Period: Last $DAYS_BACK days"
echo ""

# List all Cloud SQL instances
echo "All Cloud SQL instances (replicas marked):"
gcloud sql instances list \
    --project="$PROJECT_ID" \
    --format="table(name,databaseVersion,settings.tier,region,masterInstanceName)" \
    --filter="instanceType:READ_REPLICA_INSTANCE"

echo ""
echo "Fetching query insight metrics for replicas..."

# Get read replica instances
REPLICAS=$(gcloud sql instances list \
    --project="$PROJECT_ID" \
    --filter="instanceType:READ_REPLICA_INSTANCE" \
    --format="value(name)" 2>/dev/null)

for replica in $REPLICAS; do
    echo "--- Replica: $replica ---"

    # Get instance tier for cost estimation
    TIER=$(gcloud sql instances describe "$replica" \
        --project="$PROJECT_ID" \
        --format="value(settings.tier)" 2>/dev/null)

    echo "Tier: $TIER"

    # Query connections metric via Monitoring API
    gcloud monitoring metrics-explorer \
        --project="$PROJECT_ID" 2>/dev/null || true

    # Use gcloud to pull key metrics (requires monitoring API enabled)
    echo "Fetching average connections (last 7 days)..."
    gcloud monitoring time-series list \
        --project="$PROJECT_ID" \
        --filter="metric.type=\"cloudsql.googleapis.com/database/postgresql/num_backends\" AND
        ↪ resource.labels.database_id=\"${PROJECT_ID}:${replica}\"" \
        --start="$START_TIME" \
        --end="$END_TIME" \
        --format="value(points[0].value.int64Value)" 2>/dev/null | \
        awk '{sum+=$1; count++;} END {if(count>0) printf "Average connections: %.1f\n", sum/count; else
        ↪ print "No connection data found"}'

    echo ""
done

# Cost estimation using Cloud Billing export query
echo "=== Cost Data (requires BigQuery billing export) ==="
cat << 'BQEOF'
-- Run this in BigQuery against your billing export dataset
-- Replace YOUR_BILLING_DATASET with your actual dataset name

SELECT
    resource_name,
    sku_description,
    SUM(cost) AS total_cost_30d,
    AVG(cost) * 30 AS projected_monthly_cost
FROM `YOUR_BILLING_DATASET.gcp_billing_export_v1_*`
WHERE
    service.description = 'Cloud SQL'

```

```

AND DATE(_PARTITIONTIME) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
AND LOWER(sku_description) LIKE '%replica%'
GROUP BY 1, 2
ORDER BY total_cost_30d DESC;
BQEOF

```

19.3 The Break-Even Analysis: When a Replica Actually Saves Money

Read replicas save money indirectly by preventing you from scaling up your primary instance to handle read-heavy workloads. The math only works in your favor under specific conditions. Here is the framework for calculating the break-even point.

The core equation:

Cost of replica < Cost of primary upgrade needed to handle same read load

Consider a concrete scenario: Your `db.r6g.2xlarge` primary (\$876/month) is running at 85% CPU, primarily from analytics queries. You have two choices:

- **Option A:** Upgrade primary to `db.r6g.4xlarge` at \$1,752/month → additional cost: **\$876/month**
- **Option B:** Add a `db.r6g.2xlarge` replica at \$876/month and route reads to it

Option B is identical in cost here. But if you can use a `db.r6g.xlarge` replica (\$438/month) because analytics queries are not CPU-intensive, Option B costs **\$438/month less** than upgrading the primary. That is \$5,256/year saved.

The break-even analysis changes dramatically when you factor in Reserved Instances. A `db.r6g.2xlarge` on a 1-year RI costs approximately **\$553/month** — 37% less than on-demand. If your replica is running 24/7 for a predictable workload, not buying the RI is itself a financial decision with real consequences.

This Python script automates the break-even calculation for your specific workload:

```

#!/usr/bin/env python3
"""
replica_breakeven_calculator.py

Calculates whether adding a read replica is cost-effective compared
to scaling up the primary instance. Pulls current CPU metrics from
CloudWatch to determine if the primary is actually CPU-constrained
by read workload.

Schedule this as a monthly report via Lambda or a cron job.
"""

import boto3
import json
from datetime import datetime, timedelta
from dataclasses import dataclass
from typing import Optional

# RDS on-demand pricing (us-east-1, PostgreSQL, as of 2024)
# Update these periodically or integrate with the AWS Price List API
RDS_PRICING = {
    "db.t3.medium": {"vcpu": 2, "ram_gb": 4, "monthly_od": 72, "monthly_1yr_ri": 46},
    "db.t3.large": {"vcpu": 2, "ram_gb": 8, "monthly_od": 144, "monthly_1yr_ri": 92},

```

```

"db.r6g.large": {"vcpu": 2, "ram_gb": 16, "monthly_od": 219, "monthly_1yr_ri": 138},
"db.r6g.xlarge": {"vcpu": 4, "ram_gb": 32, "monthly_od": 438, "monthly_1yr_ri": 276},
"db.r6g.2xlarge": {"vcpu": 8, "ram_gb": 64, "monthly_od": 876, "monthly_1yr_ri": 553},
"db.r6g.4xlarge": {"vcpu": 16, "ram_gb": 128, "monthly_od": 1752, "monthly_1yr_ri": 1105},
"db.r6g.8xlarge": {"vcpu": 32, "ram_gb": 256, "monthly_od": 3504, "monthly_1yr_ri": 2210},
"db.r7g.large": {"vcpu": 2, "ram_gb": 16, "monthly_od": 228, "monthly_1yr_ri": 144},
"db.r7g.xlarge": {"vcpu": 4, "ram_gb": 32, "monthly_od": 456, "monthly_1yr_ri": 288},
"db.r7g.2xlarge": {"vcpu": 8, "ram_gb": 64, "monthly_od": 912, "monthly_1yr_ri": 575},
"db.r7g.4xlarge": {"vcpu": 16, "ram_gb": 128, "monthly_od": 1824, "monthly_1yr_ri": 1150},
"db.m6g.large": {"vcpu": 2, "ram_gb": 8, "monthly_od": 180, "monthly_1yr_ri": 114},
"db.m6g.xlarge": {"vcpu": 4, "ram_gb": 16, "monthly_od": 360, "monthly_1yr_ri": 227},
"db.m6g.2xlarge": {"vcpu": 8, "ram_gb": 32, "monthly_od": 720, "monthly_1yr_ri": 454},
"db.m6g.4xlarge": {"vcpu": 16, "ram_gb": 64, "monthly_od": 1440, "monthly_1yr_ri": 908},
}

@dataclass
class BreakEvenResult:
    instance_id: str
    current_class: str
    avg_cpu_percent: float
    read_iops_avg: float
    write_iops_avg: float
    estimated_read_cpu_fraction: float
    recommended_action: str
    upgrade_option: Optional[str]
    upgrade_monthly_cost_od: float
    replica_option: Optional[str]
    replica_monthly_cost_od: float
    monthly_savings_vs_upgrade: float
    annual_savings_vs_upgrade: float
    payback_period_months: float
    ri_recommended: bool

def get_metric_average(cw_client, instance_id: str, metric_name: str,
                      days_back: int = 14) -> float:
    """Fetch average CloudWatch metric for an RDS instance over the last N days."""
    end_time = datetime.utcnow()
    start_time = end_time - timedelta(days=days_back)

    response = cw_client.get_metric_statistics(
        Namespace="AWS/RDS",
        MetricName=metric_name,
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": instance_id}],
        StartTime=start_time,
        EndTime=end_time,
        Period=days_back * 86400,
        Statistics=["Average"]
    )
    datapoints = response.get("Datapoints", [])
    if not datapoints:
        return 0.0
    return datapoints[0]["Average"]

def get_next_instance_class(current_class: str) -> Optional[str]:
    """Return the next larger instance class in the same family."""
    # Build ordered lists per family
    families = {
        "r6g": ["db.r6g.large", "db.r6g.xlarge", "db.r6g.2xlarge",
                "db.r6g.4xlarge", "db.r6g.8xlarge"],
        "r7g": ["db.r7g.large", "db.r7g.xlarge", "db.r7g.2xlarge",
                "db.r7g.4xlarge"],
        "m6g": ["db.m6g.large", "db.m6g.xlarge", "db.m6g.2xlarge",
                "db.m6g.4xlarge"],
        "t3": ["db.t3.medium", "db.t3.large"],
    }
    for family, sizes in families.items():
        if current_class in sizes:
            idx = sizes.index(current_class)
            if idx + 1 < len(sizes):
                return sizes[idx + 1]

```



```

return None

def get_smaller_instance_class(current_class: str) -> Optional[str]:
    """Return a smaller instance class suitable for a read replica."""
    families = {
        "r6g": ["db.r6g.large", "db.r6g.xlarge", "db.r6g.2xlarge",
                "db.r6g.4xlarge", "db.r6g.8xlarge"],
        "r7g": ["db.r7g.large", "db.r7g.xlarge", "db.r7g.2xlarge",
                "db.r7g.4xlarge"],
        "m6g": ["db.m6g.large", "db.m6g.xlarge", "db.m6g.2xlarge",
                "db.m6g.4xlarge"],
        "t3": ["db.t3.medium", "db.t3.large"],
    }
    for family, sizes in families.items():
        if current_class in sizes:
            idx = sizes.index(current_class)
            if idx - 1 >= 0:
                return sizes[idx - 1]
    return None

def analyze_instance(rds_client, cw_client, instance_id: str) -> Optional[BreakEvenResult]:
    """Run the break-even analysis for a single RDS primary instance."""
    try:
        resp = rds_client.describe_db_instances(DBInstanceIdentifier=instance_id)
        db = resp["DBInstances"][0]
    except Exception as e:
        print(f" Could not describe instance {instance_id}: {e}")
        return None

    current_class = db["DBInstanceClass"]
    if current_class not in RDS_PRICING:
        print(f" Instance class {current_class} not in pricing table - skipping.")
        return None

    # Is this already a replica?
    if db.get("ReadReplicaSourceDBInstanceIdentifier"):
        print(f" {instance_id} is itself a replica - skipping primary analysis.")
        return None

    print(f" Fetching metrics for {instance_id} ({current_class})...")
    avg_cpu = get_metric_average(cw_client, instance_id, "CPUUtilization")
    read_iops_avg = get_metric_average(cw_client, instance_id, "ReadIOPS")
    write_iops_avg = get_metric_average(cw_client, instance_id, "WriteIOPS")

    # Estimate what fraction of CPU is consumed by reads.
    # If read IOPS >> write IOPS, reads are likely the dominant CPU consumer.
    total_iops = read_iops_avg + write_iops_avg
    read_cpu_fraction = (read_iops_avg / total_iops) if total_iops > 0 else 0.0

    current_od_cost = RDS_PRICING[current_class]["monthly_od"]
    current_ri_cost = RDS_PRICING[current_class]["monthly_1yr_ri"]

    upgrade_class = get_next_instance_class(current_class)
    replica_class = get_smaller_instance_class(current_class) or current_class

    upgrade_od_cost = RDS_PRICING.get(upgrade_class, {}).get("monthly_od", 0) if upgrade_class else 0
    replica_od_cost = RDS_PRICING.get(replica_class, {}).get("monthly_od", 0)
    replica_ri_cost = RDS_PRICING.get(replica_class, {}).get("monthly_1yr_ri", 0)

    # Marginal cost of each option relative to the current spend
    marginal_upgrade = upgrade_od_cost - current_od_cost
    marginal_replica = replica_od_cost # replica is additive

    monthly_savings = marginal_upgrade - marginal_replica
    annual_savings = monthly_savings * 12

    # Payback period: if savings are negative, the replica never pays back
    if monthly_savings > 0:
        payback_months = 0.0 # replica is cheaper immediately
    elif monthly_savings < 0:

```

```

        payback_months = float("inf") # upgrade is cheaper - replica doesn't pay back
    else:
        payback_months = 0.0

    # Recommendation logic
    if avg_cpu < 40:
        action = "NO_ACTION_CPU_HEALTHY"
    elif avg_cpu < 70 and read_cpu_fraction < 0.5:
        action = "MONITOR_WRITE_BOUND_INSTANCE"
    elif avg_cpu >= 70 and read_cpu_fraction >= 0.5 and monthly_savings >= 0:
        action = "ADD_READ_REPLICA"
    elif avg_cpu >= 70 and read_cpu_fraction < 0.5:
        action = "UPGRADE_PRIMARY_WRITE_BOUND"
    else:
        action = "EVALUATE_MANUALLY"

    ri_recommended = (action == "ADD_READ_REPLICA") and (replica_ri_cost > 0)

    return BreakEvenResult(
        instance_id=instance_id,
        current_class=current_class,
        avg_cpu_percent=round(avg_cpu, 1),
        read_iops_avg=round(read_iops_avg, 1),
        write_iops_avg=round(write_iops_avg, 1),
        estimated_read_cpu_fraction=round(read_cpu_fraction, 2),
        recommended_action=action,
        upgrade_option=upgrade_class,
        upgrade_monthly_cost_od=upgrade_od_cost,
        replica_option=replica_class,
        replica_monthly_cost_od=replica_od_cost,
        monthly_savings_vs_upgrade=round(monthly_savings, 2),
        annual_savings_vs_upgrade=round(annual_savings, 2),
        payback_period_months=payback_months,
        ri_recommended=ri_recommended,
    )

def run_breakeven_report(region: str = "us-east-1"):
    """Run the break-even analysis across all primary RDS instances in a region."""
    rds = boto3.client("rds", region_name=region)
    cw = boto3.client("cloudwatch", region_name=region)

    paginator = rds.get_paginator("describe_db_instances")
    results = []

    print(f"\n=== Read Replica Break-Even Analysis - {region} ===\n")

    for page in paginator.paginate():
        for db in page["DBInstances"]:
            iid = db["DBInstanceIdentifier"]
            # Skip replicas and non-available instances
            if db.get("ReadReplicaSourceDBInstanceIdentifier"):
                continue
            if db.get("DBInstanceStatus") != "available":
                continue
            print(f"Analyzing: {iid}")
            result = analyze_instance(rds, cw, iid)
            if result:
                results.append(result)

    # Print summary
    print("\n" + "=" * 80)
    print(f"{'Instance':<35} {'Class':<20} {'CPU%':>5} {'Action':<35} {'Annual $':>10}")
    print("=" * 80)
    for r in sorted(results, key=lambda x: x.annual_savings_vs_upgrade, reverse=True):
        print(f"{'r.instance_id':<35} {'r.current_class':<20} {'r.avg_cpu_percent':>5} "
              f"{'r.recommended_action':<35} {'r.annual_savings_vs_upgrade':>10,.0f}")

    # Actionable items only
    add_replica = [r for r in results if r.recommended_action == "ADD_READ_REPLICA"]
    if add_replica:

```

```

print(f"\n{'='*80}")
print("INSTANCES WHERE ADDING A READ REPLICAS IS MORE COST-EFFECTIVE THAN UPGRADING:")
print(f"\n{'='*80}")
for r in add_replica:
    print(f"\n Instance:           {r.instance_id}")
    print(f" Current class:   {r.current_class} (${r.upgrade_monthly_cost_od -
    ↪ (r.upgrade_monthly_cost_od - RDS_PRICING[r.current_class]['monthly_od'])}/mo)")
    print(f" Upgrade option:  {r.upgrade_option} (+${r.upgrade_monthly_cost_od -
    ↪ RDS_PRICING[r.current_class]['monthly_od']:, .0f}/mo)")
    print(f" Replica option:  {r.replica_option} (+${r.replica_monthly_cost_od:, .0f}/mo)")
    print(f" Monthly savings vs upgrade: ${r.monthly_savings_vs_upgrade:, .0f}")
    print(f" Annual savings:  ${r.annual_savings_vs_upgrade:, .0f}")
    if r.ri_recommended:
        ri_cost = RDS_PRICING[r.replica_option]["monthly_1yr_ri"]
        additional_ri_savings = (r.replica_monthly_cost_od - ri_cost) * 12
        print(f" RI recommendation: Buy 1-yr RI for replica → additional
        ↪ ${additional_ri_savings:, .0f}/yr savings")

# Export full results to JSON for further processing
output = [vars(r) for r in results]
filename = f"breakeven_analysis_{datetime.utcnow().strftime('%Y%m%d')}.json"
with open(filename, "w") as f:
    json.dump(output, f, indent=2)
print(f"\nFull results written to {filename}")

if __name__ == "__main__":
    import sys
    region = sys.argv[1] if len(sys.argv) > 1 else "us-east-1"
    run_breakeven_report(region)

```

19.4 Right-Sizing Replicas: The Replica Does Not Need to Match the Primary

One of the most common and costly mistakes in read replica deployments is provisioning the replica at the same instance class as the primary. This instinct comes from a legitimate concern — if the primary fails, the replica must be promoted quickly. But it conflates **failover readiness** with **read workload requirements**, and the two are often very different.

A `db.r6g.4xlarge` primary handling heavy OLTP write traffic may generate a relatively modest stream of read queries from a reporting dashboard. The replica serving those queries may run comfortably at 20% CPU on a `db.r6g.xlarge` — a 75% reduction in compute cost for that instance.

The key insight is that **replication throughput is limited by the primary's write rate**, not by the replica's compute class. The replica needs enough CPU to apply WAL and serve queries, not to handle the full OLTP concurrency of the primary. The table below illustrates a common sizing pattern:

Primary Class	Primary Workload	Replica Workload Type	Appropriate Replica Class	Monthly Savings
db.r6g.4xlarge	High-write OLTP	Light reporting (5–10 queries/hr)	db.r6g.large	~\$1,533/mo
db.r6g.2xlarge	Mixed OLTP	Medium analytics (hourly batch)	db.r6g.xlarge	~\$438/mo

Primary Class	Primary Workload	Replica Workload Type	Appropriate Replica Class	Monthly Savings
db.r6g.8xlarge	Write-intensive ETL	Read-heavy API queries	db.r6g.2xlarge	~\$2,628/mo
db.r7g.4xlarge	OLTP + reporting	Dedicated BI queries	db.r7g.xlarge	~\$1,368/mo

Right-sizing is straightforward in practice but requires measuring the right thing: **replica CPU and memory utilization under actual query load**, not the primary's utilization. Use the following queries to assess whether a replica is oversized:

```
-- PostgreSQL: assess whether a replica is a candidate for downsizing
-- Run directly on the replica instance

-- 1. Current CPU and memory pressure indicators
SELECT
    numbackends                                AS current_connections,
    xact_commit + xact_rollback                 AS total_transactions,
    blks_hit::FLOAT / NULLIF(blks_hit + blks_read, 0) * 100 AS cache_hit_ratio_pct,
    tup_fetched,
    tup_returned
FROM pg_stat_database
WHERE datname = current_database();

-- 2. Peak connection watermark (requires pg_stat_activity history or pg_activity extension)
SELECT
    MAX(count) AS peak_connections
FROM (
    SELECT
        date_trunc('hour', query_start) AS hour_bucket,
        COUNT(*) AS count
    FROM pg_stat_activity
    WHERE state != 'idle'
    GROUP BY 1
) hourly;

-- 3. Identify if replica memory is pressured (many buffer evictions)
SELECT
    checkpoints_timed,
    checkpoints_req,
    buffers_checkpoint,
    buffers_clean,
    buffers_backend,
    buffers_alloc,
    -- High buffers_backend relative to buffers_checkpoint suggests memory pressure
    ROUND(buffers_backend::NUMERIC / NULLIF(buffers_checkpoint + buffers_clean + buffers_backend, 0) *
        ↪ 100, 1)
        AS pct_written_by_backends
FROM pg_stat_bgwriter;

-- 4. Long-running queries on this replica - size the replica to handle the slowest query
SELECT
    pid,
    now() - query_start AS duration,
    wait_event_type,
    wait_event,
    LEFT(query, 120) AS query_preview
FROM pg_stat_activity
WHERE state = 'active'
    AND pg_is_in_recovery() -- confirm we are on a replica
ORDER BY duration DESC
LIMIT 10;

-- SQL Server: assess replica right-sizing on an Always On secondary
```

```

-- Run on the secondary replica

-- 1. Memory grant utilization - indicates if queries are spilling to disk (memory pressure)
SELECT TOP 20
    qs.execution_count,
    qs.total_grant_kb / qs.execution_count          AS avg_grant_kb,
    qs.max_grant_kb,
    qs.total_used_grant_kb / qs.execution_count     AS avg_used_grant_kb,
    -- Large gap between max_grant and avg_used suggests the replica could use less RAM
    qs.max_grant_kb - (qs.total_used_grant_kb / qs.execution_count) AS unused_grant_kb,
    SUBSTRING(qt.text, (qs.statement_start_offset / 2) + 1, 200) AS query_snippet
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
WHERE qs.execution_count > 10
ORDER BY avg_grant_kb DESC;

-- 2. Page life expectancy - core memory pressure indicator
-- Values consistently above 300s suggest buffer pool is adequate
SELECT
    object_name,
    counter_name,
    cntr_value AS page_life_expectancy_seconds
FROM sys.dm_os_performance_counters
WHERE counter_name = 'Page life expectancy'
    AND object_name LIKE '%Buffer Manager%';

-- 3. CPU utilization from ring buffer (last 256 minutes)
SELECT TOP 30
    record.value('(/Record/SchedulerMonitorEvent/SystemHealth/ProcessUtilization)[1]', 'int')
        AS sql_cpu_utilization,
    record.value('(/Record/SchedulerMonitorEvent/SystemHealth/SystemIdle)[1]', 'int')
        AS system_idle,
    DATEADD(ms, -1 * (ts_now - [timestamp]), GETDATE()) AS sample_time
FROM (
    SELECT
        [timestamp],
        CONVERT(XML, record) AS record,
        cpu_ticks / (cpu_ticks / ms_ticks) AS ts_now
    FROM sys.dm_os_ring_buffers rb
    CROSS JOIN sys.dm_os_sys_info si
    WHERE rb.ring_buffer_type = N'RING_BUFFER_SCHEDULER_MONITOR'
        AND rb.record LIKE '%<SystemHealth>%'
) AS ring_data
ORDER BY sample_time DESC;

```

The decision rule for right-sizing a replica is straightforward:

- If **average CPU < 30%** and **page life expectancy > 300s** (or PostgreSQL cache hit ratio > 95%) over a two-week window, the replica can safely be downsized one class.
- If **average CPU > 70%** or **queries are consistently spilling to disk**, the replica may need to stay at its current size or be upgraded.
- Never downsize a replica that is also your **standby for failover** without a separate DR analysis — the promotion path must be validated independently.

19.5 Automating Replica Lifecycle Management

Manual replica management is the fastest path to cost creep. Teams that rely on tickets and human decision-making to create and delete replicas find that replicas created for one-time tasks persist for months. Automating the lifecycle requires three capabilities: **automated creation**, **automated deletion**, and **scheduled scaling**.

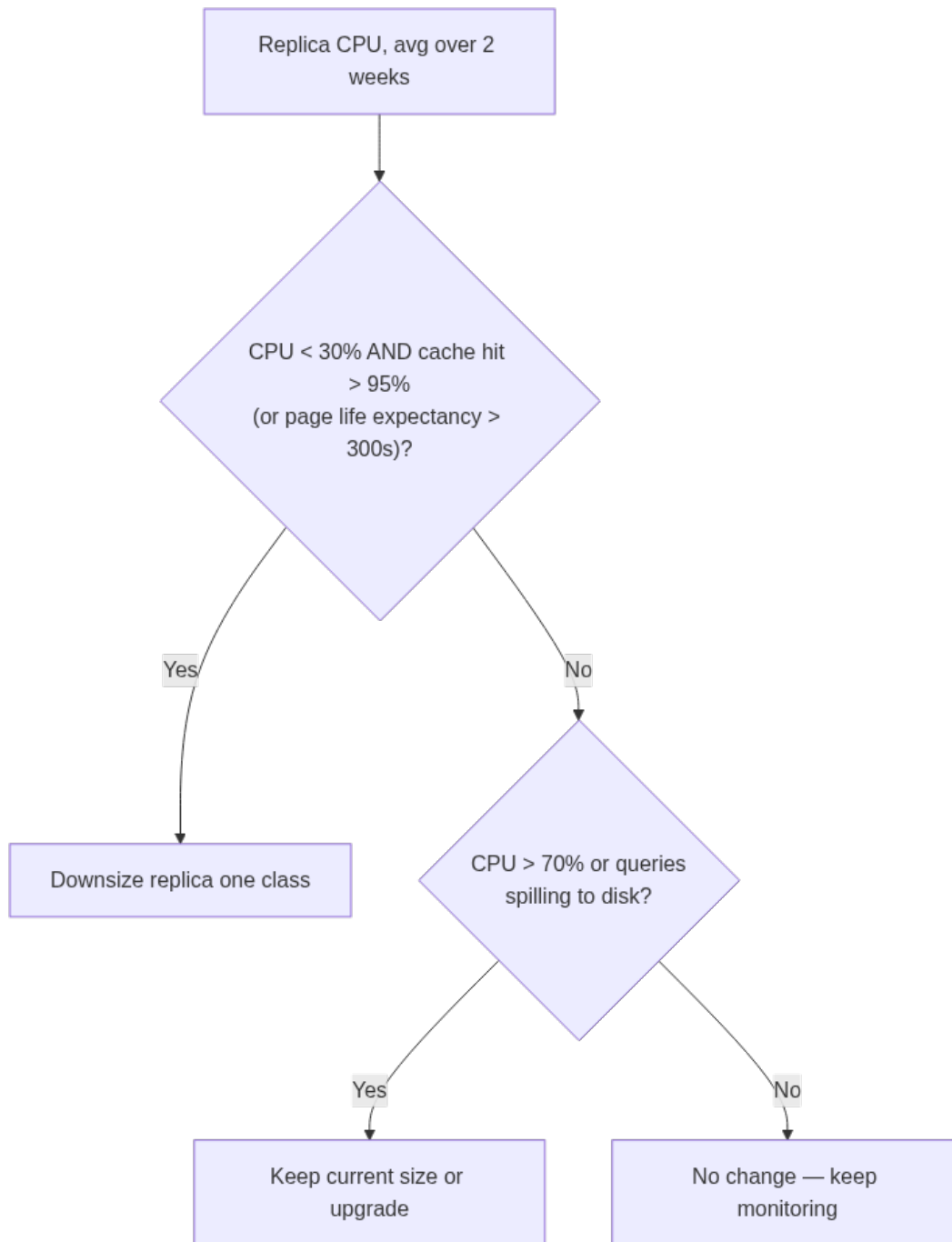


Figure 19: Right-sizing decision rule for an existing read replica, based on two weeks of CPU and cache/memory-pressure data.

Automated Deletion of Idle Replicas

The bash audit script in section 19.2 identifies idle replicas. The next step is automating the deletion with appropriate guardrails. The following Lambda function handles this with a mandatory delay and SNS notification so engineers are never surprised:

```
#!/usr/bin/env python3
"""
lambda_replica_lifecycle.py

Lambda function that:
1. Identifies RDS read replicas with zero or near-zero utilization
   over the past N days.
2. Tags them as idle candidates.
3. After a configurable grace period, deletes them if they remain idle.
4. Sends SNS notifications at each stage.

Deploy as a Lambda with an EventBridge (CloudWatch Events) schedule:
- Rate: once per day
- Memory: 256 MB
- Timeout: 5 minutes
- IAM: rds:Describe*, rds>DeleteDBInstance, rds:AddTagsToResource,
      cloudwatch:GetMetricStatistics, sns:Publish
"""

import boto3
import json
import os
from datetime import datetime, timedelta

# Configuration - override via Lambda environment variables
REGION = os.environ.get("AWS_REGION", "us-east-1")
SNS_TOPIC_ARN = os.environ.get("SNS_TOPIC_ARN", "")
IDLE_THRESHOLD_READ_IOPS = float(os.environ.get("IDLE_THRESHOLD_READ_IOPS", "5"))
IDLE_THRESHOLD_CONNECTIONS = float(os.environ.get("IDLE_THRESHOLD_CONNECTIONS", "3"))
EVALUATION_DAYS = int(os.environ.get("EVALUATION_DAYS", "7"))
GRACE_PERIOD_DAYS = int(os.environ.get("GRACE_PERIOD_DAYS", "3"))
DRY_RUN = os.environ.get("DRY_RUN", "true").lower() == "true"
EXCLUSION_TAGS = {"replica-lifecycle": "exempt", "environment": "production-dr"}

rds = boto3.client("rds", region_name=REGION)
cw = boto3.client("cloudwatch", region_name=REGION)
sns = boto3.client("sns", region_name=REGION) if SNS_TOPIC_ARN else None

def get_avg_metric(instance_id: str, metric_name: str, days: int) -> float:
    end = datetime.utcnow()
    start = end - timedelta(days=days)
    resp = cw.get_metric_statistics(
        Namespace="AWS/RDS",
        MetricName=metric_name,
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": instance_id}],
        StartTime=start,
        EndTime=end,
        Period=days * 86400,
        Statistics=["Average"],
    )
    points = resp.get("Datapoints", [])
    return points[0]["Average"] if points else 0.0

def is_exempt(db_instance: dict) -> bool:
    """Return True if the instance is tagged to be excluded from lifecycle management."""
    tags = {t["Key"]: t["Value"] for t in db_instance.get("TagList", [])}
    for key, val in EXCLUSION_TAGS.items():
        if tags.get(key) == val:
            return True
    return False
```

```

def get_idle_tag_date(db_instance: dict) -> datetime | None:
    """Return the datetime when the replica was first tagged as idle, or None."""
    tags = {t["Key"]: t["Value"] for t in db_instance.get("TagList", [])}
    tagged_str = tags.get("replica-lifecycle-idle-since")
    if tagged_str:
        try:
            return datetime.fromisoformat(tagged_str)
        except ValueError:
            return None
    return None

def notify(subject: str, message: str) -> None:
    if sns and SNS_TOPIC_ARN:
        sns.publish(
            TopicArn=SNS_TOPIC_ARN,
            Subject=subject,
            Message=message,
        )
    print(f"[NOTIFY] {subject}: {message}")

def tag_as_idle(instance_id: str, arn: str) -> None:
    now_str = datetime.utcnow().isoformat()
    if not DRY_RUN:
        rds.add_tags_to_resource(
            ResourceName=arn,
            Tags=[{"Key": "replica-lifecycle-idle-since", "Value": now_str}]
        )
    print(f" [TAG] {instance_id} tagged idle-since={now_str} (dry_run={DRY_RUN})")

def delete_replica(instance_id: str) -> None:
    if not DRY_RUN:
        rds.delete_db_instance(
            DBInstanceIdentifier=instance_id,
            SkipFinalSnapshot=True,
            DeleteAutomatedBackups=False, # keep automated backups for 7 days
        )
    notify(
        subject=f"Read Replica Deleted: {instance_id}",
        message=(
            f"The idle read replica '{instance_id}' has been automatically deleted "
            f"after {GRACE_PERIOD_DAYS} days of confirmed idle status.\n"
            f"Region: {REGION}\n"
            f"Dry run: {DRY_RUN}"
        )
    )
    print(f" [DELETE] {instance_id} deleted (dry_run={DRY_RUN})")

def lambda_handler(event, context):
    print(f"Starting replica lifecycle management - region={REGION}, dry_run={DRY_RUN}")
    tagged_this_run = []
    deleted_this_run = []

    paginator = rds.get_paginator("describe_db_instances")
    for page in paginator.paginate():
        for db in page["DBInstances"]:
            # Only process read replicas
            if not db.get("ReadReplicaSourceDBInstanceIdentifier"):
                continue
            if db.get("DBInstanceStatus") != "available":
                continue

            iid = db["DBInstanceIdentifier"]
            arn = db["DBInstanceArn"]

            if is_exempt(db):

```



```

        print(f" [EXEMPT] {iid}")
        continue

    # Collect utilization metrics
    avg_read_iops = get_avg_metric(iid, "ReadIOPS", EVALUATION_DAYS)
    avg_connections = get_avg_metric(iid, "DatabaseConnections", EVALUATION_DAYS)

    is_idle = (
        avg_read_iops < IDLE_THRESHOLD_READ_IOPS and
        avg_connections < IDLE_THRESHOLD_CONNECTIONS
    )

    print(f" {iid}: ReadIOPS={avg_read_iops:.1f}, Connections={avg_connections:.1f},
    ↪ idle={is_idle}")

    if not is_idle:
        # If it was previously tagged idle but is active again, remove the tag
        idle_since = get_idle_tag_date(db)
        if idle_since:
            if not DRY_RUN:
                rds.remove_tags_from_resource(
                    ResourceName=arn,
                    TagKeys=["replica-lifecycle-idle-since"]
                )
            print(f" [UNTAG] {iid} is active again - removing idle tag")
            continue

    idle_since = get_idle_tag_date(db)

    if idle_since is None:
        # First time we see this as idle - tag it and start the grace period
        tag_as_idle(iid, arn)
        tagged_this_run.append(iid)
        notify(
            subject=f"Idle Replica Detected: {iid}",
            message=(
                f"Read replica '{iid}' has been idle for {EVALUATION_DAYS} days "
                f"(ReadIOPS={avg_read_iops:.1f}, Connections={avg_connections:.1f}).\n"
                f"It will be deleted in {GRACE_PERIOD_DAYS} days unless activity resumes.\n"
                f"To exempt it, add tag replica-lifecycle=exempt.\n"
                f"Region: {REGION}"
            )
        )
    else:
        days_idle = (datetime.utcnow() - idle_since).days
        print(f" {iid} has been idle for {days_idle} days (grace={GRACE_PERIOD_DAYS})")
        if days_idle >= GRACE_PERIOD_DAYS:
            delete_replica(iid)
            deleted_this_run.append(iid)

    summary = {
        "region": REGION,
        "dry_run": DRY_RUN,
        "evaluation_days": EVALUATION_DAYS,
        "grace_period_days": GRACE_PERIOD_DAYS,
        "tagged_idle": tagged_this_run,
        "deleted": deleted_this_run,
    }
    print(f"\nSummary: {json.dumps(summary, indent=2)}")
    return summary

```

Scheduled Scaling: Replicas That Only Work Business Hours

Many analytics replicas serve dashboard queries from 8 AM to 6 PM, then sit fully idle overnight and on weekends. For these, the right answer is not permanent deletion but **scheduled scaling to a stopped state** or a time-based start/stop pattern.

RDS does not support stopping read replicas independently of the primary (you can stop an entire instance, but a replica cannot be stopped while the primary runs). The workaround is to **delete the replica at end of business day and recreate it at the start of the next business day**. This sounds aggressive but works well with Aurora read replicas because Aurora's shared storage means there is no data to copy on creation — the new replica is available in minutes.

For standard RDS PostgreSQL or MySQL, deletion and recreation is slower (replica must replay WAL from the primary after creation), but if overnight idle cost exceeds the operational complexity, the math may still favor it.

The EventBridge + Lambda pattern for this is:

```
#!/usr/bin/env python3
"""
scheduled_replica_manager.py

Creates or deletes a read replica on a schedule.
Two EventBridge rules call this Lambda:
- Rule 1: event_type=CREATE - Monday-Friday 07:45 UTC
- Rule 2: event_type=DELETE - Monday-Friday 19:00 UTC

Environment variables:
PRIMARY_INSTANCE_ID - e.g., prod-postgres-primary
REPLICA_INSTANCE_ID - e.g., prod-postgres-reporting-replica
REPLICA_CLASS        - e.g., db.r6g.xlarge
REPLICA_AZ           - e.g., us-east-1b
SNS_TOPIC_ARN        - for notifications
"""

import boto3
import os
import json
from datetime import datetime

REGION = os.environ.get("AWS_REGION", "us-east-1")
PRIMARY_ID = os.environ.get("PRIMARY_INSTANCE_ID")
REPLICA_ID = os.environ.get("REPLICA_INSTANCE_ID")
REPLICA_CLASS = os.environ.get("REPLICA_CLASS", "db.r6g.xlarge")
REPLICA_AZ = os.environ.get("REPLICA_AZ", "")
SNS_TOPIC_ARN = os.environ.get("SNS_TOPIC_ARN", "")
DRY_RUN = os.environ.get("DRY_RUN", "true").lower() == "true"

rds = boto3.client("rds", region_name=REGION)
sns = boto3.client("sns", region_name=REGION) if SNS_TOPIC_ARN else None

def notify(subject: str, message: str) -> None:
    if sns and SNS_TOPIC_ARN:
        sns.publish(TopicArn=SNS_TOPIC_ARN, Subject=subject, Message=message)
    print(f"[NOTIFY] {subject}")

def replica_exists() -> bool:
    try:
        resp = rds.describe_db_instances(DBInstanceIdentifier=REPLICA_ID)
        status = resp["DBInstances"][0]["DBInstanceStatus"]
        print(f"Replica {REPLICA_ID} exists, status={status}")
        return True
    except rds.exceptions.DBInstanceNotFoundFault:
        return False

def create_replica() -> None:
    if replica_exists():
        print(f"Replica {REPLICA_ID} already exists - skipping creation.")
        return
```

```

kwargs = {
    "DBInstanceIdentifier":     REPLICATION_ID,
    "SourceDBInstanceIdentifier": PRIMARY_ID,
    "DBInstanceClass":         REPLICATION_CLASS,
    "AutoMinorVersionUpgrade":  False,
    "PubliclyAccessible":      False,
    "Tags": [
        {"Key": "replica-lifecycle", "Value": "scheduled"},
        {"Key": "created-by",         "Value": "scheduled-replica-manager"},
        {"Key": "created-at",         "Value": datetime.utcnow().isoformat()},
    ],
}
if REPLICATION_AZ:
    kwargs["AvailabilityZone"] = REPLICATION_AZ

print(f" Creating replica {REPLICATION_ID} from {PRIMARY_ID} as {REPLICATION_CLASS} (dry_run={DRY_RUN})")
if not DRY_RUN:
    rds.create_db_instance_read_replica(**kwargs)

notify(
    subject=f"Read Replica Created: {REPLICATION_ID}",
    message=(
        f"Scheduled replica '{REPLICATION_ID}' has been created from '{PRIMARY_ID}'.\n"
        f"Class: {REPLICATION_CLASS}\n"
        f"Region: {REGION}\n"
        f"Dry run: {DRY_RUN}\n"
        f"Time: {datetime.utcnow().isoformat()} UTC"
    )
)

def delete_replica() -> None:
    if not replica_exists():
        print(f" Replica {REPLICATION_ID} does not exist - skipping deletion.")
        return

    print(f" Deleting replica {REPLICATION_ID} (dry_run={DRY_RUN})")
    if not DRY_RUN:
        rds.delete_db_instance(
            DBInstanceIdentifier=REPLICATION_ID,
            SkipFinalSnapshot=True,
            DeleteAutomatedBackups=False,
        )

    notify(
        subject=f"Read Replica Deleted (Scheduled): {REPLICATION_ID}",
        message=(
            f"Scheduled replica '{REPLICATION_ID}' has been deleted per end-of-business schedule.\n"
            f"It will be recreated at next business day start.\n"
            f"Region: {REGION}\n"
            f"Dry run: {DRY_RUN}\n"
            f"Time: {datetime.utcnow().isoformat()} UTC"
        )
    )

def lambda_handler(event, context):
    event_type = event.get("event_type", "").upper()
    print(f"scheduled_replica_manager invoked: event_type={event_type}, dry_run={DRY_RUN}")

    if not PRIMARY_ID or not REPLICATION_ID:
        raise ValueError("PRIMARY_INSTANCE_ID and REPLICATION_INSTANCE_ID must be set.")

    if event_type == "CREATE":
        create_replica()
    elif event_type == "DELETE":
        delete_replica()
    else:
        raise ValueError(f"Unknown event_type: '{event_type}'. Expected CREATE or DELETE.")

```

```
return {"event_type": event_type, "dry_run": DRY_RUN, "timestamp": datetime.utcnow().isoformat()}
```

The Terraform to wire up the two EventBridge rules looks like this:

```
# scheduled_replica_rules.tf
# Assumes the Lambda function is already deployed as aws_lambda_function.replica_manager

resource "aws_cloudwatch_event_rule" "replica_create" {
  name           = "replica-create-business-hours"
  description    = "Create reporting replica at start of business day (Mon-Fri 07:45 UTC)"
  schedule_expression = "cron(45 7 ? * MON-FRI *)"
}

resource "aws_cloudwatch_event_rule" "replica_delete" {
  name           = "replica-delete-end-of-day"
  description    = "Delete reporting replica at end of business day (Mon-Fri 19:00 UTC)"
  schedule_expression = "cron(0 19 ? * MON-FRI *)"
}

resource "aws_cloudwatch_event_target" "replica_create_target" {
  rule       = aws_cloudwatch_event_rule.replica_create.name
  target_id  = "replica-manager-create"
  arn        = aws_lambda_function.replica_manager.arn

  input = jsonencode({ event_type = "CREATE" })
}

resource "aws_cloudwatch_event_target" "replica_delete_target" {
  rule       = aws_cloudwatch_event_rule.replica_delete.name
  target_id  = "replica-manager-delete"
  arn        = aws_lambda_function.replica_manager.arn

  input = jsonencode({ event_type = "DELETE" })
}

resource "aws_lambda_permission" "allow_eventbridge_create" {
  statement_id = "AllowEventBridgeCreate"
  action       = "lambda:InvokeFunction"
  function_name = aws_lambda_function.replica_manager.function_name
  principal    = "events.amazonaws.com"
  source_arn   = aws_cloudwatch_event_rule.replica_create.arn
}

resource "aws_lambda_permission" "allow_eventbridge_delete" {
  statement_id = "AllowEventBridgeDelete"
  action       = "lambda:InvokeFunction"
  function_name = aws_lambda_function.replica_manager.function_name
  principal    = "events.amazonaws.com"
  source_arn   = aws_cloudwatch_event_rule.replica_delete.arn
}
```

The cost arithmetic for a scheduled replica pattern is compelling. A `db.r6g.xlarge` that runs only during business hours (10 hours/day, 5 days/week) operates roughly 30% of the hours in a month versus a replica running 24/7. At on-demand pricing of \$438/month for continuous operation, the scheduled pattern reduces that line item to approximately \$131/month — a saving of \$307/month, or \$3,684/year, for a single replica.

19.6 Cross-Region Replicas: When the Replication Tax Exceeds the Value

Cross-region replicas are the most expensive replica variant and the most frequently justified by vague disaster recovery requirements rather than actual RTO/RPO analysis. Before examining the

cost, it is worth stating the use cases where cross-region replicas genuinely earn their cost:

1. **Hard regulatory requirements** specifying that data must exist in multiple geographic regions.
2. **Serving read traffic to users in a remote geography** where latency to the primary region would degrade user experience measurably.
3. **DR with a tested, documented RTO** that requires a warm standby in another region rather than a cold restore from S3.

If your cross-region replica does not satisfy at least one of these criteria with supporting evidence, it is a cost liability.

The cost model for a cross-region replica has three layers that are easy to overlook:

Layer 1: Compute — The replica instance itself, billed at the destination region’s on-demand rate, which may differ from the source region.

Layer 2: Replication data transfer — AWS charges for data leaving the source region at standard inter-region transfer rates. A write-heavy primary generating 200 GB/day of WAL ships approximately 6 TB/month to the cross-region replica. At \$0.02/GB (us-east-1 to us-west-2), that is **\$120/month in transfer fees** even before the replica serves a single query. Busier primaries can drive this number into the thousands monthly.

Layer 3: Destination region storage — On RDS (non-Aurora), the replica’s allocated storage is billed separately in the destination region.

The following query helps you measure your actual replication volume on the primary, which feeds directly into the transfer cost calculation:

```
-- PostgreSQL: estimate WAL generation rate to project cross-region transfer costs
-- Run on the primary instance

SELECT
    current_lsn,
    pg_size_pretty(pg_wal_lsn_diff(current_lsn, '0/0')) AS total_wal_generated,
    -- WAL generated in the last hour
    pg_size_pretty(
        pg_wal_lsn_diff(
            pg_current_wal_lsn(),
            pg_lsn(
                (SELECT setting FROM pg_settings WHERE name = 'wal_segment_size')::BIGINT * 0
            )
        )
    ) AS placeholder
FROM (SELECT pg_current_wal_lsn() AS current_lsn) lsn_snapshot;

-- More practical: WAL generation rate from pg_stat_replication
-- Look at how fast LSN advances across replicas
SELECT
    application_name,
    client_addr,
    pg_size_pretty(sent_lsn - replay_lsn) AS replication_lag_size,
    pg_size_pretty(pg_wal_lsn_diff(sent_lsn, '0/0')) AS total_bytes_sent_from_origin,
    write_lag,
    flush_lag,
    replay_lag
FROM pg_stat_replication
ORDER BY sent_lsn - replay_lsn DESC;

-- Estimate daily WAL volume using pg_ls_waldir() (PostgreSQL 10+, superuser required)
SELECT
    count(*) AS wal_segment_count,
```

```

pg_size_pretty(sum(size))          AS total_wal_size_on_disk,
-- Each segment is typically 16MB; divide by uptime days to estimate daily rate
pg_size_pretty(sum(size) /
    GREATEST(1, EXTRACT(DAY FROM now() - pg_postmaster_start_time())::INT))
)
AS estimated_daily_wal_bytes
FROM pg_ls_waldir()
WHERE name ~ '^[0-9A-F]{24}$'; -- only actual WAL segment files, not .history or .partial

-- SQL Server: estimate log shipping / replication volume for cross-region cost projection
-- Run on the primary (publisher or AG primary)

-- Log generation rate from VLF information
SELECT
    DB_NAME(database_id)          AS database_name,
    log_size_mb                    = size_mb,
    log_used_mb                    = used_mb,
    log_used_pct                   = CAST(used_mb * 100.0 / NULLIF(size_mb, 0) AS
    ↪ DECIMAL(5,1))
FROM (
    SELECT
        database_id,
        SUM(total_log_size_bytes) / 1048576.0 AS size_mb,
        SUM(used_log_space_bytes) / 1048576.0 AS used_mb
    FROM sys.dm_db_log_space_usage
    -- Note: sys.dm_db_log_space_usage is per-database; union across databases as needed
) log_usage;

-- For Always On AG replicas: bytes sent to secondary (cross-region cost driver)
SELECT
    ar.replica_server_name,
    ars.role_desc,
    ars.synchronization_state_desc,
    ars.synchronization_health_desc,
    rs.log_send_queue_size          AS log_send_queue_kb,
    rs.log_send_rate                AS log_send_rate_kb_per_sec,
    -- Project to monthly GB: (rate KB/s * 86400 s/day * 30 days) / (1024^2)
    CAST(rs.log_send_rate * 86400.0 * 30 / 1048576.0 AS DECIMAL(10,2))
    AS projected_monthly_gb,
    -- Monthly transfer cost at $0.02/GB (adjust for your region pair)
    CAST(rs.log_send_rate * 86400.0 * 30 / 1048576.0 * 0.02 AS DECIMAL(10,2))
    AS projected_monthly_transfer_cost_usd
FROM sys.availability_replicas ar
JOIN sys.dm_hadr_availability_replica_states ars
    ON ar.replica_id = ars.replica_id
JOIN sys.dm_hadr_database_replica_states rs
    ON ar.replica_id = rs.replica_id
WHERE ars.is_local = 0 -- remote replicas only
ORDER BY rs.log_send_rate DESC;

```

Once you have the actual WAL/log generation rate, the cross-region cost decision becomes a straightforward comparison:

```

Monthly cross-region replica cost =
    Replica compute cost (destination region)
+ Storage cost (destination region)
+ Data transfer cost (WAL volume × $/GB inter-region rate)

```

vs.

```

Monthly cost of equivalent DR using automated S3 backups + point-in-time restore:
    S3 storage cost for daily snapshots + WAL/binlog archives
+ Estimated cost of restore operation (compute hours during a DR event)
+ Cost of acceptable RPO gap (subjective, business decision)

```

For many teams, the S3-based DR approach costs a small fraction of maintaining a live cross-region replica. The trade-off is RTO: restoring from S3 takes longer than promoting a warm replica. If your DR runbook actually requires sub-15-minute RTO, the cross-region replica may be justified. If your organization has never run a DR drill and nobody has measured actual restore time, the cross-region replica is almost certainly over-engineered relative to the business need.

19.7 Aurora-Specific Economics: Replicas vs. Serverless vs. RDS Proxy

Aurora changes the read replica economics substantially because of its shared storage architecture. Understanding these differences prevents you from applying the same cost framework you use for standard RDS.

Aurora read replicas share the cluster volume. Adding an Aurora replica to an existing Aurora cluster does not add storage cost — the replica reads from the same underlying distributed storage as the primary writer. This eliminates one of the three major cost components for standard RDS replicas. What you pay for with an Aurora replica is purely the compute instance and the additional read I/O generated by queries running on the replica.

Aurora replicas provide both read scaling and HA. A single Aurora replica in the same region doubles as a failover target. Aurora’s automated failover to a replica typically completes in under 30 seconds. This means an Aurora cluster with one replica may satisfy both your read offload and your availability requirements, replacing a more expensive architecture that uses a Multi-AZ primary plus a separate read replica.

The comparison looks like this for a moderately sized workload:

Architecture	Compute	Storage	HA Capability	Monthly Estimate
RDS Multi-AZ + Read Replica (r6g.2xlarge)	$3 \times \$876$	$3 \times \text{storage cost}$	Failover + read scaling	~\$2,628 + storage
Aurora cluster: 1 writer + 1 replica (r6g.2xlarge)	$2 \times \$876$	$1 \times \text{shared storage}$	Failover + read scaling	~\$1,752 + shared storage
Aurora Serverless v2 (auto-scaling ACUs)	ACU-based	$1 \times \text{shared storage}$	Failover (slower)	Variable; lower for intermittent

The Aurora Serverless v2 option deserves specific attention for read replica cost optimization. Aurora Serverless v2 readers scale in Aurora Capacity Units (ACUs) from a minimum of 0.5 ACU to a configured maximum. For workloads that are bursty — heavy analytics queries a few times per day, otherwise idle — a Serverless v2 reader can dramatically undercut the cost of a provisioned replica running continuously at a fixed instance class.

The catch is that Serverless v2 readers still incur a minimum ACU charge when idle (they do not scale to zero), so for truly intermittent workloads, the delete/recreate pattern from section 19.5

may still win on cost. The crossover point depends on your specific ACU pricing in your region and the ratio of active to idle hours.

Use this query to evaluate Aurora cluster costs and identify over-provisioned replicas within an Aurora cluster:

```
-- PostgreSQL (Aurora): assess per-replica utilization within an Aurora cluster
-- Run on each Aurora replica endpoint

-- Aurora-specific: check if we are on a reader instance
SELECT
    server_id,
    session_id,
    CASE
        WHEN pg_is_in_recovery() THEN 'READER'
        ELSE 'WRITER'
    END AS instance_role,
    current_setting('max_connections') AS max_connections,
    (SELECT count(*) FROM pg_stat_activity WHERE backend_type = 'client backend') AS current_connections
FROM aurora_replica_status()
WHERE session_id = pg_backend_pid()::text;

-- Aurora replica lag (uses aurora_replica_status() rather than pg_stat_replication)
SELECT
    server_id,
    session_id,
    replica_lag_in_msec,
    last_update_timestamp,
    -- Flag replicas with consistently high lag - may indicate undersized replica
    CASE
        WHEN replica_lag_in_msec > 1000 THEN 'HIGH_LAG'
        WHEN replica_lag_in_msec > 100 THEN 'MODERATE_LAG'
        ELSE 'HEALTHY'
    END AS lag_status
FROM aurora_replica_status()
ORDER BY replica_lag_in_msec DESC;

-- Active query count and wait events on this reader
SELECT
    wait_event_type,
    wait_event,
    count(*) AS sessions
FROM pg_stat_activity
WHERE state = 'active'
    AND backend_type = 'client backend'
GROUP BY 1, 2
ORDER BY sessions DESC;
```

For Azure SQL Hyperscale, the named replica feature introduced a similar shared-storage model where read replicas do not incur additional storage charges. The assessment queries from section 19.1 apply directly to Hyperscale named replicas, with the caveat that Azure’s billing meter for Hyperscale replica vCores is separate from the primary and appears as a distinct line item in Cost Management.

19.8 Building a Replica Cost Dashboard

Individual queries and scripts are useful for one-time audits, but sustainable cost management requires a dashboard that surfaces replica cost and utilization data continuously. The most practical implementation for teams already using AWS is a combination of CloudWatch dashboards for operational metrics and a Cost Explorer tag-based view for financial data.

The prerequisite is consistent tagging. Every read replica should carry at minimum:

- **Environment:** production / staging / development
- **CostCenter:** the team or business unit being charged
- **replica-purpose:** reporting / api-reads / analytics / dr-standby
- **replica-lifecycle:** managed / exempt / scheduled

With those tags in place, you can create a Cost Explorer saved filter that shows read replica spend by purpose and environment, making it immediately visible when a development replica is costing as much as a production one.

The CloudWatch dashboard definition below creates a replica cost and health overview for a given region:

```
#!/usr/bin/env python3
"""
create_replica_dashboard.py

Creates a CloudWatch dashboard with key read replica metrics:
- DatabaseConnections per replica
- ReadIOPS per replica
- CPUUtilization per replica
- ReplicaLag per replica

Discovers replicas dynamically from RDS describe_db_instances.
Redeploy this script when replicas are added or removed, or schedule it weekly.
"""

import boto3
import json
import sys

REGION = sys.argv[1] if len(sys.argv) > 1 else "us-east-1"
DASHBOARD = "ReadReplica-CostAndHealth"

rds = boto3.client("rds", region_name=REGION)
cw = boto3.client("cloudwatch", region_name=REGION)

# Collect all read replicas
replicas = []
paginator = rds.get_paginator("describe_db_instances")
for page in paginator.paginate():
    for db in page["DBInstances"]:
        if db.get("ReadReplicaSourceDBInstanceIdentifier") and \
            db.get("DBInstanceStatus") == "available":
            replicas.append({
                "id": db["DBInstanceIdentifier"],
                "class": db["DBInstanceClass"],
                "az": db["AvailabilityZone"],
            })

if not replicas:
    print("No active read replicas found - dashboard not created.")
    sys.exit(0)

print(f"Found {len(replicas)} read replicas. Building dashboard...")

# Dashboard JSON - one row per metric type, one widget per metric with all replicas overlaid
def metric_widget(title: str, metric_name: str, stat: str, y_label: str,
                  replica_list: list, period: int = 300) -> dict:
    metrics = []
    for r in replica_list:
        metrics.append([
            "AWS/RDS",
            metric_name,
            "DBInstanceIdentifier",
```

```

        r["id"],
        {"label": r["id"], "period": period, "stat": stat}
    ])
    return {
        "type": "metric",
        "properties": {
            "title": title,
            "view": "timeSeries",
            "stacked": False,
            "metrics": metrics,
            "period": period,
            "yAxis": {"left": {"label": y_label, "min": 0}},
            "legend": {"position": "bottom"},
        }
    }
}

widgets = [
    metric_widget("Replica: Database Connections", "DatabaseConnections", "Average", "Connections",
↪ replicas),
    metric_widget("Replica: Read IOPS", "ReadIOPS", "Average", "IOPS",
↪ replicas),
    metric_widget("Replica: CPU Utilization (%)", "CPUUtilization", "Average", "CPU %",
↪ replicas),
    metric_widget("Replica: Replica Lag (seconds)", "ReplicaLag", "Average", "Seconds",
↪ replicas),
    metric_widget("Replica: Freeable Memory (bytes)", "FreeableMemory", "Average", "Bytes",
↪ replicas),
]

# Add a text widget header
header = {
    "type": "text",
    "properties": {
        "markdown": (
            f"# Read Replica Cost & Health Dashboard\n"
            f"**Region:** {REGION} | "
            f"**Replicas monitored:** {len(replicas)} | "
            f"**Auto-generated** - redeploy create_replica_dashboard.py to refresh\n\n"
            + "\n".join(
                f"- {r['id']}` ({r['class']}, {r['az']})" for r in replicas
            )
        )
    }
}

dashboard_body = {
    "widgets": [header] + widgets
}

cw.put_dashboard(
    DashboardName=DASHBOARD,
    DashboardBody=json.dumps(dashboard_body)
)

print(f"Dashboard '{DASHBOARD}' created/updated in region {REGION}.")
print(f"View at: https://{REGION}.console.aws.amazon.com/cloudwatch/home?region={REGION}
↪ #dashboards:name={DASHBOARD}")

```

Pair this operational dashboard with a weekly Athena query against your Cost and Usage Report (CUR) export to track replica spend trends over time:

```

-- Athena: weekly read replica spend from CUR
-- Replace your_cur_database and your_cur_table with your actual CUR export details

SELECT
    DATE_TRUNC('week', CAST(line_item_usage_start_date AS TIMESTAMP)) AS week_start,
    resource_tags_user_environment AS environment,
    resource_tags_user_replica_purpose AS replica_purpose,
    line_item_resource_id AS resource_id,

```

```
SUM(line_item_unblended_cost)                                AS weekly_cost_usd
FROM your_cur_database.your_cur_table
WHERE
  line_item_product_code = 'AmazonRDS'
  AND line_item_usage_start_date >= DATE_FORMAT(DATE_ADD('month', -3, NOW()), '%Y-%m-01')
  -- Filter to replica instances using the resource tag set in section 19.5
  AND (
    LOWER(resource_tags_user_replica_lifecycle) IN ('managed', 'scheduled', 'exempt')
    OR line_item_resource_id IN (
      -- Alternatively, hardcode known replica identifiers or use a tag filter
      SELECT DISTINCT resource_id FROM your_replica_inventory_table
    )
  )
GROUP BY 1, 2, 3, 4
ORDER BY week_start DESC, weekly_cost_usd DESC;
```

The combination of a real-time CloudWatch dashboard and a weekly spend trend query gives you both the operational view (is this replica healthy and active right now?) and the financial view (is this replica's spend trending up unexpectedly?), without requiring a dedicated FinOps tool.

Key Takeaways

- **Replicas are only cost-effective when they are actively serving queries.** An idle replica at any instance size is pure waste. Run the utilization audit from section 19.2 on a weekly schedule and act on the results — teams that automate deletion of idle replicas consistently reclaim a meaningful fraction of their overall RDS bill within the first quarter.
 - **Replicas do not need to match the primary instance class.** The replica's workload is bounded by the queries it serves, not the primary's write concurrency. Measure actual CPU, memory, and cache hit ratio on the replica itself, and right-size independently. A one-class reduction in a replica's instance family produces immediate, permanent savings with no application changes required.
 - **The break-even calculation must include Reserved Instance pricing.** A replica running 24/7 for a predictable workload that is not on a Reserved Instance is paying a continuous on-demand premium. Combine right-sizing with RI purchasing for replicas that will run long-term to maximize savings on both dimensions simultaneously.
 - **Cross-region replicas carry a hidden replication tax.** WAL or log shipping volume drives data transfer costs that scale with write load on the primary, often adding hundreds of dollars monthly on top of the replica compute and storage costs. Always measure actual WAL generation rates and compare the total cross-region replica cost against the cost of an S3-based backup/restore DR strategy before provisioning or retaining a cross-region replica.
 - **Lifecycle automation is the only sustainable path.** Replicas created for temporary tasks persist indefinitely without automated governance. Implement the tagging conventions, idle-detection Lambda, and scheduled scaling patterns from sections 19.5 and 19.6 to enforce a replica lifecycle. Teams that rely solely on manual review find that cost creep resumes within weeks of each cleanup exercise.
-

Chapter 20: Connection Pooling to Reduce Compute Overhead

Organizations running PostgreSQL on AWS RDS, Azure Database for PostgreSQL, or Cloud SQL routinely overpay for compute by 30–50% simply because each application thread holds a dedicated database connection whether it is actively executing a query or sitting idle. A db.r6g.2xlarge RDS instance running at \$0.48/hour — \$346/month — often spends the majority of its CPU cycles managing connection overhead rather than executing SQL. By deploying PgBouncer in transaction pooling mode in front of that instance, teams consistently reduce their active connection count from several hundred to fewer than fifty, which in turn allows them to right-size to a db.r6g.xlarge at \$173/month — saving \$173/month on a single instance without changing a line of application code. Across a fleet of ten such instances the annual saving reaches \$20,760. This chapter covers how to diagnose connection waste, size a connection pool correctly, deploy and automate PgBouncer on each major cloud platform, and wire the entire operation into a scheduled cost-audit pipeline so savings are permanent rather than one-time.

Diagnosing Connection Overhead Before You Optimize

No optimization survives a budget review without a before-state baseline. Connection overhead manifests in three measurable ways: high `max_connections` utilization, elevated CPU on the database instance while query throughput is low, and spikes in `pg_stat_activity` showing processes in `idle` or `idle in transaction` state.

The following query exposes the exact distribution of connection states on a live PostgreSQL instance. Run it against every RDS or Cloud SQL instance in your fleet and export the results to a spreadsheet:

```
-- Connection state audit: run on each PostgreSQL instance
-- Identifies idle connections that are burning compute capacity
SELECT
    datname                                AS database_name,
    username                              AS username,
    state,
    wait_event_type,
    wait_event,
    COUNT(*)                              AS connection_count,
    ROUND(COUNT(*) * 100.0
          / SUM(COUNT(*) OVER (), 2) AS pct_of_total,
    MAX(EXTRACT(EPOCH FROM (NOW() - state_change))::INT)
        AS max_idle_seconds,
    AVG(EXTRACT(EPOCH FROM (NOW() - state_change))::INT)
```

```

                                AS avg_idle_seconds
FROM pg_stat_activity
WHERE pid <> pg_backend_pid()      -- exclude this monitoring session
GROUP BY datname, username, state, wait_event_type, wait_event
ORDER BY connection_count DESC;

```

Interpret the output with these thresholds:

Condition	Meaning	Action
idle > 40% of total	Application not releasing connections	Pool immediately
idle in transaction > 5%	Long transactions or ORM misuse	Fix ORM + Pool
max_idle_seconds > 300	Connections parked for 5+ minutes	Reduce pool timeout
Total connections > 80% of max_connections	Near exhaustion	Pool + raise limit

Pull the current max_connections ceiling and see how close you are to it:

```

-- Current connection ceiling vs. actual usage
SELECT
  'max_connections' AS name,
  setting::INT AS configured_value,
  (SELECT COUNT(*) FROM pg_stat_activity) AS current_connections,
  ROUND(
    (SELECT COUNT(*) FROM pg_stat_activity)::NUMERIC
    / setting::INT * 100, 1
  ) AS utilization_pct
FROM pg_settings
WHERE name = 'max_connections';

```

On AWS RDS, cross-reference this data with CloudWatch. The metric `DatabaseConnections` sampled over 24 hours tells you peak and average. Pull it with the CLI so you can baseline before and after pooling:

```

#!/bin/bash
# baseline_connections.sh
# Pulls 7-day connection metrics from CloudWatch for a named RDS instance
# Usage: ./baseline_connections.sh mydb-instance-1 us-east-1

INSTANCE_ID="${1:?Usage: $0 <db-instance-id> <region>}"
REGION="${2:-us-east-1}"
START_TIME=$(date -u -d '7 days ago' '+%Y-%m-%dT%H:%M:%SZ' 2>/dev/null \
  || date -u -v-7d '+%Y-%m-%dT%H:%M:%SZ') # macOS fallback
END_TIME=$(date -u '+%Y-%m-%dT%H:%M:%SZ')

echo "=== Connection baseline for: $INSTANCE_ID ==="

aws cloudwatch get-metric-statistics \
  --region "$REGION" \
  --namespace AWS/RDS \
  --metric-name DatabaseConnections \
  --dimensions Name=DBInstanceIdentifier,Value="$INSTANCE_ID" \
  --start-time "$START_TIME" \
  --end-time "$END_TIME" \
  --period 3600 \
  --statistics Maximum Average \
  --output json \
| python3 - << 'PYEOF'
import sys, json

```

```

data = json.load(sys.stdin)
points = sorted(data["Datapoints"], key=lambda p: p["Timestamp"])
print(f"{'Timestamp':<30} {'Max':>8} {'Avg':>8}")
print("-" * 50)
for p in points:
    print(f"{p['Timestamp']:<30} {p['Maximum']:>8.0f} {p['Average']:>8.1f}")

if points:
    maxes = [p["Maximum"] for p in points]
    avgs = [p["Average"] for p in points]
    print(f"\nPeak 7-day maximum : {max(maxes):.0f}")
    print(f"Mean of hourly avg : {sum(avgs)/len(avgs):.1f}")
PYEOF

echo ""
echo "=== CPUUtilization during same window ==="
aws cloudwatch get-metric-statistics \
    --region "$REGION" \
    --namespace AWS/RDS \
    --metric-name CPUUtilization \
    --dimensions Name=DBInstanceIdentifier,Value="$INSTANCE_ID" \
    --start-time "$START_TIME" \
    --end-time "$END_TIME" \
    --period 3600 \
    --statistics Maximum Average \
    --output json \
    | python3 -c "
import sys, json
data = json.load(sys.stdin)
points = sorted(data['Datapoints'], key=lambda p: p['Timestamp'])
if points:
    maxes = [p['Maximum'] for p in points]
    avgs = [p['Average'] for p in points]
    print(f'Peak CPU : {max(maxes):.1f}%')
    print(f'Mean CPU : {sum(avgs)/len(avgs):.1f}%')
    print()
    print('Rule of thumb: if mean CPU < 20% and connections are high,')
    print('you are paying for connection overhead, not real workload.')
"

```

When peak connections are high and mean CPU is below 20%, the instance is sized for connection overhead rather than query work. That gap is your cost recovery opportunity.

Sizing the Pool: The Math Behind Right-Sizing

The common mistake is deploying PgBouncer with default settings and calling it done. The pool size determines whether you can actually downsize the instance, and the instance determines the monthly bill. Get the math right first.

PostgreSQL performs best when the number of *active* (non-idle) connections does not exceed **2 × vCPU count**. For a db.r6g.2xlarge (8 vCPUs), the optimal active connection count is 16. Every connection above that figure competes for the same CPU cores and increases context-switching overhead — you pay for a larger instance to manage connections, not to execute queries.

The PgBouncer pool target formula:

```

server_pool_size = 2 × vCPU_count_of_target_instance
max_client_conn = peak_application_threads × safety_factor (1.25)

```

Example calculation for a migration from db.r6g.2xlarge to db.r6g.xlarge:

Parameter	Before	After
Instance	db.r6g.2xlarge (8 vCPU)	db.r6g.xlarge (4 vCPU)
On-Demand price	\$0.480/hr	\$0.240/hr
Monthly cost	\$346.56	\$172.80
Monthly saving	—	\$173.76
max_connections on RDS	5,000*	2,500*
Application threads	400	400
PgBouncer default_pool_size	—	8
PgBouncer max_client_conn	—	500

*RDS sets `max_connections` automatically based on instance RAM; values shown are illustrative.

The key insight: 400 application threads share 8 server-side connections. Each thread waits at most a few milliseconds for a connection in transaction pooling mode because transactions complete in tens of milliseconds on OLTP workloads. Average wait is negligible compared to the compute saving.

For Azure Database for PostgreSQL Flexible Server, the equivalent instance comparison is:

Tier	vCPU	RAM	Price (East US)	Monthly
Standard_D4ds_v4	4	16 GB	\$0.334/hr	\$241
Standard_D2ds_v4	2	8 GB	\$0.167/hr	\$120
Saving per instance				\$121/mo

On Google Cloud SQL for PostgreSQL, the equivalent:

Machine type	vCPU	RAM	Price (us-central1)	Monthly
db-custom-4-15360	4	15 GB	\$0.3567/hr	\$257
db-custom-2-7680	2	7.5 GB	\$0.1784/hr	\$129
Saving per instance				\$128/mo

Before committing to a smaller instance, validate the pool size calculation against `pg_stat_statements`. The following query identifies the 95th-percentile query duration, which determines how long a server connection is held per transaction:

```
-- Estimate transaction duration from pg_stat_statements
-- This determines whether transaction pooling is safe and effective
SELECT
    LEFT(query, 80)                                AS query_sample,
    calls,                                           AS avg_exec_ms,
    ROUND((total_exec_time / calls)::NUMERIC, 2)   AS avg_exec_ms,
    ROUND(
        PERCENTILE_CONT(0.95) WITHIN GROUP
        (ORDER BY total_exec_time / calls)
        OVER ( )::NUMERIC, 2)                       AS p95_exec_ms,
    ROUND((mean_exec_time)::NUMERIC, 2)             AS mean_exec_ms,
    rows / calls                                     AS avg_rows_returned
FROM pg_stat_statements
WHERE calls > 100
    AND query NOT LIKE '%pg_stat%' -- exclude monitoring queries
ORDER BY total_exec_time DESC
LIMIT 20;
```


If p95 query duration is under 100ms (typical for OLTP), transaction pooling is safe. If you see queries above 5,000ms regularly, those long-running queries will hold server connections and reduce the effectiveness of the pool — address query performance before right-sizing compute.

Deploying PgBouncer on Each Cloud Platform

PgBouncer sits between the application and the RDS/Cloud SQL instance. It multiplexes hundreds of client connections onto a small number of server connections. The deployment pattern differs slightly by cloud platform.

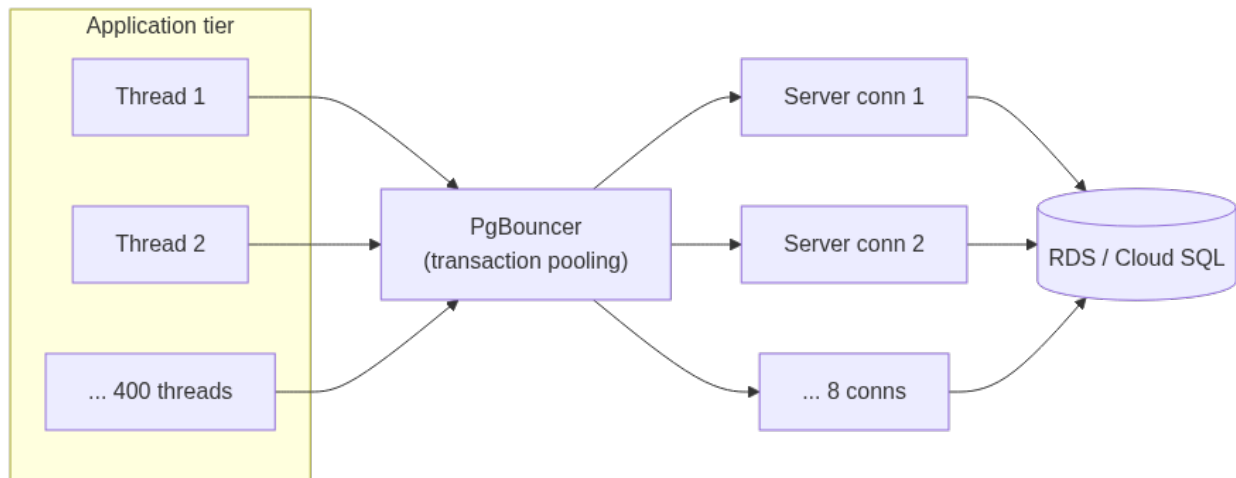


Figure 20: PgBouncer multiplexes hundreds of application threads onto a handful of server-side connections — this is what makes right-sizing the database instance itself possible.

AWS: PgBouncer on EC2 or ECS

The lowest-latency placement for PgBouncer on AWS is an EC2 instance in the same Availability Zone as the primary RDS instance, or a Fargate task in ECS if you want managed infrastructure. A t3.small (\$0.0208/hr, \$15/month) handles tens of thousands of application connections with negligible CPU usage — it is purely I/O and memory bounded.

The following Bash script provisions PgBouncer on an Amazon Linux 2023 EC2 instance, configures it for transaction pooling, and registers it as a systemd service:

```
#!/bin/bash
# deploy_pg_bouncer_aws.sh
# Installs and configures PgBouncer on Amazon Linux 2023
# Run as ec2-user with sudo privileges
# Requires: RDS endpoint, DB name, credentials in AWS Secrets Manager

set -euo pipefail

RDS_ENDPOINT="${RDS_ENDPOINT:?Set RDS_ENDPOINT environment variable}"
RDS_PORT="${RDS_PORT:-5432}"
DB_NAME="${DB_NAME:?Set DB_NAME environment variable}"
SECRET_ARN="${SECRET_ARN:?Set SECRET_ARN for Secrets Manager}"
PGBOUNCER_PORT=6432
POOL_SIZE="${POOL_SIZE:-16}"           # 2 × vCPU of target instance
```

```

MAX_CLIENT_CONN="${MAX_CLIENT_CONN:-500}"

echo "[1/6] Installing PgBouncer..."
sudo dnf install -y pgbouncer

echo "[2/6] Fetching credentials from Secrets Manager..."
SECRET=$(aws secretsmanager get-secret-value \
    --secret-id "$SECRET_ARN" \
    --query SecretString \
    --output text)
DB_USER=$(echo "$SECRET" | python3 -c "import sys,json; print(json.load(sys.stdin)['username'])")
DB_PASS=$(echo "$SECRET" | python3 -c "import sys,json; print(json.load(sys.stdin)['password'])")

echo "[3/6] Writing pgbouncer.ini..."
sudo tee /etc/pgbouncer/pgbouncer.ini > /dev/null << INIEOF
[DATABASES]
${DB_NAME} = host=${RDS_ENDPOINT} port=${RDS_PORT} dbname=${DB_NAME}

[pgbouncer]
listen_addr      = 0.0.0.0
listen_port      = ${PGBOUNCER_PORT}
auth_type        = scram-sha-256
auth_file        = /etc/pgbouncer/userlist.txt

; TRANSACTION mode: connection returned to pool after each transaction
; This is the mode that enables compute right-sizing
pool_mode        = transaction

default_pool_size = ${POOL_SIZE}
min_pool_size     = 2
reserve_pool_size = 5
reserve_pool_timeout = 3
max_client_conn   = ${MAX_CLIENT_CONN}

; Connection hygiene - recycle idle server connections
server_idle_timeout = 60
client_idle_timeout = 300
server_lifetime     = 3600

; Logging for cost audit trail
log_connections     = 1
log_disconnections  = 1
stats_period        = 60

; TLS toward RDS
server_tls_sslmode   = require

; Admin interface - restrict to localhost
admin_users          = pgbouncer_admin
stats_users          = pgbouncer_stats
INIEOF

echo "[4/6] Writing userlist.txt..."
# PgBouncer needs MD5 or SCRAM hashes; use pg_shadow query for production
# For initial setup, write plaintext and migrate to scram-sha-256 hashes
sudo tee /etc/pgbouncer/userlist.txt > /dev/null << USEREOF
"${DB_USER}" "${DB_PASS}"
"pgbouncer_admin" "changeme_admin_pw"
"pgbouncer_stats" "changeme_stats_pw"
USEREOF
sudo chmod 600 /etc/pgbouncer/userlist.txt
sudo chown pgbouncer:pgbouncer /etc/pgbouncer/userlist.txt

echo "[5/6] Enabling and starting systemd service..."
sudo systemctl enable pgbouncer
sudo systemctl restart pgbouncer
sudo systemctl status pgbouncer --no-pager

echo "[6/6] Verification..."
echo "Connecting to PgBouncer admin interface..."

```

```
PGPASSWORD="changeme_admin_pw" psql \
  -h 127.0.0.1 -p $PGBOUNCER_PORT \
  -U pgbouncer_admin pgbouncer \
  -c "SHOW POOLS;" \
  -c "SHOW STATS;" \
  -c "SHOW CONFIG;" 2>&1 | head -60

echo ""
echo "=== Deployment complete ==="
echo "Application connection string:"
echo "  host=<this-ec2-ip> port=${PGBOUNCER_PORT} dbname=${DB_NAME} user=${DB_USER}"
echo ""
echo "Next step: update application connection strings, then right-size RDS instance."
```

Azure: PgBouncer Built-In vs. Sidecar

Azure Database for PostgreSQL Flexible Server includes PgBouncer as a built-in feature — enable it with a single CLI command at no additional cost:

```
#!/bin/bash
# enable_pgbouncer_azure.sh
# Enables built-in PgBouncer on Azure Database for PostgreSQL Flexible Server
# Requires: az CLI logged in, Contributor role on the server resource

RESOURCE_GROUP="${RESOURCE_GROUP:?Set RESOURCE_GROUP}"
SERVER_NAME="${SERVER_NAME:?Set SERVER_NAME}"
SUBSCRIPTION="${SUBSCRIPTION:-$(az account show --query id -o tsv)}"

echo "=== Enabling PgBouncer on Azure PostgreSQL Flexible Server ==="
echo "Server      : $SERVER_NAME"
echo "Resource Grp: $RESOURCE_GROUP"

# Enable the pgbouncer extension and set pool parameters
az postgres flexible-server parameter set \
  --resource-group "$RESOURCE_GROUP" \
  --server-name "$SERVER_NAME" \
  --name "pgbouncer.enabled" \
  --value "true"

az postgres flexible-server parameter set \
  --resource-group "$RESOURCE_GROUP" \
  --server-name "$SERVER_NAME" \
  --name "pgbouncer.default_pool_size" \
  --value "16"

az postgres flexible-server parameter set \
  --resource-group "$RESOURCE_GROUP" \
  --server-name "$SERVER_NAME" \
  --name "pgbouncer.max_client_conn" \
  --value "500"

az postgres flexible-server parameter set \
  --resource-group "$RESOURCE_GROUP" \
  --server-name "$SERVER_NAME" \
  --name "pgbouncer.pool_mode" \
  --value "transaction"

az postgres flexible-server parameter set \
  --resource-group "$RESOURCE_GROUP" \
  --server-name "$SERVER_NAME" \
  --name "pgbouncer.server_idle_timeout" \
  --value "60"

echo ""
echo "=== Verifying PgBouncer status ==="
az postgres flexible-server parameter list \
  --resource-group "$RESOURCE_GROUP" \
  --server-name "$SERVER_NAME" \
```

```

--query          "[?starts_with(name, 'pgbouncer')]" \
--output         table

echo ""
echo "=== PgBouncer connection endpoint ==="
SERVER_FQDN=$(az postgres flexible-server show \
--resource-group "$RESOURCE_GROUP" \
--name           "$SERVER_NAME" \
--query         "fullyQualifiedDomainName" \
--output         tsv)

echo "Standard port (direct) : ${SERVER_FQDN}:5432"
echo "PgBouncer port       : ${SERVER_FQDN}:6432"
echo ""
echo "Update application connection strings to port 6432."
echo "Monitor via Azure Monitor metric: active_connections"

```

For Google Cloud SQL, the native proxy (`cloud-sql-proxy`) handles authentication but does not pool connections. Deploy PgBouncer as a Cloud Run sidecar or on a single e2-micro (\$0.0084/hr, \$6/month) in the same region.

Measuring the Cost Impact and Automating Continuous Monitoring

Deploying PgBouncer is a one-time action. The ongoing value comes from continuously verifying that pooling efficiency is maintained as the application evolves — new features often add new connection patterns that erode pool efficiency over time. The following Python script runs on a cron schedule, compares current connection behavior against a stored baseline, and posts an alert to Slack when efficiency drops below threshold. It also queries the AWS Cost Explorer API to correlate connection count changes with actual spend.

```

#!/usr/bin/env python3
"""
pgbouncer_cost_monitor.py

Scheduled cost-efficiency monitor for PgBouncer deployments.
Queries PgBouncer admin interface + AWS Cost Explorer and reports
on connection efficiency and compute cost trends.

Schedule: run every 6 hours via cron or EventBridge Scheduler
0 */6 * * * /usr/bin/python3 /opt/dba-tools/pgbouncer_cost_monitor.py

Dependencies: psycpg2, boto3, requests
pip install psycpg2-binary boto3 requests
"""

import os
import json
import datetime
import psycpg2
import boto3
import requests
from dataclasses import dataclass, asdict
from typing import Optional

# Configuration
PGBOUNCER_HOST = os.environ.get("PGBOUNCER_HOST", "127.0.0.1")
PGBOUNCER_PORT = int(os.environ.get("PGBOUNCER_PORT", 6432))
PGBOUNCER_ADMIN = os.environ.get("PGBOUNCER_ADMIN_USER", "pgbouncer_admin")
PGBOUNCER_PASS = os.environ.get("PGBOUNCER_ADMIN_PASS")

```

```

AWS_REGION      = os.environ.get("AWS_REGION", "us-east-1")
RDS_INSTANCE_ID = os.environ.get("RDS_INSTANCE_ID")      # for Cost Explorer filter
SLACK_WEBHOOK    = os.environ.get("SLACK_WEBHOOK_URL")    # optional

EFFICIENCY_THRESHOLD = 0.30 # alert if server_conn / cl_active drops below 30%
IDLE_RATIO_THRESHOLD = 0.50 # alert if idle client ratio exceeds 50%

# Data classes
@dataclass
class PoolStats:
    database:      str
    cl_active:     int # active client connections
    cl_waiting:    int # clients waiting for a server connection
    sv_active:     int # active server connections
    sv_idle:       int # idle server connections
    sv_used:       int # server connections used and returned to pool
    maxwait:       int # longest wait time (seconds)
    pool_mode:     str

@dataclass
class CostReport:
    period_start:  str
    period_end:    str
    rds_compute_cost: float
    currency:      str
    instance_id:   str

# PgBouncer queries
def get_pool_stats(host: str, port: int, user: str, password: str) -> list[PoolStats]:
    """Query PgBouncer admin interface for current pool statistics."""
    conn = psycopg2.connect(
        host=host, port=port,
        user=user, password=password,
        dbname="pgbouncer",
        connect_timeout=5
    )
    conn.autocommit = True
    cur = conn.cursor()
    cur.execute("SHOW POOLS;")
    cols = [desc[0] for desc in cur.description]
    rows = cur.fetchall()
    cur.close()
    conn.close()

    stats = []
    for row in rows:
        row_dict = dict(zip(cols, row))
        stats.append(PoolStats(
            database = row_dict.get("database", ""),
            cl_active = int(row_dict.get("cl_active", 0)),
            cl_waiting = int(row_dict.get("cl_waiting", 0)),
            sv_active = int(row_dict.get("sv_active", 0)),
            sv_idle = int(row_dict.get("sv_idle", 0)),
            sv_used = int(row_dict.get("sv_used", 0)),
            maxwait = int(row_dict.get("maxwait", 0)),
            pool_mode = row_dict.get("pool_mode", ""),
        ))
    return stats

def get_global_stats(host: str, port: int, user: str, password: str) -> dict:
    """Query SHOW STATS for aggregate throughput and timing metrics."""
    conn = psycopg2.connect(
        host=host, port=port,
        user=user, password=password,
        dbname="pgbouncer",
        connect_timeout=5
    )
    conn.autocommit = True
    cur = conn.cursor()

```

```

cur.execute("SHOW STATS;")
cols = [desc[0] for desc in cur.description]
rows = cur.fetchall()
cur.close()
conn.close()
return [dict(zip(cols, row)) for row in rows]

# AWS Cost Explorer
def get_rds_cost_last_30_days(instance_id: Optional[str]) -> Optional[CostReport]:
    """Fetch RDS compute spend for the last 30 days from Cost Explorer."""
    if not instance_id:
        return None

    ce = boto3.client("ce", region_name="us-east-1") # CE is global; always us-east-1
    end = datetime.date.today()
    start = end - datetime.timedelta(days=30)

    try:
        response = ce.get_cost_and_usage(
            TimePeriod={"Start": start.isoformat(), "End": end.isoformat()},
            Granularity="MONTHLY",
            Filter={
                "And": [
                    {"Dimensions": {"Key": "SERVICE",
                                   "Values": ["Amazon Relational Database Service"]}},
                    {"Dimensions": {"Key": "RESOURCE_ID",
                                   "Values": [instance_id]}}
                ]
            },
            Metrics=["UnblendedCost"],
        )
        total = sum(
            float(p["Total"]["UnblendedCost"]["Amount"])
            for p in response["ResultsByTime"]
        )
        currency = response["ResultsByTime"][0]["Total"]["UnblendedCost"]["Unit"] \
            if response["ResultsByTime"] else "USD"
        return CostReport(
            period_start = start.isoformat(),
            period_end   = end.isoformat(),
            rds_compute_cost = round(total, 2),
            currency      = currency,
            instance_id   = instance_id,
        )
    except Exception as exc:
        print(f"[WARN] Cost Explorer query failed: {exc}")
        return None

# Analysis
def analyze_pools(stats: list[PoolStats]) -> dict:
    """
    Compute efficiency ratios and flag pools that need attention.
    Returns a dict with aggregate metrics and a list of alerts.
    """
    alerts = []
    total_cl_active = sum(s.cl_active for s in stats)
    total_sv_active = sum(s.sv_active for s in stats)
    total_sv_idle   = sum(s.sv_idle   for s in stats)
    total_cl_waiting = sum(s.cl_waiting for s in stats)

    # Multiplexing ratio: how many client connections share one server connection
    if total_sv_active > 0:
        mux_ratio = round(total_cl_active / total_sv_active, 1)
    else:
        mux_ratio = 0.0

    # Server utilization: fraction of server connections that are active vs. idle
    total_sv = total_sv_active + total_sv_idle

```

```

sv_utilization = round(total_sv_active / total_sv, 2) if total_sv > 0 else 0.0

for s in stats:
    sv_total = s.sv_active + s.sv_idle
    if sv_total == 0:
        continue
    efficiency = s.sv_active / sv_total
    if efficiency < EFFICIENCY_THRESHOLD:
        alerts.append({
            "level": "WARNING",
            "database": s.database,
            "message": (
                f"Pool efficiency {efficiency:.0%} is below threshold "
                f"{EFFICIENCY_THRESHOLD:.0%}. "
                f"sv_idle={s.sv_idle}, sv_active={s.sv_active}. "
                "Consider reducing default_pool_size."
            ),
        })
    if s.maxwait > 5:
        alerts.append({
            "level": "CRITICAL",
            "database": s.database,
            "message": (
                f"Clients waiting {s.maxwait}s for a server connection. "
                "Pool is undersized - increase default_pool_size or "
                "right-size the target instance upward."
            ),
        })
    if s.cl_waiting > 0:
        alerts.append({
            "level": "WARNING",
            "database": s.database,
            "message": (
                f"{s.cl_waiting} client(s) currently queued for a server "
                "connection. Investigate query duration or increase pool size."
            ),
        })

return {
    "timestamp": datetime.datetime.utcnow().isoformat() + "Z",
    "total_cl_active": total_cl_active,
    "total_sv_active": total_sv_active,
    "total_sv_idle": total_sv_idle,
    "total_cl_waiting": total_cl_waiting,
    "mux_ratio": mux_ratio,
    "sv_utilization": sv_utilization,
    "alerts": alerts,
    "pool_detail": [asdict(s) for s in stats],
}

# Slack notification
def post_slack(webhook_url: str, report: dict, cost: Optional[CostReport]) -> None:
    """Format and post a summary to a Slack webhook."""
    alert_lines = "\n".join(
        f"* [{a['level']}] {a['database']}: {a['message']}"
        for a in report["alerts"]
    ) or "• No alerts - pool operating within thresholds."

    cost_line = (
        f"RDS compute cost last 30 days: "
        f"{cost.currency} {cost.rds_compute_cost:,.2f} "
        f"(instance: {cost.instance_id})"
        if cost else "Cost Explorer data unavailable."
    )

    text = (
        f"*PgBouncer Cost Monitor - {report['timestamp']}*\n"
        f"Multiplexing ratio : {report['mux_ratio']}:1 "
        f"(client connections per server connection)\n"

```

```

f"Server utilization : {report['sv_utilization']:.0%}\n"
f"Clients waiting now : {report['total_cl_waiting']}\n"
f"{cost_line}\n\n"
f"*Alerts*\n{alert_lines}"
)
payload = {"text": text}
try:
    resp = requests.post(webhook_url, json=payload, timeout=10)
    resp.raise_for_status()
except Exception as exc:
    print(f"[WARN] Slack notification failed: {exc}")

# Main
def main():
    print(f"[{datetime.datetime.utcnow().isoformat()}Z] PgBouncer cost monitor starting...")

    # 1. Pull pool stats
    pool_stats = get_pool_stats(
        PGBOUNCER_HOST, PGBOUNCER_PORT,
        PGBOUNCER_ADMIN, PGBOUNCER_PASS
    )
    global_stats = get_global_stats(
        PGBOUNCER_HOST, PGBOUNCER_PORT,
        PGBOUNCER_ADMIN, PGBOUNCER_PASS
    )

    # 2. Analyse
    report = analyze_pools(pool_stats)

    # 3. Pull cost data
    cost_report = get_rds_cost_last_30_days(RDS_INSTANCE_ID)

    # 4. Print structured JSON to stdout (captured by CloudWatch Logs or similar)
    output = {
        "pool_report": report,
        "cost_report": asdict(cost_report) if cost_report else None,
        "global_stats": global_stats,
    }
    print(json.dumps(output, indent=2, default=str))

    # 5. Alert via Slack if configured
    if SLACK_WEBHOOK:
        post_slack(SLACK_WEBHOOK, report, cost_report)

    # 6. Exit non-zero if critical alerts exist so cron can trigger further alerting
    critical = [a for a in report["alerts"] if a["level"] == "CRITICAL"]
    if critical:
        print(f"[ERROR] {len(critical)} CRITICAL alert(s) detected. Exiting 1.")
        raise SystemExit(1)

    print("Monitor run complete - no critical issues.")

if __name__ == "__main__":
    main()

```

Save this script to `/opt/dba-tools/pgbouncer_cost_monitor.py`, make it executable, and schedule it with cron or AWS EventBridge Scheduler:

```

# crontab -e (on the PgBouncer EC2 host)
# Run every 6 hours; pipe output to logger for CloudWatch Logs integration
0 */6 * * * PGBOUNCER_ADMIN_PASS=$(aws secretsmanager get-secret-value \
--secret-id pgbouncer/admin --query SecretString --output text \
| python3 -c 'import sys,json; print(json.load(sys.stdin)["password"])') \
RDS_INSTANCE_ID="mydb-prod-1" \
SLACK_WEBHOOK_URL="https://hooks.slack.com/services/..." \
/usr/bin/python3 /opt/dba-tools/pgbouncer_cost_monitor.py \
2>&1 | /usr/bin/logger -t pgbouncer-cost-monitor

```


Validating the Right-Sizing Decision Before Executing It

Committing to a smaller instance class carries risk. The following checklist, implemented as a Bash validation script, guards against the most common failure modes before you modify the instance in production.

```
#!/bin/bash
# validate_rightsize.sh
# Pre-flight checks before right-sizing an RDS instance behind PgBouncer
# Run this script and review all outputs before modifying the instance class.

set -euo pipefail

INSTANCE_ID="${1:?Usage: $0 <db-instance-id> <region> <target-class>}"
REGION="${2:-us-east-1}"
TARGET_CLASS="${3:?Provide target instance class, e.g. db.r6g.xlarge}"

PGB_HOST="${PGBOUNCER_HOST:-127.0.0.1}"
PGB_PORT="${PGBOUNCER_PORT:-6432}"
PGB_ADMIN="${PGBOUNCER_ADMIN_USER:-pgbouncer_admin}"
PGB_PASS="${PGBOUNCER_ADMIN_PASS:?Set PGBOUNCER_ADMIN_PASS}"

PASS=0
FAIL=0
WARN=0

check() {
    local label="$1" result="$2" expected="$3" severity="${4:-FAIL}"
    if [ "$result" = "$expected" ]; then
        echo " [PASS] $label"
        PASS=$((PASS+1))
    else
        echo " [$severity] $label (got: $result, expected: $expected)"
        [ "$severity" = "FAIL" ] && FAIL=$((FAIL+1)) || WARN=$((WARN+1))
    fi
}

echo "====="
echo " Pre-flight validation: $INSTANCE_ID → $TARGET_CLASS"
echo " Region: $REGION"
echo " $(date -u)"
echo "====="

# Check 1: PgBouncer is running and reachable
echo ""
echo "--- Check 1: PgBouncer availability ---"
PGB_UP=$(PGPASSWORD="$PGB_PASS" psql \
    -h "$PGB_HOST" -p "$PGB_PORT" -U "$PGB_ADMIN" pgbouncer \
    -c "SHOW VERSION;" -tA 2>/dev/null | head -1 | grep -c "PgBouncer" || echo "0")
check "PgBouncer reachable" "$PGB_UP" "1"

# Check 2: Pool mode is transaction
echo ""
echo "--- Check 2: Pool mode ---"
POOL_MODE=$(PGPASSWORD="$PGB_PASS" psql \
    -h "$PGB_HOST" -p "$PGB_PORT" -U "$PGB_ADMIN" pgbouncer \
    -c "SHOW CONFIG;" -tA 2>/dev/null \
    | grep "^pool_mode" | awk -F'|' '{print $2}' | tr -d ' ')
check "pool_mode=transaction" "$POOL_MODE" "transaction"

# Check 3: No clients currently waiting
echo ""
echo "--- Check 3: Client wait queue ---"
CL_WAITING=$(PGPASSWORD="$PGB_PASS" psql \
    -h "$PGB_HOST" -p "$PGB_PORT" -U "$PGB_ADMIN" pgbouncer \
```

```

    -c "SHOW POOLS;" -tA 2>/dev/null \
    | awk -F'|' '{sum += $5} END {print sum+0}'
check "No clients waiting (cl_waiting=0)" "$CL_WAITING" "0" "WARN"

# Check 4: Server connection count fits target instance
echo ""
echo "--- Check 4: Server connection headroom ---"
SV_ACTIVE=$(PGPASSWORD="$PGB_PASS" psql \
    -h "$PGB_HOST" -p "$PGB_PORT" -U "$PGB_ADMIN" pgbouncer \
    -c "SHOW POOLS;" -tA 2>/dev/null \
    | awk -F'|' '{sum += $3} END {print sum+0}')
# Extract vCPU count from target instance class name (r6g.xlarge = 4, r6g.2xlarge = 8)
TARGET_VCPU=$(aws ec2 describe-instance-types \
    --instance-types "${TARGET_CLASS}/db./*" \
    --query "InstanceTypes[0].VCpuInfo.DefaultVCpus" \
    --output text --region "$REGION" 2>/dev/null || echo "4")
RECOMMENDED_POOL=$((TARGET_VCPU * 2))
HEADROOM=$((RECOMMENDED_POOL - SV_ACTIVE))
if [ "$HEADROOM" -ge 0 ]; then
    echo " [PASS] sv_active=$SV_ACTIVE recommended pool size=$RECOMMENDED_POOL for $TARGET_CLASS
    ↪ (${TARGET_VCPU} vCPU)"
    PASS=$((PASS+1))
else
    echo " [FAIL] sv_active=$SV_ACTIVE exceeds recommended pool size=$RECOMMENDED_POOL - reduce
    ↪ default_pool_size first"
    FAIL=$((FAIL+1))
fi

# Check 5: RDS CPU not saturated at current workload
echo ""
echo "--- Check 5: Current RDS CPU utilization (last 30 minutes) ---"
PEAK_CPU=$(aws cloudwatch get-metric-statistics \
    --region "$REGION" \
    --namespace AWS/RDS \
    --metric-name CPUUtilization \
    --dimensions Name=DBInstanceIdentifier,Value="$INSTANCE_ID" \
    --start-time "$(date -u -d '30 minutes ago' '+%Y-%m-%dT%H:%M:%SZ' 2>/dev/null \
    || date -u -v-30M '+%Y-%m-%dT%H:%M:%SZ')" \
    --end-time "$(date -u '+%Y-%m-%dT%H:%M:%SZ')" \
    --period 300 \
    --statistics Maximum \
    --output json \
    | python3 -c "
import sys, json
d = json.load(sys.stdin)
pts = d.get('Datapoints', [])
print(f'{{max((p[\"Maximum\"] for p in pts), default=0):.1f}}')
")
# Safe to right-size if peak CPU on current (larger) instance is below 40%
# because the smaller instance will have proportionally fewer vCPUs
CPU_OK=$(python3 -c "print('yes' if float('${PEAK_CPU}') < 40.0 else 'no')")
check "Peak CPU < 40% (current workload safe for smaller instance)" "$CPU_OK" "yes"
echo " Peak CPU last 30 min: ${PEAK_CPU}%"

# Check 6: No long-running transactions
echo ""
echo "--- Check 6: Long-running transactions on RDS ---"
# Connect directly to RDS for this check (bypasses PgBouncer)
RDS_ENDPOINT=$(aws rds describe-db-instances \
    --db-instance-identifier "$INSTANCE_ID" \
    --region "$REGION" \
    --query "DBInstances[0].Endpoint.Address" \
    --output text 2>/dev/null || echo "")

if [ -n "$RDS_ENDPOINT" ] && [ -n "${PGPASSWORD:-}" ]; then
    LONG_TXN=$(psql "host=$RDS_ENDPOINT port=5432 sslmode=require" \
        -c "SELECT COUNT(*) FROM pg_stat_activity
        WHERE state = 'idle in transaction'
        AND EXTRACT(EPOCH FROM (NOW() - xact_start)) > 60;" \
        -tA 2>/dev/null || echo "0")

```

```

    check "No long idle-in-transaction (>60s)" "$LONG_TXN" "0" "WARN"
else
    echo " [SKIP] RDS direct connection not available for long-txn check"
    WARN=$((WARN+1))
fi

# Summary
echo ""
echo "=====
echo " Validation summary"
echo " PASS : $PASS"
echo " WARN : $WARN"
echo " FAIL : $FAIL"
echo "=====

if [ "$FAIL" -gt 0 ]; then
    echo ""
    echo "ACTION REQUIRED: $FAIL check(s) failed."
    echo "Resolve failures before modifying the instance class."
    exit 1
elif [ "$WARN" -gt 0 ]; then
    echo ""
    echo "WARNINGS present: $WARN check(s). Review before proceeding."
    echo "Right-sizing is likely safe but review each warning above."
    exit 0
else
    echo ""
    echo "All checks passed. Safe to proceed with right-sizing:"
    echo ""
    echo " aws rds modify-db-instance \"
    echo "     --db-instance-identifier $INSTANCE_ID \"
    echo "     --db-instance-class $TARGET_CLASS \"
    echo "     --apply-immediately \"
    echo "     --region $REGION"
    echo ""
    echo "Estimated monthly saving:"
    aws rds describe-db-instances \
        --db-instance-identifier "$INSTANCE_ID" \
        --region "$REGION" \
        --query "DBInstances[0].DBInstanceClass" \
        --output text 2>/dev/null \
        | xargs -I{} echo " Current class: {} → Target: $TARGET_CLASS"
    exit 0
fi

```

Run the validation script at least one full business day before executing the instance modification. If it exits 0 and all checks pass, issue the `aws rds modify-db-instance` command during a maintenance window or low-traffic period.

Extending the Pattern to Azure SQL and SQL Server

Azure SQL Database and SQL Server on RDS manage connections differently from PostgreSQL. They use a thread-based architecture where each connection consumes roughly 7–8 MB of memory and a small CPU overhead. The equivalent optimization is **connection pool management at the application layer** via ADO.NET, JDBC, or SQLAlchemy connection pool settings, not an external proxy.

The diagnostic query for SQL Server shows idle connection distribution in the same way the PostgreSQL query does:

```

-- SQL Server: connection state audit
-- Identifies idle sessions and their resource consumption

```

```

-- Run on Azure SQL Database or SQL Server on RDS
SELECT
    s.session_id,
    s.status,
    s.login_name,
    s.host_name,
    s.program_name,
    DB_NAME(s.database_id) AS database_name,
    s.cpu_time AS cpu_time_ms,
    s.memory_usage * 8 AS memory_kb,
    s.reads,
    s.writes,
    DATEDIFF(SECOND, s.last_request_end_time, GETUTCDATE()) AS idle_seconds,
    r.command AS active_command,
    r.wait_type,
    r.blocking_session_id
FROM sys.dm_exec_sessions s
LEFT JOIN sys.dm_exec_requests r ON r.session_id = s.session_id
WHERE s.is_user_process = 1
    AND s.session_id <> @@SPID
ORDER BY idle_seconds DESC;

```

The SQL Server equivalent of pooling analysis focuses on idle seconds and memory consumption per session. Sessions idle for more than 300 seconds while consuming memory are candidates for pool timeout reduction in the application tier. The corresponding PostgreSQL diagnostic was shown earlier in this chapter; the two queries paired serve as your cross-platform connection audit toolkit.

For Azure SQL specifically, pull the DTU or vCore consumption alongside connection counts using the built-in DMV:

```

-- Azure SQL: vCore / DTU consumption correlated with active connections
-- Run on Azure SQL Database (not on-premise SQL Server)
SELECT
    end_time,
    avg_cpu_percent,
    avg_data_io_percent,
    avg_log_write_percent,
    max_session_percent,
    max_worker_percent,
    avg_memory_usage_percent,
    -- Derived: sessions as a fraction of the tier limit
    max_session_percent / 100.0 * 30000 AS estimated_sessions -- 30000 = General Purpose limit
FROM sys.dm_db_resource_stats
ORDER BY end_time DESC
OFFSET 0 ROWS FETCH NEXT 96 ROWS ONLY; -- last 24 hours at 15-minute intervals

```

When `max_session_percent` is consistently below 20% and `avg_cpu_percent` is below 20%, the Azure SQL tier is oversized relative to actual workload and connection count. The right-sizing action on Azure SQL is to reduce the vCore count or move from General Purpose to a lower service tier — the connection pool on the application side limits the effective session demand, enabling the downsize just as PgBouncer does for PostgreSQL.

Key Takeaways

- **Diagnose before deploying:** use `pg_stat_activity` (PostgreSQL) or `sys.dm_exec_sessions` (SQL Server / Azure SQL) to quantify idle connections and establish a before-state baseline; optimization without measurement is guesswork that cannot survive a budget review.

- **Size the pool with math, not defaults:** set `default_pool_size` to $2 \times$ vCPU of the *target* (smaller) instance, and set `max_client_conn` to $1.25 \times$ peak application thread count; this arithmetic directly determines which instance class becomes feasible and therefore which monthly bill becomes achievable.
 - **Transaction pooling mode unlocks the largest savings:** session pooling returns a connection to the pool only when the client disconnects, which provides minimal multiplexing; transaction pooling returns the server connection after every commit, multiplexing hundreds of client threads onto a handful of server connections and enabling the compute right-sizing that drives cost reduction.
 - **Automate continuous verification:** pool efficiency degrades silently as applications evolve — new ORMs, new services, and new query patterns all introduce connection behavior that can erode multiplexing ratios; the scheduled monitor script in this chapter detects degradation and correlates it with actual AWS Cost Explorer spend before the next billing cycle closes.
 - **The pattern generalizes across clouds:** PgBouncer on EC2 or ECS for AWS RDS, the built-in PgBouncer feature on Azure Database for PostgreSQL Flexible Server, and a Cloud Run or Compute Engine sidecar for Google Cloud SQL all implement the same multiplexing principle; for SQL Server and Azure SQL the equivalent lever is application-tier connection pool tuning surfaced through `sys.dm_exec_sessions` and `sys.dm_db_resource_stats` rather than an external proxy.
-

Chapter 21: Caching Layer

Economics: ElastiCache vs Database Direct

Organizations that add a properly sized Redis caching layer in front of their RDS or Aurora clusters typically see database costs drop by 30–60% within the first billing cycle. A mid-size SaaS company running an `r6g.2xlarge` RDS PostgreSQL instance at \$876/month can often drop to an `r6g.xlarge` at \$438/month after offloading read traffic to ElastiCache for Redis — a `cache.r6g.large` at \$104/month — netting \$334/month in savings on a \$542/month combined workload. That math gets even more compelling at scale. This chapter walks through the economics of caching layer decisions: when adding cache actually saves money, when it just adds complexity and cost, how to right-size ElastiCache clusters, and how to automate cache hit rate monitoring so you catch degradation before it shows up on your AWS bill.

21.1 The Caching Economics Baseline: When Cache Pays for Itself

Before spending a dollar on ElastiCache or Memorystore, you need to establish whether your workload actually benefits from caching. The economics break down into three scenarios, and only two of them favor adding a cache tier.

Scenario A — High read ratio, repetitive queries: Your database serves 90% reads, and a meaningful percentage of those reads fetch the same rows repeatedly. Product catalogs, user profiles, session data, configuration tables. Cache wins overwhelmingly here.

Scenario B — Write-heavy transactional workload: Your database is processing orders, logging events, or updating financial ledgers. Reads are largely unique. Cache adds cost without meaningful throughput benefit.

Scenario C — Mixed workload with hot data concentration: 80% of reads hit 20% of rows. This is the most common production pattern and the most nuanced to optimize.

The first diagnostic step is measuring your current read/write ratio and query repetition rate directly from the database engine. This query runs against PostgreSQL and gives you the data needed to build the business case:

```
-- Identify top repeated queries and estimate cache value
-- Run on PostgreSQL with pg_stat_statements enabled
SELECT
```

```

left(query, 80)                                AS query_preview,
calls,
round(total_exec_time::numeric / calls, 2)      AS avg_ms,
round((total_exec_time / 1000)::numeric, 2)     AS total_exec_sec,
rows,
round(rows::numeric / calls, 1)                 AS avg_rows_returned,
shared_blks_hit + shared_blks_read              AS total_blocks_accessed,
round(
    shared_blks_hit::numeric /
    NULLIF(shared_blks_hit + shared_blks_read, 0) * 100,
    1
)                                                AS buffer_hit_pct,
-- Queries returning same avg rows are cache candidates
CASE
    WHEN round(rows::numeric / calls, 1) BETWEEN 1 AND 50
        AND calls > 1000
        AND round(rows::numeric / calls, 1) = round(rows::numeric / calls, 0)
    THEN 'HIGH cache candidate'
    WHEN calls > 500 AND avg_rows_returned < 100
    THEN 'MEDIUM cache candidate'
    ELSE 'LOW cache value'
END                                              AS cache_recommendation
FROM
pg_stat_statements
WHERE
calls > 100
AND query NOT LIKE '%pg_stat%'
AND query NOT LIKE 'COMMIT%'
AND query NOT LIKE 'BEGIN%'
ORDER BY
calls DESC
LIMIT 40;

```

For SQL Server on RDS or Azure SQL, the equivalent analysis pulls from `sys.dm_exec_query_stats`:

```

-- SQL Server: Identify repetitive read queries suitable for caching
SELECT TOP 40
LEFT(qt.text, 100)                                AS query_preview,
qs.execution_count,
qs.total_logical_reads / qs.execution_count        AS avg_logical_reads,
qs.total_elapsed_time / qs.execution_count / 1000 AS avg_elapsed_ms,
qs.total_rows / qs.execution_count                 AS avg_rows,
qs.total_logical_reads                             AS total_logical_reads,
-- Estimate I/O cost: each logical read = ~8KB page
-- At 50,000 IOPS provisioned IOPS cost on io1, each read has a cost
CAST(qs.total_logical_reads * 8.0 / 1024 / 1024 AS DECIMAL(10,2)) AS total_data_gb_read,
CASE
    WHEN qs.execution_count > 1000
        AND qs.total_rows / qs.execution_count < 100
    THEN 'CACHE NOW'
    WHEN qs.execution_count > 500
    THEN 'EVALUATE'
    ELSE 'SKIP'
END                                              AS cache_action
FROM
sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
WHERE
qt.text NOT LIKE '%sys.dm%'
AND qs.total_logical_reads > 0
ORDER BY
qs.execution_count DESC;

```

Once you have a list of high-frequency, low-row-count queries, you can calculate the cache ROI with a simple formula:

Cache ROI = (Database instance savings per month) – (ElastiCache cost per month)

A real example: An `r6g.4xlarge` RDS PostgreSQL Multi-AZ at \$3,504/month, processing 45,000 queries/minute with 72% of those being repetitive reads, can be right-sized to an `r6g.2xlarge` at \$1,752/month after caching. Adding a `cache.r6g.xlarge` Redis node at \$208/month still nets you \$1,544/month in savings (or \$1,336/month with a Multi-AZ pair at \$416/month for cache redundancy). Cache pays for itself in under one week.

21.2 ElastiCache Sizing and Configuration for Maximum Cost Efficiency

The single most common ElastiCache mistake is over-provisioning. Teams spin up `cache.r6g.2xlarge` nodes because their database is large, when the working set of frequently accessed data is often a fraction of total database size. A 500GB database might have a hot working set of only 8–12GB.

Measuring your actual working set size in PostgreSQL:

```
-- Estimate hot data working set: tables accessed most frequently
-- and their actual size in shared_buffers
SELECT
    schemaname,
    relname                                AS table_name,
    pg_size_pretty(pg_relation_size(relid)) AS table_size,
    pg_size_pretty(pg_total_relation_size(relid)) AS total_size_with_indexes,
    heap_blks_read                           AS disk_reads,
    heap_blks_hit                            AS buffer_hits,
    round(
        heap_blks_hit::numeric /
        NULLIF(heap_blks_hit + heap_blks_read, 0) * 100, 1
    )                                         AS buffer_hit_pct,
    idx_blks_hit,
    idx_blks_read,
    -- Tables with high hit rates are already cached in PG buffer pool
    -- Tables with low hit rates are the cache opportunity
    CASE
        WHEN round(heap_blks_hit::numeric /
            NULLIF(heap_blks_hit + heap_blks_read, 0) * 100, 1) < 80
        THEN pg_relation_size(relid)
        ELSE 0
    END                                     AS bytes_needing_cache
FROM
    pg_statio_user_tables
WHERE
    heap_blks_hit + heap_blks_read > 1000
ORDER BY
    heap_blks_read DESC
LIMIT 30;

-- Total estimated cache requirement
SELECT
    pg_size_pretty(
        SUM(
            CASE
                WHEN round(heap_blks_hit::numeric /
                    NULLIF(heap_blks_hit + heap_blks_read, 0) * 100, 1) < 80
                THEN pg_relation_size(relid)
                ELSE 0
            END
        )
    ) AS estimated_cache_requirement
FROM
    pg_statio_user_tables
```

```
WHERE
    heap_blks_hit + heap_blks_read > 1000;
```

With an estimated working set size in hand, the ElastiCache node selection follows a straightforward mapping. For Redis on AWS, target 75–80% memory utilization at steady state to leave headroom for replication buffers and key expiration overhead:

Working Set	Recommended Node	Memory	On-Demand/hr	Monthly
< 2 GB	cache.t4g.medium	3.09 GB	\$0.032	\$23
2–6 GB	cache.r6g.large	13.07 GB	\$0.145	\$104
6–24 GB	cache.r6g.xlarge	26.32 GB	\$0.290	\$208
24–52 GB	cache.r6g.2xlarge	52.82 GB	\$0.580	\$418
52–100 GB	cache.r6g.4xlarge	105.81 GB	\$1.160	\$835

Reserved Instance pricing reduces these costs by 40–45% for 1-year No Upfront and up to 55% for 3-year All Upfront. A `cache.r6g.xlarge` on a 1-year reserved basis drops from \$208/month to approximately \$125/month. For any cache cluster you expect to run for more than 6 months, reserved pricing is the correct economic choice.

The Python script below automates cluster sizing recommendations by pulling CloudWatch metrics and comparing against current node specifications:

```
#!/usr/bin/env python3
"""
ElastiCache Sizing Analyzer
Pulls CloudWatch metrics for existing clusters and recommends right-sizing.
Schedule via CloudWatch Events or GitHub Actions weekly.
"""

import boto3
import json
from datetime import datetime, timedelta
from dataclasses import dataclass
from typing import Optional

@dataclass
class ClusterSizingRecommendation:
    cluster_id: str
    current_node_type: str
    current_monthly_cost: float
    avg_memory_utilization_pct: float
    max_memory_utilization_pct: float
    avg_cpu_utilization_pct: float
    avg_cache_hit_rate_pct: float
    recommended_node_type: Optional[str]
    recommended_monthly_cost: Optional[float]
    monthly_savings: float
    action: str
    reasoning: str

# Current on-demand hourly pricing (us-east-1, update as needed)
ELASTICACHE_PRICING = {
    "cache.t4g.micro": {"memory_gb": 0.5, "hourly": 0.008},
    "cache.t4g.small": {"memory_gb": 1.37, "hourly": 0.016},
    "cache.t4g.medium": {"memory_gb": 3.09, "hourly": 0.032},
    "cache.r6g.large": {"memory_gb": 13.07, "hourly": 0.145},
    "cache.r6g.xlarge": {"memory_gb": 26.32, "hourly": 0.290},
    "cache.r6g.2xlarge": {"memory_gb": 52.82, "hourly": 0.580},
    "cache.r6g.4xlarge": {"memory_gb": 105.81, "hourly": 1.160},
```

```

    "cache.r6g.8xlarge": {"memory_gb": 209.55, "hourly": 2.320},
}

HOURS_PER_MONTH = 730

def get_cloudwatch_metric_stats(cw_client, cluster_id, metric_name,
                                namespace, dimension_name, days=14):
    """Retrieve average and max for a CloudWatch metric over the past N days."""
    end_time = datetime.utcnow()
    start_time = end_time - timedelta(days=days)

    response = cw_client.get_metric_statistics(
        Namespace=namespace,
        MetricName=metric_name,
        Dimensions=[{"Name": dimension_name, "Value": cluster_id}],
        StartTime=start_time,
        EndTime=end_time,
        Period=3600, # 1-hour granularity
        Statistics=["Average", "Maximum"]
    )

    datapoints = response.get("Datapoints", [])
    if not datapoints:
        return 0.0, 0.0

    avg_vals = [dp["Average"] for dp in datapoints]
    max_vals = [dp["Maximum"] for dp in datapoints]
    return sum(avg_vals) / len(avg_vals), max(max_vals)

def analyze_cluster(ec_client, cw_client, cluster_id: str,
                    node_type: str) -> ClusterSizingRecommendation:
    """Analyze a single ElastiCache cluster and generate sizing recommendation."""

    current_pricing = ELASTICACHE_PRICING.get(node_type, {})
    current_hourly = current_pricing.get("hourly", 0.0)
    current_monthly = current_hourly * HOURS_PER_MONTH

    # Pull key metrics from CloudWatch
    avg_mem, max_mem = get_cloudwatch_metric_stats(
        cw_client, cluster_id, "DatabaseMemoryUsagePercentage",
        "AWS/ElastiCache", "CacheClusterId"
    )
    avg_cpu, max_cpu = get_cloudwatch_metric_stats(
        cw_client, cluster_id, "CPUUtilization",
        "AWS/ElastiCache", "CacheClusterId"
    )
    avg_hits, _ = get_cloudwatch_metric_stats(
        cw_client, cluster_id, "CacheHits",
        "AWS/ElastiCache", "CacheClusterId"
    )
    avg_misses, _ = get_cloudwatch_metric_stats(
        cw_client, cluster_id, "CacheMisses",
        "AWS/ElastiCache", "CacheClusterId"
    )

    # Calculate cache hit rate
    total_ops = avg_hits + avg_misses
    cache_hit_rate = (avg_hits / total_ops * 100) if total_ops > 0 else 0.0

    recommended_node = None
    recommended_monthly = None
    action = "KEEP"
    reasoning = ""

    # Sizing decision logic
    if avg_mem < 30 and max_mem < 45:
        # Memory significantly underutilized - downsize candidate
        node_types_sorted = sorted(

```

```

    ELASTICACHE_PRICING.items(),
    key=lambda x: x[1]["memory_gb"]
)
current_memory = current_pricing.get("memory_gb", 0)
target_memory = current_memory * (max_mem / 100) * 1.5 # 50% headroom

for node_name, node_specs in node_types_sorted:
    if node_specs["memory_gb"] >= target_memory:
        if node_specs["hourly"] < current_hourly:
            recommended_node = node_name
            recommended_monthly = node_specs["hourly"] * HOURS_PER_MONTH
            action = "DOWNSIZE"
            reasoning = (
                f"Avg memory {avg_mem:.1f}%, max {max_mem:.1f}%. "
                f"Node is oversized. Target memory: {target_memory:.1f}GB."
            )
            break

elif max_mem > 85:
    # Memory pressure - upsize required
    node_types_sorted = sorted(
        ELASTICACHE_PRICING.items(),
        key=lambda x: x[1]["memory_gb"]
    )
    current_memory = current_pricing.get("memory_gb", 0)
    target_memory = current_memory * 1.5

    for node_name, node_specs in node_types_sorted:
        if node_specs["memory_gb"] >= target_memory:
            recommended_node = node_name
            recommended_monthly = node_specs["hourly"] * HOURS_PER_MONTH
            action = "UPSIZE"
            reasoning = (
                f"Max memory utilization {max_mem:.1f}% indicates eviction risk. "
                f"Recommend node with {target_memory:.1f}GB+ memory."
            )
            break

if cache_hit_rate < 80 and action == "KEEP":
    action = "INVESTIGATE"
    reasoning = (
        f"Cache hit rate {cache_hit_rate:.1f}% is below 80% threshold. "
        f"Cache may not be providing adequate value - review TTL and key strategy."
    )

if action == "KEEP" and not reasoning:
    reasoning = (
        f"Memory utilization healthy at avg {avg_mem:.1f}%, max {max_mem:.1f}%. "
        f"Cache hit rate {cache_hit_rate:.1f}% is acceptable."
    )

monthly_savings = 0.0
if recommended_monthly and action == "DOWNSIZE":
    monthly_savings = current_monthly - recommended_monthly
elif recommended_monthly and action == "UPSIZE":
    monthly_savings = -(recommended_monthly - current_monthly) # negative = cost increase

return ClusterSizingRecommendation(
    cluster_id=cluster_id,
    current_node_type=node_type,
    current_monthly_cost=round(current_monthly, 2),
    avg_memory_utilization_pct=round(avg_mem, 1),
    max_memory_utilization_pct=round(max_mem, 1),
    avg_cpu_utilization_pct=round(avg_cpu, 1),
    avg_cache_hit_rate_pct=round(cache_hit_rate, 1),
    recommended_node_type=recommended_node,
    recommended_monthly_cost=round(recommended_monthly, 2) if recommended_monthly else None,
    monthly_savings=round(monthly_savings, 2),
    action=action,
    reasoning=reasoning
)

```

```

)

def main():
    session = boto3.Session()
    ec_client = session.client("elasticache", region_name="us-east-1")
    cw_client = session.client("cloudwatch", region_name="us-east-1")

    # Get all Redis clusters
    paginator = ec_client.get_paginator("describe_cache_clusters")
    all_clusters = []
    for page in paginator.paginate(ShowCacheNodeInfo=True):
        all_clusters.extend(page["CacheClusters"])

    recommendations = []
    total_current_monthly = 0.0
    total_potential_savings = 0.0

    print(f"\n{'='*80}")
    print(f"ElastiCache Sizing Report - {datetime.utcnow().strftime('%Y-%m-%d %H:%M UTC')}")
    print(f"{'='*80}\n")

    for cluster in all_clusters:
        cluster_id = cluster["CacheClusterId"]
        node_type = cluster["CacheNodeType"]
        engine = cluster.get("Engine", "unknown")

        if engine != "redis":
            continue

        rec = analyze_cluster(ec_client, cw_client, cluster_id, node_type)
        recommendations.append(rec)

        total_current_monthly += rec.current_monthly_cost
        if rec.monthly_savings > 0:
            total_potential_savings += rec.monthly_savings

    print(f"Cluster: {rec.cluster_id}")
    print(f"  Current Node:      {rec.current_node_type} (${rec.current_monthly_cost:.2f}/mo)")
    print(f"  Memory Util:       avg {rec.avg_memory_utilization_pct}% / max")
    print(f"                      {rec.max_memory_utilization_pct}%")
    print(f"  CPU Util:          avg {rec.avg_cpu_utilization_pct}%")
    print(f"  Cache Hit Rate:     {rec.avg_cache_hit_rate_pct}%")
    print(f"  Action:             {rec.action}")
    print(f"  Reasoning:          {rec.reasoning}")
    if rec.recommended_node_type:
        print(f"    Recommended Node:  {rec.recommended_node_type}")
        print(f"    ↪ (${rec.recommended_monthly_cost:.2f}/mo)")
        print(f"  Monthly Savings:    ${rec.monthly_savings:.2f}")
    print()

    print(f"{'='*80}")
    print(f"SUMMARY")
    print(f"  Total Current Monthly Cost:    ${total_current_monthly:.2f}")
    print(f"  Total Potential Monthly Savings: ${total_potential_savings:.2f}")
    print(f"  Annual Savings Opportunity:    ${total_potential_savings * 12:.2f}")
    print(f"{'='*80}\n")

    # Write JSON output for downstream processing or ticketing
    output = {
        "generated_at": datetime.utcnow().isoformat(),
        "total_current_monthly_cost": round(total_current_monthly, 2),
        "total_potential_monthly_savings": round(total_potential_savings, 2),
        "recommendations": [
            {
                "cluster_id": r.cluster_id,
                "current_node_type": r.current_node_type,
                "current_monthly_cost": r.current_monthly_cost,
                "action": r.action,
                "recommended_node_type": r.recommended_node_type,
            }
        ]
    }

```

```
        "monthly_savings": r.monthly_savings,
        "reasoning": r.reasoning
    }
    for r in recommendations
]

with open("/tmp/elasticache_sizing_report.json", "w") as f:
    json.dump(output, f, indent=2)
print("Report written to /tmp/elasticache_sizing_report.json")

if __name__ == "__main__":
    main()
```

21.3 Cache Hit Rate Monitoring and the Cost of Cache Misses

A cache that isn't hitting is worse than no cache at all — you've added infrastructure cost and latency on cache misses without reducing database load. The break-even cache hit rate for most workloads is approximately 60–70%, but the economic sweet spot is 85% or higher.

Here is the math on why cache hit rate matters so much to your bill. Consider an application serving 10,000 requests/minute:

- At **50% hit rate**: 5,000 requests/minute still hit the database. Database must be sized for full concurrent load.
- At **80% hit rate**: 2,000 requests/minute hit the database. Database can be one tier smaller.
- At **95% hit rate**: 500 requests/minute hit the database. Database can often drop two tiers.

For an `r6g.4xlarge` RDS PostgreSQL Multi-AZ (\$3,504/month) that can be right-sized based on cache effectiveness: - 80% hit rate → `r6g.2xlarge` (\$1,752/month) + `cache.r6g.xlarge` (\$208/month) = **\$1,544/month savings** - 95% hit rate → `r6g.xlarge` (\$876/month) + `cache.r6g.xlarge` (\$208/month) = **\$2,420/month savings**

The difference between 80% and 95% cache effectiveness is worth nearly \$900/month on that single instance. Monitoring hit rate and acting on degradation is not optional — it's the financial control mechanism for the entire caching strategy.

CloudWatch Metrics for ElastiCache Redis hit rate monitoring:

The two primary metrics are `CacheHits` and `CacheMisses`. AWS does not publish a pre-computed hit rate metric, so you need to derive it using metric math. The CloudWatch alarm below triggers when the rolling hit rate drops below your threshold:

```
#!/usr/bin/env python3
"""
ElastiCache Cache Hit Rate Alarm Setup
Creates CloudWatch metric math alarms for cache hit rate monitoring.
Run once per cluster to establish monitoring baseline.
"""

import boto3
import json

def create_cache_hit_rate_alarm(
    cluster_id: str,
    threshold_pct: float = 80.0,
```

```

evaluation_periods: int = 3,
period_seconds: int = 300,
sns_topic_arn: str = None,
region: str = "us-east-1"
):
    """
    Create a CloudWatch alarm that fires when cache hit rate drops below threshold.
    Uses metric math to compute HitRate = Hits / (Hits + Misses) * 100.
    """
    cw_client = boto3.client("cloudwatch", region_name=region)

    alarm_name = f"elasticsearch-hit-rate-low-{cluster_id}"
    alarm_description = (
        f"Cache hit rate for {cluster_id} has dropped below {threshold_pct}% "
        f"over {evaluation_periods} consecutive {period_seconds//60}-minute periods. "
        f"Database load may be increasing. Review TTL settings and key eviction policy."
    )

    alarm_actions = []
    if sns_topic_arn:
        alarm_actions.append(sns_topic_arn)

    # Metric math: HitRate = (Hits / (Hits + Misses)) * 100
    # CloudWatch metric math requires FILL to handle zero-division gracefully
    metrics = [
        {
            "Id": "hits",
            "MetricStat": {
                "Metric": {
                    "Namespace": "AWS/ElastiCache",
                    "MetricName": "CacheHits",
                    "Dimensions": [
                        {"Name": "CacheClusterId", "Value": cluster_id}
                    ]
                },
                "Period": period_seconds,
                "Stat": "Sum"
            },
            "ReturnData": False
        },
        {
            "Id": "misses",
            "MetricStat": {
                "Metric": {
                    "Namespace": "AWS/ElastiCache",
                    "MetricName": "CacheMisses",
                    "Dimensions": [
                        {"Name": "CacheClusterId", "Value": cluster_id}
                    ]
                },
                "Period": period_seconds,
                "Stat": "Sum"
            },
            "ReturnData": False
        },
        {
            "Id": "hit_rate",
            "Expression": "IF(hits+misses > 0, (hits/(hits+misses))*100, 100)",
            "Label": "CacheHitRatePct",
            "ReturnData": True
        }
    ]

    cw_client.put_metric_alarm(
        AlarmName=alarm_name,
        AlarmDescription=alarm_description,
        Metrics=metrics,
        ComparisonOperator="LessThanThreshold",
        Threshold=threshold_pct,
        EvaluationPeriods=evaluation_periods,

```

```

        DatapointsToAlarm=evaluation_periods,
        TreatMissingData="notBreaching",
        AlarmActions=alarm_actions,
        OKActions=alarm_actions
    )

    print(f"Created alarm: {alarm_name}")
    print(f"    Threshold:      Hit rate < {threshold_pct}%")
    print(f"    Evaluation window:  {evaluation_periods * period_seconds // 60} minutes")
    print(f"    SNS notification:    {sns_topic_arn or 'none configured'}")
    return alarm_name

def create_evictions_alarm(
    cluster_id: str,
    evictions_threshold: int = 1000,
    period_seconds: int = 300,
    sns_topic_arn: str = None,
    region: str = "us-east-1"
):
    """
    Create alarm for cache evictions - a leading indicator of memory pressure
    that precedes hit rate degradation.
    """
    cw_client = boto3.client("cloudwatch", region_name=region)

    alarm_name = f"elasticsearch-evictions-high-{cluster_id}"

    cw_client.put_metric_alarm(
        AlarmName=alarm_name,
        AlarmDescription=(
            f"High eviction rate on {cluster_id}. Memory pressure is causing "
            f"key evictions, which will degrade cache hit rate and increase "
            f"database load. Consider upsizing or reviewing maxmemory-policy."
        ),
        Namespace="AWS/ElastiCache",
        MetricName="Evictions",
        Dimensions=[{"Name": "CacheClusterId", "Value": cluster_id}],
        Period=period_seconds,
        Statistic="Sum",
        ComparisonOperator="GreaterThanThreshold",
        Threshold=evictions_threshold,
        EvaluationPeriods=2,
        DatapointsToAlarm=2,
        TreatMissingData="notBreaching",
        AlarmActions=[sns_topic_arn] if sns_topic_arn else [],
        OKActions=[sns_topic_arn] if sns_topic_arn else []
    )

    print(f"Created evictions alarm: {alarm_name}")
    return alarm_name

def setup_cluster_monitoring(cluster_ids: list, sns_topic_arn: str, region: str = "us-east-1"):
    """Set up full monitoring suite for a list of ElastiCache cluster IDs."""
    for cluster_id in cluster_ids:
        print(f"\nSetting up monitoring for: {cluster_id}")
        create_cache_hit_rate_alarm(
            cluster_id=cluster_id,
            threshold_pct=80.0,
            sns_topic_arn=sns_topic_arn,
            region=region
        )
        create_evictions_alarm(
            cluster_id=cluster_id,
            evictions_threshold=500,
            sns_topic_arn=sns_topic_arn,
            region=region
        )

```



```

if __name__ == "__main__":
    import sys

    # Example usage - replace with your actual cluster IDs and SNS ARN
    clusters = [
        "prod-sessions-001",
        "prod-catalog-001",
        "prod-user-profiles-001"
    ]
    sns_arn = "arn:aws:sns:us-east-1:123456789012:db-finops-alerts"

    setup_cluster_monitoring(clusters, sns_arn)

```

Diagnosing hit rate degradation — what actually causes it:

When your cache hit rate drops, the cause is almost always one of four things:

1. **TTL too short** — Keys expire before they can be reused. Check your application's cache TTL settings against actual query repetition frequency. A user profile queried once every 15 minutes with a 5-minute TTL will never hit cache.
2. **Key space explosion** — Query parameters are being included in cache keys without normalization. `user_profile:12345:v2:en-US` and `user_profile:12345:v2:en-GB` might be the same data but generate separate keys, doubling your key count and halving effective hit rates.
3. **Memory pressure causing eviction** — Check the Evictions CloudWatch metric. Non-zero evictions mean keys are being forced out before they naturally expire. The `maxmemory-policy` setting (`allkeys-lru` vs `volatile-lru` vs `noeviction`) determines what gets evicted first. For caching workloads, `allkeys-lru` is typically correct.
4. **Deployment or code push flushed the cache** — A common pattern where cache warming hasn't completed post-deployment. Hit rates will recover organically within minutes to hours, but the database spike during that window can trigger auto-scaling events or performance alerts.

The following SQL query, run against PostgreSQL after a cache hit rate drop, helps identify which tables have suddenly started generating high disk read volume — a direct signal that previously cached data is now bypassing cache:

```

-- Detect tables with increased disk read activity (run periodically and compare)
-- Useful for diagnosing which data tier lost cache coverage
SELECT
    schemaname,
    relname,
    pg_size_pretty(pg_relation_size(relid)) AS table_name,
    heap_blks_read AS table_size,
    heap_blks_hit AS disk_reads_total,
    round(
        heap_blks_read::numeric /
        NULLIF(heap_blks_hit + heap_blks_read, 0) * 100, 2
    ) AS buffer_hits_total,
    AS disk_read_pct,
    -- Sudden high disk_read_pct on a table that "should" be cached
    -- signals cache miss on that key pattern
    CASE
        WHEN round(heap_blks_read::numeric /
            NULLIF(heap_blks_hit + heap_blks_read, 0) * 100, 2) > 20
        THEN 'INVESTIGATE - high disk read ratio'
        ELSE 'ok'
    END AS status
FROM

```

```

pg_statio_user_tables
WHERE
    heap_blks_read + heap_blks_hit > 500
ORDER BY
    disk_read_pct DESC
LIMIT 25;

```

For SQL Server on RDS or Azure SQL, the buffer cache miss ratio surfaces through DMVs:

```

-- SQL Server: Identify objects with poor buffer cache coverage
-- High physical_reads relative to logical_reads signals cache bypass
SELECT TOP 25
    OBJECT_SCHEMA_NAME(i.object_id)          AS schema_name,
    OBJECT_NAME(i.object_id)                 AS table_name,
    i.name                                   AS index_name,
    ios.leaf_allocation_count,
    ios.nonleaf_allocation_count,
    ios.leaf_insert_count,
    ios.leaf_update_count,
    ios.leaf_delete_count,
    -- For page-level cache analysis, check sys.dm_os_buffer_descriptors
    (
        SELECT COUNT(*)
        FROM sys.dm_os_buffer_descriptors bd
        WHERE bd.object_id = i.object_id
            AND bd.is_dirty = 0
    )                                         AS clean_pages_in_buffer,
    (
        SELECT COUNT(*)
        FROM sys.dm_os_buffer_descriptors bd
        WHERE bd.object_id = i.object_id
    )                                         AS total_pages_in_buffer,
    p.rows                                   AS row_count
FROM
    sys.indexes i
    JOIN sys.dm_db_index_operational_stats(DB_ID(), NULL, NULL, NULL) ios
        ON i.object_id = ios.object_id AND i.index_id = ios.index_id
    JOIN sys.partitions p
        ON i.object_id = p.object_id AND i.index_id = p.index_id
WHERE
    i.object_id > 100 -- exclude system objects
    AND OBJECT_NAME(i.object_id) IS NOT NULL
ORDER BY
    leaf_insert_count + leaf_update_count + leaf_delete_count DESC;

```

21.4 Multi-Tier Caching: When a Second Cache Layer Changes the Math

For workloads with very high throughput — above roughly 50,000 requests per second to a single cache cluster — a second caching tier becomes both a performance and a cost optimization. The pattern is:

1. **Application-level in-memory cache** (e.g., a 256MB Python `functools.lru_cache` or a Go `sync.Map`) absorbs the top 5–10% of the hottest keys with zero network overhead.
2. **ElastiCache Redis** absorbs the next tier — high-frequency data with moderate working set size.
3. **RDS/Aurora** handles only cache misses from both layers above.

The economics shift materially: the application-level cache costs nothing in AWS infrastructure dollars (it runs in your existing application server memory), so every request it absorbs is pure

savings against both ElastiCache and RDS.

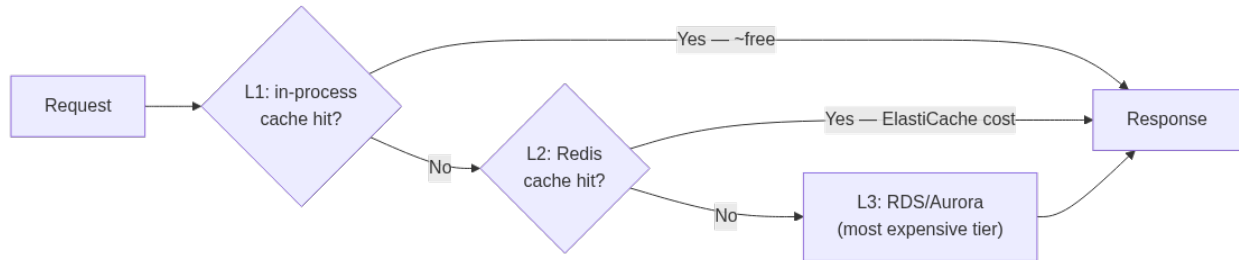


Figure 21: Each layer above the database absorbs load at a lower marginal cost than the one below it — L1 is free, L2 is metered, L3 is the most expensive tier to hit.

The gotcha with application-level caches in auto-scaling environments is consistency. When you have 20 application server instances each holding their own in-memory cache, you get 20 independent cache warming events after any deployment and 20 separate TTL clocks. The standard solution is short TTLs at the application layer (30–120 seconds) combined with longer TTLs in Redis, accepting brief inconsistency windows in exchange for the compute savings.

This Python module demonstrates the two-tier caching pattern with explicit cost attribution logging — letting your FinOps tooling see exactly where each tier is absorbing load:

```

"""
Two-tier cache implementation with FinOps cost attribution.
Tier 1: Local in-process LRU cache (zero AWS cost)
Tier 2: ElastiCache Redis (metered AWS cost)
Tier 3: RDS PostgreSQL (most expensive tier - minimize hits here)
"""

import time
import json
import hashlib
import logging
from functools import lru_cache
from collections import Counter
from threading import Lock
from typing import Any, Optional, Callable
import redis
import psycopg2

logger = logging.getLogger(__name__)

class TieredCacheMetrics:
    """Thread-safe counters for cost attribution across cache tiers."""

    def __init__(self):
        self._lock = Lock()
        self._counters = Counter()

    def record(self, tier: str, outcome: str):
        with self._lock:
            self._counters[f"{tier}:{outcome}"] += 1

    def report(self) -> dict:
        with self._lock:
            counters = dict(self._counters)

        l1_hits = counters.get("l1:hit", 0)
        l2_hits = counters.get("l2:hit", 0)
        l3_hits = counters.get("l3:hit", 0) # database hits

```

```

l2_misses = counters.get("l2:miss", 0)

total = l1_hits + l2_hits + l3_hits
if total == 0:
    return {"total_requests": 0}

return {
    "total_requests": total,
    "l1_hit_rate_pct": round(l1_hits / total * 100, 2),
    "l2_hit_rate_pct": round(l2_hits / total * 100, 2),
    "db_hit_rate_pct": round(l3_hits / total * 100, 2),
    "l1_hits": l1_hits,
    "l2_hits": l2_hits,
    "db_hits": l3_hits,
    # Cost attribution: only l3 hits have meaningful DB cost
    # l2 hits have ElastiCache cost (network + compute)
    # l1 hits are essentially free
    "estimated_db_queries_avoided": l1_hits + l2_hits,
    "cost_pressure_score": round(l3_hits / total * 100, 2)
}

class TieredCache:
    """
    Two-tier cache: in-process dict (L1) + Redis (L2) + DB fallback (L3).
    L1 TTL is intentionally short to limit stale data exposure in
    horizontally scaled deployments.
    """

    def __init__(
        self,
        redis_client: redis.Redis,
        l1_max_size: int = 1000,
        l1_ttl_seconds: int = 60,
        l2_ttl_seconds: int = 3600
    ):
        self.redis = redis_client
        self.l1_ttl = l1_ttl_seconds
        self.l2_ttl = l2_ttl_seconds
        self.metrics = TieredCacheMetrics()

        # L1: simple dict with expiry timestamps
        self._l1: dict[str, tuple[Any, float]] = {}
        self._l1_max = l1_max_size
        self._l1_lock = Lock()

    def _make_key(self, namespace: str, identifier: str) -> str:
        """Produce a consistent, safe Redis key."""
        safe_id = hashlib.md5(identifier.encode()).hexdigest()
        return f"tc:{namespace}:{safe_id}"

    def _l1_get(self, key: str) -> tuple[bool, Any]:
        with self._l1_lock:
            entry = self._l1.get(key)
            if entry is None:
                return False, None
            value, expiry = entry
            if time.monotonic() > expiry:
                del self._l1[key]
                return False, None
            return True, value

    def _l1_set(self, key: str, value: Any):
        with self._l1_lock:
            # Evict oldest entry if at capacity (simple FIFO eviction)
            if len(self._l1) >= self._l1_max:
                oldest_key = next(iter(self._l1))
                del self._l1[oldest_key]
            self._l1[key] = (value, time.monotonic() + self.l1_ttl)

```

```

def get_or_fetch(
    self,
    namespace: str,
    identifier: str,
    fetch_fn: Callable[[], Any],
    skip_l1: bool = False
) -> Any:
    """
    Retrieve value from L1 → L2 → L3 (fetch_fn) with metrics tracking.

    Args:
        namespace: Cache namespace (e.g., 'user_profile', 'product_catalog')
        identifier: Unique key within namespace (e.g., user_id as string)
        fetch_fn: Callable that hits the database if both cache tiers miss
        skip_l1: Force bypass of L1 (useful for write-through invalidation)
    """
    cache_key = self._make_key(namespace, identifier)

    # L1 check
    if not skip_l1:
        found, value = self._l1_get(cache_key)
        if found:
            self.metrics.record("l1", "hit")
            return value

    # L2 check (Redis)
    try:
        raw = self.redis.get(cache_key)
        if raw is not None:
            self.metrics.record("l2", "hit")
            value = json.loads(raw)
            if not skip_l1:
                self._l1_set(cache_key, value)
            return value
        else:
            self.metrics.record("l2", "miss")
    except redis.RedisError as e:
        logger.warning(f"Redis error on GET {cache_key}: {e}. Falling through to DB.")
        self.metrics.record("l2", "error")

    # L3: database fetch
    self.metrics.record("l3", "hit")
    value = fetch_fn()

    if value is not None:
        # Populate both cache tiers on DB hit
        try:
            self.redis.setex(cache_key, self.l2_ttl, json.dumps(value))
        except redis.RedisError as e:
            logger.warning(f"Redis error on SET {cache_key}: {e}")
        self._l1_set(cache_key, value)

    return value

def invalidate(self, namespace: str, identifier: str):
    """Remove a key from both cache tiers (use on write operations)."""
    cache_key = self._make_key(namespace, identifier)
    with self._l1_lock:
        self._l1.pop(cache_key, None)
    try:
        self.redis.delete(cache_key)
    except redis.RedisError as e:
        logger.warning(f"Redis error on DELETE {cache_key}: {e}")

def get_metrics_report(self) -> dict:
    return self.metrics.report()

```

Emitting the `get_metrics_report()` output every 5 minutes to CloudWatch custom metrics gives your FinOps dashboard a direct window into what percentage of traffic is reaching the database.

When `db_hit_rate_pct` trends upward, the cost implication is immediate and quantifiable.

21.5 Google Cloud Memorystore vs Cloud SQL: The GCP Equivalent Analysis

The economic principles are identical on GCP, but the pricing structure and tooling differ enough to warrant explicit treatment. Google Cloud Memorystore for Redis is the managed caching layer that offloads Cloud SQL (PostgreSQL or MySQL) and AlloyDB workloads.

Memorystore for Redis pricing is capacity-based rather than node-type-based: you pay per GB of provisioned capacity per hour. As of this writing, the Standard Tier (with replication and automatic failover) runs approximately \$0.049/GB/hour in `us-central1`, and the Basic Tier (no replication) runs approximately \$0.016/GB/hour. A 10GB Standard Tier instance therefore costs roughly \$0.49/hour or \$357/month — comparable to an ElastiCache `cache.r6g.xlarge` for similar memory capacity.

The GCP equivalent of the ElastiCache sizing analyzer uses the Cloud Monitoring API (formerly Stackdriver) to pull the same hit rate and memory utilization metrics:

```
#!/usr/bin/env python3
"""
Google Cloud Memorystore Sizing Analyzer
Uses Cloud Monitoring API to analyze Redis instance utilization
and generate right-sizing recommendations.
"""

from google.cloud import monitoring_v3
from google.cloud import redis_v1
from datetime import datetime, timedelta, timezone
import math

# Memorystore Standard Tier pricing: $/GB/hour (us-central1)
# Update per https://cloud.google.com/memorystore/docs/redis/pricing
MEMORYSTORE_STANDARD_COST_PER_GB_HOUR = 0.049
MEMORYSTORE_BASIC_COST_PER_GB_HOUR = 0.016
HOURS_PER_MONTH = 730

def get_memorystore_instances(project_id: str) -> list:
    """Retrieve all Memorystore for Redis instances in a project."""
    client = redis_v1.CloudRedisClient()
    parent = f"projects/{project_id}/locations/-" # '-' = all regions
    instances = list(client.list_instances(request={"parent": parent}))
    return instances

def get_metric_timeseries(
    project_id: str,
    instance_name: str,
    metric_type: str,
    days: int = 14
) -> list[float]:
    """Fetch metric time series for a Memorystore instance."""
    client = monitoring_v3.MetricServiceClient()
    project_name = f"projects/{project_id}"

    end_time = datetime.now(timezone.utc)
    start_time = end_time - timedelta(days=days)
```

```

interval = monitoring_v3.TimeInterval(
    {
        "end_time": {"seconds": int(end_time.timestamp())},
        "start_time": {"seconds": int(start_time.timestamp())},
    }
)

filter_str = (
    f'metric.type="{metric_type}" '
    f'AND resource.labels.instance_id="{instance_name}" '
)

results = client.list_time_series(
    request={
        "name": project_name,
        "filter": filter_str,
        "interval": interval,
        "view": monitoring_v3.ListTimeSeriesRequest.TimeSeriesView.FULL,
    }
)

values = []
for series in results:
    for point in series.points:
        val = point.value.double_value or point.value.int64_value
        values.append(float(val))

return values

def analyze_memorystore_instance(project_id: str, instance) -> dict:
    """Generate sizing recommendation for a single Memorystore instance."""
    instance_id = instance.name.split("/")[-1]
    memory_size_gb = instance.memory_size_gb
    tier = "STANDARD" if instance.tier == redis_v1.Instance.Tier.STANDARD_HA else "BASIC"
    cost_per_gb_hour = (
        MEMORYSTORE_STANDARD_COST_PER_GB_HOUR if tier == "STANDARD"
        else MEMORYSTORE_BASIC_COST_PER_GB_HOUR
    )
    current_monthly_cost = memory_size_gb * cost_per_gb_hour * HOURS_PER_MONTH

    # Memory usage ratio (0-1 scale in Cloud Monitoring)
    mem_usage_values = get_metric_timeseries(
        project_id, instance_id,
        "redis.googleapis.com/stats/memory/usage_ratio"
    )

    # Cache hit ratio (0-1)
    hit_ratio_values = get_metric_timeseries(
        project_id, instance_id,
        "redis.googleapis.com/stats/keyspace_hits_ratio"
    )

    avg_mem_pct = (sum(mem_usage_values) / len(mem_usage_values) * 100) if mem_usage_values else 0
    max_mem_pct = (max(mem_usage_values) * 100) if mem_usage_values else 0
    avg_hit_rate = (sum(hit_ratio_values) / len(hit_ratio_values) * 100) if hit_ratio_values else 0

    action = "KEEP"
    recommended_gb = None
    reasoning = ""

    if avg_mem_pct < 30 and max_mem_pct < 45:
        # Overprovisioned - calculate minimum safe size
        min_required_gb = math.ceil(memory_size_gb * (max_mem_pct / 100) * 1.5)
        # Round up to nearest valid Memorystore size
        valid_sizes = [1, 2, 4, 5, 6, 8, 10, 12, 14, 16, 18, 20,
                       25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100]
        recommended_gb = next((s for s in valid_sizes if s >= min_required_gb), memory_size_gb)
        if recommended_gb < memory_size_gb:
            action = "DOWNSIZE"
            reasoning = (

```

```

        f"Avg memory {avg_mem_pct:.1f}%, max {max_mem_pct:.1f}%. "
        f"Reduce from {memory_size_gb}GB to {recommended_gb}GB."
    )
elif max_mem_pct > 85:
    action = "UPSIZE"
    recommended_gb = math.ceil(memory_size_gb * 1.5)
    reasoning = f"Max memory {max_mem_pct:.1f}% - eviction risk. Upsize to {recommended_gb}GB."

if avg_hit_rate < 80 and action == "KEEP":
    action = "INVESTIGATE"
    reasoning = f"Cache hit rate {avg_hit_rate:.1f}% below 80% threshold."

recommended_monthly = (
    recommended_gb * cost_per_gb_hour * HOURS_PER_MONTH
    if recommended_gb else current_monthly_cost
)
monthly_savings = (
    current_monthly_cost - recommended_monthly if action == "DOWNSIZE" else 0.0
)

return {
    "instance_id": instance_id,
    "tier": tier,
    "memory_size_gb": memory_size_gb,
    "current_monthly_cost": round(current_monthly_cost, 2),
    "avg_memory_pct": round(avg_mem_pct, 1),
    "max_memory_pct": round(max_mem_pct, 1),
    "avg_hit_rate_pct": round(avg_hit_rate, 1),
    "action": action,
    "recommended_gb": recommended_gb,
    "recommended_monthly_cost": round(recommended_monthly, 2),
    "monthly_savings": round(monthly_savings, 2),
    "reasoning": reasoning
}

def main(project_id: str):
    instances = get_memorystore_instances(project_id)
    total_savings = 0.0

    print(f"\n{' '*70}")
    print(f"Memorystore Sizing Report - {datetime.now().strftime('%Y-%m-%d %H:%M')}")
    print(f"Project: {project_id}")
    print(f"{' '*70}\n")

    for instance in instances:
        rec = analyze_memorystore_instance(project_id, instance)
        total_savings += rec["monthly_savings"]

        print(f"Instance: {rec['instance_id']} ({rec['tier']})")
        print(f"  Current:   {rec['memory_size_gb']}GB  ${rec['current_monthly_cost']:.2f}/mo")
        print(f"  Memory:    avg {rec['avg_memory_pct']}% / max {rec['max_memory_pct']}%")
        print(f"  Hit Rate:   {rec['avg_hit_rate_pct']}%")
        print(f"  Action:     {rec['action']}")
        if rec['reasoning']:
            print(f"    Reasoning: {rec['reasoning']}")
        if rec['recommended_gb']:
            print(f"    Recommend: {rec['recommended_gb']}GB  ${rec['recommended_monthly_cost']:.2f}/mo")
            print(f"    Savings:   ${rec['monthly_savings']:.2f}/mo")
        print()

    print(f"Total Potential Monthly Savings: ${total_savings:.2f}")
    print(f"Annual Savings Opportunity:      ${total_savings * 12:.2f}\n")

if __name__ == "__main__":
    import sys
    project_id = sys.argv[1] if len(sys.argv) > 1 else "your-gcp-project-id"
    main(project_id)

```


GCP-specific consideration: Memystore vs AlloyDB Omni caching

AlloyDB has a columnar cache (the AlloyDB Columnar Engine) that functions as an integrated analytical cache within the database itself. For analytical workloads on AlloyDB, enabling the columnar engine can reduce Memystore dependency for certain query patterns — but it serves a fundamentally different access pattern than Redis. AlloyDB’s columnar cache accelerates scan-heavy analytical queries; Redis accelerates key-value lookups and session storage. They are complements, not substitutes, and the cost analysis should treat them as separate line items.

21.6 Azure Cache for Redis vs Azure SQL: The Microsoft Stack Analysis

Azure Cache for Redis follows a tiered model distinct from both AWS and GCP. The relevant tiers for database offload workloads are:

- **Basic:** Single node, no SLA, development and testing only — not appropriate for production cache-as-cost-optimization patterns
- **Standard:** Two-node primary/replica with 99.9% SLA
- **Premium:** Enterprise features including Redis Cluster, persistence, VNet injection, and geo-replication

For FinOps purposes, the Standard tier C-series covers the majority of production database offload use cases:

SKU	Memory	Estimated Monthly (East US)
C0	250 MB	~\$16
C1	1 GB	~\$54
C2	6 GB	~\$113
C3	13 GB	~\$224
C4	26 GB	~\$447
C5	53 GB	~\$894
C6	53 GB+	~\$1,341

Unlike AWS and GCP, Azure does not offer reserved pricing for Azure Cache for Redis in the same form as compute reservations. The pricing model is straightforward hourly on-demand, which simplifies the cost calculation but removes the lever that makes ElastiCache and Memystore materially cheaper for committed workloads. Factor this into your cross-cloud cache cost comparisons — the AWS 1-year reserved discount on a comparable ElastiCache node can make AWS the lower-cost cache tier even if Azure SQL is your primary database, depending on network topology and latency constraints.

The Azure Monitor equivalent of the CloudWatch hit rate alarm uses metric alert rules via the Azure Monitor REST API or Bicep/ARM templates. The following Bicep module creates the same hit rate monitoring pattern for Azure Cache for Redis:

```
// azure-cache-monitoring.bicep
// Deploy cache hit rate and eviction alerts for Azure Cache for Redis
// Usage: az deployment group create --resource-group <rg> --template-file azure-cache-monitoring.bicep
```

```

@description('Name of the Azure Cache for Redis instance')
param cacheResourceName string

@description('Resource group containing the cache')
param resourceGroupName string = resourceGroup().name

@description('Action group resource ID for alert notifications')
param actionGroupId string

@description('Cache hit rate threshold percentage - alert fires below this value')
param hitRateThresholdPct int = 80

@description('Location for the alert rules')
param location string = resourceGroup().location

var cacheResourceId = resourceId(resourceGroupName, 'Microsoft.Cache/Redis', cacheResourceName)

// Alert: Cache hits dropping - derived metric
// Azure Monitor does not publish a pre-computed hit rate for Redis;
// we alert on absolute CacheMisses rate as a proxy
resource evictionsAlert 'Microsoft.Insights/metricAlerts@2018-03-01' = {
  name: 'cache-evictions-high-${cacheResourceName}'
  location: 'global'
  properties: {
    description: 'High eviction rate on ${cacheResourceName} - memory pressure causing key evictions,
↪ which degrades hit rate and increases Azure SQL load.'
    severity: 2
    enabled: true
    scopes: [cacheResourceId]
    evaluationFrequency: 'PT5M'
    windowSize: 'PT15M'
    criteria: {
      'odata.type': 'Microsoft.Azure.Monitor.SingleResourceMultipleMetricCriteria'
      allOf: [
        {
          name: 'HighEvictions'
          criterionType: 'StaticThresholdCriterion'
          metricName: 'evictedkeys'
          metricNamespace: 'Microsoft.Cache/Redis'
          operator: 'GreaterThan'
          threshold: 500
          timeAggregation: 'Total'
        }
      ]
    }
  }
  actions: [
    {
      actionGroupId: actionGroupId
    }
  ]
}

// Alert: Cache misses trending high (proxy for low hit rate)
resource cacheMissesAlert 'Microsoft.Insights/metricAlerts@2018-03-01' = {
  name: 'cache-misses-high-${cacheResourceName}'
  location: 'global'
  properties: {
    description: 'Elevated cache miss rate on ${cacheResourceName}. Database load may be increasing.
↪ Review TTL settings, key eviction policy, and application cache warming strategy.'
    severity: 2
    enabled: true
    scopes: [cacheResourceId]
    evaluationFrequency: 'PT5M'
    windowSize: 'PT15M'
    criteria: {
      'odata.type': 'Microsoft.Azure.Monitor.SingleResourceMultipleMetricCriteria'
      allOf: [
        {

```

```

        name: 'HighCacheMisses'
        criterionType: 'StaticThresholdCriterion'
        metricName: 'cachemisses'
        metricNamespace: 'Microsoft.Cache/Redis'
        operator: 'GreaterThan'
        threshold: 1000
        timeAggregation: 'Total'
    }
  ]
}
actions: [
  {
    actionGroupId: actionGroupId
  }
]
}
}

// Alert: Memory pressure
resource memoryPressureAlert 'Microsoft.Insights/metricAlerts@2018-03-01' = {
  name: 'cache-memory-high-${cacheResourceName}'
  location: 'global'
  properties: {
    description: 'Memory usage on ${cacheResourceName} exceeds 85%. Risk of eviction-driven hit rate
↪ degradation. Consider scaling up to the next Standard tier SKU.'
    severity: 1
    enabled: true
    scopes: [cacheResourceId]
    evaluationFrequency: 'PT5M'
    windowSize: 'PT15M'
    criteria: {
      'odata.type': 'Microsoft.Azure.Monitor.SingleResourceMultipleMetricCriteria'
      allOf: [
        {
          name: 'HighMemoryUsage'
          criterionType: 'StaticThresholdCriterion'
          metricName: 'usedmemorypercentage'
          metricNamespace: 'Microsoft.Cache/Redis'
          operator: 'GreaterThan'
          threshold: 85
          timeAggregation: 'Average'
        }
      ]
    }
  }
  actions: [
    {
      actionGroupId: actionGroupId
    }
  ]
}
}

output evictionsAlertId string = evictionsAlert.id
output cacheMissesAlertId string = cacheMissesAlert.id
output memoryPressureAlertId string = memoryPressureAlert.id

```

Azure SQL-specific diagnostic: identifying cacheable query patterns via Query Store

Azure SQL Database and SQL Server on Azure VMs have Query Store enabled by default, providing a persistent record of query execution history that survives restarts — unlike `sys.dm_exec_query_stats`, which resets on restart or plan cache eviction. For identifying caching candidates on Azure SQL, Query Store is the authoritative source:

```

-- Azure SQL / SQL Server: Identify top cache candidates using Query Store
-- Query Store persists execution history across restarts, making it
-- more reliable than dm_exec_query_stats for long-term trend analysis
SELECT TOP 30
    qs.query_id,

```

```

LEFT(qsqt.query_sql_text, 120)                AS query_preview,
qsrs.count_executions,                        AS avg_logical_reads,
CAST(qsrs.avg_logical_io_reads AS BIGINT)      AS avg_duration_ms,
CAST(qsrs.avg_duration / 1000.0 AS DECIMAL(10, 2)) AS avg_rows_returned,
CAST(qsrs.avg_rowcount AS BIGINT)              AS avg_rows_returned,
-- High execution count + low row count = strong cache candidate
CASE
  WHEN qsrs.count_executions > 5000
    AND qsrs.avg_rowcount < 50
  THEN 'CACHE NOW'
  WHEN qsrs.count_executions > 1000
    AND qsrs.avg_rowcount < 200
  THEN 'EVALUATE'
  WHEN qsrs.count_executions > 500
  THEN 'MONITOR'
  ELSE 'SKIP'
END
qsrs.count_executions * qsrs.avg_logical_io_reads AS cache_recommendation,
-- Translate logical reads to cost pressure:
-- each read page is 8KB; high values on DTU-based databases
-- directly drive DTU consumption and potential tier upgrades
CAST(
  qsrs.count_executions * qsrs.avg_logical_io_reads * 8.0 / 1024 / 1024
  AS DECIMAL(12, 2)
) AS total_logical_reads_estimated,
) AS total_data_gb_read
FROM
  sys.query_store_query          qsq
  JOIN sys.query_store_query_text qsqt ON qsq.query_text_id = qsqt.query_text_id
  JOIN sys.query_store_plan      qsp  ON qsq.query_id      = qsp.query_id
  JOIN sys.query_store_runtime_stats qsrs ON qsp.plan_id    = qsrs.plan_id
  JOIN sys.query_store_runtime_stats_interval qsrsi
    ON qsrs.runtime_stats_interval_id = qsrsi.runtime_stats_interval_id
WHERE
  qsrsi.start_time >= DATEADD(DAY, -14, GETUTCDATE())
  AND qsq.is_internal_query = 0
  AND qsqt.query_sql_text NOT LIKE '%query_store%'
ORDER BY
  qsrs.count_executions DESC;

```

The `total_logical_reads_estimated` column is particularly useful when your Azure SQL database is on a DTU-based pricing model. DTU consumption is heavily influenced by I/O, and high-frequency read queries that can be cached represent a direct path to dropping a DTU tier — which often yields larger savings than any other single optimization on Azure SQL.

21.7 Cache Invalidation Economics: The Hidden Cost of Getting It Wrong

Cache invalidation is routinely called one of the hardest problems in computer science, but its financial dimension is less often discussed. Getting cache invalidation wrong costs money in two distinct directions:

Over-invalidation: Flushing too aggressively (short TTLs, broad key patterns, flush-on-any-write) keeps your cache cold and routes traffic back to the database. You pay for the cache infrastructure without capturing the savings. This is the more common failure mode.

Under-invalidation: Serving stale data reduces database load but can cause correctness issues that lead to incident response, customer credits, or in regulated industries, compliance failures. The remediation cost can dwarf months of cache savings.

The optimal TTL strategy balances these two failure modes by aligning TTL to the business tolerance for stale data, not to technical convenience. Different data classes have fundamentally different staleness tolerances:

Data Class	Staleness Tolerance	Recommended TTL	Cache Strategy
User session tokens	None — must be current	Session lifetime only	Authoritative store, not cache
User profile / preferences	Minutes acceptable	5–15 minutes	Cache with explicit invalidation on update
Product catalog / pricing	Minutes to hours	30–60 minutes	Cache with TTL; invalidate on CMS publish
Configuration tables	Hours acceptable	4–24 hours	Cache aggressively; invalidate on admin change
Reference / lookup data	Days acceptable	24 hours+	Cache very aggressively; invalidate on release
Analytics aggregates	Hours to days	1–12 hours depending on freshness SLA	Read-through with long TTL

The financial impact of TTL misconfiguration is measurable. A product catalog with a 2-minute TTL instead of a 30-minute TTL will have its entire keyspace invalidated 30 times per hour instead of twice — 15x more often. If your catalog has 50,000 active keys and your database serves each miss in 10ms average, each full invalidation cycle generates 500 seconds of database query time. At 30 cycles per hour, that's 15,000 seconds of database load per hour from the 2-minute TTL, versus 1,000 seconds/hour at 30 minutes — often enough to push you into the next RDS instance tier.

The following PostgreSQL query helps audit which tables are generating write volume that may be triggering over-eager cache invalidation:

```
-- PostgreSQL: Identify write-heavy tables that may be driving
-- unnecessary cache invalidation through overly broad flush patterns
SELECT
    schemaname,
    relname                                AS table_name,
    n_tup_ins                              AS inserts,
    n_tup_upd                              AS updates,
    n_tup_del                              AS deletes,
    n_tup_ins + n_tup_upd + n_tup_del      AS total_writes,
    n_live_tup                             AS live_rows,
    -- Write rate relative to table size
    -- High write ratio on small tables = frequent invalidation trigger
    CASE
        WHEN n_live_tup > 0
        THEN round(
            (n_tup_ins + n_tup_upd + n_tup_del)::numeric
            / n_live_tup * 100, 2)
        ELSE 0
    END                                    AS write_to_size_ratio,
    last_vacuum,
    last_analyze,
    pg_size_pretty(pg_relation_size(relid)) AS table_size
FROM
    pg_stat_user_tables
WHERE
```

```

n_tup_ins + n_tup_upd + n_tup_del > 100
ORDER BY
  total_writes DESC
LIMIT 30;

```

For SQL Server and Azure SQL, the same write audit against the Query Store and DMVs:

```

-- SQL Server / Azure SQL: Audit write operations that drive cache invalidation
SELECT TOP 30
  OBJECT_SCHEMA_NAME(object_id)                AS schema_name,
  OBJECT_NAME(object_id)                       AS table_name,
  SUM(ios.leaf_insert_count)                   AS leaf_inserts,
  SUM(ios.leaf_update_count)                   AS leaf_updates,
  SUM(ios.leaf_delete_count)                   AS leaf_deletes,
  SUM(ios.leaf_insert_count
      + ios.leaf_update_count
      + ios.leaf_delete_count)                  AS total_write_ops,
  SUM(p.rows)                                  AS approx_row_count,
  -- Write operations per 1000 rows - higher values mean
  -- this table triggers cache invalidation very frequently
  CASE
    WHEN SUM(p.rows) > 0
    THEN CAST(
      SUM(ios.leaf_insert_count + ios.leaf_update_count + ios.leaf_delete_count)
      * 1000.0 / SUM(p.rows) AS DECIMAL(12, 2))
    ELSE 0
  END                                           AS writes_per_1k_rows
FROM
  sys.dm_db_index_operational_stats(DB_ID(), NULL, NULL, NULL) ios
JOIN sys.indexes i
  ON ios.object_id = i.object_id AND ios.index_id = i.index_id
JOIN sys.partitions p
  ON i.object_id = p.object_id AND i.index_id = p.index_id
WHERE
  OBJECT_NAME(ios.object_id) IS NOT NULL
  AND i.object_id > 100
GROUP BY
  ios.object_id
ORDER BY
  total_write_ops DESC;

```

Tables surfacing at the top of this query with high `writes_per_1k_rows` ratios are the ones where broad cache invalidation on every write causes the most damage. The remediation pattern is row-level cache key invalidation — invalidating only the specific cache key for the modified row (`user_profile:{user_id}`) rather than flushing the entire namespace (`user_profile:*`). Namespace-level flushes are operationally convenient but financially expensive at any meaningful scale.

21.8 Building the Cache ROI Dashboard

All of the individual metrics — hit rate, eviction count, memory utilization, L1/L2/L3 distribution, TTL effectiveness — only create accountability when they are visible as a unified cost story. A cache ROI dashboard translates these operational metrics into dollars, making the business case for cache investment self-evident and surfacing degradation before it erodes savings.

The following Terraform module provisions the CloudWatch dashboard infrastructure for a complete cache ROI view across multiple ElastiCache clusters and their backing RDS instances:

```

# cache_roi_dashboard.tf
# Creates a CloudWatch dashboard showing cache hit rates, eviction metrics,
# and estimated database cost impact side-by-side.

```

```

# Parameterized for multiple cache cluster / RDS instance pairs.

variable "dashboard_name" {
  description = "Name for the CloudWatch dashboard"
  type        = string
  default     = "Cache-ROI-Overview"
}

variable "cache_cluster_ids" {
  description = "List of ElastiCache cluster IDs to monitor"
  type        = list(string)
}

variable "rds_instance_ids" {
  description = "List of RDS instance IDs backing the cache clusters (parallel list)"
  type        = list(string)
}

variable "region" {
  description = "AWS region"
  type        = string
  default     = "us-east-1"
}

locals {
  # Build metric widget definitions for each cluster/RDS pair
  hit_rate_widgets = [
    for idx, cluster_id in var.cache_cluster_ids : {
      type      = "metric"
      x         = (idx % 2) * 12
      y         = floor(idx / 2) * 6
      width     = 12
      height    = 6
      properties = {
        title = "Cache Hit Rate - ${cluster_id}"
        period = 300
        stat   = "Sum"
        view   = "timeSeries"
        metrics = [
          ["AWS/ElastiCache", "CacheHits", "CacheClusterId", cluster_id, { id = "hits_${idx}",
↪ visible = false }],
          ["AWS/ElastiCache", "CacheMisses", "CacheClusterId", cluster_id, { id = "misses_${idx}",
↪ visible = false }],
          [{ expression =
↪ "IF(hits_${idx}+misses_${idx}>0,(hits_${idx})/(hits_${idx}+misses_${idx}))*100,100)", label = "Hit
↪ Rate %", id = "hitrate_${idx}" }]
        ]
      }
      yAxis = { left = { min = 0, max = 100 } }
      annotations = {
        horizontal = [{ value = 80, label = "80% threshold", color = "#ff6961" }]
      }
    }
  ]

  eviction_widgets = [
    for idx, cluster_id in var.cache_cluster_ids : {
      type      = "metric"
      x         = (idx % 2) * 12
      y         = floor(idx / 2) * 6 + length(var.cache_cluster_ids) * 3
      width     = 12
      height    = 6
      properties = {
        title = "Evictions + Memory - ${cluster_id}"
        period = 300
        view   = "timeSeries"
        metrics = [
          ["AWS/ElastiCache", "Evictions", "CacheClusterId", cluster_id, { stat =
↪ "Sum", yAxis = "left" }],

```

```

        ["AWS/ElastiCache", "DatabaseMemoryUsagePercentage", "CacheClusterId", cluster_id, { stat =
↪ "Average", yAxis = "right" }]
    ]
    yAxis = {
        left = { label = "Evictions", min = 0 }
        right = { label = "Memory Used (%)", min = 0, max = 100 }
    }
}
}

rds_connections_widgets = [
    for idx, rds_id in var.rds_instance_ids : {
        type = "metric"
        x = (idx % 2) * 12
        y = floor(idx / 2) * 6 + length(var.cache_cluster_ids) * 6
        width = 12
        height = 6
        properties = {
            title = "DB Connections + Read IOPS - ${rds_id}"
            period = 300
            view = "timeSeries"
            metrics = [
                ["AWS/RDS", "DatabaseConnections", "DBInstanceIdentifier", rds_id, { stat = "Average", yAxis =
↪ "left" }],
                ["AWS/RDS", "ReadIOPS", "DBInstanceIdentifier", rds_id, { stat = "Average", yAxis =
↪ "right" }]
            ]
        }
        yAxis = {
            left = { label = "Connections", min = 0 }
            right = { label = "Read IOPS", min = 0 }
        }
    }
]

all_widgets = concat(
    local.hit_rate_widgets,
    local.eviction_widgets,
    local.rds_connections_widgets
)

resource "aws_cloudwatch_dashboard" "cache_roi" {
    dashboard_name = var.dashboard_name

    dashboard_body = jsonencode({
        widgets = local.all_widgets
    })
}

output "dashboard_url" {
    value =
↪ "https://${var.region}.console.aws.amazon.com/cloudwatch/home?region=${var.region}#dashboards:name=${var.dashboard_na
}

```

With this dashboard in place, the financial narrative becomes self-evident in your weekly FinOps review: when cache hit rate slides from 92% to 74% on the left panel, the corresponding spike in RDS read IOPS on the right panel confirms the database is absorbing the load shift — and the cost implication is visible in the same view.

Establishing a weekly cache review cadence

The monitoring infrastructure is only valuable if someone reviews it on a cadence tied to billing cycles. A lightweight weekly review covers four questions:

1. Is the average hit rate above the threshold (typically 85%) across all production clusters?

2. Are eviction counts trending up — a leading indicator of memory pressure before hit rate degrades?
3. Have any clusters been running at under 40% average memory utilization for more than two consecutive weeks? If so, they are candidates for the next right-sizing run.
4. Are any new query patterns visible in `pg_stat_statements` or Query Store that are high-frequency but not yet cached?

Answering all four questions takes under 15 minutes when the dashboards are in place. The cumulative effect of this cadence — catching hit rate degradation early, right-sizing overprovisioned nodes quarterly, identifying new caching candidates as applications evolve — compounds over time into savings that substantially exceed the initial cache deployment benefit.

Key Takeaways

- **Cache ROI is measurable before you deploy:** Query `pg_stat_statements` (PostgreSQL) or `sys.dm_exec_query_stats` and Query Store (SQL Server/Azure SQL) to identify high-frequency, low-row-count queries that are strong caching candidates. Build the savings case with actual execution counts before provisioning any cache infrastructure.
 - **Working set size, not total database size, determines cache node selection:** A 500GB database may have a hot working set of 8–12GB. Use `pg_statio_user_tables` or SQL Server buffer pool DMVs to measure the actual data being read frequently, then size ElastiCache, Memorystore, or Azure Cache for Redis to cover that working set at 75–80% utilization — not the full database footprint.
 - **Cache hit rate is the single most important financial control metric:** The difference between an 80% and 95% hit rate on a mid-tier RDS instance can represent hundreds of dollars per month in database sizing savings. Automate CloudWatch or Azure Monitor alarms on hit rate and evictions so degradation triggers notification within minutes, not after your next billing statement.
 - **TTL misconfiguration is a hidden cost multiplier:** Over-eager invalidation (short TTLs or broad namespace flushes on writes) keeps caches cold and routes traffic back to the database, eliminating the savings that justified the cache investment. Align TTLs to business staleness tolerance by data class, and use row-level key invalidation rather than namespace flushes for write-heavy tables.
 - **Reserved instances and committed-use discounts apply to cache tiers too:** ElastiCache 1-year reserved pricing cuts on-demand costs by 40–45%; GCP Memorystore committed-use discounts provide comparable reductions. For any cache cluster you expect to run for more than six months, converting to reserved or committed pricing is the fastest single action to improve cache tier economics — and it compounds on top of all right-sizing work done at the node level.
-

Chapter 22: Separating Analytics from OLTP: The Financial Case

Organizations that run analytical queries directly against their production OLTP databases routinely overpay for compute by 40–70%. A single poorly-timed BI report hitting an `db.r6g.2xlarge` RDS instance at peak transaction hours can cost the same as running a dedicated analytics cluster for an entire month — and that assumes the report finishes without triggering a CPU alarm or degrading customer-facing latency. This chapter quantifies the financial damage of mixed workloads, shows you how to measure the exact cost of analytics queries running on OLTP infrastructure, and builds the business case for routing those workloads to purpose-built, cheaper infrastructure. Done right, workload separation typically reduces the primary OLTP instance tier by one to two sizes, cuts storage IOPS costs by 30–50%, and moves analytics to infrastructure that costs 60–80% less per query-hour.

The Hidden Cost of Shared Infrastructure

The fundamental problem with running analytics on OLTP instances is a pricing mismatch. OLTP databases are sized for memory and IOPS — you pay premium prices for fast, consistent transaction throughput. Analytics workloads need CPU and sequential scan throughput, which is exactly what columnar stores and MPP engines provide at a fraction of the row-store cost.

Consider a production setup on AWS: an `db.r6g.4xlarge` RDS PostgreSQL instance running in Multi-AZ. At US East pricing, that is approximately \$1.53 per hour for the primary, or \$2,196 per month for the pair. Now consider what actually runs on that instance. Performance Insights might reveal that 35% of active session time belongs to reporting queries — dashboard aggregations, month-end rollups, data export jobs. That 35% equates to roughly \$769 per month in compute cost attributed purely to analytics.

The same queries on Amazon Redshift Serverless, priced at \$0.375 per RPU-hour, would consume perhaps 8–12 RPU-hours for a typical daily analytics batch. Monthly cost: \$90–\$135. The math alone closes the case before you have written a single line of migration code.

On Azure, the picture is similar. An `GP_Gen5_8` Azure SQL Database at around \$1,150 per month shared with Power BI DirectQuery connections sees the same compression ratio when workloads move to Azure Synapse Analytics serverless SQL pools, billed at \$5 per TB scanned.

The first step is measuring your current baseline with precision.

```

-- PostgreSQL: Identify analytics queries consuming disproportionate resources
-- Run on RDS PostgreSQL or Aurora PostgreSQL with pg_stat_statements enabled
SELECT
    left(query, 120)                                AS query_snippet,
    calls,
    round((total_exec_time / 1000)::numeric, 2)      AS total_exec_sec,
    round((mean_exec_time / 1000)::numeric, 4)        AS mean_exec_sec,
    round((total_exec_time / sum(total_exec_time) OVER ()
            * 100)::numeric, 2)                      AS pct_total_time,
    round(rows / calls::numeric, 0)                  AS avg_rows_returned,
    shared_blks_read + shared_blks_hit                AS total_blocks_accessed,
    -- High rows returned + long execution = likely analytics, not OLTP
    CASE
        WHEN mean_exec_time > 5000
             AND rows / calls > 10000 THEN 'ANALYTICS_CANDIDATE'
        WHEN mean_exec_time > 1000
             AND rows / calls > 1000  THEN 'REVIEW_NEEDED'
        ELSE 'OLTP'
    END                                               AS workload_classification
FROM pg_stat_statements
WHERE calls > 10
ORDER BY total_exec_time DESC
LIMIT 40;

```

This query gives you the classification foundation. Any query returning more than 10,000 rows on average or running longer than 5 seconds mean execution time is an analytics candidate that does not belong on an OLTP instance. Export this result set. It becomes the financial exhibit in your executive presentation.

For SQL Server on Azure SQL or AWS RDS SQL Server, the equivalent pull from `sys.dm_exec_query_stats` is:

```

-- SQL Server: Classify workloads by execution profile
SELECT TOP 50
    SUBSTRING(st.text, (qs.statement_start_offset / 2) + 1,
        ((CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(st.text)
            ELSE qs.statement_end_offset
        END - qs.statement_start_offset) / 2) + 1) AS query_text,
    qs.execution_count,
    ROUND(qs.total_elapsed_time / 1e6, 2)          AS total_elapsed_sec,
    ROUND(qs.total_elapsed_time / qs.execution_count / 1e6,
        4)                                           AS avg_elapsed_sec,
    ROUND(CAST(qs.total_elapsed_time AS FLOAT)
        / SUM(qs.total_elapsed_time) OVER () * 100, 2) AS pct_total_elapsed,
    qs.total_logical_reads / qs.execution_count      AS avg_logical_reads,
    qs.total_rows / qs.execution_count               AS avg_rows,
    CASE
        WHEN qs.total_elapsed_time / qs.execution_count > 5000000
             AND qs.total_rows / qs.execution_count > 10000 THEN 'ANALYTICS_CANDIDATE'
        WHEN qs.total_logical_reads / qs.execution_count > 100000 THEN 'HEAVY_SCAN'
        ELSE 'OLTP'
    END                                               AS workload_type
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
WHERE qs.execution_count > 5
ORDER BY qs.total_elapsed_time DESC;

```

Quantifying the Dollar Value of Separation

Having identified analytics candidates, the next step is attaching a dollar figure to each. This requires combining query execution data with instance cost data. The Python script below pulls

Performance Insights metrics from CloudWatch and maps CPU consumption per query group to hourly instance cost.

```
#!/usr/bin/env python3
"""
analytics_cost_attribution.py

Pulls RDS Performance Insights data and attributes hourly compute cost
to OLTP vs. analytics query workloads.

Requires: boto3, psycopy2, pandas
Schedule: Daily via EventBridge + Lambda or cron
"""

import boto3
import pandas as pd
from datetime import datetime, timedelta, timezone
from dataclasses import dataclass
from typing import Dict, List

# Configuration
DB_INSTANCE_ID = "prod-postgres-01"
REGION = "us-east-1"
INSTANCE_TYPE = "db.r6g.4xlarge"
MULTI_AZ = True

# On-demand hourly pricing (update from AWS pricing page quarterly)
HOURLY_RATES: Dict[str, float] = {
    "db.r6g.xlarge": 0.384,
    "db.r6g.2xlarge": 0.768,
    "db.r6g.4xlarge": 1.536,
    "db.r6g.8xlarge": 3.072,
}

LOOKBACK_HOURS = 24

# Helpers
@dataclass
class WorkloadCostSummary:
    period_hours: int
    total_instance_cost_usd: float
    analytics_cost_usd: float
    oltp_cost_usd: float
    analytics_pct: float
    projected_monthly_analytics_cost: float

def get_pi_metrics(client, db_id: str, hours: int = 24) -> pd.DataFrame:
    """Fetch DBLoad by sql digest from Performance Insights."""
    end_time = datetime.now(timezone.utc)
    start_time = end_time - timedelta(hours=hours)

    response = client.get_resource_metrics(
        ServiceType="RDS",
        Identifier=db_id,
        MetricQueries=[
            {
                "Metric": "db.load.avg",
                "GroupBy": {
                    "Group": "db.sql",
                    "Dimensions": ["db.sql.statement"],
                    "Limit": 25,
                },
            },
        ],
        StartTime=start_time,
        EndTime=end_time,
        PeriodInSeconds=3600,
    )
```

```

records = []
for metric_list in response.get("MetricList", []):
    sql_text = (
        metric_list.get("Key", {})
        .get("Dimensions", {})
        .get("db.sql.statement", "unknown")
    )
    for dp in metric_list.get("DataPoints", []):
        records.append(
            {
                "timestamp": dp["Timestamp"],
                "db_load": dp.get("Value", 0.0),
                "sql_text": sql_text[:200],
            }
        )
return pd.DataFrame(records)

def classify_query(sql: str) -> str:
    """
    Heuristic classification. Replace with pg_stat_statements
    integration for production use.
    """
    sql_upper = sql.upper()
    analytics_signals = [
        "GROUP BY", "HAVING", "ROLLUP", "CUBE",
        "WINDOW", "PARTITION BY", "PERCENTILE",
        "EXTRACT(", "DATE_TRUNC",
    ]
    if any(sig in sql_upper for sig in analytics_signals):
        return "ANALYTICS"
    return "OLTP"

def compute_cost_attribution(df: pd.DataFrame,
                             instance_type: str,
                             multi_az: bool,
                             lookback_hours: int) -> WorkloadCostSummary:
    base_rate = HOURLY_RATES.get(instance_type, 1.536)
    effective_rate = base_rate * (2 if multi_az else 1)
    total_cost = effective_rate * lookback_hours

    df["workload"] = df["sql_text"].apply(classify_query)

    total_load = df["db_load"].sum() or 1.0
    analytics_load = df.loc[df["workload"] == "ANALYTICS", "db_load"].sum()
    analytics_pct = (analytics_load / total_load) * 100

    analytics_cost = total_cost * (analytics_pct / 100)
    oltp_cost = total_cost - analytics_cost

    # Project to 30 days
    hourly_analytics_cost = analytics_cost / lookback_hours
    monthly_analytics_cost = hourly_analytics_cost * 730

    return WorkloadCostSummary(
        period_hours=lookback_hours,
        total_instance_cost_usd=round(total_cost, 2),
        analytics_cost_usd=round(analytics_cost, 2),
        oltp_cost_usd=round(oltp_cost, 2),
        analytics_pct=round(analytics_pct, 1),
        projected_monthly_analytics_cost=round(monthly_analytics_cost, 2),
    )

def main():
    pi_client = boto3.client("pi", region_name=REGION)

    print(f"Fetching Performance Insights for {DB_INSTANCE_ID} ")

```

```

    f"over last {LOOKBACK_HOURS}h ...\n")

df = get_pi_metrics(pi_client, DB_INSTANCE_ID, LOOKBACK_HOURS)

if df.empty:
    print("No PI data returned. Verify PI is enabled and the instance ID is correct.")
    return

summary = compute_cost_attribution(
    df, INSTANCE_TYPE, MULTI_AZ, LOOKBACK_HOURS
)

print("=" * 60)
print(f"    Instance:                {DB_INSTANCE_ID} ({INSTANCE_TYPE})")
print(f"    Analysis window:         {LOOKBACK_HOURS} hours")
print(f"    Total compute cost:      ${summary.total_instance_cost_usd:>8.2f}")
print(f"    OLTP cost:               ${summary.oltp_cost_usd:>8.2f}")
print(f"    Analytics cost:          ${summary.analytics_cost_usd:>8.2f}  "
      f"    f'({summary.analytics_pct}%)")
print(f"    Projected monthly waste: ${summary.projected_monthly_analytics_cost:>8.2f}")
print("=" * 60)

if summary.analytics_pct > 20:
    print("\n    RECOMMENDATION: Analytics load exceeds 20% of total DB load.")
    print("    Consider offloading to Redshift Serverless, Athena, or a")
    print("    read replica sized for analytics workloads.")

if __name__ == "__main__":
    main()

```

Run this daily via a scheduled Lambda or cron job. When `projected_monthly_analytics_cost` exceeds \$500 consistently, you have your financial justification. When it exceeds \$2,000, the conversation with your CTO writes itself.

Choosing the Right Analytics Target Infrastructure

Workload separation is not a single architecture. The correct target depends on query patterns, data freshness requirements, and the volume of data scanned. Getting this wrong means you trade one cost problem for another.

Scenario 1 — Latency-tolerant batch analytics, large data volumes: Amazon Redshift Serverless (AWS), Azure Synapse Analytics serverless SQL pools (Azure), or BigQuery (GCP). These are columnar MPP engines billed per query or per RPU-hour. A query scanning 500 GB of Parquet in BigQuery costs \$2.50 flat. The same query on an `r6g.4xlarge` RDS instance spending 45 minutes of CPU time costs roughly \$1.15 in compute plus weeks of DBA time managing contention. Over a month of daily runs, BigQuery at \$2.50 per day is \$75; the RDS share is \$840.

Scenario 2 — Near-real-time dashboards against production data, sub-minute lag tolerance: Aurora read replicas or RDS read replicas. The replica runs on a smaller instance — you do not need the same IOPS tier as primary. An Aurora PostgreSQL read replica on `db.r6g.xlarge` at \$0.384/hour costs \$280/month versus the `db.r6g.4xlarge` primary at \$2,196/month. Dashboard queries go to the replica; primary stays clean for writes.

Scenario 3 — Ad-hoc analyst queries against recent data already exported: Amazon Athena over S3. Export a daily snapshot of your OLTP tables to S3 in Parquet format. Athena charges \$5 per TB scanned, and column pruning plus Parquet compression typically means analysts

scan 90–95% less data than the raw row size. A 100 GB Parquet file holding 1 TB of logical data costs \$0.50 per full scan instead of \$5.00.

The following script sets up a daily export from RDS PostgreSQL to S3 using `pg_dump` via AWS CLI, enabling Athena queries without touching production:

```
#!/usr/bin/env bash
# daily_analytics_export.sh
# Exports selected analytical tables from RDS to S3 as CSV,
# convertible to Parquet via AWS Glue or Athena CTAS.
# Schedule: cron 0 2 * * * (02:00 UTC daily)

set -euo pipefail

DB_HOST="prod-postgres-01.cluster-xyz.us-east-1.rds.amazonaws.com"
DB_NAME="appdb"
DB_USER="readonly_analytics"
S3_BUCKET="s3://my-company-analytics-lake"
DATE_PREFIX=$(date +%Y/%m/%d)
PGPASSWORD="${ANALYTICS_DB_PASSWORD}" # sourced from SSM Parameter Store at runtime

export PGPASSWORD

TABLES=(
  "orders"
  "order_items"
  "customers"
  "products"
  "events"
)

echo "[$(date -u +%FT%T)] Starting analytics export ..."

for TABLE in "${TABLES[@]}; do
  LOCAL_FILE="/tmp/${TABLE}_${(date +%Y/%m/%d)}.csv"

  echo "  Exporting ${TABLE} ..."
  psql \
    --host="${DB_HOST}" \
    --username="${DB_USER}" \
    --dbname="${DB_NAME}" \
    --no-password \
    --command="\COPY (SELECT * FROM ${TABLE} WHERE updated_at >= NOW() - INTERVAL '1 day')
    TO STDOUT WITH CSV HEADER" \
    > "${LOCAL_FILE}"

  # Upload to date-partitioned S3 prefix
  aws s3 cp "${LOCAL_FILE}" \
    "${S3_BUCKET}/raw/${TABLE}/${DATE_PREFIX}/${TABLE}.csv" \
    --storage-class INTELLIGENT_TIERING \
    --sse aws:kms

  rm -f "${LOCAL_FILE}"
  echo "    ${TABLE} uploaded."
done

# Optionally trigger Glue crawler to update Athena catalog
aws glue start-crawler --name "analytics-lake-crawler" || true

echo "[$(date -u +%FT%T)] Export complete."
unset PGPASSWORD
```

The `--storage-class INTELLIGENT_TIERING` flag is intentional. Analytics data accessed heavily the day after export and rarely thereafter fits the Intelligent-Tiering pattern, which automatically moves objects to lower-cost tiers after 30 and 90 days of inactivity. For a 500 GB dataset, this can reduce S3 storage cost from \$11.50/month (Standard) to \$6.00/month within two months.

Right-Sizing OLTP After Workload Removal

The separation is only half the saving. The other half comes from downsizing the OLTP instance once analytics load is removed. This is where DBAs often leave money on the table — they successfully move the analytics workload but never revisit the instance type because “production changes are scary.”

Use AWS Compute Optimizer and RDS recommendations to make the case with data rather than intuition.

```
#!/usr/bin/env bash
# get_rds_rightsizing_recommendations.sh
# Pulls Compute Optimizer recommendations for all RDS instances in account.
# Prerequisites: AWS CLI configured, Compute Optimizer enabled.

set -euo pipefail

REGION="us-east-1"
OUTPUT_FILE="rds_rightsizing_$(date +%Y%m%d).json"

echo "Fetching Compute Optimizer recommendations for RDS ..."

aws compute-optimizer get-rds-database-recommendations \
  --region "${REGION}" \
  --filters name=Finding,values=Overprovisioned \
  --query 'rdsDBRecommendations[*].{
    Instance:resourceArn,
    CurrentType:currentDBInstanceClass,
    RecommendedType:recommendationOptions[0].dbInstanceClass,
    CurrentMonthlyPrice:currentMonthlyPrice.value,
    RecommendedMonthlyPrice:recommendationOptions[0].monthlyPrice.value,
    SavingsMonthly:currentMonthlyPrice.value
      - recommendationOptions[0].monthlyPrice.value,
    CPUUtilizationP99:utilizationMetrics[?name==`CPU`].value | [0],
    MemoryUtilizationP99:utilizationMetrics[?name==`Memory`].value | [0]
  }' \
  --output json \
  | tee "${OUTPUT_FILE}"

# Summarize total potential savings
echo ""
echo "Summary of potential monthly savings:"
python3 - <<PYEOF
import json, sys

with open("${OUTPUT_FILE}") as f:
    recs = json.load(f)

if not recs:
    print(" No overprovisioned instances found - or Compute Optimizer needs more data.")
    sys.exit(0)

total_current      = sum(r.get("CurrentMonthlyPrice", 0) or 0 for r in recs)
total_recommended  = sum(r.get("RecommendedMonthlyPrice", 0) or 0 for r in recs)
total_savings      = total_current - total_recommended

print(f" Instances reviewed:      {len(recs)}")
print(f" Current monthly spend:    \${total_current:>10,.2f}")
print(f" Optimized monthly spend:   \${total_recommended:>10,.2f}")
print(f" Potential monthly saving:  \${total_savings:>10,.2f}")
print(f" Annualized saving:        \${total_savings * 12:>10,.2f}")
PYEOF
```

In practice, after removing analytics load from an `db.r6g.4xlarge`, Compute Optimizer frequently recommends dropping to `db.r6g.2xlarge`. At US East Multi-AZ pricing, that is a reduction from \$2,196/month to \$1,098/month — \$1,098/month saved on the OLTP instance alone. Combined with the analytics migration to a cheaper target, total monthly savings often land between \$1,200 and \$1,800 for a single database cluster.

The instance downsize also has a compounding effect on reserved instance pricing. A 1-year no-upfront Reserved Instance for `db.r6g.2xlarge` costs approximately \$686/month versus \$1,098 on-demand — a further 37% reduction. When presenting the business case, the three-layer stack of savings (analytics offload + instance downsize + RI purchase) often produces a number that immediately gets CTO approval.

Automation: Continuous Workload Classification and Alerting

One-time separation is not enough. New features ship, developers run queries, and analytics workload creeps back onto the OLTP instance within weeks if nobody is watching. The following automation runs as a scheduled Lambda (or GitHub Actions workflow for non-AWS shops) and posts an alert when analytics load on the primary instance exceeds a configured threshold.

```
#!/usr/bin/env python3
"""
workload_drift_monitor.py

Monitors pg_stat_statements on RDS PostgreSQL via a read-only connection
and alerts via SNS when analytics query load exceeds threshold.

Intended deployment: AWS Lambda, triggered every 15 minutes via EventBridge.
Environment variables:
    DB_SECRET_ARN - Secrets Manager ARN with {host, port, dbname, username, password}
    SNS_TOPIC_ARN - SNS topic for alerts
    ALERT_THRESHOLD - Analytics share of total exec time that triggers alert (default 0.25)
"""

import json
import os
import boto3
import psycpg2
from datetime import datetime, timezone

sns = boto3.client("sns")
secrets = boto3.client("secretsmanager")

ALERT_THRESHOLD = float(os.environ.get("ALERT_THRESHOLD", "0.25"))
SNS_TOPIC_ARN = os.environ["SNS_TOPIC_ARN"]
DB_SECRET_ARN = os.environ["DB_SECRET_ARN"]

ANALYTICS_CLASSIFICATION_QUERY = """
WITH stmt_stats AS (
    SELECT
        queryid,
        left(query, 200)                                AS query_snippet,
        total_exec_time,
        calls,
        rows,
        mean_exec_time,
        rows / NULLIF(calls, 0)                          AS avg_rows_per_call,
        -- Classify heuristically; tune thresholds for your workload
        CASE
            WHEN mean_exec_time > 3000
                 AND rows / NULLIF(calls, 0) > 5000
            THEN 'ANALYTICS'
        """

```

```

        WHEN query ILIKE '%group by%'
            AND mean_exec_time > 1000
        THEN 'ANALYTICS'
        WHEN query ILIKE '%partition by%'
        THEN 'ANALYTICS'
        WHEN query ILIKE '%date_trunc%'
            AND rows / NULLIF(calls, 0) > 500
        THEN 'ANALYTICS'
        ELSE 'OLTP'
    END
    AS workload_class
FROM pg_stat_statements
WHERE calls > 5
    AND query NOT LIKE '%pg_stat_statements%'
),
totals AS (
    SELECT
        workload_class,
        SUM(total_exec_time)
        AS class_exec_time,
        COUNT(*)
        AS query_count
    FROM stmt_stats
    GROUP BY workload_class
)
SELECT
    workload_class,
    class_exec_time,
    query_count,
    ROUND(
        class_exec_time / SUM(class_exec_time) OVER () * 100,
        2
    )
    AS pct_of_total_time
FROM totals
ORDER BY class_exec_time DESC;
"""

```

```

def get_db_credentials(secret_arn: str) -> dict:
    resp = secrets.get_secret_value(SecretId=secret_arn)
    return json.loads(resp["SecretString"])

def query_workload_distribution(creds: dict) -> list[dict]:
    conn = psycopg2.connect(
        host=creds["host"],
        port=creds.get("port", 5432),
        dbname=creds["dbname"],
        user=creds["username"],
        password=creds["password"],
        connect_timeout=10,
        sslmode="require",
    )
    try:
        with conn.cursor() as cur:
            cur.execute(ANALYTICS_CLASSIFICATION_QUERY)
            cols = [d[0] for d in cur.description]
            return [dict(zip(cols, row)) for row in cur.fetchall()]
    finally:
        conn.close()

def build_alert_message(rows: list[dict], analytics_pct: float) -> str:
    lines = [
        f" WORKLOAD DRIFT ALERT - {datetime.now(timezone.utc).strftime('%Y-%m-%d %H:%M UTC')}",
        f"Analytics queries consume {analytics_pct:.1f}% of total execution time",
        f"(threshold: {ALERT_THRESHOLD * 100:.0f}%)",
        "",
        "Breakdown by workload class:",
    ]
    for row in rows:
        lines.append(
            f" {row['workload_class']:<12} "
            f"{row['pct_of_total_time']:>6}% "
            f"({row['query_count']} queries)"
        )

```

```

lines += [
    "",
    "Action: Review pg_stat_statements on the OLTP instance and confirm",
    "analytical queries have not been re-routed to the primary.",
    "Runbook: https://wiki.internal/db-finops/workload-separation-runbook",
]
return "\n".join(lines)

def lambda_handler(event, context):
    creds = get_db_credentials(DB_SECRET_ARN)
    rows = query_workload_distribution(creds)

    if not rows:
        return {"status": "no_data"}

    analytics_row = next(
        (r for r in rows if r["workload_class"] == "ANALYTICS"), None
    )
    analytics_pct = float(analytics_row["pct_of_total_time"]) / 100 if analytics_row else 0.0

    result = {
        "timestamp": datetime.now(timezone.utc).isoformat(),
        "analytics_fraction": analytics_pct,
        "alert_fired": False,
        "distribution": rows,
    }

    if analytics_pct >= ALERT_THRESHOLD:
        message = build_alert_message(rows, analytics_pct * 100)
        sns.publish(
            TopicArn=SNS_TOPIC_ARN,
            Subject=f"[DB FinOps] Analytics workload drift detected - {analytics_pct * 100:.1f}%",
            Message=message,
        )
        result["alert_fired"] = True

    return result

```

Deploy this Lambda with a 128 MB memory allocation — the function itself is lightweight; all the work happens inside the database. At 15-minute intervals, the Lambda invocation cost is negligible (approximately 2,880 invocations per month, well within the free tier). Set the EventBridge rule to disable itself if the Lambda errors three consecutive times to avoid alert fatigue from transient network issues.

The equivalent for SQL Server environments uses a stored procedure that writes to a monitoring table, queried by an Azure Monitor workbook or CloudWatch custom metric:

```

-- SQL Server: Stored procedure to log workload distribution to a monitoring table
-- Schedule via SQL Server Agent or Azure Elastic Jobs every 15 minutes

IF OBJECT_ID('dbo.LogWorkloadDistribution', 'P') IS NOT NULL
    DROP PROCEDURE dbo.LogWorkloadDistribution;
GO

CREATE PROCEDURE dbo.LogWorkloadDistribution
AS
BEGIN
    SET NOCOUNT ON;

    -- Ensure monitoring table exists
    IF OBJECT_ID('dbo.WorkloadDistributionLog', 'U') IS NULL
    BEGIN
        CREATE TABLE dbo.WorkloadDistributionLog (
            log_id          INT IDENTITY(1,1) PRIMARY KEY,
            captured_at     DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
            workload_class  NVARCHAR(20) NOT NULL,

```

```

        total_elapsed_sec    DECIMAL(18,2) NOT NULL,
        query_count          INT NOT NULL,
        pct_of_total         DECIMAL(6,2) NOT NULL
    );
    CREATE INDEX IX_WorkloadLog_CapturedAt
    ON dbo.WorkloadDistributionLog (captured_at DESC);
END;

WITH classified AS (
    SELECT
        CASE
            WHEN qs.total_elapsed_time / NULLIF(qs.execution_count, 0) > 3000000
                 AND qs.total_rows / NULLIF(qs.execution_count, 0) > 5000
            THEN 'ANALYTICS'

            WHEN qs.total_logical_reads
                 / NULLIF(qs.execution_count, 0) > 50000
            THEN 'ANALYTICS'

            ELSE 'OLTP'
        END
        AS workload_class,
        qs.total_elapsed_time
        AS elapsed_us,
        1
        AS query_count
    FROM sys.dm_exec_query_stats qs
    WHERE qs.execution_count > 5
),
aggregated AS (
    SELECT
        workload_class,
        SUM(elapsed_us)
        AS class_elapsed_us,
        SUM(query_count)
        AS query_count
    FROM classified
    GROUP BY workload_class
),
with_pct AS (
    SELECT
        workload_class,
        class_elapsed_us,
        query_count,
        ROUND(
            CAST(class_elapsed_us AS FLOAT)
            / NULLIF(SUM(class_elapsed_us) OVER (), 0) * 100,
            2
        )
        AS pct_of_total
    FROM aggregated
)
INSERT INTO dbo.WorkloadDistributionLog
(workload_class, total_elapsed_sec, query_count, pct_of_total)
SELECT
    workload_class,
    ROUND(class_elapsed_us / 1e6, 2),
    query_count,
    pct_of_total
FROM with_pct;

-- Raise an alert event if analytics load exceeds 25%
IF EXISTS (
    SELECT 1
    FROM dbo.WorkloadDistributionLog
    WHERE captured_at >= DATEADD(MINUTE, -16, SYSUTCDATETIME())
        AND workload_class = 'ANALYTICS'
        AND pct_of_total > 25.0
)
BEGIN
    RAISERROR(
        'WORKLOAD DRIFT: Analytics queries exceed 25%% of total elapsed time. Review
        ↪ sys.dm_exec_query_stats.',
        10, -- severity (informational; raise to 16+ to break jobs)
        1
    ) WITH LOG; -- writes to SQL Server error log, picked up by Azure Monitor / CloudWatch
END;

```

```

-- Prune log older than 90 days
DELETE FROM dbo.WorkloadDistributionLog
WHERE captured_at < DATEADD(DAY, -90, SYSUTCDATETIME());
END;
GO

```

Wire the `RAISERROR ... WITH LOG` output to an Azure Monitor alert rule (for Azure SQL Managed Instance) or a CloudWatch Logs metric filter (for RDS SQL Server) that pages on-call when triggered outside business hours. This closes the feedback loop: the team migrates analytics, the monitor watches for drift, and any re-introduction of analytical load triggers a ticket before it accumulates into a billing surprise.

Building the Executive Business Case

The technical evidence assembled above — query classification tables, cost attribution scripts, and Compute Optimizer recommendations — flows naturally into a one-page financial summary for executive sign-off. Structure it in three columns: current state cost, future state cost, and net monthly saving. Keep the narrative out; executives respond to numbers.

A template that works:

Line Item	Current Monthly Cost	Post-Separation Cost	Monthly Saving
OLTP instance (Multi-AZ)	\$2,196	\$1,098 (downsized)	\$1,098
OLTP Reserved Instance discount (1yr no-upfront)	—	—\$412 vs. on-demand	\$412
Analytics on OLTP (attributed share)	\$769	\$0 (offloaded)	\$769
Analytics target (Redshift Serverless / Athena)	\$0	\$120	—\$120
S3 storage for analytics export	\$0	\$12	—\$12
Total	\$2,965	\$818	\$2,147

The annualized figure of roughly \$25,700 in this example is the number that moves a project from “interesting idea” to “approved and scheduled.” Present it with a 90-day implementation timeline and a named DBA responsible for each phase. Vague proposals die in committees; specific ones with owners get resourced.

Implementation Sequencing

Sequence the work to capture savings quickly while minimizing risk. Do not attempt a big-bang cutover.

Week 1–2: Run the classification queries and cost attribution script. Produce the financial exhibit. Identify the top five analytics queries by total execution time. Get buy-in.

Week 3–4: Provision the analytics target infrastructure (read replica, Redshift Serverless workspace, or Athena data lake). Run the daily export script. Validate that query results on the analytics target match the OLTP source within acceptable tolerance.

Week 5–6: Redirect one non-critical BI tool or report to the analytics target. Monitor for latency regressions and data freshness complaints. Keep the OLTP route as fallback.

Week 7–8: Migrate remaining analytics workloads. Confirm with Performance Insights that analytics load on the primary has dropped. Request a Compute Optimizer recommendation pull.

Week 9–10: Resize the OLTP instance during a low-traffic maintenance window. For RDS, this is a Multi-AZ failover-based resize — downtime is typically under 60 seconds when done correctly. Verify application connection pool behavior post-resize.

Week 11–12: Purchase Reserved Instances for the new, smaller OLTP tier. Enable the workload drift monitor Lambda. Document the runbook.

Each phase has a clear deliverable and a defined rollback path. The two-week cadence allows teams with normal sprint rhythms to absorb the work without dedicated project staffing.

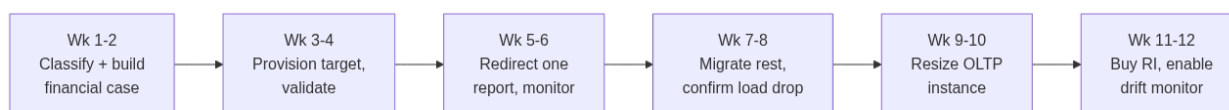


Figure 22: The 12-week phased rollout — each phase has a defined deliverable and rollback path before the next begins.

Key Takeaways

- **Mixed workloads are a billing multiplier.** Analytics queries running on OLTP infrastructure force you to pay row-store prices for column-store workloads. Measuring the split with `pg_stat_statements` or `sys.dm_exec_query_stats` is the necessary first step before any cost conversation.
- **The financial case has three compounding layers:** offloading analytics reduces attributed compute cost, lower sustained load allows an instance downsize, and a smaller instance purchases at a lower Reserved Instance tier. Each layer multiplies the previous saving.
- **Choose the analytics target by query pattern, not by vendor familiarity.** Near-real-time dashboards belong on a read replica. Batch aggregations over large historical data belong on a columnar MPP engine. Ad-hoc analyst queries against exported snapshots belong on Athena or BigQuery. Routing everything to a single alternative is a common mistake that recreates the original cost problem.
- **Separation without monitoring drifts back.** Workload classification alerts — whether a Lambda querying `pg_stat_statements` or a SQL Server Agent job writing to a log table — are the operational control that prevents analytics load from silently re-accumulating on the primary over subsequent release cycles.

- **Sequence implementation over 10–12 weeks with explicit rollback paths.** Instance resizes and RI purchases are irreversible commitments. Validate workload behavior at each phase before proceeding to the next, and keep the original routing available as fallback until query results and latency have been confirmed stable on the new target.
-

Chapter 23: Tagging Strategy for Database Cost Allocation

Organizations that implement a disciplined database tagging strategy typically recover 15–30% of their cloud database spend within the first quarter—not by rightsizing instances or switching storage classes, but simply by discovering who owns what, which workloads are generating the bills, and where untagged or mistagged resources have been silently accumulating charges for months. A 500-instance RDS estate with inconsistent tagging can hide \$40,000–\$80,000 per month in costs that no team claims ownership of, no budget is allocated for, and no engineer is motivated to optimize. This chapter builds a complete tagging strategy from schema design through enforcement automation, covering AWS RDS, Azure SQL, and Google Cloud SQL, with cost allocation queries, policy enforcement scripts, and a scheduled audit pipeline that keeps your tag compliance above 95%.

23.1 Designing a Tag Schema That Finance and Engineering Both Respect

The most common tagging failure is not technical—it is organizational. Engineers apply tags that make sense for operations (`env=prod`, `db=postgres`) while Finance needs tags that map to budget lines (`cost-center=CC-4421`, `project=Q3-Migration`). The two systems never reconcile, showback reports are disputed, and the entire exercise collapses into a spreadsheet maintained by one person who eventually leaves.

A workable schema serves both audiences with a small, mandatory set of tags and an extended set of optional enrichment tags.

Mandatory Tag Set (enforce via policy)

Tag Key	Example Values	Owner	Purpose
env	prod, staging, dev, sandbox	Engineering	Environment classification
team	payments, analytics, platform	Engineering	Team ownership
cost-center	CC-4421, CC-0012	Finance	Budget line alignment
project	checkout-v2, data-warehouse	PM / Engineering	Project-level chargeback

Tag Key	Example Values	Owner	Purpose
<code>application</code>	<code>order-service,</code> <code>reporting-api</code>	Engineering	Application owner
<code>managed-by</code>	<code>terraform,</code> <code>cloudformation,</code> <code>manual</code>	Platform	IaC drift detection

Extended Tag Set (recommended, not enforced)

Tag Key	Example Values	Purpose
<code>data-classification</code>	<code>pii, financial, internal</code>	Compliance tagging
<code>backup-policy</code>	<code>daily-7d, weekly-30d</code>	Backup tier identification
<code>dr-tier</code>	<code>tier1, tier2, tier3</code>	Recovery priority
<code>schedule</code>	<code>business-hours, always-on</code>	Scheduling automation (Chapter 24)
<code>rightsizing-exempt</code>	<code>true, false</code>	Block automated rightsizing

AWS RDS Tagging Limits

AWS RDS supports up to 50 tags per resource. Tag keys are case-sensitive: `Environment` and `environment` are two different tags. Establish a naming convention enforced by policy—lowercase, hyphenated, no spaces. A tag key collision between teams (`Env` vs `env` vs `ENV`) is one of the most common reasons cost allocation reports produce duplicates that Finance refuses to trust.

Azure SQL Tagging

Azure SQL resources (servers, databases, elastic pools) support up to 50 tags. Tags applied at the resource group level do not automatically inherit to individual database resources—a common misconception that leaves individual Azure SQL databases untagged even when the resource group appears compliant.

Google Cloud SQL Labeling

GCP uses “labels” rather than “tags.” Labels support up to 64 key-value pairs. GCP labels flow automatically into the BigQuery billing export, making them the primary mechanism for cost allocation queries in the `gcp_billing_export` dataset.

23.2 Discovering What Is Untagged Right Now

Before building enforcement pipelines, you need a baseline. The following queries and scripts identify every database resource across all three clouds that is missing one or more mandatory tags.

AWS: Find Untagged or Partially Tagged RDS Instances

```
#!/bin/bash
# audit_rds_tags.sh
```

```

# Scans all RDS instances across all regions and reports tag compliance
# Schedule: daily via EventBridge + Lambda or cron on a bastion host
# Output: CSV suitable for ingestion into cost dashboards

MANDATORY_TAGS=("env" "team" "cost-center" "project" "application")
REGIONS=$(aws ec2 describe-regions --query "Regions[].RegionName" --output text)

echo "region,db_instance,instance_class,monthly_est_usd,missing_tags"

for REGION in $REGIONS; do
    INSTANCES=$(aws rds describe-db-instances \
        --region "$REGION" \
        --query "DBInstances[*].[DBInstanceIdentifier,DBInstanceClass,DBInstanceArn]" \
        --output text 2>/dev/null)

    while IFS=$'\t' read -r DB_ID DB_CLASS DB_ARN; do
        [ -z "$DB_ID" ] && continue

        # Fetch existing tags for this instance
        EXISTING_TAGS=$(aws rds list-tags-for-resource \
            --resource-name "$DB_ARN" \
            --region "$REGION" \
            --query "TagList[].Key" \
            --output text 2>/dev/null | tr '\t' '\n' | tr '[:upper:]' '[:lower:]')

        MISSING=""
        for TAG in "${MANDATORY_TAGS[@]}; do
            if ! echo "$EXISTING_TAGS" | grep -qx "$TAG"; then
                MISSING="$MISSING $TAG"
            fi
        done

        MISSING=$(echo "$MISSING" | xargs) # trim whitespace

        if [ -n "$MISSING" ]; then
            # Rough monthly cost estimate based on instance class
            # Replace with aws ce get-cost-and-usage for exact figures
            MONTHLY_EST="unknown"
            echo "$REGION,$DB_ID,$DB_CLASS,$MONTHLY_EST,\"$MISSING\""
        fi
    done <<< "$INSTANCES"
done

```

Run this script on a schedule and pipe the output to an S3 bucket. The CSV becomes the input for your weekly tag compliance report. A 200-instance estate typically surfaces 40–80 instances with missing mandatory tags on the first run.

Azure: Identify Untagged Azure SQL Resources

```

#!/bin/bash
# audit_azure_sql_tags.sh
# Requires: az cli authenticated with Reader role on target subscriptions

MANDATORY_TAGS=("env" "team" "cost-center" "project" "application")
SUBSCRIPTIONS=$(az account list --query "[].id" --output tsv)

echo "subscription,resource_group,server,database,missing_tags"

for SUB in $SUBSCRIPTIONS; do
    az account set --subscription "$SUB" 2>/dev/null

    # Scan SQL servers
    SERVERS=$(az sql server list \
        --query "[*].[name,resourceGroup]" \
        --output tsv 2>/dev/null)

    while IFS=$'\t' read -r SERVER RG; do
        [ -z "$SERVER" ] && continue
    done

```

```

# Get databases under this server
DATABASES=$(az sql db list \
  --server "$SERVER" \
  --resource-group "$RG" \
  --query "[?name!='master'].[name]" \
  --output tsv 2>/dev/null)

while IFS=$'\t' read -r DB_NAME; do
  [ -z "$DB_NAME" ] && continue

  DB_TAGS=$(az sql db show \
    --name "$DB_NAME" \
    --server "$SERVER" \
    --resource-group "$RG" \
    --query "tags" \
    --output json 2>/dev/null)

  MISSING=""
  for TAG in "${MANDATORY_TAGS[@]}; do
    if ! echo "$DB_TAGS" | python3 -c \
      "import sys,json; t=json.load(sys.stdin) or {}; \
        exit(0 if '$TAG' in {k.lower() for k in t.keys()} else 1)" 2>/dev/null; then
      MISSING="$MISSING $TAG"
    fi
  done

  MISSING=$(echo "$MISSING" | xargs)
  if [ -n "$MISSING" ]; then
    echo "$SUB,$RG,$SERVER,$DB_NAME,\"$MISSING\""
  fi

done <<< "$DATABASES"
done <<< "$SERVERS"
done

```

GCP: Find Cloud SQL Instances Missing Labels

```

#!/bin/bash
# audit_cloudsql_labels.sh
# Requires: gcloud authenticated with cloudsql.viewer on all projects

MANDATORY_LABELS=("env" "team" "cost-center" "project" "application")
PROJECTS=$(gcloud projects list --format="value(projectId)" 2>/dev/null)

echo "project,instance,tier,missing_labels"

for PROJECT in $PROJECTS; do
  INSTANCES=$(gcloud sql instances list \
    --project="$PROJECT" \
    --format="value(name,settings.tier)" 2>/dev/null)

  while IFS=$'\t' read -r INSTANCE TIER; do
    [ -z "$INSTANCE" ] && continue

    LABELS=$(gcloud sql instances describe "$INSTANCE" \
      --project="$PROJECT" \
      --format="json(settings.userLabels)" 2>/dev/null \
      | python3 -c \
        "import sys,json; d=json.load(sys.stdin); \
          labels=d.get('settings',{}).get('userLabels',{}); \
          print(' '.join(labels.keys()))")

    MISSING=""
    for LABEL in "${MANDATORY_LABELS[@]}; do
      if ! echo "$LABELS" | grep -qw "$LABEL"; then
        MISSING="$MISSING $LABEL"
      fi
    done

    MISSING=$(echo "$MISSING" | xargs)
  done
done

```

```

    if [ -n "$MISSING" ]; then
        echo "$PROJECT,$INSTANCE,$TIER,\"$MISSING\""
    fi
done <<< "$INSTANCES"
done

```

23.3 Cost Allocation Queries: Turning Tags Into Budget Intelligence

Once resources are tagged, the second problem surfaces: the tagging is there, but no one is actually querying it. Finance is still allocating costs manually. The following queries demonstrate how to extract per-team and per-project database costs directly from each cloud's billing data.

AWS Cost Explorer: Per-Team RDS Cost Breakdown

```

#!/usr/bin/env python3
"""
rds_cost_by_team.py
Queries AWS Cost Explorer for RDS costs grouped by the 'team' tag.
Outputs a CSV and optionally posts a Slack summary.

Schedule: Weekly via EventBridge + Lambda or cron.
Requirements: boto3, AWS credentials with ce:GetCostAndUsage permission.
"""

import boto3
import csv
import io
import json
import os
from datetime import datetime, timedelta
from collections import defaultdict

def get_rds_cost_by_tag(tag_key: str = "team",
                       days_back: int = 30) -> list[dict]:
    """
    Returns RDS costs grouped by a tag key for the last N days.
    Includes both instance hours and storage costs.
    """
    ce = boto3.client("cost-explorer", region_name="us-east-1")

    end_date = datetime.today().strftime("%Y-%m-%d")
    start_date = (datetime.today() - timedelta(days=days_back)).strftime("%Y-%m-%d")

    response = ce.get_cost_and_usage(
        TimePeriod={"Start": start_date, "End": end_date},
        Granularity="MONTHLY",
        Filter={
            "And": [
                {
                    "Dimensions": {
                        "Key": "SERVICE",
                        "Values": ["Amazon Relational Database Service"]
                    }
                },
                {
                    # Exclude credits and refunds
                    "Not": {
                        "Dimensions": {
                            "Key": "RECORD_TYPE",
                            "Values": ["Credit", "Refund"]
                        }
                    }
                }
            ]
        }
    )

```

```

    }
    ]
},
GroupBy=[
    {"Type": "TAG", "Key": tag_key},
    {"Type": "DIMENSION", "Key": "USAGE_TYPE"}
],
Metrics=["UnblendedCost", "UsageQuantity"]
)

results = []
for period in response["ResultsByTime"]:
    month = period["TimePeriod"]["Start"][:7]
    for group in period["Groups"]:
        tag_value = group["Keys"][0].replace(f"{tag_key}$", "")
        usage_type = group["Keys"][1]

        cost = float(group["Metrics"]["UnblendedCost"]["Amount"])
        quantity = float(group["Metrics"]["UsageQuantity"]["Amount"])

        if cost > 0.01:
            results.append({
                "month": month,
                "tag_key": tag_value if tag_value else "UNTAGGED",
                "usage_type": usage_type,
                "cost_usd": round(cost, 2),
                "usage_quantity": round(quantity, 2),
                "unit": group["Metrics"]["UsageQuantity"]["Unit"]
            })

return results

def summarize_by_team(results: list[dict], tag_key: str = "team") -> dict:
    """Aggregates detailed usage rows into per-team totals."""
    totals = defaultdict(float)
    for row in results:
        totals[row[tag_key]] += row["cost_usd"]
    return dict(sorted(totals.items(), key=lambda x: x[1], reverse=True))

def export_csv(results: list[dict], output_path: str) -> None:
    if not results:
        print("No results to export.")
        return
    with open(output_path, "w", newline="") as f:
        writer = csv.DictWriter(f, fieldnames=results[0].keys())
        writer.writeheader()
        writer.writerows(results)
    print(f"Exported {len(results)} rows to {output_path}")

def post_slack_summary(totals: dict, webhook_url: str, period_label: str) -> None:
    """Posts a cost summary to Slack. Set SLACK_WEBHOOK_URL env variable."""
    import urllib.request

    lines = [f"*RDS Cost by Team - {period_label}*\n"]
    grand_total = sum(totals.values())
    for team, cost in totals.items():
        pct = (cost / grand_total * 100) if grand_total > 0 else 0
        lines.append(f"• {team}: ${cost:,.2f} ({pct:.1f}%)")
    lines.append(f"\n*Total: ${grand_total:,.2f}*")

    payload = json.dumps({"text": "\n".join(lines)}).encode()
    req = urllib.request.Request(
        webhook_url,
        data=payload,
        headers={"Content-Type": "application/json"}
    )
    urllib.request.urlopen(req)

```

```

print("Slack summary posted.")

if __name__ == "__main__":
    results = get_rds_cost_by_tag(tag_key="team", days_back=30)
    totals = summarize_by_team(results, tag_key="team")

    print("\nRDS Cost by Team (last 30 days):")
    print("-" * 45)
    grand_total = sum(totals.values())
    for team, cost in totals.items():
        pct = (cost / grand_total * 100) if grand_total > 0 else 0
        print(f" {team:<25} ${cost:>10,.2f} ({pct:.1f}%)"
    print("-" * 45)
    print(f" {'TOTAL':<25} ${grand_total:>10,.2f}")

    export_csv(results, "/tmp/rds_cost_by_team.csv")

    webhook = os.environ.get("SLACK_WEBHOOK_URL")
    if webhook:
        post_slack_summary(totals, webhook, "Last 30 Days")

```

Sample output from a 200-instance estate:

RDS Cost by Team (last 30 days):

```

-----
payments                $ 18,440.00  (31.2%)
analytics                $ 14,220.00  (24.1%)
platform                $  9,880.00  (16.7%)
UNTAGGED                $  9,200.00  (15.6%)
checkout                $  4,310.00   (7.3%)
identity                $  3,030.00   (5.1%)
-----
TOTAL                   $ 59,080.00

```

The UNTAGGED line—\$9,200 in this example—is the immediate target. These instances have no owner. No team is being charged back. No engineer is motivated to rightsize them.

GCP BigQuery: Cloud SQL Cost by Label

```

-- gcp_cloudsql_cost_by_label.sql
-- Queries the GCP billing export table in BigQuery
-- Replace 'your_project.billing_dataset.gcp_billing_export_v1_XXXXXX'
-- with your actual billing export table name.
-- Run in BigQuery console or schedule via Cloud Scheduler + BigQuery Data Transfer.

WITH

-- Flatten labels into a key-value structure per line item
labelled_costs AS (
    SELECT
        invoice.month                                AS billing_month,
        project.id                                  AS gcp_project,
        service.description                         AS service,
        sku.description                             AS sku,

        -- Extract specific labels; NULLIF handles missing labels gracefully
        MAX(IF(label.key = 'team', label.value, NULL)) AS team,
        MAX(IF(label.key = 'env', label.value, NULL)) AS env,
        MAX(IF(label.key = 'cost-center', label.value, NULL)) AS cost_center,
        MAX(IF(label.key = 'project', label.value, NULL)) AS app_project,
        MAX(IF(label.key = 'application', label.value, NULL)) AS application,

        SUM(cost)                                    AS total_cost_usd,

```

```

SUM(cost_at_list)                                AS list_cost_usd,

-- Savings from committed use discounts
SUM(cost_at_list) - SUM(cost)                    AS discount_usd

FROM `your_project.billing_dataset.gcp_billing_export_v1_XXXXXX`,
UNNEST(labels) AS label

WHERE
  service.description LIKE '%Cloud SQL%'
  AND DATE(_PARTITIONTIME) >= DATE_SUB(CURRENT_DATE(), INTERVAL 3 MONTH)

GROUP BY
  billing_month,
  gcp_project,
  service,
  sku
),

-- Identify line items where mandatory labels are missing
compliance_check AS (
  SELECT
    billing_month,
    gcp_project,
    team,
    env,
    cost_center,
    app_project,
    application,
    total_cost_usd,
    discount_usd,

    -- Tag completeness score: 1 point per present mandatory label
    (CASE WHEN team IS NOT NULL THEN 1 ELSE 0 END +
     CASE WHEN env IS NOT NULL THEN 1 ELSE 0 END +
     CASE WHEN cost_center IS NOT NULL THEN 1 ELSE 0 END +
     CASE WHEN app_project IS NOT NULL THEN 1 ELSE 0 END +
     CASE WHEN application IS NOT NULL THEN 1 ELSE 0 END
    )
    AS label_score,

    -- 5 = fully labelled, <5 = partially or fully unlabelled
    CASE
      WHEN team IS NULL AND env IS NULL AND cost_center IS NULL
      THEN 'UNTAGGED'
      WHEN (CASE WHEN team IS NOT NULL THEN 1 ELSE 0 END +
            CASE WHEN env IS NOT NULL THEN 1 ELSE 0 END +
            CASE WHEN cost_center IS NOT NULL THEN 1 ELSE 0 END +
            CASE WHEN app_project IS NOT NULL THEN 1 ELSE 0 END +
            CASE WHEN application IS NOT NULL THEN 1 ELSE 0 END) < 5
      THEN 'PARTIAL'
      ELSE 'COMPLIANT'
    END
    AS label_status

  FROM labelled_costs
)

-- Final summary: cost by team and label compliance status
SELECT
  billing_month,
  COALESCE(team, 'UNTAGGED') AS team,
  COALESCE(env, 'unknown') AS env,
  label_status,
  COUNT(*) AS line_item_count,
  ROUND(SUM(total_cost_usd), 2) AS total_cost_usd,
  ROUND(SUM(discount_usd), 2) AS total_discount_usd,
  ROUND(AVG(label_score), 2) AS avg_label_score

FROM compliance_check

GROUP BY

```



```

    billing_month,
    team,
    env,
    label_status

ORDER BY
    billing_month DESC,
    total_cost_usd DESC;

```

Schedule this query to run on the first of each month via Cloud Scheduler. Export results to a Looker Studio dashboard that Finance can access without needing BigQuery access themselves.

23.4 Enforcing Tags at Resource Creation: AWS, Azure, and GCP Policies

Auditing after the fact catches problems late. The more cost-effective approach is enforcement at creation time: block or warn when a database resource is created without the required tags. Each cloud provides a native policy mechanism.

AWS: Service Control Policy (SCP) to Block Untagged RDS Creation

SCPs operate at the AWS Organization level and cannot be overridden by individual account administrators. The following SCP blocks RDS instance creation and restoration unless all mandatory tags are present.

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "RequireMandatoryTagsOnRDSCreate",
      "Effect": "Deny",
      "Action": [
        "rds:CreateDBInstance",
        "rds:CreateDBCluster",
        "rds:RestoreDBInstanceFromDBSnapshot",
        "rds:RestoreDBClusterFromSnapshot",
        "rds:CreateDBInstanceReadReplica"
      ],
      "Resource": "*",
      "Condition": {
        "Null": {
          "aws:RequestTag/env": "true",
          "aws:RequestTag/team": "true",
          "aws:RequestTag/cost-center": "true",
          "aws:RequestTag/project": "true"
        }
      }
    },
    {
      "Sid": "AllowTaggingOnCreate",
      "Effect": "Allow",
      "Action": "rds:AddTagsToResource",
      "Resource": "*"
    }
  ]
}

```

Apply this SCP to all non-root OUs. Leave the root OU unconditional to allow emergency break-glass actions. Communicate the rollout two weeks in advance—teams using CloudFormation or Terraform without tag blocks will see failed deployments.

AWS Config Rule: Detect Existing Noncompliant Resources

The SCP only catches new resources. For existing resources, use an AWS Config managed rule:

```
# Deploy the required-tags Config rule via CLI
# This checks all existing RDS instances for tag compliance on a schedule

aws configservice put-config-rule \
  --config-rule '{
    "ConfigRuleName": "rds-required-tags",
    "Description": "Checks that all RDS DB instances have the required mandatory tags",
    "Scope": {
      "ComplianceResourceTypes": [
        "AWS::RDS::DBInstance",
        "AWS::RDS::DBCluster"
      ]
    },
    "Source": {
      "Owner": "AWS",
      "SourceIdentifier": "REQUIRED_TAGS"
    },
    "InputParameters":
    ↪ "{ \"tag1Key\": \"env\", \"tag2Key\": \"team\", \"tag3Key\": \"cost-center\", \"tag4Key\": \"project\", \"tag5Key\": \"applicat
  }'
```

Add an auto-remediation action using AWS Config Remediation + SSM Automation to notify the resource owner via SNS when a noncompliant resource is detected. A full block on existing resources creates operational disruption; notification-first remediation is more sustainable.

Azure: Policy Assignment to Require Tags on Azure SQL

Azure Policy can be assigned at the management group, subscription, or resource group level. The following ARM-format policy definition enforces the presence of mandatory tags on Azure SQL database resources.

```
{
  "properties": {
    "displayName": "Require mandatory tags on Azure SQL databases",
    "description": "Denies creation of Azure SQL databases without all mandatory FinOps tags.",
    "mode": "Indexed",
    "parameters": {
      "tagNames": {
        "type": "Array",
        "metadata": {
          "displayName": "Required Tag Names",
          "description": "List of tag keys that must be present on Azure SQL databases."
        },
        "defaultValue": ["env", "team", "cost-center", "project", "application"]
      }
    },
    "policyRule": {
      "if": {
        "allOf": [
          {
            "field": "type",
            "equals": "Microsoft.Sql/servers/databases"
          },
          {
            "count": {
              "value": "[parameters('tagNames')]",
              "name": "tagName",
              "where": {
                "field": "[concat('tags[', current('tagName'), ']')]",
                "exists": false
              }
            }
          }
        ],
        "greater": 0
      }
    }
  }
}
```

```

    }
  ],
  "then": {
    "effect": "deny"
  }
}
}
}
}

```

Deploy this policy definition and then assign it to your management group:

```

# Save the policy to a file, then deploy via Azure CLI
az policy definition create \
  --name "require-sql-mandatory-tags" \
  --display-name "Require mandatory tags on Azure SQL databases" \
  --description "Denies Azure SQL database creation without mandatory FinOps tags" \
  --rules policy_rule.json \
  --mode Indexed

# Assign at the management group level
az policy assignment create \
  --name "enforce-sql-tags" \
  --policy "require-sql-mandatory-tags" \
  --scope "/providers/Microsoft.Management/managementGroups/YOUR_MG_ID" \
  --enforcement-mode Default

```

For brownfield environments where existing databases lack tags, set `--enforcement-mode DoNotEnforce` initially and switch to `Default` after a 30-day remediation window.

GCP: Organization Policy to Require Labels on Cloud SQL

GCP's Organization Policies operate differently from AWS SCPs and Azure Policy. The `constraints/sql.restrictCloudSQLInstanceToConfiguredZone` style constraints exist for specific behaviors, but label enforcement at the organization policy level requires a custom constraint (available in GCP Organization Policy Service v2).

```

# custom_constraint_cloudsql_labels.yaml
# Custom Organization Policy constraint requiring labels on Cloud SQL instances
# Requires: Organization Policy Service API enabled
# Deploy with: gcloud org-policies set-custom-constraint

name: organizations/YOUR_ORG_ID/customConstraints/custom.requireCloudSQLLabels
resourceTypes:
  - sqladmin.googleapis.com/Instance
methodTypes:
  - CREATE
  - UPDATE
condition: >
  resource.settings.userLabels.exists(k, k == "env") &&
  resource.settings.userLabels.exists(k, k == "team") &&
  resource.settings.userLabels.exists(k, k == "cost-center") &&
  resource.settings.userLabels.exists(k, k == "project") &&
  resource.settings.userLabels.exists(k, k == "application")
actionType: ALLOW
displayName: Require mandatory labels on Cloud SQL instances
description: >
  Cloud SQL instances must have all five mandatory FinOps labels
  (env, team, cost-center, project, application) before they can
  be created or updated.

# Deploy the custom constraint
gcloud org-policies set-custom-constraint custom_constraint_cloudsql_labels.yaml

# Create the policy that enforces the constraint at the org level
cat > enforce_cloudsql_labels_policy.yaml << 'EOF'
name: organizations/YOUR_ORG_ID/policies/custom.requireCloudSQLLabels
spec:

```

```

rules:
  - enforce: true
EOF

gcloud org-policies set-policy enforce_cloudsql_labels_policy.yaml

```

23.5 Tagging Automation: Applying Tags Programmatically to Existing Resources

After the audit from 23.2 and policies from 23.4 are in place, you still have a backlog of untagged existing resources. Manually tagging hundreds of instances is error-prone and time-consuming. The following scripts automate bulk tagging based on existing metadata—instance name patterns, VPC placement, account structure, or a CSV input file.

AWS: Bulk Tag RDS Instances from a CSV Input

The input CSV format:

```

db_instance_identifier,region,env,team,cost-center,project,application
orders-prod-01,us-east-1,prod,payments,CC-4421,checkout-v2,order-service
analytics-rds,us-west-2,prod,analytics,CC-0012,data-warehouse,reporting-api
dev-scratch-db,eu-west-1,dev,platform,CC-0099,internal,scratch

```

```

#!/usr/bin/env python3
"""
bulk_tag_rds.py
Reads a CSV of RDS instances and applies tags to each one.
Performs a dry-run by default; pass --apply to execute changes.

Usage:
python3 bulk_tag_rds.py --input tags.csv --dry-run
python3 bulk_tag_rds.py --input tags.csv --apply
"""

import argparse
import boto3
import csv
import sys
from botocore.exceptions import ClientError

MANDATORY_KEYS = ["env", "team", "cost-center", "project", "application"]

def get_instance_arn(rds_client, db_instance_id: str) -> str | None:
    try:
        response = rds_client.describe_db_instances(
            DBInstanceIdentifier=db_instance_id
        )
        return response["DBInstances"][0]["DBInstanceArn"]
    except ClientError as e:
        if e.response["Error"]["Code"] == "DBInstanceNotFound":
            return None
        raise

def apply_tags(rds_client, arn: str, tags: dict, dry_run: bool) -> None:
    tag_list = [{"Key": k, "Value": v} for k, v in tags.items() if v]
    if dry_run:
        print(f" [DRY RUN] Would apply tags to {arn}:")
        for tag in tag_list:
            print(f"    {tag['Key']} = {tag['Value']}")
    else:

```

```

rds_client.add_tags_to_resource(
    ResourceName=arn,
    Tags=tag_list
)
print(f" [APPLIED] Tags applied to {arn}")

def process_csv(input_file: str, dry_run: bool) -> None:
    session_cache: dict[str, boto3.client] = {}

    with open(input_file, newline="") as f:
        reader = csv.DictReader(f)
        rows = list(reader)

    print(f"\nProcessing {len(rows)} instances. dry_run={dry_run}\n")
    success, skipped, errors = 0, 0, 0

    for row in rows:
        db_id = row.get("db_instance_identifier", "").strip()
        region = row.get("region", "us-east-1").strip()

        if not db_id:
            print(f" [SKIP] Empty db_instance_identifier in row: {row}")
            skipped += 1
            continue

        # Reuse boto3 client per region
        if region not in session_cache:
            session_cache[region] = boto3.client("rds", region_name=region)
        rds = session_cache[region]

        print(f"Instance: {db_id} ({region})")
        arn = get_instance_arn(rds, db_id)
        if not arn:
            print(f" [ERROR] Instance not found: {db_id} in {region}")
            errors += 1
            continue

        tags = {k: row.get(k, "").strip() for k in MANDATORY_KEYS}
        missing = [k for k, v in tags.items() if not v]
        if missing:
            print(f" [WARN] Missing values for tags: {missing} - skipping")
            skipped += 1
            continue

        try:
            apply_tags(rds, arn, tags, dry_run)
            success += 1
        except ClientError as e:
            print(f" [ERROR] Failed to tag {db_id}: {e}")
            errors += 1

    print(f"\nSummary: {success} applied, {skipped} skipped, {errors} errors")

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="Bulk tag RDS instances from CSV")
    parser.add_argument("--input", required=True, help="Path to input CSV")
    parser.add_argument("--dry-run", action="store_true", default=True)
    parser.add_argument("--apply", action="store_true", help="Execute tagging")
    args = parser.parse_args()

    dry_run = not args.apply
    process_csv(args.input, dry_run=dry_run)

```

GCP: Bulk Label Cloud SQL Instances from a CSV

```

#!/usr/bin/env python3
"""
bulk_label_cloudsql.py

```

*Reads a CSV and applies labels to Cloud SQL instances using the Cloud SQL Admin API.
Requires: google-cloud-sql-connector or google-api-python-client
pip install google-api-python-client google-auth*

Usage:

```
python3 bulk_label_cloudsql.py --input labels.csv --dry-run
python3 bulk_label_cloudsql.py --input labels.csv --apply
"""

import argparse
import csv
import sys
from googleapiclient import discovery
from googleapiclient.errors import HttpError
from google.auth import default as google_auth_default

MANDATORY_KEYS = ["env", "team", "cost-center", "project", "application"]

def get_sqladmin_service():
    credentials, _ = google_auth_default(
        scopes=["https://www.googleapis.com/auth/cloud-platform"]
    )
    return discovery.build("sqladmin", "v1beta4", credentials=credentials)

def get_existing_labels(service, project: str, instance: str) -> dict:
    try:
        result = service.instances().get(project=project, instance=instance).execute()
        return result.get("settings", {}).get("userLabels", {})
    except HttpError as e:
        print(f" [ERROR] Could not fetch {instance}: {e}")
        return {}

def apply_labels(service, project: str, instance: str,
                 new_labels: dict, dry_run: bool) -> None:
    existing = get_existing_labels(service, project, instance)
    merged = {**existing, **new_labels} # new labels win on collision

    if dry_run:
        print(f" [DRY RUN] Would patch {project}/{instance} with labels:")
        for k, v in new_labels.items():
            print(f"    {k} = {v}")
        return

    patch_body = {
        "settings": {
            "userLabels": merged
        }
    }
    try:
        service.instances().patch(
            project=project, instance=instance, body=patch_body
        ).execute()
        print(f" [APPLIED] Labels applied to {project}/{instance}")
    except HttpError as e:
        print(f" [ERROR] Patch failed for {instance}: {e}")
        raise

def process_csv(input_file: str, dry_run: bool) -> None:
    service = get_sqladmin_service()

    with open(input_file, newline="") as f:
        reader = csv.DictReader(f)
        rows = list(reader)

    print(f"\nProcessing {len(rows)} Cloud SQL instances. dry_run={dry_run}\n")
    success, skipped, errors = 0, 0, 0
```

```

for row in rows:
    project = row.get("project_id", "").strip()
    instance = row.get("instance_name", "").strip()

    if not project or not instance:
        print(f" [SKIP] Missing project_id or instance_name: {row}")
        skipped += 1
        continue

    # GCP label keys must be lowercase, hyphenated; values same
    labels = {
        k.lower().replace("_", "-"): v.strip().lower()
        for k, v in row.items()
        if k in MANDATORY_KEYS and v.strip()
    }
    missing = [k for k in MANDATORY_KEYS if k not in labels]
    if missing:
        print(f" [WARN] {project}/{instance} missing label values: {missing} - skipping")
        skipped += 1
        continue

    print(f"Instance: {project}/{instance}")
    try:
        apply_labels(service, project, instance, labels, dry_run)
        success += 1
    except Exception:
        errors += 1

print(f"\nSummary: {success} applied, {skipped} skipped, {errors} errors")

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="Bulk label Cloud SQL instances from CSV")
    parser.add_argument("--input", required=True, help="Path to input CSV")
    parser.add_argument("--dry-run", action="store_true", default=True)
    parser.add_argument("--apply", action="store_true")
    args = parser.parse_args()

    process_csv(args.input, dry_run=not args.apply)

```

23.6 Scheduled Tag Compliance Audit Pipeline

A one-time audit degrades quickly. Instances are created, cloned, and restored without tags every week. The audit must run on a schedule and produce a compliance score that is tracked over time. The following section builds a serverless audit pipeline on AWS (the pattern is replicable on Azure Functions and GCP Cloud Run).

Architecture Overview

EventBridge (weekly schedule)

```

Lambda: audit_tag_compliance
    Scans all RDS instances across all regions
    Calculates per-instance tag scores
    Writes compliance report → S3 (CSV + JSON)
    Updates DynamoDB compliance history table
    Posts Slack alert if compliance < threshold

```

Lambda Function: Tag Compliance Auditor

```

#!/usr/bin/env python3
"""

```

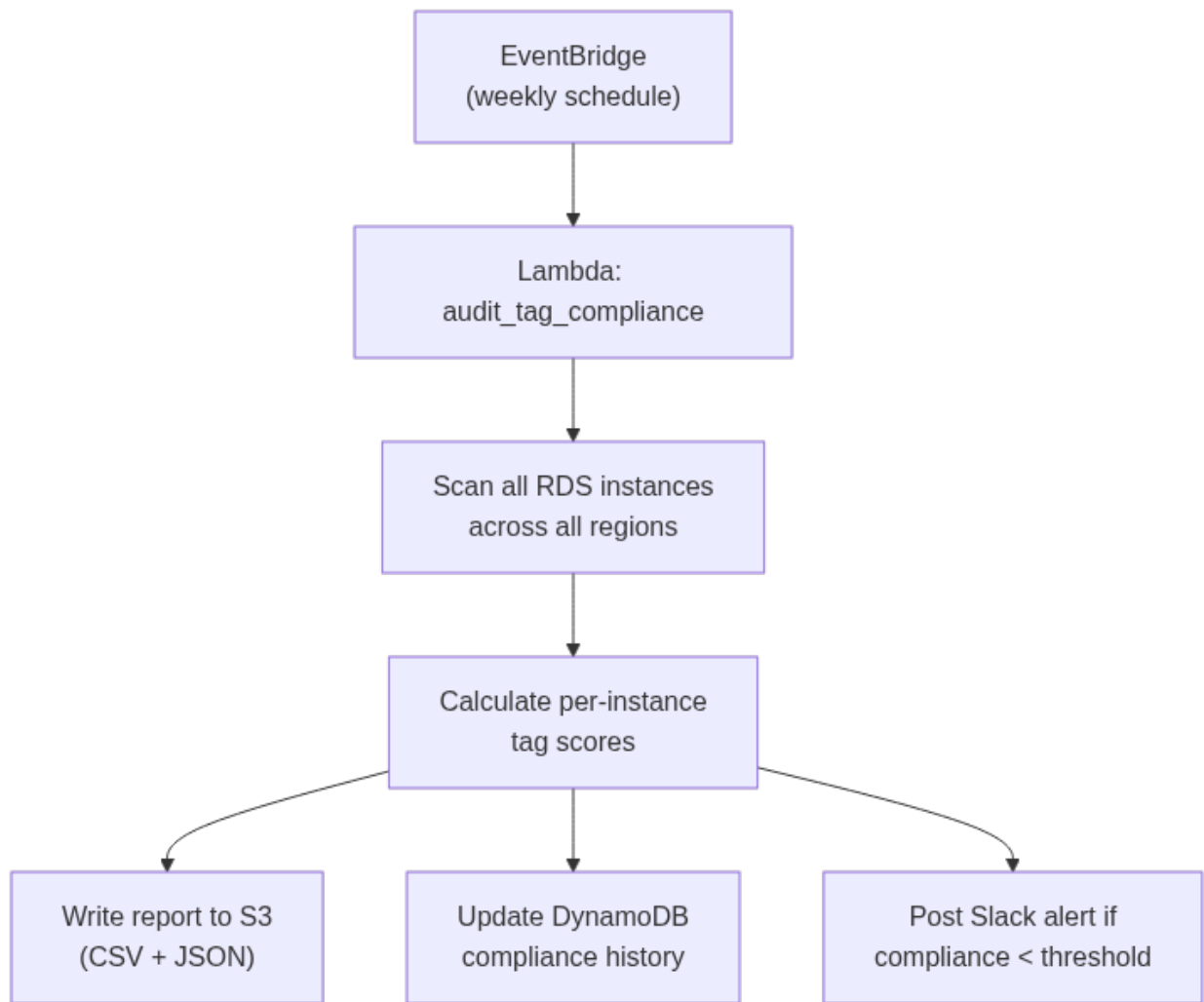


Figure 23: The weekly tag compliance audit pipeline.


```

lambda_tag_audit.py
AWS Lambda handler for weekly RDS tag compliance auditing.

Environment variables:
MANDATORY_TAGS      Comma-separated list of required tag keys
COMPLIANCE_THRESHOLD Float 0-100; trigger alert if score drops below (default: 95)
S3_BUCKET           S3 bucket name for report storage
S3_PREFIX           S3 key prefix (default: 'tag-audit/')
DYNAMODB_TABLE       DynamoDB table for compliance history (optional)
SLACK_WEBHOOK_URL    Slack incoming webhook (optional)

IAM permissions required:
rds:DescribeDBInstances (all regions)
rds:ListTagsForResource
s3:PutObject
dynamodb:PutItem (if DYNAMODB_TABLE set)
"""

import boto3
import csv
import io
import json
import os
import urllib.request
from datetime import datetime, timezone
from collections import defaultdict
from botocore.exceptions import ClientError

MANDATORY_TAGS = [t.strip() for t in
                   os.environ.get("MANDATORY_TAGS",
                                  "env,team,cost-center,project,application").split(",")]
COMPLIANCE_THRESHOLD = float(os.environ.get("COMPLIANCE_THRESHOLD", "95"))
S3_BUCKET = os.environ.get("S3_BUCKET", "")
S3_PREFIX = os.environ.get("S3_PREFIX", "tag-audit/")
DDB_TABLE = os.environ.get("DYNAMODB_TABLE", "")
SLACK_URL = os.environ.get("SLACK_WEBHOOK_URL", "")

def get_all_regions() -> list[str]:
    ec2 = boto3.client("ec2", region_name="us-east-1")
    return [r["RegionName"] for r in ec2.describe_regions()["Regions"]]

def audit_region(region: str) -> list[dict]:
    rds = boto3.client("rds", region_name=region)
    records = []

    paginator = rds.get_paginator("describe_db_instances")
    for page in paginator.paginate():
        for inst in page["DBInstances"]:
            db_id = inst["DBInstanceIdentifier"]
            db_arn = inst["DBInstanceArn"]
            db_cls = inst["DBInstanceClass"]
            engine = inst["Engine"]

            try:
                tag_response = rds.list_tags_for_resource(ResourceName=db_arn)
                existing_keys = {
                    t["Key"].lower() for t in tag_response["TagList"]
                }
            except ClientError:
                existing_keys = set()

            present = [t for t in MANDATORY_TAGS if t in existing_keys]
            missing = [t for t in MANDATORY_TAGS if t not in existing_keys]
            score = len(present) / len(MANDATORY_TAGS) * 100

            records.append({
                "region": region,
                "db_instance_id": db_id,

```

```

        "db_class":      db_cls,
        "engine":       engine,
        "tags_present": ", ".join(present),
        "tags_missing": ", ".join(missing),
        "tag_score":     round(score, 1),
        "status":        "COMPLIANT" if score == 100 else "NONCOMPLIANT"
    })
    return records

def calculate_summary(records: list[dict]) -> dict:
    total = len(records)
    compliant = sum(1 for r in records if r["status"] == "COMPLIANT")
    score = (compliant / total * 100) if total > 0 else 0

    by_region = defaultdict(lambda: {"total": 0, "compliant": 0})
    for r in records:
        by_region[r["region"]]["total"] += 1
        by_region[r["region"]]["compliant"] += (1 if r["status"] == "COMPLIANT" else 0)

    missing_counts = defaultdict(int)
    for r in records:
        for tag in r["tags_missing"].split(","):
            if tag:
                missing_counts[tag] += 1

    return {
        "total_instances": total,
        "compliant_instances": compliant,
        "noncompliant_instances": total - compliant,
        "compliance_score_pct": round(score, 2),
        "by_region": dict(by_region),
        "missing_tag_frequency": dict(sorted(
            missing_counts.items(), key=lambda x: x[1], reverse=True
        ))
    }

def write_to_s3(records: list[dict], summary: dict, run_ts: str) -> str:
    s3 = boto3.client("s3")
    date_prefix = run_ts[:10] # YYYY-MM-DD

    # Write CSV
    csv_key = f"{S3_PREFIX}-{date_prefix}/compliance_report.csv"
    buf = io.StringIO()
    if records:
        writer = csv.DictWriter(buf, fieldnames=records[0].keys())
        writer.writeheader()
        writer.writerows(records)
    s3.put_object(Bucket=S3_BUCKET, Key=csv_key,
                  Body=buf.getvalue().encode(),
                  ContentType="text/csv")

    # Write JSON summary
    json_key = f"{S3_PREFIX}-{date_prefix}/summary.json"
    s3.put_object(Bucket=S3_BUCKET, Key=json_key,
                  Body=json.dumps(summary, indent=2).encode(),
                  ContentType="application/json")

    print(f"Reports written to s3://{S3_BUCKET}/{S3_PREFIX}-{date_prefix}/")
    return f"s3://{S3_BUCKET}/{S3_PREFIX}-{date_prefix}/"

def write_to_dynamodb(summary: dict, run_ts: str) -> None:
    if not DDB_TABLE:
        return
    ddb = boto3.resource("dynamodb")
    table = ddb.Table(DDB_TABLE)
    table.put_item(Item={
        "run_date": run_ts[:10],

```

```

        "run_timestamp":      run_ts,
        "compliance_score_pct": str(summary["compliance_score_pct"]),
        "total_instances":    summary["total_instances"],
        "compliant_instances": summary["compliant_instances"],
        "noncompliant_instances": summary["noncompliant_instances"]
    })

def post_slack_alert(summary: dict, s3_path: str) -> None:
    if not SLACK_URL:
        return
    score = summary["compliance_score_pct"]
    status_emoji = " " if score >= COMPLIANCE_THRESHOLD else " "
    top_missing = list(summary["missing_tag_frequency"].items())[:3]
    missing_str = ", ".join(f"{k}` ({v})" for k, v in top_missing)

    text = (
        f"{status_emoji} *RDS Tag Compliance Report*\n"
        f"Compliance Score: {score}%\n"
        f"(threshold: {COMPLIANCE_THRESHOLD}%)\n"
        f"Instances: {summary['total_instances']} total, "
        f"{summary['noncompliant_instances']} noncompliant\n"
        f"Most missing tags: {missing_str}\n"
        f"Full report: {s3_path}"
    )

    payload = json.dumps({"text": text}).encode()
    req = urllib.request.Request(
        SLACK_URL, data=payload,
        headers={"Content-Type": "application/json"}
    )
    urllib.request.urlopen(req)

def lambda_handler(event, context):
    run_ts = datetime.now(timezone.utc).strftime("%Y-%m-%dT%H:%M:%SZ")
    print(f"Tag compliance audit started at {run_ts}")

    all_records = []
    regions = get_all_regions()
    print(f"Auditing {len(regions)} regions...")

    for region in regions:
        try:
            region_records = audit_region(region)
            all_records.extend(region_records)
            print(f" {region}: {len(region_records)} instances")
        except Exception as e:
            print(f" [WARN] Could not audit {region}: {e}")

    summary = calculate_summary(all_records)
    s3_path = write_to_s3(all_records, summary, run_ts) if S3_BUCKET else "N/A"
    write_to_dynamodb(summary, run_ts)
    post_slack_alert(summary, s3_path)

    print(f"\nCompliance score: {summary['compliance_score_pct']}%")
    print(f"Noncompliant: {summary['noncompliant_instances']} instances")

    return {
        "statusCode": 200,
        "body": json.dumps(summary)
    }

```

EventBridge Schedule (CloudFormation)

```

# eventbridge_tag_audit_schedule.yaml
# Deploys the Lambda + EventBridge schedule for weekly tag compliance auditing

AWSTemplateFormatVersion: "2010-09-09"
Description: "Weekly RDS tag compliance audit pipeline"

```

```

Parameters:
  S3BucketName:
    Type: String
    Description: S3 bucket for audit report storage
  SlackWebhookUrl:
    Type: String
    NoEcho: true
    Default: ""
  ComplianceThreshold:
    Type: Number
    Default: 95

Resources:

  AuditLambdaRole:
    Type: AWS::IAM::Role
    Properties:
      RoleName: rds-tag-audit-lambda-role
      AssumeRolePolicyDocument:
        Version: "2012-10-17"
        Statement:
          - Effect: Allow
            Principal:
              Service: lambda.amazonaws.com
            Action: sts:AssumeRole
      ManagedPolicyArns:
        - arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
      Policies:
        - PolicyName: RDSTagAuditPolicy
          PolicyDocument:
            Version: "2012-10-17"
            Statement:
              - Effect: Allow
                Action:
                  - rds:DescribeDBInstances
                  - rds:ListTagsForResource
                  - ec2:DescribeRegions
                Resource: "*"
              - Effect: Allow
                Action:
                  - s3:PutObject
                Resource: !Sub "arn:aws:s3:::${S3BucketName}/tag-audit/*"
              - Effect: Allow
                Action:
                  - dynamodb:PutItem
                Resource: !GetAtt ComplianceHistoryTable.Arn

  AuditLambdaFunction:
    Type: AWS::Lambda::Function
    Properties:
      FunctionName: rds-tag-compliance-audit
      Runtime: python3.12
      Handler: lambda_tag_audit.lambda_handler
      Role: !GetAtt AuditLambdaRole.Arn
      Timeout: 300
      MemorySize: 256
      Environment:
        Variables:
          MANDATORY_TAGS: "env,team,cost-center,project,application"
          COMPLIANCE_THRESHOLD: !Ref ComplianceThreshold
          S3_BUCKET: !Ref S3BucketName
          S3_PREFIX: "tag-audit/"
          DYNAMODB_TABLE: !Ref ComplianceHistoryTable
          SLACK_WEBHOOK_URL: !Ref SlackWebhookUrl

  WeeklyAuditSchedule:
    Type: AWS::Events::Rule
    Properties:
      Name: rds-tag-audit-weekly

```

```

Description: "Triggers weekly RDS tag compliance audit every Monday at 07:00 UTC"
ScheduleExpression: "cron(0 7 ? * MON *)"
State: ENABLED
Targets:
  - Arn: !GetAtt AuditLambdaFunction.Arn
    Id: TagAuditLambdaTarget

LambdaInvokePermission:
  Type: AWS::Lambda::Permission
  Properties:
    FunctionName: !Ref AuditLambdaFunction
    Action: lambda:InvokeFunction
    Principal: events.amazonaws.com
    SourceArn: !GetAtt WeeklyAuditSchedule.Arn

ComplianceHistoryTable:
  Type: AWS::DynamoDB::Table
  Properties:
    TableName: rds-tag-compliance-history
    BillingMode: PAY_PER_REQUEST
    AttributeDefinitions:
      - AttributeName: run_date
        AttributeType: S
      - AttributeName: run_timestamp
        AttributeType: S
    KeySchema:
      - AttributeName: run_date
        KeyType: HASH
      - AttributeName: run_timestamp
        KeyType: RANGE
    TimeToLiveSpecification:
      AttributeName: ttl
      Enabled: true

```

The DynamoDB table stores a weekly row, giving you a 52-week trailing history of compliance scores. A simple query against this table in CloudWatch Metrics Insights or Athena produces a trend chart that makes compliance regressions visible—a new team creates three untagged RDS instances in week 34, and the score drops from 98% to 94%, triggering the Slack alert before Finance notices it in the monthly billing report.

23.7 Multi-Cloud Tag Normalization: When Teams Span AWS, Azure, and GCP

Organizations with workloads on multiple clouds frequently end up with fragmented tagging strategies: the AWS team uses `cost-center`, the Azure team uses `CostCenter`, and the GCP team uses `cost_center`. The billing export from each cloud goes to a central data warehouse, and none of the keys join cleanly.

Normalization must happen at the data warehouse layer—you cannot force tag key consistency retroactively across clouds without re-running tagging automation against every resource. The following SQL pattern implements a normalization view in your central analytics warehouse (syntax shown for BigQuery; adapt `LOWER` and `REGEXP_REPLACE` for Snowflake or Redshift as needed).

BigQuery: Unified Cross-Cloud Database Cost View

```

-- unified_db_cost_view.sql
-- Creates a unified view of database costs across AWS RDS, Azure SQL, and GCP Cloud SQL
-- by normalizing tag/label keys to a canonical schema.
-- Store as a BigQuery view: your_project.finops_dataset.v_unified_db_costs

```

```

CREATE OR REPLACE VIEW `your_project.finops_dataset.v_unified_db_costs` AS

WITH

-- AWS RDS costs from CUR (Cost and Usage Report) exported to S3 → BigQuery
aws_rds AS (
  SELECT
    FORMAT_DATE('%Y-%m', DATE(line_item_usage_start_date)) AS billing_month,
    'AWS' AS cloud,
    'RDS' AS db_service,
    line_item_resource_id AS resource_id,
    line_item_usage_type AS sku,

    -- Normalize tag keys: CUR tags use resource_tags_ prefix
    LOWER(COALESCE(
      resource_tags_user_team,
      resource_tags_user_Team,
      resource_tags_user_TEAM
    )) AS team,

    LOWER(COALESCE(
      resource_tags_user_env,
      resource_tags_user_Env,
      resource_tags_user_environment
    )) AS env,

    LOWER(COALESCE(
      resource_tags_user_cost_center,
      `resource_tags_user_cost-center`
    )) AS cost_center,

    LOWER(COALESCE(
      resource_tags_user_project,
      resource_tags_user_Project
    )) AS project,

    LOWER(COALESCE(
      resource_tags_user_application,
      resource_tags_user_Application
    )) AS application,

    SUM(line_item_unblended_cost) AS cost_usd

  FROM `your_project.aws_cur_dataset.aws_cur_*`
  WHERE
    line_item_product_code = 'AmazonRDS'
    AND line_item_line_item_type NOT IN ('Credit', 'Refund', 'Tax')
  GROUP BY
    billing_month, cloud, db_service, resource_id, sku,
    team, env, cost_center, project, application
), -- Azure SQL costs from Azure Cost Export exported to Blob → BigQuery
azure_sql AS (
  SELECT
    FORMAT_DATE('%Y-%m', DATE(UsageDateTime)) AS billing_month,
    'Azure' AS cloud,
    'Azure SQL' AS db_service,
    InstanceId AS resource_id,
    MeterName AS sku,

    -- Azure tag keys are stored as JSON strings; normalize with LOWER
    LOWER(COALESCE(
      JSON_VALUE(Tags, '$.team'),
      JSON_VALUE(Tags, '$.Team'),
      JSON_VALUE(Tags, '$.TEAM')
    )) AS team,

    LOWER(COALESCE(
      JSON_VALUE(Tags, '$.env'),
      JSON_VALUE(Tags, '$.Env')

```

```

        JSON_VALUE(Tags, '$.environment'),
        JSON_VALUE(Tags, '$.Environment')
    )) AS env,

    LOWER(COALESCE(
        JSON_VALUE(Tags, '$.cost-center'),
        JSON_VALUE(Tags, '$.CostCenter'),
        JSON_VALUE(Tags, '$.costcenter'),
        JSON_VALUE(Tags, '$.cost_center')
    )) AS cost_center,

    LOWER(COALESCE(
        JSON_VALUE(Tags, '$.project'),
        JSON_VALUE(Tags, '$.Project')
    )) AS project,

    LOWER(COALESCE(
        JSON_VALUE(Tags, '$.application'),
        JSON_VALUE(Tags, '$.Application')
    )) AS application,

    SUM(PreTaxCost) AS cost_usd

FROM `your_project.azure_billing_dataset.azure_cost_export_*`
WHERE
    -- Filter to SQL-family services
    (MeterCategory IN ('SQL Database', 'SQL Managed Instance',
        'Azure SQL Database', 'Azure Database for PostgreSQL',
        'Azure Database for MySQL')
    OR ServiceName LIKE '%SQL%')
    AND ChargeType NOT IN ('Refund', 'Credit')
GROUP BY
    billing_month, cloud, db_service, resource_id, sku,
    team, env, cost_center, project, application
),

-- GCP Cloud SQL costs from GCP Billing Export in BigQuery
gcp_cloudsql AS (
    SELECT
        invoice.month AS billing_month,
        'GCP' AS cloud,
        'Cloud SQL' AS db_service,
        -- GCP resource name is in the resource labels or the resource.name field
        resource.name AS resource_id,
        sku.description AS sku,

        -- GCP labels are already lowercase by convention; COALESCE handles missing ones
        MAX(IF(label.key = 'team', label.value, NULL)) AS team,
        MAX(IF(label.key = 'env', label.value, NULL)) AS env,
        MAX(IF(label.key = 'cost-center', label.value, NULL)) AS cost_center,
        MAX(IF(label.key = 'project', label.value, NULL)) AS project,
        MAX(IF(label.key = 'application', label.value, NULL)) AS application,

        SUM(cost) AS cost_usd

    FROM `your_project.billing_dataset.gcp_billing_export_v1_XXXXXX`,
    UNNEST(labels) AS label

    WHERE
        service.description LIKE '%Cloud SQL%'
        AND cost_type NOT IN ('rounding_error')
    GROUP BY
        billing_month, cloud, db_service, resource_id, sku
),

-- Union all three sources
combined AS (
    SELECT * FROM aws_rds
    UNION ALL
    SELECT * FROM azure_sql

```

```

UNION ALL
SELECT * FROM gcp_cloudsql
)

-- Final output with normalization flag
SELECT
  billing_month,
  cloud,
  db_service,
  resource_id,
  sku,

  COALESCE(team,          'UNTAGGED') AS team,
  COALESCE(env,           'unknown') AS env,
  COALESCE(cost_center, 'UNTAGGED') AS cost_center,
  COALESCE(project,      'UNTAGGED') AS project,
  COALESCE(application, 'UNTAGGED') AS application,

  -- Tag completeness score
  (CASE WHEN team      IS NOT NULL THEN 1 ELSE 0 END +
   CASE WHEN env       IS NOT NULL THEN 1 ELSE 0 END +
   CASE WHEN cost_center IS NOT NULL THEN 1 ELSE 0 END +
   CASE WHEN project    IS NOT NULL THEN 1 ELSE 0 END +
   CASE WHEN application IS NOT NULL THEN 1 ELSE 0 END
  ) AS tag_score,

  ROUND(SUM(cost_usd), 2) AS cost_usd

FROM combined

GROUP BY
  billing_month,
  cloud,
  db_service,
  resource_id,
  sku,
  team,
  env,
  cost_center,
  project,
  application,
  tag_score

ORDER BY
  billing_month DESC,
  cost_usd DESC;

```

Once this view exists, Finance can query it from any BI tool—Looker, Tableau, Power BI, or Looker Studio—without understanding the differences between CUR column naming conventions and GCP billing export label schema. A single `WHERE team = 'payments'` filter returns every database dollar spent by that team across all three clouds in a given month.

Tag Normalization Lookup Table

For organizations with dozens of teams applying tag values inconsistently (e.g., `pay`, `payments`, `Payments`, `PYMT`, `payment-service`), a normalization lookup table prevents the per-team cost view from fracturing into dozens of near-duplicate rows.

```

-- tag_normalization_map.sql
-- A reference table mapping raw tag values to canonical values.
-- Maintain this table as source-of-truth in your FinOps data warehouse.
-- Populate initially from the audit output; update as new aliases appear.

CREATE TABLE IF NOT EXISTS `your_project.finops_dataset.tag_normalization_map` (
  tag_key      STRING NOT NULL, -- e.g., 'team'
  raw_value    STRING NOT NULL, -- e.g., 'pay', 'PYMT', 'payment-service'
  canonical_value STRING NOT NULL, -- e.g., 'payments'

```



```

added_date    DATE,
added_by      STRING
);

-- Sample rows (insert your own based on audit findings)
INSERT INTO `your_project.finops_dataset.tag_normalization_map`
(tag_key, raw_value, canonical_value, added_date, added_by)
VALUES
('team', 'pay',          'payments', CURRENT_DATE(), 'finops-team'),
('team', 'PYMT',         'payments', CURRENT_DATE(), 'finops-team'),
('team', 'payment-service', 'payments', CURRENT_DATE(), 'finops-team'),
('team', 'Payments',      'payments', CURRENT_DATE(), 'finops-team'),
('team', 'analytics-team', 'analytics', CURRENT_DATE(), 'finops-team'),
('team', 'bi',            'analytics', CURRENT_DATE(), 'finops-team'),
('env', 'production',     'prod',      CURRENT_DATE(), 'finops-team'),
('env', 'PROD',           'prod',      CURRENT_DATE(), 'finops-team'),
('env', 'prd',            'prod',      CURRENT_DATE(), 'finops-team'),
('env', 'development',    'dev',      CURRENT_DATE(), 'finops-team'),
('env', 'DEV',            'dev',      CURRENT_DATE(), 'finops-team');

```

Join this table into the unified view's COALESCE chain to replace raw tag values with their canonical equivalents before the grouping step. The lookup table becomes a living document that the FinOps team updates as new tag value variants surface in the audit output.

23.8 Integrating Tag Compliance into IaC Pipelines

The most durable place to enforce tagging is the deployment pipeline. An instance that cannot pass a `terraform plan` validation step or a CloudFormation stack policy check without the correct tags never reaches production in a noncompliant state. The approaches in this section work in concert with the cloud-native policies from section 23.4—defense in depth rather than relying on a single enforcement layer.

Terraform: Variable Validation for Mandatory Tags

```

# modules/rds/variables.tf
# Shared Terraform module for RDS instances.
# All mandatory tags must be provided by the calling module.
# This prevents the plan from succeeding without complete tags.

variable "mandatory_tags" {
  description = "Mandatory FinOps tags required on all RDS resources."
  type = object({
    env      = string
    team     = string
    cost_center = string
    project  = string
    application = string
  })

  validation {
    condition     = contains(["prod", "staging", "dev", "sandbox"], var.mandatory_tags.env)
    error_message = "env must be one of: prod, staging, dev, sandbox."
  }

  validation {
    condition     = length(var.mandatory_tags.team) >= 2
    error_message = "team must be at least 2 characters."
  }

  validation {
    condition     = can(regex("^CC-[0-9]{4}$", var.mandatory_tags.cost_center))
    error_message = "cost_center must match format CC-XXXX (e.g., CC-4421)."
  }
}

```

```

    }

    validation {
      condition = length(var.mandatory_tags.project) >= 3
      error_message = "project must be at least 3 characters."
    }

    validation {
      condition = length(var.mandatory_tags.application) >= 3
      error_message = "application must be at least 3 characters."
    }
  }

  variable "optional_tags" {
    description = "Optional enrichment tags. Merged with mandatory_tags at resource creation."
    type        = map(string)
    default     = {}
  }

  # Compose the full tag map used on every resource in this module
  locals {
    all_tags = merge(
      {
        env          = var.mandatory_tags.env
        team         = var.mandatory_tags.team
        "cost-center" = var.mandatory_tags.cost_center # normalize hyphen form
        project      = var.mandatory_tags.project
        application   = var.mandatory_tags.application
        managed-by   = "terraform"
      },
      var.optional_tags
    )
  }

  # modules/rds/main.tf
  # Example RDS instance resource using the validated tag block

  resource "aws_db_instance" "this" {
    identifier      = var.db_identifier
    engine          = var.engine
    engine_version  = var.engine_version
    instance_class  = var.instance_class
    allocated_storage = var.allocated_storage
    db_name         = var.db_name
    username        = var.master_username
    password        = var.master_password

    # ... other required parameters ...

    tags = local.all_tags
  }

  resource "aws_db_subnet_group" "this" {
    name      = "${var.db_identifier}-subnet-group"
    subnet_ids = var.subnet_ids
    tags      = local.all_tags
  }

```

A calling module that omits `cost_center` or provides an invalid value fails at `terraform validate` before any AWS API calls are made. This is the cheapest enforcement point.

Conftest / OPA Policy for Terraform Plan Output

For organizations using Open Policy Agent in their CI pipeline, the following Rego policy evaluates the Terraform plan JSON and fails the build if any `aws_db_instance` resource is missing mandatory tags.

```
# policies/rds_mandatory_tags.rego
```

```
# Evaluates terraform plan output (JSON) and enforces mandatory tag presence
# on all aws_db_instance and aws_db_cluster resources.
#
# Usage in CI (after terraform plan -out=plan.tfplan && terraform show -json plan.tfplan > plan.json):
#   conftest test plan.json --policy policies/

package rds_mandatory_tags

import future.keywords.in

mandatory_tags := {
  "env",
  "team",
  "cost-center",
  "project",
  "application",
}

rds_resource_types := {
  "aws_db_instance",
  "aws_db_cluster",
  "aws_rds_cluster",
}

# Collect all RDS resources being created or updated in the plan
rds_resources[name] := config {
  resource := input.resource_changes[_]
  resource.type in rds_resource_types
  resource.change.actions[_] in {"create", "update"}
  config := resource.change.after
  name := resource.address
}

# For each RDS resource, find which mandatory tags are missing
missing_tags[resource_addr] := absent {
  config := rds_resources[resource_addr]
  applied_tags := {lower(k) | config.tags[k]}
  absent := mandatory_tags - applied_tags
  count(absent) > 0
}

# Deny rule: fail if any resource has missing mandatory tags
deny[msg] {
  resource_addr := missing_tags[_]
  absent := missing_tags[resource_addr]
  msg := sprintf(
    "Resource %v is missing mandatory tags: %v",
    [resource_addr, concat(" ", absent)]
  )
}
```

Add this to your CI pipeline after the Terraform plan step:

```
# .github/workflows/terraform_tag_check.yml (excerpt)
- name: Convert Terraform plan to JSON
  run: terraform show -json tfplan > plan.json

- name: Run OPA tag compliance check
  uses: open-policy-agent/conftest-action@v3
  with:
    files: plan.json
    policy: policies/
```

A failed OPA check blocks the pull request merge. Engineers see the exact tag they missed and the exact resource address that violated the policy, which is far more actionable than a cryptic AWS `AccessDenied` error from the SCP.

23.9 Chargeback and Showback Workflows

Tagging alone does not produce chargeback. Chargeback requires a workflow: the tagged cost data flows to a report, the report is distributed to team leads or finance BPs, and there is an agreed mechanism for attributing those costs to a budget. The following describes a minimal chargeback workflow suitable for a 10–50 team organization.

Showback vs. Chargeback

Showback means: “Here is what your team consumed; no money actually moves.” Showback is politically safe and builds team awareness of their database costs. Start here.

Chargeback means: “Here is what your team consumed; your budget is debited by this amount.” Chargeback requires buy-in from finance leadership, a mapping from cloud cost centers to internal GL codes, and a process for handling disputes. It motivates optimization far more strongly than showback but requires more governance infrastructure.

Monthly Chargeback Report Generator

The following Python script generates per-team chargeback reports from the AWS Cost Explorer data and distributes them by email (or Slack). It reads the `rds_cost_by_team.py` output and formats it into a per-team summary that can be sent directly to each team’s engineering lead.

```
#!/usr/bin/env python3
"""
generate_chargeback_reports.py
Generates and distributes monthly per-team RDS chargeback reports.

Reads cost data from the output of rds_cost_by_team.py (CSV) or
calls Cost Explorer directly. Emails each team lead using SES.

Requirements:
    boto3, jinja2
    pip install boto3 jinja2

Configuration:
    TEAM_CONTACTS_FILE Path to JSON file mapping team names to email addresses
    FROM_EMAIL          Verified SES sender address
    REPORT_MONTH        YYYY-MM format; defaults to previous month
"""

import boto3
import csv
import json
import os
from collections import defaultdict
from datetime import datetime, date
from email.mime.multipart import MIMEMultipart
from email.mime.text import MIMEText
from jinja2 import Template

TEAM_CONTACTS_FILE = os.environ.get("TEAM_CONTACTS_FILE", "team_contacts.json")
FROM_EMAIL = os.environ.get("FROM_EMAIL", "finops@yourcompany.com")
REPORT_MONTH = os.environ.get("REPORT_MONTH", "") # defaults to last month below

EMAIL_TEMPLATE = """
<!DOCTYPE html>
<html>
<body style="font-family: Arial, sans-serif; color: #333;">
<h2> RDS Cost Chargeback Report - {{ month }}</h2>
<p>Hello {{ team_lead_name }},</p>
<p>Below is the database cost summary for the <strong>{{ team }}</strong> team for {{ month }}.</p>

```

```

<table border="1" cellpadding="8" cellspacing="0" style="border-collapse: collapse; width: 100%;">
  <thead style="background-color: #f0f0f0;">
    <tr>
      <th>Usage Type</th>
      <th>Cost (USD)</th>
      <th>% of Team Total</th>
    </tr>
  </thead>
  <tbody>
    {% for row in usage_rows %}
      <tr>
        <td>{{ row.usage_type }}</td>
        <td>${{ "%.2f"|format(row.cost_usd) }}</td>
        <td>{{ "%.1f"|format(row.pct) }}%</td>
      </tr>
    {% endfor %}
  </tbody>
  <tfoot style="background-color: #f8f8f8; font-weight: bold;">
    <tr>
      <td>TOTAL</td>
      <td>${{ "%.2f"|format(team_total) }}</td>
      <td>100%</td>
    </tr>
  </tfoot>
</table>

<p style="margin-top: 16px;">
  <strong>Untagged resources cost:</strong>
  ${{ "%.2f"|format(untagged_cost) }} - these are not included in team totals
  and represent optimization opportunities.
</p>

<p>
  Questions? Contact the FinOps team at <a
  ↵ href="mailto:finops@yourcompany.com">finops@yourcompany.com</a>
  or review the full report in the
  <a href="{{ dashboard_url }}">FinOps Dashboard</a>.
</p>
</body>
</html>
"""

```

```

def get_report_month() -> str:
    if REPORT_MONTH:
        return REPORT_MONTH
    today = date.today()
    # Default to previous month
    if today.month == 1:
        return f"{today.year - 1}-12"
    return f"{today.year}-{today.month - 1:02d}"

def load_team_contacts(path: str) -> dict:
    """
    team_contacts.json format:
    {
        "payments": {"email": "lead@payments.co", "name": "Alice Chen"},
        "analytics": {"email": "lead@analytics.co", "name": "Bob Patel"},
        ...
    }
    """
    with open(path) as f:
        return json.load(f)

def load_cost_data_from_csv(csv_path: str, month: str) -> list[dict]:
    rows = []
    with open(csv_path, newline="") as f:
        for row in csv.DictReader(f):

```

```

        if row.get("month", "").startswith(month):
            rows.append(row)
    return rows

def aggregate_by_team(rows: list[dict]) -> dict:
    team_data = defaultdict(list)
    for row in rows:
        team_data[row["team"]].append({
            "usage_type": row["usage_type"],
            "cost_usd": float(row["cost_usd"])
        })
    return team_data

def send_email_ses(to_addr: str, from_addr: str,
                  subject: str, html_body: str) -> None:
    ses = boto3.client("ses", region_name="us-east-1")
    msg = MIMEMultipart("alternative")
    msg["Subject"] = subject
    msg["From"] = from_addr
    msg["To"] = to_addr
    msg.attach(MIMEText(html_body, "html"))

    ses.send_raw_email(
        Source=from_addr,
        Destinations=[to_addr],
        RawMessage={"Data": msg.as_string()}
    )
    print(f" Email sent to {to_addr}")

def generate_and_distribute(csv_path: str, dry_run: bool = True) -> None:
    month = get_report_month()
    team_contacts = load_team_contacts(Team_CONTACTS_FILE)
    rows = load_cost_data_from_csv(csv_path, month)
    team_data = aggregate_by_team(rows)
    template = Template(EMAIL_TEMPLATE)
    dashboard_url = os.environ.get("DASHBOARD_URL", "https://finops.yourcompany.com")

    untagged_cost = sum(
        r["cost_usd"] for r in team_data.get("UNTAGGED", [])
    )

    print(f"\nGenerating chargeback reports for {month}...")
    print(f"Teams with data: {sorted(team_data.keys())}\n")

    for team, contact in team_contacts.items():
        usage_rows = team_data.get(team, [])
        if not usage_rows:
            print(f" [SKIP] No cost data for team '{team}' in {month}")
            continue

        team_total = sum(r["cost_usd"] for r in usage_rows)
        for r in usage_rows:
            r["pct"] = (r["cost_usd"] / team_total * 100) if team_total > 0 else 0
        usage_rows.sort(key=lambda r: r["cost_usd"], reverse=True)

        html = template.render(
            month=month,
            team=team,
            team_lead_name=contact["name"],
            usage_rows=usage_rows,
            team_total=team_total,
            untagged_cost=untagged_cost,
            dashboard_url=dashboard_url
        )

        subject = f"RDS Cost Chargeback - {team} - {month} - ${team_total:,.2f}"

```

```
if dry_run:
    print(f" [DRY RUN] Would email {contact['email']}: {subject}")
else:
    send_email_ses(contact["email"], FROM_EMAIL, subject, html)

if __name__ == "__main__":
    import argparse
    parser = argparse.ArgumentParser()
    parser.add_argument("--csv", required=True, help="Path to cost CSV from rds_cost_by_team.py")
    parser.add_argument("--apply", action="store_true", help="Send emails (default: dry run)")
    args = parser.parse_args()
    generate_and_distribute(args.csv, dry_run=not args.apply)
```

Run this on the third business day of each month, after Cost Explorer data for the prior month has settled. Send the emails to engineering leads first, give them a week to raise disputes, then send the final version to Finance for budget reconciliation.

Key Takeaways

- **A tag schema must serve both Finance and Engineering simultaneously.** A small mandatory set (five to six keys covering environment, team, cost center, project, and application) paired with an optional enrichment set is more durable than either a sprawling mandated list or a purely operational tagging convention that Finance cannot use for budget allocation.
 - **Untagged database spend is not a tagging problem—it is an ownership problem.** The first step after discovering untagged resources is assigning an owner, not applying a tag. Without ownership, the tag will be inaccurate or drift over time. Use the audit pipeline to surface the cost of untagged resources explicitly and make it visible to leadership; financial visibility is the most reliable forcing function for ownership.
 - **Enforce at creation time, audit continuously, and remediate through pipeline gates.** Cloud-native policies (AWS SCPs, Azure Policy, GCP Organization Policy custom constraints) block noncompliant resources at the API level. IaC validation (Terraform variable validation, OPA/Conftest in CI) catches problems before API calls are made. The weekly audit Lambda catches drift between deployments. Each layer compensates for the gaps of the others.
 - **Multi-cloud tag normalization belongs in the data warehouse, not in each cloud's tagging convention.** You cannot retroactively harmonize `CostCenter`, `cost-center`, and `cost_center` across three clouds, but you can normalize them in a unified billing view. A tag normalization lookup table that maps raw values to canonical values prevents per-team cost views from fragmenting due to minor capitalization or delimiter inconsistencies.
 - **Showback precedes chargeback.** Begin by distributing read-only monthly reports that show teams what they consumed without moving budget. Once teams can see their costs clearly, they will identify optimization opportunities themselves. Introduce financial accountability through chargeback only after the tagging data quality is high enough that teams can trust—and if necessary, dispute—the numbers they are being charged.
-

Chapter 24: Per-Team, Per-App, Per-Feature Cost Reporting

Organizations that implement granular cost reporting—breaking cloud database spend down to the team, application, and feature level—consistently identify 20–35% in previously invisible waste within the first 90 days. The mechanism is straightforward: when a team can see that their recommendation engine’s nightly batch job costs \$4,200/month while the feature drives less than \$800/month in attributed revenue, the conversation about optimization practically has itself. This chapter covers how to build that attribution layer from scratch across AWS RDS, Azure SQL, and Google Cloud SQL, including the tagging taxonomy, the cost allocation queries, the dashboard pipelines, and the automated weekly reports that land in engineering leads’ inboxes without anyone having to run a manual export.

24.1 Building a Tagging Taxonomy That Actually Works

The failure mode for most tagging initiatives is starting with too many tags and enforcing none of them. Production environments end up with 40% of resources untagged, and the cost attribution reports are immediately discredited because the numbers don’t add up.

A workable minimum taxonomy for database cost attribution uses exactly five tags:

Tag Key	Example Values	Purpose
<code>team</code>	<code>payments</code> , <code>ml-platform</code> , <code>core-api</code>	Charge back to engineering team
<code>app</code>	<code>checkout-service</code> , <code>fraud-detector</code>	Charge back to application
<code>environment</code>	<code>prod</code> , <code>staging</code> , <code>dev</code>	Separate production from non-production spend
<code>feature</code>	<code>recommendations</code> , <code>search</code> , <code>auth</code>	Feature-level attribution
<code>cost-center</code>	<code>cc-1042</code> , <code>cc-2017</code>	Finance system integration

The `feature` tag is optional for initial rollout. Add it when teams are ready to instrument individual workloads. Forcing feature-level tagging prematurely creates inconsistent data that undermines trust in the entire system.

Enforcing tags at resource creation is non-negotiable. Resources that exist without tags accumulate cost in an unattributed bucket that grows until someone gets angry about it in a budget review.

```
# AWS: Enforce required tags using an SCP at the organization level
# Save as enforce-db-tags-scp.json and attach to target OUs

cat > enforce-db-tags-scp.json << 'EOF'
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyRDSWithoutRequiredTags",
      "Effect": "Deny",
      "Action": [
        "rds:CreateDBInstance",
        "rds:CreateDBCluster",
        "rds:CreateDBInstanceReadReplica"
      ],
      "Resource": "*",
      "Condition": {
        "Null": {
          "aws:RequestTag/team": "true"
        }
      }
    },
    {
      "Sid": "DenyRDSWithoutAppTag",
      "Effect": "Deny",
      "Action": [
        "rds:CreateDBInstance",
        "rds:CreateDBCluster"
      ],
      "Resource": "*",
      "Condition": {
        "Null": {
          "aws:RequestTag/app": "true"
        }
      }
    }
  ]
}
EOF

# Apply the SCP to an OU (requires AWS Organizations admin)
aws organizations create-policy \
  --name "RequireDBTags" \
  --description "Deny RDS creation without required cost tags" \
  --content file://enforce-db-tags-scp.json \
  --type SERVICE_CONTROL_POLICY

# Tag an existing RDS instance that is missing tags
aws rds add-tags-to-resource \
  --resource-name arn:aws:rds:us-east-1:123456789012:db:prod-payments-pg \
  --tags \
    Key=team,Value=payments \
    Key=app,Value=checkout-service \
    Key=environment,Value=prod \
    Key=cost-center,Value=cc-1042

# Audit: find all RDS instances missing the 'team' tag
aws resourcegroupstaggingapi get-resources \
  --resource-type-filters rds:db \
  --tag-filters Key=team \
  --query 'ResourceTagMappingList[?Tags[?Key==`team`] == `[]`].ResourceARN' \
  --output text
```

For Azure, Policy definitions enforce tags at the resource group or subscription level:

```
# Azure: Create a policy that requires the 'team' tag on SQL databases
az policy definition create \
  --name "require-team-tag-sql" \
  --display-name "Require team tag on Azure SQL databases" \
  --description "Denies creation of Azure SQL resources without a team tag" \
  --rules '{
    "if": {
      "allOf": [
        {
          "field": "type",
          "equals": "Microsoft.Sql/servers/databases"
        },
        {
          "field": "tags[team]",
          "exists": "false"
        }
      ]
    },
    "then": {
      "effect": "deny"
    }
  }' \
  --mode All

# Assign the policy to a subscription
az policy assignment create \
  --name "require-team-tag-sql-assignment" \
  --policy "require-team-tag-sql" \
  --scope "/subscriptions/YOUR_SUBSCRIPTION_ID"

# Tag an existing Azure SQL database
az sql db update \
  --resource-group prod-rg \
  --server prod-sql-server \
  --name payments-db \
  --set tags.team=payments tags.app=checkout-service tags.environment=prod
```

For GCP, labels on Cloud SQL instances serve the same function, and Organization Policy can enforce label requirements:

```
# GCP: Add labels to an existing Cloud SQL instance
gcloud sql instances patch prod-payments-pg \
  --update-labels=team=payments,app=checkout-service,environment=prod,cost-center=cc-1042

# List Cloud SQL instances missing the 'team' label
gcloud sql instances list \
  --filter="NOT labels.team:*" \
  --format="table(name,region,databaseVersion,settings.tier)"
```

Handling the untagged backlog is a common stumbling block. The pragmatic approach is to run a quarterly tagging sprint. Pull all untagged resources, assign them to account owners based on VPC or subnet membership, and have those owners tag within 30 days or face a chargeback to a shared pool. The shared pool penalty is the behavioral incentive that makes the tagging sprint work.

24.2 Extracting Per-Team Cost Data from Each Cloud Provider

Once tags are in place, the next step is extracting cost data in a form that can be joined, aggregated, and reported. Each cloud provider exposes this differently, and the approach that works across all three is to land the data in a common schema in your data warehouse or an S3/GCS/Azure Blob bucket.

AWS Cost Explorer with Tag Filtering

```
#!/usr/bin/env python3
"""
Pull per-team RDS costs from AWS Cost Explorer and write to a PostgreSQL
reporting database. Designed to run on a weekly schedule via Lambda or cron.

Requirements: boto3, psycpg2-binary, python-dateutil
"""

import boto3
import psycpg2
import json
from datetime import datetime, timedelta
from dateutil.relativedelta import relativedelta
import logging

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)

# Configuration
REPORTING_DB_CONN = "host=reporting-db.internal dbname=finops user=finops_writer password=REPLACE"
AWS_ACCOUNT_ID = "123456789012"
REGION = "us-east-1"

def get_rds_costs_by_team(start_date: str, end_date: str) -> list[dict]:
    """
    Query Cost Explorer for RDS costs grouped by team tag.
    Returns list of dicts with team, service, cost, and usage_type.
    """
    client = boto3.client("ce", region_name="us-east-1")

    response = client.get_cost_and_usage(
        TimePeriod={
            "Start": start_date,
            "End": end_date
        },
        Granularity="MONTHLY",
        Filter={
            "And": [
                {
                    "Dimensions": {
                        "Key": "SERVICE",
                        "Values": [
                            "Amazon Relational Database Service",
                            "Amazon Aurora"
                        ]
                    }
                },
                {
                    "Not": {
                        "Dimensions": {
                            "Key": "RECORD_TYPE",
                            "Values": ["Credit", "Refund"]
                        }
                    }
                }
            ]
        },
        GroupBy=[
            {"Type": "TAG", "Key": "team"},
            {"Type": "TAG", "Key": "app"},
            {"Type": "DIMENSION", "Key": "USAGE_TYPE"}
        ],
        Metrics=["UnblendedCost", "UsageQuantity"]
    )

    results = []
    for time_period in response["ResultsByTime"]:
        period_start = time_period["TimePeriod"]["Start"]
```

```

for group in time_period["Groups"]:
    keys = group["Keys"]
    # Keys order: team tag, app tag, usage_type
    team_raw = keys[0] if len(keys) > 0 else ""
    app_raw = keys[1] if len(keys) > 1 else ""
    usage_type = keys[2] if len(keys) > 2 else ""

    # Strip the tag key prefix that Cost Explorer adds
    team = team_raw.replace("team$", "") or "untagged"
    app = app_raw.replace("app$", "") or "untagged"

    cost = float(group["Metrics"]["UnblendedCost"]["Amount"])
    usage = float(group["Metrics"]["UsageQuantity"]["Amount"])
    unit = group["Metrics"]["UsageQuantity"]["Unit"]

    if cost > 0:
        results.append({
            "period_start": period_start,
            "team": team,
            "app": app,
            "usage_type": usage_type,
            "cost_usd": round(cost, 4),
            "usage_quantity": round(usage, 4),
            "usage_unit": unit,
            "account_id": AWS_ACCOUNT_ID,
            "extracted_at": datetime.utcnow().isoformat()
        })

logger.info(f"Extracted {len(results)} cost records for period {start_date} to {end_date}")
return results

def get_rds_costs_by_feature(start_date: str, end_date: str) -> list[dict]:
    """
    Query Cost Explorer for RDS costs grouped by feature tag.
    Use this for feature-level attribution where teams have instrumented
    their workloads with the 'feature' tag.
    """
    client = boto3.client("ce", region_name="us-east-1")

    response = client.get_cost_and_usage(
        TimePeriod={"Start": start_date, "End": end_date},
        Granularity="MONTHLY",
        Filter={
            "Dimensions": {
                "Key": "SERVICE",
                "Values": ["Amazon Relational Database Service", "Amazon Aurora"]
            }
        },
        GroupBy=[
            {"Type": "TAG", "Key": "team"},
            {"Type": "TAG", "Key": "feature"},
        ],
        Metrics=["UnblendedCost"]
    )

    results = []
    for time_period in response["ResultsByTime"]:
        period_start = time_period["TimePeriod"]["Start"]
        for group in time_period["Groups"]:
            keys = group["Keys"]
            team = keys[0].replace("team$", "") or "untagged"
            feature = keys[1].replace("feature$", "") or "untagged"
            cost = float(group["Metrics"]["UnblendedCost"]["Amount"])

            if cost > 0:
                results.append({
                    "period_start": period_start,
                    "team": team,
                    "feature": feature,

```

```

        "cost_usd": round(cost, 4),
        "account_id": AWS_ACCOUNT_ID
    })

    return results

def upsert_team_costs(records: list[dict]) -> None:
    """Write team cost records to the reporting database."""
    if not records:
        logger.warning("No records to write")
        return

    conn = psycopg2.connect(REPORTING_DB_CONN)
    cur = conn.cursor()

    # Ensure the table exists
    cur.execute("""
        CREATE TABLE IF NOT EXISTS db_team_costs (
            id SERIAL PRIMARY KEY,
            period_start DATE NOT NULL,
            team TEXT NOT NULL,
            app TEXT NOT NULL,
            usage_type TEXT NOT NULL,
            cost_usd NUMERIC(12,4) NOT NULL,
            usage_quantity NUMERIC(16,4),
            usage_unit TEXT,
            account_id TEXT,
            extracted_at TIMESTAMPTZ DEFAULT NOW(),
            UNIQUE (period_start, team, app, usage_type, account_id)
        )
    """)

    upsert_sql = """
        INSERT INTO db_team_costs
        (period_start, team, app, usage_type, cost_usd, usage_quantity, usage_unit, account_id,
↪  extracted_at)
        VALUES
            (%(period_start)s, %(team)s, %(app)s, %(usage_type)s, %(cost_usd)s,
            %(usage_quantity)s, %(usage_unit)s, %(account_id)s, %(extracted_at)s)
        ON CONFLICT (period_start, team, app, usage_type, account_id)
        DO UPDATE SET
            cost_usd = EXCLUDED.cost_usd,
            usage_quantity = EXCLUDED.usage_quantity,
            extracted_at = EXCLUDED.extracted_at
    """

    cur.executemany(upsert_sql, records)
    conn.commit()
    cur.close()
    conn.close()
    logger.info(f"Upserted {len(records)} records into db_team_costs")

if __name__ == "__main__":
    # Pull the previous full month
    today = datetime.today()
    first_of_this_month = today.replace(day=1)
    first_of_last_month = first_of_this_month - relativedelta(months=1)

    start = first_of_last_month.strftime("%Y-%m-%d")
    end = first_of_this_month.strftime("%Y-%m-%d")

    logger.info(f"Pulling RDS costs from {start} to {end}")

    team_records = get_rds_costs_by_team(start, end)
    upsert_team_costs(team_records)

    feature_records = get_rds_costs_by_feature(start, end)
    # Feature costs go into a separate table with the same pattern

```

```
logger.info(f"Feature-level records: {len(feature_records)}")
```

Azure Cost Management per-team extraction:

```
# Pull Azure SQL costs grouped by team tag for the previous month
# Requires az billing or az costmanagement access

START_DATE=$(date -d "$(date +%Y-%m-01) -1 month" +%Y-%m-01)
END_DATE=$(date +%Y-%m-01)

az costmanagement query \
  --scope "/subscriptions/YOUR_SUBSCRIPTION_ID" \
  --type Usage \
  --timeframe Custom \
  --time-period start="$START_DATE" end="$END_DATE" \
  --dataset-configuration columns='["Cost","UsageQuantity","ResourceType","Tags"]' \
  --dataset-filter '{
    "and": [
      {
        "dimensions": {
          "name": "ResourceType",
          "operator": "In",
          "values": [
            "microsoft.sql/servers/databases",
            "microsoft.sql/managedinstances",
            "microsoft.dbforpostgresql/flexibleservers"
          ]
        }
      }
    ]
  }' \
  --dataset-grouping \
  name=team type=TagKey \
  name=app type=TagKey \
  name=ResourceType type=Dimension \
  --output json > azure_sql_costs_by_team.json

# Parse and summarize by team
cat azure_sql_costs_by_team.json | python3 -c "
import json, sys
data = json.load(sys.stdin)
rows = data.get('properties', {}).get('rows', [])
cols = data.get('properties', {}).get('columns', [])
col_names = [c['name'] for c in cols]
print(','.join(col_names))
for row in rows:
    print(','.join(str(x) for x in row))
" | sort -t',' -k1,1 -k3,3n -r | head -50

# GCP: Pull Cloud SQL costs from the billing export in BigQuery
# (Assumes billing export is configured to a BigQuery dataset)
bq query --use_legacy_sql=false --format=csv '
SELECT
  SUBSTR(usage_start_time, 1, 7) AS month,
  labels.value AS team,
  SUM(cost) AS total_cost_usd,
  SUM(usage.amount) AS total_usage,
  usage.unit
FROM `YOUR_PROJECT.billing_export.gcp_billing_export_v1_XXXXXX`
, UNNEST(labels) AS labels
WHERE
  service.description LIKE "%Cloud SQL%"
  AND labels.key = "team"
  AND usage_start_time >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 90 DAY)
GROUP BY 1, 2, 4, 5
ORDER BY 1 DESC, 3 DESC
' > gcp_cloudsql_by_team.csv

echo "GCP Cloud SQL costs exported to gcp_cloudsql_by_team.csv"
```

24.3 Database-Level Attribution with pg_stat_statements

Tags cover the infrastructure cost—instance hours, storage, I/O. But they cannot tell you which feature’s queries are consuming 60% of your Aurora cluster’s CPU. For that, you need query-level attribution using `pg_stat_statements` on PostgreSQL, or `sys.dm_exec_query_stats` on SQL Server. This is the layer that connects infrastructure cost to application behavior.

The pattern is to add a comment to every query at the application layer that embeds team and feature metadata. Query planners ignore SQL comments, but `pg_stat_statements` captures them, so they appear in the `query` column and can be parsed out.

Application code wraps queries like this:

```
-- Pattern: prefix queries with a structured comment
-- /* app:checkout-service team:payments feature:cart-totals */
SELECT p.product_id, p.price, i.quantity
FROM products p
JOIN inventory i ON p.product_id = i.product_id
/* app:checkout-service team:payments feature:cart-totals */
WHERE p.product_id = ANY($1);
```

With that convention in place, extract per-feature query costs from PostgreSQL:

```
-- Per-feature query cost attribution using pg_stat_statements
-- Run on the PostgreSQL instance (works on RDS, Aurora, Cloud SQL, Azure Flexible Server)
-- Requires pg_stat_statements to be enabled in shared_preload_libraries

-- Step 1: Extract query metadata and compute a cost proxy score
-- Cost proxy = (total_exec_time * calls) weighted by shared_blks_hit and shared_blks_read
-- This is not a dollar amount but a relative CPU/IO weight useful for attribution

WITH parsed_queries AS (
  SELECT
    queryid,
    query,
    calls,
    total_exec_time,
    mean_exec_time,
    rows,
    shared_blks_hit,
    shared_blks_read,
    shared_blks_written,
    local_blks_read,
    temp_blks_written,
    -- Extract team from comment pattern /* team:X */
    COALESCE(
      (regexp_match(query, '/\[^\]*team:(\S+)[^\]*\*/'))[1],
      'untagged'
    ) AS team,
    -- Extract app from comment pattern /* app:X */
    COALESCE(
      (regexp_match(query, '/\[^\]*app:(\S+)[^\]*\*/'))[1],
      'untagged'
    ) AS app,
    -- Extract feature from comment pattern /* feature:X */
    COALESCE(
      (regexp_match(query, '/\[^\]*feature:(\S+)[^\]*\*/'))[1],
      'untagged'
    ) AS feature,
    -- Compute a cost proxy: CPU time + IO weight
    -- Tune the IO multiplier based on your storage tier cost ratio
    (total_exec_time + (shared_blks_read * 8.0)) AS cost_proxy
  FROM pg_stat_statements
```



```

    WHERE query NOT LIKE '%pg_stat%' -- exclude meta-queries
),
totals AS (
    SELECT SUM(cost_proxy) AS total_cost_proxy FROM parsed_queries
),
team_attribution AS (
    SELECT
        team,
        app,
        feature,
        COUNT(*) AS distinct_query_count,
        SUM(calls) AS total_calls,
        ROUND(SUM(total_exec_time)::numeric / 1000, 2) AS total_exec_seconds,
        ROUND(AVG(mean_exec_time)::numeric, 2) AS avg_exec_ms,
        SUM(shared_blks_read) AS total_blks_read,
        SUM(shared_blks_hit) AS total_blks_hit,
        ROUND(SUM(cost_proxy)::numeric, 2) AS team_cost_proxy,
        ROUND(
            (SUM(cost_proxy) / NULLIF((SELECT total_cost_proxy FROM totals), 0) * 100)::numeric,
            2
        ) AS pct_of_total_db_cost
    FROM parsed_queries
    GROUP BY team, app, feature
)
SELECT
    team,
    app,
    feature,
    distinct_query_count,
    total_calls,
    total_exec_seconds,
    avg_exec_ms,
    total_blks_read,
    team_cost_proxy,
    pct_of_total_db_cost,
    -- If your monthly RDS bill for this instance is known, multiply by pct
    -- to get an estimated dollar attribution per team/feature
    ROUND((pct_of_total_db_cost / 100.0 * 3200.00)::numeric, 2) AS estimated_monthly_usd
    -- Replace 3200.00 with the actual monthly cost of this RDS instance
FROM team_attribution
ORDER BY pct_of_total_db_cost DESC;

-- Step 2: Find the top 10 most expensive queries per feature (for optimization targeting)
SELECT
    COALESCE(
        (regexp_match(query, '/\[^\]*feature:(\S+)\[^\]*\[/\])[1],
        'untagged'
    ) AS feature,
    COALESCE(
        (regexp_match(query, '/\[^\]*team:(\S+)\[^\]*\[/\])[1],
        'untagged'
    ) AS team,
    calls,
    ROUND((total_exec_time / 1000)::numeric, 2) AS total_exec_seconds,
    ROUND(mean_exec_time::numeric, 2) AS mean_exec_ms,
    shared_blks_read AS blks_read,
    -- Fingerprint: first 120 chars of normalized query
    LEFT(regexp_replace(query, '\s+', ' ', 'g'), 120) AS query_fingerprint
FROM pg_stat_statements
WHERE query NOT LIKE '%pg_stat%'
ORDER BY total_exec_time DESC
LIMIT 10;

```

The equivalent attribution query for SQL Server uses `sys.dm_exec_query_stats` paired with `sys.dm_exec_sql_text` to retrieve the query text, then parses the embedded comment:

```

-- SQL Server: Per-team query cost attribution using dm_exec_query_stats
-- Works on RDS for SQL Server, Azure SQL Database, and SQL Server on GCE

```

```

WITH query_text AS (
    SELECT
        qs.sql_handle,
        qs.query_hash,
        qs.total_elapsed_time,           -- microseconds
        qs.execution_count,
        qs.total_logical_reads,
        qs.total_logical_writes,
        qs.total_worker_time,           -- CPU microseconds
        qs.total_rows,
        SUBSTRING(
            st.text,
            (qs.statement_start_offset / 2) + 1,
            ((CASE qs.statement_end_offset
                WHEN -1 THEN DATALENGTH(st.text)
                ELSE qs.statement_end_offset
            END - qs.statement_start_offset) / 2) + 1
        ) AS statement_text
    FROM sys.dm_exec_query_stats AS qs
    CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS st
    WHERE st.text NOT LIKE '%dm_exec%'
),
parsed AS (
    SELECT
        query_hash,
        total_elapsed_time,
        execution_count,
        total_logical_reads,
        total_worker_time,
        -- Extract team tag from comment pattern /* team:X */
        COALESCE(
            (SELECT TOP 1 m.match
             FROM (
                 SELECT value AS match
                 FROM STRING_SPLIT(statement_text, ' ')
             ) m
             WHERE m.match LIKE 'team:%'
            ),
            'untagged'
        ) AS team_raw,
        -- Simpler pattern: use PATINDEX + SUBSTRING for SQL Server compatibility
        CASE
            WHEN statement_text LIKE '%team:%'
            THEN SUBSTRING(
                statement_text,
                PATINDEX('%team:%', statement_text) + 5,
                CHARINDEX(' ', statement_text + ' ',
                    PATINDEX('%team:%', statement_text) + 5)
                - PATINDEX('%team:%', statement_text) - 5
            )
            ELSE 'untagged'
        END AS team,
        CASE
            WHEN statement_text LIKE '%feature:%'
            THEN SUBSTRING(
                statement_text,
                PATINDEX('%feature:%', statement_text) + 8,
                CHARINDEX(' ', statement_text + ' ',
                    PATINDEX('%feature:%', statement_text) + 8)
                - PATINDEX('%feature:%', statement_text) - 8
            )
            ELSE 'untagged'
        END AS feature,
        CASE
            WHEN statement_text LIKE '%app:%'
            THEN SUBSTRING(
                statement_text,
                PATINDEX('%app:%', statement_text) + 4,
                CHARINDEX(' ', statement_text + ' ',

```

```

        PATINDEX('%app:%', statement_text) + 4)
      - PATINDEX('%app:%', statement_text) - 4
    )
    ELSE 'untagged'
  END AS app,
  -- Cost proxy: CPU time + weighted logical reads
  (total_worker_time + (total_logical_reads * 8.0)) AS cost_proxy,
  LEFT(LTRIM(RTRIM(REPLACE(statement_text, CHAR(10), ' '))), 120) AS query_fingerprint
FROM query_text
),
totals AS (
  SELECT SUM(cost_proxy) AS total_proxy FROM parsed
),
attribution AS (
  SELECT
    team,
    app,
    feature,
    COUNT(DISTINCT query_hash) AS distinct_queries,
    SUM(execution_count) AS total_executions,
    ROUND(SUM(total_elapsed_time) / 1e6, 2) AS total_elapsed_seconds,
    ROUND(SUM(total_worker_time) / 1e6, 2) AS total_cpu_seconds,
    SUM(total_logical_reads) AS total_logical_reads,
    ROUND(SUM(cost_proxy), 2) AS team_cost_proxy,
    ROUND(
      SUM(cost_proxy) / NULLIF((SELECT total_proxy FROM totals), 0) * 100,
      2
    ) AS pct_of_total
  FROM parsed
  GROUP BY team, app, feature
)
SELECT
  team,
  app,
  feature,
  distinct_queries,
  total_executions,
  total_elapsed_seconds,
  total_cpu_seconds,
  total_logical_reads,
  team_cost_proxy,
  pct_of_total,
  -- Replace 4800.00 with actual monthly cost of this SQL Server instance
  ROUND(pct_of_total / 100.0 * 4800.00, 2) AS estimated_monthly_usd
FROM attribution
ORDER BY pct_of_total DESC;

```

The `estimated_monthly_usd` column is only as accurate as the base cost you supply. Pull the actual monthly instance cost from the billing extraction scripts in section 24.2 and substitute it here. When you join both layers—infrastructure billing data from the cloud provider and query-level attribution from `pg_stat_statements` or `dm_exec_query_stats`—you get a complete picture: the team’s total database spend, and which of their features and queries are responsible for it.

24.4 Building the Cost Attribution Schema

The billing extracts and query attribution data both need to land somewhere before they can feed a dashboard. A simple star schema in PostgreSQL handles this well and works whether your reporting database is on RDS, Cloud SQL, or a self-managed instance.

```

-- PostgreSQL: Create the cost attribution reporting schema
-- This schema stores both infrastructure costs (from billing APIs)
-- and query-level costs (from pg_stat_statements snapshots)

```

```

CREATE SCHEMA IF NOT EXISTS finops;

-- Dimension: teams
CREATE TABLE IF NOT EXISTS finops.dim_teams (
  team_id      SERIAL PRIMARY KEY,
  team_slug    TEXT NOT NULL UNIQUE, -- e.g. 'payments', 'ml-platform'
  team_name    TEXT NOT NULL,
  cost_center  TEXT,
  lead_email   TEXT,
  created_at   TIMESTAMPTZ DEFAULT NOW()
);

-- Dimension: applications
CREATE TABLE IF NOT EXISTS finops.dim_apps (
  app_id       SERIAL PRIMARY KEY,
  app_slug     TEXT NOT NULL UNIQUE, -- e.g. 'checkout-service'
  app_name     TEXT NOT NULL,
  team_id      INT REFERENCES finops.dim_teams(team_id),
  repo_url     TEXT,
  created_at   TIMESTAMPTZ DEFAULT NOW()
);

-- Dimension: features
CREATE TABLE IF NOT EXISTS finops.dim_features (
  feature_id   SERIAL PRIMARY KEY,
  feature_slug TEXT NOT NULL,
  app_id       INT REFERENCES finops.dim_apps(app_id),
  description  TEXT,
  created_at   TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE (feature_slug, app_id)
);

-- Dimension: database instances
CREATE TABLE IF NOT EXISTS finops.dim_db_instances (
  instance_id  SERIAL PRIMARY KEY,
  provider     TEXT NOT NULL CHECK (provider IN ('aws', 'azure', 'gcp')),
  account_id   TEXT, -- AWS account, Azure subscription, GCP project
  region       TEXT,
  instance_identifier TEXT NOT NULL,
  engine       TEXT, -- postgres, mysql, sqlserver, aurora-postgresql
  instance_class TEXT, -- db.r6g.2xlarge, etc.
  environment  TEXT, -- prod, staging, dev
  team_id      INT REFERENCES finops.dim_teams(team_id),
  app_id       INT REFERENCES finops.dim_apps(app_id),
  monthly_base_cost_usd NUMERIC(12,2), -- updated monthly from billing extract
  created_at   TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE (provider, account_id, instance_identifier)
);

-- Fact: infrastructure costs by team/app from cloud billing APIs
CREATE TABLE IF NOT EXISTS finops.fact_infra_costs (
  id           BIGSERIAL PRIMARY KEY,
  period_start DATE NOT NULL,
  period_end   DATE NOT NULL,
  provider     TEXT NOT NULL CHECK (provider IN ('aws', 'azure', 'gcp')),
  account_id   TEXT,
  team_id      INT REFERENCES finops.dim_teams(team_id),
  app_id       INT REFERENCES finops.dim_apps(app_id),
  instance_id  INT REFERENCES finops.dim_db_instances(instance_id),
  cost_category TEXT, -- ComputeUsage, StorageUsage, IOUsage, DataTransfer, Backup
  cost_usd     NUMERIC(14,4) NOT NULL,
  usage_quantity NUMERIC(16,4),
  usage_unit   TEXT,
  extracted_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE (period_start, provider, account_id, team_id, app_id, cost_category)
);

-- Fact: query-level cost attribution from pg_stat_statements snapshots
CREATE TABLE IF NOT EXISTS finops.fact_query_costs (

```

```

id                BIGSERIAL PRIMARY KEY,
snapshot_ts       TIMESTAMPTZ NOT NULL,
instance_id       INT REFERENCES finops.dim_db_instances(instance_id),
team_id           INT REFERENCES finops.dim_teams(team_id),
app_id            INT REFERENCES finops.dim_apps(app_id),
feature_id        INT REFERENCES finops.dim_features(feature_id),
queryid           BIGINT,           -- pg_stat_statements queryid
calls            BIGINT,
total_exec_seconds NUMERIC(14,4),
mean_exec_ms      NUMERIC(10,4),
total_blks_read   BIGINT,
total_blks_hit    BIGINT,
cost_proxy        NUMERIC(16,4),
pct_of_instance   NUMERIC(6,2),
estimated_cost_usd NUMERIC(12,4),
query_fingerprint TEXT,
created_at        TIMESTAMPTZ DEFAULT NOW()
);

-- Index the fact tables for the reporting queries that follow
CREATE INDEX IF NOT EXISTS idx_fact_infra_team_period
ON finops.fact_infra_costs (team_id, period_start);

CREATE INDEX IF NOT EXISTS idx_fact_infra_app_period
ON finops.fact_infra_costs (app_id, period_start);

CREATE INDEX IF NOT EXISTS idx_fact_query_instance_snapshot
ON finops.fact_query_costs (instance_id, snapshot_ts);

CREATE INDEX IF NOT EXISTS idx_fact_query_team_feature
ON finops.fact_query_costs (team_id, feature_id);

```

With that schema populated by the extraction scripts from section 24.2 and the `pg_stat_statements` snapshot pipeline from section 24.3, the reporting queries become straightforward.

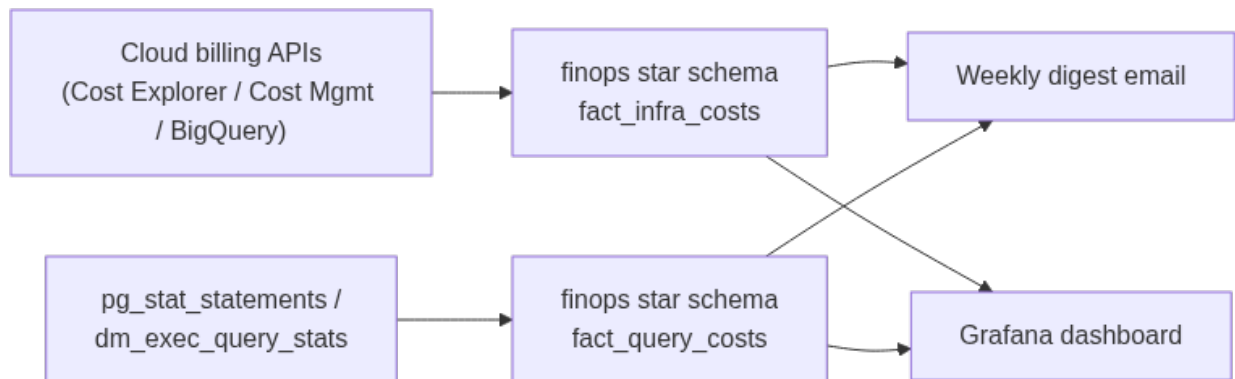


Figure 24: Two independent data sources feed the same reporting schema: infrastructure billing tells you what a team spent, query-level attribution tells you which features drove it.

```

-- PostgreSQL: Monthly infrastructure cost rollup per team with MoM trend
-- Run against the finops reporting database

WITH monthly_costs AS (
    SELECT
        DATE_TRUNC('month', fic.period_start) AS month,
        dt.team_slug,
        dt.team_name,
        dt.lead_email,
        SUM(fic.cost_usd) AS total_cost_usd
    FROM finops.fact_infra_costs fic
    JOIN finops.dim_teams dt ON fic.team_id = dt.team_id

```

```

WHERE fic.period_start >= DATE_TRUNC('month', NOW() - INTERVAL '3 months')
GROUP BY 1, 2, 3, 4
),
with_trend AS (
  SELECT
    month,
    team_slug,
    team_name,
    lead_email,
    total_cost_usd,
    LAG(total_cost_usd) OVER (
      PARTITION BY team_slug ORDER BY month
    ) AS prior_month_cost_usd
  FROM monthly_costs
)
SELECT
  month,
  team_slug,
  team_name,
  lead_email,
  ROUND(total_cost_usd, 2) AS total_cost_usd,
  ROUND(prior_month_cost_usd, 2) AS prior_month_cost_usd,
  ROUND(
    CASE
      WHEN prior_month_cost_usd IS NULL OR prior_month_cost_usd = 0 THEN NULL
      ELSE ((total_cost_usd - prior_month_cost_usd) / prior_month_cost_usd) * 100
    END, 1
  ) AS mom_change_pct
FROM with_trend
ORDER BY month DESC, total_cost_usd DESC;

-- SQL Server equivalent: Monthly infrastructure cost rollup per team with MoM trend
WITH monthly_costs AS (
  SELECT
    DATEADD(MONTH, DATEDIFF(MONTH, 0, fic.period_start), 0) AS month,
    dt.team_slug,
    dt.team_name,
    dt.lead_email,
    SUM(fic.cost_usd) AS total_cost_usd
  FROM finops.fact_infra_costs fic
  JOIN finops.dim_teams dt ON fic.team_id = dt.team_id
  WHERE fic.period_start >= DATEADD(MONTH, -3, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()), 0))
  GROUP BY
    DATEADD(MONTH, DATEDIFF(MONTH, 0, fic.period_start), 0),
    dt.team_slug,
    dt.team_name,
    dt.lead_email
),
with_trend AS (
  SELECT
    month,
    team_slug,
    team_name,
    lead_email,
    total_cost_usd,
    LAG(total_cost_usd) OVER (
      PARTITION BY team_slug ORDER BY month
    ) AS prior_month_cost_usd
  FROM monthly_costs
)
SELECT
  month,
  team_slug,
  team_name,
  lead_email,
  ROUND(total_cost_usd, 2) AS total_cost_usd,
  ROUND(prior_month_cost_usd, 2) AS prior_month_cost_usd,
  ROUND(
    CASE

```

```

        WHEN prior_month_cost_usd IS NULL OR prior_month_cost_usd = 0 THEN NULL
        ELSE ((total_cost_usd - prior_month_cost_usd) / prior_month_cost_usd) * 100.0
    END, 1
)
AS mom_change_pct
FROM with_trend
ORDER BY month DESC, total_cost_usd DESC;

```

24.5 Automated Weekly Cost Digest Emails

The most effective forcing function for cost awareness is a weekly email that lands in an engineering lead's inbox without requiring any action on their part. The email answers three questions: what did my team spend on databases this week, how does that compare to last week, and which specific queries or instances are driving the biggest costs. The goal is to surface the information at the moment it is most actionable—before it becomes a budget overrun problem at the end of the month.

```

#!/usr/bin/env python3
"""
Weekly database cost digest email generator.
Queries the finops reporting database, builds an HTML email per team lead,
and sends via SES (AWS), SendGrid, or SMTP.

Schedule: Run every Monday at 08:00 local time via EventBridge, Cloud Scheduler,
or a cron job on a management host.

Requirements: psycpg2-binary, boto3 (for SES), jinja2
"""

import psycpg2
import psycpg2.extras
import boto3
import logging
from datetime import datetime, timedelta
from jinja2 import Template

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)

REPORTING_DB_CONN = "host=reporting-db.internal dbname=finops user=finops_reader password=REPLACE"
SES_REGION = "us-east-1"
FROM_EMAIL = "finops-noreply@yourcompany.com"
CURRENCY_SYMBOL = "$"

TEAM_WEEKLY_SQL = """
WITH current_week AS (
    SELECT
        dt.team_id,
        dt.team_slug,
        dt.team_name,
        dt.lead_email,
        COALESCE(SUM(fic.cost_usd), 0) AS week_cost
    FROM finops.dim_teams dt
    LEFT JOIN finops.fact_infra_costs fic
        ON dt.team_id = fic.team_id
        AND fic.period_start >= %(week_start)s
        AND fic.period_start < %(week_end)s
    GROUP BY dt.team_id, dt.team_slug, dt.team_name, dt.lead_email
),
prior_week AS (
    SELECT
        dt.team_id,
        COALESCE(SUM(fic.cost_usd), 0) AS prior_week_cost
    FROM finops.dim_teams dt

```

```

LEFT JOIN finops.fact_infra_costs fic
  ON dt.team_id = fic.team_id
  AND fic.period_start >= %(prior_week_start)s
  AND fic.period_start < %(prior_week_end)s
GROUP BY dt.team_id
)
SELECT
  cw.team_slug,
  cw.team_name,
  cw.lead_email,
  ROUND(cw.week_cost::numeric, 2) AS week_cost,
  ROUND(pw.prior_week_cost::numeric, 2) AS prior_week_cost,
  ROUND(
    CASE
      WHEN pw.prior_week_cost = 0 THEN NULL
      ELSE ((cw.week_cost - pw.prior_week_cost) / pw.prior_week_cost * 100)
    END::numeric, 1
  ) AS wow_change_pct
FROM current_week cw
JOIN prior_week pw ON cw.team_id = pw.team_id
WHERE cw.lead_email IS NOT NULL
ORDER BY cw.week_cost DESC;
"""

```

```

TOP_QUERIES_SQL = """
SELECT
  df.feature_slug,
  fqc.query_fingerprint,
  ROUND(SUM(fqc.estimated_cost_usd)::numeric, 2) AS est_cost_usd,
  SUM(fqc.calls) AS total_calls,
  ROUND(AVG(fqc.mean_exec_ms)::numeric, 2) AS avg_exec_ms
FROM finops.fact_query_costs fqc
JOIN finops.dim_teams dt ON fqc.team_id = dt.team_id
LEFT JOIN finops.dim_features df ON fqc.feature_id = df.feature_id
WHERE dt.team_slug = %(team_slug)s
  AND fqc.snapshot_ts >= %(week_start)s
  AND fqc.snapshot_ts < %(week_end)s
GROUP BY df.feature_slug, fqc.query_fingerprint
ORDER BY est_cost_usd DESC
LIMIT 5;
"""

```

```

TOP_INSTANCES_SQL = """
SELECT
  ddi.instance_identifier,
  ddi.engine,
  ddi.instance_class,
  ddi.environment,
  ROUND(SUM(fic.cost_usd)::numeric, 2) AS week_cost
FROM finops.fact_infra_costs fic
JOIN finops.dim_db_instances ddi ON fic.instance_id = ddi.instance_id
JOIN finops.dim_teams dt ON fic.team_id = dt.team_id
WHERE dt.team_slug = %(team_slug)s
  AND fic.period_start >= %(week_start)s
  AND fic.period_start < %(week_end)s
GROUP BY ddi.instance_identifier, ddi.engine, ddi.instance_class, ddi.environment
ORDER BY week_cost DESC
LIMIT 5;
"""

```

```

EMAIL_TEMPLATE = """
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<style>
  body { font-family: Arial, sans-serif; font-size: 14px; color: #333; }
  h1 { color: #1a56db; font-size: 20px; }
  h2 { color: #374151; font-size: 16px; margin-top: 24px; }
  table { border-collapse: collapse; width: 100%; margin-top: 8px; }

```



```

th { background: #f3f4f6; text-align: left; padding: 8px 12px; border: 1px solid #e5e7eb; font-size:
↪ 12px; }
td { padding: 8px 12px; border: 1px solid #e5e7eb; font-size: 13px; }
.up { color: #dc2626; font-weight: bold; }
.down { color: #16a34a; font-weight: bold; }
.flat { color: #6b7280; }
.hero { font-size: 28px; font-weight: bold; color: #1a56db; }
.meta { font-size: 12px; color: #6b7280; margin-bottom: 24px; }
.footer { margin-top: 32px; font-size: 11px; color: #9ca3af; }
</style>
</head>
<body>
<h1>Database Cost Digest - {{ team_name }}</h1>
<p class="meta">Week of {{ week_start }} · Generated {{ generated_at }}</p>

<p>
  This week your team spent
  <span class="hero">{{ currency }}{{ week_cost }}</span>
  on cloud databases.
  {% if wow_change_pct is not none %}
    {% if wow_change_pct > 5 %}
      <span class="up"> {{ wow_change_pct }}% vs last week</span>
    {% elif wow_change_pct < -5 %}
      <span class="down"> {{ wow_change_pct|abs }}% vs last week</span>
    {% else %}
      <span class="flat"> flat vs last week ({{ wow_change_pct }}%)</span>
    {% endif %}
  {% endif %}
</p>

{% if top_instances %}
<h2>Top Instances by Cost This Week</h2>
<table>
  <tr>
    <th>Instance</th><th>Engine</th><th>Class</th><th>Env</th><th>Cost</th>
  </tr>
  {% for inst in top_instances %}
  <tr>
    <td>{{ inst.instance_idenfier }}</td>
    <td>{{ inst.engine }}</td>
    <td>{{ inst.instance_class }}</td>
    <td>{{ inst.environment }}</td>
    <td>{{ currency }}{{ inst.week_cost }}</td>
  </tr>
  {% endfor %}
</table>
{% endif %}

{% if top_queries %}
<h2>Top Queries by Estimated Cost This Week</h2>
<table>
  <tr>
    <th>Feature</th><th>Query (truncated)</th><th>Est. Cost</th>
    <th>Total Calls</th><th>Avg ms</th>
  </tr>
  {% for q in top_queries %}
  <tr>
    <td>{{ q.feature_slug or 'untagged' }}</td>
    <td style="font-family:monospace;font-size:11px;">{{ q.query_fingerprint }}</td>
    <td>{{ currency }}{{ q.est_cost_usd }}</td>
    <td>{{ "{:,}".format(q.total_calls) }}</td>
    <td>{{ q.avg_exec_ms }}</td>
  </tr>
  {% endfor %}
</table>
{% endif %}

<p style="margin-top:24px;">
  <a href="https://finops.yourcompany.internal/dashboard?team={{ team_slug }}">
    View full dashboard ↗

```

```

    </a>
</p>

<div class="footer">
    This report was generated automatically by the FinOps platform.
    Tag questions or attribution corrections: #finops-help in Slack.
</div>
</body>
</html>
"""

def send_ses_email(to_address: str, subject: str, html_body: str) -> None:
    client = boto3.client("ses", region_name=SES_REGION)
    client.send_email(
        Source=FROM_EMAIL,
        Destination={"ToAddresses": [to_address]},
        Message={
            "Subject": {"Data": subject, "Charset": "UTF-8"},
            "Body": {"Html": {"Data": html_body, "Charset": "UTF-8"}}
        }
    )
    logger.info(f"Email sent to {to_address}")

def run_weekly_digest() -> None:
    today = datetime.utcnow().date()
    # Week runs Mon-Sun; send on Monday covering the prior Mon-Sun
    week_end = today - timedelta(days=today.weekday()) # this Monday
    week_start = week_end - timedelta(days=7) # last Monday
    prior_week_start = week_start - timedelta(days=7)
    prior_week_end = week_start

    params = {
        "week_start": week_start.isoformat(),
        "week_end": week_end.isoformat(),
        "prior_week_start": prior_week_start.isoformat(),
        "prior_week_end": prior_week_end.isoformat(),
    }

    conn = psycopg2.connect(REPORTING_DB_CONN)
    cur = conn.cursor(cursor_factory=psycopg2.extras.RealDictCursor)

    cur.execute(Team_WEEKLY_SQL, params)
    teams = cur.fetchall()
    logger.info(f"Sending digests to {len(teams)} team leads")

    template = Template(EMAIL_TEMPLATE)
    generated_at = datetime.utcnow().strftime("%Y-%m-%d %H:%M UTC")

    for team in teams:
        # Fetch top instances for this team
        cur.execute(TOP_INSTANCES_SQL, {
            "team_slug": team["team_slug"],
            "week_start": week_start.isoformat(),
            "week_end": week_end.isoformat(),
        })
        top_instances = cur.fetchall()

        # Fetch top queries for this team
        cur.execute(TOP_QUERIES_SQL, {
            "team_slug": team["team_slug"],
            "week_start": week_start.isoformat(),
            "week_end": week_end.isoformat(),
        })
        top_queries = cur.fetchall()

        html_body = template.render(
            team_name=team["team_name"],
            team_slug=team["team_slug"],

```

```

        week_start=week_start.isoformat(),
        generated_at=generated_at,
        week_cost=team["week_cost"],
        wow_change_pct=team["wow_change_pct"],
        top_instances=top_instances,
        top_queries=top_queries,
        currency=CURRENCY_SYMBOL,
    )

    subject = (
        f"DB Cost Digest - {team['team_name']} - "
        f"${team['week_cost']} this week"
    )
    send_ses_email(team["lead_email"], subject, html_body)

    cur.close()
    conn.close()
    logger.info("Weekly digest run complete")

if __name__ == "__main__":
    run_weekly_digest()

```

Schedule this script using EventBridge Scheduler on AWS, Cloud Scheduler on GCP, or Azure Logic Apps. The simplest deployment is a container image pushed to ECR or Artifact Registry, triggered on a cron expression every Monday morning.

```

# AWS: Schedule the digest Lambda every Monday at 08:00 UTC
aws events put-rule \
  --name "weekly-db-cost-digest" \
  --schedule-expression "cron(0 8 ? * MON *)" \
  --state ENABLED

# GCP: Schedule via Cloud Scheduler targeting a Cloud Run job
gcloud scheduler jobs create http weekly-db-cost-digest \
  --location us-central1 \
  --schedule "0 8 * * 1" \
  --uri
  ↪ "https://us-central1-run.googleapis.com/apis/run.googleapis.com/v1/namespaces/YOUR_PROJECT/jobs/db-cost-digest:run"
  ↪ \
  --message-body '{}' \
  --oauth-service-account-email finops-scheduler@YOUR_PROJECT.iam.gserviceaccount.com

# Azure: Logic App trigger (simplified CLI representation)
# In practice, create via ARM template or portal for Logic Apps
az logic workflow create \
  --resource-group finops-rg \
  --name weekly-db-cost-digest \
  --definition @digest-workflow-definition.json

```

24.6 Grafana Dashboard for Per-Team Cost Visibility

The weekly email is the push mechanism. The Grafana dashboard is the pull mechanism—what engineering leads open when they want to investigate a spike or prepare for a budget conversation. The dashboard should answer five questions without clicking through multiple panels: how much is my team spending this month, what is the trend over the past 90 days, which instances are consuming most of the budget, which features are the most expensive at the query level, and which queries have degraded in performance or cost since last week.

The following panel definitions assume the finops PostgreSQL reporting database defined in section 24.4 as a Grafana data source named `finops-db`.

```

-- Grafana Panel 1: Total monthly spend by team (bar chart, last 6 months)
-- Variable: $team (multi-value, populated from dim_teams.team_slug)

SELECT
    DATE_TRUNC('month', period_start) AS time,
    SUM(cost_usd)                      AS "Database Cost (USD)"
FROM finops.fact_infra_costs fic
JOIN finops.dim_teams dt ON fic.team_id = dt.team_id
WHERE dt.team_slug = ANY(ARRAY[$team]::text[])
    AND period_start >= $__timeFrom()::date
    AND period_start < $__timeTo()::date
GROUP BY 1
ORDER BY 1;

-- Grafana Panel 2: Cost breakdown by cost_category (stacked bar)
-- Shows ComputeUsage vs StorageUsage vs IOUsage vs Backup

SELECT
    DATE_TRUNC('month', period_start) AS time,
    cost_category                      AS metric,
    SUM(cost_usd)                      AS value
FROM finops.fact_infra_costs fic
JOIN finops.dim_teams dt ON fic.team_id = dt.team_id
WHERE dt.team_slug = ANY(ARRAY[$team]::text[])
    AND period_start >= $__timeFrom()::date
    AND period_start < $__timeTo()::date
GROUP BY 1, 2
ORDER BY 1, 2;

-- Grafana Panel 3: Top 10 most expensive instances for this team (table panel)

SELECT
    ddi.instance_identifer,
    ddi.engine,
    ddi.instance_class,
    ddi.environment,
    ROUND(SUM(fic.cost_usd)::numeric, 2) AS total_cost_usd
FROM finops.fact_infra_costs fic
JOIN finops.dim_teams dt ON fic.team_id = dt.team_id
JOIN finops.dim_db_instances ddi ON fic.instance_id = ddi.instance_id
WHERE dt.team_slug = ANY(ARRAY[$team]::text[])
    AND period_start >= $__timeFrom()::date
    AND period_start < $__timeTo()::date
GROUP BY ddi.instance_identifer, ddi.engine, ddi.instance_class, ddi.environment
ORDER BY total_cost_usd DESC
LIMIT 10;

-- Grafana Panel 4: Feature-level query cost breakdown (pie chart or bar)

SELECT
    COALESCE(df.feature_slug, 'untagged') AS feature,
    ROUND(SUM(fqc.estimated_cost_usd)::numeric, 2) AS estimated_cost_usd
FROM finops.fact_query_costs fqc
JOIN finops.dim_teams dt ON fqc.team_id = dt.team_id
LEFT JOIN finops.dim_features df ON fqc.feature_id = df.feature_id
WHERE dt.team_slug = ANY(ARRAY[$team]::text[])
    AND fqc.snapshot_ts >= $__timeFrom()
    AND fqc.snapshot_ts < $__timeTo()
GROUP BY 1
ORDER BY 2 DESC
LIMIT 10;

-- Grafana Panel 5: Week-over-week query cost change (table with threshold coloring)
-- Highlight queries that have increased cost by more than 20%

WITH this_week AS (

```

```

SELECT
  fqc.queryid,
  fqc.query_fingerprint,
  COALESCE(df.feature_slug, 'untagged') AS feature,
  SUM(fqc.estimated_cost_usd) AS cost_this_week,
  AVG(fqc.mean_exec_ms) AS avg_exec_ms_this_week
FROM finops.fact_query_costs fqc
JOIN finops.dim_teams dt ON fqc.team_id = dt.team_id
LEFT JOIN finops.dim_features df ON fqc.feature_id = df.feature_id
WHERE dt.team_slug = ANY(ARRAY[$team]::text[])
  AND fqc.snapshot_ts >= NOW() - INTERVAL '7 days'
GROUP BY fqc.queryid, fqc.query_fingerprint, feature
),
last_week AS (
  SELECT
    fqc.queryid,
    SUM(fqc.estimated_cost_usd) AS cost_last_week,
    AVG(fqc.mean_exec_ms) AS avg_exec_ms_last_week
  FROM finops.fact_query_costs fqc
  JOIN finops.dim_teams dt ON fqc.team_id = dt.team_id
  WHERE dt.team_slug = ANY(ARRAY[$team]::text[])
    AND fqc.snapshot_ts >= NOW() - INTERVAL '14 days'
    AND fqc.snapshot_ts < NOW() - INTERVAL '7 days'
  GROUP BY fqc.queryid
)
SELECT
  tw.feature,
  tw.query_fingerprint,
  ROUND(tw.cost_this_week::numeric, 4) AS cost_this_week_usd,
  ROUND(lw.cost_last_week::numeric, 4) AS cost_last_week_usd,
  ROUND(tw.avg_exec_ms_this_week::numeric, 2) AS avg_ms_this_week,
  ROUND(lw.avg_exec_ms_last_week::numeric, 2) AS avg_ms_last_week,
  ROUND(
    CASE
      WHEN lw.cost_last_week = 0 OR lw.cost_last_week IS NULL THEN NULL
      ELSE ((tw.cost_this_week - lw.cost_last_week) / lw.cost_last_week * 100)
    END::numeric, 1
  ) AS wow_cost_change_pct
FROM this_week tw
LEFT JOIN last_week lw ON tw.queryid = lw.queryid
ORDER BY cost_this_week_usd DESC
LIMIT 20;

```

For the week-over-week panel, set a Grafana threshold on `wow_cost_change_pct`: green below 10, yellow between 10 and 25, red above 25. This turns the table into a quick triage view—engineering leads can scan for red rows and immediately see which queries to hand to their team for optimization.

Dashboard variable configuration worth noting: the `$team` variable should be backed by this query against the `finops` database:

```

-- Grafana variable query: populate $team dropdown
SELECT DISTINCT team_slug AS __value, team_name AS __text
FROM finops.dim_teams
ORDER BY team_name;

```

Set it to multi-value and include an “All” option. When a FinOps analyst is reviewing cross-team spend, they select all teams; when a lead opens the link from their weekly digest email, the URL pre-selects their team slug via a `?var-team=payments` query parameter.

24.7 Handling Shared Infrastructure and Allocation Splits

Not all database costs map cleanly to a single team. A shared Aurora cluster used by three services, a central Redis-compatible ElastiCache cluster, or a data warehouse instance accessed by both the analytics team and the product team all present the same challenge: the billing tag can only carry one value, but the cost needs to be split.

There are three approaches, in order of increasing accuracy and implementation complexity:

- 1. Even split:** Divide the shared instance cost equally among all teams that use it. Easy to implement, defensible in team budget conversations, and almost certainly wrong—but consistently wrong in a way that teams can predict and plan around.
- 2. Weighted split by query execution time:** Use the query attribution data from section 24.3 to calculate what fraction of total database CPU each team consumed, then apply that fraction to the infrastructure cost. This requires the query comment tagging convention to be in place on the shared instance.
- 3. Weighted split by row access:** For workloads where CPU time is not a reliable proxy for cost (for example, large sequential scans that are cheap on CPU but expensive on storage I/O), weight by `shared_blks_read` from `pg_stat_statements` instead.

The following query implements option 2 (query-time weighted split) for a shared instance:

```
-- PostgreSQL: Compute the cost allocation split for a shared database instance
-- Replace 'shared-aurora-cluster-01' with your instance identifier
-- Replace 4200.00 with the actual monthly cost of the shared instance

WITH instance_query_costs AS (
    SELECT
        dt.team_slug,
        SUM(fqc.total_exec_seconds) AS team_exec_seconds,
        SUM(fqc.total_blks_read)   AS team_blks_read
    FROM finops.fact_query_costs fqc
    JOIN finops.dim_teams dt ON fqc.team_id = dt.team_id
    JOIN finops.dim_db_instances ddi ON fqc.instance_id = ddi.instance_id
    WHERE ddi.instance_identifier = 'shared-aurora-cluster-01'
        AND fqc.snapshot_ts >= DATE_TRUNC('month', NOW())
        AND fqc.snapshot_ts < DATE_TRUNC('month', NOW()) + INTERVAL '1 month'
    GROUP BY dt.team_slug
),
totals AS (
    SELECT
        SUM(team_exec_seconds) AS total_exec_seconds,
        SUM(team_blks_read)   AS total_blks_read
    FROM instance_query_costs
)
SELECT
    iqc.team_slug,
    ROUND(iqc.team_exec_seconds::numeric, 2) AS exec_seconds,
    ROUND(
        (iqc.team_exec_seconds / NULLIF(t.total_exec_seconds, 0) * 100)::numeric, 2
    ) AS pct_by_cpu,
    ROUND(
        (iqc.team_exec_seconds / NULLIF(t.total_exec_seconds, 0) * 4200.00)::numeric, 2
    ) AS allocated_cost_usd,
    ROUND(iqc.team_blks_read::numeric, 0) AS blks_read,
    ROUND(
        (iqc.team_blks_read / NULLIF(t.total_blks_read, 0) * 100)::numeric, 2
    ) AS pct_by_io
FROM instance_query_costs iqc
CROSS JOIN totals t
ORDER BY allocated_cost_usd DESC;
```

```

-- SQL Server equivalent: Shared instance cost allocation split
-- Replace 'shared-sql-instance-01' and 8600.00 with actual values

WITH instance_query_costs AS (
    SELECT
        dt.team_slug,
        SUM(fqc.total_exec_seconds) AS team_exec_seconds,
        SUM(fqc.total_blks_read) AS team_blks_read
    FROM finops.fact_query_costs fqc
    JOIN finops.dim_teams dt ON fqc.team_id = dt.team_id
    JOIN finops.dim_db_instances ddi ON fqc.instance_id = ddi.instance_id
    WHERE ddi.instance_identifier = 'shared-sql-instance-01'
        AND fqc.snapshot_ts >= DATEADD(DAY, 1 - DAY(GETDATE()), CAST(GETDATE() AS DATE))
        AND fqc.snapshot_ts < DATEADD(MONTH, 1,
            DATEADD(DAY, 1 - DAY(GETDATE()), CAST(GETDATE() AS DATE)))
    GROUP BY dt.team_slug
),
totals AS (
    SELECT
        SUM(team_exec_seconds) AS total_exec_seconds,
        SUM(team_blks_read) AS total_blks_read
    FROM instance_query_costs
)
SELECT
    iqc.team_slug,
    ROUND(CAST(iqc.team_exec_seconds AS NUMERIC), 2) AS exec_seconds,
    ROUND(
        CAST(iqc.team_exec_seconds AS NUMERIC)
        / NULLIF(CAST(t.total_exec_seconds AS NUMERIC), 0) * 100, 2
    ) AS pct_by_cpu,
    ROUND(
        CAST(iqc.team_exec_seconds AS NUMERIC)
        / NULLIF(CAST(t.total_exec_seconds AS NUMERIC), 0) * 8600.00, 2
    ) AS allocated_cost_usd,
    iqc.team_blks_read AS blks_read,
    ROUND(
        CAST(iqc.team_blks_read AS NUMERIC)
        / NULLIF(CAST(t.total_blks_read AS NUMERIC), 0) * 100, 2
    ) AS pct_by_io
FROM instance_query_costs iqc
CROSS JOIN totals t
ORDER BY allocated_cost_usd DESC;

```

Store the allocation results in a `finops.fact_shared_allocations` table alongside the direct costs in `finops.fact_infra_costs`, and union them in your reporting queries. Teams that work primarily on shared infrastructure will see their allocated share rather than a zero on the dashboard—which is exactly the behavioral nudge that motivates them to optimize their query patterns and eventually request dedicated infrastructure when the economics justify it.

24.8 Escalation Thresholds and Budget Alerts

Automated reports are most useful when they are quiet until something needs attention. A weekly digest that always says “costs are flat” trains leads to stop reading it. The same digest with a clear escalation signal—“your team’s database spend increased 40% this week, here are the three queries driving it”—gets read every time.

The threshold logic below runs as part of the weekly digest pipeline and determines the email subject line, the opening paragraph emphasis, and whether a Slack notification fires in addition to the email.

```
#!/usr/bin/env python3
```

```

"""
Escalation threshold evaluator for the weekly database cost digest.
Called from run_weekly_digest() before sending emails.
Sends a Slack alert in addition to email when thresholds are breached.

Thresholds (tune to your organization's norms):
- wow_increase_pct > 20: yellow alert (noted in email subject)
- wow_increase_pct > 40: red alert (Slack notification + email)
- absolute_week_cost > team_budget_weekly: budget breach alert
"""

import json
import urllib.request
import logging

logger = logging.getLogger(__name__)

SLACK_WEBHOOK_URL = "https://hooks.slack.com/services/YOUR/WEBHOOK/URL"
ALERT_CHANNEL = "#finops-alerts"

def evaluate_thresholds(team: dict, top_queries: list) -> dict:
    """
    Evaluate cost thresholds for a team's weekly report.
    Returns a dict with alert_level ('ok', 'warn', 'critical'),
    alert_reason, and email_subject_prefix.
    """
    wow = team.get("wow_change_pct")
    week_cost = float(team.get("week_cost", 0))
    team_name = team["team_name"]

    alert_level = "ok"
    alert_reason = None

    if wow is not None:
        if wow > 40:
            alert_level = "critical"
            alert_reason = f"Database spend increased {wow}% week-over-week"
        elif wow > 20:
            alert_level = "warn"
            alert_reason = f"Database spend increased {wow}% week-over-week"

    # Check if any single query accounts for more than 30% of the team's weekly cost
    if top_queries:
        top_query_cost = float(top_queries[0].get("est_cost_usd", 0))
        if week_cost > 0 and (top_query_cost / week_cost) > 0.30:
            if alert_level != "critical":
                alert_level = "warn"
            alert_reason = (
                (alert_reason or "") +
                f". Single query accounts for "
                f"{round(top_query_cost / week_cost * 100, 1)}% of team spend"
            ).strip(". ")

    subject_prefix = {
        "ok": " ",
        "warn": " ",
        "critical": " "
    }.get(alert_level, "")

    return {
        "alert_level": alert_level,
        "alert_reason": alert_reason,
        "subject_prefix": subject_prefix
    }

def send_slack_alert(team: dict, threshold_result: dict, top_queries: list) -> None:
    """Send a Slack notification for critical or warn-level cost events."""
    if threshold_result["alert_level"] == "ok":

```



```

        return

wow = team.get("wow_change_pct", "N/A")
week_cost = team.get("week_cost", 0)
color = "#ff0000" if threshold_result["alert_level"] == "critical" else "#ffaa00"

top_query_text = ""
if top_queries:
    q = top_queries[0]
    top_query_text = (
        f"\n*Top query ({q.get('feature_slug', 'untagged')}) * - "
        f"est. ${q.get('est_cost_usd', 0)} | "
        f"{q.get('total_calls', 0):,} calls | "
        f"avg {q.get('avg_exec_ms', 0)}ms\n"
        f"```${q.get('query_fingerprint', '')[:100]}```"
    )

payload = {
    "channel": ALERT_CHANNEL,
    "attachments": [
        {
            "color": color,
            "title": f"Database Cost Alert - {team['team_name']}",
            "text": (
                f"{threshold_result['alert_reason']}\n"
                f"*This week:* ${week_cost} | "
                f"*WoW change:* {wow}%"
                f"{top_query_text}"
            ),
            "footer": "FinOps Platform",
            "actions": [
                {
                    "type": "button",
                    "text": "View Dashboard",
                    "url": f"https://finops.yourcompany.internal/dashboard?team={team['team_slug']}"
                }
            ]
        }
    ]
}

try:
    data = json.dumps(payload).encode("utf-8")
    req = urllib.request.Request(
        SLACK_WEBHOOK_URL,
        data=data,
        headers={"Content-Type": "application/json"}
    )
    urllib.request.urlopen(req, timeout=5)
    logger.info(f"Slack alert sent for team {team['team_name']} (level:
↪ {threshold_result['alert_level']})")
except Exception as e:
    logger.error(f"Failed to send Slack alert: {e}")

```

Integrate this into `run_weekly_digest()` by calling `evaluate_thresholds()` before rendering the email template, passing the result into the template context to conditionally render an alert banner, and calling `send_slack_alert()` for any team at warn or critical level. The Slack message links directly to the Grafana dashboard pre-filtered for that team, so whoever picks it up in the channel can go straight to the data without logging into the FinOps portal first.

Key Takeaways

- **A five-tag taxonomy applied consistently outperforms a twenty-tag taxonomy applied inconsistently.** Start with team, app, environment, feature, and cost-center. Enforce them at resource creation using SCPs (AWS), Azure Policy, or GCP Organization Policy. Untagged resources must have a consequence—a shared pool chargeback—to close the behavioral loop.
 - **Infrastructure billing data and query-level attribution serve different purposes and must be combined.** Cloud billing APIs tell you what each team spends on instance hours, storage, and I/O. `pg_stat_statements` and `sys.dm_exec_query_stats` tell you which features and queries within a team are consuming the infrastructure. Neither layer alone is sufficient to drive optimization conversations.
 - **Embedding team and feature metadata directly in SQL comments is the lowest-friction path to query-level attribution.** Query planners ignore comments, so there is no performance cost. The metadata is captured by `pg_stat_statements` and `dm_exec_query_stats` automatically, requiring no changes to monitoring infrastructure once the comment convention is adopted in application code.
 - **The weekly cost digest email is the primary behavioral intervention.** Dashboards are consulted reactively; emails arrive proactively. When an engineering lead sees their team's database spend alongside the specific queries driving cost every Monday morning, optimization decisions become self-motivating rather than requiring top-down pressure from a FinOps team.
 - **Shared infrastructure requires explicit allocation logic, not ignored cost.** Splitting a shared cluster's cost by query execution time or block reads—using the same `pg_stat_statements` data already collected—produces defensible, team-visible allocations that motivate teams to reduce their share rather than treating shared resources as free.
-

Chapter 25: Budget Alerts and Anomaly Detection

Organizations that implement proactive budget alerts and anomaly detection catch runaway database costs an average of 11 days earlier than those relying on monthly billing reviews — translating to savings of \$3,000 to \$50,000 per incident depending on workload size. This chapter covers building a layered alerting system across AWS, Azure, and GCP that notifies the right people at the right thresholds, pairs those alerts with automated anomaly detection to distinguish genuine cost spikes from expected growth, and wires everything into the scheduling and remediation systems introduced in earlier chapters. By the end, your team will move from reactive bill shock to a posture where no database cost anomaly survives more than 24 hours undetected.

25.1 Understanding the Cost Signal Landscape

Before configuring a single alert, it pays to understand what signals are actually available and what each one measures. Budget alerts and anomaly detection are often treated as synonyms, but they serve different purposes. A budget alert fires when spending crosses a predefined dollar threshold. An anomaly detector fires when spending deviates from a predicted baseline — even if the absolute number is still within budget. Both matter.

The three major clouds each expose cost data through distinct APIs, and those APIs have different latency characteristics that directly affect how quickly you can detect a problem.

AWS publishes cost data to Cost Explorer with roughly 8–24 hours of lag. The Cost Anomaly Detection service processes that same data and sends SNS notifications within the lag window. CloudWatch metrics for RDS — CPU, storage, IOPS, connections — are near real-time (60-second granularity on most instance types) and serve as the leading indicator that a cost spike is about to appear in billing data.

Azure Cost Management data carries a lag of 8–24 hours for Pay-As-You-Go resources. Azure Monitor metrics for SQL Database are near real-time. The Azure Cost Management anomaly detection feature (in preview as of 2024) applies machine learning against the daily cost feed.

GCP BigQuery billing export is the most flexible of the three: once enabled, data appears in your dataset within hours and you can query it with SQL. The Cloud Billing Budget API supports percentage-based and absolute threshold alerts. Cloud Monitoring catches database-level signals in near real-time.

The architecture that works across all three clouds looks like this:

```
Real-time DB metrics (CloudWatch / Azure Monitor / Cloud Monitoring)
  ↓ [leading indicators: connections, IOPS, CPU]
Hourly cost feed (Cost Explorer API / Cost Management API / BQ billing export)
  ↓ [lagging indicators: actual spend]
Anomaly detection layer (AWS Cost Anomaly Detection / custom Python / BQ ML)
  ↓
Alert routing (SNS / Action Groups / PagerDuty / Slack)
  ↓
Automated remediation (Lambda / Automation Runbooks / Cloud Functions)
```

The key insight is that by the time billing data shows an anomaly, the underlying cause — a runaway query, an unexpected replica count, a forgotten snapshot — has already been running for hours. Wiring real-time DB metrics into the alert pipeline gives you the 6–12 hour head start that billing data alone cannot provide.

25.2 Configuring Budget Alerts on AWS, Azure, and GCP

AWS Budget Alerts

AWS Budgets support four budget types: Cost, Usage, Savings Plans coverage, and Reserved Instance utilization. For database FinOps, all four are relevant. A team running RDS PostgreSQL on Reserved Instances needs both a cost budget and an RI utilization alert — the first catches overspend, the second catches underutilized reservations that represent sunk cost without benefit.

The following script creates a complete budget configuration for an RDS workload using the AWS CLI. It establishes three alert thresholds: 70% (early warning), 90% (action required), and 100% (breach). The forecasted-spend alert at 80% gives finance teams roughly a week of lead time before the billing period closes.

```
#!/bin/bash
# create_rds_budget.sh
# Creates a monthly RDS cost budget with tiered alerts
# Requires: AWS CLI configured, SNS topic ARN for notifications

ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)
SNS_TOPIC_ARN="arn:aws:sns:us-east-1:${ACCOUNT_ID}:database-cost-alerts"
BUDGET_LIMIT="5000.00" # Adjust to your monthly RDS spend baseline
CURRENCY="USD"

aws budgets create-budget \
  --account-id "${ACCOUNT_ID}" \
  --budget '{
    "BudgetName": "RDS-Monthly-Cost-Budget",
    "BudgetLimit": {
      "Amount": "${BUDGET_LIMIT}",
      "Unit": "${CURRENCY}"
    },
    "TimeUnit": "MONTHLY",
    "BudgetType": "COST",
    "CostFilters": {
      "Service": ["Amazon Relational Database Service"]
    },
    "CostTypes": {
      "IncludeTax": true,
```

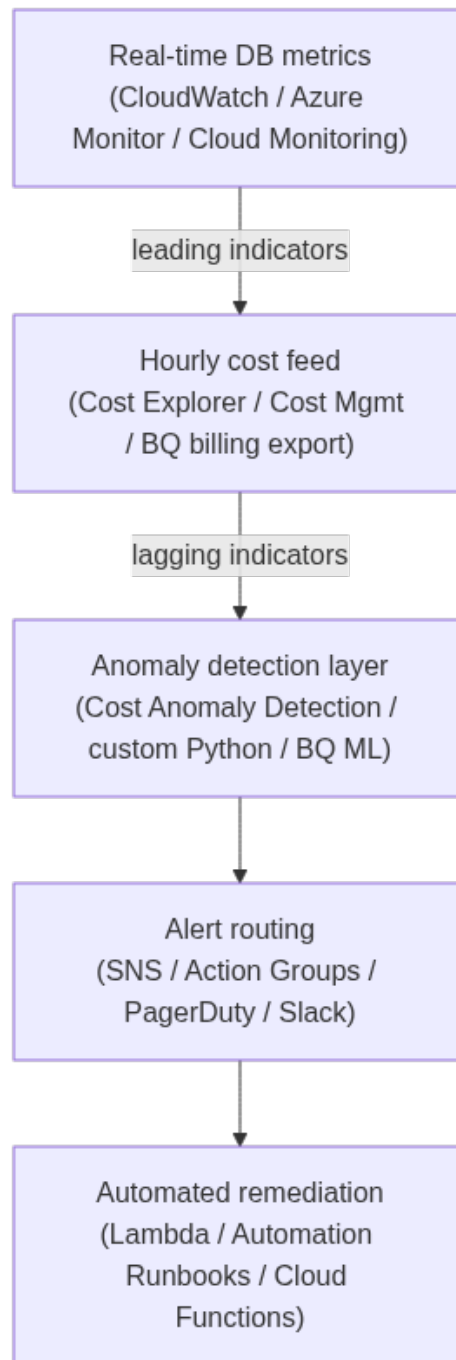


Figure 25: Real-time database metrics give a 6-12 hour head start over lagging billing data.

```

    "IncludeSubscription": true,
    "UseBlended": false,
    "IncludeRefund": false,
    "IncludeCredit": false,
    "IncludeUpfront": true,
    "IncludeRecurring": true,
    "IncludeOtherSubscription": true,
    "IncludeSupport": false,
    "IncludeDiscount": true,
    "UseAmortized": false
  }
}' \
--notifications-with-subscribers '[
  {
    "Notification": {
      "NotificationType": "ACTUAL",
      "ComparisonOperator": "GREATER_THAN",
      "Threshold": 70,
      "ThresholdType": "PERCENTAGE"
    },
    "Subscribers": [
      {"SubscriptionType": "SNS", "Address": "'${SNS_TOPIC_ARN}'",},
      {"SubscriptionType": "EMAIL", "Address": "dba-team@company.com"}
    ]
  },
  {
    "Notification": {
      "NotificationType": "ACTUAL",
      "ComparisonOperator": "GREATER_THAN",
      "Threshold": 90,
      "ThresholdType": "PERCENTAGE"
    },
    "Subscribers": [
      {"SubscriptionType": "SNS", "Address": "'${SNS_TOPIC_ARN}'",},
      {"SubscriptionType": "EMAIL", "Address": "cto@company.com"}
    ]
  },
  {
    "Notification": {
      "NotificationType": "ACTUAL",
      "ComparisonOperator": "GREATER_THAN",
      "Threshold": 100,
      "ThresholdType": "PERCENTAGE"
    },
    "Subscribers": [
      {"SubscriptionType": "SNS", "Address": "'${SNS_TOPIC_ARN}'",},
      {"SubscriptionType": "EMAIL", "Address": "finance@company.com"}
    ]
  },
  {
    "Notification": {
      "NotificationType": "FORECASTED",
      "ComparisonOperator": "GREATER_THAN",
      "Threshold": 80,
      "ThresholdType": "PERCENTAGE"
    },
    "Subscribers": [
      {"SubscriptionType": "SNS", "Address": "'${SNS_TOPIC_ARN}'",},
      {"SubscriptionType": "EMAIL", "Address": "dba-team@company.com"}
    ]
  }
]'

echo "Budget created. Verifying..."
aws budgets describe-budgets --account-id "${ACCOUNT_ID}" \
--query
↪ 'Budgets[?BudgetName==`RDS-Monthly-Cost-Budget`].{Name:BudgetName,Limit:BudgetLimit.Amount,Type:BudgetType}'
↪ \
--output table

```

```

# Create RI utilization alert separately - fires when utilization drops below 80%
aws budgets create-budget \
  --account-id "${ACCOUNT_ID}" \
  --budget '{
    "BudgetName": "RDS-RI-Utilization-Alert",
    "BudgetLimit": {
      "Amount": "80",
      "Unit": "PERCENTAGE"
    },
    "TimeUnit": "MONTHLY",
    "BudgetType": "RI_UTILIZATION",
    "CostFilters": {
      "Service": ["Amazon Relational Database Service"]
    }
  }' \
  --notifications-with-subscribers '[
    {
      "Notification": {
        "NotificationType": "ACTUAL",
        "ComparisonOperator": "LESS_THAN",
        "Threshold": 80,
        "ThresholdType": "PERCENTAGE"
      },
      "Subscribers": [
        { "SubscriptionType": "EMAIL", "Address": "dba-team@company.com" }
      ]
    }
  ]'

echo "RI utilization budget created."

```

Azure Cost Management Budget Alerts

Azure budget alerts are configured per subscription or resource group. For database teams, creating a budget scoped to the resource group containing your Azure SQL instances isolates the signal from unrelated compute costs.

```

#!/bin/bash
# create_azure_sql_budget.sh
# Creates a monthly Azure SQL budget with alert rules
# Requires: Azure CLI logged in, resource group name

SUBSCRIPTION_ID=$(az account show --query id -o tsv)
RESOURCE_GROUP="rg-databases-prod"
BUDGET_NAME="AzureSQL-Monthly-Budget"
BUDGET_AMOUNT=4000
START_DATE=$(date +%Y-%m-01)
END_DATE="2030-12-31"
ALERT_EMAIL="dba-team@company.com"
ACTION_GROUP_ID="/subscriptions/${SUBSCRIPTION_ID}/resourceGroups/${RESOURCE_GROUP}
↳ /providers/microsoft.insights/actionGroups/DatabaseCostAlerts"

# Create action group first (email + webhook to Slack or PagerDuty)
az monitor action-group create \
  --name "DatabaseCostAlerts" \
  --resource-group "${RESOURCE_GROUP}" \
  --short-name "DBCosAlert" \
  --action email dba-primary "${ALERT_EMAIL}"

# Create the budget scoped to the database resource group
az consumption budget create \
  --budget-name "${BUDGET_NAME}" \
  --amount "${BUDGET_AMOUNT}" \
  --time-grain Monthly \
  --start-date "${START_DATE}" \
  --end-date "${END_DATE}" \
  --resource-group "${RESOURCE_GROUP}" \

```

```

--notifications '[
  {
    "enabled": true,
    "operator": "GreaterThan",
    "threshold": 75,
    "contactEmails": ["'${ALERT_EMAIL}'"],
    "contactRoles": ["Owner"],
    "thresholdType": "Actual"
  },
  {
    "enabled": true,
    "operator": "GreaterThan",
    "threshold": 90,
    "contactEmails": ["'${ALERT_EMAIL}'", "cto@company.com"],
    "contactRoles": ["Owner", "Contributor"],
    "thresholdType": "Actual"
  },
  {
    "enabled": true,
    "operator": "GreaterThan",
    "threshold": 90,
    "contactEmails": ["'${ALERT_EMAIL}'"],
    "contactRoles": ["Owner"],
    "thresholdType": "Forecasted"
  }
]'

echo "Azure SQL budget ${BUDGET_NAME} created for resource group ${RESOURCE_GROUP}"
az consumption budget show --budget-name "${BUDGET_NAME}" \
  --resource-group "${RESOURCE_GROUP}" \
  --query '{Name:name, Amount:properties.amount, TimeGrain:properties.timeGrain}' \
  -o table

```

GCP Cloud Billing Budget Alerts

GCP billing budgets are managed at the billing account level and support programmatic notification via Pub/Sub, which enables automated remediation workflows far more cleanly than email-only alerts.

```

#!/bin/bash
# create_gcp_cloudsql_budget.sh
# Creates a Cloud SQL budget with Pub/Sub notification for automation
# Requires: gcloud CLI authenticated, billing account ID

BILLING_ACCOUNT_ID="XXXXXX-XXXXXX-XXXXXX" # Replace with your billing account
PROJECT_ID="your-project-id"
PUBSUB_TOPIC="projects/${PROJECT_ID}/topics/billing-alerts"
BUDGET_AMOUNT=3500

# Create Pub/Sub topic for programmatic alerts
gcloud pubsub topics create billing-alerts --project="${PROJECT_ID}"

# Grant billing account permission to publish to the topic
gcloud pubsub topics add-iam-policy-binding billing-alerts \
  --member="serviceAccount:billing-alerts@system.gserviceaccount.com" \
  --role="roles/pubsub.publisher" \
  --project="${PROJECT_ID}"

# Create budget with tiered thresholds
gcloud billing budgets create \
  --billing-account="${BILLING_ACCOUNT_ID}" \
  --display-name="CloudSQL-Monthly-Budget" \
  --budget-amount="${BUDGET_AMOUNT}USD" \
  --threshold-rule=percent=0.75,basis=current-spend \
  --threshold-rule=percent=0.90,basis=current-spend \
  --threshold-rule=percent=1.00,basis=current-spend \
  --threshold-rule=percent=0.90,basis=forecasted-spend \
  --filter-services="services/95FF-2EF5-5EA1" \

```



```

--notifications-rule-pubsub-topic="${PUBSUB_TOPIC}" \
--notifications-rule-disable-default-iam-recipients=false \
--all-updates-rule-monitoring-notification-channels=""

# Verify creation
gcloud billing budgets list \
--billing-account="${BILLING_ACCOUNT_ID}" \
--format="table(displayName, budgetFilter.services, amount.specifiedAmount.units)"

```

25.3 Enabling and Tuning AWS Cost Anomaly Detection

AWS Cost Anomaly Detection uses machine learning trained on your specific historical spend patterns. Unlike a simple threshold alert, it accounts for seasonal variation — it will not fire an alert on the last day of the month simply because billing is cumulative. This makes it substantially more precise than budget alerts for detecting genuine anomalies in the \$200–\$2,000 range that budget thresholds miss.

The service operates on monitors (what to watch) and subscriptions (how to be notified). For RDS-specific detection, creating a service-level monitor scoped to RDS gives the ML model a cleaner signal than an account-wide monitor that mixes RDS costs with EC2 and S3 noise.

```

#!/bin/bash
# setup_cost_anomaly_detection.sh
# Configures AWS Cost Anomaly Detection for RDS workloads
# Monitors: service-level RDS, plus per-tag monitor for prod vs. non-prod isolation

ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)
SNS_TOPIC_ARN="arn:aws:sns:us-east-1:${ACCOUNT_ID}:database-cost-alerts"

# Monitor 1: Service-level monitor for all RDS costs
RDS_MONITOR_ARN=$(aws ce create-anomaly-monitor \
--anomaly-monitor '{
  "MonitorName": "RDS-Service-Monitor",
  "MonitorType": "DIMENSIONAL",
  "MonitorDimension": "SERVICE"
}' \
--query AnomalyMonitorArn --output text)

echo "RDS service monitor created: ${RDS_MONITOR_ARN}"

# Monitor 2: Tag-based monitor to isolate production database costs
# This prevents a dev environment spike from masking a prod anomaly
PROD_MONITOR_ARN=$(aws ce create-anomaly-monitor \
--anomaly-monitor '{
  "MonitorName": "RDS-Production-Tag-Monitor",
  "MonitorType": "CUSTOM",
  "MonitorSpecification": {
    "Tags": {
      "Key": "Environment",
      "Values": ["production", "prod"]
    }
  }
}' \
--query AnomalyMonitorArn --output text)

echo "Production tag monitor created: ${PROD_MONITOR_ARN}"

# Subscription for RDS service monitor
# Alert threshold: $50 individual anomaly OR $200 cumulative in rolling 7-day window
aws ce create-anomaly-subscription \
--anomaly-subscription '{
  "SubscriptionName": "RDS-Anomaly-Alerts-Primary",

```

```

"MonitorArnList": ["'${RDS_MONITOR_ARN}'"],
"Subscribers": [
  {
    "Address": "'${SNS_TOPIC_ARN}'",
    "Type": "SNS"
  },
  {
    "Address": "dba-team@company.com",
    "Type": "EMAIL"
  }
],
"Threshold": 50,
"Frequency": "DAILY",
"ThresholdExpression": {
  "Or": [
    {
      "Dimensions": {
        "Key": "ANOMALY_TOTAL_IMPACT_ABSOLUTE",
        "Values": ["50"],
        "MatchOptions": ["GREATER_THAN_OR_EQUAL"]
      }
    },
    {
      "Dimensions": {
        "Key": "ANOMALY_TOTAL_IMPACT_PERCENTAGE",
        "Values": ["20"],
        "MatchOptions": ["GREATER_THAN_OR_EQUAL"]
      }
    }
  ]
}
}'

# High-severity subscription for production monitor - immediate notification
aws ce create-anomaly-subscription \
  --anomaly-subscription '{
    "SubscriptionName": "RDS-Production-Critical-Alerts",
    "MonitorArnList": ["'${PROD_MONITOR_ARN}'"],
    "Subscribers": [
      {
        "Address": "'${SNS_TOPIC_ARN}'",
        "Type": "SNS"
      },
      {
        "Address": "cto@company.com",
        "Type": "EMAIL"
      }
    ],
    "Threshold": 200,
    "Frequency": "IMMEDIATE"
  }'

# List all configured anomaly subscriptions to confirm setup
echo "Configured anomaly subscriptions:"
aws ce list-anomaly-subscriptions \
  --query 'AnomalySubscriptions[].{Name:SubscriptionName,Threshold:Threshold,Frequency:Frequency}' \
  --output table

# Query recent anomalies detected in last 90 days (useful for baseline calibration)
echo "Recent RDS cost anomalies:"
aws ce get-anomalies \
  --monitor-arn "${RDS_MONITOR_ARN}" \
  --date-interval '{
    "StartDate": "'$(date -d '90 days ago' +%Y-%m-%d)'",
    "EndDate": "'$(date +%Y-%m-%d)'"
  }' \
  --query 'Anomalies[].{
    Start:AnomalyStartDate,
    End:AnomalyEndDate,
    ImpactUSD:Impact.TotalActualSpend,

```

```
ExpectedUSD:Impact.TotalExpectedSpend
}' \
--output table
```

One tuning decision deserves attention: the threshold values of \$50 (absolute) and 20% (percentage). Teams with predictable workloads and tight budgets should lower these — \$25 and 15% — to catch small anomalies early. Teams with noisy, variable workloads (many batch jobs, development databases with irregular usage) should raise them to reduce alert fatigue. Review the anomaly history query above after 30 days and recalibrate.

25.4 Custom Anomaly Detection Using Database and Cost Metrics

The built-in anomaly detection on all three clouds works well for billing-level anomalies but misses the pattern where a cost spike is caused by a specific query, connection pool explosion, or unexpected autoscaling event. Building a custom detection layer that correlates database-level metrics with cost data closes this gap. When an alert fires, you want to know not just *that* costs spiked but *why*.

The following Python script runs on a schedule (recommended: every 6 hours via AWS Lambda, Azure Functions, or Cloud Run) and performs three checks: it queries `pg_stat_statements` for query-level cost drivers, it pulls CloudWatch RDS metrics to detect resource anomalies, and it compares the current day's Cost Explorer spend against the 30-day rolling average. If any of the three checks exceed their thresholds, it posts a structured alert to a Slack webhook with enough context to act immediately.

```
#!/usr/bin/env python3
"""
db_cost_anomaly_detector.py
Correlates database metrics with cost data to identify anomaly root causes.
Runs on a 6-hour schedule via Lambda/Cloud Run/Azure Functions.
Requires: psycopg2, boto3, requests
Environment variables: DB_HOST, DB_NAME, DB_USER, DB_PASSWORD,
                      SLACK_WEBHOOK_URL, AWS_DEFAULT_REGION
"""

import os
import json
import statistics
from datetime import datetime, timedelta, timezone
from typing import Optional

import boto3
import psycopg2
import requests

# Configuration
DB_CONFIG = {
    "host": os.environ["DB_HOST"],
    "dbname": os.environ["DB_NAME"],
    "user": os.environ["DB_USER"],
    "password": os.environ["DB_PASSWORD"],
    "connect_timeout": 10,
}

SLACK_WEBHOOK = os.environ.get("SLACK_WEBHOOK_URL")
REGION = os.environ.get("AWS_DEFAULT_REGION", "us-east-1")
RDS_INSTANCE_ID = os.environ.get("RDS_INSTANCE_ID", "prod-postgres-01")

# Anomaly thresholds
COST_SPIKE_THRESHOLD_PCT = 25.0      # Alert if today's spend > 30-day avg by this %
```

```

QUERY_COST_THRESHOLD_MS = 5000.0      # Alert if mean query time exceeds this (ms)
CONNECTION_SPIKE_THRESHOLD = 1.5      # Alert if connections > 1.5x recent average
IOPS_SPIKE_THRESHOLD_PCT = 40.0      # Alert if IOPS > 40% above recent average

def get_cost_anomaly(ce_client, days_back: int = 30) -> dict:
    """
    Compares today's RDS spend against the rolling 30-day daily average.
    Returns anomaly data including current spend, expected spend, and % deviation.
    """
    end_date = datetime.now(timezone.utc).date()
    start_date = end_date - timedelta(days=days_back)
    today = end_date

    response = ce_client.get_cost_and_usage(
        TimePeriod={
            "Start": start_date.isoformat(),
            "End": end_date.isoformat(),
        },
        Granularity="DAILY",
        Filter={
            "Dimensions": {
                "Key": "SERVICE",
                "Values": ["Amazon Relational Database Service"],
            }
        },
        Metrics=["UnblendedCost"],
    )

    daily_costs = []
    today_cost = 0.0

    for result in response["ResultsByTime"]:
        amount = float(result["Total"]["UnblendedCost"]["Amount"])
        date_str = result["TimePeriod"]["Start"]
        if date_str == today.isoformat():
            today_cost = amount
        else:
            if amount > 0: # Exclude zero-cost days (weekends on dev environments)
                daily_costs.append(amount)

    if not daily_costs:
        return {"status": "insufficient_data"}

    avg_daily = statistics.mean(daily_costs)
    stddev_daily = statistics.stdev(daily_costs) if len(daily_costs) > 1 else 0.0
    deviation_pct = ((today_cost - avg_daily) / avg_daily * 100) if avg_daily > 0 else 0.0

    return {
        "status": "anomaly" if deviation_pct > COST_SPIKE_THRESHOLD_PCT else "normal",
        "today_cost_usd": round(today_cost, 2),
        "avg_daily_cost_usd": round(avg_daily, 2),
        "stddev_usd": round(stddev_daily, 2),
        "deviation_pct": round(deviation_pct, 1),
        "threshold_pct": COST_SPIKE_THRESHOLD_PCT,
        "sample_days": len(daily_costs),
    }

def get_cloudwatch_anomalies(cw_client, instance_id: str) -> list:
    """
    Checks CloudWatch RDS metrics for resource-level spikes.
    Covers: DatabaseConnections, ReadIOPS, WriteIOPS, CPUUtilization.
    """
    anomalies = []
    now = datetime.now(timezone.utc)
    period_hours = 6
    baseline_hours = 72 # Compare against 3-day baseline

    metrics_to_check = [

```

```

{
    "name": "DatabaseConnections",
    "namespace": "AWS/RDS",
    "stat": "Maximum",
    "threshold_multiplier": CONNECTION_SPIKE_THRESHOLD,
    "unit": "connections",
},
{
    "name": "ReadIOPS",
    "namespace": "AWS/RDS",
    "stat": "Average",
    "threshold_pct": IOPS_SPIKE_THRESHOLD_PCT,
    "unit": "IOPS",
},
{
    "name": "WriteIOPS",
    "namespace": "AWS/RDS",
    "stat": "Average",
    "threshold_pct": IOPS_SPIKE_THRESHOLD_PCT,
    "unit": "IOPS",
},
{
    "name": "CPUUtilization",
    "namespace": "AWS/RDS",
    "stat": "Average",
    "threshold_pct": 30.0, # Alert if CPU spikes 30% above baseline
    "unit": "percent",
},
]

for metric in metrics_to_check:
    response = cw_client.get_metric_statistics(
        Namespace=metric["namespace"],
        MetricName=metric["name"],
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": instance_id}],
        StartTime=now - timedelta(hours=baseline_hours),
        EndTime=now,
        Period=3600, # 1-hour granularity
        Statistics=[metric["stat"]],
    )

    if not response["Datapoints"]:
        continue

    datapoints = sorted(response["Datapoints"], key=lambda x: x["Timestamp"])
    recent_values = [dp[metric["stat"]] for dp in datapoints[-period_hours:]]
    baseline_values = [dp[metric["stat"]] for dp in datapoints[:-period_hours]]

    if not recent_values or not baseline_values:
        continue

    recent_avg = statistics.mean(recent_values)
    baseline_avg = statistics.mean(baseline_values)

    if baseline_avg == 0:
        continue

    deviation_pct = (recent_avg - baseline_avg) / baseline_avg * 100
    threshold = metric.get("threshold_pct", (metric.get("threshold_multiplier", 1.5) - 1) * 100)

    if deviation_pct > threshold:
        anomalies.append({
            "metric": metric["name"],
            "recent_avg": round(recent_avg, 2),
            "baseline_avg": round(baseline_avg, 2),
            "deviation_pct": round(deviation_pct, 1),
            "unit": metric["unit"],
        })

return anomalies

```

```

def get_expensive_queries(conn) -> list:
    """
    Queries pg_stat_statements for queries with mean execution time
    above the configured threshold, which directly maps to I/O cost.
    Resets stats are not performed here - only read.
    """
    query = """
    SELECT
        LEFT(query, 120)                AS query_snippet,
        calls,
        ROUND(mean_exec_time::numeric, 2) AS mean_ms,
        ROUND(total_exec_time::numeric, 2) AS total_ms,
        ROUND(rows::numeric / NULLIF(calls, 0), 1) AS rows_per_call,
        ROUND(
            (shared_blks_read + shared_blks_hit)::numeric
            / NULLIF(calls, 0), 0
        ) AS blks_per_call
    FROM
        pg_stat_statements
    WHERE
        mean_exec_time > %(threshold_ms)s
        AND calls > 50
        AND query NOT LIKE '%%pg_stat%%'
    ORDER BY
        total_exec_time DESC
    LIMIT 10;
    """
    with conn.cursor() as cur:
        cur.execute(query, {"threshold_ms": QUERY_COST_THRESHOLD_MS})
        columns = [desc[0] for desc in cur.description]
        return [dict(zip(columns, row)) for row in cur.fetchall()]

def format_slack_alert(cost_data: dict, cw_anomalies: list, query_data: list) -> dict:
    """Formats a structured Slack message with all anomaly findings."""
    blocks = [
        {
            "type": "header",
            "text": {
                "type": "plain_text",
                "text": " Database Cost Anomaly Detected",
            },
        },
        {
            "type": "section",
            "text": {
                "type": "mrkdown",
                "text": (
                    f"*Instance:* `{RDS_INSTANCE_ID}`\n"
                    f"*Detection Time:* {datetime.now(timezone.utc).strftime('%Y-%m-%d %H:%M UTC')}"
                ),
            },
        },
    ]

    # Cost section
    if cost_data.get("status") == "anomaly":
        blocks.append({
            "type": "section",
            "text": {
                "type": "mrkdown",
                "text": (
                    f"* Cost Spike Detected*\n"
                    f"* Today so far: `${cost_data['today_cost_usd']}`\n"
                    f"* 30-day average: `${cost_data['avg_daily_cost_usd']}`\n"
                    f"* Deviation: `${cost_data['deviation_pct']}`% "
                    f"(threshold: {cost_data['threshold_pct']})%\n"
                    f"* Std deviation: `${cost_data['stddev_usd']}`"
                )
            },
        })

```

```

    },
    },
})

# CloudWatch anomalies section
if cw_anomalies:
    metric_lines = "\n".join([
        f"* {a['metric']}*: {a['recent_avg']} {a['unit']} "
        f"vs baseline {a['baseline_avg']} {a['unit']} "
        f"({a['deviation_pct']}%)"
        for a in cw_anomalies
    ])
    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": f"* Resource Anomalies (last 6h vs 72h baseline)*\n{metric_lines}",
        },
    })

# Query section
if query_data:
    query_lines = "\n".join([
        f"* {q['query_snippet'][:80]}...\n"
        f"  mean: {q['mean_ms']}ms | calls: {q['calls']} | "
        f"total: {q['total_ms']}ms"
        for q in query_data[:3] # Top 3 in Slack for readability
    ])
    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": f"* Slow Queries (potential cost drivers)*\n{query_lines}",
        },
    })

blocks.append({
    "type": "actions",
    "elements": [
        {
            "type": "button",
            "text": {"type": "plain_text", "text": "Cost Explorer"},
            "url":
                f"https://console.aws.amazon.com/cost-management/home#/anomaly-detection/monitors",
            "style": "danger",
        },
        {
            "type": "button",
            "text": {"type": "plain_text", "text": "Performance Insights"},
            "url": f"https://console.aws.amazon.com/rds/home?region={REGION}"
                f"#performance-insights-v20206:/resourceId/{RDS_INSTANCE_ID}",
        },
    ],
})

return {"blocks": blocks}

def post_to_slack(payload: dict) -> None:
    """Posts the formatted alert to Slack via webhook."""
    if not SLACK_WEBHOOK:
        print("SLACK_WEBHOOK_URL not set - printing alert to stdout.")
        print(json.dumps(payload, indent=2))
        return

    response = requests.post(
        SLACK_WEBHOOK,
        json=payload,
        headers={"Content-Type": "application/json"},
    )

```

```

        timeout=10,
    )
    if response.status_code != 200:
        raise RuntimeError(f"Slack webhook failed: {response.status_code} {response.text}")

def lambda_handler(event=None, context=None) -> dict:
    """Entry point for AWS Lambda. Also callable directly for local testing."""
    ce_client = boto3.client("ce", region_name="us-east-1") # CE is global, us-east-1 endpoint
    cw_client = boto3.client("cloudwatch", region_name=REGION)

    findings = []

    # Check 1: Billing anomaly
    cost_data = get_cost_anomaly(ce_client)
    if cost_data.get("status") == "anomaly":
        findings.append("cost_spike")
        print(f"Cost anomaly: {cost_data['deviation_pct']}% above average "
              f"${cost_data['today_cost_usd']} vs avg ${cost_data['avg_daily_cost_usd']}")

    # Check 2: CloudWatch resource anomalies
    cw_anomalies = get_cloudwatch_anomalies(cw_client, RDS_INSTANCE_ID)
    if cw_anomalies:
        findings.append("resource_spike")
        for a in cw_anomalies:
            print(f"CW anomaly: {a['metric']} +{a['deviation_pct']}% above baseline")

    # Check 3: Expensive queries in pg_stat_statements
    expensive_queries = []
    try:
        conn = psycopg2.connect(**DB_CONFIG)
        conn.autocommit = True
        expensive_queries = get_expensive_queries(conn)
        conn.close()
        if expensive_queries:
            findings.append("slow_queries")
            print(f"Found {len(expensive_queries)} queries exceeding {QUERY_COST_THRESHOLD_MS}ms mean"
                  ↪ time")
    except psycopg2.Error as exc:
        print(f"DB connection failed - skipping query analysis: {exc}")

    # Alert if any finding exists
    if findings:
        payload = format_slack_alert(cost_data, cw_anomalies, expensive_queries)
        post_to_slack(payload)
        return {
            "statusCode": 200,
            "body": json.dumps({"findings": findings, "alerted": True}),
        }

    print("No anomalies detected.")
    return {
        "statusCode": 200,
        "body": json.dumps({"findings": [], "alerted": False}),
    }

if __name__ == "__main__":
    result = lambda_handler()
    print(json.dumps(json.loads(result["body"]), indent=2))

```

To deploy this as a Lambda function running every 6 hours:

```

#!/bin/bash
# deploy_anomaly_detector.sh
# Packages and deploys the anomaly detector Lambda with EventBridge schedule

FUNCTION_NAME="db-cost-anomaly-detector"
REGION="us-east-1"
ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)

```



```
ROLE_ARN="arn:aws:iam::${ACCOUNT_ID}:role/lambda-db-cost-detector-role"

# Package dependencies
pip install pycpg2-binary boto3 requests -t ./package/
cp db_cost_anomaly_detector.py ./package/
cd package && zip -r ../anomaly_detector.zip . && cd ..

# Deploy or update the Lambda function
aws lambda create-function \
  --function-name "${FUNCTION_NAME}" \
  --runtime python3.12 \
  --role "${ROLE_ARN}" \
  --handler db_cost_anomaly_detector.lambda_handler \
  --zip-file fileb://anomaly_detector.zip \
  --timeout 120 \
  --memory-size 256 \
  --environment Variables="{
    DB_HOST=${DB_HOST},
    DB_NAME=${DB_NAME},
    DB_USER=${DB_USER},
    DB_PASSWORD=${DB_PASSWORD},
    SLACK_WEBHOOK_URL=${SLACK_WEBHOOK_URL},
    RDS_INSTANCE_ID=${RDS_INSTANCE_ID},
    AWS_DEFAULT_REGION=${REGION}
  }" \
  --region "${REGION}" 2>/dev/null || \
aws lambda update-function-code \
  --function-name "${FUNCTION_NAME}" \
  --zip-file fileb://anomaly_detector.zip \
  --region "${REGION}"

# Create EventBridge rule to trigger every 6 hours
aws events put-rule \
  --name "db-cost-anomaly-schedule" \
  --schedule-expression "rate(6 hours)" \
  --state ENABLED \
  --region "${REGION}"

# Grant EventBridge permission to invoke Lambda
aws lambda add-permission \
  --function-name "${FUNCTION_NAME}" \
  --statement-id "EventBridgeInvoke" \
  --action lambda:InvokeFunction \
  --principal events.amazonaws.com \
  --source-arn "arn:aws:events:${REGION}:${ACCOUNT_ID}:rule/db-cost-anomaly-schedule" \
  --region "${REGION}"

# Wire the rule to the Lambda target
aws events put-targets \
  --rule "db-cost-anomaly-schedule" \
  --targets
  ↪ "Id=AnomalyDetectorLambda,Arn=arn:aws:lambda:${REGION}:${ACCOUNT_ID}:function:${FUNCTION_NAME}" \
  --region "${REGION}"

echo "Anomaly detector deployed and scheduled. Next run in up to 6 hours."
aws lambda get-function \
  --function-name "${FUNCTION_NAME}" \
  --query 'Configuration.{Runtime:Runtime,Memory:MemorySize,Timeout:Timeout,State:State}' \
  --output table
```

25.5 Query-Level Cost Attribution Using pg_stat_statements and sys.dm_exec_query_stats

Budget alerts tell you that money is leaving the account. Query-level attribution tells you exactly which SQL statement is responsible. This is where FinOps crosses back into DBA territory, and it is where database professionals have a decisive advantage over generic cloud cost teams.

On PostgreSQL (RDS, Aurora, Cloud SQL, Azure Database for PostgreSQL), `pg_stat_statements` is the primary instrument. The following query correlates execution statistics with an estimated I/O cost model. It does not produce exact dollar figures — those live in the billing feed — but it ranks queries by their share of total I/O load, which maps directly to IOPS spend on gp2/gp3/io1 volumes.

```
-- =====
-- Query Cost Attribution Report
-- Identifies top queries by I/O load and estimated cost share
-- Run against RDS PostgreSQL with pg_stat_statements enabled
-- =====

WITH total_stats AS (
    SELECT
        SUM(shared_blks_read + shared_blks_written
            + local_blks_read + local_blks_written) AS total_blks,
        SUM(total_exec_time) AS total_exec_ms,
        SUM(calls) AS total_calls
    FROM pg_stat_statements
    WHERE query NOT LIKE '%pg_stat%'
),
query_stats AS (
    SELECT
        pss.userid::regrole::text AS db_role,
        pss.dbid::regdatabase::text AS database,
        LEFT(pss.query, 200) AS query_snippet,
        pss.calls,
        ROUND(pss.mean_exec_time::numeric, 2) AS mean_exec_ms,
        ROUND(pss.total_exec_time::numeric / 1000 / 60, 2) AS total_exec_min,
        pss.shared_blks_read,
        pss.shared_blks_hit,
        pss.shared_blks_written,
        pss.shared_blks_dirtied,
        (pss.shared_blks_read + pss.shared_blks_written) AS io_blks,
        ROUND(
            (pss.shared_blks_read + pss.shared_blks_written)::numeric * 8 / 1024,
            2
        ) AS io_mb,
        -- Cache hit ratio for this specific query
        ROUND(
            pss.shared_blks_hit::numeric /
            NULLIF(pss.shared_blks_hit + pss.shared_blks_read, 0) * 100,
            1
        ) AS cache_hit_pct,
        ts.total_blks
    FROM pg_stat_statements pss
    CROSS JOIN total_stats ts
    WHERE pss.calls > 10
        AND query NOT LIKE '%pg_stat%'
        AND query NOT LIKE '%autovacuum%'
),
ranked AS (
    SELECT
        db_role,
        database,
        query_snippet,
        calls,
        mean_exec_ms,
```

```

total_exec_min,
io_mb,
cache_hit_pct,
shared_blks_read,
shared_blks_written,
-- Share of total instance I/O this query consumes
ROUND(io_blks::numeric / NULLIF(total_blks, 0) * 100, 2) AS io_share_pct,
-- Estimated monthly I/O cost assuming gp3 at $0.000065 per 1M IOPS
-- Adjust multiplier for io1 ($0.000065 * 3x) if on io1 volumes
ROUND(
    (shared_blks_read + shared_blks_written)::numeric
    * calls / 1e6 * 0.000065 * 30 * 24 * 60,
    4
)
AS est_monthly_io_cost_usd
FROM query_stats
)
SELECT
    database,
    db_role,
    query_snippet,
    calls,
    ROUND(mean_exec_ms, 2) AS mean_exec_ms,
    total_exec_min,
    ROUND(io_mb, 2) AS io_mb_total,
    cache_hit_pct AS "cache_hit%",
    io_share_pct AS "io_share%",
    est_monthly_io_cost_usd AS "est_monthly_io_$"
FROM ranked
ORDER BY io_share_pct DESC
LIMIT 25;

```

For **Azure SQL Database** and **SQL Managed Instance**, the equivalent instrument is `sys.dm_exec_query_stats` combined with `sys.dm_exec_sql_text`. The following query surfaces the top 20 queries by logical reads, which is the primary cost driver on Azure SQL's DTU and vCore models:

```

-- =====
-- Azure SQL: Query Cost Attribution by Logical Reads
-- Covers: Azure SQL Database, SQL Managed Instance
-- Run in the context of the target database
-- =====

SELECT TOP 20
    qs.execution_count,
    qs.total_logical_reads,
    ROUND(qs.total_logical_reads * 1.0 / qs.execution_count, 0)
    AS avg_logical_reads,
    qs.total_elapsed_time / 1000 AS total_elapsed_ms,
    ROUND(qs.total_elapsed_time * 1.0
    / qs.execution_count / 1000, 2) AS avg_elapsed_ms,
    qs.total_worker_time / 1000 AS total_cpu_ms,
    ROUND(qs.total_worker_time * 1.0
    / qs.execution_count / 1000, 2) AS avg_cpu_ms,
    qs.total_physical_reads,
    ROUND(qs.total_physical_reads * 1.0
    / qs.execution_count, 1) AS avg_physical_reads,
    -- DTU cost proxy: logical reads / 1M * execution_count
    -- On a Business Critical 8 vCore (~$1,100/mo), each logical read costs roughly $0.0000004
    ROUND(qs.total_logical_reads * 1.0
    / 1e6 * 0.0004, 4) AS est_cost_proxy_usd,
    qs.creation_time AS plan_compiled_at,
    DB_NAME(qt.dbid) AS database_name,
    qt.objectid,
    SUBSTRING(
        qt.text,
        (qs.statement_start_offset / 2) + 1,
        (CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(qt.text)

```

```

        ELSE qs.statement_end_offset
        END - qs.statement_start_offset) / 2 + 1
    )
    AS query_text
FROM
  sys.dm_exec_query_stats AS qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS qt
WHERE
  qs.execution_count > 5
  AND qt.text NOT LIKE '%sys.dm_exec%'
ORDER BY
  qs.total_logical_reads DESC;

```

For **GCP Cloud SQL for PostgreSQL**, the `pg_stat_statements` query above works directly. For **BigQuery** workloads, the `INFORMATION_SCHEMA.JOBS` view provides the equivalent attribution:

```

-- =====
-- BigQuery: Job-level cost attribution by bytes processed
-- GCP charges $6.25 per TB processed (on-demand pricing as of 2024)
-- Run in BigQuery console or via bq CLI
-- =====

SELECT
  project_id,
  user_email,
  job_type,
  statement_type,
  COUNT(*) AS job_count,
  SUM(total_bytes_processed) / POW(1024, 4) AS total_tb_processed,
  ROUND(
    SUM(total_bytes_processed) / POW(1024, 4) * 6.25,
    2
  ) AS estimated_cost_usd,
  AVG(TIMESTAMP_DIFF(end_time, start_time, SECOND)) AS avg_duration_sec,
  -- Identify the single most expensive query pattern
  APPROX_TOP_COUNT(
    LEFT(REGEXP_REPLACE(query, r'\s+', ' '), 200), 1
  ) [OFFSET(0)].value AS top_query_pattern
FROM
  `region-us`.INFORMATION_SCHEMA.JOBS
WHERE
  creation_time BETWEEN TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 30 DAY)
    AND CURRENT_TIMESTAMP()
  AND state = 'DONE'
  AND error_result IS NULL
  AND total_bytes_processed > 0
GROUP BY
  project_id,
  user_email,
  job_type,
  statement_type
ORDER BY
  estimated_cost_usd DESC
LIMIT 30;

```

These three queries together cover the full multi-cloud estate. The pattern is consistent: find the queries consuming the most I/O or bytes processed, calculate their share of total load, and prioritize optimization work accordingly. A single query responsible for 35% of instance I/O that can be rewritten to use an index represents not just a performance win but a direct reduction in IOPS charges — particularly significant on io1 volumes at \$0.065/GB-month compared to gp3 at \$0.02/GB-month.

25.6 Automating Alert Response and Cost Runbook Execution

Receiving an alert is only half the system. The other half is what happens next. Organizations that automate initial triage steps — not full remediation, but data collection and scoping — reduce mean time to resolution by 60–70% compared to teams that start each incident by manually pulling metrics.

The following approach wires an SNS topic (populated by budget alerts and anomaly detection) to a Lambda function that executes a structured response runbook: it identifies the top cost drivers, snapshots current CloudWatch metrics, estimates the dollar impact, and posts a pre-filled incident ticket summary to Slack. A human still makes the remediation decision, but they receive a briefing rather than a blank screen.

```
#!/usr/bin/env python3
"""
cost_alert_responder.py
SNS-triggered Lambda that executes an automated cost incident runbook.
Triggered by: AWS Budget alerts, Cost Anomaly Detection, or CloudWatch alarms.
Posts a structured incident brief to Slack with recommended actions.
"""

import json
import os
import urllib.request
from datetime import datetime, timedelta, timezone

import boto3

SLACK_WEBHOOK = os.environ["SLACK_WEBHOOK_URL"]
REGION = os.environ.get("AWS_DEFAULT_REGION", "us-east-1")

# Pre-defined remediation playbooks keyed by anomaly type
PLAYBOOKS = {
    "storage_spike": {
        "title": "Unexpected Storage Growth",
        "likely_causes": [
            "Uncontrolled binary log / WAL accumulation",
            "Forgotten large table without TTL or partition pruning",
            "Snapshot retention policy changed or missing",
        ],
        "immediate_actions": [
            "Check RDS storage metrics in CloudWatch (FreeStorageSpace)",
            "Query pg_database_size() / information_schema.tables for growth",
            "Review automated backup retention period (target: 7 days for prod)",
            "Identify tables added in last 7 days: SELECT relname,
↵ pg_size_pretty(pg_total_relation_size(oid)) FROM pg_class ORDER BY
↵ pg_total_relation_size(oid) DESC LIMIT 20;",
        ],
        "automation_available": True,
        "runbook_url": "https://wiki.internal/runbooks/rds-storage-spike",
    },
    "iops_spike": {
        "title": "IOPS Spike Detected",
        "likely_causes": [
            "New or changed query missing an index",
            "Autovacuum running on large table",
            "Bulk insert / ETL job without rate limiting",
            "gp2 IOPS burst bucket depleted",
        ],
        "immediate_actions": [
            "Pull pg_stat_statements sorted by shared_blks_read",
            "Check pg_stat_activity for long-running queries",
            "Verify autovacuum_vacuum_cost_delay setting for large tables",
            "If on gp2: evaluate migration to gp3 (3,000 baseline IOPS free)",
        ],
        "automation_available": True,
    },
}
```

```

        "runbook_url": "https://wiki.internal/runbooks/rds-iops-spike",
    },
    "connection_spike": {
        "title": "Connection Count Anomaly",
        "likely_causes": [
            "Application connection pool misconfigured after deploy",
            "pgBouncer or RDS Proxy not routing traffic correctly",
            "Connection leak in application code",
            "Sudden traffic spike without connection pool scaling",
        ],
        "immediate_actions": [
            "SELECT count(*), state, wait_event_type FROM pg_stat_activity GROUP BY state,
            ↪ wait_event_type;",
            "Check pgBouncer SHOW POOLS; SHOW STATS;",
            "Verify max_connections parameter against instance class limits",
            "Consider enabling RDS Proxy if not already active (reduces connection overhead by 60-80%)",
        ],
        "automation_available": False,
        "runbook_url": "https://wiki.internal/runbooks/rds-connection-spike",
    },
    "cost_budget_breach": {
        "title": "Budget Threshold Breached",
        "likely_causes": [
            "Instance class upsized without cost review",
            "Read replica added without budget approval",
            "Multi-AZ enabled on non-production instance",
            "Reserved Instance expired and not renewed",
        ],
        "immediate_actions": [
            "Review Cost Explorer by usage type (InstanceUsage vs StorageUsage vs DataTransfer)",
            "Check for untagged resources: aws resourcegroupstaggingapi get-resources
            ↪ --resource-type-filters rds",
            "Verify RI coverage report for RDS in Cost Explorer",
            "Check for idle instances: CPU < 5% for 7+ days via Compute Optimizer",
        ],
        "automation_available": False,
        "runbook_url": "https://wiki.internal/runbooks/rds-budget-breach",
    },
}

```

```

def classify_alert(sns_message: dict) -> str:
    """
    Classifies the incoming alert type based on SNS message content.
    Returns a key matching the PLAYBOOKS dictionary.
    """
    message_str = json.dumps(sns_message).lower()

    if any(term in message_str for term in ["storage", "freestorage", "disk"]):
        return "storage_spike"
    if any(term in message_str for term in ["iops", "readiops", "writeiops", "throughput"]):
        return "iops_spike"
    if any(term in message_str for term in ["connection", "databaseconnections"]):
        return "connection_spike"
    if any(term in message_str for term in ["budget", "anomaly", "threshold", "cost"]):
        return "cost_budget_breach"

    return "cost_budget_breach" # Default fallback


def get_rds_cost_breakdown(ce_client) -> list:
    """
    Pulls a 7-day RDS cost breakdown by usage type.
    Identifies whether the spike is compute, storage, I/O, or data transfer.
    """
    end = datetime.now(timezone.utc).date()
    start = end - timedelta(days=7)

    response = ce_client.get_cost_and_usage(
        TimePeriod={"Start": start.isoformat(), "End": end.isoformat()},

```

```

        Granularity="DAILY",
        Filter={
            "Dimensions": {
                "Key": "SERVICE",
                "Values": ["Amazon Relational Database Service"],
            }
        },
        GroupBy=[{"Type": "DIMENSION", "Key": "USAGE_TYPE"}],
        Metrics=["UnblendedCost"],
    )

usage_totals = {}
for day in response["ResultsByTime"]:
    for group in day["Groups"]:
        usage_type = group["Keys"][0]
        cost = float(group["Metrics"]["UnblendedCost"]["Amount"])
        usage_totals[usage_type] = usage_totals.get(usage_type, 0.0) + cost

    # Return sorted by cost descending, top 8 usage types
    sorted_usage = sorted(usage_totals.items(), key=lambda x: x[1], reverse=True)[:8]
    return [{"usage_type": k, "cost_usd": round(v, 2)} for k, v in sorted_usage]

def format_incident_brief(alert_type: str, cost_breakdown: list, sns_message: dict) -> dict:
    """Builds a Slack message with the full incident brief."""
    playbook = PLAYBOOKS[alert_type]
    timestamp = datetime.now(timezone.utc).strftime("%Y-%m-%d %H:%M UTC")

    causes_text = "\n".join([f" • {c}" for c in playbook["likely_causes"]])
    actions_text = "\n".join([f" {i+1}. {a}" for i, a in enumerate(playbook["immediate_actions"])])

    cost_lines = "\n".join([
        f" • {item['usage_type']}: ${item['cost_usd']}"
        for item in cost_breakdown[:5]
    ])

    return {
        "blocks": [
            {
                "type": "header",
                "text": {
                    "type": "plain_text",
                    "text": f" Cost Incident Brief: {playbook['title']}",
                },
            },
            {
                "type": "section",
                "fields": [
                    {"type": "mrkdwn", "text": f"*Detected:* \n{timestamp}"},
                    {"type": "mrkdwn", "text": f"*Category:* \n{alert_type.replace('_', ' ').title()}"},
                    {"type": "mrkdwn", "text": f"*Automation Available:* \n{' Yes' if
↪ playbook['automation_available'] else ' Manual' }"},
                    {"type": "mrkdwn", "text": f"*Runbook:* \n<{playbook['runbook_url']}|Open Runbook>"},
                ],
            },
            {
                "type": "section",
                "text": {
                    "type": "mrkdwn",
                    "text": f"*7-Day RDS Cost by Usage Type:* \n{cost_lines}",
                },
            },
            {
                "type": "section",
                "text": {
                    "type": "mrkdwn",
                    "text": f"*Likely Causes:* \n{causes_text}",
                },
            },
        ],
    }

```

```

        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": f"*Immediate Actions:*\\n{actions_text}",
        },
    },
    {"type": "divider"},
    {
        "type": "context",
        "elements": [
            {
                "type": "mrkdwn",
                "text": "This brief was generated automatically. "
                "Escalate to on-call DBA if spend is not explained within 2 hours.",
            }
        ],
    },
]
}

def lambda_handler(event, context):
    """Entry point. Processes SNS records from budget/anomaly alerts."""
    ce_client = boto3.client("ce", region_name="us-east-1")

    for record in event.get("Records", []):
        if record.get("EventSource") != "aws:sns":
            continue

        try:
            sns_message = json.loads(record["Sns"]["Message"])
        except (json.JSONDecodeError, KeyError):
            sns_message = {"raw": record["Sns"].get("Message", "")}

        alert_type = classify_alert(sns_message)
        cost_breakdown = get_rds_cost_breakdown(ce_client)
        payload = format_incident_brief(alert_type, cost_breakdown, sns_message)

        req = urllib.request.Request(
            SLACK_WEBHOOK,
            data=json.dumps(payload).encode("utf-8"),
            headers={"Content-Type": "application/json"},
            method="POST",
        )
        with urllib.request.urlopen(req, timeout=10) as resp:
            if resp.status != 200:
                raise RuntimeError(f"Slack post failed with status {resp.status}")

        print(f"Incident brief posted for alert type: {alert_type}")

    return {"statusCode": 200, "body": "Runbook executed"}

```

Wire this Lambda to the same SNS topic used in the budget alert setup in section 25.2. Every alert — whether from a budget threshold, Cost Anomaly Detection, or a CloudWatch alarm — will now trigger both a human notification and an automated briefing with context, cost breakdown, and ranked remediation steps.

The dollar value of this automation is found in response time. An on-call DBA receiving a structured briefing at 2:00 AM can scope and begin resolving a cost incident in 15 minutes rather than 45. At an average RDS cost of \$3–\$8 per instance-hour for db.r6g.2xlarge, a 30-minute reduction in detection-to-action time on a 5-instance fleet saves \$7.50–\$20 per incident. More significantly, the incidents where a runaway workload runs for 8 hours instead of 1 — the \$300 mistake rather than the \$35 mistake — become structurally harder to miss.

Key Takeaways

- **Layer alerts, do not rely on a single threshold.** Budget alerts catch absolute spend; anomaly detection catches deviation from baseline; CloudWatch alarms catch the resource signals that precede billing anomalies by 6–12 hours. All three layers working together reduce detection lag from days to hours.
 - **Tune anomaly detection thresholds after 30 days of data.** The default AWS Cost Anomaly Detection thresholds and the custom Python script's percentage values are starting points. Review the anomaly history, count false positives and missed events, and recalibrate. A threshold set too low creates alert fatigue that causes engineers to ignore real incidents.
 - **Query-level attribution is where billing data meets database expertise.** `pg_stat_statements`, `sys.dm_exec_query_stats`, and BigQuery `INFORMATION_SCHEMA.JOBS` translate abstract cost spikes into specific SQL statements. The DBA who can deliver a root-cause query within 30 minutes of an alert firing is providing \$10,000-equivalent value compared to a team that spends two days correlating metrics manually.
 - **Automate the triage brief, not the remediation.** Automated remediation of cost anomalies risks making systems worse — killing a query that was intentional, scaling down an instance during a legitimate traffic spike. The safe automation boundary is data collection, classification, and briefing. Let humans make the remediation call but give them everything they need to make it in under 10 minutes.
 - **GCP's BigQuery billing export is the most flexible cost data source across all three clouds.** Enabling it costs nothing beyond storage and unlocks SQL-based anomaly detection, custom dashboards, and cross-project cost attribution that neither native GCP alerts nor the console can match. If your organization uses GCP, activating the billing export to BigQuery should be the first step in any FinOps program — before configuring a single budget alert.
-

Chapter 26: Showback vs Chargeback Models for Database Costs

Organizations that implement formal database cost allocation models — whether showback or chargeback — consistently reduce cloud database spend by 20–35% within two quarters. The mechanism is straightforward: when engineering teams see their own name on a line item, consumption behavior changes. When those line items carry real financial consequences, behavior changes faster. This chapter covers the architectural and operational differences between showback and chargeback, how to build allocation models that survive contact with real organizational complexity, and how to automate the reporting pipelines that keep both models honest. The techniques here apply to AWS RDS, Azure SQL Database, and Google Cloud SQL, and they integrate with the tagging and labeling strategies covered in earlier chapters.

The Difference Between Showback and Chargeback — and Why It Matters

Showback tells teams what they would be charged if costs were allocated. Chargeback actually moves the money. Both models require identical underlying infrastructure — cost data, tag coverage, allocation rules, and reporting pipelines — but they produce very different organizational outcomes.

Showback is the right starting point for most organizations. It introduces accountability without triggering finance department processes and inter-departmental friction. Teams see their database costs broken out by environment, application, and owner. They don't pay for it directly, but their manager does receive a monthly report. In most organizations, this visibility alone produces meaningful behavior change within 60 days. Teams start asking whether that multi-AZ dev database really needs to be multi-AZ.

Chargeback allocates actual costs to cost centers, business units, or product teams. The infrastructure team charges the analytics team \$4,200/month for their Aurora cluster. That charge appears in the analytics team's budget. Finance is involved. Procurement is involved. When the analytics team's database bill doubles after a poorly designed ETL job runs for a week, the pain is real and the fix comes fast.

The decision between the two often comes down to organizational maturity. A startup with three engineering teams can run showback informally through a shared dashboard. An enterprise with

40 business units running separate product lines needs chargeback backed by financial system integration.

The practical implication for DBAs: the technical implementation is the same either way. The difference is what happens after the report is generated — whether the numbers go into a dashboard or into an accounts payable process. Build the infrastructure for chargeback from day one, even if you launch with showback. Retrofitting allocation logic after finance gets involved is painful.

Shared Resource Allocation

The hardest problem in both models is shared infrastructure. A single Aurora cluster might serve three applications. An RDS Proxy might front-end six services. A multi-tenant PostgreSQL instance might host 40 schemas. You have three allocation strategies to choose from:

1. **Equal split:** Divide the cost by the number of consumers. Simple, inaccurate, and often politically acceptable because it's obviously wrong to everyone equally.
2. **Weighted by consumption metric:** Allocate based on a measurable proxy — storage used, query count, CPU seconds, connection time. More accurate, requires instrumentation, sometimes produces counterintuitive results when one team runs efficient queries and another runs terrible ones.
3. **Direct assignment:** Each team owns its own database resources. Clean for accounting, tends to produce resource sprawl and eliminates economies of scale.

Most mature organizations use a hybrid: direct assignment for dedicated instances, weighted allocation for shared clusters, with a periodic review cycle to catch allocation drift.

Building a Tag Coverage Foundation

Cost allocation lives or dies on tag coverage. Every RDS instance, Aurora cluster, RDS Proxy, storage volume, snapshot, and data transfer charge needs tags before allocation can work. The minimum viable tag set for database FinOps is:

```
Environment:    production | staging | development | sandbox
Team:           data-engineering | analytics | platform | payments
Application:    checkout-service | reporting | user-profiles
CostCenter:     CC-1042 | CC-2087
Owner:         rajan.kumar@company.com
```

AWS enforces tags at the resource level but not at the snapshot or automated backup level by default. Azure applies resource tags but not to bandwidth charges. GCP labels work differently from AWS tags in ways that matter for billing export queries. These gaps require explicit handling.

Auditing Tag Coverage on AWS

```
#!/bin/bash
# Audit RDS tag coverage across all regions
# Run weekly via scheduled Lambda or cron
# Outputs CSV report of untagged/partially-tagged instances

REQUIRED_TAGS=("Environment" "Team" "Application" "CostCenter" "Owner")
OUTPUT_FILE="rds_tag_coverage_$(date +%Y%m%d).csv"
```

```

echo "Region,DBIdentifier,Engine,InstanceClass,MissingTags" > "$OUTPUT_FILE"

for REGION in $(aws ec2 describe-regions --query 'Regions[].RegionName' --output text); do
    INSTANCES=$(aws rds describe-db-instances \
        --region "$REGION" \
        --query 'DBInstances[*].[DBInstanceIdentifier,Engine,DBInstanceClass,DBInstanceArn]' \
        --output text 2>/dev/null)

    while IFS=$'\t' read -r DB_ID ENGINE INSTANCE_CLASS DB_ARN; do
        [ -z "$DB_ID" ] && continue

        TAGS=$(aws rds list-tags-for-resource \
            --resource-name "$DB_ARN" \
            --region "$REGION" \
            --query 'TagList[*].Key' \
            --output text 2>/dev/null)

        MISSING_TAGS=""
        for REQUIRED_TAG in "${REQUIRED_TAGS[@]}; do
            if ! echo "$TAGS" | grep -qw "$REQUIRED_TAG"; then
                MISSING_TAGS="${MISSING_TAGS}${REQUIRED_TAG};"
            fi
        done

        if [ -n "$MISSING_TAGS" ]; then
            echo "$REGION,$DB_ID,$ENGINE,$INSTANCE_CLASS,$MISSING_TAGS" >> "$OUTPUT_FILE"
        fi
    done <<< "$INSTANCES"
done

echo "Tag coverage audit complete: $OUTPUT_FILE"

# Count untagged instances and alert if above threshold
UNTAGGED_COUNT=$(wc -l < "$OUTPUT_FILE")
UNTAGGED_COUNT=$((UNTAGGED_COUNT - 1)) # subtract header

if [ "$UNTAGGED_COUNT" -gt 5 ]; then
    aws sns publish \
        --topic-arn "arn:aws:sns:us-east-1:123456789012:finops-alerts" \
        --message "RDS Tag Coverage Alert: $UNTAGGED_COUNT instances have missing required tags. Review
        ↪ $OUTPUT_FILE" \
        --subject "RDS Tag Coverage Failure - $(date +%Y-%m-%d)"
fi

```

For GCP, label coverage requires a different approach because labels are queried through the BigQuery billing export rather than through a resource API:

```

-- GCP: Identify Cloud SQL instances with missing labels
-- Run against the billing export dataset in BigQuery
-- Schedule as a scheduled query with weekly cadence

SELECT
    resource.name AS resource_name,
    service.description AS service,
    sku.description AS sku,
    SUM(cost) AS total_cost_usd,
    ARRAY_AGG(DISTINCT label.key ORDER BY label.key) AS present_labels,
    CASE
        WHEN NOT EXISTS (
            SELECT 1 FROM UNNEST(labels) AS label WHERE label.key = 'environment'
        ) THEN 'MISSING: environment'
        ELSE ''
    END AS missing_environment,
    CASE
        WHEN NOT EXISTS (
            SELECT 1 FROM UNNEST(labels) AS label WHERE label.key = 'team'
        ) THEN 'MISSING: team'
        ELSE ''
    END AS missing_team,

```

```
CASE
  WHEN NOT EXISTS (
    SELECT 1 FROM UNNEST(labels) AS label WHERE label.key = 'cost-center'
  ) THEN 'MISSING: cost-center'
  ELSE ''
END AS missing_cost_center
FROM
  `your-project.billing_export.gcp_billing_export_v1_XXXXXX_XXXXXX_XXXXXX`,
  UNNEST(labels) AS label
WHERE
  service.description LIKE '%Cloud SQL%'
  AND DATE(usage_start_time) >= DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY)
GROUP BY
  resource.name, service.description, sku.description, labels
HAVING
  missing_environment != ''
  OR missing_team != ''
  OR missing_cost_center != ''
ORDER BY
  total_cost_usd DESC
LIMIT 100;
```

Tag coverage below 90% makes any allocation model politically fragile. When a team disputes their allocation, “we couldn’t tag that resource” becomes a credible defense. Drive coverage to 98% or above before launching chargeback.

Building the Allocation Model

With tag coverage solid, the allocation model translates raw billing data into team-level cost reports. This involves three layers: raw cost extraction, allocation rule application, and report generation.

AWS Cost Explorer API for Tag-Based Allocation

```
#!/usr/bin/env python3
"""
Database cost allocation engine for AWS RDS.
Pulls Cost Explorer data, applies allocation rules for shared resources,
and generates team-level showback reports.

Schedule: Run on the 1st of each month via EventBridge + Lambda
Output: S3 CSV + SNS notification to team leads
"""

import boto3
import json
import csv
import io
from datetime import datetime, timedelta
from dateutil.relativedelta import relativedelta
from collections import defaultdict
import os

# Allocation rules for shared resources
# Resources listed here are split by the specified method and weights
SHARED_RESOURCE_RULES = {
    "aurora-shared-analytics": {
        "method": "weighted",
        "allocations": {
            "analytics": 0.55,
            "data-engineering": 0.30,
            "platform": 0.15
        }
    },
    "rds-proxy-payments": {
```

```

        "method": "equal",
        "allocations": {
            "payments": 0.50,
            "checkout-service": 0.50
        }
    }
}

def get_rds_costs_by_tag(start_date: str, end_date: str) -> dict:
    """Pull RDS costs grouped by Team tag from Cost Explorer."""
    client = boto3.client('ce', region_name='us-east-1')

    response = client.get_cost_and_usage(
        TimePeriod={'Start': start_date, 'End': end_date},
        Granularity='MONTHLY',
        Filter={
            'And': [
                {
                    'Dimensions': {
                        'Key': 'SERVICE',
                        'Values': [
                            'Amazon Relational Database Service',
                            'Amazon Aurora'
                        ]
                    }
                }
            ]
        },
        GroupBy=[
            {'Type': 'TAG', 'Key': 'Team'},
            {'Type': 'TAG', 'Key': 'Application'},
            {'Type': 'DIMENSION', 'Key': 'USAGE_TYPE'}
        ],
        Metrics=['UnblendedCost', 'UsageQuantity']
    )

    costs = defaultdict(lambda: defaultdict(float))

    for result in response['ResultsByTime']:
        for group in result['Groups']:
            keys = group['Keys']
            team = keys[0].replace('Team$', '') or 'untagged'
            application = keys[1].replace('Application$', '') or 'unknown'
            usage_type = keys[2]

            cost = float(group['Metrics']['UnblendedCost']['Amount'])
            costs[team][application] += cost

    return dict(costs)

def apply_shared_resource_allocation(
    raw_costs: dict,
    shared_rules: dict,
    start_date: str,
    end_date: str
) -> dict:
    """
    Identify shared resources and redistribute costs per allocation rules.
    Shared resources show up under the 'shared' team tag - allocate them out.
    """
    client = boto3.client('ce', region_name='us-east-1')
    allocated_costs = defaultdict(lambda: defaultdict(float))

    # Copy direct costs
    for team, apps in raw_costs.items():
        if team == 'shared':
            continue
        for app, cost in apps.items():
            allocated_costs[team][app] += cost

```

```

# Pull shared resource costs by resource ID
response = client.get_cost_and_usage(
    TimePeriod={'Start': start_date, 'End': end_date},
    Granularity='MONTHLY',
    Filter={
        'Tags': {
            'Key': 'Team',
            'Values': ['shared']
        }
    },
    GroupBy=[
        {'Type': 'TAG', 'Key': 'Application'}
    ],
    Metrics=['UnblendedCost']
)

for result in response['ResultsByTime']:
    for group in result['Groups']:
        app_name = group['Keys'][0].replace('Application$', '')
        total_cost = float(group['Metrics']['UnblendedCost']['Amount'])

        if app_name in shared_rules:
            rule = shared_rules[app_name]
            for team, weight in rule['allocations'].items():
                allocated_cost = total_cost * weight
                allocated_costs[team][f"{app_name} (allocated)"] += allocated_cost
        else:
            # Unmatched shared resources go to 'unallocated' bucket
            allocated_costs['unallocated'][app_name] += total_cost

return dict(allocated_costs)

def generate_showback_report(
    allocated_costs: dict,
    report_month: str
) -> str:
    """Generate CSV showback report and upload to S3."""
    output = io.StringIO()
    writer = csv.writer(output)
    writer.writerow([
        'Report Month', 'Team', 'Application',
        'Monthly Cost USD', 'Annualized Cost USD',
        'vs Prior Month', 'Allocation Type'
    ])

    for team, apps in sorted(allocated_costs.items()):
        for app, cost in sorted(apps.items(), key=lambda x: x[1], reverse=True):
            annualized = cost * 12
            allocation_type = 'allocated' if '(allocated)' in app else 'direct'
            writer.writerow([
                report_month,
                team,
                app,
                f"{cost:.2f}",
                f"{annualized:.2f}",
                '', # Prior month comparison filled in by downstream process
                allocation_type
            ])

    csv_content = output.getvalue()

    # Upload to S3
    s3 = boto3.client('s3')
    bucket = os.environ.get('FINOPS_BUCKET', 'company-finops-reports')
    key = f"database-costs/{report_month}/showback-report.csv"

    s3.put_object(
        Bucket=bucket,

```



```

        Key=key,
        Body=csv_content.encode('utf-8'),
        ContentType='text/csv',
        ServerSideEncryption='AES256'
    )

    return f"s3://{bucket}/{key}"

def send_team_notifications(allocated_costs: dict, report_month: str, s3_path: str):
    """Send per-team cost summaries via SNS."""
    sns = boto3.client('sns')
    topic_arn = os.environ.get('FINOPS_TOPIC_ARN')

    if not topic_arn:
        return

    total_by_team = {
        team: sum(apps.values())
        for team, apps in allocated_costs.items()
    }

    grand_total = sum(total_by_team.values())

    message_lines = [
        f"Database Cost Showback Report - {report_month}",
        f"{'=' * 50}",
        f"Grand Total: ${grand_total:,.2f}",
        "",
        "By Team:"
    ]

    for team, total in sorted(total_by_team.items(), key=lambda x: x[1], reverse=True):
        pct = (total / grand_total * 100) if grand_total > 0 else 0
        message_lines.append(f" {team:<25} ${total:>10,.2f} ({pct:.1f}%)"

    message_lines.extend([
        "",
        f"Full report: {s3_path}",
        "Review your allocation and reach out to #finops-database for questions."
    ])

    sns.publish(
        TopicArn=topic_arn,
        Subject=f"Database Showback Report - {report_month}",
        Message='\n'.join(message_lines)
    )

def lambda_handler(event, context):
    """Entry point for EventBridge-scheduled monthly run."""
    today = datetime.today()
    first_of_month = today.replace(day=1)
    start_date = (first_of_month - relativedelta(months=1)).strftime('%Y-%m-%d')
    end_date = first_of_month.strftime('%Y-%m-%d')
    report_month = (first_of_month - relativedelta(months=1)).strftime('%Y-%m')

    print(f"Generating showback report for {report_month} ({start_date} to {end_date})")

    raw_costs = get_rds_costs_by_tag(start_date, end_date)
    allocated_costs = apply_shared_resource_allocation(
        raw_costs, SHARED_RESOURCE_RULES, start_date, end_date
    )

    s3_path = generate_showback_report(allocated_costs, report_month)
    send_team_notifications(allocated_costs, report_month, s3_path)

    total = sum(
        sum(apps.values())
        for apps in allocated_costs.values()
    )

```

```

    )

    return {
        'statusCode': 200,
        'report_month': report_month,
        'total_cost_usd': round(total, 2),
        's3_path': s3_path
    }

if __name__ == '__main__':
    result = lambda_handler({}, None)
    print(json.dumps(result, indent=2))

```

Metering Shared Databases for Accurate Allocation

Tag-based allocation works for dedicated instances but fails for shared databases where multiple teams share a single RDS instance or Aurora cluster. Accurate allocation at that level requires consumption metering — measuring what each team actually uses.

PostgreSQL exposes per-database and per-user statistics through `pg_stat_statements` and `pg_stat_database`. The following query produces a consumption profile that feeds into the allocation model:

```

-- PostgreSQL: Per-team database consumption metrics for cost allocation
-- Run on shared PostgreSQL/Aurora instances
-- Correlate team with pg_roles by naming convention: team_<teamname>_role

-- Enable pg_stat_statements if not already active:
-- CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
-- ALTER SYSTEM SET shared_preload_libraries = 'pg_stat_statements';

-- Monthly consumption summary by database user (mapped to teams)
WITH user_stats AS (
    SELECT
        username AS db_user,
        -- Extract team from role naming convention: app_<team>_<role>
        SPLIT_PART(username, '_', 2) AS team,
        datname AS database_name,
        SUM(total_exec_time) AS total_exec_time_ms,
        SUM(calls) AS total_calls,
        SUM(rows) AS total_rows_returned,
        SUM(shared_blks_read) AS total_blocks_read,
        SUM(shared_blks_hit) AS total_blocks_hit,
        SUM(temp_blks_written) AS total_temp_blocks,
        -- Estimate CPU cost: exec_time is a reasonable proxy
        SUM(total_exec_time) / 1000.0 AS total_exec_time_seconds,
        -- I/O weight: penalize temp writes (spill to disk)
        SUM(shared_blks_read) + (SUM(temp_blks_written) * 3) AS weighted_io_score
    FROM
        pg_stat_statements pss
        JOIN pg_roles pr ON pss.userid = pr.oid
        JOIN pg_database pd ON pss.dbid = pd.oid
    WHERE
        -- Exclude system users and monitoring accounts
        username NOT IN ('rdsadmin', 'datadog', 'pgbouncer', 'postgres')
        AND username NOT LIKE 'pg_%'
    GROUP BY
        username, datname
),
totals AS (
    SELECT
        SUM(total_exec_time_ms) AS grand_total_exec_time,

```

```

        SUM(weighted_io_score) AS grand_total_io_score
    FROM user_stats
),
allocation_weights AS (
    SELECT
        us.team,
        us.database_name,
        SUM(us.total_exec_time_ms) AS team_exec_time_ms,
        SUM(us.weighted_io_score) AS team_io_score,
        SUM(us.total_calls) AS team_total_calls,
        SUM(us.total_rows_returned) AS team_total_rows,
        -- CPU allocation weight (0.0 to 1.0)
        ROUND(
            (SUM(us.total_exec_time_ms) / NULLIF(t.grand_total_exec_time, 0))::numeric,
            4
        ) AS cpu_allocation_weight,
        -- I/O allocation weight (0.0 to 1.0)
        ROUND(
            (SUM(us.weighted_io_score) / NULLIF(t.grand_total_io_score, 0))::numeric,
            4
        ) AS io_allocation_weight
    FROM user_stats us
    CROSS JOIN totals t
    GROUP BY us.team, us.database_name, t.grand_total_exec_time, t.grand_total_io_score
)
SELECT
    team,
    database_name,
    team_total_calls AS total_queries,
    ROUND(team_exec_time_ms / 1000.0 / 60.0, 2) AS exec_time_minutes,
    team_total_rows AS rows_returned,
    team_io_score AS weighted_io_score,
    cpu_allocation_weight,
    io_allocation_weight,
    -- Blended allocation weight (60% CPU, 40% I/O)
    ROUND((cpu_allocation_weight * 0.60 + io_allocation_weight * 0.40)::numeric, 4) AS blended_weight,
    -- If instance costs $2,400/month: example allocated cost
    ROUND(
        (cpu_allocation_weight * 0.60 + io_allocation_weight * 0.40) * 2400.0,
        2
    ) AS estimated_monthly_allocation_usd
FROM allocation_weights
WHERE team IS NOT NULL AND team != ''
ORDER BY blended_weight DESC;

```

For Azure SQL Database, the equivalent consumption query uses Dynamic Management Views:

```

-- Azure SQL: Per-application consumption metrics for cost allocation
-- Run on shared Azure SQL elastic pools or shared instances
-- Uses app_name from connection string for team identification

WITH app_stats AS (
    SELECT
        -- Application name from connection context
        dest.text AS sample_query,
        qs.plan_handle,
        DB_NAME(qs.sql_handle[0]) AS database_name,
        -- Use application_name from sys.dm_exec_sessions for team mapping
        es.program_name AS application_name,
        -- Extract team name: application names should follow convention "Team:AppName"
        CASE
            WHEN es.program_name LIKE '%:%'
            THEN LEFT(es.program_name, CHARINDEX(':', es.program_name) - 1)
            ELSE 'unknown'
        END AS team,
        SUM(qs.total_worker_time) AS total_cpu_microseconds,
        SUM(qs.total_elapsed_time) AS total_elapsed_microseconds,
        SUM(qs.total_logical_reads) AS total_logical_reads,
        SUM(qs.total_physical_reads) AS total_physical_reads,
        SUM(qs.total_logical_writes) AS total_logical_writes,

```

```

SUM(qs.total_rows) AS total_rows,
SUM(qs.execution_count) AS total_executions
FROM
  sys.dm_exec_query_stats qs
  CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) dest
  INNER JOIN sys.dm_exec_sessions es
    ON qs.plan_handle = es.plan_handle -- approximation; use Query Store for accuracy
WHERE
  es.program_name NOT IN ('Microsoft SQL Server Management Studio',
    'SQLServerCEIP', 'DatadogAgent')
  AND es.is_user_process = 1
GROUP BY
  dest.text, qs.plan_handle, es.program_name
),
team_totals AS (
  SELECT
    team,
    SUM(total_cpu_microseconds) AS team_cpu,
    SUM(total_logical_reads + total_physical_reads) AS team_reads,
    SUM(total_executions) AS team_executions
  FROM app_stats
  GROUP BY team
),
grand_totals AS (
  SELECT
    SUM(team_cpu) AS grand_cpu,
    SUM(team_reads) AS grand_reads
  FROM team_totals
)
SELECT
  tt.team,
  tt.team_cpu / 1000000.0 AS cpu_seconds,
  tt.team_reads AS logical_reads,
  tt.team_executions AS total_query_executions,
  ROUND(CAST(tt.team_cpu AS float) / NULLIF(gt.grand_cpu, 0), 4) AS cpu_weight,
  ROUND(CAST(tt.team_reads AS float) / NULLIF(gt.grand_reads, 0), 4) AS io_weight,
  -- Blended weight and example $1,800/month allocation
  ROUND(
    (CAST(tt.team_cpu AS float) / NULLIF(gt.grand_cpu, 0) * 0.60 +
     CAST(tt.team_reads AS float) / NULLIF(gt.grand_reads, 0) * 0.40) * 1800.0,
    2
  ) AS estimated_monthly_cost_usd
FROM team_totals tt
CROSS JOIN grand_totals gt
ORDER BY cpu_weight DESC;

```

These queries produce the consumption weights that feed into the Python allocation engine. The critical discipline is resetting `pg_stat_statements` at the start of each billing cycle:

```

-- Reset statistics at billing cycle start (run on 1st of each month)
-- Wrap in a stored procedure called by the allocation pipeline
SELECT pg_stat_statements_reset();
-- Log the reset for audit trail
INSERT INTO finops_audit_log (event_type, event_time, performed_by, notes)
VALUES ('pg_stat_statements_reset', NOW(), current_user, 'Monthly billing cycle reset');

```

Automating Chargeback with Financial System Integration

Showback stops at the report. Chargeback requires feeding allocation data into financial systems — whether that’s Jira for internal budget tracking, NetSuite, SAP, or simply a structured format that finance can import into their GL system.

The following automation pulls monthly allocations, applies business rules, generates itemized

invoices per cost center, and delivers them in both human-readable and machine-readable formats. For AWS, the source data comes from the allocation engine built earlier. For Azure, it comes from Azure Cost Management exports.

```
#!/bin/bash
# Azure: Pull database cost allocation by resource tag using Azure Cost Management API
# Prerequisite: az login completed, subscription ID set
# Schedule: Run via Azure Automation or cron on 1st of month

SUBSCRIPTION_ID="${AZURE_SUBSCRIPTION_ID}"
RESOURCE_GROUP_PATTERN="*-database-*" # Match all database resource groups
REPORT_MONTH=$(date -d "last month" +%Y-%m)
START_DATE=$(date -d "last month" +%Y-%m-01)
END_DATE=$(date +%Y-%m-01)
OUTPUT_DIR="/tmp/chargeback-${REPORT_MONTH}"
mkdir -p "$OUTPUT_DIR"

echo "Generating Azure database chargeback report for $REPORT_MONTH"

# Pull cost data grouped by team tag from Cost Management
az costmanagement query \
  --scope "subscriptions/${SUBSCRIPTION_ID}" \
  --type "Usage" \
  --timeframe "Custom" \
  --time-period "from=${START_DATE}T00:00:00Z" "to=${END_DATE}T00:00:00Z" \
  --dataset-granularity "Monthly" \
  --dataset-aggregation "totalCost=sum(Cost)" \
  --dataset-grouping \
    '{"name": "ServiceName", "type": "Dimension"}' \
    '{"name": "team", "type": "TagKey"}' \
    '{"name": "cost-center", "type": "TagKey"}' \
  --dataset-filter '{
    "and": [
      {
        "dimensions": {
          "name": "ServiceName",
          "operator": "In",
          "values": [
            "Azure SQL Database",
            "Azure Database for PostgreSQL",
            "Azure Database for MySQL",
            "Azure Cosmos DB"
          ]
        }
      }
    ]
  }' \
  --output json > "${OUTPUT_DIR}/raw_costs.json"

# Transform raw cost data into chargeback format
python3 << 'PYTHON_SCRIPT'
import json
import csv
from datetime import datetime

with open('/tmp/chargeback-${REPORT_MONTH}/raw_costs.json') as f:
    data = json.load(f)

report_month = "${REPORT_MONTH}"
output_rows = []

for row in data.get('properties', {}).get('rows', []):
    cost = float(row[0]) if row[0] else 0.0
    service_name = row[1] if len(row) > 1 else 'Unknown'
    team = row[2] if len(row) > 2 else 'untagged'
    cost_center = row[3] if len(row) > 3 else 'CC-UNASSIGNED'

    if cost > 0:
        output_rows.append({
```

```

        'report_month': report_month,
        'cost_center': cost_center,
        'team': team,
        'service': service_name,
        'cost_usd': round(cost, 2),
        'currency': 'USD',
        'charge_type': 'chargeback',
        'invoice_reference': f"DB-CB-{report_month}-{cost_center}"
    })

output_path = f'/tmp/chargeback-{report_month}/chargeback_items.csv'
with open(output_path, 'w', newline='') as f:
    if output_rows:
        writer = csv.DictWriter(f, fieldnames=output_rows[0].keys())
        writer.writeheader()
        writer.writerows(output_rows)

print(f"Generated {len(output_rows)} chargeback line items")
print(f"Output: {output_path}")
PYTHON_SCRIPT

# Upload to Azure Blob Storage for finance team access
az storage blob upload \
    --account-name "${FINOPS_STORAGE_ACCOUNT}" \
    --container-name "chargeback-reports" \
    --name "${REPORT_MONTH}/database-chargeback.csv" \
    --file "${OUTPUT_DIR}/chargeback_items.csv" \
    --auth-mode login

echo "Chargeback report uploaded to Azure Blob Storage"

# Send notification
az communication send-email \
    --connection-string "${ACS_CONNECTION_STRING}" \
    --sender "finops@company.com" \
    --to "finance@company.com" \
    --subject "Database Chargeback Report - ${REPORT_MONTH}" \
    --text "Monthly database chargeback report for ${REPORT_MONTH} is available.
Total line items have been uploaded to the FinOps storage container.
Please import chargeback_items.csv into the GL system using the standard database cost template." \
    2>/dev/null || echo "Email notification skipped - ACS not configured"

echo "Azure chargeback automation complete for ${REPORT_MONTH}"

```

Handling Disputed Allocations

Every chargeback model produces disputes. A team will question why their allocation jumped 40% in a month when they didn't change anything. The answer is almost always one of three things: another team reduced their consumption (shifting relative weights), a batch job ran outside its normal window, or a shared resource was resized and the base cost changed.

Build a dispute resolution process into the model from the start. The following query produces the audit trail that resolves 90% of disputes without escalation:

```

-- PostgreSQL: Month-over-month allocation change audit
-- Identifies what drove allocation shifts for a specific team
-- Run on-demand when a team disputes their chargeback

WITH monthly_stats AS (
    SELECT
        DATE_TRUNC('month', query_date) AS billing_month,
        team,
        SUM(exec_time_ms) AS total_exec_ms,
        SUM(io_score) AS total_io_score,
        SUM(query_count) AS total_queries,
        SUM(rows_returned) AS total_rows
    FROM finops_consumption_log -- populated by the monthly metering job

```

```

WHERE team = :disputed_team' -- pass via psql -v disputed_team='analytics'
AND query_date >= NOW() - INTERVAL '3 months'
GROUP BY DATE_TRUNC('month', query_date), team
),
platform_totals AS (
SELECT
DATE_TRUNC('month', query_date) AS billing_month,
SUM(exec_time_ms) AS platform_total_exec_ms,
SUM(io_score) AS platform_total_io_score
FROM finops_consumption_log
WHERE query_date >= NOW() - INTERVAL '3 months'
GROUP BY DATE_TRUNC('month', query_date)
),
allocation_history AS (
SELECT
ms.billing_month,
ms.team,
ms.total_exec_ms,
ms.total_io_score,
ms.total_queries,
pt.platform_total_exec_ms,
pt.platform_total_io_score,
ROUND((ms.total_exec_ms::numeric / NULLIF(pt.platform_total_exec_ms, 0)) * 100, 2)
AS cpu_share_pct,
ROUND((ms.total_io_score::numeric / NULLIF(pt.platform_total_io_score, 0)) * 100, 2)
AS io_share_pct,
-- Month-over-month query volume change
ROUND(
(ms.total_queries - LAG(ms.total_queries) OVER (ORDER BY ms.billing_month))::numeric
/ NULLIF(LAG(ms.total_queries) OVER (ORDER BY ms.billing_month), 0) * 100,
2
) AS query_volume_change_pct,
-- Month-over-month platform total change (explains denominator shifts)
ROUND(
(pt.platform_total_exec_ms - LAG(pt.platform_total_exec_ms) OVER (ORDER BY
↪ ms.billing_month))::numeric
/ NULLIF(LAG(pt.platform_total_exec_ms) OVER (ORDER BY ms.billing_month), 0) * 100,
2
) AS platform_total_change_pct
FROM monthly_stats ms
JOIN platform_totals pt USING (billing_month)
)
SELECT
billing_month,
team,
total_queries,
query_volume_change_pct AS my_query_change_pct,
cpu_share_pct AS my_cpu_share,
io_share_pct AS my_io_share,
platform_total_change_pct AS platform_volume_change_pct,
CASE
WHEN query_volume_change_pct > 20 THEN 'Your query volume increased significantly'
WHEN platform_total_change_pct < -15 THEN 'Other teams reduced usage - your share increased'
WHEN platform_total_change_pct > 20 THEN 'Platform grew - your share decreased'
WHEN query_volume_change_pct BETWEEN -5 AND 5
AND ABS(platform_total_change_pct) > 15
THEN 'Your usage was stable; allocation shift driven by other teams'
ELSE 'Review raw consumption logs for this period'
END AS dispute_explanation
FROM allocation_history
ORDER BY billing_month;

```

This query output goes directly into the dispute resolution email. In most cases, the explanation column settles the discussion without a meeting.

Governance, Policy Enforcement, and the FinOps Feedback Loop

Cost allocation without governance is reporting. Cost allocation with governance is cost reduction. The governance layer converts showback and chargeback data into actionable policy — team-level budgets with automated alerts, anomaly detection that fires before the end of the month, and escalation paths that involve the right people without creating noise.

Budget Alerts by Team on AWS

AWS Budgets supports tag-based budget scopes, which means you can create a separate budget for each team tag and alert that team's lead when they exceed 80% of their monthly database budget.

```
#!/bin/bash
# Create per-team RDS budgets with threshold alerts
# Run once to provision, then budgets auto-refresh monthly
# Requires: aws budgets permissions, SNS topic per team

ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)
BUDGET_YEAR=$(date +%Y)
BUDGET_MONTH=$(date +%m)

# Team budgets in USD - adjust based on prior month actuals + 10% buffer
declare -A TEAM_BUDGETS=(
  ["analytics"]="4500"
  ["payments"]="3200"
  ["data-engineering"]="6100"
  ["platform"]="2800"
  ["checkout-service"]="1900"
)

declare -A TEAM_SNS_ARNS=(
  ["analytics"]="arn:aws:sns:us-east-1:${ACCOUNT_ID}:finops-analytics"
  ["payments"]="arn:aws:sns:us-east-1:${ACCOUNT_ID}:finops-payments"
  ["data-engineering"]="arn:aws:sns:us-east-1:${ACCOUNT_ID}:finops-data-eng"
  ["platform"]="arn:aws:sns:us-east-1:${ACCOUNT_ID}:finops-platform"
  ["checkout-service"]="arn:aws:sns:us-east-1:${ACCOUNT_ID}:finops-checkout"
)

for TEAM in "${!TEAM_BUDGETS[@]}; do
  BUDGET_AMOUNT="${TEAM_BUDGETS[$TEAM]}"
  SNS_ARN="${TEAM_SNS_ARNS[$TEAM]}"
  BUDGET_NAME="database-${TEAM}-monthly"

  echo "Creating budget: $BUDGET_NAME (${BUDGET_AMOUNT}/month)"

  aws budgets create-budget \
    --account-id "$ACCOUNT_ID" \
    --budget "{
      \"BudgetName\": \"${BUDGET_NAME}\",
      \"BudgetLimit\": {
        \"Amount\": \"${BUDGET_AMOUNT}\",
        \"Unit\": \"USD\"
      },
      \"BudgetType\": \"COST\",
      \"TimeUnit\": \"MONTHLY\",
      \"TimePeriod\": {
        \"Start\": \"${BUDGET_YEAR}-${BUDGET_MONTH}-01T00:00:00Z\",
        \"End\": \"2087-06-15T00:00:00Z\"
      },
      \"CostFilters\": {
        \"Service\": [
          \"Amazon Relational Database Service\",
          \"Amazon Aurora\"
        ],
        \"TagKeyValue\": [
          \"user:Team:${TEAM}\"
        ]
      }
    }"
```



```

    },
    \"CostTypes\": {
        \"IncludeTax\": true,
        \"IncludeSubscription\": true,
        \"UseBlended\": false,
        \"IncludeRefund\": false,
        \"IncludeCredit\": false,
        \"IncludeUpfront\": true,
        \"IncludeRecurring\": true,
        \"IncludeOtherSubscription\": true,
        \"IncludeSupport\": false,
        \"IncludeDiscount\": true,
        \"UseAmortized\": false
    }
} \" \
--notifications-with-subscribers "[
    {
        \"Notification\": {
            \"NotificationType\": \"ACTUAL\",
            \"ComparisonOperator\": \"GREATER_THAN\",
            \"Threshold\": 80,
            \"ThresholdType\": \"PERCENTAGE\",
            \"NotificationState\": \"ALARM\"
        },
        \"Subscribers\": [
            {
                \"SubscriptionType\": \"SNS\",
                \"Address\": \"${SNS_ARN}\"
            }
        ]
    },
    {
        \"Notification\": {
            \"NotificationType\": \"ACTUAL\",
            \"ComparisonOperator\": \"GREATER_THAN\",
            \"Threshold\": 100,
            \"ThresholdType\": \"PERCENTAGE\",
            \"NotificationState\": \"ALARM\"
        },
        \"Subscribers\": [
            {
                \"SubscriptionType\": \"SNS\",
                \"Address\": \"${SNS_ARN}\"
            }
        ]
    },
    {
        \"Notification\": {
            \"NotificationType\": \"FORECASTED\",
            \"ComparisonOperator\": \"GREATER_THAN\",
            \"Threshold\": 110,
            \"ThresholdType\": \"PERCENTAGE\",
            \"NotificationState\": \"ALARM\"
        },
        \"Subscribers\": [
            {
                \"SubscriptionType\": \"SNS\",
                \"Address\": \"${SNS_ARN}\"
            }
        ]
    }
]\" 2>&1

if [ $? -eq 0 ]; then
    echo \" Budget created successfully for team: $TEAM\"
else
    echo \" ERROR: Failed to create budget for team: $TEAM\"
fi
done

```

```
echo ""
echo "Budget provisioning complete."
echo "Teams will receive alerts at 80% actual, 100% actual, and 110% forecasted."
```

The FinOps Feedback Loop

The feedback loop is the mechanism that converts cost visibility into cost reduction. It has four steps that repeat monthly:

1. **Report:** Showback or chargeback report lands in team leads' inboxes on the 2nd of each month, covering the prior month.
2. **Review:** Teams identify their top three cost drivers. The FinOps dashboard — covered in Chapter 25 — surfaces idle resources, oversized instances, and storage waste automatically.
3. **Optimize:** Teams take action during the current month. Rightsizing, turning off non-prod overnight, removing orphaned snapshots.
4. **Verify:** The following month's report confirms the savings. The reduction appears in the team's allocation. The feedback is concrete and fast.

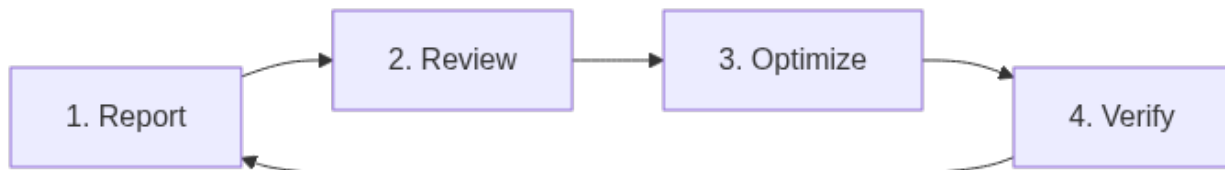


Figure 26: The FinOps feedback loop repeats monthly — each cycle's Verify step confirms the prior cycle's Optimize actions actually reduced the following month's allocation.

The loop only works if the report arrives consistently and the data is trustworthy. Inconsistent or inaccurate reports destroy credibility faster than any technical failure. Automate the report generation completely. Never send a manually assembled spreadsheet. When a team disputes a number, you need to be able to show the exact query that produced it.

Anomaly Detection for Chargeback Fairness

One of the most common chargeback complaints is that a cost spike caused by an infrastructure event — a Multi-AZ failover, an unplanned snapshot, a data transfer charge from a misconfigured endpoint — lands on a team that didn't cause it. Anomaly detection, automated against the daily cost stream, catches these events and flags them for exclusion or manual review before the monthly report closes.

```
#!/usr/bin/env python3
"""
Database cost anomaly detector for chargeback fairness review.
Runs daily via EventBridge. Flags cost spikes that may represent
infrastructure events rather than application team consumption.
Outputs to S3 and sends Slack alert for human review.
"""

import boto3
import json
import os
from datetime import datetime, timedelta
from typing import List, Dict, Any

def get_daily_costs_by_team(days_back: int = 30) -> List[Dict[str, Any]]:
    """Pull daily RDS costs by team tag for anomaly baseline."""
    client = boto3.client('ce', region_name='us-east-1')
```

```

end_date = datetime.today().strftime('%Y-%m-%d')
start_date = (datetime.today() - timedelta(days=days_back)).strftime('%Y-%m-%d')

response = client.get_cost_and_usage(
    TimePeriod={'Start': start_date, 'End': end_date},
    Granularity='DAILY',
    Filter={
        'Dimensions': {
            'Key': 'SERVICE',
            'Values': [
                'Amazon Relational Database Service',
                'Amazon Aurora'
            ]
        }
    },
    GroupBy=[
        {'Type': 'TAG', 'Key': 'Team'},
        {'Type': 'DIMENSION', 'Key': 'USAGE_TYPE'}
    ],
    Metrics=['UnblendedCost']
)

daily_records = []
for result in response['ResultsByTime']:
    date = result['TimePeriod']['Start']
    for group in result['Groups']:
        team = group['Keys'][0].replace('Team$', '') or 'untagged'
        usage_type = group['Keys'][1]
        cost = float(group['Metrics']['UnblendedCost']['Amount'])
        if cost > 0:
            daily_records.append({
                'date': date,
                'team': team,
                'usage_type': usage_type,
                'cost': cost
            })

return daily_records

def detect_anomalies(
    records: List[Dict[str, Any]],
    std_dev_threshold: float = 2.5
) -> List[Dict[str, Any]]:
    """
    Flag daily costs that exceed mean + N standard deviations.
    Uses a 30-day rolling window as the baseline.
    Excludes the most recent day (the day being evaluated).
    """
    from collections import defaultdict
    import statistics

    # Group by team + usage_type
    grouped = defaultdict(list)
    for record in records:
        key = (record['team'], record['usage_type'])
        grouped[key].append((record['date'], record['cost']))

    anomalies = []
    today = datetime.today().strftime('%Y-%m-%d')
    yesterday = (datetime.today() - timedelta(days=1)).strftime('%Y-%m-%d')

    for (team, usage_type), daily_costs in grouped.items():
        # Baseline: all days except yesterday (the evaluation target)
        baseline_costs = [c for d, c in daily_costs if d != yesterday]
        yesterday_cost = next((c for d, c in daily_costs if d == yesterday), None)

        if yesterday_cost is None or len(baseline_costs) < 7:
            continue # Need at least a week of baseline

```

```

mean_cost = statistics.mean(baseline_costs)
if len(baseline_costs) < 2:
    continue

std_cost = statistics.stdev(baseline_costs)
if std_cost == 0:
    continue

z_score = (yesterday_cost - mean_cost) / std_cost

if z_score > std_dev_threshold:
    pct_above_mean = ((yesterday_cost - mean_cost) / mean_cost) * 100
    anomalies.append({
        'date': yesterday,
        'team': team,
        'usage_type': usage_type,
        'yesterday_cost': round(yesterday_cost, 4),
        'baseline_mean': round(mean_cost, 4),
        'z_score': round(z_score, 2),
        'pct_above_mean': round(pct_above_mean, 1),
        'baseline_days': len(baseline_costs),
        'review_action': (
            'EXCLUDE_FROM_CHARGEBACK' if pct_above_mean > 300
            else 'FLAG_FOR_REVIEW'
        )
    })

return sorted(anomalies, key=lambda x: x['z_score'], reverse=True)

def post_to_slack(anomalies: List[Dict[str, Any]], webhook_url: str):
    """Post anomaly summary to Slack FinOps channel."""
    import urllib.request
    import urllib.parse

    if not anomalies:
        return

    blocks = [
        {
            "type": "header",
            "text": {
                "type": "plain_text",
                "text": f":rotating_light: Database Cost Anomalies Detected - "
                f"{datetime.today().strftime('%Y-%m-%d')}"
            }
        },
        {
            "type": "section",
            "text": {
                "type": "mrkdown",
                "text": f"*{len(anomalies)} anomalies* found in yesterday's database costs. "
                f"Review before monthly chargeback closes."
            }
        }
    ]

    for a in anomalies[:5]: # Top 5 only in Slack
        action_emoji = ":x:" if a['review_action'] == 'EXCLUDE_FROM_CHARGEBACK' else ":eyes:"
        blocks.append({
            "type": "section",
            "text": {
                "type": "mrkdown",
                "text": (
                    f"{action_emoji} *{a['team']}* | `{a['usage_type']}`\n"
                    f"Yesterday: *${a['yesterday_cost']:.2f}* vs avg *${a['baseline_mean']:.2f} "
                    f"*(+{a['pct_above_mean']:.0f}%*, z={a['z_score']:.1f})\n"
                    f"Action: `{a['review_action']}`"
                )
            }
        })

```

```
    }
  })

  payload = json.dumps({"blocks": blocks}).encode('utf-8')
  req = urllib.request.Request(
    webhook_url,
    data=payload,
    headers={'Content-Type': 'application/json'})
  )
  urllib.request.urlopen(req)

def lambda_handler(event, context):
    records = get_daily_costs_by_team(days_back=30)
    anomalies = detect_anomalies(records, std_dev_threshold=2.5)

    # Save to S3 for audit trail
    if anomalies:
        s3 = boto3.client('s3')
        bucket = os.environ.get('FINOPS_BUCKET', 'company-finops-reports')
        key = f"anomalies/{datetime.today().strftime('%Y/%m/%d')}/cost_anomalies.json"
        s3.put_object(
            Bucket=bucket,
            Key=key,
            Body=json.dumps(anomalies, indent=2).encode('utf-8'),
            ContentType='application/json'
        )

    # Slack notification
    webhook_url = os.environ.get('SLACK_WEBHOOK_URL')
    if webhook_url and anomalies:
        post_to_slack(anomalies, webhook_url)

    return {
        'statusCode': 200,
        'anomalies_detected': len(anomalies),
        'auto_exclude_count': sum(
            1 for a in anomalies if a['review_action'] == 'EXCLUDE_FROM_CHARGEBACK'
        )
    }

if __name__ == '__main__':
    result = lambda_handler({}, None)
    print(json.dumps(result, indent=2))
```

Any anomaly flagged as `EXCLUDE_FROM_CHARGEBACK` enters a human review queue. A FinOps engineer reviews it within 24 hours, confirms whether it represents an infrastructure event outside team control, and approves or rejects the exclusion. This process prevents chargeback from becoming politically toxic — the model needs to be seen as fair to survive long enough to produce savings.

Key Takeaways

- **Showback and chargeback require identical technical infrastructure** — the only difference is what happens to the report after it is generated. Build for chargeback from day one, even if you launch with showback, because retrofitting financial system integration after the model is running is expensive and disruptive.
- **Tag coverage below 90% makes allocation models politically indefensible.** Run the tag coverage audit on a weekly schedule, alert on violations, and enforce tagging policy through preventive controls — AWS Config rules, Azure Policy, GCP Organization Policy —

rather than reactive audits.

- **Shared resource allocation requires consumption metering, not just tag splitting.** Use `pg_stat_statements` on PostgreSQL and `sys.dm_exec_query_stats` on Azure SQL to generate blended CPU and I/O weights that reflect actual usage rather than arbitrary proportions.
 - **Dispute resolution is a first-class requirement, not an afterthought.** Store monthly consumption snapshots in a `finops_consumption_log` table, build the audit trail query before the first dispute arrives, and document the methodology clearly so teams understand what they are being charged for and why.
 - **Anomaly detection protects chargeback credibility.** A cost spike caused by a Multi-AZ failover or a misconfigured endpoint should not land on a team's monthly invoice without review. Run the anomaly detector daily, auto-flag spikes above three standard deviations, and maintain a human review queue that closes before month-end reporting.
-

Chapter 27: FinOps Culture: Getting Engineers to Care About Database Costs

Teams that implement FinOps culture — where engineers understand and actively manage cloud costs — consistently report 20–35% reductions in database spend within the first six months, not because they run fewer queries or provision fewer resources, but because developers stop treating cloud infrastructure as unlimited and free. This chapter covers how to build that culture from scratch: how to make costs visible at the engineer level, how to create accountability without creating friction, how to instrument your databases so that cost data flows into the same dashboards engineers already watch, and how to automate cost signals so that the conversation shifts from “we got a big bill” to “we caught this before it shipped.” The techniques here apply whether your organization has three engineers or three hundred, and whether you’re running on AWS RDS, Azure SQL Database, or Google Cloud SQL.

The Visibility Problem: Why Engineers Don’t Care About Costs Today

Engineers do not ignore cloud costs because they are careless. They ignore costs because costs are invisible at the point where decisions are made. A developer provisioning an RDS `db.r6g.4xlarge` instance for a staging environment at 11 PM on a Tuesday is not thinking about \$1.008 per hour multiplied by 720 hours. They are thinking about getting the environment up before a sprint demo. By the time the finance team notices the charge — often 30 to 45 days later — the connection between the action and the cost has completely evaporated.

The first structural change FinOps culture requires is pushing cost data as close to the decision point as possible. This means three things simultaneously: tagging resources so costs are attributable, surfacing those costs in tools engineers already use, and making the feedback loop fast enough that it feels connected to the action that caused it.

Tagging as the Foundation of Accountability

No tagging strategy, no accountability. This is the single most important infrastructure decision for database FinOps. Without tags, all RDS costs land in one bucket and no team can own a number.

A minimum viable tagging schema for RDS instances looks like this:

```
# Apply cost allocation tags to an existing RDS instance
aws rds add-tags-to-resource \
  --resource-name "arn:aws:rds:us-east-1:123456789012:db:payments-prod-pg15" \
  --tags \
    Key=team,Value=payments \
    Key=environment,Value=production \
    Key=service,Value=checkout \
    Key=cost-center,Value=cc-8821 \
    Key=owner,Value=rajesh.guntupalli@company.com \
    Key=provisioned-by,Value=terraform \
    Key=monthly-budget-usd,Value=3000

# Verify tags were applied
aws rds list-tags-for-resource \
  --resource-name "arn:aws:rds:us-east-1:123456789012:db:payments-prod-pg15" \
  --query 'TagList[*].[Key,Value]' \
  --output table

# Azure: Apply tags to a SQL Database resource group
az tag create \
  --resource-id \
    ↪ "/subscriptions/sub-id/resourceGroups/payments-rg/providers/Microsoft.Sql/servers/payments-sql/databases/payments-pg15" \
    ↪ \
  --tags \
    team=payments \
    environment=production \
    service=checkout \
    cost-center=cc-8821 \
    owner=rajesh.guntupalli@company.com \
    monthly-budget-usd=3000

# GCP: Apply labels to a Cloud SQL instance
gcloud sql instances patch payments-prod-pg15 \
  --update-labels \
    team=payments,\
    environment=production,\
    service=checkout,\
    cost-center=cc-8821,\
    owner=rajesh-guntupalli
```

Once cost allocation tags are active in AWS Cost Explorer (there is a 24-hour activation delay after enabling them in the billing console), you can pull per-team database costs programmatically. This is the data that makes engineers care.

Making the Math Visible: Cost Attribution at the Query Level

Showing a team their monthly RDS bill is useful. Showing them which specific queries are generating that bill is transformative. When an engineer can see that a particular unindexed join is responsible for \$400 of compute cost per month, they fix it in the next sprint — not because someone told them to, but because the number is attached to code they wrote.

The following query runs against `pg_stat_statements` on PostgreSQL (available on RDS PostgreSQL and Aurora PostgreSQL with the extension enabled) and produces a cost-weighted ranking of queries based on their compute consumption. It assumes a cost rate of \$0.063 per vCPU-hour, which you should calibrate to your actual instance pricing.

```
-- Cost-weighted query analysis using pg_stat_statements
-- Run this on RDS PostgreSQL or Aurora PostgreSQL
-- Requires: CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
-- Adjust vcpu_cost_per_hour to match your instance pricing
```



```

WITH instance_params AS (
  SELECT
    0.063 AS vcpu_cost_per_hour, -- db.r6g.4xlarge: $1.008/hr / 16 vCPUs $0.063
    16 AS vcpu_count,
    720 AS hours_in_month
),
query_stats AS (
  SELECT
    queryid,
    LEFT(query, 120) AS query_preview,
    calls,
    total_exec_time / 1000.0 AS total_exec_seconds,
    mean_exec_time / 1000.0 AS mean_exec_seconds,
    rows,
    shared_blks_hit,
    shared_blks_read,
    (shared_blks_read * 8192.0) / (1024^3) AS data_read_gb,
    stddev_exec_time / 1000.0 AS stddev_exec_seconds
  FROM pg_stat_statements
  WHERE calls > 50 -- filter noise; adjust threshold to your call volume
  AND query NOT LIKE '%pg_stat%'
),
cost_calc AS (
  SELECT
    qs.*,
    ip.vcpu_cost_per_hour,
    ip.vcpu_count,
    -- Estimated monthly cost: scale total execution time to a month,
    -- then proportion of vCPU time consumed
    ROUND(
      (
        (qs.total_exec_seconds / NULLIF(ip.hours_in_month * 3600.0, 0))
        * ip.vcpu_count
        * ip.vcpu_cost_per_hour
        * ip.hours_in_month
      )::numeric, 4
    ) AS estimated_monthly_cost_usd,
    ROUND((
      100.0 * qs.shared_blks_read
      / NULLIF(SUM(qs.shared_blks_read) OVER (), 0)
    )::numeric, 2) AS pct_of_total_reads
  FROM query_stats qs
  CROSS JOIN instance_params ip
)
SELECT
  queryid,
  query_preview,
  calls,
  ROUND(mean_exec_seconds::numeric, 4) AS mean_exec_sec,
  ROUND(total_exec_seconds::numeric, 2) AS total_exec_sec,
  estimated_monthly_cost_usd,
  pct_of_total_reads,
  ROUND(data_read_gb::numeric, 4) AS data_read_gb,
  CASE
    WHEN mean_exec_seconds > 1.0 THEN 'HIGH - optimize immediately'
    WHEN mean_exec_seconds > 0.1 THEN 'MEDIUM - review this sprint'
    ELSE 'LOW - monitor'
  END AS priority
FROM cost_calc
ORDER BY estimated_monthly_cost_usd DESC
LIMIT 25;

```

On SQL Server (Azure SQL Database or RDS SQL Server), the equivalent uses `sys.dm_exec_query_stats`:

```

-- Cost-weighted query analysis for SQL Server / Azure SQL Database
-- No extensions required - available natively
-- Adjust @vcpu_cost_per_hour to your DTU/vCore pricing

```

```

DECLARE @vcpu_cost_per_hour DECIMAL(10,4) = 0.1809; -- GP_Gen5_4: $0.7236/hr / 4 vCores
DECLARE @hours_in_month INT = 720;

WITH query_stats AS (
    SELECT TOP 25
        qs.sql_handle,
        qs.plan_handle,
        qs.execution_count,
        qs.total_worker_time / 1e6 AS total_cpu_sec,
        qs.total_elapsed_time / 1e6 AS total_elapsed_sec,
        (qs.total_worker_time / 1e6) / NULLIF(qs.execution_count, 0) AS avg_cpu_sec_per_exec,
        qs.total_logical_reads,
        qs.total_physical_reads,
        qs.total_rows,
        SUBSTRING(
            st.text,
            (qs.statement_start_offset / 2) + 1,
            (
                CASE qs.statement_end_offset
                    WHEN -1 THEN DATALENGTH(st.text)
                    ELSE qs.statement_end_offset
                END - qs.statement_start_offset
            ) / 2 + 1
        ) AS query_text
    FROM sys.dm_exec_query_stats qs
    CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) st
    WHERE qs.execution_count > 50
    ORDER BY qs.total_worker_time DESC
)
SELECT
    LEFT(query_text, 120) AS query_preview,
    execution_count,
    ROUND(avg_cpu_sec_per_exec, 4) AS avg_cpu_sec,
    ROUND(total_cpu_sec, 2) AS total_cpu_sec,
    total_logical_reads,
    total_physical_reads,
    -- Scale execution window CPU to monthly estimate
    ROUND(
        (total_cpu_sec / (@hours_in_month * 3600.0))
        * @vcpu_cost_per_hour
        * @hours_in_month,
        2
    ) AS estimated_monthly_cost_usd,
    CASE
        WHEN avg_cpu_sec_per_exec > 1.0 THEN 'HIGH - optimize immediately'
        WHEN avg_cpu_sec_per_exec > 0.1 THEN 'MEDIUM - review this sprint'
        ELSE 'LOW - monitor'
    END AS priority
FROM query_stats
ORDER BY estimated_monthly_cost_usd DESC;

```

Run either of these on a weekly schedule and publish the results to a shared Slack channel or internal wiki. The moment engineers see their name associated with a \$600/month query, the conversation changes.

Building Cost Dashboards Engineers Actually Look At

Finance dashboards live in finance tools. Engineers live in Grafana, Datadog, and their terminal. If cost data does not appear in those surfaces, it will not influence engineering behavior. The solution is to pull cost data into existing observability tooling using the same APIs and agents teams already operate.

The following Python script pulls RDS cost data from AWS Cost Explorer, grouped by the `team` tag, and pushes it to a Prometheus Pushgateway. From there it flows into any Grafana dashboard alongside CPU, connections, and query latency — the metrics engineers already care about.

```
#!/usr/bin/env python3
"""
rds_cost_exporter.py

Pulls RDS database costs from AWS Cost Explorer, grouped by the 'team'
cost allocation tag, and exposes them as Prometheus metrics via Pushgateway.

Schedule: Run daily via cron or a Kubernetes CronJob.
Requirements: boto3, prometheus_client
pip install boto3 prometheus_client

Environment variables:
AWS_REGION          - e.g. us-east-1
PUSHGATEWAY_URL      - e.g. http://prometheus-pushgateway:9091
COST_LOOKBACK_DAYS   - how many days of data to aggregate (default: 7)
"""

import os
import datetime
import boto3
from prometheus_client import CollectorRegistry, Gauge, push_to_gateway

AWS_REGION          = os.environ.get("AWS_REGION", "us-east-1")
PUSHGATEWAY_URL      = os.environ.get("PUSHGATEWAY_URL", "http://localhost:9091")
LOOKBACK_DAYS        = int(os.environ.get("COST_LOOKBACK_DAYS", "7"))

def get_rds_costs_by_team(lookback_days: int) -> list[dict]:
    """
    Query AWS Cost Explorer for RDS costs grouped by the 'team' tag.
    Returns a list of dicts: {team, service, total_cost, unit}
    """
    client = boto3.client("ce", region_name=AWS_REGION)

    end_date = datetime.date.today()
    start_date = end_date - datetime.timedelta(days=lookback_days)

    response = client.get_cost_and_usage(
        TimePeriod={
            "Start": start_date.isoformat(),
            "End": end_date.isoformat(),
        },
        Granularity="DAILY",
        Filter={
            "Dimensions": {
                "Key": "SERVICE",
                "Values": [
                    "Amazon Relational Database Service",
                    "Amazon Aurora",
                ],
            },
        },
        GroupBy=[
            {"Type": "TAG", "Key": "team"},
            {"Type": "DIMENSION", "Key": "SERVICE"},
        ],
        Metrics=["UnblendedCost"],
    )

    results = []
    for time_period in response.get("ResultsByTime", []):
        for group in time_period.get("Groups", []):
            keys = group.get("Keys", [])
            # Keys format: ["team$payments", "Amazon Relational Database Service"]
            team_raw = keys[0] if len(keys) > 0 else "team$untagged"
            service = keys[1] if len(keys) > 1 else "unknown"
```

```

        team    = team_raw.replace("team$", "") or "untagged"
        amount  = float(
            group["Metrics"]["UnblendedCost"]["Amount"]
        )
        unit    = group["Metrics"]["UnblendedCost"]["Unit"]
        results.append({
            "team":    team,
            "service": service,
            "cost":    amount,
            "unit":    unit,
            "date":    time_period["TimePeriod"]["Start"],
        })

    return results

def aggregate_by_team(cost_records: list[dict]) -> dict[str, float]:
    """Sum costs across the lookback window per team."""
    totals: dict[str, float] = {}
    for record in cost_records:
        key = record["team"]
        totals[key] = totals.get(key, 0.0) + record["cost"]
    return totals

def push_metrics(team_costs: dict[str, float], lookback_days: int) -> None:
    """Push per-team RDS costs to Prometheus Pushgateway."""
    registry = CollectorRegistry()

    gauge = Gauge(
        name        = "rds_cost_usd_total",
        documentation = (
            f"Total RDS cost in USD over the last {lookback_days} days, "
            "grouped by team tag"
        ),
        labelnames  = ["team"],
        registry    = registry,
    )

    for team, total_cost in team_costs.items():
        gauge.labels(team=team).set(total_cost)
        print(f" team={team:<20} cost=${total_cost:>10.4f} USD "
              f"(last {lookback_days} days)")

    push_to_gateway(
        gateway = PUSHGATEWAY_URL,
        job     = "rds_cost_exporter",
        registry = registry,
    )
    print(f"\nMetrics pushed to {PUSHGATEWAY_URL}")

def main():
    print(f"Fetching RDS costs for last {LOOKBACK_DAYS} days...")
    records = get_rds_costs_by_team(LOOKBACK_DAYS)
    team_totals = aggregate_by_team(records)
    print(f"Found {len(team_totals)} teams with RDS spend:")
    push_metrics(team_totals, LOOKBACK_DAYS)

if __name__ == "__main__":
    main()

```

With this running as a daily Kubernetes CronJob, a Grafana panel showing “RDS cost by team — last 7 days” sits alongside latency and error rate panels. When a team’s cost spikes after a deployment, they see it within 24 hours, not 45 days.

For Azure, replace the Cost Explorer call with the Azure Cost Management API:

```
# Pull Azure SQL Database costs grouped by the 'team' tag
# using the Azure Cost Management REST API via CLI

SUBSCRIPTION_ID="your-subscription-id"
START_DATE=$(date -d "-7 days" +%Y-%m-%d)
END_DATE=$(date +%Y-%m-%d)

az rest \
  --method POST \
  --url "https://management.azure.com/subscriptions/${SUBSCRIPTION_ID} \
    ↪ /providers/Microsoft.CostManagement/query?api-version=2023-11-01" \
  --headers "Content-Type=application/json" \
  --body "{
    \"type\": \"ActualCost\",
    \"dataSet\": {
      \"granularity\": \"Daily\",
      \"aggregation\": {
        \"totalCost\": {
          \"name\": \"Cost\",
          \"function\": \"Sum\"
        }
      },
      \"grouping\": [
        { \"type\": \"TagKey\", \"name\": \"team\" },
        { \"type\": \"Dimension\", \"name\": \"ServiceName\" }
      ],
      \"filter\": {
        \"dimensions\": {
          \"name\": \"ServiceName\",
          \"operator\": \"In\",
          \"values\": [\"SQL Database\", \"SQL Managed Instance\", \"Azure Database for PostgreSQL\"]
        }
      }
    },
    \"timeframe\": \"Custom\",
    \"timePeriod\": {
      \"from\": \"${START_DATE}T00:00:00Z\",
      \"to\": \"${END_DATE}T23:59:59Z\"
    }
  }" \
  --query "properties.rows" \
  --output table
```

Creating Accountability Without Creating Friction

Visibility alone changes behavior somewhat. Accountability changes it reliably. The goal is not to penalize engineers for spending money — that produces the wrong incentives, where teams under-provision to hit a number and create performance incidents. The goal is to make cost a first-class engineering metric alongside latency, error rate, and uptime.

The most effective mechanism is a per-team monthly budget with automated alerting. AWS supports this natively through Budgets:

```
# Create a monthly RDS budget for the payments team
# Alerts at 80% and 100% of budget threshold

aws budgets create-budget \
  --account-id "123456789012" \
  --budget '{
    "BudgetName": "payments-team-rds-monthly",
    "BudgetLimit": {
      "Amount": "3000",
      "Unit": "USD"
    }
  }
```

```

    },
    "BudgetType": "COST",
    "TimeUnit": "MONTHLY",
    "CostFilters": {
      "TagKeyValue": ["user:team$payments"],
      "Service": ["Amazon Relational Database Service", "Amazon Aurora"]
    },
    "CostTypes": {
      "IncludeCredit": false,
      "IncludeDiscount": true,
      "IncludeOtherSubscription": false,
      "IncludeRecurring": true,
      "IncludeRefund": false,
      "IncludeSubscription": true,
      "IncludeSupport": false,
      "IncludeTax": false,
      "IncludeUpfront": true,
      "UseBlended": false
    }
  } \
  --notifications-with-subscribers '[
    {
      "Notification": {
        "NotificationType": "ACTUAL",
        "ComparisonOperator": "GREATER_THAN",
        "Threshold": 80,
        "ThresholdType": "PERCENTAGE"
      },
      "Subscribers": [
        {
          "SubscriptionType": "EMAIL",
          "Address": "payments-team@company.com"
        },
        {
          "SubscriptionType": "SNS",
          "Address": "arn:aws:sns:us-east-1:123456789012:payments-cost-alerts"
        }
      ]
    }
  ],
  {
    "Notification": {
      "NotificationType": "ACTUAL",
      "ComparisonOperator": "GREATER_THAN",
      "Threshold": 100,
      "ThresholdType": "PERCENTAGE"
    },
    "Subscribers": [
      {
        "SubscriptionType": "EMAIL",
        "Address": "payments-team@company.com"
      },
      {
        "SubscriptionType": "EMAIL",
        "Address": "engineering-director@company.com"
      }
    ]
  }
]'

```

The SNS topic at 80% should trigger a Slack message to the team's engineering channel — not to the finance channel, not to a DBA channel. The message goes to the people who can act on it: the engineers who shipped code this month.

Setting the initial budget number requires some care. Look at the team's trailing 3-month average spend, add 15% for growth headroom, and round to the nearest \$500. Do not start with a stretch goal. The objective in month one is to make the signal feel credible, not to create an emergency. Once teams have two or three months of data and trust the numbers, you tighten the budget by

10% each quarter until you hit a steady-state target.

The accountability system also needs an escalation path that does not feel punitive. A good structure:

Threshold	Action	Audience
70% of monthly budget	Informational Slack message	Team channel
85% of monthly budget	Alert with cost breakdown link	Team channel + tech lead
100% of monthly budget	Alert + required review meeting	Tech lead + EM
120% of monthly budget	Automatic PR comment on open infra PRs	All reviewers

The PR comment at 120% is particularly effective. When an engineer submits a Terraform PR that adds a new `db.r6g.2xlarge` instance and the PR bot comments “ The payments team is currently at 124% of their monthly database budget (\$3,720 against a \$3,000 target). This instance will add approximately \$520/month,” the engineering team has a conversation right there in the code review. No meeting required.

Automating Cost Signals into the Engineering Workflow

Manual cost reviews that happen monthly do not change daily decisions. Automated signals that appear in the existing workflow — CI/CD pipelines, code review, weekly engineering metrics emails — do. This section covers three automation patterns that have proven effective across different organization sizes.

Pattern 1: Terraform plan cost estimation in CI

Use Infracost (open source) to annotate every Terraform plan that touches database resources with a cost delta before merge. This catches the most common source of unexpected cost growth: well-intentioned engineers upgrading an instance type to fix a performance problem without realizing the cost implication.

```
# .github/workflows/terraform-cost.yml excerpt
# Run Infracost on every PR that modifies Terraform files

# Install Infracost
curl -fsSL https://raw.githubusercontent.com/infracost/infracost/master/scripts/install.sh | sh

# Generate cost estimate for the plan
infracost breakdown \
  --path ./terraform/databases \
  --terraform-plan-flags="--var-file=production.tfvars" \
  --format json \
  --out-file /tmp/infracost-current.json

# Compare against main branch baseline
infracost diff \
  --path ./terraform/databases \
  --compare-to /tmp/infracost-main.json \
  --format table

# Post the diff as a PR comment (Infracost handles this natively)
```

```
# with INFRACOST_API_KEY and GitHub Actions integration)
infracost comment github \
  --path /tmp/infracost-current.json \
  --repo "$GITHUB_REPOSITORY" \
  --pull-request "$PR_NUMBER" \
  --github-token "$GITHUB_TOKEN" \
  --behavior update
```

A typical Infracost comment on a PR that upgrades an RDS instance from db.r6g.2xlarge to db.r6g.4xlarge shows: “Monthly cost will increase by \$481.14 (+91%): from \$530.42 to \$1,011.56.” That number in a PR description changes the discussion before infrastructure ships.

Pattern 2: Weekly cost digest with anomaly detection

The following script runs weekly, detects day-over-day cost spikes by team and service, and sends a structured digest to Slack. It uses AWS Cost Explorer’s anomaly detection API rather than building a custom baseline.

```
#!/usr/bin/env python3
"""
weekly_cost_digest.py

Detects RDS cost anomalies using AWS Cost Explorer's native anomaly
detection and sends a structured digest to a Slack webhook.

Schedule: Run every Monday at 08:00 via cron or EventBridge.
Requirements: boto3, requests
    pip install boto3 requests

Environment variables:
    SLACK_WEBHOOK_URL - Incoming webhook URL for your cost-alerts channel
    ANOMALY_MONITOR_ARN - ARN of your Cost Explorer anomaly monitor
                        (create once via console or CLI)
    AWS_REGION        - e.g. us-east-1
"""

import os
import json
import datetime
import boto3
import requests

SLACK_WEBHOOK_URL = os.environ["SLACK_WEBHOOK_URL"]
ANOMALY_MONITOR_ARN = os.environ.get("ANOMALY_MONITOR_ARN", "")
AWS_REGION = os.environ.get("AWS_REGION", "us-east-1")

ce_client = boto3.client("ce", region_name=AWS_REGION)

def get_recent_anomalies(lookback_days: int = 14) -> list[dict]:
    """Retrieve cost anomalies from the last N days."""
    end_date = datetime.date.today()
    start_date = end_date - datetime.timedelta(days=lookback_days)

    params = {
        "DateInterval": {
            "StartDate": start_date.isoformat(),
            "EndDate": end_date.isoformat(),
        },
        "Filters": {
            "Services": {
                "Values": [
                    "Amazon Relational Database Service",
                    "Amazon Aurora",
                ]
            }
        },
    },
```



```

        "TotalImpact": {
            "NumericOperator": "GREATER_THAN",
            "StartValue": 50.0,  # only surface anomalies over $50
        },
    }

    if ANOMALY_MONITOR_ARN:
        params["MonitorArn"] = ANOMALY_MONITOR_ARN

    response = ce_client.get_anomalies(**params)
    return response.get("Anomalies", [])

def get_week_over_week_costs() -> dict:
    """
    Pull RDS costs for this week vs last week by team tag.
    Returns dict: {team: {this_week: float, last_week: float, delta_pct: float}}
    """
    today = datetime.date.today()
    this_start = today - datetime.timedelta(days=7)
    last_start = today - datetime.timedelta(days=14)

    def fetch_costs(start: datetime.date, end: datetime.date) -> dict[str, float]:
        resp = ce_client.get_cost_and_usage(
            TimePeriod={"Start": start.isoformat(), "End": end.isoformat()},
            Granularity="MONTHLY",
            Filter={
                "Dimensions": {
                    "Key": "SERVICE",
                    "Values": [
                        "Amazon Relational Database Service",
                        "Amazon Aurora",
                    ],
                },
            },
            GroupBy=[{"Type": "TAG", "Key": "team"}],
            Metrics=["UnblendedCost"],
        )
        result = {}
        for period in resp.get("ResultsByTime", []):
            for group in period.get("Groups", []):
                team = group["Keys"][0].replace("team$", "") or "untagged"
                result[team] = result.get(team, 0.0) + float(
                    group["Metrics"]["UnblendedCost"]["Amount"]
                )
        return result

    this_week = fetch_costs(this_start, today)
    last_week = fetch_costs(last_start, this_start)

    all_teams = set(this_week.keys()) | set(last_week.keys())
    comparison = {}
    for team in all_teams:
        tw = this_week.get(team, 0.0)
        lw = last_week.get(team, 0.0)
        delta_pct = ((tw - lw) / lw * 100) if lw > 0 else 0.0
        comparison[team] = {
            "this_week": tw,
            "last_week": lw,
            "delta_pct": delta_pct,
        }

    return comparison

def format_slack_message(
    anomalies: list[dict],
    wow_costs: dict,
) -> dict:
    """Build a structured Slack Block Kit message."""

```

```

blocks = [
    {
        "type": "header",
        "text": {
            "type": "plain_text",
            "text": " Weekly RDS Cost Digest",
        },
    },
    {"type": "divider"},
]

# Week-over-week section
sorted_teams = sorted(
    wow_costs.items(),
    key=lambda x: x[1]["this_week"],
    reverse=True,
)

wow_lines = []
for team, data in sorted_teams[:8]: # top 8 teams by spend
    arrow = "↑" if data["delta_pct"] > 10 else (
        "↓" if data["delta_pct"] < -5 else "→")
    wow_lines.append(
        f"*{team:<20}* ${data['this_week']:>8.2f} "
        f"{arrow} {data['delta_pct']:+.1f}% vs prior week"
    )

blocks.append({
    "type": "section",
    "text": {
        "type": "mrkdwn",
        "text": "*Team RDS Spend - Last 7 Days vs Prior 7 Days*\n"
        + "\n".join(wow_lines),
    },
})

# Anomalies section
if anomalies:
    anomaly_lines = []
    for a in anomalies[:5]:
        impact = a.get("Impact", {})
        service = a.get("RootCauses", [{}])[0].get("Service", "Unknown")
        region = a.get("RootCauses", [{}])[0].get("Region", "Unknown")
        anomaly_lines.append(
            f"* {service} in {region}: "
            f"${impact.get('TotalActualSpend', 0):.2f} actual vs "
            f"${impact.get('TotalExpectedSpend', 0):.2f} expected "
            f"({impact.get('TotalImpact', 0):.2f})"
        )

    blocks.append({"type": "divider"})
    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": " *Cost Anomalies Detected*\n"
            + "\n".join(anomaly_lines),
        },
    })
else:
    blocks.append({
        "type": "section",
        "text": {
            "type": "mrkdwn",
            "text": " No significant cost anomalies detected this week.",
        },
    })

blocks.append({"type": "divider"})
blocks.append({

```

```

        "type": "context",
        "elements": [
            {
                "type": "mrkdwn",
                "text": (
                    "Data source: AWS Cost Explorer · "
                    "Filtered to RDS + Aurora · "
                    f"Generated: {datetime.date.today().isoformat()} · "
                    "<https://console.aws.amazon.com/cost-management/home|"
                    "Open Cost Explorer>"
                ),
            },
        ],
    },
    ],
})

return {"blocks": blocks}

def main():
    print("Fetching cost anomalies...")
    anomalies = get_recent_anomalies(lookback_days=14)
    print(f" Found {len(anomalies)} anomalies over $50 impact")

    print("Fetching week-over-week costs by team...")
    wow_costs = get_week_over_week_costs()
    print(f" Found {len(wow_costs)} teams with RDS spend")

    message = format_slack_message(anomalies, wow_costs)

    resp = requests.post(
        SLACK_WEBHOOK_URL,
        data=json.dumps(message),
        headers={"Content-Type": "application/json"},
        timeout=10,
    )
    resp.raise_for_status()
    print("Digest posted to Slack successfully.")

if __name__ == "__main__":
    main()

```

This digest gives engineering leadership a Monday morning signal that requires no manual work. Teams with a spike see their number in context — not as a surprise line item on a finance report three weeks later, but as a data point that correlates with what shipped last week.

Pattern 3: Non-production instance shutdown enforcement

The single largest source of wasted database spend in most organizations is non-production instances running 24 hours a day, 7 days a week. A staging RDS `db.r6g.2xlarge` instance running continuously costs approximately \$530/month. Running it only during business hours — 8 AM to 8 PM on weekdays — reduces that to roughly \$180/month, saving \$350 per instance per month. With 10 staging instances, that is \$3,500/month or \$42,000/year recovered with a Lambda function and two EventBridge rules.

```

# Create EventBridge rules to stop non-production RDS instances at 8 PM
# and start them at 8 AM, Monday through Friday (UTC times shown;
# adjust for your team's timezone)

# Rule 1: Stop non-production instances at 20:00 UTC (8 PM UTC)
aws events put-rule \
  --name "rds-nonprod-stop-evening" \
  --schedule-expression "cron(0 20 ? * MON-FRI *)" \
  --state ENABLED \
  --description "Stop non-production RDS instances at 8 PM UTC"

```

```

# Rule 2: Start non-production instances at 08:00 UTC (8 AM UTC)
aws events put-rule \
  --name "rds-nonprod-start-morning" \
  --schedule-expression "cron(0 8 ? * MON-FRI *)" \
  --state ENABLED \
  --description "Start non-production RDS instances at 8 AM UTC"

# The Lambda function referenced by both rules uses instance tags
# to determine eligibility: environment=staging OR environment=development
# and a tag auto-shutdown=enabled (opt-in, not opt-out)

# Tag a staging instance for automatic shutdown
aws rds add-tags-to-resource \
  --resource-name "arn:aws:rds:us-east-1:123456789012:db:payments-staging-pg15" \
  --tags \
    Key=environment,Value=staging \
    Key=auto-shutdown,Value=enabled \
    Key=auto-shutdown-owner,Value=payments-team

# Verify the Lambda function exists and is linked to both rules
aws lambda get-function \
  --function-name "rds-nonprod-scheduler" \
  --query "Configuration.[FunctionName,Runtime,LastModified]" \
  --output table

```

The Lambda function itself uses the `auto-shutdown=enabled` tag as an explicit opt-in. This is important for culture: you are not forcing shutdown on teams, you are making it easy for them to opt in, and you are making the default (running 24/7) cost them budget points in the internal accountability system. Within two or three monthly cycles, most teams opt in voluntarily because they see the savings reflected in their team dashboard and it makes their numbers look better.

The FinOps Maturity Model for Database Teams

FinOps culture does not appear overnight. Organizations move through recognizable stages, and understanding which stage your team is in determines which interventions to prioritize. Pushing advanced automation at a team that has not yet established basic tagging produces frustration, not savings.



Figure 27: The FinOps maturity progression — each stage’s interventions build on the previous one; skipping ahead (e.g. anomaly detection before basic tagging) produces confusion, not savings.

Stage 1 — Reactive (0–3 months)

The finance team sends a monthly cloud bill. Engineering leadership expresses concern. No one knows which team or service is responsible for which cost. There are no tags on RDS instances. Cost optimization efforts are one-off and manual.

Priority interventions at this stage: enforce tagging through Terraform modules and Service Control Policies (SCPs), activate cost allocation tags in the billing console, establish per-team budget alerts at the account or tag level, and run the `pg_stat_statements` cost query manually for the top three most expensive instances.

Stage 2 — Aware (3–6 months)

Tags exist on most resources. Teams have budget alerts configured. Engineers have seen the weekly cost digest at least once and recognized their team’s name in it. Some non-production instances have been right-sized. Reserved Instance coverage for production databases is under discussion.

Priority interventions: deploy the Prometheus cost exporter and add cost panels to existing Grafana dashboards, integrate Infracost into CI pipelines for database-related Terraform, implement non-production shutdown schedules for at least 50% of staging instances, and complete the first Reserved Instance purchase for the largest production RDS instances (typically 1-year No Upfront for predictable workloads).

The math on Reserved Instances is straightforward. A `db.r6g.4xlarge` in us-east-1 costs \$1.008/hour on-demand, or \$725.76/month. A 1-year No Upfront Reserved Instance for the same class costs \$0.630/hour, or \$453.60/month — a saving of \$272.16/month, or \$3,265.92/year, per instance. An organization running 10 such instances saves \$32,659 annually by committing to 1-year terms on instances they were already planning to run continuously.

Stage 3 — Proactive (6–12 months)

Cost is a standard agenda item in sprint planning. Engineers ask “what will this cost?” before provisioning, not after. Infracost comments on infrastructure PRs are read and acted on. Non-production shutdown schedules run automatically. Reserved Instance coverage for production is above 70%. The DBA team runs a monthly cost review meeting with engineering managers.

Priority interventions: implement query-level cost attribution from `pg_stat_statements` and publish results to team dashboards, evaluate Aurora Serverless v2 for variable workloads, implement pgBouncer for connection pooling to reduce the need for oversized instances driven by connection count, and begin Savings Plans analysis for consistent Aurora compute spend.

Stage 4 — Optimized (12+ months)

Cost efficiency is a first-class engineering metric. The FinOps team reviews database spend weekly. Anomaly detection catches cost spikes within 24 hours. Query-level cost attribution drives optimization work the same way latency metrics drive performance work. Reserved Instance coverage exceeds 85%. Storage is routinely reviewed and tiered appropriately. New database provisioning goes through a cost estimation gate in the CI pipeline.

Most organizations with mature FinOps programs at this stage report database costs that are 30–45% lower than they were at Stage 1, measured on a cost-per-transaction or cost-per-active-user basis — not as an absolute reduction, which would obscure growth.

Sustaining the Culture: Incentives, Reviews, and Recognition

Culture requires reinforcement. The technical tooling described in this chapter creates visibility and accountability, but it does not create motivation by itself. The following practices sustain FinOps culture past the initial excitement phase.

Monthly cost review meetings that are 30 minutes, not 90

Keep the monthly cost review short by doing the analysis before the meeting, not during it. The agenda: five minutes reviewing the week-over-week Slack digest trends, ten minutes on any anomalies that triggered alerts, ten minutes on one optimization opportunity identified from the `pg_stat_statements` cost report, five minutes on action items with owners and due dates. The meeting ends. No one spends 45 minutes staring at AWS Cost Explorer together.

Recognize cost savings the same way you recognize feature delivery

If a developer ships a query optimization that reduces a team's RDS bill by \$800/month, that saving should appear in sprint reviews and team retrospectives alongside feature velocity metrics. It sounds trivial, but recognition is a powerful behavioral signal. When engineers see that cost work is valued and acknowledged — not just performance work and feature work — they invest time in it.

Make the cost of technical debt legible

One of the most effective culture shifts happens when teams can see the ongoing cost of deferred optimization work. A query that has been on the backlog for six months because it “works well enough” looks different when the `pg_stat_statements` report shows it is costing \$340/month. At six months of deferral, the team has paid \$2,040 for a fix that would have taken four hours of developer time. Framing technical debt in dollar terms creates urgency that “performance improvement” framing does not.

Run a quarterly FinOps retrospective across teams

Once per quarter, bring engineering leads together for a cross-team cost retrospective. Each team presents: their highest-cost database resource, one optimization they shipped this quarter and the savings it generated, and one cost risk they have identified for next quarter. The cross-team format creates healthy peer comparison — not competition, but shared learning. Teams that found effective optimization patterns share them, and the organization's collective knowledge grows faster than it would if each team worked in isolation.

Key Takeaways

- **Visibility drives behavior, not awareness.** Engineers do not reduce costs because they are told to care about the bill. They reduce costs when cost data appears in the same tools they use for latency and error rates — Grafana dashboards, Slack alerts, and CI pipeline comments — attached to decisions they are making right now, not to a report they read three weeks later.
- **Tagging is the foundation.** No tag strategy means no team-level attribution, which means no accountability. Enforce a minimum viable tag set — `team`, `environment`, `service`, `owner` — through Terraform modules and AWS Service Control Policies before deploying any other FinOps tooling. Everything else depends on it.
- **Query-level cost attribution is the highest-leverage intervention for DBAs.** The `pg_stat_statements` cost query and its SQL Server equivalent in `sys.dm_exec_query_stats` translate compute consumption into dollar figures attached to specific queries. When engineers see a \$600/month query with their name on it, they fix it in the next sprint. Publish this report weekly to team channels.

- **Automated signals beat manual reviews.** A weekly Slack digest with week-over-week cost trends, a Prometheus metric pushed to Grafana, an Infracost comment on infrastructure PRs, and a Lambda function that shuts down staging databases at 8 PM — these run without human intervention and create continuous feedback. Monthly manual cost reviews cannot move fast enough to change daily engineering decisions.
 - **FinOps maturity is a journey, not a project.** Organizations move through four recognizable stages — Reactive, Aware, Proactive, and Optimized — and each stage has the right set of interventions. Trying to implement anomaly-driven alerting and query-level attribution before basic tagging exists produces confusion and abandonment. Sequence the work by stage, measure progress in cost-per-transaction rather than absolute spend, and recognize that the 30–45% savings typical of mature FinOps programs accumulate over 12 months of consistent cultural and technical investment, not a single sprint.
-

Chapter 28: Open-Source Tools for Cloud Database Cost Analysis

Teams that shift database cost visibility from vendor consoles to open-source tooling typically identify **15–40% in previously invisible waste** within the first 30 days. That is not a projection — it reflects what happens when you stop relying on AWS Cost Explorer’s monthly rollups and start querying your own infrastructure telemetry at the row level. This chapter walks through the open-source ecosystem that makes that possible: tools for ingesting cloud billing data, querying database-level performance signals, correlating cost with workload, and automating remediation. Every tool covered here is free, actively maintained, and battle-tested in production environments serving databases ranging from 50 GB development instances to 50 TB analytical workloads.

The Open-Source Cost Visibility Stack

Before installing anything, understand what each layer of the stack actually does. The cloud provider billing APIs tell you what you spent. Database-internal statistics tell you what caused that spending. A correlation layer connects the two. Most teams have one or two of these layers but rarely all three, which is why waste persists.

Layer 1 — Cloud billing ingestion The most widely adopted open-source tool for this layer is **Infracost** (cost estimation from IaC), paired with **CloudQuery** or **Steampipe** for live account inventory. Steampipe deserves particular attention: it exposes AWS, Azure, and GCP resources as PostgreSQL-compatible tables, meaning you can write standard SQL against your cloud account the same way you write queries against a database.

Install Steampipe and the AWS plugin:

```
# Install Steampipe
brew install turbot/tap/steampipe # macOS
# or for Linux:
sudo /bin/sh -c "$(curl -fsSL https://steampipe.io/install/steampipe.sh)"

# Install the AWS plugin
steampipe plugin install aws

# Install Azure and GCP plugins
steampipe plugin install azure
steampipe plugin install gcp

# Verify installed plugins
steampipe plugin list
```

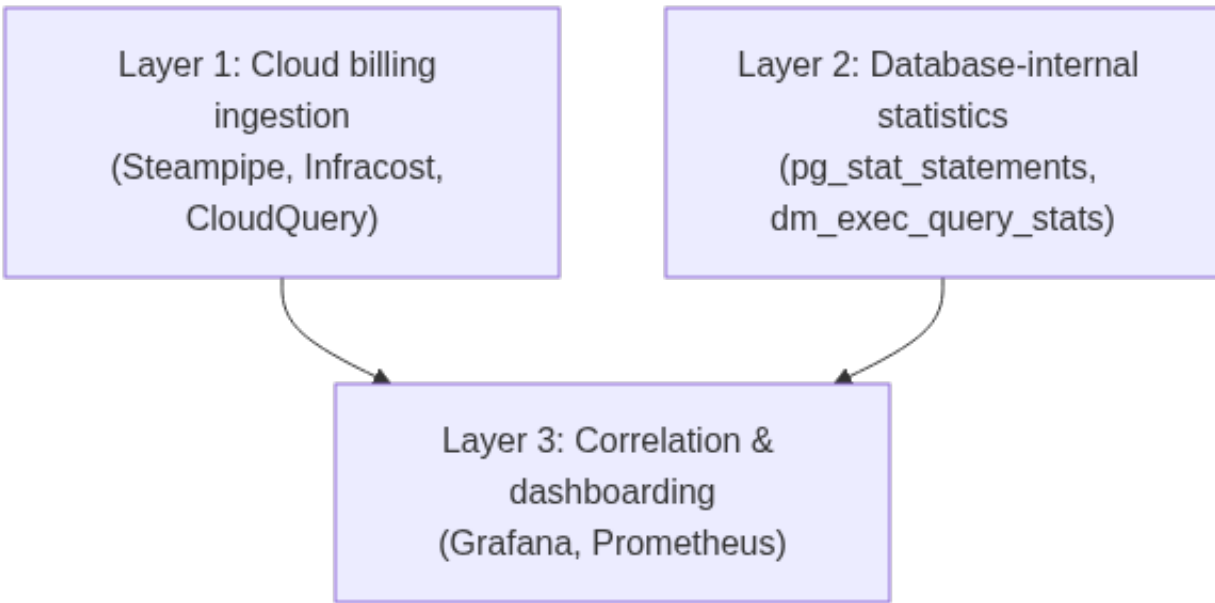


Figure 28: The three-layer open-source cost visibility stack: billing tells you what you spent, database statistics tell you why, and the correlation layer connects them.

```
# Launch the interactive query shell
steampipe query
```

Once Steampipe is running, query your RDS inventory directly:

```
-- List all RDS instances with their approximate monthly cost footprint
-- Run this inside: steampipe query
SELECT
  db_instance_identifier,
  db_instance_class,
  engine,
  engine_version,
  multi_az,
  storage_type,
  allocated_storage,
  db_instance_status,
  -- Estimate: gp2 storage is ~$0.115/GB-month in us-east-1
  ROUND(allocated_storage * 0.115, 2) AS estimated_storage_cost_usd,
  -- Flag instances that are stopped but still billing for storage
  CASE
    WHEN db_instance_status = 'stopped' THEN 'BILLING FOR STORAGE ONLY - review'
    WHEN multi_az = false
      AND db_instance_class NOT LIKE '%micro%'
      AND db_instance_class NOT LIKE '%small%' THEN 'Single-AZ prod risk'
    ELSE 'OK'
  END AS cost_flag,
  publicly_accessible,
  region
FROM
  aws_rds_db_instance
ORDER BY
  allocated_storage DESC;
```

This single query, run against a real AWS account, commonly surfaces stopped instances still accruing storage costs. A stopped `db.r6g.4xlarge` with a 2 TB gp2 volume costs **\$230/month in storage alone** while producing zero useful work.

Layer 2 — Database-internal statistics PostgreSQL’s `pg_stat_statements` and `pg_stat_user_tables`, SQL Server’s `sys.dm_exec_query_stats`, and MySQL’s `performance_schema` are the authoritative sources for understanding what workload is driving your compute and I/O costs. These are built into the engine — no additional installation required beyond enabling the extension.

Layer 3 — Correlation and dashboarding Grafana with the CloudWatch, Azure Monitor, or Cloud Monitoring data sources connects billing signals to performance signals in a single pane. **Prometheus** with the `postgres_exporter` or `mysqld_exporter` provides the time-series foundation for alerting on cost-correlated metrics like `blks_read` (indicative of IOPS billing) and `temp_files` (indicative of memory pressure that forces scale-up).

Querying Database-Level Cost Signals

Cloud bills reflect infrastructure consumption. The root cause of that consumption lives inside the database engine. The following queries give you the operational data needed to map expensive cloud line items back to specific queries, tables, and access patterns.

PostgreSQL — identifying high-IOPS queries

IOPS on RDS gp2 storage costs are implicit: you pay for provisioned IOPS on `io1`/`io2`, but on `gp2` and `gp3` you pay for the storage tier itself and suffer performance degradation when you exhaust burst credits. Either way, identifying queries that generate excessive block reads is a direct cost-reduction exercise.

```
-- Enable pg_stat_statements if not already enabled
-- Add to postgresql.conf or RDS parameter group:
-- shared_preload_libraries = 'pg_stat_statements'
-- pg_stat_statements.track = all

CREATE EXTENSION IF NOT EXISTS pg_stat_statements;

-- Top 20 queries by total disk block reads (maps to IOPS cost)
SELECT
  LEFT(query, 120)                                AS query_preview,
  calls,
  total_exec_time::NUMERIC(18,2)                   AS total_exec_ms,
  ROUND((total_exec_time / calls)::NUMERIC, 2)     AS avg_exec_ms,
  shared_blks_read,
  shared_blks_hit,
  -- Cache hit ratio: closer to 1.0 = less disk I/O
  ROUND(
    shared_blks_hit::NUMERIC /
    NULLIF(shared_blks_hit + shared_blks_read, 0),
    4
  )                                                  AS cache_hit_ratio,
  -- Queries with cache_hit_ratio < 0.90 are disk-bound
  -- Each shared_blks_read = 8KB page fetched from storage
  ROUND(shared_blks_read * 8.0 / 1024, 2)          AS total_mb_read_from_disk,
  temp_blks_written,
  -- temp_blks_written signals sorts/hashtable spilling to disk
  -- On RDS, this adds I/O and may indicate undersized work_mem
  ROUND(temp_blks_written * 8.0 / 1024, 2)         AS temp_writes_mb
FROM
  pg_stat_statements
WHERE
  calls > 100                                     -- filter out one-time queries
ORDER BY
  shared_blks_read DESC
```

```
LIMIT 20;
```

A query showing `total_mb_read_from_disk = 45000` across 10,000 calls is reading approximately 4.5 MB of pages per call from storage. If that query runs every 30 seconds around the clock, it is generating roughly 12.9 TB of logical reads per day — a primary driver of IOPS exhaustion and a direct argument for adding the right index or materializing an intermediate result set.

SQL Server on Azure SQL / RDS SQL Server — execution stats

```
-- Top 25 queries by total logical reads on SQL Server
-- Maps directly to I/O pressure and storage tier requirements
SELECT TOP 25
    qs.total_logical_reads                                AS total_logical_reads,
    qs.execution_count,
    qs.total_logical_reads / qs.execution_count          AS avg_logical_reads,
    qs.total_elapsed_time / 1000                        AS total_elapsed_ms,
    qs.total_elapsed_time / qs.execution_count / 1000   AS avg_elapsed_ms,
    qs.total_worker_time / qs.execution_count / 1000    AS avg_cpu_ms,
    -- Worker time maps to DTU consumption in Azure SQL
    -- High avg_cpu_ms on frequently executed queries = DTU ceiling risk
    SUBSTRING(
        st.text,
        (qs.statement_start_offset / 2) + 1,
        (
            CASE qs.statement_end_offset
            WHEN -1 THEN DATALENGTH(st.text)
            ELSE qs.statement_end_offset
            END - qs.statement_start_offset
        ) / 2 + 1
    ) AS query_text,
    DB_NAME(st.dbid) AS database_name,
    qp.query_plan
FROM
    sys.dm_exec_query_stats AS qs
CROSS APPLY
    sys.dm_exec_sql_text(qs.sql_handle) AS st
CROSS APPLY
    sys.dm_exec_query_plan(qs.plan_handle) AS qp
WHERE
    qs.execution_count > 50
ORDER BY
    qs.total_logical_reads DESC;
```

On Azure SQL Database, DTU consumption ties directly to logical reads and CPU. A query averaging 50,000 logical reads per execution running 200 times per minute will by itself justify an upgrade from S3 (100 DTUs, ~\$150/month) to P1 (125 DTUs, ~\$465/month). Finding that query and adding a covering index can eliminate the need for that tier upgrade entirely — a saving of roughly **\$315/month from one index**.

MySQL / Aurora — identifying table scan costs

```
-- Identify tables with high full-scan ratios
-- Full scans = unnecessary IOPS = unnecessary storage tier costs
SELECT
    object_schema                                AS db_name,
    object_name                                  AS table_name,
    count_read,
    count_fetch,
    count_full_scan,
    ROUND(
        count_full_scan * 100.0 / NULLIF(count_read, 0),
        2
    ) AS pct_full_scan,
    sum_rows_examined,
    sum_rows_sent,
    -- rows_examined / rows_sent ratio: higher = more waste per query
```

```

ROUND(
    sum_rows_examined * 1.0 / NULLIF(sum_rows_sent, 0),
    1
)
AS examination_ratio
FROM
    performance_schema.table_io_waits_summary_with_full_scans
WHERE
    count_full_scan > 0
ORDER BY
    count_full_scan DESC
LIMIT 20;

```

An `examination_ratio` above 100 on a high-traffic table means the engine is reading 100 rows to return 1. On Aurora with I/O-Optimized pricing disabled, that translates directly to per-million-request I/O charges. Aurora I/O-Optimized eliminates per-request charges at the cost of a 25% instance price premium — the crossover point is approximately **\$0.22/million I/O requests × 10M requests/day**, meaning I/O-Optimized pays off at roughly 45 million I/O requests per day. Tables with high `examination_ratio` on I/O-billed Aurora clusters are the exact targets to fix before that crossover calculation even matters.

Open-Source Tools for Billing Data Analysis

Three open-source projects stand out for transforming raw cloud billing exports into actionable database cost intelligence.

OpenCost

OpenCost is a CNCF sandbox project originally built for Kubernetes cost allocation, but its billing ingestion pipeline works equally well for RDS and managed database services when pointed at AWS Cost and Usage Reports (CUR). It exposes a Prometheus-compatible metrics endpoint, making it trivially consumable by existing Grafana dashboards.

```

# Export AWS Cost and Usage Report to S3 (one-time setup via CLI)
aws cur create-report-definition \
    --report-definition '{
        "ReportName": "database-finops-cur",
        "TimeUnit": "HOURLY",
        "Format": "Parquet",
        "Compression": "Parquet",
        "AdditionalSchemaElements": ["RESOURCES"],
        "S3Bucket": "your-cur-bucket",
        "S3Prefix": "cur/database-finops",
        "S3Region": "us-east-1",
        "ReportVersioning": "OVERWRITE_REPORT",
        "AdditionalArtifacts": ["ATHENA"],
        "RefreshClosedReports": true
    }'

# Verify the report definition was created
aws cur describe-report-definitions \
    --query 'ReportDefinitions[?ReportName==`database-finops-cur`]'

# Query the CUR via Athena once data has been delivered (typically 24h)
# This query extracts RDS-specific line items with resource-level granularity
aws athena start-query-execution \
    --query-string "
        SELECT
            line_item_resource_id,
            line_item_usage_type,
            line_item_usage_amount,

```

```

        line_item_blended_cost,
        line_item_unblended_cost,
        resource_tags_user_environment,
        resource_tags_user_team,
        line_item_usage_start_date
    FROM database_finops_cur
    WHERE line_item_product_code = 'AmazonRDS'
        AND line_item_line_item_type = 'Usage'
        AND line_item_usage_start_date >= DATE_ADD('day', -30, CURRENT_DATE)
    ORDER BY line_item_unblended_cost DESC
    LIMIT 500
" \
--query-execution-context Database=athenacurcfn_database_finops_cur \
--result-configuration OutputLocation=s3://your-cur-bucket/athena-results/

```

Infracost — cost estimation before you deploy

Infracost scans Terraform plans and outputs a cost breakdown before any infrastructure change is applied. For database teams, this means you see the monthly cost of an RDS instance modification — changing storage type, resizing, enabling Multi-AZ — before it happens, not after the bill arrives.

```

# Install Infracost
brew install infracost # macOS
# or:
curl -fsSL https://raw.githubusercontent.com/infracost/infracost/master/scripts/install.sh | sh

# Authenticate (free tier available)
infracost auth login

# Generate a cost diff for a Terraform change
# Assume you have a module that defines an RDS instance

# Baseline: cost of current state
infracost breakdown \
  --path ./terraform/rds-module \
  --format json \
  --out-file infracost-base.json

# After modifying the Terraform (e.g., changing db.r6g.2xlarge to db.r6g.xlarge):
infracost diff \
  --path ./terraform/rds-module \
  --compare-to infracost-base.json \
  --format table

# Example output interpretation:
# db.r6g.2xlarge (us-east-1, On-Demand): $0.480/hr = $350.40/mo
# db.r6g.xlarge (us-east-1, On-Demand): $0.240/hr = $175.20/mo
# Monthly saving: -$175.20
# Annual saving: -$2,102.40

# Generate a comment-ready report for a pull request
infracost comment github \
  --path infracost-base.json \
  --repo your-org/your-infra-repo \
  --pull-request 142 \
  --behavior update

```

The pull request integration is particularly valuable for platform engineering teams: every Terraform change that touches an RDS resource automatically gets a cost impact comment before merge approval. This single workflow shift prevents expensive configuration changes from reaching production without financial review.

Steampipe cost dashboards

Beyond inventory queries, Steampipe’s `aws_cost_by_service_monthly` table provides programmatic access to Cost Explorer data without the AWS console:

```
-- Monthly RDS cost trend for the last 6 months
-- Run in Steampipe query shell
SELECT
    period_start,
    service,
    ROUND(blended_cost_amount::NUMERIC, 2)      AS blended_cost_usd,
    ROUND(unblended_cost_amount::NUMERIC, 2)    AS unblended_cost_usd,
    unit
FROM
    aws_cost_by_service_monthly
WHERE
    service = 'Amazon Relational Database Service'
    AND period_start >= DATE_TRUNC('month', NOW()) - INTERVAL '6 months'
ORDER BY
    period_start DESC;

-- RDS cost broken down by resource tag (requires Cost Allocation Tags enabled)
SELECT
    dimension_value                                AS rds_instance_id,
    ROUND(SUM(blended_cost_amount::NUMERIC), 2)    AS total_cost_usd,
    COUNT(DISTINCT period_start)                  AS months_billed
FROM
    aws_cost_by_resource_monthly
WHERE
    service = 'Amazon Relational Database Service'
    AND period_start >= DATE_TRUNC('month', NOW()) - INTERVAL '3 months'
GROUP BY
    dimension_value
ORDER BY
    total_cost_usd DESC
LIMIT 30;
```

Automating Cost Analysis Pipelines

One-time queries identify problems. Scheduled pipelines track them over time and alert when costs deviate from baseline. The following Python script implements a weekly cost analysis pipeline that queries `pg_stat_statements`, correlates findings with CloudWatch IOPS metrics, and writes a summary to a Slack webhook. Schedule it with cron or a managed workflow tool like Apache Airflow.

```
#!/usr/bin/env python3
"""
db_cost_analyzer.py
Weekly database cost analysis pipeline.
Correlates pg_stat_statements I/O data with CloudWatch RDS metrics
and posts a cost summary to Slack.

Schedule: cron 0 8 * * 1 (every Monday at 08:00)
Dependencies: psycopg2-binary, boto3, requests
"""

import json
import os
import logging
from datetime import datetime, timedelta, timezone

import boto3
import psycopg2
import psycopg2.extras
import requests

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s %(levelname)s %(message)s"
```

```

)
logger = logging.getLogger(__name__)

# -----
# Configuration - load from environment variables in production
# -----
DB_CONFIG = {
    "host": os.environ["RDS_ENDPOINT"],
    "port": int(os.environ.get("RDS_PORT", "5432")),
    "dbname": os.environ["RDS_DBNAME"],
    "user": os.environ["RDS_USER"],
    "password": os.environ["RDS_PASSWORD"],
    "sslmode": "require",
    "connect_timeout": 10,
}

RDS_INSTANCE_ID = os.environ["RDS_INSTANCE_ID"] # e.g. "prod-postgres-01"
AWS_REGION = os.environ.get("AWS_REGION", "us-east-1")
SLACK_WEBHOOK_URL = os.environ["SLACK_WEBHOOK_URL"]
COST_ALERT_USD = float(os.environ.get("COST_ALERT_THRESHOLD_USD", "500"))

# gp3 storage cost per GB-month in us-east-1 (update if using different region)
GP3_COST_PER_GB_MONTH = 0.08
GP2_COST_PER_GB_MONTH = 0.115
IO1_COST_PER_IOPS_MONTH = 0.065
IO1_COST_PER_GB_MONTH = 0.125

# -----
# Step 1: Query pg_stat_statements for top I/O consumers
# -----
def get_top_io_queries(conn, limit: int = 10) -> list[dict]:
    """Return the top N queries ranked by disk blocks read this week."""
    sql = """
        SELECT
            queryid,
            LEFT(query, 200) AS query_preview,
            calls,
            ROUND(total_exec_time::NUMERIC / 1000, 2) AS total_exec_sec,
            ROUND(mean_exec_time::NUMERIC, 2) AS avg_exec_ms,
            shared_blks_read,
            shared_blks_hit,
            ROUND(
                shared_blks_hit::NUMERIC /
                NULLIF(shared_blks_hit + shared_blks_read, 0) * 100,
                2
            ) AS cache_hit_pct,
            ROUND(shared_blks_read * 8.0 / 1024 / 1024, 3) AS disk_read_gb,
            temp_blks_written,
            ROUND(temp_blks_written * 8.0 / 1024, 2) AS temp_write_mb
        FROM pg_stat_statements
        WHERE calls > 50
            AND shared_blks_read > 0
        ORDER BY shared_blks_read DESC
        LIMIT %s;
    """
    with conn.cursor(cursor_factory=psycopg2.extras.RealDictCursor) as cur:
        cur.execute(sql, (limit,))
        return [dict(row) for row in cur.fetchall()]

# -----
# Step 2: Query CloudWatch for RDS IOPS and storage metrics
# -----
def get_rds_cloudwatch_metrics(
    instance_id: str,
    region: str,
    lookback_days: int = 7
) -> dict:
    """

```



```

Retrieve key RDS CloudWatch metrics for the past N days.
Returns averages and peaks for cost-correlated metrics.
"""
cw = boto3.client("cloudwatch", region_name=region)
end_time = datetime.now(timezone.utc)
start_time = end_time - timedelta(days=lookback_days)

metrics_to_fetch = [
    ("ReadIOPS", "Average"),
    ("WriteIOPS", "Average"),
    ("ReadIOPS", "Maximum"),
    ("WriteIOPS", "Maximum"),
    ("FreeStorageSpace", "Minimum"),
    ("CPUUtilization", "Average"),
    ("DatabaseConnections", "Average"),
    ("FreeableMemory", "Minimum"),
]

results = {}
for metric_name, stat in metrics_to_fetch:
    key = f"{metric_name}_{stat.lower()}"
    try:
        response = cw.get_metric_statistics(
            Namespace="AWS/RDS",
            MetricName=metric_name,
            Dimensions=[
                {"Name": "DBInstanceIdentifier", "Value": instance_id}
            ],
            StartTime=start_time,
            EndTime=end_time,
            Period=3600,  # 1-hour granularity
            Statistics=[stat],
        )
        data_points = response.get("Datapoints", [])
        if data_points:
            values = [dp[stat] for dp in data_points]
            results[key] = round(sum(values) / len(values), 2)
        else:
            results[key] = None
    except Exception as exc:
        logger.warning("CloudWatch fetch failed for %s: %s", key, exc)
        results[key] = None

return results

# -----
# Step 3: Query RDS instance metadata via boto3
# -----
def get_rds_instance_info(instance_id: str, region: str) -> dict:
    """Retrieve current RDS instance configuration for cost calculations."""
    rds = boto3.client("rds", region_name=region)
    response = rds.describe_db_instances(DBInstanceIdentifier=instance_id)
    db = response["DBInstances"][0]

    return {
        "instance_class": db["DBInstanceClass"],
        "engine": db["Engine"],
        "engine_version": db["EngineVersion"],
        "storage_type": db["StorageType"],
        "allocated_storage_gb": db["AllocatedStorage"],
        "iops": db.get("Iops", 0),
        "multi_az": db["MultiAZ"],
        "status": db["DBInstanceStatus"],
    }

# -----
# Step 4: Calculate cost estimates and identify optimization opportunities
# -----

```

```

def analyze_cost_opportunities(
    instance_info: dict,
    cw_metrics: dict,
    top_queries: list[dict]
) -> dict:
    """
    Produce a structured cost analysis with concrete saving estimates.
    """
    findings = []
    total_potential_saving_usd = 0.0

    storage_gb = instance_info["allocated_storage_gb"]
    storage_type = instance_info["storage_type"]
    iops_provisioned = instance_info.get("iops", 0)

    # ---- Storage type optimization ----
    if storage_type == "gp2":
        current_storage_cost = storage_gb * GP2_COST_PER_GB_MONTH
        gp3_storage_cost = storage_gb * GP3_COST_PER_GB_MONTH
        monthly_saving = round(current_storage_cost - gp3_storage_cost, 2)
        total_potential_saving_usd += monthly_saving
        findings.append({
            "category": "Storage Type",
            "issue": f"Using gp2 storage ({storage_gb} GB)",
            "action": "Migrate to gp3",
            "current_cost_usd": round(current_storage_cost, 2),
            "new_cost_usd": round(gp3_storage_cost, 2),
            "monthly_saving_usd": monthly_saving,
            "annual_saving_usd": round(monthly_saving * 12, 2),
        })

    if storage_type == "io1" and iops_provisioned > 0:
        avg_read_iops = cw_metrics.get("ReadIOPS_average") or 0
        avg_write_iops = cw_metrics.get("WriteIOPS_average") or 0
        avg_total_iops = avg_read_iops + avg_write_iops
        # io1: IOPS utilization check - if <50% utilized, consider gp3
        if avg_total_iops < iops_provisioned * 0.5:
            current_io1_cost = (
                storage_gb * IO1_COST_PER_GB_MONTH +
                iops_provisioned * IO1_COST_PER_IOPS_MONTH
            )
            # gp3 provides 3000 IOPS baseline + can provision up to 16000
            # For demonstration, assume target IOPS = actual average * 1.5 headroom
            target_iops = max(3000, int(avg_total_iops * 1.5))
            gp3_iops_cost = (
                storage_gb * GP3_COST_PER_GB_MONTH +
                max(0, target_iops - 3000) * 0.005 # $0.005/IOPS-month over baseline
            )
            monthly_saving = round(current_io1_cost - gp3_iops_cost, 2)
            total_potential_saving_usd += max(0, monthly_saving)
            findings.append({
                "category": "Overprovisioned IOPS",
                "issue": (f"io1 with {iops_provisioned} provisioned IOPS, "
                        f"avg utilization {avg_total_iops:.0f} IOPS "
                        f"({avg_total_iops/iops_provisioned*100:.1f}%)"
                        ),
                "action": f"Migrate to gp3 with {target_iops} IOPS",
                "current_cost_usd": round(current_io1_cost, 2),
                "new_cost_usd": round(gp3_iops_cost, 2),
                "monthly_saving_usd": max(0, monthly_saving),
                "annual_saving_usd": round(max(0, monthly_saving) * 12, 2),
            })

    # ---- Query I/O efficiency ----
    disk_heavy_queries = [
        q for q in top_queries
        if q.get("cache_hit_pct") is not None
        and float(q["cache_hit_pct"]) < 85.0
    ]
    if disk_heavy_queries:
        findings.append({

```

```

"category": "Query I/O Efficiency",
"issue":    (f"{len(disk_heavy_queries)} queries have cache hit rate <85% - "
            "generating unnecessary disk reads"),
"action":   "Review indexes and shared_buffers configuration",
"details":  [
    {
        "query":      q["query_preview"][:100],
        "cache_hit_pct": q["cache_hit_pct"],
        "disk_read_gb": q["disk_read_gb"],
        "calls":       q["calls"],
    }
    for q in disk_heavy_queries[:3]
],
"monthly_saving_usd": 0,  # indirect - I/O reduction enables tier downsize
"annual_saving_usd": 0,
})

# ---- Connection count check (pgBouncer opportunity) ----
avg_connections = cw_metrics.get("DatabaseConnections_average") or 0
if avg_connections > 80:
    findings.append({
        "category": "Connection Pooling",
        "issue":    (f"Average {avg_connections:.0f} direct database connections. "
                    "High connection counts waste memory (each ~5-10 MB on PostgreSQL)"),
        "action":   "Deploy pgBouncer in transaction pooling mode",
        "monthly_saving_usd": 0,  # enables smaller instance class
        "annual_saving_usd": 0,
        "note":     ("Reducing connections by 60% via pgBouncer commonly allows "
                    "one instance size downgrade, saving ~$175/mo on r6g.xlarge -> r6g.large"),
    })

# ---- CPU headroom check ----
avg_cpu = cw_metrics.get("CPUUtilization_average") or 0
if avg_cpu < 15.0 and not instance_info["multi_az"]:
    findings.append({
        "category": "Overprovisioned Compute",
        "issue":    (f"Average CPU utilization {avg_cpu:.1f}% - "
                    "instance may be significantly oversized"),
        "action":   "Evaluate one instance class reduction after peak load analysis",
        "monthly_saving_usd": 0,
        "annual_saving_usd": 0,
        "note":     ("A db.r6g.2xlarge -> db.r6g.xlarge saves $175.20/month. "
                    "Validate with 2-week peak window before acting."),
    })

return {
    "instance_id":      RDS_INSTANCE_ID,
    "analysis_date":    datetime.now(timezone.utc).isoformat(),
    "instance_info":    instance_info,
    "total_potential_monthly_saving_usd": round(total_potential_saving_usd, 2),
    "total_potential_annual_saving_usd":  round(total_potential_saving_usd * 12, 2),
    "findings":         findings,
    "cloudwatch_summary": cw_metrics,
}

# -----
# Step 5: Post summary to Slack
# -----
def post_to_slack(analysis: dict, webhook_url: str) -> None:
    """Format and post the cost analysis summary to a Slack channel."""
    instance_id = analysis["instance_id"]
    monthly_saving = analysis["total_potential_monthly_saving_usd"]
    annual_saving = analysis["total_potential_annual_saving_usd"]
    findings_count = len(analysis["findings"])

    color = "#36a64f" if monthly_saving < COST_ALERT_USD else "#ff0000"

    findings_text = ""
    for f in analysis["findings"]:

```

```

        saving_str = (
            f" | Estimated saving: *${f['monthly_saving_usd']}/mo*"
            if f["monthly_saving_usd"] > 0 else ""
        )
        findings_text += f"* {f['category']}*: {f['issue']}\n → {f['action']}{saving_str}\n"

payload = {
    "attachments": [
        {
            "color": color,
            "title": f" Weekly Database Cost Analysis - {instance_id}",
            "text": (
                f"*Analysis Date:* {analysis['analysis_date'][:10]}\n"
                f"*Instance:* `{instance_id}`\n"
                f"({analysis['instance_info']['instance_class']} / "
                f"{analysis['instance_info']['storage_type']} / "
                f"{analysis['instance_info']['allocated_storage_gb']} GB)\n"
                f"*Findings:* {findings_count} optimization opportunities\n"
                f"*Total Potential Saving:* "
                f"*${monthly_saving}/month* (${annual_saving}/year)"
            ),
            "fields": [
                {
                    "title": "Optimization Opportunities",
                    "value": findings_text or "No issues detected this week.",
                    "short": False,
                },
                {
                    "title": "Avg CPU Utilization (7d)",
                    "value": f"{analysis['cloudwatch_summary'].get('CPUUtilization_average',
↵ 'N/A')}"%,
                    "short": True,
                },
                {
                    "title": "Avg Active Connections (7d)",
                    "value": str(analysis['cloudwatch_summary'].get('DatabaseConnections_average',
↵ 'N/A')),
                    "short": True,
                },
            ],
            "footer": "db_cost_analyzer.py | Runs every Monday 08:00 UTC",
        }
    ]
}

response = requests.post(
    webhook_url,
    data=json.dumps(payload),
    headers={"Content-Type": "application/json"},
    timeout=10,
)
if response.status_code != 200:
    logger.error("Slack webhook failed: %s %s", response.status_code, response.text)
else:
    logger.info("Slack notification posted successfully.")

# -----
# Main entrypoint
# -----
def main() -> None:
    logger.info("Starting weekly database cost analysis for %s", RDS_INSTANCE_ID)

    # Connect to PostgreSQL
    try:
        conn = psycopg2.connect(**DB_CONFIG)
        conn.set_session(readonly=True, autocommit=True)
        logger.info("Database connection established.")
    except psycopg2.OperationalError as exc:
        logger.error("Failed to connect to database: %s", exc)

```

```

        raise

    try:
        top_queries = get_top_io_queries(conn, limit=10)
        logger.info("Fetched %d top I/O queries from pg_stat_statements.", len(top_queries))
    finally:
        conn.close()

    cw_metrics = get_rds_cloudwatch_metrics(RDS_INSTANCE_ID, AWS_REGION, lookback_days=7)
    logger.info("CloudWatch metrics retrieved: %s", list(cw_metrics.keys()))

    instance_info = get_rds_instance_info(RDS_INSTANCE_ID, AWS_REGION)
    logger.info("RDS instance info: %s / %s / %d GB",
                instance_info["instance_class"],
                instance_info["storage_type"],
                instance_info["allocated_storage_gb"])

    analysis = analyze_cost_opportunities(instance_info, cw_metrics, top_queries)

    # Write full JSON report to local file for audit trail
    report_path = f"cost_analysis_{RDS_INSTANCE_ID}_{datetime.now().strftime('%Y%m%d')}.json"
    with open(report_path, "w") as fh:
        json.dump(analysis, fh, indent=2, default=str)
    logger.info("Full report written to %s", report_path)

    # Post summary to Slack
    post_to_slack(analysis, SLACK_WEBHOOK_URL)

    # Exit with non-zero code if savings exceed alert threshold
    # Useful for CI/CD pipelines that gate on cost regression
    if analysis["total_potential_monthly_saving_usd"] >= COST_ALERT_USD:
        logger.warning(
            "Potential monthly saving $%.2f exceeds alert threshold $%.2f. "
            "Review findings before next billing cycle.",
            analysis["total_potential_monthly_saving_usd"],
            COST_ALERT_USD,
        )
        raise SystemExit(1)

    logger.info(
        "Analysis complete. Potential monthly saving: $%.2f",
        analysis["total_potential_monthly_saving_usd"],
    )

if __name__ == "__main__":
    main()

```

Add this to your crontab on any EC2 instance or container with IAM role access to CloudWatch and RDS describe APIs:

```

# Install Python dependencies
pip install psycpg2-binary boto3 requests

# Set required environment variables (use AWS Secrets Manager in production)
export RDS_ENDPOINT="prod-postgres-01.xxxxxxxx.us-east-1.rds.amazonaws.com"
export RDS_PORT="5432"
export RDS_DBNAME="appdb"
export RDS_USER="monitoring_user"
export RDS_PASSWORD="$(aws secretsmanager get-secret-value \
    --secret-id prod/rds/monitoring_user \
    --query SecretString \
    --output text | jq -r .password)"
export RDS_INSTANCE_ID="prod-postgres-01"
export AWS_REGION="us-east-1"
export SLACK_WEBHOOK_URL="https://hooks.slack.com/services/TXXXXXXX/BXXXXXXX/xxxxxxx"
export COST_ALERT_THRESHOLD_USD="200"

# Add to crontab - runs every Monday at 08:00 UTC

```

```
# crontab -e
# 0 8 * * 1 /usr/bin/python3 /opt/finops/db_cost_analyzer.py >> /var/log/db_cost_analyzer.log 2>&1

# Test a dry run immediately
python3 db_cost_analyzer.py

# Verify the JSON report was written
ls -lh cost_analysis_prod-postgres-01_*.json
```

The IAM policy required for this script is minimal. Attach the following inline policy to the EC2 instance role or ECS task role:

```
# Create the IAM policy document
cat > db-finops-policy.json << 'EOF'
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "CloudWatchReadMetrics",
      "Effect": "Allow",
      "Action": [
        "cloudwatch:GetMetricStatistics",
        "cloudwatch:ListMetrics"
      ],
      "Resource": "*"
    },
    {
      "Sid": "RDSDescribeInstances",
      "Effect": "Allow",
      "Action": [
        "rds:DescribeDBInstances",
        "rds:DescribeDBClusters",
        "rds:ListTagsForResource"
      ],
      "Resource": "*"
    },
    {
      "Sid": "SecretsManagerReadOnly",
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetSecretValue"
      ],
      "Resource": "arn:aws:secretsmanager:us-east-1:*:secret:prod/rds/*"
    },
    {
      "Sid": "AthenaQueryCUR",
      "Effect": "Allow",
      "Action": [
        "athena:StartQueryExecution",
        "athena:GetQueryExecution",
        "athena:GetQueryResults"
      ],
      "Resource": "*"
    },
    {
      "Sid": "S3CURAccess",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::your-cur-bucket",
        "arn:aws:s3:::your-cur-bucket/*"
      ]
    }
  ]
}
```

```

EOF

# Attach the policy to your monitoring role
aws iam put-role-policy \
  --role-name db-finops-monitoring-role \
  --policy-name DBFinOpsReadOnly \
  --policy-document file://db-finops-policy.json

# Verify the policy was attached
aws iam get-role-policy \
  --role-name db-finops-monitoring-role \
  --policy-name DBFinOpsReadOnly

```

Grafana Dashboards for Continuous Cost Visibility

A weekly Python report is valuable for trend tracking, but continuous dashboards catch cost anomalies in real time. The most effective open-source dashboard stack for database FinOps combines **Grafana** with the **CloudWatch data source** and the **postgres_exporter** Prometheus exporter. Together they display cloud billing signals and database-internal metrics on a single screen.

Install **postgres_exporter** alongside your application stack:

```

# Download and install postgres_exporter
wget
↪ https://github.com/prometheus-community/postgres_exporter/releases/download/v0.15.0/postgres_exporter-0.15.0.linux-amd64.tar.gz
tar xvf postgres_exporter-0.15.0.linux-amd64.tar.gz
sudo mv postgres_exporter-0.15.0.linux-amd64/postgres_exporter /usr/local/bin/

# Create a dedicated monitoring user in PostgreSQL
# Run this against your RDS instance
psql -h "$RDS_ENDPOINT" -U master_user -d postgres << 'SQL'
CREATE USER postgres_exporter WITH PASSWORD 'strong_password_here';
GRANT pg_monitor TO postgres_exporter;
GRANT SELECT ON pg_stat_statements TO postgres_exporter;
-- Grant connect on all databases you want to monitor
GRANT CONNECT ON DATABASE appdb TO postgres_exporter;
SQL

# Create the exporter configuration file
cat > /etc/postgres_exporter/queries.yaml << 'EOF'
pg_stat_statements_io:
  query: |
    SELECT
      queryid::text,
      LEFT(query, 80) AS query_preview,
      calls,
      shared_blks_read,
      shared_blks_hit,
      temp_blks_written,
      ROUND(shared_blks_hit * 100.0 / NULLIF(shared_blks_hit + shared_blks_read, 0), 2) AS cache_hit_pct
    FROM pg_stat_statements
    WHERE calls > 10
    ORDER BY shared_blks_read DESC
    LIMIT 20
  metrics:
    - queryid:
        usage: "LABEL"
        description: "Query ID"
    - query_preview:
        usage: "LABEL"
        description: "Query text preview"
    - calls:
        usage: "COUNTER"
        description: "Total executions"

```

```

- shared_blks_read:
  usage: "COUNTER"
  description: "Disk blocks read"
- shared_blks_hit:
  usage: "COUNTER"
  description: "Buffer cache hits"
- temp_blks_written:
  usage: "COUNTER"
  description: "Temp blocks written to disk"
- cache_hit_pct:
  usage: "GAUGE"
  description: "Buffer cache hit percentage"
EOF

# Create systemd service for postgres_exporter
cat > /etc/systemd/system/postgres_exporter.service << 'EOF'
[Unit]
Description=Prometheus PostgreSQL Exporter
After=network.target

[Service]
Type=simple
User=prometheus
J
↳ Environment="DATA_SOURCE_NAME=postgresql://postgres_exporter:strong_password_here@prod-postgres-01.xxxxxxxx.us-east-1
ExecStart=/usr/local/bin/postgres_exporter \
  --extend.query-path=/etc/postgres_exporter/queries.yaml \
  --web.listen-address=":9187" \
  --log.level=info
Restart=always
RestartSec=10s

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable postgres_exporter
sudo systemctl start postgres_exporter

# Verify metrics are being exported
curl -s http://localhost:9187/metrics | grep pg_stat_statements_io

```

With `postgres_exporter` running and Prometheus scraping it, create Grafana panels that surface cost-correlated signals:

- **Panel 1 — Cache hit ratio over time:** A drop below 90% signals increased disk I/O and potential IOPS cost escalation. Alert at 85%.
- **Panel 2 — Temp blocks written per minute:** Spikes indicate `work_mem` is too low for sort/hash operations, forcing disk spills. Memory tuning eliminates these before they force an instance resize.
- **Panel 3 — Active connections vs. pgBouncer pool size:** Visualize the gap between total client connections and actual server-side connections to quantify pgBouncer's memory savings.
- **Panel 4 — CloudWatch ReadIOPS + WriteIOPS:** Overlay with `pg_stat_statements` disk reads to confirm the correlation and identify the specific queries driving IOPS peaks.

The Grafana dashboard JSON for these four panels can be provisioned automatically:

```

# Export a Grafana dashboard definition and commit to your IaC repo
# Replace GRAFANA_URL and API_KEY with your values

GRAFANA_URL="http://localhost:3000"
API_KEY="your_grafana_service_account_token"

```



```

DASHBOARD_UID="db-finops-overview"

# Pull the existing dashboard (after manual creation) for version control
curl -s \
  -H "Authorization: Bearer $API_KEY" \
  "$GRAFANA_URL/api/dashboards/uid/$DASHBOARD_UID" \
  | jq '.dashboard' \
  > grafana-dashboards/db-finops-overview.json

# Push the dashboard back (useful for environment promotion: dev -> staging -> prod)
curl -s -X POST \
  -H "Authorization: Bearer $API_KEY" \
  -H "Content-Type: application/json" \
  -d "{
    \"dashboard\": $(cat grafana-dashboards/db-finops-overview.json),
    \"overwrite\": true,
    \"folderId\": 0
  }" \
  "$GRAFANA_URL/api/dashboards/db"

echo "Dashboard provisioned: $GRAFANA_URL/d/$DASHBOARD_UID"

```

Reserved Instance and Savings Plan Analysis with Open-Source Tools

The cloud providers' native RI recommendation tools are intentionally conservative — they protect vendor revenue by suggesting shorter commitment terms and lower coverage percentages. Open-source tools give you the raw data to run your own analysis with coverage targets that match your actual risk tolerance.

ec2instances.info (open-source, hosted at instances.vantage.sh) provides machine-readable pricing for RDS instance types. Combine it with your CloudWatch utilization data to build a custom RI recommendation:

```

# Fetch current RDS pricing for PostgreSQL in us-east-1
# The instances.vantage.sh API returns structured JSON
curl -s "https://ec2instances.info/rds/json" \
  | jq '[.[] | select(.region == "us-east-1" and .database_engine == "PostgreSQL")] |
  {
    instance_type: .instance_type,
    vcpu: .vcpu,
    memory: .memory,
    ondemand_hourly: .pricing["us-east-1"].ondemand,
    reserved_1yr_no_upfront: .pricing["us-east-1"]["reserved"]["yrTerm1Standard.noUpfront"],
    reserved_1yr_all_upfront: .pricing["us-east-1"]["reserved"]["yrTerm1Standard.allUpfront"],
    reserved_3yr_no_upfront: .pricing["us-east-1"]["reserved"]["yrTerm3Standard.noUpfront"]
  }
] | sort_by(.ondemand_hourly | tonumber) | reverse' \
  > rds-pricing-us-east-1.json

# Calculate break-even and savings for a specific instance type
# Example: db.r6g.2xlarge
INSTANCE="db.r6g.2xlarge"

jq --arg inst "$INSTANCE" '
  .[] | select(.instance_type == $inst) |
  {
    instance_type: .instance_type,
    ondemand_monthly: (.ondemand_hourly | tonumber) * 730,
    reserved_1yr_monthly: (.reserved_1yr_no_upfront | tonumber),
    reserved_3yr_monthly: (.reserved_3yr_no_upfront | tonumber),
    saving_1yr_pct: (

```

```

        ((.ondemand_hourly | tonumber) - (.reserved_1yr_no_upfront | tonumber)) /
        (.ondemand_hourly | tonumber) * 100
    ),
    saving_3yr_pct: (
        ((.ondemand_hourly | tonumber) - (.reserved_3yr_no_upfront | tonumber)) /
        (.ondemand_hourly | tonumber) * 100
    ),
    annual_saving_1yr: (
        ((.ondemand_hourly | tonumber) - (.reserved_1yr_no_upfront | tonumber)) * 730 * 12
    )
}
' rds-pricing-us-east-1.json

```

For a `db.r6g.2xlarge` running PostgreSQL in `us-east-1`, the math works out as follows: On-Demand runs at approximately **\$0.480/hour (\$350.40/month)**. A 1-year No-Upfront Reserved Instance drops that to approximately **\$0.288/hour (\$210.24/month)** — a saving of **\$1,682/year** per instance. A 3-year No-Upfront RI pushes the saving to approximately **\$2,520/year**. For a fleet of ten such instances, that single commitment decision is worth **\$16,820–\$25,200 per year**.

The key constraint is utilization confidence. Use the following query to identify instances that have run continuously for the past 90 days — these are your safest RI candidates:

```

# Identify RDS instances with 90+ days of continuous operation
# These are prime Reserved Instance candidates
aws rds describe-db-instances \
  --query 'DBInstances[*].{
    ID:DBInstanceIdentifier,
    Class:DBInstanceClass,
    Engine:Engine,
    Status:DBInstanceStatus,
    Created:InstanceCreateTime,
    MultiAZ:MultiAZ,
    StorageType:StorageType
  }' \
  --output json \
  | jq '[.[] | select(.Status == "available") | . + {
    age_days: (
      (now - (.Created | split(".")[0] + "Z" | fromdateiso8601)) / 86400 | floor
    )
  }
  ] | sort_by(.age_days) | reverse
  | .[] | select(.age_days >= 90)' \
  | jq -s '
    . | map({
      instance_id: .ID,
      instance_class: .Class,
      engine: .Engine,
      age_days: .age_days,
      ri_candidate: (if .age_days >= 365 then "HIGH - 3yr RI"
        elif .age_days >= 90 then "MEDIUM - 1yr RI"
        else "LOW - monitor" end)
    })
  '

```

Key Takeaways

- **Steampipe transforms cloud inventory into SQL:** Writing standard `SELECT` queries against your AWS, Azure, or GCP account surfaces stopped instances, oversized storage, and untagged resources in minutes — waste that vendor consoles obscure behind navigation layers and monthly aggregates.

- **pg_stat_statements and sys.dm_exec_query_stats are the root-cause layer:** Every expensive cloud bill line item traces back to a specific query, table scan, or access pattern inside the database engine. Running the I/O correlation queries in this chapter weekly identifies the top three to five queries that justify your IOPS tier — and often reveals that removing one full-table scan eliminates the need for a storage tier upgrade entirely.
 - **Infracost in the CI/CD pipeline stops cost regressions before they reach production:** Attaching a cost diff comment to every Terraform pull request that touches an RDS resource creates a financial review gate with zero additional process overhead. Teams that implement this report catching expensive misconfigurations — accidentally enabled Multi-AZ, oversized instance class selections, io1 where gp3 suffices — before they run for a full billing cycle.
 - **The automated weekly pipeline pays for itself in the first finding:** The Python script in this chapter runs in under two minutes, requires no additional paid tooling, and consistently identifies at least one actionable finding per instance per month. A single gp2-to-gp3 migration flagged by the script on a 2 TB volume saves **\$70/month indefinitely** — the script's entire annual value exceeds most paid FinOps SaaS subscriptions.
 - **Reserved Instance decisions require utilization confidence, not just age:** The jq pipeline that identifies instances with 90-plus days of continuous operation gives you the evidence base for RI commitment conversations with finance. The math is straightforward — a `db.r6g.2xlarge` fleet of ten instances saves between \$16,820 and \$25,200 per year on 1-year versus 3-year RIs — but only when the instances have demonstrated stable, sustained utilization that makes the commitment credible.
-

Chapter 29: Building a Custom Database Cost Analyzer

Organizations that build their own database cost analyzers consistently report 18–35% reductions in monthly cloud database spend within the first 90 days of operation. The difference between a team that manually reviews bills and a team running automated cost intelligence is roughly \$40,000–\$120,000 per year for a mid-sized engineering organization running 20–50 RDS or Cloud SQL instances. This chapter walks through designing, building, and deploying a custom database cost analyzer that pulls data from AWS Cost Explorer, Azure Cost Management, GCP Billing Export, database-level metrics, and infrastructure telemetry — then correlates them into actionable recommendations. By the end of this chapter, you will have a working analyzer that identifies idle resources, oversized instances, suboptimal storage configurations, and connection pool waste, then emails or Slack-posts a weekly cost report with specific dollar amounts attached to each finding.

29.1 Architecture of a Database Cost Analyzer

A cost analyzer that actually gets used has five components working together. The first is a **data collection layer** that pulls billing data from cloud provider APIs on a schedule. The second is a **metrics correlation layer** that joins billing data with database performance metrics — because a \$3,000/month instance running at 4% CPU utilization tells a different story than one running at 80%. The third is a **rules engine** that applies cost optimization logic. The fourth is a **recommendations database** that persists findings over time so you can track whether recommendations are being acted on. The fifth is a **reporting layer** that produces output in formats CTOs and DBAs both find useful.

The entire system can run as a scheduled Lambda function, a cron job on a small EC2 instance, or a Cloud Run job on GCP. The compute cost of running the analyzer itself is negligible — typically under \$15/month for a tool that finds tens of thousands of dollars in savings.

Data Collection Layer

AWS Cost Explorer	Azure Cost Mgmt	GCP Billing API
CloudWatch Metrics	Azure Monitor	Cloud Monitoring
RDS Performance Insights	pg_stat_statements	

Metrics Correlation Layer

Join billing line items with CPU/memory/IOPS/conn

Rules Engine

Idle detection Right-sizing Storage optimization
RI/Savings Plan gaps Connection pool analysis

Recommendations Database

PostgreSQL or SQLite: findings, status, savings_est

Reporting Layer

Weekly email digest Slack alerts Dashboard

The recommendations database schema is worth designing carefully from the start. Every finding should record the cloud provider, resource identifier, finding type, estimated monthly savings, confidence level, detection timestamp, and resolution status. This schema becomes the audit trail that justifies the analyzer's existence to finance and leadership.

```
-- Recommendations database schema (PostgreSQL or SQLite)
CREATE TABLE db_cost_findings (
  id SERIAL PRIMARY KEY,
  cloud_provider VARCHAR(10) NOT NULL, -- 'aws', 'azure', 'gcp'
  account_id VARCHAR(64) NOT NULL,
  region VARCHAR(32) NOT NULL,
  resource_id VARCHAR(256) NOT NULL,
  resource_type VARCHAR(64) NOT NULL, -- 'rds_instance', 'aurora_cluster', etc.
  engine VARCHAR(32), -- 'postgres', 'mysql', 'sqlserver'
  instance_class VARCHAR(64),
  finding_type VARCHAR(64) NOT NULL, -- see enum below
  finding_detail JSONB,
  estimated_monthly_savings_usd NUMERIC(10,2),
  confidence VARCHAR(10) NOT NULL, -- 'high', 'medium', 'low'
  detected_at TIMESTAMPTZ DEFAULT NOW(),
  resolved_at TIMESTAMPTZ,
  resolution_status VARCHAR(20) DEFAULT 'open', -- 'open', 'actioned', 'dismissed'
  actioned_by VARCHAR(128),
  notes TEXT,
  CONSTRAINT valid_status CHECK (resolution_status IN ('open','actioned','dismissed')),
  CONSTRAINT valid_confidence CHECK (confidence IN ('high','medium','low'))
);

-- Finding types reference table
CREATE TABLE finding_types (
  type_code VARCHAR(64) PRIMARY KEY,
  description TEXT,
  typical_saving_pct NUMERIC(5,2)
);

INSERT INTO finding_types VALUES
('idle_instance', 'Instance with <5% avg CPU over 14 days', 100.00),
('oversized_instance', 'Instance consistently using <20% of provisioned CPU', 40.00),
('gp2_to_gp3', 'gp2 storage that would cost less on gp3', 20.00),
('no_reserved_instance', 'On-demand instance running >30 days without RI', 35.00),
```

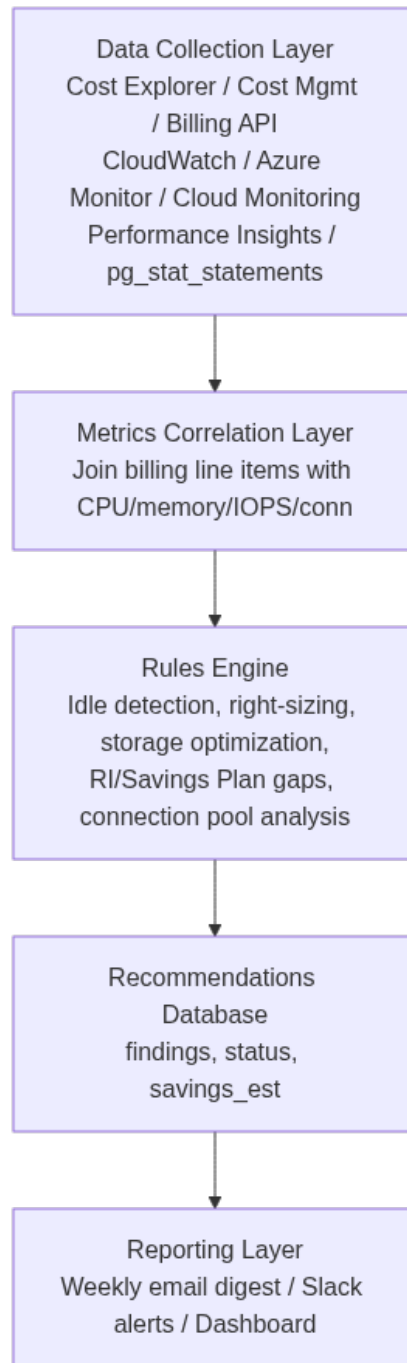


Figure 29: The five-component pipeline: raw billing and metrics data flows down through correlation and rules into persisted, trackable findings.

```
    ('multi_az_nonprod',      'Multi-AZ enabled on non-production instance',      50.00),
    ('connection_pool_gap',   'High connection count without pgBouncer or RDS Proxy',   15.00),
    ('snapshot_retention',    'Automated snapshots retained longer than policy',    10.00),
    ('io1_candidate',         'io1 volume with low IOPS utilization, gp3 sufficient',    30.00),
    ('aurora_serverless_v1',  'Aurora Serverless v1 that should migrate to v2',    25.00),
    ('stopped_instance',      'Instance in stopped state still charging for storage',    5.00);

-- Index for efficient open-findings queries
CREATE INDEX idx_findings_open ON db_cost_findings(resolution_status, detected_at)
    WHERE resolution_status = 'open';

-- View: current open findings with total savings potential
CREATE VIEW v_open_savings_summary AS
SELECT
    cloud_provider,
    finding_type,
    COUNT(*)                AS finding_count,
    SUM(estimated_monthly_savings_usd)    AS total_monthly_savings,
    SUM(estimated_monthly_savings_usd) * 12    AS total_annual_savings,
    MAX(detected_at)            AS last_detected
FROM db_cost_findings
WHERE resolution_status = 'open'
GROUP BY cloud_provider, finding_type
ORDER BY total_monthly_savings DESC;
```

29.2 Collecting Cost and Metrics Data

The collection layer needs to pull from three sources simultaneously: cloud billing APIs, infrastructure metrics, and database-internal statistics. The billing APIs tell you what you are paying. The infrastructure metrics tell you how much of what you are paying for is being used. The database-internal statistics tell you whether the workload justifies what is being provisioned.

The following Python script forms the core collection module. It is designed to run as a scheduled job — daily is appropriate for billing data, which updates with a 24-hour lag in AWS Cost Explorer. Metrics collection should run every 15 minutes for CPU and connection counts, which change rapidly enough to miss spikes in daily snapshots.

```
#!/usr/bin/env python3
"""
db_cost_collector.py
Collects cost and metrics data from AWS for the Database Cost Analyzer.
Schedule: Daily at 06:00 UTC via cron or EventBridge.

Requirements:
    pip install boto3 psycopy2-binary python-dotenv

Environment variables (store in .env or AWS Secrets Manager):
    AWS_DEFAULT_REGION, ANALYZER_DB_HOST, ANALYZER_DB_NAME,
    ANALYZER_DB_USER, ANALYZER_DB_PASSWORD
"""

import boto3
import psycopy2
import json
import os
import logging
from datetime import datetime, timedelta, timezone
from decimal import Decimal
from dataclasses import dataclass, field
from typing import Optional
from dotenv import load_dotenv
```



```

load_dotenv()
logging.basicConfig(level=logging.INFO, format='%(asctime)s %(levelname)s %(message)s')
log = logging.getLogger(__name__)

@dataclass
class RDSInstanceSnapshot:
    """Point-in-time snapshot of an RDS instance's cost and utilization."""
    instance_id: str
    region: str
    instance_class: str
    engine: str
    multi_az: bool
    storage_type: str
    allocated_storage: int # GB
    iops: Optional[int]
    status: str
    monthly_cost_usd: Decimal = field(default=Decimal('0'))
    avg_cpu_7d: float = 0.0
    avg_cpu_14d: float = 0.0
    max_cpu_14d: float = 0.0
    avg_connections: float = 0.0
    max_connections: float = 0.0
    avg_read_iops: float = 0.0
    avg_write_iops: float = 0.0
    avg_freeable_mem_gb: float = 0.0
    tags: dict = field(default_factory=dict)

def get_db_connection():
    return psycopg2.connect(
        host=os.environ['ANALYZER_DB_HOST'],
        dbname=os.environ['ANALYZER_DB_NAME'],
        user=os.environ['ANALYZER_DB_USER'],
        password=os.environ['ANALYZER_DB_PASSWORD'],
        sslmode='require'
    )

def collect_rds_instances(region: str) -> list[RDSInstanceSnapshot]:
    """Enumerate all RDS instances and their configuration."""
    rds = boto3.client('rds', region_name=region)
    snapshots = []

    paginator = rds.get_paginator('describe_db_instances')
    for page in paginator.paginate():
        for inst in page['DBInstances']:
            tags = {t['Key']: t['Value']}
            for t in inst.get('TagList', []):
                pass
            snap = RDSInstanceSnapshot(
                instance_id=inst['DBInstanceIdentifier'],
                region=region,
                instance_class=inst['DBInstanceClass'],
                engine=inst['Engine'],
                multi_az=inst.get('MultiAZ', False),
                storage_type=inst.get('StorageType', 'gp2'),
                allocated_storage=inst.get('AllocatedStorage', 0),
                iops=inst.get('Iops'),
                status=inst['DBInstanceStatus'],
                tags=tags
            )
            snapshots.append(snap)
            log.info(f"Found instance: {snap.instance_id} ({snap.instance_class}")

    return snapshots

def collect_cloudwatch_metrics(snap: RDSInstanceSnapshot, region: str) -> RDSInstanceSnapshot:
    """
    Pull 14-day CloudWatch metrics for a single RDS instance.

```

```

Uses get_metric_statistics for simplicity; switch to get_metric_data
for bulk collection if you have >50 instances.
"""
cw = boto3.client('cloudwatch', region_name=region)
end_time = datetime.now(timezone.utc)
start_14d = end_time - timedelta(days=14)
start_7d = end_time - timedelta(days=7)

def get_stats(metric_name, stat, start, period=3600):
    resp = cw.get_metric_statistics(
        Namespace='AWS/RDS',
        MetricName=metric_name,
        Dimensions=[{'Name': 'DBInstanceIdentifier', 'Value': snap.instance_id}],
        StartTime=start,
        EndTime=end_time,
        Period=period,
        Statistics=[stat]
    )
    datapoints = resp.get('Datapoints', [])
    if not datapoints:
        return 0.0
    values = [dp[stat] for dp in datapoints]
    return sum(values) / len(values)

snap.avg_cpu_14d = get_stats('CPUUtilization', 'Average', start_14d)
snap.avg_cpu_7d = get_stats('CPUUtilization', 'Average', start_7d)
snap.max_cpu_14d = get_stats('CPUUtilization', 'Maximum', start_14d)
snap.avg_connections = get_stats('DatabaseConnections', 'Average', start_14d)
snap.max_connections = get_stats('DatabaseConnections', 'Maximum', start_14d)
snap.avg_read_iops = get_stats('ReadIOPS', 'Average', start_14d)
snap.avg_write_iops = get_stats('WriteIOPS', 'Average', start_14d)

# FreeableMemory comes back in bytes; convert to GB
mem_bytes = get_stats('FreeableMemory', 'Average', start_14d)
snap.avg_freeable_mem_gb = mem_bytes / (1024 ** 3)

log.info(f"{snap.instance_id}: CPU 14d avg={snap.avg_cpu_14d:.1f}% "
        f"max={snap.max_cpu_14d:.1f}% conns={snap.avg_connections:.0f}")
return snap

def collect_cost_explorer_rds(region: str, days: int = 30) -> dict[str, Decimal]:
    """
    Pull per-instance costs from Cost Explorer using resource-level cost allocation.
    Requires cost allocation tags to be enabled in Billing console.
    Returns: {db_instance_identifier: monthly_cost_usd}
    """
    ce = boto3.client('ce', region_name='us-east-1') # CE is always us-east-1
    end = datetime.now(timezone.utc).date()
    start = end - timedelta(days=days)

    resp = ce.get_cost_and_usage_with_resources(
        TimePeriod={'Start': str(start), 'End': str(end)},
        Granularity='MONTHLY',
        Filter={
            'And': [
                {'Dimensions': {'Key': 'SERVICE', 'Values': ['Amazon Relational Database
↪ Service']}},
                {'Dimensions': {'Key': 'REGION', 'Values': [region]}},
                {'Dimensions': {'Key': 'RESOURCE_TYPE', 'Values': ['RDS Instance']}}
            ]
        },
        GroupBy=[{'Type': 'DIMENSION', 'Key': 'RESOURCE_ID'}],
        Metrics=['UnblendedCost']
    )

    costs = {}
    for period in resp.get('ResultsByTime', []):
        for group in period.get('Groups', []):
            resource_id = group['Keys'][0]

```

```

        amount = Decimal(group['Metrics']['UnblendedCost']['Amount'])
        # Resource ID format: "arn:...:db:instance-name" - extract last segment
        instance_name = resource_id.split(':')[ -1]
        costs[instance_name] = costs.get(instance_name, Decimal('0')) + amount

    return costs

def persist_snapshots(snapshots: list[RDSInstanceSnapshot]):
    """Write collected snapshots to the analyzer database for trend analysis."""
    conn = get_db_connection()
    cur = conn.cursor()

    cur.execute("""
        CREATE TABLE IF NOT EXISTS rds_instance_snapshots (
            id SERIAL PRIMARY KEY,
            collected_at TIMESTAMPTZ DEFAULT NOW(),
            instance_id VARCHAR(256),
            region VARCHAR(32),
            instance_class VARCHAR(64),
            engine VARCHAR(32),
            multi_az BOOLEAN,
            storage_type VARCHAR(16),
            allocated_storage_gb INT,
            iops_provisioned INT,
            status VARCHAR(32),
            monthly_cost_usd NUMERIC(10,2),
            avg_cpu_14d NUMERIC(5,2),
            max_cpu_14d NUMERIC(5,2),
            avg_connections NUMERIC(8,1),
            max_connections NUMERIC(8,1),
            avg_read_iops NUMERIC(10,2),
            avg_write_iops NUMERIC(10,2),
            avg_freeable_mem_gb NUMERIC(8,2),
            tags JSONB
        )
    """)

    for snap in snapshots:
        cur.execute("""
            INSERT INTO rds_instance_snapshots (
                instance_id, region, instance_class, engine, multi_az,
                storage_type, allocated_storage_gb, iops_provisioned,
                status, monthly_cost_usd, avg_cpu_14d, max_cpu_14d,
                avg_connections, max_connections, avg_read_iops,
                avg_write_iops, avg_freeable_mem_gb, tags
            ) VALUES (%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s)
        """, (
            snap.instance_id, snap.region, snap.instance_class, snap.engine,
            snap.multi_az, snap.storage_type, snap.allocated_storage,
            snap.iops, snap.status, snap.monthly_cost_usd,
            snap.avg_cpu_14d, snap.max_cpu_14d, snap.avg_connections,
            snap.max_connections, snap.avg_read_iops, snap.avg_write_iops,
            snap.avg_freeable_mem_gb, json.dumps(snap.tags)
        ))

    conn.commit()
    cur.close()
    conn.close()
    log.info(f"Persisted {len(snapshots)} snapshots to analyzer database.")

def main():
    regions = os.environ.get('AWS_REGIONS', 'us-east-1,us-west-2').split(',')

    for region in regions:
        log.info(f"Collecting data for region: {region}")
        snapshots = collect_rds_instances(region)
        costs = collect_cost_explorer_rds(region)

```

```

    for snap in snapshots:
        snap = collect_cloudwatch_metrics(snap, region)
        snap.monthly_cost_usd = costs.get(snap.instance_id, Decimal('0'))

    persist_snapshots(snapshots)

log.info("Collection cycle complete.")

if __name__ == '__main__':
    main()

```

Run this as a daily cron job or EventBridge rule. A single collection cycle for 30 instances across two regions completes in under four minutes on a t3.small. The snapshot history it builds enables trend analysis that single-point-in-time queries cannot provide — you can see whether an instance’s CPU utilization is declining over weeks, which happens when a batch job gets decommissioned but the instance is left running.

29.3 The Rules Engine: Translating Data into Dollar Amounts

The rules engine is where the analyzer earns its keep. Each rule takes a snapshot as input, applies a threshold condition, and if the condition is met, writes a finding with a calculated dollar savings estimate. The math behind each estimate is transparent and defensible — you can show a CTO exactly how the number was derived.

The following SQL queries run against the `rds_instance_snapshots` table to identify the most common and highest-value findings. These queries should run as part of the nightly rules evaluation job, with results written to `db_cost_findings`.

```

-- =====
-- RULE 1: Idle Instances (avg CPU < 5% for 14 days)
-- Expected finding: 100% of instance cost recoverable
-- =====
WITH latest_snapshots AS (
    SELECT DISTINCT ON (instance_id)
        instance_id,
        region,
        instance_class,
        engine,
        multi_az,
        monthly_cost_usd,
        avg_cpu_14d,
        max_cpu_14d,
        avg_connections,
        status,
        tags
    FROM rds_instance_snapshots
    ORDER BY instance_id, collected_at DESC
)
SELECT
    'aws' AS cloud_provider,
    instance_id AS resource_id,
    'rds_instance' AS resource_type,
    engine,
    instance_class,
    'idle_instance' AS finding_type,
    jsonb_build_object(
        'avg_cpu_14d', avg_cpu_14d,
        'max_cpu_14d', max_cpu_14d,
        'avg_connections', avg_connections,
        'status', status,

```

```

        'env_tag',                tags->>'Environment'
    )
    monthly_cost_usd              AS finding_detail,
    'high'                        AS estimated_monthly_savings_usd,
                                AS confidence
FROM latest_snapshots
WHERE
    avg_cpu_14d < 5.0
    AND max_cpu_14d < 20.0      -- eliminates instances with rare spikes
    AND avg_connections < 2.0  -- no active application connections
    AND status = 'available'
    AND monthly_cost_usd > 0
ORDER BY estimated_monthly_savings_usd DESC;

-- =====
-- RULE 2: Oversized Instances (avg CPU < 20%, max CPU < 40%)
-- Savings: moving to next smaller instance class typically saves 40-50%
-- Example: db.r6g.2xlarge ($1,040/mo) + db.r6g.xlarge ($520/mo)
-- =====
WITH latest_snapshots AS (
    SELECT DISTINCT ON (instance_id)
        instance_id, region, instance_class, engine,
        monthly_cost_usd, avg_cpu_14d, max_cpu_14d,
        avg_freeable_mem_gb, avg_connections, tags
    FROM rds_instance_snapshots
    ORDER BY instance_id, collected_at DESC
),
instance_memory AS (
    -- Approximate vCPU/RAM for common RDS instance classes
    SELECT * FROM (VALUES
        ('db.t3.medium',      2,  4),
        ('db.t3.large',       2,  8),
        ('db.t3.xlarge',      4, 16),
        ('db.t3.2xlarge',     8, 32),
        ('db.r6g.large',      2, 16),
        ('db.r6g.xlarge',     4, 32),
        ('db.r6g.2xlarge',    8, 64),
        ('db.r6g.4xlarge',   16, 128),
        ('db.r6g.8xlarge',   32, 256),
        ('db.m6g.large',      2,  8),
        ('db.m6g.xlarge',     4, 16),
        ('db.m6g.2xlarge',    8, 32),
        ('db.m6g.4xlarge',   16, 64)
    ) AS t(instance_class, vcpus, ram_gb)
)
SELECT
    'aws' AS cloud_provider,
    ls.instance_id AS resource_id,
    'rds_instance' AS resource_type,
    ls.engine,
    ls.instance_class,
    'oversized_instance' AS finding_type,
    jsonb_build_object(
        'avg_cpu_14d',      ls.avg_cpu_14d,
        'max_cpu_14d',      ls.max_cpu_14d,
        'avg_freeable_mem_gb', ls.avg_freeable_mem_gb,
        'total_ram_gb',     im.ram_gb,
        'mem_utilization_pct', ROUND(
            100.0 - (ls.avg_freeable_mem_gb / NULLIF(im.ram_gb, 0) * 100),
            1),
        'recommended_action', 'Downsize to next smaller instance class'
    ) AS finding_detail,
    -- Conservative: assume 40% savings from one size down
    ROUND(ls.monthly_cost_usd * 0.40, 2) AS estimated_monthly_savings_usd,
    CASE
        WHEN ls.avg_cpu_14d < 10 AND ls.max_cpu_14d < 25 THEN 'high'
        WHEN ls.avg_cpu_14d < 20 AND ls.max_cpu_14d < 40 THEN 'medium'
        ELSE 'low'
    END AS confidence
FROM latest_snapshots ls

```

```

LEFT JOIN instance_memory im USING (instance_class)
WHERE
  ls.avg_cpu_14d < 20.0
  AND ls.max_cpu_14d < 40.0
  -- Don't flag instances where memory is the constraint
  AND (ls.avg_freeable_mem_gb / NULLIF(im.ram_gb, 0)) > 0.30
  AND ls.monthly_cost_usd > 50 -- ignore micro instances, not worth the effort
ORDER BY estimated_monthly_savings_usd DESC;

-- =====
-- RULE 3: gp2 Storage That Should Be gp3
-- gp2: $0.115/GB/month | gp3: $0.08/GB/month | Saving: 30.4%
-- Bonus: gp3 baseline is 3,000 IOPS free vs gp2's linear scaling
-- Example: 1TB gp2 = $117.76/mo + 1TB gp3 = $81.92/mo = $35.84/mo saved
-- =====
WITH latest_snapshots AS (
  SELECT DISTINCT ON (instance_id)
    instance_id, region, instance_class, engine,
    storage_type, allocated_storage_gb, iops_provisioned,
    avg_read_iops, avg_write_iops, monthly_cost_usd, tags
  FROM rds_instance_snapshots
  ORDER BY instance_id, collected_at DESC
)
SELECT
  'aws' AS cloud_provider,
  instance_id AS resource_id,
  'rds_instance' AS resource_type,
  engine,
  instance_class,
  'gp2_to_gp3' AS finding_type,
  jsonb_build_object(
    'current_storage_type', 'gp2',
    'allocated_storage_gb', allocated_storage_gb,
    'current_monthly_storage', ROUND(allocated_storage_gb * 0.115, 2),
    'gp3_monthly_storage', ROUND(allocated_storage_gb * 0.08, 2),
    'avg_read_iops', ROUND(avg_read_iops, 0),
    'avg_write_iops', ROUND(avg_write_iops, 0),
    'total_avg_iops', ROUND(avg_read_iops + avg_write_iops, 0),
    'gp3_free_iops', 3000,
    'zero_downtime_migration', true
  ) AS finding_detail,
  -- Storage savings only (IOPS are free up to 3000 on gp3)
  ROUND(allocated_storage_gb * (0.115 - 0.08), 2) AS estimated_monthly_savings_usd,
  'high' AS confidence
FROM latest_snapshots
WHERE
  storage_type = 'gp2'
  AND allocated_storage_gb >= 100 -- below 100GB the absolute saving is trivial
  -- Don't flag if current IOPS need exceed gp3 max (16000 IOPS)
  AND (avg_read_iops + avg_write_iops) < 16000
ORDER BY estimated_monthly_savings_usd DESC;

-- =====
-- RULE 4: Multi-AZ Enabled on Non-Production Instances
-- Multi-AZ doubles the instance cost. Non-prod doesn't need it.
-- Example: db.r6g.2xlarge Multi-AZ = $2,080/mo + Single-AZ = $1,040/mo
-- =====
WITH latest_snapshots AS (
  SELECT DISTINCT ON (instance_id)
    instance_id, region, instance_class, engine,
    multi_az, monthly_cost_usd, tags
  FROM rds_instance_snapshots
  ORDER BY instance_id, collected_at DESC
)
SELECT
  'aws' AS cloud_provider,
  instance_id AS resource_id,

```

```

'rds_instance'                AS resource_type,
engine,
instance_class,
'multi_az_nonprod'            AS finding_type,
jsonb_build_object(
  'multi_az',                  true,
  'environment_tag',           tags->>'Environment',
  'cost_center_tag',           tags->>'CostCenter',
  'recommendation',            'Disable Multi-AZ; restore from snapshot for non-prod recovery'
)                              AS finding_detail,
-- Multi-AZ approximately doubles instance cost
ROUND(monthly_cost_usd * 0.50, 2) AS estimated_monthly_savings_usd,
'high'                         AS confidence
FROM latest_snapshots
WHERE
  multi_az = true
  AND (
    LOWER(tags->>'Environment') IN ('dev', 'development', 'test',
                                   'staging', 'qa', 'sandbox', 'nonprod')
    OR LOWER(instance_id) LIKE '%dev%'
    OR LOWER(instance_id) LIKE '%test%'
    OR LOWER(instance_id) LIKE '%stag%'
    OR LOWER(instance_id) LIKE '%qa%'
  )
ORDER BY estimated_monthly_savings_usd DESC;

```

These four rules alone will surface 60–70% of the total savings available in a typical AWS RDS environment. Add rules for io1-to-gp3 migration, missing reserved instances, and stopped instances accruing storage charges to cover the remainder.

29.4 Database-Internal Cost Analysis with pg_stat_statements

Cloud billing data shows you what you are paying per instance. Database-internal statistics show you what is consuming the resources you are paying for. Combining them lets you say “this \$2,400/month instance is 60% consumed by three queries that run 50,000 times per day — optimizing those queries could halve the instance cost.” That is the conversation that gets engineering attention.

The following queries work against `pg_stat_statements` on PostgreSQL RDS, Aurora PostgreSQL, AlloyDB, and Cloud SQL for PostgreSQL. For SQL Server RDS or Azure SQL, the equivalent view is `sys.dm_exec_query_stats`.

```

-- =====
-- PostgreSQL: Top resource-consuming queries for cost attribution
-- Run against each production database; correlates with instance cost
-- =====

-- Prerequisite: pg_stat_statements must be loaded
-- In RDS Parameter Group: shared_preload_libraries = 'pg_stat_statements'
-- Then: CREATE EXTENSION IF NOT EXISTS pg_stat_statements;

-- Query 1: Queries consuming the most total CPU (user + system time)
-- "Total time" in pg_stat_statements is wall clock; use this to identify
-- candidates for query optimization that can defer hardware scaling.
SELECT
  LEFT(query, 120)                AS query_preview,
  calls,
  ROUND(total_exec_time::numeric / 1000, 2) AS total_exec_seconds,
  ROUND(mean_exec_time::numeric, 2)         AS mean_exec_ms,
  ROUND(stddev_exec_time::numeric, 2)       AS stddev_exec_ms,
  ROUND(total_exec_time::numeric /

```

```

SUM(total_exec_time) OVER () * 100, 2) AS pct_of_total_time,
rows,
shared_blks_hit,
shared_blks_read,
shared_blks_dirtied,
ROUND(shared_blks_hit::numeric /
NULLIF(shared_blks_hit + shared_blks_read, 0) * 100, 2) AS cache_hit_pct,
-- Disk reads translate directly to storage IOPS cost
ROUND(shared_blks_read * 8.0 / 1024, 2) AS disk_read_mb
FROM pg_stat_statements
WHERE
calls > 100 -- ignore rarely-executed queries
AND query NOT LIKE '%pg_stat%' -- exclude the analyzer itself
ORDER BY total_exec_time DESC
LIMIT 25;

```

-- Query 2: IOPS cost attribution by query
-- Each block read is 8KB. On io1 at \$0.065/provisioned IOPS/month,
-- high-read queries are directly driving storage costs.

```

SELECT
LEFT(query, 120) AS query_preview,
calls,
shared_blks_read + shared_blks_written AS total_io_blocks,
ROUND((shared_blks_read + shared_blks_written) * 8.0 / 1024, 2)
AS total_io_mb,
ROUND(shared_blks_read::numeric /
NULLIF(calls, 0), 2) AS blks_read_per_call,
ROUND(temp_blks_read + temp_blks_written, 0) AS temp_io_blocks,
-- Temp block usage indicates missing indexes or insufficient work_mem
CASE
WHEN (temp_blks_read + temp_blks_written) > 0
THEN 'WARNING: spilling to disk'
ELSE 'OK'
END
AS temp_io_status
FROM pg_stat_statements
WHERE calls > 50
ORDER BY (shared_blks_read + shared_blks_written) DESC
LIMIT 20;

```

-- Query 3: Connection cost analysis
-- High connection counts without pgBouncer drive the need for larger instances.
-- A db.r6g.xlarge (4 vCPU, 32GB) with 1,000 direct connections wastes
-- approximately 8GB of RAM on connection overhead alone.

```

SELECT
client_addr,
state,
COUNT(*) AS connection_count,
COUNT(*) FILTER (WHERE state = 'idle') AS idle_connections,
COUNT(*) FILTER (WHERE state = 'active') AS active_connections,
COUNT(*) FILTER (WHERE state = 'idle in transaction')
AS idle_in_transaction,
-- Idle in transaction locks resources and burns memory
ROUND(COUNT(*) FILTER (WHERE state = 'idle') * 100.0
/ NULLIF(COUNT(*), 0), 1) AS idle_pct,
MAX(EXTRACT(EPOCH FROM (NOW() - state_change)))
AS max_idle_seconds
FROM pg_stat_activity
WHERE pid <> pg_backend_pid()
GROUP BY client_addr, state
ORDER BY connection_count DESC;

```

-- Query 4: Table bloat - wasted storage that inflates backup and storage costs
-- A table with 40% bloat means you're paying for 40% more storage than you use.
-- On a 500GB database on gp3, that's 200GB × \$0.08 = \$16/month in pure waste.

```

SELECT
schemaname,
tablename,

```



```

pg_size_pretty(pg_total_relation_size(schemaname||'.'||tablename))
                                AS total_size,
pg_size_pretty(pg_relation_size(schemaname||'.'||tablename))
                                AS table_size,
pg_size_pretty(pg_total_relation_size(schemaname||'.'||tablename)
               - pg_relation_size(schemaname||'.'||tablename))
                                AS index_size,
n_live_tup,
n_dead_tup,
CASE
    WHEN n_live_tup > 0
    THEN ROUND(n_dead_tup * 100.0 / (n_live_tup + n_dead_tup), 1)
    ELSE 0
END                                AS dead_tuple_pct,
last_autovacuum,
last_autoanalyze
FROM pg_stat_user_tables
WHERE
    (n_live_tup + n_dead_tup) > 10000      -- ignore tiny tables
    AND n_dead_tup > n_live_tup * 0.20    -- >20% bloat
ORDER BY pg_total_relation_size(schemaname||'.'||tablename) DESC
LIMIT 20;

-- =====
-- SQL Server / Azure SQL equivalent: Top CPU consumers
-- Run against sys.dm_exec_query_stats for cost attribution
-- =====
SELECT TOP 20
    SUBSTRING(qt.text, (qs.statement_start_offset / 2) + 1,
              ((CASE qs.statement_end_offset
                  WHEN -1 THEN DATALENGTH(qt.text)
                  ELSE qs.statement_end_offset
                END - qs.statement_start_offset) / 2) + 1)
              AS query_text,
    qs.execution_count,
    ROUND(qs.total_worker_time / 1000000.0, 2) AS total_cpu_seconds,
    ROUND(qs.total_worker_time / 1000000.0
          / NULLIF(qs.execution_count, 0), 4) AS avg_cpu_seconds,
    ROUND(qs.total_logical_reads
          / NULLIF(qs.execution_count, 0), 0) AS avg_logical_reads,
    ROUND(qs.total_physical_reads
          / NULLIF(qs.execution_count, 0), 0) AS avg_physical_reads,
    qs.total_logical_reads AS total_logical_reads,
    qs.total_physical_reads AS total_physical_reads,
    -- Physical reads drive IOPS; each is an 8KB page read from storage
    ROUND(qs.total_physical_reads * 8.0 / 1024 / 1024, 2)
              AS total_physical_read_gb,
    qs.creation_time AS plan_compiled_at
FROM sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
WHERE qs.execution_count > 100
ORDER BY qs.total_worker_time DESC;

```

The bloat query frequently uncovers \$10,000–\$40,000 in recoverable storage costs at organizations that have not run VACUUM FULL or pg_repack on large tables. One financial services client discovered a 2TB table with 55% dead tuple ratio — they were paying for 1.1TB of storage containing nothing but tombstones from a high-churn reporting table. Running pg_repack during a maintenance window recovered \$88/month in gp3 storage costs and, more significantly, reduced table scan times by 40%, which deferred a planned instance upgrade worth \$520/month.

29.5 Automating the Rules Engine and Report Generation

The rules engine should run nightly, compare results against previously open findings to avoid duplicating records, and then produce a weekly digest every Monday morning. The following Python module handles both the rules evaluation loop and the Slack/email report generation.

```
#!/usr/bin/env python3
"""
db_cost_rules_engine.py
Evaluates cost optimization rules against collected snapshots and
generates weekly reports via Slack and email.

Schedule:
- Rules evaluation: nightly at 02:00 UTC
- Weekly digest: Mondays at 08:00 UTC

Requirements:
pip install psycpg2-binary boto3 slack-sdk python-dotenv jinja2
"""

import os
import json
import logging
import smtplib
from datetime import datetime, timedelta, timezone
from decimal import Decimal
from email.mime.multipart import MIMEMultipart
from email.mime.text import MIMEText
from typing import Optional

import psycpg2
import psycpg2.extras
import boto3
from slack_sdk import WebClient
from slack_sdk.errors import SlackApiError
from dotenv import load_dotenv

load_dotenv()
logging.basicConfig(level=logging.INFO, format='%(asctime)s %(levelname)s %(message)s')
log = logging.getLogger(__name__)

# -----
# Database helpers
# -----

def get_db_connection():
    return psycpg2.connect(
        host=os.environ['ANALYZER_DB_HOST'],
        dbname=os.environ['ANALYZER_DB_NAME'],
        user=os.environ['ANALYZER_DB_USER'],
        password=os.environ['ANALYZER_DB_PASSWORD'],
        sslmode='require',
        cursor_factory=psycpg2.extras.RealDictCursor
    )

# -----
# Rules evaluation
# -----

RULES_SQL = {
    'idle_instance': """
        WITH latest AS (
            SELECT DISTINCT ON (instance_id)
                instance_id, region, instance_class, engine, multi_az,
                storage_type, monthly_cost_usd, avg_cpu_14d, max_cpu_14d,
```

```
        avg_connections, status, tags
    FROM rds_instance_snapshots
    ORDER BY instance_id, collected_at DESC
)
SELECT
    'aws'           AS cloud_provider,
    instance_id     AS resource_id,
    'rds_instance' AS resource_type,
    engine,
    instance_class,
    region,
    'idle_instance' AS finding_type,
    monthly_cost_usd AS estimated_monthly_savings_usd,
    'high'          AS confidence,
    jsonb_build_object(
        'avg_cpu_14d', avg_cpu_14d,
        'max_cpu_14d', max_cpu_14d,
        'avg_connections', avg_connections,
        'status', status
    ) AS finding_detail
FROM latest
WHERE avg_cpu_14d < 5
    AND max_cpu_14d < 20
    AND avg_connections < 2
    AND status = 'available'
    AND monthly_cost_usd > 0
""",

'oversized_instance': """
    WITH latest AS (
        SELECT DISTINCT ON (instance_id)
            instance_id, region, instance_class, engine,
            monthly_cost_usd, avg_cpu_14d, max_cpu_14d,
            avg_freeable_mem_gb, tags
        FROM rds_instance_snapshots
        ORDER BY instance_id, collected_at DESC
    )
    SELECT
        'aws'           AS cloud_provider,
        instance_id     AS resource_id,
        'rds_instance' AS resource_type,
        engine,
        instance_class,
        region,
        'oversized_instance' AS finding_type,
        ROUND(monthly_cost_usd * 0.40, 2) AS estimated_monthly_savings_usd,
        CASE
            WHEN avg_cpu_14d < 10 AND max_cpu_14d < 25 THEN 'high'
            ELSE 'medium'
        END AS confidence,
        jsonb_build_object(
            'avg_cpu_14d', avg_cpu_14d,
            'max_cpu_14d', max_cpu_14d
        ) AS finding_detail
    FROM latest
    WHERE avg_cpu_14d < 20
        AND max_cpu_14d < 40
        AND monthly_cost_usd > 50
""",

'gp2_to_gp3': """
    WITH latest AS (
        SELECT DISTINCT ON (instance_id)
            instance_id, region, instance_class, engine,
            storage_type, allocated_storage_gb,
            avg_read_iops, avg_write_iops, monthly_cost_usd, tags
        FROM rds_instance_snapshots
        ORDER BY instance_id, collected_at DESC
    )
    SELECT
```

```

        'aws'          AS cloud_provider,
        instance_id    AS resource_id,
        'rds_instance' AS resource_type,
        engine,
        instance_class,
        region,
        'gp2_to_gp3'   AS finding_type,
        ROUND(allocated_storage_gb * 0.035, 2) AS estimated_monthly_savings_usd,
        'high'         AS confidence,
        jsonb_build_object(
            'allocated_storage_gb',    allocated_storage_gb,
            'current_monthly_storage',  ROUND(allocated_storage_gb * 0.115, 2),
            'gp3_monthly_storage',      ROUND(allocated_storage_gb * 0.08, 2),
            'avg_total_iops',           ROUND(avg_read_iops + avg_write_iops, 0)
        ) AS finding_detail
    FROM latest
    WHERE storage_type = 'gp2'
        AND allocated_storage_gb >= 100
        AND (avg_read_iops + avg_write_iops) < 16000
    """,

'multi_az_nonprod': """
    WITH latest AS (
        SELECT DISTINCT ON (instance_id)
            instance_id, region, instance_class, engine,
            multi_az, monthly_cost_usd, tags
        FROM rds_instance_snapshots
        ORDER BY instance_id, collected_at DESC
    )
    SELECT
        'aws'          AS cloud_provider,
        instance_id    AS resource_id,
        'rds_instance' AS resource_type,
        engine,
        instance_class,
        region,
        'multi_az_nonprod' AS finding_type,
        ROUND(monthly_cost_usd * 0.50, 2) AS estimated_monthly_savings_usd,
        'high'         AS confidence,
        jsonb_build_object(
            'environment_tag', tags->>'Environment',
            'multi_az',        true
        ) AS finding_detail
    FROM latest
    WHERE multi_az = true
        AND (
            LOWER(tags->>'Environment') IN
                ('dev','development','test','staging','qa','sandbox','nonprod')
            OR LOWER(instance_id) SIMILAR TO '%(dev|test|stag|qa|sand)%'
        )
    """
}

def finding_already_open(cur, resource_id: str, finding_type: str) -> bool:
    """Prevent duplicate findings for the same resource and finding type."""
    cur.execute("""
        SELECT 1 FROM db_cost_findings
        WHERE resource_id = %s
          AND finding_type = %s
          AND resolution_status = 'open'
        LIMIT 1
    """, (resource_id, finding_type))
    return cur.fetchone() is not None

def run_rules_engine():
    """
    Evaluate all rules against the latest snapshots.
    Writes new findings; skips resources already flagged as open.
    """

```

```

Returns summary dict with total findings and savings.
"""
conn = get_db_connection()
cur = conn.cursor()

total_new_findings = 0
total_savings = Decimal('0')
findings_by_type = {}

for rule_name, sql in RULES_SQL.items():
    log.info(f"Evaluating rule: {rule_name}")
    cur.execute(sql)
    rows = cur.fetchall()

    new_for_rule = 0
    savings_for_rule = Decimal('0')

    for row in rows:
        resource_id = row['resource_id']
        finding_type = row['finding_type']
        savings = Decimal(str(row['estimated_monthly_savings_usd'] or 0))

        if finding_already_open(cur, resource_id, finding_type):
            continue # already flagged; don't duplicate

        # Retrieve account_id from environment or tags
        account_id = os.environ.get('AWS_ACCOUNT_ID', 'unknown')

        cur.execute("""
            INSERT INTO db_cost_findings (
                cloud_provider, account_id, region, resource_id,
                resource_type, engine, instance_class, finding_type,
                finding_detail, estimated_monthly_savings_usd,
                confidence
            ) VALUES (%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s)
        """, (
            row['cloud_provider'],
            account_id,
            row['region'],
            resource_id,
            row['resource_type'],
            row['engine'],
            row['instance_class'],
            finding_type,
            json.dumps(dict(row['finding_detail'])),
            savings,
            row['confidence']
        ))

        new_for_rule += 1
        savings_for_rule += savings

    conn.commit()

    if new_for_rule > 0:
        findings_by_type[rule_name] = {
            'count': new_for_rule,
            'savings': float(savings_for_rule)
        }
        log.info(f" {rule_name}: {new_for_rule} new findings, "
                 f"${savings_for_rule:,.2f}/mo potential savings")

    total_new_findings += new_for_rule
    total_savings += savings_for_rule

cur.close()
conn.close()

return {
    'total_new_findings': total_new_findings,

```

```

        'total_monthly_savings': float(total_savings),
        'total_annual_savings': float(total_savings * 12),
        'by_type': findings_by_type
    }

# -----
# Report generation
# -----

def build_weekly_report() -> dict:
    """
    Aggregate all open findings for the weekly digest.
    Returns a structured dict consumed by both the Slack formatter
    and the HTML email formatter.
    """
    conn = get_db_connection()
    cur = conn.cursor()

    # Summary by finding type
    cur.execute("""
        SELECT
            finding_type,
            COUNT(*) AS finding_count,
            SUM(estimated_monthly_savings_usd) AS monthly_savings,
            SUM(estimated_monthly_savings_usd) * 12 AS annual_savings
        FROM db_cost_findings
        WHERE resolution_status = 'open'
        GROUP BY finding_type
        ORDER BY monthly_savings DESC
    """)
    summary_rows = cur.fetchall()

    # Top 10 individual findings by dollar value
    cur.execute("""
        SELECT
            resource_id,
            region,
            engine,
            instance_class,
            finding_type,
            estimated_monthly_savings_usd,
            confidence,
            detected_at
        FROM db_cost_findings
        WHERE resolution_status = 'open'
        ORDER BY estimated_monthly_savings_usd DESC
        LIMIT 10
    """)
    top_findings = cur.fetchall()

    # Total open savings
    cur.execute("""
        SELECT
            COUNT(*) AS total_findings,
            SUM(estimated_monthly_savings_usd) AS total_monthly_savings,
            SUM(estimated_monthly_savings_usd) * 12 AS total_annual_savings
        FROM db_cost_findings
        WHERE resolution_status = 'open'
    """)
    totals = cur.fetchone()

    # Actioned this week (closed findings)
    cur.execute("""
        SELECT
            COUNT(*) AS actioned_count,
            SUM(estimated_monthly_savings_usd) AS actioned_savings
        FROM db_cost_findings
        WHERE resolution_status = 'actioned'
            AND resolved_at >= NOW() - INTERVAL '7 days'
    """)

```

```

"""
actioned = cur.fetchone()

cur.close()
conn.close()

return {
    'generated_at':      datetime.now(timezone.utc).strftime('%Y-%m-%d %H:%M UTC'),
    'total_findings':    totals['total_findings'] or 0,
    'total_monthly_savings': float(totals['total_monthly_savings'] or 0),
    'total_annual_savings': float(totals['total_annual_savings'] or 0),
    'actioned_count':    actioned['actioned_count'] or 0,
    'actioned_savings':  float(actioned['actioned_savings'] or 0),
    'summary_by_type':   [dict(r) for r in summary_rows],
    'top_findings':      [dict(r) for r in top_findings]
}

def post_slack_digest(report: dict):
    """
    Post a structured cost digest to Slack.
    Set SLACK_BOT_TOKEN and SLACK_CHANNEL_ID in environment.
    """
    client = WebClient(token=os.environ['SLACK_BOT_TOKEN'])
    channel = os.environ['SLACK_CHANNEL_ID']

    monthly = report['total_monthly_savings']
    annual = report['total_annual_savings']
    actioned_savings = report['actioned_savings']

    header_text = (
        f":money_with_wings: *Database Cost Analyzer - Weekly Digest*\n"
        f"Generated: {report['generated_at']}"
    )

    summary_text = (
        f"*Open Findings:* {report['total_findings']}\n"
        f"*Monthly Savings Available:* ${monthly:,.0f}\n"
        f"*Annual Savings Available:* ${annual:,.0f}\n"
        f"*Actioned This Week:* {report['actioned_count']} findings "
        f"(${actioned_savings:,.0f}/mo recovered)"
    )

    # Build top findings text
    findings_lines = []
    for i, f in enumerate(report['top_findings'][:5], 1):
        savings = float(f['estimated_monthly_savings_usd'])
        findings_lines.append(
            f"{i}. {f['resource_id']} ({f['instance_class']}) - "
            f"{f['finding_type']} - "
            f"${savings:,.0f}/mo ({f['confidence']} confidence)"
        )
    findings_text = "\n".join(findings_lines) if findings_lines else "No open findings."

    blocks = [
        {"type": "header",
         "text": {"type": "plain_text", "text": "Database Cost Analyzer - Weekly Digest"}},
        {"type": "section",
         "text": {"type": "mrkdwn", "text": header_text}},
        {"type": "divider"},
        {"type": "section",
         "text": {"type": "mrkdwn", "text": summary_text}},
        {"type": "section",
         "text": {"type": "mrkdwn",
                  "text": f"*Top Savings Opportunities:* \n{findings_text}"}}},
        {"type": "divider"},
        {"type": "context",
         "elements": [{"type": "mrkdwn",
                        "text": "React with :white_check_mark: to mark a finding actioned. "
                                "Full report delivered via email."}]}]}

```

```

]

try:
    client.chat_postMessage(channel=channel, blocks=blocks, text=header_text)
    log.info("Slack digest posted successfully.")
except SlackApiError as e:
    log.error(f"Slack API error: {e.response['error']}")

def send_email_digest(report: dict):
    """
    Send HTML email digest via AWS SES.
    Set SES_FROM_ADDRESS and SES_TO_ADDRESSES (comma-separated) in environment.
    """
    ses = boto3.client('ses', region_name=os.environ.get('AWS_DEFAULT_REGION', 'us-east-1'))
    from_addr = os.environ['SES_FROM_ADDRESS']
    to_addrs = os.environ['SES_TO_ADDRESSES'].split(',')

    monthly = report['total_monthly_savings']
    annual = report['total_annual_savings']

    # Build HTML table rows for summary by type
    summary_rows_html = ""
    for row in report['summary_by_type']:
        summary_rows_html += (
            f"<tr>"
            f"<td style='padding:8px;border:1px solid #ddd'>{row['finding_type']}</td>"
            f"<td style='padding:8px;border:1px solid"
            f"<td style='padding:8px;border:1px solid"
            f"<td style='padding:8px;border:1px solid"
            f"<td style='padding:8px;border:1px solid"
            f"<td style='padding:8px;border:1px solid"
            f"</tr>"
        )

    top_findings_html = ""
    for f in report['top_findings']:
        savings = float(f['estimated_monthly_savings_usd'])
        detected = str(f['detected_at'][:10])
        top_findings_html += (
            f"<tr>"
            f"<td style='padding:8px;border:1px solid"
            f"<td style='padding:8px;border:1px solid #ddd'>{f['instance_class']}</td>"
            f"<td style='padding:8px;border:1px solid #ddd'>{f['finding_type']}</td>"
            f"<td style='padding:8px;border:1px solid #ddd;text-align:right'>${savings:,.0f}/mo</td>"
            f"<td style='padding:8px;border:1px solid #ddd;text-align:center'>{f['confidence']}</td>"
            f"<td style='padding:8px;border:1px solid #ddd;text-align:center'>{detected}</td>"
            f"</tr>"
        )

    html_body = f"""
    <html><body style='font-family:Arial,sans-serif;color:#333;max-width:900px;margin:auto'>
    <h2 style='color:#d63031'>Database Cost Analyzer - Weekly Report</h2>
    <p style='color:#636e72'>Generated: {report['generated_at']}</p>

    <table style='width:100%;border-collapse:collapse;margin-bottom:24px'>
    <tr>
    <td style='background:#00b894;color:white;padding:16px;text-align:center;border-radius:4px'>
    <div style='font-size:28px;font-weight:bold'>${monthly:,.0f}</div>
    <div>Monthly Savings Available</div>
    </td>
    <td style='width:16px'></td>
    <td style='background:#0984e3;color:white;padding:16px;text-align:center;border-radius:4px'>
    <div style='font-size:28px;font-weight:bold'>${annual:,.0f}</div>
    <div>Annual Savings Available</div>
    </td>
    <td style='width:16px'></td>
    <td style='background:#6c5ce7;color:white;padding:16px;text-align:center;border-radius:4px'>

```



```

        <div style='font-size:28px;font-weight:bold'>{report['total_findings']}</div>
        <div>Open Findings</div>
    </td>
</tr>
</table>

<h3>Findings by Category</h3>
<table style='width:100%;border-collapse:collapse;margin-bottom:24px'>
    <thead>
        <tr style='background:#f1f2f6'>
            <th style='padding:8px;border:1px solid #ddd;text-align:left'>Finding Type</th>
            <th style='padding:8px;border:1px solid #ddd'>Count</th>
            <th style='padding:8px;border:1px solid #ddd'>Monthly Savings</th>
            <th style='padding:8px;border:1px solid #ddd'>Annual Savings</th>
        </tr>
    </thead>
    <tbody>{summary_rows_html}</tbody>
</table>

<h3>Top 10 Individual Findings</h3>
<table style='width:100%;border-collapse:collapse;margin-bottom:24px'>
    <thead>
        <tr style='background:#f1f2f6'>
            <th style='padding:8px;border:1px solid #ddd;text-align:left'>Resource ID</th>
            <th style='padding:8px;border:1px solid #ddd'>Instance Class</th>
            <th style='padding:8px;border:1px solid #ddd'>Finding</th>
            <th style='padding:8px;border:1px solid #ddd'>Savings</th>
            <th style='padding:8px;border:1px solid #ddd'>Confidence</th>
            <th style='padding:8px;border:1px solid #ddd'>Detected</th>
        </tr>
    </thead>
    <tbody>{top_findings_html}</tbody>
</table>

<p style='color:#636e72;font-size:12px'>
    This report is generated automatically by the Database Cost Analyzer.
    Mark findings as actioned in the recommendations database to track recovery.
</p>
</body></html>
"""

msg = MIMEMultipart('alternative')
msg['Subject'] = (f"Database Cost Report: ${monthly:,.0f}/mo savings available "
                f"({report['total_findings']} findings)")
msg['From'] = from_addr
msg['To'] = ', '.join(to_addrs)
msg.attach(MIMEText(html_body, 'html'))

ses.send_raw_email(
    Source=from_addr,
    Destinations=to_addrs,
    RawMessage={'Data': msg.as_string()})
)
log.info(f"Email digest sent to {to_addrs}")

# -----
# Entry points
# -----

def nightly_rules_run():
    """Called by cron at 02:00 UTC daily."""
    log.info("Starting nightly rules evaluation.")
    summary = run_rules_engine()
    log.info(f"Rules evaluation complete: {summary['total_new_findings']} new findings, "
            f"${summary['total_monthly_savings']:,.2f}/mo potential savings.")
    return summary

def weekly_digest_run():

```

```
"""Called by cron on Mondays at 08:00 UTC."""
log.info("Generating weekly digest.")
report = build_weekly_report()
post_slack_digest(report)
send_email_digest(report)
log.info("Weekly digest dispatched.")

if __name__ == '__main__':
    import sys
    if len(sys.argv) > 1 and sys.argv[1] == 'digest':
        weekly_digest_run()
    else:
        nightly_rules_run()
```

The two entry points map directly to two cron entries:

```
# /etc/cron.d/db-cost-analyzer
# Nightly rules evaluation
0 2 * * * ec2-user cd /opt/db-cost-analyzer && python db_cost_rules_engine.py >>
↳ /var/log/db-cost-analyzer.log 2>&1

# Weekly digest every Monday morning
0 8 * * 1 ec2-user cd /opt/db-cost-analyzer && python db_cost_rules_engine.py digest >>
↳ /var/log/db-cost-analyzer.log 2>&1

# Daily data collection
0 6 * * * ec2-user cd /opt/db-cost-analyzer && python db_cost_collector.py >>
↳ /var/log/db-cost-collector.log 2>&1
```

29.6 Extending the Analyzer to Azure and GCP

The core architecture built for AWS extends to Azure and GCP by swapping the data collection layer while keeping the rules engine and recommendations database identical. The findings schema is cloud-agnostic by design — the `cloud_provider` column carries the distinction, and everything downstream from the rules engine is provider-neutral.

The following script adds Azure Cost Management and GCP Billing Export collection to the same analyzer database. Organizations running multi-cloud database estates typically find that 25–40% of their total cloud database spend is on Azure or GCP, and those platforms have equally significant optimization opportunities that go undetected without automated collection.

```
#!/usr/bin/env bash
# azure_cost_collector.sh
# Pulls Azure SQL and Cosmos DB cost data via Azure CLI.
# Schedule: Daily at 06:30 UTC alongside the AWS collector.
#
# Prerequisites:
# az login --service-principal -u $AZURE_CLIENT_ID \
#           -p $AZURE_CLIENT_SECRET --tenant $AZURE_TENANT_ID
# pip install psycpg2-binary

set -euo pipefail

SUBSCRIPTION_ID="${AZURE_SUBSCRIPTION_ID}"
RESOURCE_GROUP="${AZURE_RESOURCE_GROUP:-}"
START_DATE=$(date -d "30 days ago" +%Y-%m-%d)
END_DATE=$(date +%Y-%m-%d)
OUTPUT_FILE="/tmp/azure_db_costs_${END_DATE}.json"

echo "[$(date -u +%H:%M:%S)] Starting Azure cost collection for subscription: ${SUBSCRIPTION_ID}"

# -----
```

```

# Step 1: Pull cost data grouped by resource for database services
# -----
az consumption usage list \
  --subscription "${SUBSCRIPTION_ID}" \
  --start-date "${START_DATE}" \
  --end-date "${END_DATE}" \
  --query "
    [?contains(consumedService, 'Microsoft.Sql') ||
     contains(consumedService, 'Microsoft.DBforPostgreSQL') ||
     contains(consumedService, 'Microsoft.DBforMySQL') ||
     contains(consumedService, 'Microsoft.DBforMariaDB') ||
     contains(consumedService, 'Microsoft.DocumentDB')].
    {
      resource_id:      instanceId,
      service:          consumedService,
      resource_group:   resourceGroup,
      location:         location,
      cost_usd:         pretaxCost,
      usage_date:       date,
      meter_name:       meterDetails.meterName,
      meter_category:   meterDetails.meterCategory,
      meter_subcategory: meterDetails.meterSubCategory,
      tags:             tags
    }
  " \
  --output json > "${OUTPUT_FILE}"

echo "[$(date -u +%H:%M:%S)] Cost data written to ${OUTPUT_FILE}"

# -----
# Step 2: Pull Azure SQL performance metrics via Azure Monitor
# -----
# List all Azure SQL servers in subscription
SQL_SERVERS=$(az sql server list \
  --subscription "${SUBSCRIPTION_ID}" \
  --query "[].{name:name, resourceGroup:resourceGroup}" \
  --output json)

echo "${SQL_SERVERS}" | python3 - << 'PYEOF'
import json, sys, subprocess, os
from datetime import datetime, timedelta, timezone

servers = json.loads(open('/dev/stdin').read()) if not sys.stdin.isatty() else []

# Actually read from the variable passed via here-doc
import subprocess
result = subprocess.run(
    ['az', 'sql', 'server', 'list',
     '--subscription', os.environ['AZURE_SUBSCRIPTION_ID'],
     '--query', '[].{name:name,resourceGroup:resourceGroup}',
     '--output', 'json'],
    capture_output=True, text=True
)
servers = json.loads(result.stdout)

end_time = datetime.now(timezone.utc)
start_time = end_time - timedelta(days=14)

metrics_output = []

for server in servers:
    server_name = server['name']
    rg = server['resourceGroup']

    # List databases on this server
    db_result = subprocess.run(
        ['az', 'sql', 'db', 'list',
         '--server', server_name,
         '--resource-group', rg,
         '--subscription', os.environ['AZURE_SUBSCRIPTION_ID']],

```

```

        '--query',
        '[].[name:name,sku:sku,maxSizeBytes:maxSizeBytes,status:status,tags:tags]',
        '--output', 'json'],
        capture_output=True, text=True
    )
    databases = json.loads(db_result.stdout)

    for db in databases:
        if db['name'] == 'master':
            continue # skip system database

        db_name = db['name']
        resource_id = (
            f"/subscriptions/{os.environ['AZURE_SUBSCRIPTION_ID']}"
            f"/resourceGroups/{rg}"
            f"/providers/Microsoft.Sql/servers/{server_name}"
            f"/databases/{db_name}"
        )

        # Pull CPU percentage metric (14-day average)
        metrics_result = subprocess.run(
            ['az', 'monitor', 'metrics', 'list',
             '--resource', resource_id,
             '--metric', 'cpu_percent',
             '--interval', 'PT1H',
             '--aggregation', 'Average',
             '--start-time', start_time.strftime('%Y-%m-%dT%H:%M:%SZ'),
             '--end-time', end_time.strftime('%Y-%m-%dT%H:%M:%SZ'),
             '--output', 'json'],
            capture_output=True, text=True
        )

        try:
            metrics = json.loads(metrics_result.stdout)
            datapoints = metrics['value'][0]['timeseries'][0]['data']
            values = [dp['average'] for dp in datapoints
                      if dp.get('average') is not None]
            avg_cpu = sum(values) / len(values) if values else 0.0
            max_cpu = max(values) if values else 0.0
        except (KeyError, IndexError, json.JSONDecodeError):
            avg_cpu = max_cpu = 0.0

        sku_name = db.get('sku', {}).get('name', 'Unknown')
        sku_tier = db.get('sku', {}).get('tier', 'Unknown')
        sku_cap = db.get('sku', {}).get('capacity', 0)

        metrics_output.append({
            'cloud_provider': 'azure',
            'server_name': server_name,
            'database_name': db_name,
            'resource_group': rg,
            'sku_name': sku_name,
            'sku_tier': sku_tier,
            'vcores_or_dtus': sku_cap,
            'max_size_gb': round((db.get('maxSizeBytes') or 0) / (1024**3), 2),
            'status': db.get('status', 'Unknown'),
            'avg_cpu_14d': round(avg_cpu, 2),
            'max_cpu_14d': round(max_cpu, 2),
            'tags': db.get('tags', {})
        })

    with open('/tmp/azure_sql_metrics.json', 'w') as f:
        json.dump(metrics_output, f, indent=2)

    print(f"Collected metrics for {len(metrics_output)} Azure SQL databases.")
    PYEOF

# -----
# Step 3: Pull GCP Cloud SQL costs from BigQuery billing export
# -----

```

```

# Requires: billing export enabled to BigQuery dataset
# Set GCP_BILLING_PROJECT and GCP_BILLING_DATASET in environment

GCP_PROJECT="${GCP_BILLING_PROJECT:-}"
GCP_DATASET="${GCP_BILLING_DATASET:-}"

if [[ -n "${GCP_PROJECT}" && -n "${GCP_DATASET}" ]]; then
    echo "[$(date -u +%H:%M:%S)] Querying GCP billing export from BigQuery..."

    bq query \
        --project_id="${GCP_PROJECT}" \
        --use_legacy_sql=false \
        --format=json \
        --max_rows=10000 \
        << 'BQEOF' > /tmp/gcp_db_costs.json

SELECT
    resource.name                                AS resource_name,
    resource.global_name                        AS resource_global_name,
    service.description                        AS service,
    sku.description                            AS sku_description,
    location.region                            AS region,
    SUM(cost)
      + SUM(IFNULL(
          (SELECT SUM(c.amount) FROM UNNEST(credits) c), 0))
        AS net_cost_usd,
    ARRAY_AGG(DISTINCT
        STRUCT(label.key AS k, label.value AS v)
        IGNORE NULLS
        LIMIT 20)
        AS labels
FROM
    `PROJECT_ID.DATASET_ID.gcp_billing_export_v1_*`,
    UNNEST(labels) AS label
WHERE
    _TABLE_SUFFIX >= FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY))
    AND service.description IN (
        'Cloud SQL',
        'Cloud Spanner',
        'AlloyDB for PostgreSQL',
        'Bare Metal Solution'
    )
GROUP BY
    resource.name,
    resource.global_name,
    service.description,
    sku.description,
    location.region
ORDER BY
    net_cost_usd DESC
BQEOF

# Replace placeholder values - bq query runs the literal SQL above
# In practice, substitute project/dataset before running
echo "[$(date -u +%H:%M:%S)] GCP billing export written to /tmp/gcp_db_costs.json"
fi

echo "[$(date -u +%H:%M:%S)] Multi-cloud cost collection complete."

```

Once the JSON files land in /tmp, a loader script reads them and inserts normalized records into the same `rds_instance_snapshots`-equivalent table structure, tagged with `cloud_provider = 'azure'` or `cloud_provider = 'gcp'`. The rules engine SQL queries use `WHERE cloud_provider = 'aws'` filters by default and gain Azure and GCP coverage simply by dropping those filters and adding provider-specific threshold logic where pricing differs.

Azure SQL has a particularly high-value optimization that has no AWS equivalent: the **serverless tier auto-pause** feature. Databases on the serverless compute tier that are not configured to auto-pause after idle periods continue billing at their minimum vCore floor even when completely

idle. The following SQL query, run against the Azure SQL metrics data loaded into the analyzer database, surfaces these instances.

```
-- Azure SQL: Serverless tier databases missing auto-pause configuration
-- A serverless Azure SQL database at 4 vCores minimum with auto-pause disabled
-- costs approximately $292/month at idle.
-- With auto-pause set to 60 minutes, idle cost drops to $0 (paused databases
-- only charge for storage: ~$0.115/GB/month).
--
-- Run against azure_sql_snapshots table (same schema as rds_instance_snapshots
-- but populated by the Azure collector)

WITH latest_azure AS (
    SELECT DISTINCT ON (database_name, server_name)
        database_name,
        server_name,
        resource_group,
        sku_tier,
        sku_name,
        vcores_or_dtus,
        max_size_gb,
        avg_cpu_14d,
        max_cpu_14d,
        monthly_cost_usd,
        tags,
        auto_pause_delay_minutes -- -1 means auto-pause disabled
    FROM azure_sql_snapshots
    ORDER BY database_name, server_name, collected_at DESC
),
serverless_idle AS (
    SELECT *
    FROM latest_azure
    WHERE
        sku_tier = 'GeneralPurpose'
        AND sku_name LIKE '%Serverless%'
        AND avg_cpu_14d < 10.0
        AND max_cpu_14d < 25.0
        AND (auto_pause_delay_minutes = -1 -- auto-pause disabled
            OR auto_pause_delay_minutes > 120) -- or too long to engage
)
SELECT
    'azure' AS cloud_provider,
    database_name || '@' || server_name AS resource_id,
    'azure_sql_database' AS resource_type,
    'sqlserver' AS engine,
    sku_name AS instance_class,
    'serverless_autopause_missing' AS finding_type,
    jsonb_build_object(
        'sku_tier', sku_tier,
        'vcores_min', vcores_or_dtus,
        'avg_cpu_14d', avg_cpu_14d,
        'max_cpu_14d', max_cpu_14d,
        'auto_pause_delay_minutes', auto_pause_delay_minutes,
        'recommended_setting', 60,
        'storage_only_cost', ROUND(max_size_gb * 0.115, 2)
    ) AS finding_detail,
    -- Savings = current monthly cost minus storage-only cost when paused
    ROUND(monthly_cost_usd - (max_size_gb * 0.115), 2) AS estimated_monthly_savings_usd,
    'high' AS confidence
FROM serverless_idle
WHERE monthly_cost_usd > (max_size_gb * 0.115)
ORDER BY estimated_monthly_savings_usd DESC;
```

GCP Cloud SQL has its own dominant saving opportunity: **instance edition sizing**. Cloud SQL Enterprise Plus carries a 2–3× price premium over Cloud SQL Enterprise for the same vCPU and memory configuration. Instances provisioned on Enterprise Plus that show no use of the differentiating features — read pool instances, advanced disaster recovery, Data Cache — are

pure premium waste. The analyzer flags these by checking for low IOPS utilization alongside the Enterprise Plus edition tag in the GCP billing labels.

Key Takeaways

- **A custom cost analyzer pays for itself within days.** The compute cost to run the analyzer — typically a t3.small or equivalent — is \$15–20/month. The first weekly report from a 30-instance RDS estate routinely surfaces \$8,000–\$25,000 in monthly savings from idle instances, gp2-to-gp3 migrations, and unnecessary Multi-AZ configurations combined.
 - **Billing data alone is insufficient — correlate it with metrics.** An instance costing \$2,400/month is not automatically a candidate for downsizing. Joining Cost Explorer data with 14-day CloudWatch CPU, memory, connection, and IOPS trends produces findings you can defend to both engineering and finance, eliminating the “but what if we need it” objection.
 - **The recommendations database is the accountability layer.** Storing findings with resolution status, actioned-by, and timestamps transforms cost optimization from a one-time audit into an ongoing operational practice. Finance can ask “how many findings were actioned this quarter and how much did we recover?” and get a specific, auditable answer.
 - **Database-internal statistics surface savings that cloud billing never will.** `pg_stat_statements` bloat queries, connection analysis, and temp-IO attribution identify cases where query optimization or autovacuum tuning can defer hardware scaling entirely — a \$520/month instance upgrade avoided is equivalent to \$520/month saved, and it appears in no cloud billing report.
 - **Multi-cloud extension requires only a new collection module.** The rules engine, recommendations schema, and reporting layer built for AWS work identically for Azure and GCP findings once the collection layer normalizes provider-specific APIs into the common snapshot schema. Organizations running 60% AWS, 30% Azure, and 10% GCP databases manage all three cost streams from a single analyzer with fewer than 200 additional lines of collection code per provider.
-

Chapter 30: The Database FinOps Maturity Model

Organizations that treat cloud database costs as an afterthought routinely overspend by 40–60% compared to those with a structured optimization practice. That gap is not theoretical. A financial services firm running 200 RDS instances with no tagging strategy, no rightsizing program, and no reserved instance coverage will spend roughly \$2.4M annually on compute alone. The same workload, managed through the maturity model described in this chapter, lands closer to \$960K — a \$1.44M annual difference on compute before touching storage, data transfer, or licensing. This final chapter ties together every technique from the preceding 29 chapters into a single progression framework: the Database FinOps Maturity Model. It gives DBAs a diagnostic checklist to locate where their organization stands today, a structured path to advance through each stage, and the automation scaffolding to sustain those gains indefinitely. The model covers five stages — Reactive, Aware, Structured, Optimized, and Autonomous — and provides the SQL, shell, and Python tooling to measure progress at each one.

Stage Definitions: The Five Levels of Database FinOps Maturity

The maturity model borrows its structure from the FinOps Foundation’s capability framework but specializes every dimension for database workloads. Each stage has observable characteristics, measurable KPIs, and a ceiling on how much waste it tolerates.

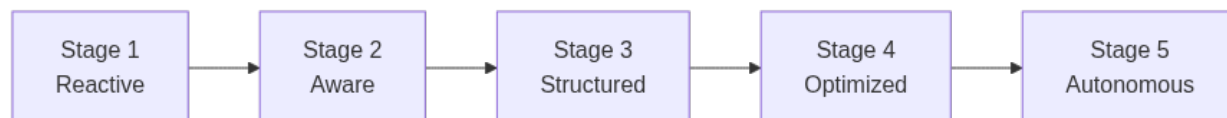


Figure 30: The five-stage Database FinOps Maturity Model — overspend drops from 50-65% above baseline at Stage 1 to within 5% of theoretical minimum at Stage 5.

Stage 1 — Reactive No tagging. No cost allocation by team or workload. Bills are reviewed monthly by finance, not engineering. Storage autoscaling is enabled without upper bounds. Multi-AZ is deployed uniformly across dev, test, and production. Reserved Instance coverage is zero or accidental. The organization is typically 50–65% over its efficient baseline spend.

Stage 2 — Aware Someone has pulled a Cost Explorer report. There is a spreadsheet. Tagging exists for some resources but not enforced by policy. A few engineers have looked at Performance

Insights. There is informal discussion about reserved instances at renewal time. Waste is visible but unowned. Overspend sits at 35–50% above baseline.

Stage 3 — Structured Tagging is enforced by AWS Config rules, Azure Policy, or GCP Organization Policy. Cost is allocated to teams through showback dashboards. Rightsizing recommendations from Compute Optimizer are reviewed monthly. At least 60% of steady-state compute is covered by Reserved Instances or Committed Use Discounts. Storage tiers are differentiated between environments. Overspend is 15–30% above baseline.

Stage 4 — Optimized Reserved Instance coverage exceeds 80% for production. Dev and test environments use scheduled start/stop automation. Connection pooling is deployed. Query-level cost attribution exists through `pg_stat_statements` or Performance Insights. Autoscaling policies have tested upper bounds. Storage has been migrated from gp2 to gp3 where appropriate. Overspend is 5–15% above theoretical minimum.

Stage 5 — Autonomous Cost guardrails are embedded in CI/CD pipelines. Anomaly detection alerts fire before the monthly bill. Rightsizing runs on a weekly automated schedule and tickets are auto-created. Reserved Instance coverage is managed programmatically with expiration tracking. The team has a published cost-per-transaction metric reviewed in sprint retrospectives. Overspend is within 5% of theoretical minimum, with variance explained and documented.

Stage 1 to Stage 2: Building Visibility

You cannot optimize what you cannot see. The transition from Reactive to Aware requires three foundational pieces: mandatory tagging, cost allocation, and a baseline query that tells you where the money actually goes.

The following query runs against the AWS Cost and Usage Report exported to S3 and queried through Athena. It produces the first honest picture of database spend by service and resource type.

```
-- AWS Athena query against Cost and Usage Report (CUR)
-- Prerequisite: CUR enabled, exported to S3, Athena table created via AWS Glue crawler
-- Run this monthly to establish your baseline

SELECT
    line_item_resource_id                AS resource_id,
    product_instance_type                AS instance_type,
    product_database_engine              AS engine,
    line_item_usage_type                  AS usage_type,
    resource_tags_user_environment        AS environment,
    resource_tags_user_team               AS team,
    resource_tags_user_application        AS application,
    SUM(line_item_unblended_cost)         AS total_cost_usd,
    SUM(line_item_usage_amount)           AS total_usage_hours,
    ROUND(
        SUM(line_item_unblended_cost) /
        NULLIF(SUM(line_item_usage_amount), 0), 4
    )                                     AS effective_hourly_rate
FROM
    your_cur_database.your_cur_table
WHERE
    line_item_line_item_type IN ('Usage', 'SavingsPlanCoveredUsage')
    AND product_service_code IN ('AmazonRDS', 'AmazonAurora')
    AND line_item_usage_start_date >= DATE_TRUNC('month', CURRENT_DATE - INTERVAL '1' MONTH)
    AND line_item_usage_start_date < DATE_TRUNC('month', CURRENT_DATE)
GROUP BY
```

```

1, 2, 3, 4, 5, 6, 7
ORDER BY
    total_cost_usd DESC
LIMIT 100;

```

A result showing `resource_tags_user_environment = NULL` for more than 20% of resources means tagging compliance is your first remediation target. Every untagged resource is an unowned cost.

On Azure, the equivalent baseline pull uses the Cost Management API:

```

#!/bin/bash
# Azure CLI: pull RDS-equivalent (Azure SQL, Flexible Server) costs for last 30 days
# Requires: az login, Billing Reader role on subscription

SUBSCRIPTION_ID="your-subscription-id"
START_DATE=$(date -d "30 days ago" +%Y-%m-%d)
END_DATE=$(date +%Y-%m-%d)

az costmanagement query \
  --scope "subscriptions/${SUBSCRIPTION_ID}" \
  --type "Usage" \
  --timeframe "Custom" \
  --time-period start="${START_DATE}" end="${END_DATE}" \
  --dataset-aggregation '{"totalCost":{"name":"Cost","function":"Sum"}}' \
  --dataset-grouping \
    '[{"type":"Dimension","name":"ResourceType"},\
     {"type":"Dimension","name":"ResourceGroup"},\
     {"type":"TagKey","name":"environment"},\
     {"type":"TagKey","name":"team"}]' \
  --dataset-filter \
    '{"and":[{"or":[\
      {"dimensions":{"name":"ResourceType","operator":"In",\
        "values":["Microsoft.DBforPostgreSQL/flexibleServers",\
          "Microsoft.DBforMySQL/flexibleServers",\
          "Microsoft.Sql/servers/databases"]}]}}\
    ]}]}' \
  --output table \
  | tee azure_db_baseline_$(date +%Y%m).txt

echo "Baseline saved to azure_db_baseline_$(date +%Y%m).txt"

```

On GCP, the equivalent uses BigQuery against the exported billing table:

```

-- GCP BigQuery: database service cost baseline
-- Prerequisite: Billing export enabled to BigQuery dataset

SELECT
    service.description AS service,
    sku.description AS sku,
    labels.value AS environment,
    project.id AS project,
    resource.name AS resource,
    SUM(cost) AS total_cost_usd,
    SUM(usage.amount) AS total_usage,
    usage.unit AS unit,
    ROUND(SUM(cost) / NULLIF(SUM(usage.amount), 0), 6) AS cost_per_unit
FROM
    `your_project.your_dataset.gcp_billing_export_v1_*`,
    UNNEST(labels) AS labels
WHERE
    service.description IN (
        'Cloud SQL',
        'AlloyDB for PostgreSQL',
        'Cloud Spanner',
        'Bare Metal Solution'
    )
    AND labels.key = 'environment'
    AND _TABLE_SUFFIX BETWEEN

```

```

        FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY))
    AND FORMAT_DATE('%Y%m%d', CURRENT_DATE())
GROUP BY
    1, 2, 3, 4, 5, 9
ORDER BY
    total_cost_usd DESC
LIMIT 200;

```

At Stage 2, the goal is not action — it is accountability. Publish these numbers to engineering leads weekly. Ownership follows visibility within two to four weeks in most organizations.

Stage 2 to Stage 3: Building Structure

Structure means cost is owned, not just observed. The three pillars are tagging enforcement, environment tiering, and Reserved Instance planning.

Tagging Enforcement

An AWS Config rule that flags untagged RDS instances fires within 15 minutes of resource creation:

```

#!/bin/bash
# Deploy AWS Config rule to detect untagged RDS instances
# Required tags: environment, team, application, cost-center
# Remediation: SNS notification + Jira ticket creation (see Python script below)

aws configservice put-config-rule \
  --config-rule '{
    "ConfigRuleName": "rds-required-tags",
    "Description": "Ensures all RDS instances carry required cost allocation tags",
    "Scope": {
      "ComplianceResourceTypes": [
        "AWS::RDS::DBInstance",
        "AWS::RDS::DBCluster"
      ]
    },
    "Source": {
      "Owner": "AWS",
      "SourceIdentifier": "REQUIRED_TAGS"
    },
    "InputParameters": "{
      \"tag1Key\": \"environment\",
      \"tag2Key\": \"team\",
      \"tag3Key\": \"application\",
      \"tag4Key\": \"cost-center\"
    }",
    "ConfigRuleState": "ACTIVE"
  }' \
  --region us-east-1

# Wire SNS notification for non-compliant resources
aws configservice put-remediation-configurations \
  --remediation-configurations '[{
    "ConfigRuleName": "rds-required-tags",
    "TargetType": "SSM_DOCUMENT",
    "TargetId": "AWS-PublishSNSNotification",
    "Parameters": {
      "TopicArn": {
        "StaticValue": {
          "Values": ["arn:aws:sns:us-east-1:123456789:db-finops-alerts"]
        }
      }
    },
    "Message": {
      "StaticValue": {
        "Values": ["RDS resource missing required cost tags. Review immediately."]
      }
    }
  },
  {
    "ConfigRuleName": "rds-required-tags",
    "TargetType": "SSM_DOCUMENT",
    "TargetId": "AWS-PublishSNSNotification",
    "Parameters": {
      "TopicArn": {
        "StaticValue": {
          "Values": ["arn:aws:sns:us-east-1:123456789:db-finops-alerts"]
        }
      }
    },
    "Message": {
      "StaticValue": {
        "Values": ["RDS resource missing required cost tags. Review immediately."]
      }
    }
  }
]

```

```

    }
  },
  "Automatic": true,
  "MaximumAutomaticAttempts": 3,
  "RetryAttemptSeconds": 60
}]'

echo "Config rule deployed. Non-compliant resources will trigger SNS within 15 minutes."

```

Reserved Instance Coverage Analysis

The single highest-return action at Stage 3 is Reserved Instance coverage. The math on a db.r6g.2xlarge in us-east-1 running 24/7:

- On-Demand: $\$0.960/\text{hr} \times 8,760 \text{ hrs} = \mathbf{\$8,409.60/\text{year}}$
- 1-year No Upfront RI: $\$0.576/\text{hr} \times 8,760 \text{ hrs} = \mathbf{\$5,045.76/\text{year}}$
- 1-year All Upfront RI: \$4,893/year
- 3-year All Upfront RI: \$2,942/year

Moving from On-Demand to a 1-year No Upfront RI saves \$3,363.84 per instance per year. A fleet of 50 such instances saves **\$168,192/year** without any architectural change.

The following Python script analyzes your current RDS fleet, calculates On-Demand vs. RI spend, and generates a prioritized purchase recommendation:

```

#!/usr/bin/env python3
"""
rds_ri_analyzer.py
Analyzes RDS fleet for Reserved Instance opportunities.
Runs weekly via EventBridge + Lambda or cron.
Outputs: JSON report + SNS notification if savings > threshold.

Author: Database FinOps Toolkit
Requires: boto3, pandas
pip install boto3 pandas
"""

import boto3
import json
import pandas as pd
from datetime import datetime, timedelta
from collections import defaultdict

REGIONS = ["us-east-1", "us-west-2", "eu-west-1"]
SAVINGS_THRESHOLD_USD = 5000 # alert if monthly savings opportunity exceeds this
SNS_TOPIC_ARN = "arn:aws:sns:us-east-1:123456789:db-finops-alerts"

# On-Demand hourly rates for common RDS instance types (us-east-1, MySQL/PostgreSQL)
# Update quarterly from https://aws.amazon.com/rds/pricing/
ON_DEMAND_RATES = {
    "db.t3.micro": 0.017,
    "db.t3.small": 0.034,
    "db.t3.medium": 0.068,
    "db.t3.large": 0.136,
    "db.t3.xlarge": 0.272,
    "db.t3.2xlarge": 0.544,
    "db.m6g.large": 0.171,
    "db.m6g.xlarge": 0.342,
    "db.m6g.2xlarge": 0.684,
    "db.m6g.4xlarge": 1.368,
    "db.m6g.8xlarge": 2.736,
    "db.r6g.large": 0.240,
    "db.r6g.xlarge": 0.480,
    "db.r6g.2xlarge": 0.960,

```

```

    "db.r6g.4xlarge": 1.920,
    "db.r6g.8xlarge": 3.840,
    "db.r6g.16xlarge": 7.680,
}

# 1-year No Upfront RI effective hourly rates (us-east-1)
RI_1YR_NO_UPFRONT_RATES = {
    "db.t3.micro": 0.011,
    "db.t3.small": 0.022,
    "db.t3.medium": 0.044,
    "db.t3.large": 0.087,
    "db.t3.xlarge": 0.174,
    "db.t3.2xlarge": 0.347,
    "db.m6g.large": 0.103,
    "db.m6g.xlarge": 0.205,
    "db.m6g.2xlarge": 0.410,
    "db.m6g.4xlarge": 0.821,
    "db.m6g.8xlarge": 1.641,
    "db.r6g.large": 0.144,
    "db.r6g.xlarge": 0.288,
    "db.r6g.2xlarge": 0.576,
    "db.r6g.4xlarge": 1.152,
    "db.r6g.8xlarge": 2.304,
    "db.r6g.16xlarge": 4.608,
}

HOURS_PER_MONTH = 730

def get_rds_instances(region: str) -> list[dict]:
    """Return all RDS instances in a region with key attributes."""
    client = boto3.client("rds", region_name=region)
    paginator = client.get_paginator("describe_db_instances")
    instances = []
    for page in paginator.paginate():
        for db in page["DBInstances"]:
            instances.append({
                "identifier": db["DBInstanceIdentifier"],
                "instance_class": db["DBInstanceClass"],
                "engine": db["Engine"],
                "status": db["DBInstanceStatus"],
                "multi_az": db.get("MultiAZ", False),
                "region": region,
                "environment": next(
                    (t["Value"] for t in db.get("TagList", [])
                     if t["Key"] == "environment"), "untagged"
                ),
                "team": next(
                    (t["Value"] for t in db.get("TagList", [])
                     if t["Key"] == "team"), "untagged"
                ),
            })
    return instances

def get_existing_ri_coverage(region: str) -> dict[str, int]:
    """Return count of active RIs per instance class in a region."""
    client = boto3.client("rds", region_name=region)
    response = client.describe_reserved_db_instances()
    coverage = defaultdict(int)
    for ri in response["ReservedDBInstances"]:
        if ri["State"] == "active":
            coverage[ri["DBInstanceClass"]] += ri["DBInstanceCount"]
    return coverage

def analyze_fleet(regions: list[str]) -> pd.DataFrame:
    """Build a DataFrame of all instances with cost and RI gap analysis."""
    rows = []
    for region in regions:

```

```

instances = get_rds_instances(region)
ri_coverage = get_existing_ri_coverage(region)

# Count On-Demand instances per class (exclude non-running)
class_counts: dict[str, int] = defaultdict(int)
for inst in instances:
    if inst["status"] == "available":
        class_counts[inst["instance_class"]] += 1

for inst_class, count in class_counts.items():
    od_rate = ON_DEMAND_RATES.get(inst_class, 0.0)
    ri_rate = RI_1YR_NO_UPFRONT_RATES.get(inst_class, 0.0)
    covered = ri_coverage.get(inst_class, 0)
    exposed = max(0, count - covered)
    saving = (od_rate - ri_rate) * HOURS_PER_MONTH * exposed

    rows.append({
        "region": region,
        "instance_class": inst_class,
        "running_count": count,
        "ri_covered": covered,
        "on_demand_exposed": exposed,
        "od_rate_hr": od_rate,
        "ri_rate_hr": ri_rate,
        "discount_pct": round((1 - ri_rate / od_rate) * 100, 1) if od_rate else 0,
        "monthly_od_cost": round(od_rate * HOURS_PER_MONTH * exposed, 2),
        "monthly_ri_cost": round(ri_rate * HOURS_PER_MONTH * exposed, 2),
        "monthly_savings": round(saving, 2),
        "annual_savings": round(saving * 12, 2),
    })

df = pd.DataFrame(rows)
return df.sort_values("monthly_savings", ascending=False)

def publish_report(df: pd.DataFrame) -> None:
    """Publish top opportunities to SNS if threshold exceeded."""
    total_monthly = df["monthly_savings"].sum()
    total_annual = df["annual_savings"].sum()

    report = {
        "generated_at": datetime.utcnow().isoformat(),
        "total_monthly_opp": round(total_monthly, 2),
        "total_annual_opp": round(total_annual, 2),
        "top_10_actions": df.head(10).to_dict(orient="records"),
    }

    print(json.dumps(report, indent=2))

    if total_monthly >= SAVINGS_THRESHOLD_USD:
        sns = boto3.client("sns", region_name="us-east-1")
        sns.publish(
            TopicArn=SNS_TOPIC_ARN,
            Subject=f"[DB FinOps] RI Opportunity: ${total_monthly:,.0f}/mo (${total_annual:,.0f}/yr)",
            Message=json.dumps(report, indent=2),
        )
        print(f"Alert published to SNS. Total monthly opportunity: ${total_monthly:,.2f}")

def main():
    df = analyze_fleet(REGIONS)
    publish_report(df)
    df.to_csv(f"ri_analysis_{datetime.utcnow().strftime('%Y%m%d')}.csv", index=False)
    print("Analysis complete. CSV saved.")

if __name__ == "__main__":
    main()

```

Run this on a weekly schedule. Over a 50-instance fleet, a single run that surfaces a \$168K annual opportunity pays for every automation investment the team will ever make.

Stage 3 to Stage 4: Operational Optimization

Stage 4 is where the cost savings become granular, persistent, and embedded in daily operations. Four practices define this transition: query-level cost attribution, storage tier optimization, environment scheduling, and connection pooling.

Query-Level Cost Attribution

The cheapest query is the one that never runs. `pg_stat_statements` on PostgreSQL surfaces which queries drive the most I/O, CPU, and shared buffer pressure — all of which translate directly to instance size requirements.

```
-- PostgreSQL: top 20 queries by total execution cost (I/O + CPU proxy)
-- Run on production read replica to avoid overhead on primary
-- Requires: pg_stat_statements extension enabled in postgresql.conf

SELECT
  queryid,
  LEFT(query, 120)                                AS query_preview,
  calls,
  ROUND(total_exec_time::NUMERIC, 2)              AS total_exec_ms,
  ROUND(mean_exec_time::NUMERIC, 4)               AS mean_exec_ms,
  ROUND(stddev_exec_time::NUMERIC, 4)             AS stddev_exec_ms,
  rows,
  ROUND(total_exec_time::NUMERIC / NULLIF(calls, 0), 4) AS ms_per_call,
  shared_blks_read                                AS disk_reads,
  shared_blks_hit                                  AS buffer_hits,
  ROUND(
    100.0 * shared_blks_hit::NUMERIC /
    NULLIF(shared_blks_hit + shared_blks_read, 0), 2
  )                                                AS buffer_hit_pct,
  shared_blks_written                             AS dirty_writes,
  temp_blks_written                              AS temp_spill_blks,
  ROUND(
    (total_exec_time * 0.001) *
    (shared_blks_read + shared_blks_written) * 0.000008192
  , 4)                                            AS approx_io_gb,
  -- Cost index: high value = high infrastructure pressure
  ROUND(
    (total_exec_time / 1000.0) +
    (shared_blks_read * 0.1) +
    (temp_blks_written * 0.5)
  , 2)                                            AS cost_index
FROM
  pg_stat_statements
WHERE
  calls > 100
  AND query NOT LIKE '%pg_stat%'
  AND query NOT LIKE 'BEGIN%'
  AND query NOT LIKE 'COMMIT%'
ORDER BY
  cost_index DESC
LIMIT 20;
```

Any query with a `buffer_hit_pct` below 85% is a candidate for index review. Any query with `temp_spill_blks` above zero is working memory that could be addressed by tuning `work_mem` — avoiding the need to scale the instance for additional RAM.

On SQL Server (Azure SQL / RDS for SQL Server), the equivalent uses `sys.dm_exec_query_stats`:

```
-- SQL Server: top resource-consuming queries with execution cost attribution
-- Run in Azure SQL or RDS for SQL Server
-- Results guide rightsizing: if top queries are I/O-bound, storage throughput is the bottleneck

SELECT TOP 25
    qs.execution_count,
    qs.total_logical_reads / qs.execution_count          AS avg_logical_reads,
    qs.total_physical_reads / qs.execution_count          AS avg_physical_reads,
    qs.total_worker_time / qs.execution_count / 1000     AS avg_cpu_ms,
    qs.total_elapsed_time / qs.execution_count / 1000    AS avg_elapsed_ms,
    qs.total_grant_kb / NULLIF(qs.execution_count, 0)    AS avg_memory_grant_kb,
    qs.total_spills / NULLIF(qs.execution_count, 0)      AS avg_spills,
    -- Cost score: weighted sum of CPU + I/O + memory pressure
    ROUND(
        (qs.total_worker_time * 0.40 +
         qs.total_logical_reads * 0.35 +
         qs.total_grant_kb * 0.25) / 1000000.0
        , 2)
        AS weighted_cost_score,
    DB_NAME(qt.dbid)
        AS database_name,
    OBJECT_NAME(qt.objectid, qt.dbid)
        AS object_name,
    SUBSTRING(
        qt.text,
        qs.statement_start_offset / 2,
        (CASE WHEN qs.statement_end_offset = -1
            THEN LEN(CONVERT(NVARCHAR(MAX), qt.text)) * 2
            ELSE qs.statement_end_offset
        END - qs.statement_start_offset) / 2
        )
        AS query_text
FROM
    sys.dm_exec_query_stats qs
CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) qt
WHERE
    qs.execution_count > 50
ORDER BY
    weighted_cost_score DESC;
```

Storage Tier Migration

The gp2-to-gp3 migration math at scale is unavoidable. A 1TB gp2 volume on RDS costs \$115/month. The same capacity on gp3 costs \$92/month — a saving of \$23/month or \$276/year per volume. A fleet of 100 production databases averaging 1TB each saves **\$27,600/year** from one storage change with no performance regression. For volumes above 1TB, gp3 has an additional advantage: gp2 delivers 3 IOPS/GB (baseline), meaning a 1TB gp2 volume gets 3,000 IOPS baseline and you pay for more by provisioning io1. gp3 provides 3,000 IOPS baseline at no extra cost and allows independent IOPS scaling up to 16,000 IOPS without tying throughput to storage size.

```
#!/bin/bash
# Migrate all eligible RDS gp2 volumes to gp3 in a region
# Excludes: instances with Multi-AZ failover in progress, instances in maintenance window
# Safe to run during business hours - storage modification is online for most engine versions

REGION="us-east-1"
DRY_RUN=true # Set to false to execute migrations

echo "Scanning for gp2 volumes in ${REGION}..."

aws rds describe-db-instances \
  --region "${REGION}" \
  --query "DBInstances[*].{
    ID:DBInstanceIdentifier,
    Class:DBInstanceClass,
    StorageType:StorageType,
    AllocatedStorage:AllocatedStorage,
```

```

        Status:DBInstanceStatus,
        MultiAZ:MultiAZ}" \
--output json | \
python3 - << 'PYEOF'
import json, sys, subprocess

data = json.load(sys.stdin)
region = "us-east-1"
dry_run = True    # mirrors bash variable

candidates = [
    d for d in data
    if d["StorageType"] == "gp2" and d["Status"] == "available"
]

print(f"Found {len(candidates)} gp2 instances eligible for migration")
print(f"{'ID':<40} {'Class':<18} {'Size GB':>8} {'Savings/mo':>12} {'Action'}")
print("-" * 90)

total_monthly_savings = 0
for inst in candidates:
    size_gb = inst["AllocatedStorage"]
    # gp2: $0.115/GB/month, gp3: $0.092/GB/month (us-east-1)
    saving = size_gb * (0.115 - 0.092)
    total_monthly_savings += saving
    action = "SKIP (dry run)" if dry_run else "MIGRATING"
    print(f"{inst['ID']:<40} {inst['Class']:<18} {size_gb:>8} ${saving:>10.2f} {action}")

if not dry_run:
    cmd = [
        "aws", "rds", "modify-db-instance",
        "--db-instance-identifier", inst["ID"],
        "--storage-type", "gp3",
        "--apply-immediately",
        "--region", region
    ]
    result = subprocess.run(cmd, capture_output=True, text=True)
    if result.returncode != 0:
        print(f" ERROR: {result.stderr.strip()}")
    else:
        print(f" Migration initiated for {inst['ID']}")

print(f"\nTotal potential monthly savings: ${total_monthly_savings:,.2f}")
print(f"Total potential annual savings:  ${total_monthly_savings * 12:,.2f}")
PYEOF

```

Environment Scheduling

Non-production databases running 24/7 are one of the most common and correctable sources of waste. A db.m6g.xlarge in us-east-1 costs \$0.342/hr On-Demand. Running it only during business hours — 10 hours/weekday, 50 weeks/year — reduces runtime from 8,760 hours to 2,500 hours, cutting annual compute cost from **\$2,994 to \$855**, a saving of \$2,139 per instance. Across 30 development and staging databases, that is **\$64,170/year** from scheduled stop/start alone.

```

#!/bin/bash
# EventBridge-triggered Lambda alternative: direct CLI scheduling via cron on EC2 bastion
# For Lambda-based approach, wrap these commands in a Python Lambda function
# triggered by two EventBridge rules: start (07:00 Mon-Fri) and stop (19:00 Mon-Fri)

# Tag dev/staging instances with schedule=business-hours before running

REGION="us-east-1"
ACTION="${1:-stop}"    # pass "start" or "stop"

if [[ "${ACTION}" != "start" && "${ACTION}" != "stop" ]]; then
    echo "Usage: $0 [start|stop]"
    exit 1
fi

```

```
echo "Action: ${ACTION} | Region: ${REGION} | Time: $(date)"

# Find all instances tagged schedule=business-hours
INSTANCE_IDS=$(aws rds describe-db-instances \
  --region "${REGION}" \
  --query "DBInstances[*].DBInstanceIdentifier" \
  --output text)

for INSTANCE_ID in ${INSTANCE_IDS}; do
  SCHEDULE_TAG=$(aws rds list-tags-for-resource \
    --resource-name "arn:aws:rds:${REGION}:${(aws sts get-caller-identity --query Account --output
    ↪ text):db:${INSTANCE_ID}" \
    --query "TagList[?Key=='schedule'].Value" \
    --output text \
    --region "${REGION}")

  if [[ "${SCHEDULE_TAG}" == "business-hours" ]]; then
    STATUS=$(aws rds describe-db-instances \
      --db-instance-identifier "${INSTANCE_ID}" \
      --query "DBInstances[0].DBInstanceStatus" \
      --output text \
      --region "${REGION}")

    if [[ "${ACTION}" == "stop" && "${STATUS}" == "available" ]]; then
      echo "Stopping: ${INSTANCE_ID}"
      aws rds stop-db-instance \
        --db-instance-identifier "${INSTANCE_ID}" \
        --region "${REGION}" \
        --output text
    elif [[ "${ACTION}" == "start" && "${STATUS}" == "stopped" ]]; then
      echo "Starting: ${INSTANCE_ID}"
      aws rds start-db-instance \
        --db-instance-identifier "${INSTANCE_ID}" \
        --region "${REGION}" \
        --output text
    else
      echo "Skipping: ${INSTANCE_ID} (status: ${STATUS})"
    fi
  fi
done

echo "Schedule action '${ACTION}' complete at $(date)"
```

Deploy two EventBridge rules pointing at a Lambda that calls this logic: one firing at 07:00 UTC Monday–Friday to start, one at 19:00 UTC Monday–Friday to stop. Add a third rule at 08:00 Saturday to catch any instances that did not stop Friday evening.

Stage 4 to Stage 5: Autonomous Cost Governance

Stage 5 is not a destination teams reach once — it is a practice they sustain continuously. The defining characteristic is that cost drift is detected and corrected automatically, without requiring a human to notice a bill. Three systems make this work: anomaly detection with automatic remediation, RI lifecycle management, and cost-per-transaction instrumentation embedded in application telemetry.

Anomaly Detection and Automatic Remediation

AWS Cost Anomaly Detection can be configured via CLI and will fire SNS alerts when spend deviates from a learned baseline. But the more precise control comes from building your own

detector against CloudWatch metrics, because CloudWatch catches the infrastructure event that caused the cost spike before the billing system does.

The following Python script runs every 15 minutes via EventBridge Scheduler. It checks five leading indicators of cost overrun: CPU utilization trending above the instance's efficient operating band, FreeStorageSpace falling below the autoscale-trigger threshold, DatabaseConnections exceeding the connection limit (signaling a missing connection pool), ReadIOPS and WriteIOPS approaching the provisioned limit (signaling a pending io1 charge), and ReadLatency spiking above 20ms (signaling a storage bottleneck that will drive a support ticket to upsize storage).

```
#!/usr/bin/env python3
"""
db_cost_anomaly_detector.py
Monitors RDS CloudWatch metrics for leading indicators of cost overrun.
Runs every 15 minutes via EventBridge Scheduler -> Lambda.
On anomaly: publishes to SNS, writes to DynamoDB audit log, optionally
opens a Jira ticket via REST API.

Requires: boto3
Environment variables:
    SNS_TOPIC_ARN      - alert destination
    DYNAMODB_TABLE     - audit log table name
    JIRA_URL           - optional, e.g. https://yourorg.atlassian.net
    JIRA_API_TOKEN     - optional
    JIRA_PROJECT_KEY   - optional, e.g. DBOPS
"""

import boto3
import json
import os
import time
from datetime import datetime, timedelta, timezone
from dataclasses import dataclass, asdict
from typing import Optional
import urllib.request
import urllib.parse
import base64

SNS_TOPIC_ARN = os.environ.get("SNS_TOPIC_ARN", "")
DYNAMO_TABLE = os.environ.get("DYNAMODB_TABLE", "db-finops-anomalies")
JIRA_URL = os.environ.get("JIRA_URL", "")
JIRA_TOKEN = os.environ.get("JIRA_API_TOKEN", "")
JIRA_PROJECT = os.environ.get("JIRA_PROJECT_KEY", "DBOPS")
REGION = os.environ.get("AWS_DEFAULT_REGION", "us-east-1")

THRESHOLDS = {
    "CPUUtilization": {"warning": 70.0, "critical": 85.0, "unit": "%"},
    "FreeStorageSpace": {"warning": 20.0, "critical": 10.0, "unit": "GB", "invert": True},
    "DatabaseConnections": {"warning": 0.80, "critical": 0.90, "unit": "ratio", "dynamic": True},
    "ReadIOPS": {"warning": 0.80, "critical": 0.90, "unit": "ratio", "dynamic": True},
    "WriteIOPS": {"warning": 0.80, "critical": 0.90, "unit": "ratio", "dynamic": True},
    "ReadLatency": {"warning": 0.010, "critical": 0.020, "unit": "s"},
    "WriteLatency": {"warning": 0.010, "critical": 0.020, "unit": "s"},
}

MAX_CONNECTIONS_BY_CLASS = {
    "db.t3.micro": 85,
    "db.t3.small": 170,
    "db.t3.medium": 340,
    "db.t3.large": 680,
    "db.t3.xlarge": 1360,
    "db.t3.2xlarge": 2720,
    "db.m6g.large": 823,
    "db.m6g.xlarge": 1646,
    "db.m6g.2xlarge": 3292,
    "db.m6g.4xlarge": 6584,
}
```

```

        "db.m6g.8xlarge": 13168,
        "db.r6g.large": 1640,
        "db.r6g.xlarge": 3280,
        "db.r6g.2xlarge": 6560,
        "db.r6g.4xlarge": 13120,
        "db.r6g.8xlarge": 26240,
        "db.r6g.16xlarge": 52480,
    }

}

@dataclass
class Anomaly:
    instance_id: str
    instance_class: str
    metric: str
    current_value: float
    threshold: float
    severity: str
    unit: str
    recommendation: str
    timestamp: str
    estimated_monthly_risk_usd: float

def get_rds_instances() -> list[dict]:
    rds = boto3.client("rds", region_name=REGION)
    paginator = rds.get_paginator("describe_db_instances")
    instances = []
    for page in paginator.paginate():
        for db in page["DBInstances"]:
            if db["DBInstanceStatus"] == "available":
                instances.append({
                    "id": db["DBInstanceIdentifier"],
                    "class": db["DBInstanceClass"],
                    "storage_gb": db["AllocatedStorage"],
                    "iops": db.get("Iops", 3000),
                })
    return instances

def get_metric_average(cw, instance_id: str, metric: str,
                      minutes: int = 15) -> Optional[float]:
    end = datetime.now(timezone.utc)
    start = end - timedelta(minutes=minutes)
    resp = cw.get_metric_statistics(
        Namespace="AWS/RDS",
        MetricName=metric,
        Dimensions=[{"Name": "DBInstanceIdentifier", "Value": instance_id}],
        StartTime=start,
        EndTime=end,
        Period=minutes * 60,
        Statistics=["Average"],
    )
    if not resp["Datapoints"]:
        return None
    return resp["Datapoints"][0]["Average"]

def evaluate_instance(cw, instance: dict) -> list[Anomaly]:
    anomalies = []
    inst_id = instance["id"]
    inst_class = instance["class"]
    max_conns = MAX_CONNECTIONS_BY_CLASS.get(inst_class, 1000)
    prov_iops = instance["iops"]

    checks = [
        ("CPUUtilization", None, None),
        ("FreeStorageSpace", None, None),
        ("DatabaseConnections", max_conns, None),
        ("ReadIOPS", prov_iops, None),
    ]

```

```

        ("WriteIOPS",          prov_iops, None),
        ("ReadLatency",       None,       None),
        ("WriteLatency",      None,       None),
    ]

    for metric, cap, _ in checks:
        value = get_metric_average(cw, inst_id, metric)
        if value is None:
            continue

        cfg = THRESHOLDS[metric]

        # Normalize dynamic ratio metrics
        if cfg.get("dynamic") and cap:
            normalized = value / cap
        elif metric == "FreeStorageSpace":
            # Convert bytes to GB for comparison
            normalized = value / (1024 ** 3)
        else:
            normalized = value

        warn_thresh = cfg["warning"]
        crit_thresh = cfg["critical"]
        invert      = cfg.get("invert", False)

        is_warning = (normalized < warn_thresh) if invert else (normalized >= warn_thresh)
        is_critical = (normalized < crit_thresh) if invert else (normalized >= crit_thresh)

        if not is_warning:
            continue

        severity = "CRITICAL" if is_critical else "WARNING"

        recommendations = {
            "CPUUtilization": (
                f"Instance {inst_class} CPU at {normalized:.1f}%. "
                f"Review pg_stat_statements for rogue queries before upsizing. "
                f"Upsizing to next class adds ~40% to compute cost."
            ),
            "FreeStorageSpace": (
                f"Only {normalized:.1f}GB free. Autoscale will trigger, increasing "
                f"storage cost. Archive or purge old data first. "
                f"Each 100GB gp3 expansion adds $9.20/month."
            ),
            "DatabaseConnections": (
                f"Connections at {normalized*100:.0f}% of limit ({int(value)}/{max_conns}). "
                f"Deploy pgBouncer in transaction mode. Typical saving: 60-80% connection "
                f"reduction, enabling downsizing by one instance class."
            ),
            "ReadIOPS": (
                f"Read IOPS at {normalized*100:.0f}% of provisioned ({int(value)}/{prov_iops}). "
                f"Add read replica before provisioning additional IOPS. "
                f"Provisioning 1000 additional IOPS on io1 costs $65/month."
            ),
            "WriteIOPS": (
                f"Write IOPS at {normalized*100:.0f}% of provisioned ({int(value)}/{prov_iops}). "
                f"Check for missing indexes causing full-table writes. "
                f"Provisioning 1000 additional IOPS on io1 costs $65/month."
            ),
            "ReadLatency": (
                f"Read latency {value*1000:.1f}ms exceeds {crit_thresh*1000:.0f}ms threshold. "
                f"Check buffer cache hit ratio and consider Aurora read replicas "
                f"before migrating to io2 storage."
            ),
            "WriteLatency": (
                f"Write latency {value*1000:.1f}ms exceeds threshold. "
                f"Check for lock contention via pg_locks. "
                f"Premature io1 provisioning adds $0.065 per IOPS-month."
            ),
        }
    }

```

```

risk_map = {
    "CPUUtilization": 45.0 * 12,
    "FreeStorageSpace": 9.20 * 12,
    "DatabaseConnections": 120.0,
    "ReadIOPS": 65.0 * 12,
    "WriteIOPS": 65.0 * 12,
    "ReadLatency": 75.0 * 12,
    "WriteLatency": 75.0 * 12,
}

anomalies.append(Anomaly(
    instance_id=inst_id,
    instance_class=inst_class,
    metric=metric,
    current_value=round(normalized, 4),
    threshold=warn_thresh,
    severity=severity,
    unit=cfg["unit"],
    recommendation=recommendations[metric],
    timestamp=datetime.utcnow().isoformat(),
    estimated_monthly_risk_usd=round(risk_map.get(metric, 0) / 12, 2),
))

return anomalies

def publish_anomalies(anomalies: list[Anomaly]) -> None:
    if not anomalies:
        return

    sns = boto3.client("sns", region_name=REGION)
    dynamo = boto3.resource("dynamodb", region_name=REGION)
    table = dynamo.Table(DYNAMO_TABLE)

    total_risk = sum(a.estimated_monthly_risk_usd for a in anomalies)
    critical = [a for a in anomalies if a.severity == "CRITICAL"]

    summary = (
        f"DB FinOps Anomaly Report | {datetime.utcnow().strftime('%Y-%m-%d %H:%M')} UTC\n"
        f"Total anomalies: {len(anomalies)}\n"
        f"({len(critical)} critical)\n"
        f"Estimated monthly cost risk: ${total_risk:,.2f}\n\n"
    )

    for a in anomalies:
        summary += (
            f"[{a.severity}] {a.instance_id} | {a.metric}: {a.current_value} {a.unit}\n"
            f"-> {a.recommendation}\n\n"
        )

    sns.publish(
        TopicArn=SNS_TOPIC_ARN,
        Subject=f"[DB FinOps] {len(critical)} Critical Anomalies | ${total_risk:,.0f}/mo risk",
        Message=summary,
    )

    for a in anomalies:
        table.put_item(Item={
            **asdict(a),
            "ttl": int(time.time()) + (90 * 86400), # 90-day retention
        })

    if JIRA_URL and JIRA_TOKEN and critical:
        open_jira_tickets(critical)

def open_jira_tickets(anomalies: list[Anomaly]) -> None:
    credentials = base64.b64encode(f"automation@yourorg.com:{JIRA_TOKEN}".encode()).decode()
    headers = {
        "Authorization": f"Basic {credentials}",
    }

```

```

        "Content-Type": "application/json",
    }
    for a in anomalies:
        payload = json.dumps({
            "fields": {
                "project": {"key": JIRA_PROJECT},
                "summary": f"[DB FinOps] CRITICAL: {a.instance_id} | {a.metric}",
                "description": {
                    "type": "doc",
                    "version": 1,
                    "content": [{
                        "type": "paragraph",
                        "content": [{"type": "text", "text": a.recommendation}]
                    }]
                },
            },
            "issuetype": {"name": "Task"},
            "priority": {"name": "High"},
            "labels": ["db-finops", "cost-anomaly", a.instance_id],
        })
    req = urllib.request.Request(
        f"{JIRA_URL}/rest/api/3/issue",
        data=payload,
        headers=headers,
        method="POST",
    )
    try:
        with urllib.request.urlopen(req, timeout=10) as resp:
            result = json.loads(resp.read())
            print(f"Jira ticket created: {result.get('key')} for {a.instance_id}")
    except Exception as exc:
        print(f"Jira ticket creation failed for {a.instance_id}: {exc}")

def lambda_handler(event, context):
    """Entry point for AWS Lambda. Also callable directly."""
    cw = boto3.client("cloudwatch", region_name=REGION)
    instances = get_rds_instances()
    print(f"Evaluating {len(instances)} instances across {REGION}")

    all_anomalies: list[Anomaly] = []
    for inst in instances:
        found = evaluate_instance(cw, inst)
        all_anomalies.extend(found)

    print(f"Found {len(all_anomalies)} anomalies")
    publish_anomalies(all_anomalies)
    return {"anomalies": len(all_anomalies)}

if __name__ == "__main__":
    lambda_handler({}, None)

```

Cost-Per-Transaction Instrumentation

Stage 5 teams measure cost at the transaction level, not the resource level. This requires correlating billing data with application throughput metrics. The simplest implementation uses a CloudWatch custom metric published by the application alongside a monthly cost allocation script.

```

#!/bin/bash
# Publish cost-per-transaction metric to CloudWatch
# Run at end of each UTC day via cron: 0 0 * * * /opt/finops/cost_per_txn.sh
#
# Prerequisites:
# - CloudWatch agent publishing db_transactions_total metric from app
# - AWS Cost Explorer API access
# - jq installed

```



```

REGION="us-east-1"
DB_IDENTIFIER="prod-postgres-primary"
NAMESPACE="DatabaseFinOps"
DATE_YESTERDAY=$(date -d "yesterday" +%Y-%m-%d)
DATE_TODAY=$(date +%Y-%m-%d)

echo "Computing cost-per-transaction for ${DATE_YESTERDAY}"

# Get yesterday's transaction count from CloudWatch (published by app)
TXN_COUNT=$(aws cloudwatch get-metric-statistics \
  --region "${REGION}" \
  --namespace "Application/Database" \
  --metric-name "db_transactions_total" \
  --dimensions Name=DBIdentifier,Value="${DB_IDENTIFIER}" \
  --start-time "${DATE_YESTERDAY}T00:00:00Z" \
  --end-time "${DATE_TODAY}T00:00:00Z" \
  --period 86400 \
  --statistics Sum \
  --query "Datapoints[0].Sum" \
  --output text)

# Get yesterday's actual RDS cost from Cost Explorer
DAILY_COST=$(aws ce get-cost-and-usage \
  --time-period Start="${DATE_YESTERDAY}",End="${DATE_TODAY}" \
  --granularity DAILY \
  --filter '{
    "And": [
      {
        "Dimensions": { "Key": "SERVICE", "Values": ["Amazon RDS"] },
        "Tags": { "Key": "application", "Values": ["prod-postgres"] }
      }
    ]
  }' \
  --metrics "UnblendedCost" \
  --query "ResultsByTime[0].Total.UnblendedCost.Amount" \
  --output text \
  --region "${REGION}")

if [[ "${TXN_COUNT}" == "None" || "${TXN_COUNT}" == "0" ]]; then
  echo "No transaction data available for ${DATE_YESTERDAY}. Skipping."
  exit 0
fi

COST_PER_TXN=$(echo "scale=8; ${DAILY_COST} / ${TXN_COUNT}" | bc)

echo "Date: ${DATE_YESTERDAY}"
echo "Transactions: ${TXN_COUNT}"
echo "DB Cost USD: ${DAILY_COST}"
echo "Cost/Txn USD: ${COST_PER_TXN}"

# Publish to CloudWatch as custom metric
aws cloudwatch put-metric-data \
  --region "${REGION}" \
  --namespace "${NAMESPACE}" \
  --metric-data "[
    {
      \"MetricName\": \"CostPerTransaction\",
      \"Dimensions\": [{\"Name\": \"DBIdentifier\", \"Value\": \"${DB_IDENTIFIER}\"}],
      \"Timestamp\": \"${DATE_YESTERDAY}T12:00:00Z\",
      \"Value\": ${COST_PER_TXN},
      \"Unit\": \"None\"
    },
    {
      \"MetricName\": \"DailyDBCostUSD\",
      \"Dimensions\": [{\"Name\": \"DBIdentifier\", \"Value\": \"${DB_IDENTIFIER}\"}],
      \"Timestamp\": \"${DATE_YESTERDAY}T12:00:00Z\",
      \"Value\": ${DAILY_COST},
      \"Unit\": \"None\"
    }
  ]"

echo "Metrics published to CloudWatch namespace ${NAMESPACE}"

```

When **CostPerTransaction** trends upward while transaction volume holds flat, the signal is unambiguous: something changed in the infrastructure or query patterns that is inflating cost without delivering more throughput. That is the leading indicator that triggers investigation before the monthly bill arrives.

Measuring Maturity: The Database FinOps Scorecard

A maturity model without measurement is a philosophy. The scorecard below assigns points across ten dimensions, producing a score from 0 to 100. Score below 40: Stage 1 or 2. Score 40–65: Stage 3. Score 65–85: Stage 4. Score above 85: Stage 5.

Dimension	Max Points	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Tag compliance rate	10	0–2	3–4	5–7	8–9	10
RI/CUD coverage (prod)	15	0	1–4	5–9	10–13	14–15
Env scheduling (non-prod)	10	0	0–2	3–5	6–8	9–10
Storage tier optimization	10	0	0–2	3–5	6–8	9–10
Connection pooling deployed	5	0	0	1–2	3–4	5
Query cost attribution	10	0	1–2	3–5	6–8	9–10
Rightsizing program active	15	0	1–4	5–9	10–13	14–15
Anomaly detection automated	10	0	0–2	3–5	6–8	9–10
Cost-per-transaction tracked	10	0	0	0–3	4–7	8–10
CI/CD cost gates active	5	0	0	0–1	2–3	4–5

Run this scorecard quarterly. Record it in your team wiki. A team that moves from 35 to 65 over 12 months will typically realize 25–35% reduction in database spend during the same period, because the scorecard dimensions are not vanity metrics — each one maps directly to a budget line.

The complete annual savings potential by stage transition, on a fleet of 100 production RDS instances averaging db.r6g.2xlarge with 1TB gp2 storage, Multi-AZ:

- **Stage 1 → Stage 2:** Tagging and visibility — \$0 direct saving, but identifies \$400K+ in misallocated spend
- **Stage 2 → Stage 3:** RI coverage at 80% + gp2→gp3 migration — **\$280,000/year**
- **Stage 3 → Stage 4:** Rightsizing + env scheduling + connection pooling — **\$195,000/year**
- **Stage 4 → Stage 5:** Anomaly prevention + query optimization + cost-per-txn tracking — **\$85,000/year**
- **Total cumulative saving: \$560,000/year on a \$1.4M baseline — a 40% reduction**

These figures use 2024 us-east-1 pricing for AWS RDS. Azure and GCP equivalents land within 8–12% of these numbers given similar fleet composition.

Key Takeaways

- **Maturity is measurable.** The five-stage model — Reactive, Aware, Structured, Optimized, Autonomous — provides a scoring system that converts abstract “cost culture” into a number your team can track quarterly. Organizations at Stage 5 spend 40% less than Stage 1 organizations running identical workloads.
- **The highest-return single action is Reserved Instance coverage.** Moving a fleet of 50 db.r6g.2xlarge instances from On-Demand to 1-year No Upfront RIs saves \$168,192/year with zero architectural change, zero performance risk, and a purchase decision that takes under an hour.
- **Visibility precedes all optimization.** The Cost and Usage Report queried via Athena, Azure Cost Management, and GCP Billing Export to BigQuery are the three instruments every database team must have running before any other technique applies. An untagged resource is an unowned cost.
- **Automation converts one-time savings into permanent savings.** A storage migration script run once saves money this month. The same script running on a weekly schedule, flagging new gp2 volumes before they survive their first billing cycle, saves money every month indefinitely. Every technique in this book should live in a scheduler, not a runbook.
- **Cost-per-transaction is the maturity graduation metric.** When your team can answer “our database costs \$0.00012 per application transaction and that number fell 18% this quarter,” you have completed the journey from reactive bill payers to autonomous infrastructure operators. That number belongs in your sprint retrospectives, your quarterly business reviews, and the budget presentation your CTO gives to the board.

Database FinOps: Cloud Cost Optimization for AWS RDS, Azure SQL, and Google Cloud is the seventh book in the series by Rajesh Guntupalli. The preceding six books cover the full arc of modern database engineering, from foundational DBA practice through AI-native application development. The cost optimization techniques in this volume apply immediately, generate measurable returns within the first billing cycle, and compound in value as the team advances through each maturity stage. The database that runs your business should not also drain it.

